

My Project

0.1

Generated by Doxygen 1.9.1

1 Main Page	1
1.0.1 Procedure for editing any file -	1
2 Hierarchical Index	3
2.1 Class Hierarchy	3
3 Class Index	13
3.1 Class List	13
4 File Index	27
4.1 File List	27
5 Class Documentation	37
5.1 __mavlink_actuator_control_target_t Struct Reference	37
5.2 __mavlink_actuator_output_status_t Struct Reference	37
5.3 __mavlink_adsb_vehicle_t Struct Reference	38
5.4 __mavlink_airspeed_t Struct Reference	38
5.5 __mavlink_ais_vessel_t Struct Reference	38
5.6 __mavlink_altitude_t Struct Reference	39
5.7 __mavlink_att_pos_mocap_t Struct Reference	39
5.8 __mavlink_attitude_quaternion_cov_t Struct Reference	40
5.9 __mavlink_attitude_quaternion_t Struct Reference	40
5.10 __mavlink_attitude_t Struct Reference	40
5.11 __mavlink_attitude_target_t Struct Reference	41
5.12 __mavlink_auth_key_t Struct Reference	41
5.13 __mavlink_autopilot_version_t Struct Reference	41
5.14 __mavlink_available_modes_monitor_t Struct Reference	42
5.15 __mavlink_available_modes_t Struct Reference	42
5.16 __mavlink_battery_info_t Struct Reference	42
5.17 __mavlink_bitfield Union Reference	43
5.18 __mavlink_button_change_t Struct Reference	43
5.19 __mavlink_camera_fov_status_t Struct Reference	43
5.20 __mavlink_camera_image_captured_t Struct Reference	44
5.21 __mavlink_camera_information_t Struct Reference	44
5.22 __mavlink_camera_thermal_range_t Struct Reference	45
5.23 __mavlink_camera_tracking_geo_status_t Struct Reference	45
5.24 __mavlink_camera_tracking_image_status_t Struct Reference	46
5.25 __mavlink_camera_trigger_t Struct Reference	46
5.26 __mavlink_can_filter_modify_t Struct Reference	46
5.27 __mavlink_can_frame_t Struct Reference	47
5.28 __mavlink_canfd_frame_t Struct Reference	47
5.29 __mavlink_cellular_config_t Struct Reference	47
5.30 __mavlink_cellular_status_t Struct Reference	48
5.31 __mavlink_change_operator_control_ack_t Struct Reference	48

5.32	__mavlink_change_operator_control_t Struct Reference	48
5.33	__mavlink_collision_t Struct Reference	49
5.34	__mavlink_command_ack_t Struct Reference	49
5.35	__mavlink_command_cancel_t Struct Reference	49
5.36	__mavlink_command_int_t Struct Reference	50
5.37	__mavlink_command_long_t Struct Reference	50
5.38	__mavlink_component_information_basic_t Struct Reference	51
5.39	__mavlink_component_information_t Struct Reference	51
5.40	__mavlink_component_metadata_t Struct Reference	51
5.41	__mavlink_control_system_state_t Struct Reference	52
5.42	__mavlink_current_event_sequence_t Struct Reference	52
5.43	__mavlink_current_mode_t Struct Reference	52
5.44	__mavlink_data_stream_t Struct Reference	53
5.45	__mavlink_data_transmission_handshake_t Struct Reference	53
5.46	__mavlink_debug_float_array_t Struct Reference	53
5.47	__mavlink_debug_t Struct Reference	54
5.48	__mavlink_debug_vect_t Struct Reference	54
5.49	__mavlink_encapsulated_data_t Struct Reference	54
5.50	__mavlink_esc_info_t Struct Reference	55
5.51	__mavlink_esc_status_t Struct Reference	55
5.52	__mavlink_estimator_status_t Struct Reference	55
5.53	__mavlink_event_t Struct Reference	56
5.54	__mavlink_extended_sys_state_t Struct Reference	56
5.55	__mavlink_fence_status_t Struct Reference	56
5.56	__mavlink_field_info Struct Reference	57
5.57	__mavlink_figure_eight_execution_status_t Struct Reference	57
5.58	__mavlink_file_transfer_protocol_t Struct Reference	57
5.59	__mavlink_flight_information_t Struct Reference	58
5.60	__mavlink_follow_target_t Struct Reference	58
5.61	__mavlink_fuel_status_t Struct Reference	58
5.62	__mavlink_generator_status_t Struct Reference	59
5.63	__mavlink_gimbal_device_attitude_status_t Struct Reference	59
5.64	__mavlink_gimbal_device_information_t Struct Reference	60
5.65	__mavlink_gimbal_device_set_attitude_t Struct Reference	60
5.66	__mavlink_gimbal_manager_information_t Struct Reference	61
5.67	__mavlink_gimbal_manager_set_attitude_t Struct Reference	61
5.68	__mavlink_gimbal_manager_set_manual_control_t Struct Reference	61
5.69	__mavlink_gimbal_manager_set_pitchyaw_t Struct Reference	62
5.70	__mavlink_gimbal_manager_status_t Struct Reference	62
5.71	__mavlink_global_position_int_cov_t Struct Reference	62
5.72	__mavlink_global_position_int_t Struct Reference	63
5.73	__mavlink_global_vision_position_estimate_t Struct Reference	63

5.74 __mavlink_gps2_rtk_t Struct Reference	63
5.75 __mavlink_gps_inject_data_t Struct Reference	64
5.76 __mavlink_gps_rtcn_data_t Struct Reference	64
5.77 __mavlink_gps_rtk_t Struct Reference	64
5.78 __mavlink_gps_status_t Struct Reference	65
5.79 __mavlink_heartbeat_t Struct Reference	65
5.80 __mavlink_high_latency2_t Struct Reference	65
5.81 __mavlink_high_latency_t Struct Reference	66
5.82 __mavlink_highres_imu_t Struct Reference	67
5.83 __mavlink_hil_actuator_controls_t Struct Reference	67
5.84 __mavlink_hil_controls_t Struct Reference	67
5.85 __mavlink_hil_optical_flow_t Struct Reference	68
5.86 __mavlink_hil_rc_inputs_raw_t Struct Reference	68
5.87 __mavlink_hil_sensor_t Struct Reference	69
5.88 __mavlink_hil_state_quaternion_t Struct Reference	69
5.89 __mavlink_hil_state_t Struct Reference	70
5.90 __mavlink_hygrometer_sensor_t Struct Reference	70
5.91 __mavlink_illuminator_status_t Struct Reference	70
5.92 __mavlink_isbd_link_status_t Struct Reference	71
5.93 __mavlink_link_node_status_t Struct Reference	71
5.94 __mavlink_local_position_ned_cov_t Struct Reference	71
5.95 __mavlink_local_position_ned_system_global_offset_t Struct Reference	72
5.96 __mavlink_local_position_ned_t Struct Reference	72
5.97 __mavlink_log_data_t Struct Reference	72
5.98 __mavlink_log_entry_t Struct Reference	73
5.99 __mavlink_log_erase_t Struct Reference	73
5.100 __mavlink_log_request_data_t Struct Reference	73
5.101 __mavlink_log_request_end_t Struct Reference	73
5.102 __mavlink_log_request_list_t Struct Reference	74
5.103 __mavlink_logging_ack_t Struct Reference	74
5.104 __mavlink_logging_data_acked_t Struct Reference	74
5.105 __mavlink_logging_data_t Struct Reference	75
5.106 __mavlink_manual_setpoint_t Struct Reference	75
5.107 __mavlink_memory_vect_t Struct Reference	75
5.108 __mavlink_message_info Struct Reference	76
5.109 __mavlink_message_interval_t Struct Reference	76
5.110 __mavlink_mission_ack_t Struct Reference	76
5.111 __mavlink_mission_clear_all_t Struct Reference	77
5.112 __mavlink_mission_item_int_t Struct Reference	77
5.113 __mavlink_mission_item_reached_t Struct Reference	77
5.114 __mavlink_mission_item_t Struct Reference	78
5.115 __mavlink_mission_request_int_t Struct Reference	78

5.116	__mavlink_mission_request_list_t Struct Reference	78
5.117	__mavlink_mission_request_partial_list_t Struct Reference	79
5.118	__mavlink_mission_request_t Struct Reference	79
5.119	__mavlink_mission_set_current_t Struct Reference	79
5.120	__mavlink_mission_write_partial_list_t Struct Reference	80
5.121	__mavlink_mount_orientation_t Struct Reference	80
5.122	__mavlink_msg_entry Struct Reference	80
5.123	__mavlink_named_value_float_t Struct Reference	81
5.124	__mavlink_named_value_int_t Struct Reference	81
5.125	__mavlink_nav_controller_output_t Struct Reference	81
5.126	__mavlink_odometry_t Struct Reference	82
5.127	__mavlink_onboard_computer_status_t Struct Reference	82
5.128	__mavlink_open_drone_id_arm_status_t Struct Reference	83
5.129	__mavlink_open_drone_id_authentication_t Struct Reference	83
5.130	__mavlink_open_drone_id_basic_id_t Struct Reference	83
5.131	__mavlink_open_drone_id_location_t Struct Reference	84
5.132	__mavlink_open_drone_id_message_pack_t Struct Reference	84
5.133	__mavlink_open_drone_id_operator_id_t Struct Reference	85
5.134	__mavlink_open_drone_id_self_id_t Struct Reference	85
5.135	__mavlink_open_drone_id_system_t Struct Reference	85
5.136	__mavlink_open_drone_id_system_update_t Struct Reference	86
5.137	__mavlink_optical_flow_rad_t Struct Reference	86
5.138	__mavlink_orbit_execution_status_t Struct Reference	86
5.139	__mavlink_param_error_t Struct Reference	87
5.140	__mavlink_param_ext_ack_t Struct Reference	87
5.141	__mavlink_param_ext_request_list_t Struct Reference	87
5.142	__mavlink_param_ext_request_read_t Struct Reference	87
5.143	__mavlink_param_ext_set_t Struct Reference	88
5.144	__mavlink_param_ext_value_t Struct Reference	88
5.145	__mavlink_param_map_rc_t Struct Reference	88
5.146	__mavlink_param_request_list_t Struct Reference	89
5.147	__mavlink_param_request_read_t Struct Reference	89
5.148	__mavlink_param_set_t Struct Reference	89
5.149	__mavlink_param_value_t Struct Reference	89
5.150	__mavlink_ping_t Struct Reference	90
5.151	__mavlink_play_tune_t Struct Reference	90
5.152	__mavlink_play_tune_v2_t Struct Reference	90
5.153	__mavlink_position_target_global_int_t Struct Reference	91
5.154	__mavlink_position_target_local_ned_t Struct Reference	91
5.155	__mavlink_power_status_t Struct Reference	92
5.156	__mavlink_protocol_version_t Struct Reference	92
5.157	__mavlink_radio_status_t Struct Reference	92

5.158	__mavlink_raw_pressure_t Struct Reference	93
5.159	__mavlink_raw_rpm_t Struct Reference	93
5.160	__mavlink_rc_channels_override_t Struct Reference	93
5.161	__mavlink_rc_channels_raw_t Struct Reference	94
5.162	__mavlink_rc_channels_scaled_t Struct Reference	94
5.163	__mavlink_rc_channels_t Struct Reference	95
5.164	__mavlink_relay_status_t Struct Reference	95
5.165	__mavlink_request_data_stream_t Struct Reference	96
5.166	__mavlink_request_event_t Struct Reference	96
5.167	__mavlink_resource_request_t Struct Reference	96
5.168	__mavlink_response_event_error_t Struct Reference	97
5.169	__mavlink_safety_allowed_area_t Struct Reference	97
5.170	__mavlink_safety_set_allowed_area_t Struct Reference	97
5.171	__mavlink_scaled_imu2_t Struct Reference	98
5.172	__mavlink_scaled_imu3_t Struct Reference	98
5.173	__mavlink_scaled_imu_t Struct Reference	99
5.174	__mavlink_scaled_pressure2_t Struct Reference	99
5.175	__mavlink_scaled_pressure3_t Struct Reference	99
5.176	__mavlink_scaled_pressure_t Struct Reference	100
5.177	__mavlink_serial_control_t Struct Reference	100
5.178	__mavlink_set_actuator_control_target_t Struct Reference	100
5.179	__mavlink_set_mode_t Struct Reference	101
5.180	__mavlink_set_position_target_global_int_t Struct Reference	101
5.181	__mavlink_set_position_target_local_ned_t Struct Reference	102
5.182	__mavlink_setup_signing_t Struct Reference	102
5.183	__mavlink_signing Struct Reference	103
5.184	__mavlink_signing_streams::__mavlink_signing_stream Struct Reference	103
5.185	__mavlink_signing_streams Struct Reference	104
5.186	__mavlink_sim_state_t Struct Reference	105
5.187	__mavlink_status Struct Reference	106
5.188	__mavlink_storage_information_t Struct Reference	107
5.189	__mavlink_supported_tunes_t Struct Reference	107
5.190	__mavlink_system_time_t Struct Reference	108
5.191	__mavlink_terrain_check_t Struct Reference	108
5.192	__mavlink_terrain_data_t Struct Reference	108
5.193	__mavlink_terrain_report_t Struct Reference	108
5.194	__mavlink_terrain_request_t Struct Reference	109
5.195	__mavlink_time_estimate_to_target_t Struct Reference	109
5.196	__mavlink_timesync_t Struct Reference	109
5.197	__mavlink_trajectory_representation_bezier_t Struct Reference	110
5.198	__mavlink_trajectory_representation_waypoints_t Struct Reference	110
5.199	__mavlink_tunnel_t Struct Reference	110

5.200	__mavlink_uavcan_node_info_t Struct Reference	111
5.201	__mavlink_uavcan_node_status_t Struct Reference	111
5.202	__mavlink_utm_global_position_t Struct Reference	111
5.203	__mavlink_v2_extension_t Struct Reference	112
5.204	__mavlink_vfr_hud_t Struct Reference	112
5.205	__mavlink_vibration_t Struct Reference	113
5.206	__mavlink_vicon_position_estimate_t Struct Reference	113
5.207	__mavlink_video_stream_information_t Struct Reference	113
5.208	__mavlink_video_stream_status_t Struct Reference	114
5.209	__mavlink_vision_position_estimate_t Struct Reference	114
5.210	__mavlink_vision_speed_estimate_t Struct Reference	114
5.211	__mavlink_wheel_distance_t Struct Reference	115
5.212	__mavlink_wifi_config_ap_t Struct Reference	115
5.213	__mavlink_winch_status_t Struct Reference	115
5.214	__mavlink_wind_cov_t Struct Reference	116
5.215	mavlink::common::msg::ACTUATOR_CONTROL_TARGET Struct Reference	116
5.215.1	Detailed Description	117
5.215.2	Member Function Documentation	118
5.215.2.1	deserialize()	118
5.215.2.2	get_name()	118
5.215.2.3	serialize()	118
5.216	mavlink::common::msg::ACTUATOR_OUTPUT_STATUS Struct Reference	119
5.216.1	Detailed Description	120
5.216.2	Member Function Documentation	120
5.216.2.1	deserialize()	120
5.216.2.2	get_name()	120
5.216.2.3	serialize()	121
5.217	mavlink::common::msg::ADSB_VEHICLE Struct Reference	121
5.217.1	Detailed Description	122
5.217.2	Member Function Documentation	122
5.217.2.1	deserialize()	122
5.217.2.2	get_name()	123
5.217.2.3	serialize()	123
5.218	mavlink::common::msg::AIRSPEED Struct Reference	123
5.218.1	Detailed Description	125
5.218.2	Member Function Documentation	125
5.218.2.1	deserialize()	125
5.218.2.2	get_name()	125
5.218.2.3	serialize()	125
5.219	mavlink::common::msg::AIS_VESSEL Struct Reference	126
5.219.1	Detailed Description	127
5.219.2	Member Function Documentation	127

5.219.2.1 deserialize()	128
5.219.2.2 get_name()	129
5.219.2.3 serialize()	129
5.220 mavlink::common::msg::ALTITUDE Struct Reference	129
5.220.1 Detailed Description	131
5.220.2 Member Function Documentation	131
5.220.2.1 deserialize()	131
5.220.2.2 get_name()	131
5.220.2.3 serialize()	132
5.221 ATLAS_HC_ARM_DISARM_ACK Struct Reference	132
5.222 ATLAS_HC_HEARTBEAT_BC Struct Reference	132
5.223 ATLAS_HC_SYS_STAT Struct Reference	133
5.224 ATLAS_HC_TIMESYNC_RESP Struct Reference	133
5.225 mavlink::common::msg::ATT_POS_MOCAP Struct Reference	133
5.225.1 Detailed Description	134
5.225.2 Member Function Documentation	135
5.225.2.1 deserialize()	135
5.225.2.2 get_name()	135
5.225.2.3 serialize()	135
5.226 mavlink::common::msg::ATTITUDE Struct Reference	136
5.226.1 Detailed Description	137
5.226.2 Member Function Documentation	137
5.226.2.1 deserialize()	137
5.226.2.2 get_name()	137
5.226.2.3 serialize()	138
5.227 mavlink::common::msg::ATTITUDE_QUATERNION Struct Reference	138
5.227.1 Detailed Description	140
5.227.2 Member Function Documentation	140
5.227.2.1 deserialize()	140
5.227.2.2 get_name()	140
5.227.2.3 serialize()	140
5.228 mavlink::common::msg::ATTITUDE_QUATERNION_COV Struct Reference	141
5.228.1 Detailed Description	142
5.228.2 Member Function Documentation	142
5.228.2.1 deserialize()	142
5.228.2.2 get_name()	143
5.228.2.3 serialize()	143
5.229 mavlink::common::msg::ATTITUDE_TARGET Struct Reference	143
5.229.1 Detailed Description	145
5.229.2 Member Function Documentation	145
5.229.2.1 deserialize()	145
5.229.2.2 get_name()	145

5.229.2.3 serialize()	146
5.230 mavlink::common::msg::AUTH_KEY Struct Reference	146
5.230.1 Detailed Description	147
5.230.2 Member Function Documentation	147
5.230.2.1 deserialize()	147
5.230.2.2 get_name()	148
5.230.2.3 serialize()	148
5.231 mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE Struct Reference	148
5.231.1 Detailed Description	150
5.231.2 Member Function Documentation	150
5.231.2.1 deserialize()	150
5.231.2.2 get_name()	151
5.231.2.3 serialize()	151
5.232 mavlink::standard::msg::AUTOPILOT_VERSION Struct Reference	151
5.232.1 Detailed Description	153
5.232.2 Member Function Documentation	153
5.232.2.1 deserialize()	153
5.232.2.2 get_name()	153
5.232.2.3 serialize()	154
5.233 mavlink::common::msg::AVAILABLE_MODES Struct Reference	154
5.233.1 Detailed Description	156
5.233.2 Member Function Documentation	156
5.233.2.1 deserialize()	156
5.233.2.2 get_name()	156
5.233.2.3 serialize()	156
5.234 mavlink::common::msg::AVAILABLE_MODES_MONITOR Struct Reference	157
5.234.1 Detailed Description	158
5.234.2 Member Function Documentation	158
5.234.2.1 deserialize()	158
5.234.2.2 get_name()	159
5.234.2.3 serialize()	159
5.235 mavlink::common::msg::BATTERY_INFO Struct Reference	159
5.235.1 Detailed Description	161
5.235.2 Member Function Documentation	161
5.235.2.1 deserialize()	161
5.235.2.2 get_name()	162
5.235.2.3 serialize()	162
5.236 mavlink::common::msg::BATTERY_STATUS Struct Reference	162
5.236.1 Detailed Description	164
5.236.2 Member Function Documentation	164
5.236.2.1 deserialize()	164
5.236.2.2 get_name()	165

5.236.2.3 serialize()	165
5.237 mavlink::common::msg::BUTTON_CHANGE Struct Reference	165
5.237.1 Detailed Description	166
5.237.2 Member Function Documentation	167
5.237.2.1 deserialize()	167
5.237.2.2 get_name()	167
5.237.2.3 serialize()	167
5.238 mavlink::common::msg::CAMERA_CAPTURE_STATUS Struct Reference	168
5.238.1 Detailed Description	169
5.238.2 Member Function Documentation	169
5.238.2.1 deserialize()	169
5.238.2.2 get_name()	169
5.238.2.3 serialize()	170
5.239 mavlink::common::msg::CAMERA_FOV_STATUS Struct Reference	170
5.239.1 Detailed Description	172
5.239.2 Member Function Documentation	172
5.239.2.1 deserialize()	172
5.239.2.2 get_name()	172
5.239.2.3 serialize()	172
5.240 mavlink::common::msg::CAMERA_IMAGE_CAPTURED Struct Reference	173
5.240.1 Detailed Description	174
5.240.2 Member Function Documentation	174
5.240.2.1 deserialize()	175
5.240.2.2 get_name()	176
5.240.2.3 serialize()	176
5.241 mavlink::common::msg::CAMERA_INFORMATION Struct Reference	176
5.241.1 Detailed Description	178
5.241.2 Member Function Documentation	178
5.241.2.1 deserialize()	178
5.241.2.2 get_name()	179
5.241.2.3 serialize()	179
5.242 mavlink::common::msg::CAMERA_SETTINGS Struct Reference	179
5.242.1 Detailed Description	180
5.242.2 Member Function Documentation	181
5.242.2.1 deserialize()	181
5.242.2.2 get_name()	181
5.242.2.3 serialize()	181
5.243 mavlink::common::msg::CAMERA_THERMAL_RANGE Struct Reference	182
5.243.1 Detailed Description	183
5.243.2 Member Function Documentation	183
5.243.2.1 deserialize()	183
5.243.2.2 get_name()	184

5.243.2.3 serialize()	184
5.244 mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS Struct Reference	184
5.244.1 Detailed Description	186
5.244.2 Member Function Documentation	186
5.244.2.1 deserialize()	186
5.244.2.2 get_name()	186
5.244.2.3 serialize()	186
5.245 mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS Struct Reference	187
5.245.1 Detailed Description	188
5.245.2 Member Function Documentation	188
5.245.2.1 deserialize()	188
5.245.2.2 get_name()	189
5.245.2.3 serialize()	189
5.246 mavlink::common::msg::CAMERA_TRIGGER Struct Reference	189
5.246.1 Detailed Description	191
5.246.2 Member Function Documentation	191
5.246.2.1 deserialize()	191
5.246.2.2 get_name()	191
5.246.2.3 serialize()	191
5.247 mavlink::common::msg::CAN_FILTER_MODIFY Struct Reference	192
5.247.1 Detailed Description	193
5.247.2 Member Function Documentation	193
5.247.2.1 deserialize()	193
5.247.2.2 get_name()	194
5.247.2.3 serialize()	194
5.248 mavlink::common::msg::CAN_FRAME Struct Reference	194
5.248.1 Detailed Description	196
5.248.2 Member Function Documentation	196
5.248.2.1 deserialize()	196
5.248.2.2 get_name()	196
5.248.2.3 serialize()	196
5.249 mavlink::common::msg::CANFD_FRAME Struct Reference	197
5.249.1 Detailed Description	198
5.249.2 Member Function Documentation	198
5.249.2.1 deserialize()	198
5.249.2.2 get_name()	199
5.249.2.3 serialize()	199
5.250 mavlink::common::msg::CELLULAR_CONFIG Struct Reference	199
5.250.1 Detailed Description	201
5.250.2 Member Function Documentation	201
5.250.2.1 deserialize()	201
5.250.2.2 get_name()	201

5.250.2.3 serialize()	202
5.251 mavlink::common::msg::CELLULAR_STATUS Struct Reference	202
5.251.1 Detailed Description	204
5.251.2 Member Function Documentation	204
5.251.2.1 deserialize()	204
5.251.2.2 get_name()	204
5.251.2.3 serialize()	204
5.252 mavlink::common::msg::CHANGE_OPERATOR_CONTROL Struct Reference	205
5.252.1 Detailed Description	206
5.252.2 Member Function Documentation	206
5.252.2.1 deserialize()	206
5.252.2.2 get_name()	207
5.252.2.3 serialize()	207
5.253 mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK Struct Reference	207
5.253.1 Detailed Description	209
5.253.2 Member Function Documentation	209
5.253.2.1 deserialize()	209
5.253.2.2 get_name()	209
5.253.2.3 serialize()	209
5.254 mavlink::common::msg::COLLISION Struct Reference	210
5.254.1 Detailed Description	211
5.254.2 Member Function Documentation	211
5.254.2.1 deserialize()	211
5.254.2.2 get_name()	212
5.254.2.3 serialize()	212
5.255 mavlink::common::msg::COMMAND_ACK Struct Reference	212
5.255.1 Detailed Description	214
5.255.2 Member Function Documentation	214
5.255.2.1 deserialize()	214
5.255.2.2 get_name()	214
5.255.2.3 serialize()	214
5.256 mavlink::common::msg::COMMAND_CANCEL Struct Reference	215
5.256.1 Detailed Description	216
5.256.2 Member Function Documentation	216
5.256.2.1 deserialize()	216
5.256.2.2 get_name()	217
5.256.2.3 serialize()	217
5.257 mavlink::common::msg::COMMAND_INT Struct Reference	217
5.257.1 Detailed Description	219
5.257.2 Member Function Documentation	219
5.257.2.1 deserialize()	219
5.257.2.2 get_name()	220

5.257.2.3 serialize()	220
5.258 mavlink::common::msg::COMMAND_LONG Struct Reference	220
5.258.1 Detailed Description	222
5.258.2 Member Function Documentation	222
5.258.2.1 deserialize()	222
5.258.2.2 get_name()	223
5.258.2.3 serialize()	223
5.259 mavlink::common::msg::COMPONENT_INFORMATION Struct Reference	223
5.259.1 Detailed Description	224
5.259.2 Member Function Documentation	225
5.259.2.1 deserialize()	225
5.259.2.2 get_name()	225
5.259.2.3 serialize()	225
5.260 mavlink::common::msg::COMPONENT_INFORMATION_BASIC Struct Reference	226
5.260.1 Detailed Description	227
5.260.2 Member Function Documentation	227
5.260.2.1 deserialize()	227
5.260.2.2 get_name()	227
5.260.2.3 serialize()	228
5.261 mavlink::common::msg::COMPONENT_METADATA Struct Reference	228
5.261.1 Detailed Description	229
5.261.2 Member Function Documentation	230
5.261.2.1 deserialize()	230
5.261.2.2 get_name()	230
5.261.2.3 serialize()	230
5.262 mavlink::common::msg::CONTROL_SYSTEM_STATE Struct Reference	231
5.262.1 Detailed Description	232
5.262.2 Member Function Documentation	232
5.262.2.1 deserialize()	232
5.262.2.2 get_name()	233
5.262.2.3 serialize()	233
5.263 mavlink::common::msg::CURRENT_EVENT_SEQUENCE Struct Reference	233
5.263.1 Detailed Description	235
5.263.2 Member Function Documentation	235
5.263.2.1 deserialize()	235
5.263.2.2 get_name()	235
5.263.2.3 serialize()	235
5.264 mavlink::common::msg::CURRENT_MODE Struct Reference	236
5.264.1 Detailed Description	237
5.264.2 Member Function Documentation	237
5.264.2.1 deserialize()	237
5.264.2.2 get_name()	238

5.264.2.3 serialize()	238
5.265 mavlink::common::msg::DATA_STREAM Struct Reference	238
5.265.1 Detailed Description	240
5.265.2 Member Function Documentation	240
5.265.2.1 deserialize()	240
5.265.2.2 get_name()	240
5.265.2.3 serialize()	240
5.266 mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE Struct Reference	241
5.266.1 Detailed Description	242
5.266.2 Member Function Documentation	242
5.266.2.1 deserialize()	242
5.266.2.2 get_name()	243
5.266.2.3 serialize()	243
5.267 mavlink::common::msg::DEBUG Struct Reference	243
5.267.1 Detailed Description	245
5.267.2 Member Function Documentation	245
5.267.2.1 deserialize()	245
5.267.2.2 get_name()	245
5.267.2.3 serialize()	245
5.268 mavlink::common::msg::DEBUG_FLOAT_ARRAY Struct Reference	246
5.268.1 Detailed Description	247
5.268.2 Member Function Documentation	247
5.268.2.1 deserialize()	247
5.268.2.2 get_name()	248
5.268.2.3 serialize()	248
5.269 mavlink::common::msg::DEBUG_VECT Struct Reference	248
5.269.1 Detailed Description	250
5.269.2 Member Function Documentation	250
5.269.2.1 deserialize()	250
5.269.2.2 get_name()	250
5.269.2.3 serialize()	250
5.270 mavlink::common::msg::DISTANCE_SENSOR Struct Reference	251
5.270.1 Detailed Description	252
5.270.2 Member Function Documentation	252
5.270.2.1 deserialize()	252
5.270.2.2 get_name()	253
5.270.2.3 serialize()	253
5.271 mavlink::common::msg::EFI_STATUS Struct Reference	253
5.271.1 Detailed Description	255
5.271.2 Member Function Documentation	255
5.271.2.1 deserialize()	255
5.271.2.2 get_name()	256

5.271.2.3 serialize()	256
5.272 mavlink::common::msg::ENCAPSULATED_DATA Struct Reference	256
5.272.1 Detailed Description	258
5.272.2 Member Function Documentation	258
5.272.2.1 deserialize()	258
5.272.2.2 get_name()	258
5.272.2.3 serialize()	258
5.273 mavlink::common::msg::ESC_INFO Struct Reference	259
5.273.1 Detailed Description	260
5.273.2 Member Function Documentation	260
5.273.2.1 deserialize()	260
5.273.2.2 get_name()	261
5.273.2.3 serialize()	261
5.274 mavlink::common::msg::ESC_STATUS Struct Reference	261
5.274.1 Detailed Description	263
5.274.2 Member Function Documentation	263
5.274.2.1 deserialize()	263
5.274.2.2 get_name()	263
5.274.2.3 serialize()	264
5.275 mavlink::common::msg::ESTIMATOR_STATUS Struct Reference	265
5.275.1 Detailed Description	266
5.275.2 Member Function Documentation	266
5.275.2.1 deserialize()	267
5.275.2.2 get_name()	268
5.275.2.3 serialize()	268
5.276 mavlink::common::msg::EVENT Struct Reference	268
5.276.1 Detailed Description	270
5.276.2 Member Function Documentation	270
5.276.2.1 deserialize()	270
5.276.2.2 get_name()	270
5.276.2.3 serialize()	271
5.277 mavlink::common::msg::EXTENDED_SYS_STATE Struct Reference	271
5.277.1 Detailed Description	272
5.277.2 Member Function Documentation	273
5.277.2.1 deserialize()	273
5.277.2.2 get_name()	273
5.277.2.3 serialize()	273
5.278 mavlink::common::msg::FENCE_STATUS Struct Reference	274
5.278.1 Detailed Description	275
5.278.2 Member Function Documentation	275
5.278.2.1 deserialize()	275
5.278.2.2 get_name()	275

5.278.2.3 serialize()	276
5.279 mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS Struct Reference	276
5.279.1 Detailed Description	278
5.279.2 Member Function Documentation	278
5.279.2.1 deserialize()	278
5.279.2.2 get_name()	278
5.279.2.3 serialize()	278
5.280 mavlink::common::msg::FILE_TRANSFER_PROTOCOL Struct Reference	279
5.280.1 Detailed Description	280
5.280.2 Member Function Documentation	280
5.280.2.1 deserialize()	280
5.280.2.2 get_name()	281
5.280.2.3 serialize()	281
5.281 mavlink::common::msg::FLIGHT_INFORMATION Struct Reference	281
5.281.1 Detailed Description	283
5.281.2 Member Function Documentation	283
5.281.2.1 deserialize()	283
5.281.2.2 get_name()	283
5.281.2.3 serialize()	284
5.282 mavlink::common::msg::FOLLOW_TARGET Struct Reference	285
5.282.1 Detailed Description	286
5.282.2 Member Function Documentation	286
5.282.2.1 deserialize()	286
5.282.2.2 get_name()	287
5.282.2.3 serialize()	287
5.283 mavlink::common::msg::FUEL_STATUS Struct Reference	287
5.283.1 Detailed Description	289
5.283.2 Member Function Documentation	289
5.283.2.1 deserialize()	289
5.283.2.2 get_name()	290
5.283.2.3 serialize()	290
5.284 mavlink::common::msg::GENERATOR_STATUS Struct Reference	290
5.284.1 Detailed Description	292
5.284.2 Member Function Documentation	292
5.284.2.1 deserialize()	292
5.284.2.2 get_name()	292
5.284.2.3 serialize()	293
5.285 mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS Struct Reference	293
5.285.1 Detailed Description	295
5.285.2 Member Function Documentation	295
5.285.2.1 deserialize()	295
5.285.2.2 get_name()	295

5.285.2.3 serialize()	296
5.286 mavlink::common::msg::GIMBAL_DEVICE_INFORMATION Struct Reference	296
5.286.1 Detailed Description	298
5.286.2 Member Function Documentation	298
5.286.2.1 deserialize()	298
5.286.2.2 get_name()	298
5.286.2.3 serialize()	299
5.287 mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE Struct Reference	300
5.287.1 Detailed Description	301
5.287.2 Member Function Documentation	301
5.287.2.1 deserialize()	302
5.287.2.2 get_name()	303
5.287.2.3 serialize()	303
5.288 mavlink::common::msg::GIMBAL_MANAGER_INFORMATION Struct Reference	303
5.288.1 Detailed Description	305
5.288.2 Member Function Documentation	305
5.288.2.1 deserialize()	305
5.288.2.2 get_name()	305
5.288.2.3 serialize()	306
5.289 mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE Struct Reference	306
5.289.1 Detailed Description	308
5.289.2 Member Function Documentation	308
5.289.2.1 deserialize()	308
5.289.2.2 get_name()	308
5.289.2.3 serialize()	308
5.290 mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL Struct Reference	309
5.290.1 Detailed Description	310
5.290.2 Member Function Documentation	310
5.290.2.1 deserialize()	310
5.290.2.2 get_name()	311
5.290.2.3 serialize()	311
5.291 mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW Struct Reference	311
5.291.1 Detailed Description	313
5.291.2 Member Function Documentation	313
5.291.2.1 deserialize()	313
5.291.2.2 get_name()	314
5.291.2.3 serialize()	314
5.292 mavlink::common::msg::GIMBAL_MANAGER_STATUS Struct Reference	314
5.292.1 Detailed Description	316
5.292.2 Member Function Documentation	316
5.292.2.1 deserialize()	316
5.292.2.2 get_name()	316

5.292.2.3 serialize()	316
5.293 mavlink::standard::msg::GLOBAL_POSITION_INT Struct Reference	317
5.293.1 Detailed Description	318
5.293.2 Member Function Documentation	318
5.293.2.1 deserialize()	318
5.293.2.2 get_name()	319
5.293.2.3 serialize()	319
5.294 mavlink::common::msg::GLOBAL_POSITION_INT_COV Struct Reference	319
5.294.1 Detailed Description	321
5.294.2 Member Function Documentation	321
5.294.2.1 deserialize()	321
5.294.2.2 get_name()	321
5.294.2.3 serialize()	322
5.295 mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE Struct Reference	322
5.295.1 Detailed Description	324
5.295.2 Member Function Documentation	324
5.295.2.1 deserialize()	324
5.295.2.2 get_name()	324
5.295.2.3 serialize()	324
5.296 mavlink::common::msg::GPS2_RAW Struct Reference	325
5.296.1 Detailed Description	326
5.296.2 Member Function Documentation	327
5.296.2.1 deserialize()	327
5.296.2.2 get_name()	327
5.296.2.3 serialize()	327
5.297 mavlink::common::msg::GPS2_RTK Struct Reference	328
5.297.1 Detailed Description	329
5.297.2 Member Function Documentation	329
5.297.2.1 deserialize()	329
5.297.2.2 get_name()	330
5.297.2.3 serialize()	330
5.298 mavlink::common::msg::GPS_GLOBAL_ORIGIN Struct Reference	330
5.298.1 Detailed Description	331
5.298.2 Member Function Documentation	332
5.298.2.1 deserialize()	332
5.298.2.2 get_name()	332
5.298.2.3 serialize()	332
5.299 mavlink::common::msg::GPS_INJECT_DATA Struct Reference	333
5.299.1 Detailed Description	334
5.299.2 Member Function Documentation	334
5.299.2.1 deserialize()	334
5.299.2.2 get_name()	334

5.299.2.3 serialize()	335
5.300 mavlink::common::msg::GPS_INPUT Struct Reference	335
5.300.1 Detailed Description	336
5.300.2 Member Function Documentation	337
5.300.2.1 deserialize()	337
5.300.2.2 get_name()	337
5.300.2.3 serialize()	337
5.301 mavlink::common::msg::GPS_RAW_INT Struct Reference	338
5.301.1 Detailed Description	339
5.301.2 Member Function Documentation	339
5.301.2.1 deserialize()	339
5.301.2.2 get_name()	340
5.301.2.3 serialize()	340
5.302 mavlink::common::msg::GPS_RTCM_DATA Struct Reference	340
5.302.1 Detailed Description	342
5.302.2 Member Function Documentation	342
5.302.2.1 deserialize()	342
5.302.2.2 get_name()	342
5.302.2.3 serialize()	342
5.303 mavlink::common::msg::GPS_RTK Struct Reference	343
5.303.1 Detailed Description	344
5.303.2 Member Function Documentation	344
5.303.2.1 deserialize()	344
5.303.2.2 get_name()	345
5.303.2.3 serialize()	345
5.304 mavlink::common::msg::GPS_STATUS Struct Reference	345
5.304.1 Detailed Description	347
5.304.2 Member Function Documentation	347
5.304.2.1 deserialize()	347
5.304.2.2 get_name()	347
5.304.2.3 serialize()	348
5.305 HC_ATLAS_ARM_DISARM_CMD Struct Reference	349
5.306 HC_ATLAS_HEARTBEAT_BC Struct Reference	349
5.307 HC_ATLAS_MANUAL_CONTROL_BC Struct Reference	349
5.308 HC_ATLAS_TIMESYNC_REQ Struct Reference	350
5.309 HC_RADIO_STATUS Struct Reference	350
5.310 mavlink::minimal::msg::HEARTBEAT Struct Reference	350
5.310.1 Detailed Description	352
5.310.2 Member Function Documentation	352
5.310.2.1 deserialize()	352
5.310.2.2 get_name()	352
5.310.2.3 serialize()	352

5.311 mavlink::common::msg::HIGH_LATENCY Struct Reference	353
5.311.1 Detailed Description	354
5.311.2 Member Function Documentation	355
5.311.2.1 deserialize()	355
5.311.2.2 get_name()	355
5.311.2.3 serialize()	355
5.312 mavlink::common::msg::HIGH_LATENCY2 Struct Reference	356
5.312.1 Detailed Description	357
5.312.2 Member Function Documentation	357
5.312.2.1 deserialize()	357
5.312.2.2 get_name()	358
5.312.2.3 serialize()	358
5.313 mavlink::common::msg::HIGHRES_IMU Struct Reference	358
5.313.1 Detailed Description	360
5.313.2 Member Function Documentation	360
5.313.2.1 deserialize()	360
5.313.2.2 get_name()	361
5.313.2.3 serialize()	361
5.314 mavlink::common::msg::HIL_ACTUATOR_CONTROLS Struct Reference	361
5.314.1 Detailed Description	362
5.314.2 Member Function Documentation	363
5.314.2.1 deserialize()	363
5.314.2.2 get_name()	363
5.314.2.3 serialize()	363
5.315 mavlink::common::msg::HIL_CONTROLS Struct Reference	364
5.315.1 Detailed Description	365
5.315.2 Member Function Documentation	365
5.315.2.1 deserialize()	365
5.315.2.2 get_name()	366
5.315.2.3 serialize()	366
5.316 mavlink::common::msg::HIL_GPS Struct Reference	366
5.316.1 Detailed Description	368
5.316.2 Member Function Documentation	368
5.316.2.1 deserialize()	368
5.316.2.2 get_name()	368
5.316.2.3 serialize()	369
5.317 mavlink::common::msg::HIL_OPTICAL_FLOW Struct Reference	370
5.317.1 Detailed Description	371
5.317.2 Member Function Documentation	371
5.317.2.1 deserialize()	371
5.317.2.2 get_name()	372
5.317.2.3 serialize()	372

5.318 mavlink::common::msg::HIL_RC_INPUTS_RAW Struct Reference	372
5.318.1 Detailed Description	374
5.318.2 Member Function Documentation	374
5.318.2.1 deserialize()	374
5.318.2.2 get_name()	375
5.318.2.3 serialize()	375
5.319 mavlink::common::msg::HIL_SENSOR Struct Reference	375
5.319.1 Detailed Description	377
5.319.2 Member Function Documentation	377
5.319.2.1 deserialize()	377
5.319.2.2 get_name()	377
5.319.2.3 serialize()	377
5.320 mavlink::common::msg::HIL_STATE Struct Reference	378
5.320.1 Detailed Description	379
5.320.2 Member Function Documentation	379
5.320.2.1 deserialize()	380
5.320.2.2 get_name()	381
5.320.2.3 serialize()	381
5.321 mavlink::common::msg::HIL_STATE_QUATERNION Struct Reference	381
5.321.1 Detailed Description	383
5.321.2 Member Function Documentation	383
5.321.2.1 deserialize()	383
5.321.2.2 get_name()	384
5.321.2.3 serialize()	384
5.322 mavlink::common::msg::HOME_POSITION Struct Reference	384
5.322.1 Detailed Description	386
5.322.2 Member Function Documentation	386
5.322.2.1 deserialize()	386
5.322.2.2 get_name()	386
5.322.2.3 serialize()	386
5.323 mavlink::common::msg::HYGROMETER_SENSOR Struct Reference	387
5.323.1 Detailed Description	388
5.323.2 Member Function Documentation	388
5.323.2.1 deserialize()	388
5.323.2.2 get_name()	389
5.323.2.3 serialize()	389
5.324 mavlink::common::msg::ILLUMINATOR_STATUS Struct Reference	389
5.324.1 Detailed Description	391
5.324.2 Member Function Documentation	391
5.324.2.1 deserialize()	391
5.324.2.2 get_name()	391
5.324.2.3 serialize()	392

5.325 mavlink::Message::Info Struct Reference	392
5.326 mavlink::common::msg::ISBD_LINK_STATUS Struct Reference	392
5.326.1 Detailed Description	394
5.326.2 Member Function Documentation	394
5.326.2.1 deserialize()	394
5.326.2.2 get_name()	394
5.326.2.3 serialize()	395
5.327 mavlink::common::msg::LANDING_TARGET Struct Reference	395
5.327.1 Detailed Description	396
5.327.2 Member Function Documentation	396
5.327.2.1 deserialize()	396
5.327.2.2 get_name()	397
5.327.2.3 serialize()	397
5.328 mavlink::common::msg::LINK_NODE_STATUS Struct Reference	397
5.328.1 Detailed Description	399
5.328.2 Member Function Documentation	399
5.328.2.1 deserialize()	399
5.328.2.2 get_name()	399
5.328.2.3 serialize()	400
5.329 mavlink::common::msg::LOCAL_POSITION_NED Struct Reference	400
5.329.1 Detailed Description	402
5.329.2 Member Function Documentation	402
5.329.2.1 deserialize()	402
5.329.2.2 get_name()	402
5.329.2.3 serialize()	402
5.330 mavlink::common::msg::LOCAL_POSITION_NED_COV Struct Reference	403
5.330.1 Detailed Description	404
5.330.2 Member Function Documentation	404
5.330.2.1 deserialize()	404
5.330.2.2 get_name()	405
5.330.2.3 serialize()	405
5.331 mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET Struct Reference	405
5.331.1 Detailed Description	407
5.331.2 Member Function Documentation	407
5.331.2.1 deserialize()	407
5.331.2.2 get_name()	407
5.331.2.3 serialize()	408
5.332 mavlink::common::msg::LOG_DATA Struct Reference	408
5.332.1 Detailed Description	409
5.332.2 Member Function Documentation	410
5.332.2.1 deserialize()	410
5.332.2.2 get_name()	410

5.332.2.3 serialize()	410
5.333 mavlink::common::msg::LOG_ENTRY Struct Reference	411
5.333.1 Detailed Description	412
5.333.2 Member Function Documentation	412
5.333.2.1 deserialize()	412
5.333.2.2 get_name()	412
5.333.2.3 serialize()	413
5.334 mavlink::common::msg::LOG_ERASE Struct Reference	413
5.334.1 Detailed Description	414
5.334.2 Member Function Documentation	415
5.334.2.1 deserialize()	415
5.334.2.2 get_name()	415
5.334.2.3 serialize()	415
5.335 mavlink::common::msg::LOG_REQUEST_DATA Struct Reference	416
5.335.1 Detailed Description	417
5.335.2 Member Function Documentation	417
5.335.2.1 deserialize()	417
5.335.2.2 get_name()	417
5.335.2.3 serialize()	418
5.336 mavlink::common::msg::LOG_REQUEST_END Struct Reference	418
5.336.1 Detailed Description	419
5.336.2 Member Function Documentation	420
5.336.2.1 deserialize()	420
5.336.2.2 get_name()	420
5.336.2.3 serialize()	420
5.337 mavlink::common::msg::LOG_REQUEST_LIST Struct Reference	421
5.337.1 Detailed Description	422
5.337.2 Member Function Documentation	422
5.337.2.1 deserialize()	422
5.337.2.2 get_name()	422
5.337.2.3 serialize()	423
5.338 mavlink::common::msg::LOGGING_ACK Struct Reference	423
5.338.1 Detailed Description	424
5.338.2 Member Function Documentation	425
5.338.2.1 deserialize()	425
5.338.2.2 get_name()	425
5.338.2.3 serialize()	425
5.339 mavlink::common::msg::LOGGING_DATA Struct Reference	426
5.339.1 Detailed Description	427
5.339.2 Member Function Documentation	427
5.339.2.1 deserialize()	427
5.339.2.2 get_name()	427

5.339.2.3 serialize()	428
5.340 mavlink::common::msg::LOGGING_DATA_ACKED Struct Reference	428
5.340.1 Detailed Description	429
5.340.2 Member Function Documentation	430
5.340.2.1 deserialize()	430
5.340.2.2 get_name()	430
5.340.2.3 serialize()	430
5.341 mavlink::common::msg::MAG_CAL_REPORT Struct Reference	431
5.341.1 Detailed Description	432
5.341.2 Member Function Documentation	432
5.341.2.1 deserialize()	432
5.341.2.2 get_name()	433
5.341.2.3 serialize()	433
5.342 mavlink::common::msg::MANUAL_CONTROL Struct Reference	433
5.342.1 Detailed Description	435
5.342.2 Member Function Documentation	435
5.342.2.1 deserialize()	435
5.342.2.2 get_name()	436
5.342.2.3 serialize()	436
5.343 mavlink::common::msg::MANUAL_SETPOINT Struct Reference	436
5.343.1 Detailed Description	438
5.343.2 Member Function Documentation	438
5.343.2.1 deserialize()	438
5.343.2.2 get_name()	438
5.343.2.3 serialize()	438
5.344 mavlink_sha256_ctx Struct Reference	439
5.345 mavlink::common::msg::MEMORY_VECT Struct Reference	439
5.345.1 Detailed Description	440
5.345.2 Member Function Documentation	441
5.345.2.1 deserialize()	441
5.345.2.2 get_name()	441
5.345.2.3 serialize()	441
5.346 mavlink::Message Struct Reference	442
5.346.1 Detailed Description	444
5.346.2 Member Function Documentation	444
5.346.2.1 deserialize()	444
5.346.2.2 get_name()	446
5.346.2.3 serialize()	447
5.347 mavlink::common::msg::MESSAGE_INTERVAL Struct Reference	449
5.347.1 Detailed Description	451
5.347.2 Member Function Documentation	451
5.347.2.1 deserialize()	451

5.347.2.2 get_name()	451
5.347.2.3 serialize()	452
5.348 message_sender Class Reference	452
5.348.1 Detailed Description	452
5.348.2 Constructor & Destructor Documentation	452
5.348.2.1 message_sender()	452
5.348.3 Member Function Documentation	453
5.348.3.1 buffer_arm_disarm_cmd()	453
5.348.3.2 buffer_heartbeat()	453
5.348.3.3 buffer_manual_control()	453
5.349 mavlink::common::msg::MISSION_ACK Struct Reference	454
5.349.1 Detailed Description	455
5.349.2 Member Function Documentation	456
5.349.2.1 deserialize()	456
5.349.2.2 get_name()	456
5.349.2.3 serialize()	456
5.350 mavlink::common::msg::MISSION_CLEAR_ALL Struct Reference	457
5.350.1 Detailed Description	458
5.350.2 Member Function Documentation	458
5.350.2.1 deserialize()	458
5.350.2.2 get_name()	458
5.350.2.3 serialize()	459
5.351 mavlink::common::msg::MISSION_COUNT Struct Reference	459
5.351.1 Detailed Description	460
5.351.2 Member Function Documentation	460
5.351.2.1 deserialize()	460
5.351.2.2 get_name()	461
5.351.2.3 serialize()	461
5.352 mavlink::common::msg::MISSION_CURRENT Struct Reference	461
5.352.1 Detailed Description	463
5.352.2 Member Function Documentation	463
5.352.2.1 deserialize()	463
5.352.2.2 get_name()	463
5.352.2.3 serialize()	464
5.353 mavlink::common::msg::MISSION_ITEM Struct Reference	464
5.353.1 Detailed Description	466
5.353.2 Member Function Documentation	466
5.353.2.1 deserialize()	466
5.353.2.2 get_name()	466
5.353.2.3 serialize()	467
5.354 mavlink::common::msg::MISSION_ITEM_INT Struct Reference	467
5.354.1 Detailed Description	468

5.354.2 Member Function Documentation	469
5.354.2.1 deserialize()	469
5.354.2.2 get_name()	469
5.354.2.3 serialize()	469
5.355 mavlink::common::msg::MISSION_ITEM_REACHED Struct Reference	470
5.355.1 Detailed Description	471
5.355.2 Member Function Documentation	471
5.355.2.1 deserialize()	471
5.355.2.2 get_name()	471
5.355.2.3 serialize()	472
5.356 mavlink::common::msg::MISSION_REQUEST Struct Reference	472
5.356.1 Detailed Description	473
5.356.2 Member Function Documentation	473
5.356.2.1 deserialize()	473
5.356.2.2 get_name()	474
5.356.2.3 serialize()	474
5.357 mavlink::common::msg::MISSION_REQUEST_INT Struct Reference	474
5.357.1 Detailed Description	476
5.357.2 Member Function Documentation	476
5.357.2.1 deserialize()	476
5.357.2.2 get_name()	476
5.357.2.3 serialize()	477
5.358 mavlink::common::msg::MISSION_REQUEST_LIST Struct Reference	478
5.358.1 Detailed Description	479
5.358.2 Member Function Documentation	479
5.358.2.1 deserialize()	479
5.358.2.2 get_name()	480
5.358.2.3 serialize()	480
5.359 mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST Struct Reference	480
5.359.1 Detailed Description	482
5.359.2 Member Function Documentation	482
5.359.2.1 deserialize()	482
5.359.2.2 get_name()	482
5.359.2.3 serialize()	482
5.360 mavlink::common::msg::MISSION_SET_CURRENT Struct Reference	483
5.360.1 Detailed Description	484
5.360.2 Member Function Documentation	484
5.360.2.1 deserialize()	484
5.360.2.2 get_name()	485
5.360.2.3 serialize()	485
5.361 mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST Struct Reference	485
5.361.1 Detailed Description	487

5.361.2 Member Function Documentation	487
5.361.2.1 deserialize()	487
5.361.2.2 get_name()	487
5.361.2.3 serialize()	487
5.362 mavlink::common::msg::MOUNT_ORIENTATION Struct Reference	488
5.362.1 Detailed Description	489
5.362.2 Member Function Documentation	489
5.362.2.1 deserialize()	489
5.362.2.2 get_name()	490
5.362.2.3 serialize()	490
5.363 mavlink::MsgMap Class Reference	490
5.363.1 Detailed Description	491
5.364 mavlink::common::msg::NAMED_VALUE_FLOAT Struct Reference	491
5.364.1 Detailed Description	492
5.364.2 Member Function Documentation	493
5.364.2.1 deserialize()	493
5.364.2.2 get_name()	493
5.364.2.3 serialize()	493
5.365 mavlink::common::msg::NAMED_VALUE_INT Struct Reference	494
5.365.1 Detailed Description	495
5.365.2 Member Function Documentation	495
5.365.2.1 deserialize()	495
5.365.2.2 get_name()	495
5.365.2.3 serialize()	496
5.366 mavlink::common::msg::NAV_CONTROLLER_OUTPUT Struct Reference	496
5.366.1 Detailed Description	498
5.366.2 Member Function Documentation	498
5.366.2.1 deserialize()	498
5.366.2.2 get_name()	498
5.366.2.3 serialize()	498
5.367 mavlink::common::msg::OBSTACLE_DISTANCE Struct Reference	499
5.367.1 Detailed Description	500
5.367.2 Member Function Documentation	500
5.367.2.1 deserialize()	500
5.367.2.2 get_name()	501
5.367.2.3 serialize()	501
5.368 mavlink::common::msg::ODOMETRY Struct Reference	501
5.368.1 Detailed Description	503
5.368.2 Member Function Documentation	503
5.368.2.1 deserialize()	503
5.368.2.2 get_name()	504
5.368.2.3 serialize()	504

5.369 mavlink::common::msg::ONBOARD_COMPUTER_STATUS Struct Reference	504
5.369.1 Detailed Description	506
5.369.2 Member Function Documentation	506
5.369.2.1 deserialize()	506
5.369.2.2 get_name()	507
5.369.2.3 serialize()	507
5.370 mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS Struct Reference	507
5.370.1 Detailed Description	509
5.370.2 Member Function Documentation	509
5.370.2.1 deserialize()	509
5.370.2.2 get_name()	509
5.370.2.3 serialize()	509
5.371 mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION Struct Reference	510
5.371.1 Detailed Description	511
5.371.2 Member Function Documentation	511
5.371.2.1 deserialize()	511
5.371.2.2 get_name()	512
5.371.2.3 serialize()	512
5.372 mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID Struct Reference	512
5.372.1 Detailed Description	514
5.372.2 Member Function Documentation	514
5.372.2.1 deserialize()	514
5.372.2.2 get_name()	514
5.372.2.3 serialize()	515
5.373 mavlink::common::msg::OPEN_DRONE_ID_LOCATION Struct Reference	515
5.373.1 Detailed Description	517
5.373.2 Member Function Documentation	517
5.373.2.1 deserialize()	517
5.373.2.2 get_name()	517
5.373.2.3 serialize()	517
5.374 mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK Struct Reference	518
5.374.1 Detailed Description	519
5.374.2 Member Function Documentation	519
5.374.2.1 deserialize()	519
5.374.2.2 get_name()	520
5.374.2.3 serialize()	520
5.375 mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID Struct Reference	520
5.375.1 Detailed Description	522
5.375.2 Member Function Documentation	522
5.375.2.1 deserialize()	522
5.375.2.2 get_name()	522
5.375.2.3 serialize()	522

5.376 mavlink::common::msg::OPEN_DRONE_ID_SELF_ID Struct Reference	523
5.376.1 Detailed Description	524
5.376.2 Member Function Documentation	524
5.376.2.1 deserialize()	524
5.376.2.2 get_name()	525
5.376.2.3 serialize()	525
5.377 mavlink::common::msg::OPEN_DRONE_ID_SYSTEM Struct Reference	525
5.377.1 Detailed Description	527
5.377.2 Member Function Documentation	527
5.377.2.1 deserialize()	527
5.377.2.2 get_name()	528
5.377.2.3 serialize()	528
5.378 mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE Struct Reference	528
5.378.1 Detailed Description	530
5.378.2 Member Function Documentation	530
5.378.2.1 deserialize()	530
5.378.2.2 get_name()	530
5.378.2.3 serialize()	530
5.379 mavlink::common::msg::OPTICAL_FLOW Struct Reference	531
5.379.1 Detailed Description	532
5.379.2 Member Function Documentation	532
5.379.2.1 deserialize()	532
5.379.2.2 get_name()	533
5.379.2.3 serialize()	533
5.380 mavlink::common::msg::OPTICAL_FLOW_RAD Struct Reference	533
5.380.1 Detailed Description	535
5.380.2 Member Function Documentation	535
5.380.2.1 deserialize()	535
5.380.2.2 get_name()	535
5.380.2.3 serialize()	536
5.381 mavlink::common::msg::ORBIT_EXECUTION_STATUS Struct Reference	536
5.381.1 Detailed Description	537
5.381.2 Member Function Documentation	538
5.381.2.1 deserialize()	538
5.381.2.2 get_name()	538
5.381.2.3 serialize()	538
5.382 packet_receiver Class Reference	539
5.382.1 Detailed Description	539
5.382.2 Member Function Documentation	539
5.382.2.1 receive_ack()	539
5.382.2.2 receive_heartbeat()	539
5.382.2.3 receive_radio_status()	540

5.382.2.4 receive_sys_status()	540
5.382.2.5 receive_timesync()	540
5.383 mavlink::common::msg::PARAM_ERROR Struct Reference	541
5.383.1 Detailed Description	542
5.383.2 Member Function Documentation	542
5.383.2.1 deserialize()	542
5.383.2.2 get_name()	542
5.383.2.3 serialize()	543
5.384 mavlink::common::msg::PARAM_EXT_ACK Struct Reference	543
5.384.1 Detailed Description	544
5.384.2 Member Function Documentation	545
5.384.2.1 deserialize()	545
5.384.2.2 get_name()	545
5.384.2.3 serialize()	545
5.385 mavlink::common::msg::PARAM_EXT_REQUEST_LIST Struct Reference	546
5.385.1 Detailed Description	547
5.385.2 Member Function Documentation	547
5.385.2.1 deserialize()	547
5.385.2.2 get_name()	547
5.385.2.3 serialize()	548
5.386 mavlink::common::msg::PARAM_EXT_REQUEST_READ Struct Reference	548
5.386.1 Detailed Description	549
5.386.2 Member Function Documentation	550
5.386.2.1 deserialize()	550
5.386.2.2 get_name()	550
5.386.2.3 serialize()	550
5.387 mavlink::common::msg::PARAM_EXT_SET Struct Reference	551
5.387.1 Detailed Description	552
5.387.2 Member Function Documentation	552
5.387.2.1 deserialize()	552
5.387.2.2 get_name()	552
5.387.2.3 serialize()	553
5.388 mavlink::common::msg::PARAM_EXT_VALUE Struct Reference	553
5.388.1 Detailed Description	554
5.388.2 Member Function Documentation	555
5.388.2.1 deserialize()	555
5.388.2.2 get_name()	555
5.388.2.3 serialize()	555
5.389 mavlink::common::msg::PARAM_MAP_RC Struct Reference	556
5.389.1 Detailed Description	557
5.389.2 Member Function Documentation	557
5.389.2.1 deserialize()	557

5.389.2.2	get_name()	557
5.389.2.3	serialize()	558
5.390	mavlink::common::msg::PARAM_REQUEST_LIST Struct Reference	558
5.390.1	Detailed Description	559
5.390.2	Member Function Documentation	560
5.390.2.1	deserialize()	560
5.390.2.2	get_name()	560
5.390.2.3	serialize()	560
5.391	mavlink::common::msg::PARAM_REQUEST_READ Struct Reference	561
5.391.1	Detailed Description	562
5.391.2	Member Function Documentation	562
5.391.2.1	deserialize()	562
5.391.2.2	get_name()	562
5.391.2.3	serialize()	563
5.392	mavlink::common::msg::PARAM_SET Struct Reference	563
5.392.1	Detailed Description	565
5.392.2	Member Function Documentation	565
5.392.2.1	deserialize()	565
5.392.2.2	get_name()	565
5.392.2.3	serialize()	565
5.393	mavlink::common::msg::PARAM_VALUE Struct Reference	566
5.393.1	Detailed Description	567
5.393.2	Member Function Documentation	567
5.393.2.1	deserialize()	567
5.393.2.2	get_name()	568
5.393.2.3	serialize()	568
5.394	mavlink::common::msg::PING Struct Reference	568
5.394.1	Detailed Description	570
5.394.2	Member Function Documentation	570
5.394.2.1	deserialize()	570
5.394.2.2	get_name()	570
5.394.2.3	serialize()	571
5.395	mavlink::common::msg::PLAY_TUNE Struct Reference	572
5.395.1	Detailed Description	573
5.395.2	Member Function Documentation	573
5.395.2.1	deserialize()	573
5.395.2.2	get_name()	574
5.395.2.3	serialize()	574
5.396	mavlink::common::msg::PLAY_TUNE_V2 Struct Reference	574
5.396.1	Detailed Description	576
5.396.2	Member Function Documentation	576
5.396.2.1	deserialize()	576

5.396.2.2 <code>get_name()</code>	576
5.396.2.3 <code>serialize()</code>	576
5.397 <code>mavlink::common::msg::POSITION_TARGET_GLOBAL_INT</code> Struct Reference	577
5.397.1 Detailed Description	578
5.397.2 Member Function Documentation	578
5.397.2.1 <code>deserialize()</code>	578
5.397.2.2 <code>get_name()</code>	579
5.397.2.3 <code>serialize()</code>	579
5.398 <code>mavlink::common::msg::POSITION_TARGET_LOCAL_NED</code> Struct Reference	579
5.398.1 Detailed Description	581
5.398.2 Member Function Documentation	581
5.398.2.1 <code>deserialize()</code>	581
5.398.2.2 <code>get_name()</code>	582
5.398.2.3 <code>serialize()</code>	582
5.399 <code>mavlink::common::msg::POWER_STATUS</code> Struct Reference	582
5.399.1 Detailed Description	583
5.399.2 Member Function Documentation	584
5.399.2.1 <code>deserialize()</code>	584
5.399.2.2 <code>get_name()</code>	584
5.399.2.3 <code>serialize()</code>	584
5.400 <code>mavlink::common::msg::PROTOCOL_VERSION</code> Struct Reference	585
5.400.1 Detailed Description	586
5.400.2 Member Function Documentation	586
5.400.2.1 <code>deserialize()</code>	586
5.400.2.2 <code>get_name()</code>	586
5.400.2.3 <code>serialize()</code>	587
5.401 <code>mavlink::common::msg::RADIO_STATUS</code> Struct Reference	587
5.401.1 Detailed Description	589
5.401.2 Member Function Documentation	589
5.401.2.1 <code>deserialize()</code>	589
5.401.2.2 <code>get_name()</code>	589
5.401.2.3 <code>serialize()</code>	589
5.402 <code>mavlink::common::msg::RAW_IMU</code> Struct Reference	590
5.402.1 Detailed Description	591
5.402.2 Member Function Documentation	591
5.402.2.1 <code>deserialize()</code>	591
5.402.2.2 <code>get_name()</code>	592
5.402.2.3 <code>serialize()</code>	592
5.403 <code>mavlink::common::msg::RAW_PRESSURE</code> Struct Reference	592
5.403.1 Detailed Description	594
5.403.2 Member Function Documentation	594
5.403.2.1 <code>deserialize()</code>	594

5.403.2.2 get_name()	594
5.403.2.3 serialize()	594
5.404 mavlink::common::msg::RAW_RPM Struct Reference	595
5.404.1 Detailed Description	596
5.404.2 Member Function Documentation	596
5.404.2.1 deserialize()	596
5.404.2.2 get_name()	597
5.404.2.3 serialize()	597
5.405 mavlink::common::msg::RC_CHANNELS Struct Reference	597
5.405.1 Detailed Description	599
5.405.2 Member Function Documentation	599
5.405.2.1 deserialize()	599
5.405.2.2 get_name()	600
5.405.2.3 serialize()	600
5.406 mavlink::common::msg::RC_CHANNELS_OVERRIDE Struct Reference	600
5.406.1 Detailed Description	602
5.406.2 Member Function Documentation	602
5.406.2.1 deserialize()	602
5.406.2.2 get_name()	603
5.406.2.3 serialize()	603
5.407 mavlink::common::msg::RC_CHANNELS_RAW Struct Reference	603
5.407.1 Detailed Description	605
5.407.2 Member Function Documentation	605
5.407.2.1 deserialize()	605
5.407.2.2 get_name()	605
5.407.2.3 serialize()	606
5.408 mavlink::common::msg::RC_CHANNELS_SCALED Struct Reference	606
5.408.1 Detailed Description	608
5.408.2 Member Function Documentation	608
5.408.2.1 deserialize()	608
5.408.2.2 get_name()	608
5.408.2.3 serialize()	608
5.409 mavlink::common::msg::RELAY_STATUS Struct Reference	609
5.409.1 Detailed Description	610
5.409.2 Member Function Documentation	610
5.409.2.1 deserialize()	610
5.409.2.2 get_name()	611
5.409.2.3 serialize()	611
5.410 mavlink::common::msg::REQUEST_DATA_STREAM Struct Reference	611
5.410.1 Detailed Description	613
5.410.2 Member Function Documentation	613
5.410.2.1 deserialize()	613

5.410.2.2 <code>get_name()</code>	613
5.410.2.3 <code>serialize()</code>	613
5.411 <code>mavlink::common::msg::REQUEST_EVENT</code> Struct Reference	614
5.411.1 Detailed Description	615
5.411.2 Member Function Documentation	615
5.411.2.1 <code>deserialize()</code>	615
5.411.2.2 <code>get_name()</code>	616
5.411.2.3 <code>serialize()</code>	616
5.412 <code>mavlink::common::msg::RESOURCE_REQUEST</code> Struct Reference	616
5.412.1 Detailed Description	618
5.412.2 Member Function Documentation	618
5.412.2.1 <code>deserialize()</code>	618
5.412.2.2 <code>get_name()</code>	618
5.412.2.3 <code>serialize()</code>	618
5.413 <code>mavlink::common::msg::RESPONSE_EVENT_ERROR</code> Struct Reference	619
5.413.1 Detailed Description	620
5.413.2 Member Function Documentation	620
5.413.2.1 <code>deserialize()</code>	620
5.413.2.2 <code>get_name()</code>	621
5.413.2.3 <code>serialize()</code>	621
5.414 <code>mavlink::common::msg::SAFETY_ALLOWED_AREA</code> Struct Reference	621
5.414.1 Detailed Description	623
5.414.2 Member Function Documentation	623
5.414.2.1 <code>deserialize()</code>	623
5.414.2.2 <code>get_name()</code>	623
5.414.2.3 <code>serialize()</code>	624
5.415 <code>mavlink::common::msg::SAFETY_SET_ALLOWED_AREA</code> Struct Reference	624
5.415.1 Detailed Description	626
5.415.2 Member Function Documentation	626
5.415.2.1 <code>deserialize()</code>	626
5.415.2.2 <code>get_name()</code>	626
5.415.2.3 <code>serialize()</code>	626
5.416 <code>mavlink::common::msg::SCALED_IMU</code> Struct Reference	627
5.416.1 Detailed Description	628
5.416.2 Member Function Documentation	628
5.416.2.1 <code>deserialize()</code>	628
5.416.2.2 <code>get_name()</code>	629
5.416.2.3 <code>serialize()</code>	629
5.417 <code>mavlink::common::msg::SCALED_IMU2</code> Struct Reference	629
5.417.1 Detailed Description	631
5.417.2 Member Function Documentation	631
5.417.2.1 <code>deserialize()</code>	631

5.417.2.2 get_name()	631
5.417.2.3 serialize()	632
5.418 mavlink::common::msg::SCALED_IMU3 Struct Reference	632
5.418.1 Detailed Description	634
5.418.2 Member Function Documentation	634
5.418.2.1 deserialize()	634
5.418.2.2 get_name()	634
5.418.2.3 serialize()	634
5.419 mavlink::common::msg::SCALED_PRESSURE Struct Reference	635
5.419.1 Detailed Description	636
5.419.2 Member Function Documentation	636
5.419.2.1 deserialize()	636
5.419.2.2 get_name()	637
5.419.2.3 serialize()	637
5.420 mavlink::common::msg::SCALED_PRESSURE2 Struct Reference	637
5.420.1 Detailed Description	639
5.420.2 Member Function Documentation	639
5.420.2.1 deserialize()	639
5.420.2.2 get_name()	639
5.420.2.3 serialize()	639
5.421 mavlink::common::msg::SCALED_PRESSURE3 Struct Reference	640
5.421.1 Detailed Description	641
5.421.2 Member Function Documentation	641
5.421.2.1 deserialize()	641
5.421.2.2 get_name()	642
5.421.2.3 serialize()	642
5.422 mavlink::common::msg::SERIAL_CONTROL Struct Reference	642
5.422.1 Detailed Description	644
5.422.2 Member Function Documentation	644
5.422.2.1 deserialize()	644
5.422.2.2 get_name()	644
5.422.2.3 serialize()	645
5.423 mavlink::common::msg::SERVO_OUTPUT_RAW Struct Reference	645
5.423.1 Detailed Description	647
5.423.2 Member Function Documentation	647
5.423.2.1 deserialize()	647
5.423.2.2 get_name()	647
5.423.2.3 serialize()	648
5.424 mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET Struct Reference	649
5.424.1 Detailed Description	650
5.424.2 Member Function Documentation	650
5.424.2.1 deserialize()	650

5.424.2.2 <code>get_name()</code>	651
5.424.2.3 <code>serialize()</code>	651
5.425 <code>mavlink::common::msg::SET_ATTITUDE_TARGET</code> Struct Reference	651
5.425.1 Detailed Description	653
5.425.2 Member Function Documentation	653
5.425.2.1 <code>deserialize()</code>	653
5.425.2.2 <code>get_name()</code>	653
5.425.2.3 <code>serialize()</code>	654
5.426 <code>mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN</code> Struct Reference	654
5.426.1 Detailed Description	655
5.426.2 Member Function Documentation	656
5.426.2.1 <code>deserialize()</code>	656
5.426.2.2 <code>get_name()</code>	656
5.426.2.3 <code>serialize()</code>	656
5.427 <code>mavlink::common::msg::SET_HOME_POSITION</code> Struct Reference	657
5.427.1 Detailed Description	658
5.427.2 Member Function Documentation	658
5.427.2.1 <code>deserialize()</code>	658
5.427.2.2 <code>get_name()</code>	659
5.427.2.3 <code>serialize()</code>	659
5.428 <code>mavlink::common::msg::SET_MODE</code> Struct Reference	659
5.428.1 Detailed Description	661
5.428.2 Member Function Documentation	661
5.428.2.1 <code>deserialize()</code>	661
5.428.2.2 <code>get_name()</code>	661
5.428.2.3 <code>serialize()</code>	661
5.429 <code>mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT</code> Struct Reference	662
5.429.1 Detailed Description	663
5.429.2 Member Function Documentation	663
5.429.2.1 <code>deserialize()</code>	664
5.429.2.2 <code>get_name()</code>	665
5.429.2.3 <code>serialize()</code>	665
5.430 <code>mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED</code> Struct Reference	665
5.430.1 Detailed Description	667
5.430.2 Member Function Documentation	667
5.430.2.1 <code>deserialize()</code>	667
5.430.2.2 <code>get_name()</code>	668
5.430.2.3 <code>serialize()</code>	668
5.431 <code>mavlink::common::msg::SETUP_SIGNING</code> Struct Reference	668
5.431.1 Detailed Description	670
5.431.2 Member Function Documentation	670
5.431.2.1 <code>deserialize()</code>	670

5.431.2.2 get_name()	670
5.431.2.3 serialize()	670
5.432 mavlink::common::msg::SIM_STATE Struct Reference	671
5.432.1 Detailed Description	672
5.432.2 Member Function Documentation	673
5.432.2.1 deserialize()	673
5.432.2.2 get_name()	673
5.432.2.3 serialize()	673
5.433 mavlink::common::msg::SMART_BATTERY_INFO Struct Reference	674
5.433.1 Detailed Description	675
5.433.2 Member Function Documentation	675
5.433.2.1 deserialize()	675
5.433.2.2 get_name()	676
5.433.2.3 serialize()	676
5.434 mavlink::common::msg::STATUSTEXT Struct Reference	676
5.434.1 Detailed Description	678
5.434.2 Member Function Documentation	678
5.434.2.1 deserialize()	678
5.434.2.2 get_name()	678
5.434.2.3 serialize()	679
5.435 mavlink::common::msg::STORAGE_INFORMATION Struct Reference	680
5.435.1 Detailed Description	681
5.435.2 Member Function Documentation	681
5.435.2.1 deserialize()	681
5.435.2.2 get_name()	682
5.435.2.3 serialize()	682
5.436 mavlink::common::msg::SUPPORTED_TUNES Struct Reference	682
5.436.1 Detailed Description	684
5.436.2 Member Function Documentation	684
5.436.2.1 deserialize()	684
5.436.2.2 get_name()	684
5.436.2.3 serialize()	684
5.437 mavlink::common::msg::SYS_STATUS Struct Reference	685
5.437.1 Detailed Description	686
5.437.2 Member Function Documentation	686
5.437.2.1 deserialize()	687
5.437.2.2 get_name()	688
5.437.2.3 serialize()	688
5.438 mavlink::common::msg::SYSTEM_TIME Struct Reference	688
5.438.1 Detailed Description	690
5.438.2 Member Function Documentation	690
5.438.2.1 deserialize()	690

5.438.2.2 <code>get_name()</code>	690
5.438.2.3 <code>serialize()</code>	691
5.439 <code>mavlink::common::msg::TERRAIN_CHECK</code> Struct Reference	691
5.439.1 Detailed Description	692
5.439.2 Member Function Documentation	692
5.439.2.1 <code>deserialize()</code>	692
5.439.2.2 <code>get_name()</code>	693
5.439.2.3 <code>serialize()</code>	693
5.440 <code>mavlink::common::msg::TERRAIN_DATA</code> Struct Reference	693
5.440.1 Detailed Description	695
5.440.2 Member Function Documentation	695
5.440.2.1 <code>deserialize()</code>	695
5.440.2.2 <code>get_name()</code>	695
5.440.2.3 <code>serialize()</code>	695
5.441 <code>mavlink::common::msg::TERRAIN_REPORT</code> Struct Reference	696
5.441.1 Detailed Description	697
5.441.2 Member Function Documentation	697
5.441.2.1 <code>deserialize()</code>	697
5.441.2.2 <code>get_name()</code>	698
5.441.2.3 <code>serialize()</code>	698
5.442 <code>mavlink::common::msg::TERRAIN_REQUEST</code> Struct Reference	698
5.442.1 Detailed Description	700
5.442.2 Member Function Documentation	700
5.442.2.1 <code>deserialize()</code>	700
5.442.2.2 <code>get_name()</code>	700
5.442.2.3 <code>serialize()</code>	700
5.443 <code>thumbstickControl</code> Struct Reference	701
5.443.1 Detailed Description	701
5.444 <code>mavlink::common::msg::TIME_ESTIMATE_TO_TARGET</code> Struct Reference	701
5.444.1 Detailed Description	702
5.444.2 Member Function Documentation	703
5.444.2.1 <code>deserialize()</code>	703
5.444.2.2 <code>get_name()</code>	703
5.444.2.3 <code>serialize()</code>	703
5.445 <code>mavlink::common::msg::TIMESYNC</code> Struct Reference	704
5.445.1 Detailed Description	705
5.445.2 Member Function Documentation	705
5.445.2.1 <code>deserialize()</code>	705
5.445.2.2 <code>get_name()</code>	706
5.445.2.3 <code>serialize()</code>	706
5.446 <code>mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER</code> Struct Reference	706
5.446.1 Detailed Description	708

5.446.2 Member Function Documentation	708
5.446.2.1 deserialize()	708
5.446.2.2 get_name()	708
5.446.2.3 serialize()	708
5.447 mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS Struct Reference	709
5.447.1 Detailed Description	710
5.447.2 Member Function Documentation	710
5.447.2.1 deserialize()	710
5.447.2.2 get_name()	711
5.447.2.3 serialize()	711
5.448 mavlink::common::msg::TUNNEL Struct Reference	711
5.448.1 Detailed Description	713
5.448.2 Member Function Documentation	713
5.448.2.1 deserialize()	713
5.448.2.2 get_name()	713
5.448.2.3 serialize()	714
5.449 mavlink::common::msg::UAVCAN_NODE_INFO Struct Reference	715
5.449.1 Detailed Description	716
5.449.2 Member Function Documentation	716
5.449.2.1 deserialize()	716
5.449.2.2 get_name()	717
5.449.2.3 serialize()	717
5.450 mavlink::common::msg::UAVCAN_NODE_STATUS Struct Reference	717
5.450.1 Detailed Description	719
5.450.2 Member Function Documentation	719
5.450.2.1 deserialize()	719
5.450.2.2 get_name()	719
5.450.2.3 serialize()	720
5.451 mavlink::impl::UIntBuffer< _T > Struct Template Reference	721
5.452 mavlink::impl::UIntBufferHelper< _N > Struct Template Reference	721
5.453 mavlink::impl::UIntBufferHelper< 1 > Struct Reference	721
5.454 mavlink::impl::UIntBufferHelper< 2 > Struct Reference	721
5.455 mavlink::impl::UIntBufferHelper< 4 > Struct Reference	722
5.456 mavlink::impl::UIntBufferHelper< 8 > Struct Reference	722
5.457 mavlink::common::msg::UTM_GLOBAL_POSITION Struct Reference	722
5.457.1 Detailed Description	724
5.457.2 Member Function Documentation	724
5.457.2.1 deserialize()	724
5.457.2.2 get_name()	724
5.457.2.3 serialize()	724
5.458 mavlink::common::msg::V2_EXTENSION Struct Reference	725
5.458.1 Detailed Description	726

5.458.2 Member Function Documentation	726
5.458.2.1 deserialize()	726
5.458.2.2 get_name()	727
5.458.2.3 serialize()	727
5.459 mavlink::common::msg::VFR_HUD Struct Reference	727
5.459.1 Detailed Description	729
5.459.2 Member Function Documentation	729
5.459.2.1 deserialize()	729
5.459.2.2 get_name()	729
5.459.2.3 serialize()	729
5.460 mavlink::common::msg::VIBRATION Struct Reference	730
5.460.1 Detailed Description	731
5.460.2 Member Function Documentation	731
5.460.2.1 deserialize()	731
5.460.2.2 get_name()	732
5.460.2.3 serialize()	732
5.461 mavlink::common::msg::VICON_POSITION_ESTIMATE Struct Reference	732
5.461.1 Detailed Description	734
5.461.2 Member Function Documentation	734
5.461.2.1 deserialize()	734
5.461.2.2 get_name()	734
5.461.2.3 serialize()	735
5.462 mavlink::common::msg::VIDEO_STREAM_INFORMATION Struct Reference	735
5.462.1 Detailed Description	737
5.462.2 Member Function Documentation	737
5.462.2.1 deserialize()	737
5.462.2.2 get_name()	737
5.462.2.3 serialize()	737
5.463 mavlink::common::msg::VIDEO_STREAM_STATUS Struct Reference	738
5.463.1 Detailed Description	739
5.463.2 Member Function Documentation	739
5.463.2.1 deserialize()	739
5.463.2.2 get_name()	740
5.463.2.3 serialize()	740
5.464 mavlink::common::msg::VISION_POSITION_ESTIMATE Struct Reference	740
5.464.1 Detailed Description	742
5.464.2 Member Function Documentation	742
5.464.2.1 deserialize()	742
5.464.2.2 get_name()	742
5.464.2.3 serialize()	743
5.465 mavlink::common::msg::VISION_SPEED_ESTIMATE Struct Reference	743
5.465.1 Detailed Description	744

5.465.2 Member Function Documentation	745
5.465.2.1 deserialize()	745
5.465.2.2 get_name()	745
5.465.2.3 serialize()	745
5.466 mavlink::common::msg::WHEEL_DISTANCE Struct Reference	746
5.466.1 Detailed Description	747
5.466.2 Member Function Documentation	747
5.466.2.1 deserialize()	747
5.466.2.2 get_name()	747
5.466.2.3 serialize()	748
5.467 mavlink::common::msg::WIFI_CONFIG_AP Struct Reference	748
5.467.1 Detailed Description	749
5.467.2 Member Function Documentation	749
5.467.2.1 deserialize()	749
5.467.2.2 get_name()	750
5.467.2.3 serialize()	750
5.468 mavlink::common::msg::WINCH_STATUS Struct Reference	750
5.468.1 Detailed Description	752
5.468.2 Member Function Documentation	752
5.468.2.1 deserialize()	752
5.468.2.2 get_name()	752
5.468.2.3 serialize()	753
5.469 mavlink::common::msg::WIND_COV Struct Reference	753
5.469.1 Detailed Description	754
5.469.2 Member Function Documentation	754
5.469.2.1 deserialize()	754
5.469.2.2 get_name()	755
5.469.2.3 serialize()	755
6 File Documentation	757
6.1 single_hand_controller/displayHandler.cpp File Reference	757
6.1.1 Detailed Description	758
6.1.2 Function Documentation	758
6.1.2.1 updateDisplay()	758
6.2 single_hand_controller/include/definitions.h File Reference	759
6.2.1 Detailed Description	760
6.3 single_hand_controller/include/displayHandler.hpp File Reference	761
6.3.1 Detailed Description	762
6.3.2 Function Documentation	762
6.3.2.1 setupDisplay()	762
6.3.2.2 updateDisplay()	762
6.4 single_hand_controller/include/includes.h File Reference	763

6.4.1 Detailed Description	763
6.5 single_hand_controller/include/IOhandler.hpp File Reference	763
6.5.1 Detailed Description	764
6.5.2 Function Documentation	764
6.5.2.1 getXY()	765
6.5.2.2 getXY_raw()	766
6.6 single_hand_controller/include/mavlink/common/common.h File Reference	766
6.6.1 Detailed Description	797
6.6.2 Macro Definition Documentation	797
6.6.2.1 HAVE_ENUM_ACTUATOR_CONFIGURATION	798
6.6.2.2 HAVE_ENUM_ACTUATOR_OUTPUT_FUNCTION	798
6.6.2.3 HAVE_ENUM_ADSB_FLAGS	798
6.6.2.4 HAVE_ENUM_AIS_FLAGS	798
6.6.2.5 HAVE_ENUM_AUTOTUNE_AXIS	798
6.6.2.6 HAVE_ENUM_COMP_METADATA_TYPE	798
6.6.2.7 HAVE_ENUM_FENCE_TYPE	799
6.6.2.8 HAVE_ENUM_GIMBAL_MANAGER_CAP_FLAGS	799
6.6.2.9 HAVE_ENUM_MAV_BATTERY_FAULT	799
6.6.2.10 HAVE_ENUM_MAV_BATTERY_MODE	799
6.6.2.11 HAVE_ENUM_MAV_CMD	799
6.6.2.12 HAVE_ENUM_MAV_DATA_STREAM	800
6.6.2.13 HAVE_ENUM_MAV_FRAME	800
6.6.2.14 HAVE_ENUM_MAV_FTP_ERR	800
6.6.2.15 HAVE_ENUM_MAV_FTP_OPCODE	800
6.6.2.16 HAVE_ENUM_MAV_FUEL_TYPE	801
6.6.2.17 HAVE_ENUM_MAV_GENERATOR_STATUS_FLAG	801
6.6.2.18 HAVE_ENUM_MAV_MODE	801
6.6.2.19 HAVE_ENUM_MAV_MOUNT_MODE	801
6.6.2.20 HAVE_ENUM_MAV_ROI	801
6.6.2.21 HAVE_ENUM_MAV_SEVERITY	801
6.6.2.22 HAVE_ENUM_MAV_STANDARD_MODE	802
6.6.2.23 HAVE_ENUM_MISSION_STATE	802
6.6.2.24 HAVE_ENUM_PARACHUTE_ACTION	802
6.6.2.25 HAVE_ENUM_POSITION_TARGET_TYEMASK	802
6.6.2.26 HAVE_ENUM_PREFLIGHT_STORAGE_MISSION_ACTION	802
6.6.2.27 HAVE_ENUM_PREFLIGHT_STORAGE_PARAMETER_ACTION	803
6.6.2.28 HAVE_ENUM_RC_SUB_TYPE	803
6.6.2.29 HAVE_ENUM_RC_TYPE	803
6.7 single_hand_controller/include/mavlink/common/common.hpp File Reference	803
6.7.1 Detailed Description	829
6.7.2 Enumeration Type Documentation	829
6.7.2.1 ACTUATOR_CONFIGURATION	829

6.7.2.2 ACTUATOR_OUTPUT_FUNCTION	830
6.7.2.3 ADSB_FLAGS	830
6.7.2.4 AIS_FLAGS	830
6.7.2.5 AUTOTUNE_AXIS	830
6.7.2.6 COMP_METADATA_TYPE	830
6.7.2.7 FENCE_TYPE	830
6.7.2.8 GIMBAL_MANAGER_CAP_FLAGS	831
6.7.2.9 MAV_BATTERY_FAULT	831
6.7.2.10 MAV_BATTERY_MODE	831
6.7.2.11 MAV_CMD	831
6.7.2.12 MAV_DATA_STREAM	831
6.7.2.13 MAV_FRAME	832
6.7.2.14 MAV_FTP_ERR	832
6.7.2.15 MAV_FTP_OPCODE	832
6.7.2.16 MAV_FUEL_TYPE	832
6.7.2.17 MAV_GENERATOR_STATUS_FLAG	833
6.7.2.18 MAV_MODE	833
6.7.2.19 MAV_MOUNT_MODE	833
6.7.2.20 MAV_ROI	833
6.7.2.21 MAV_SEVERITY	833
6.7.2.22 MAV_STANDARD_MODE	833
6.7.2.23 MISSION_STATE	834
6.7.2.24 PARACHUTE_ACTION	834
6.7.2.25 POSITION_TARGET_TYPEMASK	834
6.7.2.26 PREFLIGHT_STORAGE_MISSION_ACTION	834
6.7.2.27 PREFLIGHT_STORAGE_PARAMETER_ACTION	834
6.7.2.28 RC_SUB_TYPE	835
6.7.2.29 RC_TYPE	835
6.8 single_hand_controller/include/mavlink/common/gtestsuite.hpp File Reference	835
6.8.1 Detailed Description	839
6.9 single_hand_controller/include/mavlink/minimal/gtestsuite.hpp File Reference	840
6.9.1 Detailed Description	840
6.10 single_hand_controller/include/mavlink/standard/gtestsuite.hpp File Reference	840
6.10.1 Detailed Description	841
6.11 single_hand_controller/include/mavlink/common/mavlink.h File Reference	841
6.11.1 Detailed Description	842
6.12 single_hand_controller/include/mavlink/minimal/mavlink.h File Reference	842
6.12.1 Detailed Description	842
6.13 single_hand_controller/include/mavlink/standard/mavlink.h File Reference	843
6.13.1 Detailed Description	843
6.14 single_hand_controller/include/mavlink/common/testsuite.h File Reference	844
6.14.1 Detailed Description	844

6.15 single_hand_controller/include/mavlink/minimal/testsuite.h File Reference	844
6.15.1 Detailed Description	845
6.16 single_hand_controller/include/mavlink/standard/testsuite.h File Reference	845
6.16.1 Detailed Description	846
6.17 single_hand_controller/include/mavlink/common/version.h File Reference	846
6.17.1 Detailed Description	847
6.18 single_hand_controller/include/mavlink/minimal/version.h File Reference	847
6.18.1 Detailed Description	848
6.19 single_hand_controller/include/mavlink/standard/version.h File Reference	848
6.19.1 Detailed Description	849
6.20 single_hand_controller/include/mavlink/mavlink_conversions.h File Reference	849
6.20.1 Detailed Description	851
6.20.2 Function Documentation	851
6.20.2.1 mavlink_dcm_to_euler()	851
6.20.2.2 mavlink_dcm_to_quaternion()	852
6.20.2.3 mavlink_euler_to_dcm()	852
6.20.2.4 mavlink_euler_to_quaternion()	852
6.20.2.5 mavlink_quaternion_to_dcm()	853
6.20.2.6 mavlink_quaternion_to_euler()	853
6.21 single_hand_controller/include/mavlink/minimal/minimal.h File Reference	853
6.21.1 Detailed Description	857
6.21.2 Macro Definition Documentation	857
6.21.2.1 HAVE_ENUM_MAV_AUTOPILOT	858
6.21.2.2 HAVE_ENUM_MAV_COMPONENT	858
6.21.2.3 HAVE_ENUM_MAV_MODE_FLAG_DECODE_POSITION	858
6.21.2.4 HAVE_ENUM_MAV_TYPE	858
6.22 single_hand_controller/include/mavlink/minimal/minimal.hpp File Reference	859
6.22.1 Detailed Description	862
6.22.2 Enumeration Type Documentation	862
6.22.2.1 MAV_AUTOPILOT	862
6.22.2.2 MAV_COMPONENT	862
6.22.2.3 MAV_MODE_FLAG_DECODE_POSITION	863
6.22.2.4 MAV_TYPE	863
6.23 single_hand_controller/include/mavlink/standard/standard.h File Reference	863
6.23.1 Detailed Description	865
6.23.2 Macro Definition Documentation	865
6.23.2.1 HAVE_ENUM_FIRMWARE_VERSION_TYPE	865
6.23.2.2 HAVE_ENUM_MAV_PROTOCOL_CAPABILITY	865
6.24 single_hand_controller/include/mavlink/standard/standard.hpp File Reference	866
6.24.1 Detailed Description	867
6.24.2 Enumeration Type Documentation	867
6.24.2.1 FIRMWARE_VERSION_TYPE	868

6.24.2.2 MAV_PROTOCOL_CAPABILITY	868
6.25 single_hand_controller/include/message_sender.hpp File Reference	868
6.25.1 Detailed Description	869
6.26 single_hand_controller/include/message_structs.h File Reference	869
6.26.1 Detailed Description	870
6.27 single_hand_controller/include/packetHandler.h File Reference	871
6.27.1 Detailed Description	872
6.27.2 Function Documentation	872
6.27.2.1 heartbeat_timed_out()	872
6.27.2.2 setupSigning()	872
6.28 single_hand_controller/include/setupFunctions.h File Reference	873
6.28.1 Detailed Description	874
6.28.2 Function Documentation	874
6.28.2.1 identifyControllerDrift()	874
6.28.2.2 initMAVLink()	874
6.28.2.3 initUGV_state()	874
6.28.2.4 setupDisplay()	875
6.29 single_hand_controller/include/stateHandler.hpp File Reference	875
6.29.1 Detailed Description	876
6.29.2 Function Documentation	876
6.29.2.1 setUGV_state()	876
6.30 single_hand_controller/include/timer.hpp File Reference	877
6.30.1 Detailed Description	877
6.30.2 Function Documentation	878
6.30.2.1 end_OFP_timer()	878
6.30.2.2 timeup()	878
6.31 single_hand_controller/IOhandler.cpp File Reference	878
6.31.1 Detailed Description	879
6.31.2 Function Documentation	879
6.31.2.1 getXY()	879
6.31.2.2 getXY_raw()	880
6.32 single_hand_controller/message_sender.cpp File Reference	880
6.32.1 Detailed Description	880
6.33 single_hand_controller/packet_receiver.cpp File Reference	881
6.33.1 Detailed Description	881
6.34 single_hand_controller/packetHandler.cpp File Reference	881
6.34.1 Detailed Description	882
6.34.2 Function Documentation	883
6.34.2.1 heartbeat_timed_out()	883
6.34.2.2 setupSigning()	883
6.35 single_hand_controller/setupFunctions.cpp File Reference	883
6.35.1 Detailed Description	884

6.35.2 Function Documentation	884
6.35.2.1 identifyControllerDrift()	884
6.35.2.2 initMAVLink()	884
6.35.2.3 setupDisplay()	885
6.36 single_hand_controller/stateHandler.cpp File Reference	885
6.36.1 Detailed Description	885
6.36.2 Function Documentation	886
6.36.2.1 setUGV_state()	886
6.37 single_hand_controller/timer.cpp File Reference	886
6.37.1 Detailed Description	887
6.37.2 Function Documentation	887
6.37.2.1 end_OFP_timer()	887
6.37.2.2 timeup()	887
Index	889

Chapter 1

Main Page

Application entry point is in `single_hand_controller/single_hand_controller.ino`

All header files are defined in `single_hand_controller/include/includes.h`

1.0.1 Procedure for editing any file -

1. Review documentation for existing functionality, in order to ensure new functionality being added doesn't clash with existing functionality.
2. Check ammendment history of the given file, in case the similar functionality existed in a previous version, then verify why it was removed, and the functionality can be brought back from git commit history
3. Edit file and track all the edits made to the given file
4. Update file header comments (date, author, version number, description, and ammendment history. Add the history in case it doesn't exist)

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

__mavlink_actuator_control_target_t	37
__mavlink_actuator_output_status_t	37
__mavlink_adsb_vehicle_t	38
__mavlink_airspeed_t	38
__mavlink_ais_vessel_t	38
__mavlink_altitude_t	39
__mavlink_att_pos_mocap_t	39
__mavlink_attitude_quaternion_cov_t	40
__mavlink_attitude_quaternion_t	40
__mavlink_attitude_t	40
__mavlink_attitude_target_t	41
__mavlink_auth_key_t	41
__mavlink_autopilot_version_t	41
__mavlink_available_modes_monitor_t	42
__mavlink_available_modes_t	42
__mavlink_battery_info_t	42
__mavlink_bitfield	43
__mavlink_button_change_t	43
__mavlink_camera_fov_status_t	43
__mavlink_camera_image_captured_t	44
__mavlink_camera_information_t	44
__mavlink_camera_thermal_range_t	45
__mavlink_camera_tracking_geo_status_t	45
__mavlink_camera_tracking_image_status_t	46
__mavlink_camera_trigger_t	46
__mavlink_can_filter_modify_t	46
__mavlink_can_frame_t	47
__mavlink_canfd_frame_t	47
__mavlink_cellular_config_t	47
__mavlink_cellular_status_t	48
__mavlink_change_operator_control_ack_t	48
__mavlink_change_operator_control_t	48
__mavlink_collision_t	49
__mavlink_command_ack_t	49
__mavlink_command_cancel_t	49

__mavlink_command_int_t	50
__mavlink_command_long_t	50
__mavlink_component_information_basic_t	51
__mavlink_component_information_t	51
__mavlink_component_metadata_t	51
__mavlink_control_system_state_t	52
__mavlink_current_event_sequence_t	52
__mavlink_current_mode_t	52
__mavlink_data_stream_t	53
__mavlink_data_transmission_handshake_t	53
__mavlink_debug_float_array_t	53
__mavlink_debug_t	54
__mavlink_debug_vect_t	54
__mavlink_encapsulated_data_t	54
__mavlink_esc_info_t	55
__mavlink_esc_status_t	55
__mavlink_estimator_status_t	55
__mavlink_event_t	56
__mavlink_extended_sys_state_t	56
__mavlink_fence_status_t	56
__mavlink_field_info	57
__mavlink_figure_eight_execution_status_t	57
__mavlink_file_transfer_protocol_t	57
__mavlink_flight_information_t	58
__mavlink_follow_target_t	58
__mavlink_fuel_status_t	58
__mavlink_generator_status_t	59
__mavlink_gimbal_device_attitude_status_t	59
__mavlink_gimbal_device_information_t	60
__mavlink_gimbal_device_set_attitude_t	60
__mavlink_gimbal_manager_information_t	61
__mavlink_gimbal_manager_set_attitude_t	61
__mavlink_gimbal_manager_set_manual_control_t	61
__mavlink_gimbal_manager_set_pitchyaw_t	62
__mavlink_gimbal_manager_status_t	62
__mavlink_global_position_int_cov_t	62
__mavlink_global_position_int_t	63
__mavlink_global_vision_position_estimate_t	63
__mavlink_gps2_rtk_t	63
__mavlink_gps_inject_data_t	64
__mavlink_gps_rtcn_data_t	64
__mavlink_gps_rtk_t	64
__mavlink_gps_status_t	65
__mavlink_heartbeat_t	65
__mavlink_high_latency2_t	65
__mavlink_high_latency_t	66
__mavlink_highres_imu_t	67
__mavlink_hil_actuator_controls_t	67
__mavlink_hil_controls_t	67
__mavlink_hil_optical_flow_t	68
__mavlink_hil_rc_inputs_raw_t	68
__mavlink_hil_sensor_t	69
__mavlink_hil_state_quaternion_t	69
__mavlink_hil_state_t	70
__mavlink_hygrometer_sensor_t	70
__mavlink_illuminator_status_t	70
__mavlink_isbd_link_status_t	71
__mavlink_link_node_status_t	71

__mavlink_local_position_ned_cov_t	71
__mavlink_local_position_ned_system_global_offset_t	72
__mavlink_local_position_ned_t	72
__mavlink_log_data_t	72
__mavlink_log_entry_t	73
__mavlink_log_erase_t	73
__mavlink_log_request_data_t	73
__mavlink_log_request_end_t	73
__mavlink_log_request_list_t	74
__mavlink_logging_ack_t	74
__mavlink_logging_data_acked_t	74
__mavlink_logging_data_t	75
__mavlink_manual_setpoint_t	75
__mavlink_memory_vect_t	75
__mavlink_message_info	76
__mavlink_message_interval_t	76
__mavlink_mission_ack_t	76
__mavlink_mission_clear_all_t	77
__mavlink_mission_item_int_t	77
__mavlink_mission_item_reached_t	77
__mavlink_mission_item_t	78
__mavlink_mission_request_int_t	78
__mavlink_mission_request_list_t	78
__mavlink_mission_request_partial_list_t	79
__mavlink_mission_request_t	79
__mavlink_mission_set_current_t	79
__mavlink_mission_write_partial_list_t	80
__mavlink_mount_orientation_t	80
__mavlink_msg_entry	80
__mavlink_named_value_float_t	81
__mavlink_named_value_int_t	81
__mavlink_nav_controller_output_t	81
__mavlink_odometry_t	82
__mavlink_onboard_computer_status_t	82
__mavlink_open_drone_id_arm_status_t	83
__mavlink_open_drone_id_authentication_t	83
__mavlink_open_drone_id_basic_id_t	83
__mavlink_open_drone_id_location_t	84
__mavlink_open_drone_id_message_pack_t	84
__mavlink_open_drone_id_operator_id_t	85
__mavlink_open_drone_id_self_id_t	85
__mavlink_open_drone_id_system_t	85
__mavlink_open_drone_id_system_update_t	86
__mavlink_optical_flow_rad_t	86
__mavlink_orbit_execution_status_t	86
__mavlink_param_error_t	87
__mavlink_param_ext_ack_t	87
__mavlink_param_ext_request_list_t	87
__mavlink_param_ext_request_read_t	87
__mavlink_param_ext_set_t	88
__mavlink_param_ext_value_t	88
__mavlink_param_map_rc_t	88
__mavlink_param_request_list_t	89
__mavlink_param_request_read_t	89
__mavlink_param_set_t	89
__mavlink_param_value_t	89
__mavlink_ping_t	90
__mavlink_play_tune_t	90

__mavlink_play_tune_v2_t	90
__mavlink_position_target_global_int_t	91
__mavlink_position_target_local_ned_t	91
__mavlink_power_status_t	92
__mavlink_protocol_version_t	92
__mavlink_radio_status_t	92
__mavlink_raw_pressure_t	93
__mavlink_raw_rpm_t	93
__mavlink_rc_channels_override_t	93
__mavlink_rc_channels_raw_t	94
__mavlink_rc_channels_scaled_t	94
__mavlink_rc_channels_t	95
__mavlink_relay_status_t	95
__mavlink_request_data_stream_t	96
__mavlink_request_event_t	96
__mavlink_resource_request_t	96
__mavlink_response_event_error_t	97
__mavlink_safety_allowed_area_t	97
__mavlink_safety_set_allowed_area_t	97
__mavlink_scaled_imu2_t	98
__mavlink_scaled_imu3_t	98
__mavlink_scaled_imu_t	99
__mavlink_scaled_pressure2_t	99
__mavlink_scaled_pressure3_t	99
__mavlink_scaled_pressure_t	100
__mavlink_serial_control_t	100
__mavlink_set_actuator_control_target_t	100
__mavlink_set_mode_t	101
__mavlink_set_position_target_global_int_t	101
__mavlink_set_position_target_local_ned_t	102
__mavlink_setup_signing_t	102
__mavlink_signing	103
__mavlink_signing_streams::__mavlink_signing_stream	103
__mavlink_signing_streams	104
__mavlink_sim_state_t	105
__mavlink_status	106
__mavlink_storage_information_t	107
__mavlink_supported_tunes_t	107
__mavlink_system_time_t	108
__mavlink_terrain_check_t	108
__mavlink_terrain_data_t	108
__mavlink_terrain_report_t	108
__mavlink_terrain_request_t	109
__mavlink_time_estimate_to_target_t	109
__mavlink_timesync_t	109
__mavlink_trajectory_representation_bezier_t	110
__mavlink_trajectory_representation_waypoints_t	110
__mavlink_tunnel_t	110
__mavlink_uavcan_node_info_t	111
__mavlink_uavcan_node_status_t	111
__mavlink_utm_global_position_t	111
__mavlink_v2_extension_t	112
__mavlink_vfr_hud_t	112
__mavlink_vibration_t	113
__mavlink_vicon_position_estimate_t	113
__mavlink_video_stream_information_t	113
__mavlink_video_stream_status_t	114
__mavlink_vision_position_estimate_t	114

__mavlink_vision_speed_estimate_t	114
__mavlink_wheel_distance_t	115
__mavlink_wifi_config_ap_t	115
__mavlink_winch_status_t	115
__mavlink_wind_cov_t	116
ATLAS_HC_ARM_DISARM_ACK	132
ATLAS_HC_HEARTBEAT_BC	132
ATLAS_HC_SYS_STAT	133
ATLAS_HC_TIMESYNC_RESP	133
HC_ATLAS_ARM_DISARM_CMD	349
HC_ATLAS_HEARTBEAT_BC	349
HC_ATLAS_MANUAL_CONTROL_BC	349
HC_ATLAS_TIMESYNC_REQ	350
HC_RADIO_STATUS	350
mavlink::Message::Info	392
mavlink_sha256_ctx	439
mavlink::Message	442
mavlink::common::msg::ACTUATOR_CONTROL_TARGET	116
mavlink::common::msg::ACTUATOR_OUTPUT_STATUS	119
mavlink::common::msg::ADSB_VEHICLE	121
mavlink::common::msg::AIRSPEED	123
mavlink::common::msg::AIS_VESSEL	126
mavlink::common::msg::ALTITUDE	129
mavlink::common::msg::ATTITUDE	136
mavlink::common::msg::ATTITUDE_QUATERNION	138
mavlink::common::msg::ATTITUDE_QUATERNION_COV	141
mavlink::common::msg::ATTITUDE_TARGET	143
mavlink::common::msg::ATT_POS_MOCAP	133
mavlink::common::msg::AUTH_KEY	146
mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE	148
mavlink::common::msg::AVAILABLE_MODES	154
mavlink::common::msg::AVAILABLE_MODES_MONITOR	157
mavlink::common::msg::BATTERY_INFO	159
mavlink::common::msg::BATTERY_STATUS	162
mavlink::common::msg::BUTTON_CHANGE	165
mavlink::common::msg::CAMERA_CAPTURE_STATUS	168
mavlink::common::msg::CAMERA_FOV_STATUS	170
mavlink::common::msg::CAMERA_IMAGE_CAPTURED	173
mavlink::common::msg::CAMERA_INFORMATION	176
mavlink::common::msg::CAMERA_SETTINGS	179
mavlink::common::msg::CAMERA_THERMAL_RANGE	182
mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS	184
mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS	187
mavlink::common::msg::CAMERA_TRIGGER	189
mavlink::common::msg::CANFD_FRAME	197
mavlink::common::msg::CAN_FILTER_MODIFY	192
mavlink::common::msg::CAN_FRAME	194
mavlink::common::msg::CELLULAR_CONFIG	199
mavlink::common::msg::CELLULAR_STATUS	202
mavlink::common::msg::CHANGE_OPERATOR_CONTROL	205
mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK	207
mavlink::common::msg::COLLISION	210
mavlink::common::msg::COMMAND_ACK	212
mavlink::common::msg::COMMAND_CANCEL	215
mavlink::common::msg::COMMAND_INT	217
mavlink::common::msg::COMMAND_LONG	220
mavlink::common::msg::COMPONENT_INFORMATION	223
mavlink::common::msg::COMPONENT_INFORMATION_BASIC	226

mavlink::common::msg::COMPONENT_METADATA	228
mavlink::common::msg::CONTROL_SYSTEM_STATE	231
mavlink::common::msg::CURRENT_EVENT_SEQUENCE	233
mavlink::common::msg::CURRENT_MODE	236
mavlink::common::msg::DATA_STREAM	238
mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE	241
mavlink::common::msg::DEBUG	243
mavlink::common::msg::DEBUG_FLOAT_ARRAY	246
mavlink::common::msg::DEBUG_VECT	248
mavlink::common::msg::DISTANCE_SENSOR	251
mavlink::common::msg::EFI_STATUS	253
mavlink::common::msg::ENCAPSULATED_DATA	256
mavlink::common::msg::ESC_INFO	259
mavlink::common::msg::ESC_STATUS	261
mavlink::common::msg::ESTIMATOR_STATUS	265
mavlink::common::msg::EVENT	268
mavlink::common::msg::EXTENDED_SYS_STATE	271
mavlink::common::msg::FENCE_STATUS	274
mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS	276
mavlink::common::msg::FILE_TRANSFER_PROTOCOL	279
mavlink::common::msg::FLIGHT_INFORMATION	281
mavlink::common::msg::FOLLOW_TARGET	285
mavlink::common::msg::FUEL_STATUS	287
mavlink::common::msg::GENERATOR_STATUS	290
mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS	293
mavlink::common::msg::GIMBAL_DEVICE_INFORMATION	296
mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE	300
mavlink::common::msg::GIMBAL_MANAGER_INFORMATION	303
mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE	306
mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL	309
mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW	311
mavlink::common::msg::GIMBAL_MANAGER_STATUS	314
mavlink::common::msg::GLOBAL_POSITION_INT_COV	319
mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE	322
mavlink::common::msg::GPS2_RAW	325
mavlink::common::msg::GPS2_RTK	328
mavlink::common::msg::GPS_GLOBAL_ORIGIN	330
mavlink::common::msg::GPS_INJECT_DATA	333
mavlink::common::msg::GPS_INPUT	335
mavlink::common::msg::GPS_RAW_INT	338
mavlink::common::msg::GPS_RTCM_DATA	340
mavlink::common::msg::GPS_RTK	343
mavlink::common::msg::GPS_STATUS	345
mavlink::common::msg::HIGHRES_IMU	358
mavlink::common::msg::HIGH_LATENCY	353
mavlink::common::msg::HIGH_LATENCY2	356
mavlink::common::msg::HIL_ACTUATOR_CONTROLS	361
mavlink::common::msg::HIL_CONTROLS	364
mavlink::common::msg::HIL_GPS	366
mavlink::common::msg::HIL_OPTICAL_FLOW	370
mavlink::common::msg::HIL_RC_INPUTS_RAW	372
mavlink::common::msg::HIL_SENSOR	375
mavlink::common::msg::HIL_STATE	378
mavlink::common::msg::HIL_STATE_QUATERNION	381
mavlink::common::msg::HOME_POSITION	384
mavlink::common::msg::HYGROMETER_SENSOR	387
mavlink::common::msg::ILLUMINATOR_STATUS	389
mavlink::common::msg::ISBD_LINK_STATUS	392

mavlink::common::msg::LANDING_TARGET	395
mavlink::common::msg::LINK_NODE_STATUS	397
mavlink::common::msg::LOCAL_POSITION_NED	400
mavlink::common::msg::LOCAL_POSITION_NED_COV	403
mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET	405
mavlink::common::msg::LOGGING_ACK	423
mavlink::common::msg::LOGGING_DATA	426
mavlink::common::msg::LOGGING_DATA_ACKED	428
mavlink::common::msg::LOG_DATA	408
mavlink::common::msg::LOG_ENTRY	411
mavlink::common::msg::LOG_ERASE	413
mavlink::common::msg::LOG_REQUEST_DATA	416
mavlink::common::msg::LOG_REQUEST_END	418
mavlink::common::msg::LOG_REQUEST_LIST	421
mavlink::common::msg::MAG_CAL_REPORT	431
mavlink::common::msg::MANUAL_CONTROL	433
mavlink::common::msg::MANUAL_SETPOINT	436
mavlink::common::msg::MEMORY_VECT	439
mavlink::common::msg::MESSAGE_INTERVAL	449
mavlink::common::msg::MISSION_ACK	454
mavlink::common::msg::MISSION_CLEAR_ALL	457
mavlink::common::msg::MISSION_COUNT	459
mavlink::common::msg::MISSION_CURRENT	461
mavlink::common::msg::MISSION_ITEM	464
mavlink::common::msg::MISSION_ITEM_INT	467
mavlink::common::msg::MISSION_ITEM_REACHED	470
mavlink::common::msg::MISSION_REQUEST	472
mavlink::common::msg::MISSION_REQUEST_INT	474
mavlink::common::msg::MISSION_REQUEST_LIST	478
mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST	480
mavlink::common::msg::MISSION_SET_CURRENT	483
mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST	485
mavlink::common::msg::MOUNT_ORIENTATION	488
mavlink::common::msg::NAMED_VALUE_FLOAT	491
mavlink::common::msg::NAMED_VALUE_INT	494
mavlink::common::msg::NAV_CONTROLLER_OUTPUT	496
mavlink::common::msg::OBSTACLE_DISTANCE	499
mavlink::common::msg::ODOMETRY	501
mavlink::common::msg::ONBOARD_COMPUTER_STATUS	504
mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS	507
mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION	510
mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID	512
mavlink::common::msg::OPEN_DRONE_ID_LOCATION	515
mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK	518
mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID	520
mavlink::common::msg::OPEN_DRONE_ID_SELF_ID	523
mavlink::common::msg::OPEN_DRONE_ID_SYSTEM	525
mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE	528
mavlink::common::msg::OPTICAL_FLOW	531
mavlink::common::msg::OPTICAL_FLOW_RAD	533
mavlink::common::msg::ORBIT_EXECUTION_STATUS	536
mavlink::common::msg::PARAM_ERROR	541
mavlink::common::msg::PARAM_EXT_ACK	543
mavlink::common::msg::PARAM_EXT_REQUEST_LIST	546
mavlink::common::msg::PARAM_EXT_REQUEST_READ	548
mavlink::common::msg::PARAM_EXT_SET	551
mavlink::common::msg::PARAM_EXT_VALUE	553
mavlink::common::msg::PARAM_MAP_RC	556

mavlink::common::msg::PARAM_REQUEST_LIST	558
mavlink::common::msg::PARAM_REQUEST_READ	561
mavlink::common::msg::PARAM_SET	563
mavlink::common::msg::PARAM_VALUE	566
mavlink::common::msg::PING	568
mavlink::common::msg::PLAY_TUNE	572
mavlink::common::msg::PLAY_TUNE_V2	574
mavlink::common::msg::POSITION_TARGET_GLOBAL_INT	577
mavlink::common::msg::POSITION_TARGET_LOCAL_NED	579
mavlink::common::msg::POWER_STATUS	582
mavlink::common::msg::PROTOCOL_VERSION	585
mavlink::common::msg::RADIO_STATUS	587
mavlink::common::msg::RAW_IMU	590
mavlink::common::msg::RAW_PRESSURE	592
mavlink::common::msg::RAW_RPM	595
mavlink::common::msg::RC_CHANNELS	597
mavlink::common::msg::RC_CHANNELS_OVERRIDE	600
mavlink::common::msg::RC_CHANNELS_RAW	603
mavlink::common::msg::RC_CHANNELS_SCALED	606
mavlink::common::msg::RELAY_STATUS	609
mavlink::common::msg::REQUEST_DATA_STREAM	611
mavlink::common::msg::REQUEST_EVENT	614
mavlink::common::msg::RESOURCE_REQUEST	616
mavlink::common::msg::RESPONSE_EVENT_ERROR	619
mavlink::common::msg::SAFETY_ALLOWED_AREA	621
mavlink::common::msg::SAFETY_SET_ALLOWED_AREA	624
mavlink::common::msg::SCALED_IMU	627
mavlink::common::msg::SCALED_IMU2	629
mavlink::common::msg::SCALED_IMU3	632
mavlink::common::msg::SCALED_PRESSURE	635
mavlink::common::msg::SCALED_PRESSURE2	637
mavlink::common::msg::SCALED_PRESSURE3	640
mavlink::common::msg::SERIAL_CONTROL	642
mavlink::common::msg::SERVO_OUTPUT_RAW	645
mavlink::common::msg::SETUP_SIGNING	668
mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET	649
mavlink::common::msg::SET_ATTITUDE_TARGET	651
mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN	654
mavlink::common::msg::SET_HOME_POSITION	657
mavlink::common::msg::SET_MODE	659
mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT	662
mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED	665
mavlink::common::msg::SIM_STATE	671
mavlink::common::msg::SMART_BATTERY_INFO	674
mavlink::common::msg::STATUSTEXT	676
mavlink::common::msg::STORAGE_INFORMATION	680
mavlink::common::msg::SUPPORTED_TUNES	682
mavlink::common::msg::SYSTEM_TIME	688
mavlink::common::msg::SYS_STATUS	685
mavlink::common::msg::TERRAIN_CHECK	691
mavlink::common::msg::TERRAIN_DATA	693
mavlink::common::msg::TERRAIN_REPORT	696
mavlink::common::msg::TERRAIN_REQUEST	698
mavlink::common::msg::TIMESYNC	704
mavlink::common::msg::TIME_ESTIMATE_TO_TARGET	701
mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER	706
mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS	709
mavlink::common::msg::TUNNEL	711

mavlink::common::msg::UAVCAN_NODE_INFO	715
mavlink::common::msg::UAVCAN_NODE_STATUS	717
mavlink::common::msg::UTM_GLOBAL_POSITION	722
mavlink::common::msg::V2_EXTENSION	725
mavlink::common::msg::VFR_HUD	727
mavlink::common::msg::VIBRATION	730
mavlink::common::msg::VICON_POSITION_ESTIMATE	732
mavlink::common::msg::VIDEO_STREAM_INFORMATION	735
mavlink::common::msg::VIDEO_STREAM_STATUS	738
mavlink::common::msg::VISION_POSITION_ESTIMATE	740
mavlink::common::msg::VISION_SPEED_ESTIMATE	743
mavlink::common::msg::WHEEL_DISTANCE	746
mavlink::common::msg::WIFI_CONFIG_AP	748
mavlink::common::msg::WINCH_STATUS	750
mavlink::common::msg::WIND_COV	753
mavlink::minimal::msg::HEARTBEAT	350
mavlink::standard::msg::AUTOPILOT_VERSION	151
mavlink::standard::msg::GLOBAL_POSITION_INT	317
message_sender	452
mavlink::MsgMap	490
packet_receiver	539
thumbstickControl	701
mavlink::impl::UIntBuffer< _T >	721
mavlink::impl::UIntBufferHelper< _N >	721
mavlink::impl::UIntBufferHelper< 1 >	721
mavlink::impl::UIntBufferHelper< 2 >	721
mavlink::impl::UIntBufferHelper< 4 >	722
mavlink::impl::UIntBufferHelper< 8 >	722

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

__mavlink_actuator_control_target_t	37
__mavlink_actuator_output_status_t	37
__mavlink_adsb_vehicle_t	38
__mavlink_airspeed_t	38
__mavlink_ais_vessel_t	38
__mavlink_altitude_t	39
__mavlink_att_pos_mocap_t	39
__mavlink_attitude_quaternion_cov_t	40
__mavlink_attitude_quaternion_t	40
__mavlink_attitude_t	40
__mavlink_attitude_target_t	41
__mavlink_auth_key_t	41
__mavlink_autopilot_version_t	41
__mavlink_available_modes_monitor_t	42
__mavlink_available_modes_t	42
__mavlink_battery_info_t	42
__mavlink_bitfield	43
__mavlink_button_change_t	43
__mavlink_camera_fov_status_t	43
__mavlink_camera_image_captured_t	44
__mavlink_camera_information_t	44
__mavlink_camera_thermal_range_t	45
__mavlink_camera_tracking_geo_status_t	45
__mavlink_camera_tracking_image_status_t	46
__mavlink_camera_trigger_t	46
__mavlink_can_filter_modify_t	46
__mavlink_can_frame_t	47
__mavlink_canfd_frame_t	47
__mavlink_cellular_config_t	47
__mavlink_cellular_status_t	48
__mavlink_change_operator_control_ack_t	48
__mavlink_change_operator_control_t	48
__mavlink_collision_t	49
__mavlink_command_ack_t	49
__mavlink_command_cancel_t	49

__mavlink_command_int_t	50
__mavlink_command_long_t	50
__mavlink_component_information_basic_t	51
__mavlink_component_information_t	51
__mavlink_component_metadata_t	51
__mavlink_control_system_state_t	52
__mavlink_current_event_sequence_t	52
__mavlink_current_mode_t	52
__mavlink_data_stream_t	53
__mavlink_data_transmission_handshake_t	53
__mavlink_debug_float_array_t	53
__mavlink_debug_t	54
__mavlink_debug_vect_t	54
__mavlink_encapsulated_data_t	54
__mavlink_esc_info_t	55
__mavlink_esc_status_t	55
__mavlink_estimator_status_t	55
__mavlink_event_t	56
__mavlink_extended_sys_state_t	56
__mavlink_fence_status_t	56
__mavlink_field_info	57
__mavlink_figure_eight_execution_status_t	57
__mavlink_file_transfer_protocol_t	57
__mavlink_flight_information_t	58
__mavlink_follow_target_t	58
__mavlink_fuel_status_t	58
__mavlink_generator_status_t	59
__mavlink_gimbal_device_attitude_status_t	59
__mavlink_gimbal_device_information_t	60
__mavlink_gimbal_device_set_attitude_t	60
__mavlink_gimbal_manager_information_t	61
__mavlink_gimbal_manager_set_attitude_t	61
__mavlink_gimbal_manager_set_manual_control_t	61
__mavlink_gimbal_manager_set_pitchyaw_t	62
__mavlink_gimbal_manager_status_t	62
__mavlink_global_position_int_cov_t	62
__mavlink_global_position_int_t	63
__mavlink_global_vision_position_estimate_t	63
__mavlink_gps2_rtk_t	63
__mavlink_gps_inject_data_t	64
__mavlink_gps_rtcn_data_t	64
__mavlink_gps_rtk_t	64
__mavlink_gps_status_t	65
__mavlink_heartbeat_t	65
__mavlink_high_latency2_t	65
__mavlink_high_latency_t	66
__mavlink_highres_imu_t	67
__mavlink_hil_actuator_controls_t	67
__mavlink_hil_controls_t	67
__mavlink_hil_optical_flow_t	68
__mavlink_hil_rc_inputs_raw_t	68
__mavlink_hil_sensor_t	69
__mavlink_hil_state_quaternion_t	69
__mavlink_hil_state_t	70
__mavlink_hygrometer_sensor_t	70
__mavlink_illuminator_status_t	70
__mavlink_isbd_link_status_t	71
__mavlink_link_node_status_t	71

__mavlink_local_position_ned_cov_t	71
__mavlink_local_position_ned_system_global_offset_t	72
__mavlink_local_position_ned_t	72
__mavlink_log_data_t	72
__mavlink_log_entry_t	73
__mavlink_log_erase_t	73
__mavlink_log_request_data_t	73
__mavlink_log_request_end_t	73
__mavlink_log_request_list_t	74
__mavlink_logging_ack_t	74
__mavlink_logging_data_acked_t	74
__mavlink_logging_data_t	75
__mavlink_manual_setpoint_t	75
__mavlink_memory_vect_t	75
__mavlink_message_info	76
__mavlink_message_interval_t	76
__mavlink_mission_ack_t	76
__mavlink_mission_clear_all_t	77
__mavlink_mission_item_int_t	77
__mavlink_mission_item_reached_t	77
__mavlink_mission_item_t	78
__mavlink_mission_request_int_t	78
__mavlink_mission_request_list_t	78
__mavlink_mission_request_partial_list_t	79
__mavlink_mission_request_t	79
__mavlink_mission_set_current_t	79
__mavlink_mission_write_partial_list_t	80
__mavlink_mount_orientation_t	80
__mavlink_msg_entry	80
__mavlink_named_value_float_t	81
__mavlink_named_value_int_t	81
__mavlink_nav_controller_output_t	81
__mavlink_odometry_t	82
__mavlink_onboard_computer_status_t	82
__mavlink_open_drone_id_arm_status_t	83
__mavlink_open_drone_id_authentication_t	83
__mavlink_open_drone_id_basic_id_t	83
__mavlink_open_drone_id_location_t	84
__mavlink_open_drone_id_message_pack_t	84
__mavlink_open_drone_id_operator_id_t	85
__mavlink_open_drone_id_self_id_t	85
__mavlink_open_drone_id_system_t	85
__mavlink_open_drone_id_system_update_t	86
__mavlink_optical_flow_rad_t	86
__mavlink_orbit_execution_status_t	86
__mavlink_param_error_t	87
__mavlink_param_ext_ack_t	87
__mavlink_param_ext_request_list_t	87
__mavlink_param_ext_request_read_t	87
__mavlink_param_ext_set_t	88
__mavlink_param_ext_value_t	88
__mavlink_param_map_rc_t	88
__mavlink_param_request_list_t	89
__mavlink_param_request_read_t	89
__mavlink_param_set_t	89
__mavlink_param_value_t	89
__mavlink_ping_t	90
__mavlink_play_tune_t	90

__mavlink_play_tune_v2_t	90
__mavlink_position_target_global_int_t	91
__mavlink_position_target_local_ned_t	91
__mavlink_power_status_t	92
__mavlink_protocol_version_t	92
__mavlink_radio_status_t	92
__mavlink_raw_pressure_t	93
__mavlink_raw_rpm_t	93
__mavlink_rc_channels_override_t	93
__mavlink_rc_channels_raw_t	94
__mavlink_rc_channels_scaled_t	94
__mavlink_rc_channels_t	95
__mavlink_relay_status_t	95
__mavlink_request_data_stream_t	96
__mavlink_request_event_t	96
__mavlink_resource_request_t	96
__mavlink_response_event_error_t	97
__mavlink_safety_allowed_area_t	97
__mavlink_safety_set_allowed_area_t	97
__mavlink_scaled_imu2_t	98
__mavlink_scaled_imu3_t	98
__mavlink_scaled_imu_t	99
__mavlink_scaled_pressure2_t	99
__mavlink_scaled_pressure3_t	99
__mavlink_scaled_pressure_t	100
__mavlink_serial_control_t	100
__mavlink_set_actuator_control_target_t	100
__mavlink_set_mode_t	101
__mavlink_set_position_target_global_int_t	101
__mavlink_set_position_target_local_ned_t	102
__mavlink_setup_signing_t	102
__mavlink_signing	103
__mavlink_signing_streams::__mavlink_signing_stream	103
__mavlink_signing_streams	104
__mavlink_sim_state_t	105
__mavlink_status	106
__mavlink_storage_information_t	107
__mavlink_supported_tunes_t	107
__mavlink_system_time_t	108
__mavlink_terrain_check_t	108
__mavlink_terrain_data_t	108
__mavlink_terrain_report_t	108
__mavlink_terrain_request_t	109
__mavlink_time_estimate_to_target_t	109
__mavlink_timesync_t	109
__mavlink_trajectory_representation_bezier_t	110
__mavlink_trajectory_representation_waypoints_t	110
__mavlink_tunnel_t	110
__mavlink_uavcan_node_info_t	111
__mavlink_uavcan_node_status_t	111
__mavlink_utm_global_position_t	111
__mavlink_v2_extension_t	112
__mavlink_vfr_hud_t	112
__mavlink_vibration_t	113
__mavlink_vicon_position_estimate_t	113
__mavlink_video_stream_information_t	113
__mavlink_video_stream_status_t	114
__mavlink_vision_position_estimate_t	114

__mavlink_vision_speed_estimate_t	114
__mavlink_wheel_distance_t	115
__mavlink_wifi_config_ap_t	115
__mavlink_winch_status_t	115
__mavlink_wind_cov_t	116
mavlink::common::msg::ACTUATOR_CONTROL_TARGET	
ACTUATOR_CONTROL_TARGET message	116
mavlink::common::msg::ACTUATOR_OUTPUT_STATUS	
ACTUATOR_OUTPUT_STATUS message	119
mavlink::common::msg::ADSB_VEHICLE	
ADSB_VEHICLE message	121
mavlink::common::msg::AIRSPEED	
AIRSPEED message	123
mavlink::common::msg::AIS_VESSEL	
AIS_VESSEL message	126
mavlink::common::msg::ALTITUDE	
ALTITUDE message	129
ATLAS_HC_ARM_DISARM_ACK	132
ATLAS_HC_HEARTBEAT_BC	132
ATLAS_HC_SYS_STAT	133
ATLAS_HC_TIMESYNC_RESP	133
mavlink::common::msg::ATT_POS_MOCAP	
ATT_POS_MOCAP message	133
mavlink::common::msg::ATTITUDE	
ATTITUDE message	136
mavlink::common::msg::ATTITUDE_QUATERNION	
ATTITUDE_QUATERNION message	138
mavlink::common::msg::ATTITUDE_QUATERNION_COV	
ATTITUDE_QUATERNION_COV message	141
mavlink::common::msg::ATTITUDE_TARGET	
ATTITUDE_TARGET message	143
mavlink::common::msg::AUTH_KEY	
AUTH_KEY message	146
mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE	
AUTOPILOT_STATE_FOR_GIMBAL_DEVICE message	148
mavlink::standard::msg::AUTOPILOT_VERSION	
AUTOPILOT_VERSION message	151
mavlink::common::msg::AVAILABLE_MODES	
AVAILABLE_MODES message	154
mavlink::common::msg::AVAILABLE_MODES_MONITOR	
AVAILABLE_MODES_MONITOR message	157
mavlink::common::msg::BATTERY_INFO	
BATTERY_INFO message	159
mavlink::common::msg::BATTERY_STATUS	
BATTERY_STATUS message	162
mavlink::common::msg::BUTTON_CHANGE	
BUTTON_CHANGE message	165
mavlink::common::msg::CAMERA_CAPTURE_STATUS	
CAMERA_CAPTURE_STATUS message	168
mavlink::common::msg::CAMERA_FOV_STATUS	
CAMERA_FOV_STATUS message	170
mavlink::common::msg::CAMERA_IMAGE_CAPTURED	
CAMERA_IMAGE_CAPTURED message	173
mavlink::common::msg::CAMERA_INFORMATION	
CAMERA_INFORMATION message	176
mavlink::common::msg::CAMERA_SETTINGS	
CAMERA_SETTINGS message	179

mavlink::common::msg::CAMERA_THERMAL_RANGE	
CAMERA_THERMAL_RANGE message	182
mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS	
CAMERA_TRACKING_GEO_STATUS message	184
mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS	
CAMERA_TRACKING_IMAGE_STATUS message	187
mavlink::common::msg::CAMERA_TRIGGER	
CAMERA_TRIGGER message	189
mavlink::common::msg::CAN_FILTER_MODIFY	
CAN_FILTER_MODIFY message	192
mavlink::common::msg::CAN_FRAME	
CAN_FRAME message	194
mavlink::common::msg::CANFD_FRAME	
CANFD_FRAME message	197
mavlink::common::msg::CELLULAR_CONFIG	
CELLULAR_CONFIG message	199
mavlink::common::msg::CELLULAR_STATUS	
CELLULAR_STATUS message	202
mavlink::common::msg::CHANGE_OPERATOR_CONTROL	
CHANGE_OPERATOR_CONTROL message	205
mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK	
CHANGE_OPERATOR_CONTROL_ACK message	207
mavlink::common::msg::COLLISION	
COLLISION message	210
mavlink::common::msg::COMMAND_ACK	
COMMAND_ACK message	212
mavlink::common::msg::COMMAND_CANCEL	
COMMAND_CANCEL message	215
mavlink::common::msg::COMMAND_INT	
COMMAND_INT message	217
mavlink::common::msg::COMMAND_LONG	
COMMAND_LONG message	220
mavlink::common::msg::COMPONENT_INFORMATION	
COMPONENT_INFORMATION message	223
mavlink::common::msg::COMPONENT_INFORMATION_BASIC	
COMPONENT_INFORMATION_BASIC message	226
mavlink::common::msg::COMPONENT_METADATA	
COMPONENT_METADATA message	228
mavlink::common::msg::CONTROL_SYSTEM_STATE	
CONTROL_SYSTEM_STATE message	231
mavlink::common::msg::CURRENT_EVENT_SEQUENCE	
CURRENT_EVENT_SEQUENCE message	233
mavlink::common::msg::CURRENT_MODE	
CURRENT_MODE message	236
mavlink::common::msg::DATA_STREAM	
DATA_STREAM message	238
mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE	
DATA_TRANSMISSION_HANDSHAKE message	241
mavlink::common::msg::DEBUG	
DEBUG message	243
mavlink::common::msg::DEBUG_FLOAT_ARRAY	
DEBUG_FLOAT_ARRAY message	246
mavlink::common::msg::DEBUG_VECT	
DEBUG_VECT message	248
mavlink::common::msg::DISTANCE_SENSOR	
DISTANCE_SENSOR message	251
mavlink::common::msg::EFI_STATUS	
EFI_STATUS message	253

mavlink::common::msg::ENCAPSULATED_DATA	
ENCAPSULATED_DATA message	256
mavlink::common::msg::ESC_INFO	
ESC_INFO message	259
mavlink::common::msg::ESC_STATUS	
ESC_STATUS message	261
mavlink::common::msg::ESTIMATOR_STATUS	
ESTIMATOR_STATUS message	265
mavlink::common::msg::EVENT	
EVENT message	268
mavlink::common::msg::EXTENDED_SYS_STATE	
EXTENDED_SYS_STATE message	271
mavlink::common::msg::FENCE_STATUS	
FENCE_STATUS message	274
mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS	
FIGURE_EIGHT_EXECUTION_STATUS message	276
mavlink::common::msg::FILE_TRANSFER_PROTOCOL	
FILE_TRANSFER_PROTOCOL message	279
mavlink::common::msg::FLIGHT_INFORMATION	
FLIGHT_INFORMATION message	281
mavlink::common::msg::FOLLOW_TARGET	
FOLLOW_TARGET message	285
mavlink::common::msg::FUEL_STATUS	
FUEL_STATUS message	287
mavlink::common::msg::GENERATOR_STATUS	
GENERATOR_STATUS message	290
mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS	
GIMBAL_DEVICE_ATTITUDE_STATUS message	293
mavlink::common::msg::GIMBAL_DEVICE_INFORMATION	
GIMBAL_DEVICE_INFORMATION message	296
mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE	
GIMBAL_DEVICE_SET_ATTITUDE message	300
mavlink::common::msg::GIMBAL_MANAGER_INFORMATION	
GIMBAL_MANAGER_INFORMATION message	303
mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE	
GIMBAL_MANAGER_SET_ATTITUDE message	306
mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL	
GIMBAL_MANAGER_SET_MANUAL_CONTROL message	309
mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW	
GIMBAL_MANAGER_SET_PITCHYAW message	311
mavlink::common::msg::GIMBAL_MANAGER_STATUS	
GIMBAL_MANAGER_STATUS message	314
mavlink::standard::msg::GLOBAL_POSITION_INT	
GLOBAL_POSITION_INT message	317
mavlink::common::msg::GLOBAL_POSITION_INT_COV	
GLOBAL_POSITION_INT_COV message	319
mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE	
GLOBAL_VISION_POSITION_ESTIMATE message	322
mavlink::common::msg::GPS2_RAW	
GPS2_RAW message	325
mavlink::common::msg::GPS2_RTK	
GPS2_RTK message	328
mavlink::common::msg::GPS_GLOBAL_ORIGIN	
GPS_GLOBAL_ORIGIN message	330
mavlink::common::msg::GPS_INJECT_DATA	
GPS_INJECT_DATA message	333
mavlink::common::msg::GPS_INPUT	
GPS_INPUT message	335

mavlink::common::msg::GPS_RAW_INT	
GPS_RAW_INT message	338
mavlink::common::msg::GPS_RTCM_DATA	
GPS_RTCM_DATA message	340
mavlink::common::msg::GPS_RTK	
GPS_RTK message	343
mavlink::common::msg::GPS_STATUS	
GPS_STATUS message	345
HC_ATLAS_ARM_DISARM_CMD	349
HC_ATLAS_HEARTBEAT_BC	349
HC_ATLAS_MANUAL_CONTROL_BC	349
HC_ATLAS_TIMESYNC_REQ	350
HC_RADIO_STATUS	350
mavlink::minimal::msg::HEARTBEAT	
HEARTBEAT message	350
mavlink::common::msg::HIGH_LATENCY	
HIGH_LATENCY message	353
mavlink::common::msg::HIGH_LATENCY2	
HIGH_LATENCY2 message	356
mavlink::common::msg::HIGHRES_IMU	
HIGHRES_IMU message	358
mavlink::common::msg::HIL_ACTUATOR_CONTROLS	
HIL_ACTUATOR_CONTROLS message	361
mavlink::common::msg::HIL_CONTROLS	
HIL_CONTROLS message	364
mavlink::common::msg::HIL_GPS	
HIL_GPS message	366
mavlink::common::msg::HIL_OPTICAL_FLOW	
HIL_OPTICAL_FLOW message	370
mavlink::common::msg::HIL_RC_INPUTS_RAW	
HIL_RC_INPUTS_RAW message	372
mavlink::common::msg::HIL_SENSOR	
HIL_SENSOR message	375
mavlink::common::msg::HIL_STATE	
HIL_STATE message	378
mavlink::common::msg::HIL_STATE_QUATERNION	
HIL_STATE_QUATERNION message	381
mavlink::common::msg::HOME_POSITION	
HOME_POSITION message	384
mavlink::common::msg::HYGROMETER_SENSOR	
HYGROMETER_SENSOR message	387
mavlink::common::msg::ILLUMINATOR_STATUS	
ILLUMINATOR_STATUS message	389
mavlink::Message::Info	392
mavlink::common::msg::ISBD_LINK_STATUS	
ISBD_LINK_STATUS message	392
mavlink::common::msg::LANDING_TARGET	
LANDING_TARGET message	395
mavlink::common::msg::LINK_NODE_STATUS	
LINK_NODE_STATUS message	397
mavlink::common::msg::LOCAL_POSITION_NED	
LOCAL_POSITION_NED message	400
mavlink::common::msg::LOCAL_POSITION_NED_COV	
LOCAL_POSITION_NED_COV message	403
mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET	
LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET message	405
mavlink::common::msg::LOG_DATA	
LOG_DATA message	408

mavlink::common::msg::LOG_ENTRY	
LOG_ENTRY message	411
mavlink::common::msg::LOG_ERASE	
LOG_ERASE message	413
mavlink::common::msg::LOG_REQUEST_DATA	
LOG_REQUEST_DATA message	416
mavlink::common::msg::LOG_REQUEST_END	
LOG_REQUEST_END message	418
mavlink::common::msg::LOG_REQUEST_LIST	
LOG_REQUEST_LIST message	421
mavlink::common::msg::LOGGING_ACK	
LOGGING_ACK message	423
mavlink::common::msg::LOGGING_DATA	
LOGGING_DATA message	426
mavlink::common::msg::LOGGING_DATA_ACKED	
LOGGING_DATA_ACKED message	428
mavlink::common::msg::MAG_CAL_REPORT	
MAG_CAL_REPORT message	431
mavlink::common::msg::MANUAL_CONTROL	
MANUAL_CONTROL message	433
mavlink::common::msg::MANUAL_SETPOINT	
MANUAL_SETPOINT message	436
mavlink_sha256_ctx	439
mavlink::common::msg::MEMORY_VECT	
MEMORY_VECT message	439
mavlink::Message	
MAVLink Message base class	442
mavlink::common::msg::MESSAGE_INTERVAL	
MESSAGE_INTERVAL message	449
message_sender	
Pack data and write MAVLink packets to buffer	452
mavlink::common::msg::MISSION_ACK	
MISSION_ACK message	454
mavlink::common::msg::MISSION_CLEAR_ALL	
MISSION_CLEAR_ALL message	457
mavlink::common::msg::MISSION_COUNT	
MISSION_COUNT message	459
mavlink::common::msg::MISSION_CURRENT	
MISSION_CURRENT message	461
mavlink::common::msg::MISSION_ITEM	
MISSION_ITEM message	464
mavlink::common::msg::MISSION_ITEM_INT	
MISSION_ITEM_INT message	467
mavlink::common::msg::MISSION_ITEM_REACHED	
MISSION_ITEM_REACHED message	470
mavlink::common::msg::MISSION_REQUEST	
MISSION_REQUEST message	472
mavlink::common::msg::MISSION_REQUEST_INT	
MISSION_REQUEST_INT message	474
mavlink::common::msg::MISSION_REQUEST_LIST	
MISSION_REQUEST_LIST message	478
mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST	
MISSION_REQUEST_PARTIAL_LIST message	480
mavlink::common::msg::MISSION_SET_CURRENT	
MISSION_SET_CURRENT message	483
mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST	
MISSION_WRITE_PARTIAL_LIST message	485

mavlink::common::msg::MOUNT_ORIENTATION	
MOUNT_ORIENTATION message	488
mavlink::MsgMap	
Serialization helper wrapper for mavlink_message_t	490
mavlink::common::msg::NAMED_VALUE_FLOAT	
NAMED_VALUE_FLOAT message	491
mavlink::common::msg::NAMED_VALUE_INT	
NAMED_VALUE_INT message	494
mavlink::common::msg::NAV_CONTROLLER_OUTPUT	
NAV_CONTROLLER_OUTPUT message	496
mavlink::common::msg::OBSTACLE_DISTANCE	
OBSTACLE_DISTANCE message	499
mavlink::common::msg::ODOMETRY	
ODOMETRY message	501
mavlink::common::msg::ONBOARD_COMPUTER_STATUS	
ONBOARD_COMPUTER_STATUS message	504
mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS	
OPEN_DRONE_ID_ARM_STATUS message	507
mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION	
OPEN_DRONE_ID_AUTHENTICATION message	510
mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID	
OPEN_DRONE_ID_BASIC_ID message	512
mavlink::common::msg::OPEN_DRONE_ID_LOCATION	
OPEN_DRONE_ID_LOCATION message	515
mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK	
OPEN_DRONE_ID_MESSAGE_PACK message	518
mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID	
OPEN_DRONE_ID_OPERATOR_ID message	520
mavlink::common::msg::OPEN_DRONE_ID_SELF_ID	
OPEN_DRONE_ID_SELF_ID message	523
mavlink::common::msg::OPEN_DRONE_ID_SYSTEM	
OPEN_DRONE_ID_SYSTEM message	525
mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE	
OPEN_DRONE_ID_SYSTEM_UPDATE message	528
mavlink::common::msg::OPTICAL_FLOW	
OPTICAL_FLOW message	531
mavlink::common::msg::OPTICAL_FLOW_RAD	
OPTICAL_FLOW_RAD message	533
mavlink::common::msg::ORBIT_EXECUTION_STATUS	
ORBIT_EXECUTION_STATUS message	536
packet_receiver	
Methods within this class is meant to be invoked by the packet handler class, in order to process all received packets	539
mavlink::common::msg::PARAM_ERROR	
PARAM_ERROR message	541
mavlink::common::msg::PARAM_EXT_ACK	
PARAM_EXT_ACK message	543
mavlink::common::msg::PARAM_EXT_REQUEST_LIST	
PARAM_EXT_REQUEST_LIST message	546
mavlink::common::msg::PARAM_EXT_REQUEST_READ	
PARAM_EXT_REQUEST_READ message	548
mavlink::common::msg::PARAM_EXT_SET	
PARAM_EXT_SET message	551
mavlink::common::msg::PARAM_EXT_VALUE	
PARAM_EXT_VALUE message	553
mavlink::common::msg::PARAM_MAP_RC	
PARAM_MAP_RC message	556

mavlink::common::msg::PARAM_REQUEST_LIST	
PARAM_REQUEST_LIST message	558
mavlink::common::msg::PARAM_REQUEST_READ	
PARAM_REQUEST_READ message	561
mavlink::common::msg::PARAM_SET	
PARAM_SET message	563
mavlink::common::msg::PARAM_VALUE	
PARAM_VALUE message	566
mavlink::common::msg::PING	
PING message	568
mavlink::common::msg::PLAY_TUNE	
PLAY_TUNE message	572
mavlink::common::msg::PLAY_TUNE_V2	
PLAY_TUNE_V2 message	574
mavlink::common::msg::POSITION_TARGET_GLOBAL_INT	
POSITION_TARGET_GLOBAL_INT message	577
mavlink::common::msg::POSITION_TARGET_LOCAL_NED	
POSITION_TARGET_LOCAL_NED message	579
mavlink::common::msg::POWER_STATUS	
POWER_STATUS message	582
mavlink::common::msg::PROTOCOL_VERSION	
PROTOCOL_VERSION message	585
mavlink::common::msg::RADIO_STATUS	
RADIO_STATUS message	587
mavlink::common::msg::RAW_IMU	
RAW_IMU message	590
mavlink::common::msg::RAW_PRESSURE	
RAW_PRESSURE message	592
mavlink::common::msg::RAW_RPM	
RAW_RPM message	595
mavlink::common::msg::RC_CHANNELS	
RC_CHANNELS message	597
mavlink::common::msg::RC_CHANNELS_OVERRIDE	
RC_CHANNELS_OVERRIDE message	600
mavlink::common::msg::RC_CHANNELS_RAW	
RC_CHANNELS_RAW message	603
mavlink::common::msg::RC_CHANNELS_SCALED	
RC_CHANNELS_SCALED message	606
mavlink::common::msg::RELAY_STATUS	
RELAY_STATUS message	609
mavlink::common::msg::REQUEST_DATA_STREAM	
REQUEST_DATA_STREAM message	611
mavlink::common::msg::REQUEST_EVENT	
REQUEST_EVENT message	614
mavlink::common::msg::RESOURCE_REQUEST	
RESOURCE_REQUEST message	616
mavlink::common::msg::RESPONSE_EVENT_ERROR	
RESPONSE_EVENT_ERROR message	619
mavlink::common::msg::SAFETY_ALLOWED_AREA	
SAFETY_ALLOWED_AREA message	621
mavlink::common::msg::SAFETY_SET_ALLOWED_AREA	
SAFETY_SET_ALLOWED_AREA message	624
mavlink::common::msg::SCALED_IMU	
SCALED_IMU message	627
mavlink::common::msg::SCALED_IMU2	
SCALED_IMU2 message	629
mavlink::common::msg::SCALED_IMU3	
SCALED_IMU3 message	632

mavlink::common::msg::SCALED_PRESSURE	
SCALED_PRESSURE message	635
mavlink::common::msg::SCALED_PRESSURE2	
SCALED_PRESSURE2 message	637
mavlink::common::msg::SCALED_PRESSURE3	
SCALED_PRESSURE3 message	640
mavlink::common::msg::SERIAL_CONTROL	
SERIAL_CONTROL message	642
mavlink::common::msg::SERVO_OUTPUT_RAW	
SERVO_OUTPUT_RAW message	645
mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET	
SET_ACTUATOR_CONTROL_TARGET message	649
mavlink::common::msg::SET_ATTITUDE_TARGET	
SET_ATTITUDE_TARGET message	651
mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN	
SET_GPS_GLOBAL_ORIGIN message	654
mavlink::common::msg::SET_HOME_POSITION	
SET_HOME_POSITION message	657
mavlink::common::msg::SET_MODE	
SET_MODE message	659
mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT	
SET_POSITION_TARGET_GLOBAL_INT message	662
mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED	
SET_POSITION_TARGET_LOCAL_NED message	665
mavlink::common::msg::SETUP_SIGNING	
SETUP_SIGNING message	668
mavlink::common::msg::SIM_STATE	
SIM_STATE message	671
mavlink::common::msg::SMART_BATTERY_INFO	
SMART_BATTERY_INFO message	674
mavlink::common::msg::STATUSTEXT	
STATUSTEXT message	676
mavlink::common::msg::STORAGE_INFORMATION	
STORAGE_INFORMATION message	680
mavlink::common::msg::SUPPORTED_TUNES	
SUPPORTED_TUNES message	682
mavlink::common::msg::SYS_STATUS	
SYS_STATUS message	685
mavlink::common::msg::SYSTEM_TIME	
SYSTEM_TIME message	688
mavlink::common::msg::TERRAIN_CHECK	
TERRAIN_CHECK message	691
mavlink::common::msg::TERRAIN_DATA	
TERRAIN_DATA message	693
mavlink::common::msg::TERRAIN_REPORT	
TERRAIN_REPORT message	696
mavlink::common::msg::TERRAIN_REQUEST	
TERRAIN_REQUEST message	698
thumbstickControl	
Thumbstick control is used within this library	701
mavlink::common::msg::TIME_ESTIMATE_TO_TARGET	
TIME_ESTIMATE_TO_TARGET message	701
mavlink::common::msg::TIMESYNC	
TIMESYNC message	704
mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER	
TRAJECTORY_REPRESENTATION_BEZIER message	706
mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS	
TRAJECTORY_REPRESENTATION_WAYPOINTS message	709

mavlink::common::msg::TUNNEL	
TUNNEL message	711
mavlink::common::msg::UAVCAN_NODE_INFO	
UAVCAN_NODE_INFO message	715
mavlink::common::msg::UAVCAN_NODE_STATUS	
UAVCAN_NODE_STATUS message	717
mavlink::impl::UIntBuffer<_T>	721
mavlink::impl::UIntBufferHelper<_N>	721
mavlink::impl::UIntBufferHelper< 1 >	721
mavlink::impl::UIntBufferHelper< 2 >	721
mavlink::impl::UIntBufferHelper< 4 >	722
mavlink::impl::UIntBufferHelper< 8 >	722
mavlink::common::msg::UTM_GLOBAL_POSITION	
UTM_GLOBAL_POSITION message	722
mavlink::common::msg::V2_EXTENSION	
V2_EXTENSION message	725
mavlink::common::msg::VFR_HUD	
VFR_HUD message	727
mavlink::common::msg::VIBRATION	
VIBRATION message	730
mavlink::common::msg::VICON_POSITION_ESTIMATE	
VICON_POSITION_ESTIMATE message	732
mavlink::common::msg::VIDEO_STREAM_INFORMATION	
VIDEO_STREAM_INFORMATION message	735
mavlink::common::msg::VIDEO_STREAM_STATUS	
VIDEO_STREAM_STATUS message	738
mavlink::common::msg::VISION_POSITION_ESTIMATE	
VISION_POSITION_ESTIMATE message	740
mavlink::common::msg::VISION_SPEED_ESTIMATE	
VISION_SPEED_ESTIMATE message	743
mavlink::common::msg::WHEEL_DISTANCE	
WHEEL_DISTANCE message	746
mavlink::common::msg::WIFI_CONFIG_AP	
WIFI_CONFIG_AP message	748
mavlink::common::msg::WINCH_STATUS	
WINCH_STATUS message	750
mavlink::common::msg::WIND_COV	
WIND_COV message	753

Chapter 4

File Index

4.1 File List

Here is a list of all documented files with brief descriptions:

single_hand_controller/displayHandler.cpp	757
single_hand_controller/IOhandler.cpp	878
single_hand_controller/message_sender.cpp	880
single_hand_controller/packet_receiver.cpp	881
single_hand_controller/packetHandler.cpp	881
single_hand_controller/setupFunctions.cpp	883
single_hand_controller/stateHandler.cpp	885
single_hand_controller/timer.cpp	886
single_hand_controller/include/definitions.h	759
single_hand_controller/include/displayHandler.hpp	761
single_hand_controller/include/includes.h	763
single_hand_controller/include/IOhandler.hpp	763
single_hand_controller/include/message_sender.hpp	868
single_hand_controller/include/message_structs.h	
This header file is part of the Single Hand Controller for Scout project; Defines all message structs being sent and received by the Single Hand Controller	869
single_hand_controller/include/packet_receiver.hpp	??
single_hand_controller/include/packetHandler.h	871
single_hand_controller/include/setupFunctions.h	
Defines all functions required during setup phase of hand controller	873
single_hand_controller/include/stateHandler.hpp	875
single_hand_controller/include/timer.hpp	877
single_hand_controller/include/mavlink/checksum.h	??
single_hand_controller/include/mavlink/mavlink_conversions.h	
These conversion functions follow the NASA rotation standards definition file available online	849
single_hand_controller/include/mavlink/mavlink_get_info.h	??
single_hand_controller/include/mavlink/mavlink_helpers.h	??
single_hand_controller/include/mavlink/mavlink_sha256.h	??
single_hand_controller/include/mavlink/mavlink_types.h	??
single_hand_controller/include/mavlink/message.hpp	??
single_hand_controller/include/mavlink/msgmap.hpp	??
single_hand_controller/include/mavlink/protocol.h	??
single_hand_controller/include/mavlink/common/common.h	
MAVLink comm protocol generated from common.xml	766
single_hand_controller/include/mavlink/common/common.hpp	
MAVLink comm protocol generated from common.xml	803

single_hand_controller/include/mavlink/common/gtestsuite.hpp	
MAVLink comm testsuite protocol generated from common.xml	835
single_hand_controller/include/mavlink/common/mavlink.h	
MAVLink comm protocol built from common.xml	841
single_hand_controller/include/mavlink/common/mavlink_msg_actuator_control_target.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_actuator_control_target.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_actuator_output_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_actuator_output_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_adsb_vehicle.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_adsb_vehicle.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_airspeed.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_airspeed.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_ais_vessel.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_ais_vessel.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_altitude.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_altitude.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_att_pos_mocap.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_att_pos_mocap.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion_cov.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion_cov.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude_target.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_attitude_target.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_auth_key.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_auth_key.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_autopilot_state_for_gimbal_device.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_autopilot_state_for_gimbal_device.h	??
hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_available_modes.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_available_modes.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_available_modes_monitor.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_available_modes_monitor.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_battery_info.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_battery_info.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_battery_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_battery_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_button_change.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_button_change.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_capture_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_capture_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_fov_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_fov_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_image_captured.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_image_captured.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_information.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_information.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_settings.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_settings.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_thermal_range.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_thermal_range.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_geo_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_geo_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_image_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_image_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_camera_trigger.h	??

single_hand_controller/include/mavlink/common/mavlink_msg_camera_trigger.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_can_filter_modify.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_can_filter_modify.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_can_frame.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_can_frame.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_canfd_frame.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_canfd_frame.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_cellular_config.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_cellular_config.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_cellular_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_cellular_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control_ack.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control_ack.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_collision.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_collision.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_ack.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_ack.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_cancel.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_cancel.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_int.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_int.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_long.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_command_long.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_component_information.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_component_information.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_component_information_basic.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_component_information_basic.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_component_metadata.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_component_metadata.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_control_system_state.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_control_system_state.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_current_event_sequence.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_current_event_sequence.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_current_mode.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_current_mode.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_data_stream.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_data_stream.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_data_transmission_handshake.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_data_transmission_handshake.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_debug.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_debug.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_debug_float_array.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_debug_float_array.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_debug_vect.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_debug_vect.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_distance_sensor.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_distance_sensor.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_efi_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_efi_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_encapsulated_data.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_encapsulated_data.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_esc_info.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_esc_info.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_esc_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_esc_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_estimator_status.h	??

single_hand_controller/include/mavlink/common/mavlink_msg_estimator_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_event.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_event.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_extended_sys_state.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_extended_sys_state.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_fence_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_fence_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_figure_eight_execution_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_figure_eight_execution_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_file_transfer_protocol.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_file_transfer_protocol.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_flight_information.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_flight_information.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_follow_target.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_follow_target.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_fuel_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_fuel_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_generator_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_generator_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_attitude_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_attitude_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_information.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_information.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_set_attitude.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_set_attitude.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_information.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_information.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_attitude.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_attitude.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_manual ↵ control.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_manual ↵ control.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_pitchyaw.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_pitchyaw.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_global_position_int_cov.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_global_position_int_cov.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_global_vision_position_estimate.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_global_vision_position_estimate.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps2_raw.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps2_raw.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps2_rtk.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps2_rtk.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_global_origin.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_global_origin.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_inject_data.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_inject_data.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_input.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_input.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_raw_int.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_raw_int.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtcn_data.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtcn_data.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtk.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtk.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_gps_status.h	??

single_hand_controller/include/mavlink/common/mavlink_msg_gps_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_high_latency.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_high_latency.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_high_latency2.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_high_latency2.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_highres_imu.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_highres_imu.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_actuator_controls.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_actuator_controls.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_controls.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_controls.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_gps.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_gps.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_optical_flow.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_optical_flow.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_rc_inputs_raw.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_rc_inputs_raw.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_sensor.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_sensor.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_state.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_state.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_state_quaternion.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hil_state_quaternion.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_home_position.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_home_position.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_hygrometer_sensor.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_hygrometer_sensor.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_illuminator_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_illuminator_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_isbd_link_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_isbd_link_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_landing_target.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_landing_target.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_link_node_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_link_node_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_cov.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_cov.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_system_global← _offset.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_system_global← _offset.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_data.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_data.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_entry.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_entry.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_erase.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_erase.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_request_data.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_request_data.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_request_end.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_request_end.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_request_list.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_log_request_list.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_logging_ack.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_logging_ack.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_logging_data.h	??

single_hand_controller/include/mavlink/common/mavlink_msg_logging_data.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_logging_data_acked.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_logging_data_acked.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mag_cal_report.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mag_cal_report.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_manual_control.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_manual_control.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_manual_setpoint.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_manual_setpoint.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_memory_vect.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_memory_vect.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_message_interval.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_message_interval.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_ack.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_ack.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_clear_all.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_clear_all.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_count.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_count.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_current.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_current.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_item.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_item.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_int.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_int.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_reached.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_reached.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_int.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_int.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_list.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_list.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_partial_list.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_partial_list.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_set_current.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_set_current.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_write_partial_list.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mission_write_partial_list.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_mount_orientation.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_mount_orientation.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_named_value_float.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_named_value_float.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_named_value_int.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_named_value_int.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_nav_controller_output.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_nav_controller_output.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_obstacle_distance.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_obstacle_distance.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_odometry.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_odometry.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_onboard_computer_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_onboard_computer_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_arm_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_arm_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_authentication.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_authentication.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_basic_id.h	??

single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_basic_id.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_location.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_location.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_message_pack.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_message_pack.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_operator_id.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_operator_id.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_self_id.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_self_id.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system_update.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system_update.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_optical_flow.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_optical_flow.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_optical_flow_rad.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_optical_flow_rad.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_orbit_execution_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_orbit_execution_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_error.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_error.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_ack.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_ack.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_list.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_list.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_read.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_read.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_set.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_set.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_value.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_value.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_map_rc.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_map_rc.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_request_list.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_request_list.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_request_read.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_request_read.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_set.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_set.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_value.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_param_value.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_ping.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_ping.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_play_tune.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_play_tune.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_play_tune_v2.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_play_tune_v2.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_position_target_global_int.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_position_target_global_int.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_position_target_local_ned.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_position_target_local_ned.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_power_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_power_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_protocol_version.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_protocol_version.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_radio_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_radio_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_raw_imu.h	??

single_hand_controller/include/mavlink/common/mavlink_msg_raw_imu.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_raw_pressure.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_raw_pressure.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_raw_rpm.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_raw_rpm.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_override.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_override.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_raw.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_raw.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_scaled.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_scaled.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_relay_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_relay_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_request_data_stream.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_request_data_stream.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_request_event.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_request_event.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_resource_request.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_resource_request.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_response_event_error.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_response_event_error.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_safety_allowed_area.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_safety_allowed_area.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_safety_set_allowed_area.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_safety_set_allowed_area.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu2.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu2.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu3.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu3.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure2.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure2.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure3.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure3.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_serial_control.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_serial_control.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_servo_output_raw.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_servo_output_raw.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_actuator_control_target.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_actuator_control_target.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_attitude_target.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_attitude_target.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_gps_global_origin.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_gps_global_origin.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_home_position.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_home_position.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_mode.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_mode.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_global_int.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_global_int.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_local_ned.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_local_ned.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_setup_signing.h	??

single_hand_controller/include/mavlink/common/mavlink_msg_setup_signing.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_sim_state.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_sim_state.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_smart_battery_info.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_smart_battery_info.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_statustext.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_statustext.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_storage_information.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_storage_information.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_supported_tunes.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_supported_tunes.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_sys_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_sys_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_system_time.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_system_time.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_check.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_check.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_data.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_data.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_report.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_report.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_request.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_terrain_request.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_time_estimate_to_target.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_time_estimate_to_target.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_timesync.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_timesync.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_bezier.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_bezier.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_waypoints.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_waypoints.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_waypoints.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_tunnel.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_tunnel.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_info.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_info.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_utm_global_position.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_utm_global_position.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_v2_extension.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_v2_extension.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_vfr_hud.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_vfr_hud.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_vibration.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_vibration.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_vicon_position_estimate.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_vicon_position_estimate.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_information.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_information.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_status.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_status.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_vision_position_estimate.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_vision_position_estimate.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_vision_speed_estimate.h	??
single_hand_controller/include/mavlink/common/mavlink_msg_vision_speed_estimate.hpp	??
single_hand_controller/include/mavlink/common/mavlink_msg_wheel_distance.h	??

single_hand_controller/include/mavlink/common/ mavlink_msg_wheel_distance.hpp	??
single_hand_controller/include/mavlink/common/ mavlink_msg_wifi_config_ap.h	??
single_hand_controller/include/mavlink/common/ mavlink_msg_wifi_config_ap.hpp	??
single_hand_controller/include/mavlink/common/ mavlink_msg_winch_status.h	??
single_hand_controller/include/mavlink/common/ mavlink_msg_winch_status.hpp	??
single_hand_controller/include/mavlink/common/ mavlink_msg_wind_cov.h	??
single_hand_controller/include/mavlink/common/ mavlink_msg_wind_cov.hpp	??
single_hand_controller/include/mavlink/common/ testsuite.h	
MAVLink comm protocol testsuite generated from common.xml	844
single_hand_controller/include/mavlink/common/ version.h	
MAVLink comm protocol built from common.xml	846
single_hand_controller/include/mavlink/minimal/ gtestsuite.hpp	
MAVLink comm testsuite protocol generated from minimal.xml	840
single_hand_controller/include/mavlink/minimal/ mavlink.h	
MAVLink comm protocol built from minimal.xml	842
single_hand_controller/include/mavlink/minimal/ mavlink_msg_heartbeat.h	??
single_hand_controller/include/mavlink/minimal/ mavlink_msg_heartbeat.hpp	??
single_hand_controller/include/mavlink/minimal/ minimal.h	
MAVLink comm protocol generated from minimal.xml	853
single_hand_controller/include/mavlink/minimal/ minimal.hpp	
MAVLink comm protocol generated from minimal.xml	859
single_hand_controller/include/mavlink/minimal/ testsuite.h	
MAVLink comm protocol testsuite generated from minimal.xml	844
single_hand_controller/include/mavlink/minimal/ version.h	
MAVLink comm protocol built from minimal.xml	847
single_hand_controller/include/mavlink/standard/ gtestsuite.hpp	
MAVLink comm testsuite protocol generated from standard.xml	840
single_hand_controller/include/mavlink/standard/ mavlink.h	
MAVLink comm protocol built from standard.xml	843
single_hand_controller/include/mavlink/standard/ mavlink_msg_autopilot_version.h	??
single_hand_controller/include/mavlink/standard/ mavlink_msg_autopilot_version.hpp	??
single_hand_controller/include/mavlink/standard/ mavlink_msg_global_position_int.h	??
single_hand_controller/include/mavlink/standard/ mavlink_msg_global_position_int.hpp	??
single_hand_controller/include/mavlink/standard/ standard.h	
MAVLink comm protocol generated from standard.xml	863
single_hand_controller/include/mavlink/standard/ standard.hpp	
MAVLink comm protocol generated from standard.xml	866
single_hand_controller/include/mavlink/standard/ testsuite.h	
MAVLink comm protocol testsuite generated from standard.xml	845
single_hand_controller/include/mavlink/standard/ version.h	
MAVLink comm protocol built from standard.xml	848

Chapter 5

Class Documentation

5.1 `__mavlink_actuator_control_target_t` Struct Reference

Public Attributes

- `uint64_t` **time_usec**
- `float` **controls** [8]
- `uint8_t` **group_mlx**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_actuator_control_target.h`

5.2 `__mavlink_actuator_output_status_t` Struct Reference

Public Attributes

- `uint64_t` **time_usec**
- `uint32_t` **active**
- `float` **actuator** [32]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_actuator_output_status.h`

5.3 `__mavlink_adsb_vehicle_t` Struct Reference

Public Attributes

- `uint32_t ICAO_address`
- `int32_t lat`
- `int32_t lon`
- `int32_t altitude`
- `uint16_t heading`
- `uint16_t hor_velocity`
- `int16_t ver_velocity`
- `uint16_t flags`
- `uint16_t squawk`
- `uint8_t altitude_type`
- `char callsign [9]`
- `uint8_t emitter_type`
- `uint8_t tslc`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_adsb_vehicle.h`

5.4 `__mavlink_airspeed_t` Struct Reference

Public Attributes

- `float airspeed`
- `float raw_press`
- `int16_t temperature`
- `uint8_t id`
- `uint8_t flags`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_airspeed.h`

5.5 `__mavlink_ais_vessel_t` Struct Reference

Public Attributes

- `uint32_t MMSI`
- `int32_t lat`
- `int32_t lon`
- `uint16_t COG`
- `uint16_t heading`
- `uint16_t velocity`
- `uint16_t dimension_bow`
- `uint16_t dimension_stern`

- `uint16_t` **tslc**
- `uint16_t` **flags**
- `int8_t` **turn_rate**
- `uint8_t` **navigational_status**
- `uint8_t` **type**
- `uint8_t` **dimension_port**
- `uint8_t` **dimension_starboard**
- `char` **callsign** [7]
- `char` **name** [20]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_ais_vessel.h`

5.6 `__mavlink_altitude_t` Struct Reference

Public Attributes

- `uint64_t` **time_usec**
- `float` **altitude_monotonic**
- `float` **altitude_amsl**
- `float` **altitude_local**
- `float` **altitude_relative**
- `float` **altitude_terrain**
- `float` **bottom_clearance**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_altitude.h`

5.7 `__mavlink_att_pos_mocap_t` Struct Reference

Public Attributes

- `uint64_t` **time_usec**
- `float` **q** [4]
- `float` **x**
- `float` **y**
- `float` **z**
- `float` **covariance** [21]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_att_pos_mocap.h`

5.8 `__mavlink_attitude_quaternion_cov_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float q [4]`
- `float rollspeed`
- `float pitchspeed`
- `float yawspeed`
- `float covariance [9]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion_cov.h`

5.9 `__mavlink_attitude_quaternion_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float q1`
- `float q2`
- `float q3`
- `float q4`
- `float rollspeed`
- `float pitchspeed`
- `float yawspeed`
- `float repr_offset_q [4]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion.h`

5.10 `__mavlink_attitude_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float roll`
- `float pitch`
- `float yaw`
- `float rollspeed`
- `float pitchspeed`
- `float yawspeed`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_attitude.h`

5.11 __mavlink_attitude_target_t Struct Reference

Public Attributes

- uint32_t **time_boot_ms**
- float **q** [4]
- float **body_roll_rate**
- float **body_pitch_rate**
- float **body_yaw_rate**
- float **thrust**
- uint8_t **type_mask**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_attitude_target.h

5.12 __mavlink_auth_key_t Struct Reference

Public Attributes

- char **key** [32]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_auth_key.h

5.13 __mavlink_autopilot_version_t Struct Reference

Public Attributes

- uint64_t **capabilities**
- uint64_t **uid**
- uint32_t **flight_sw_version**
- uint32_t **middleware_sw_version**
- uint32_t **os_sw_version**
- uint32_t **board_version**
- uint16_t **vendor_id**
- uint16_t **product_id**
- uint8_t **flight_custom_version** [8]
- uint8_t **middleware_custom_version** [8]
- uint8_t **os_custom_version** [8]
- uint8_t **uid2** [18]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/standard/mavlink_msg_autopilot_version.h

5.14 `__mavlink_available_modes_monitor_t` Struct Reference

Public Attributes

- `uint8_t seq`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_available_modes_monitor.h`

5.15 `__mavlink_available_modes_t` Struct Reference

Public Attributes

- `uint32_t custom_mode`
- `uint32_t properties`
- `uint8_t number_modes`
- `uint8_t mode_index`
- `uint8_t standard_mode`
- `char mode_name [35]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_available_modes.h`

5.16 `__mavlink_battery_info_t` Struct Reference

Public Attributes

- `float discharge_minimum_voltage`
- `float charging_minimum_voltage`
- `float resting_minimum_voltage`
- `float charging_maximum_voltage`
- `float charging_maximum_current`
- `float nominal_voltage`
- `float discharge_maximum_current`
- `float discharge_maximum_burst_current`
- `float design_capacity`
- `float full_charge_capacity`
- `uint16_t cycle_count`
- `uint16_t weight`
- `uint8_t id`
- `uint8_t battery_function`
- `uint8_t type`
- `uint8_t state_of_health`
- `uint8_t cells_in_series`
- `char manufacture_date [9]`
- `char serial_number [32]`
- `char name [50]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_battery_info.h`

5.17 `__mavlink_bitfield` Union Reference

Public Attributes

- `uint8_t uint8`
- `int8_t int8`
- `uint16_t uint16`
- `int16_t int16`
- `uint32_t uint32`
- `int32_t int32`

The documentation for this union was generated from the following file:

- `single_hand_controller/include/mavlink/mavlink_helpers.h`

5.18 `__mavlink_button_change_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `uint32_t last_change_ms`
- `uint8_t state`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_button_change.h`

5.19 `__mavlink_camera_fov_status_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `int32_t lat_camera`
- `int32_t lon_camera`
- `int32_t alt_camera`
- `int32_t lat_image`
- `int32_t lon_image`
- `int32_t alt_image`
- `float q [4]`
- `float hfov`
- `float vfov`
- `uint8_t camera_device_id`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_fov_status.h`

5.20 __mavlink_camera_image_captured_t Struct Reference

Public Attributes

- uint64_t **time_utc**
- uint32_t **time_boot_ms**
- int32_t **lat**
- int32_t **lon**
- int32_t **alt**
- int32_t **relative_alt**
- float **q** [4]
- int32_t **image_index**
- uint8_t **camera_id**
- int8_t **capture_result**
- char **file_url** [205]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_camera_image_captured.h

5.21 __mavlink_camera_information_t Struct Reference

Public Attributes

- uint32_t **time_boot_ms**
- uint32_t **firmware_version**
- float **focal_length**
- float **sensor_size_h**
- float **sensor_size_v**
- uint32_t **flags**
- uint16_t **resolution_h**
- uint16_t **resolution_v**
- uint16_t **cam_definition_version**
- uint8_t **vendor_name** [32]
- uint8_t **model_name** [32]
- uint8_t **lens_id**
- char **cam_definition_uri** [140]
- uint8_t **gimbal_device_id**
- uint8_t **camera_device_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_camera_information.h

5.22 `__mavlink_camera_thermal_range_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float max`
- `float max_point_x`
- `float max_point_y`
- `float min`
- `float min_point_x`
- `float min_point_y`
- `uint8_t stream_id`
- `uint8_t camera_device_id`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_thermal_range.h`

5.23 `__mavlink_camera_tracking_geo_status_t` Struct Reference

Public Attributes

- `int32_t lat`
- `int32_t lon`
- `float alt`
- `float h_acc`
- `float v_acc`
- `float vel_n`
- `float vel_e`
- `float vel_d`
- `float vel_acc`
- `float dist`
- `float hdg`
- `float hdg_acc`
- `uint8_t tracking_status`
- `uint8_t camera_device_id`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_geo_status.h`

5.24 `__mavlink_camera_tracking_image_status_t` Struct Reference

Public Attributes

- float **point_x**
- float **point_y**
- float **radius**
- float **rec_top_x**
- float **rec_top_y**
- float **rec_bottom_x**
- float **rec_bottom_y**
- uint8_t **tracking_status**
- uint8_t **tracking_mode**
- uint8_t **target_data**
- uint8_t **camera_device_id**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_image_status.h`

5.25 `__mavlink_camera_trigger_t` Struct Reference

Public Attributes

- uint64_t **time_usec**
- uint32_t **seq**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_trigger.h`

5.26 `__mavlink_can_filter_modify_t` Struct Reference

Public Attributes

- uint16_t **ids** [16]
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **bus**
- uint8_t **operation**
- uint8_t **num_ids**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_can_filter_modify.h`

5.27 `__mavlink_can_frame_t` Struct Reference

Public Attributes

- `uint32_t id`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t bus`
- `uint8_t len`
- `uint8_t data [8]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_can_frame.h`

5.28 `__mavlink_canfd_frame_t` Struct Reference

Public Attributes

- `uint32_t id`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t bus`
- `uint8_t len`
- `uint8_t data [64]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_canfd_frame.h`

5.29 `__mavlink_cellular_config_t` Struct Reference

Public Attributes

- `uint8_t enable_lte`
- `uint8_t enable_pin`
- `char pin [16]`
- `char new_pin [16]`
- `char apn [32]`
- `char puk [16]`
- `uint8_t roaming`
- `uint8_t response`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_cellular_config.h`

5.30 `__mavlink_cellular_status_t` Struct Reference

Public Attributes

- `uint16_t mcc`
- `uint16_t mnc`
- `uint16_t lac`
- `uint8_t status`
- `uint8_t failure_reason`
- `uint8_t type`
- `uint8_t quality`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_cellular_status.h`

5.31 `__mavlink_change_operator_control_ack_t` Struct Reference

Public Attributes

- `uint8_t gcs_system_id`
- `uint8_t control_request`
- `uint8_t ack`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control_ack.h`

5.32 `__mavlink_change_operator_control_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t control_request`
- `uint8_t version`
- `char passkey [25]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control.h`

5.33 `__mavlink_collision_t` Struct Reference

Public Attributes

- `uint32_t id`
- `float time_to_minimum_delta`
- `float altitude_minimum_delta`
- `float horizontal_minimum_delta`
- `uint8_t src`
- `uint8_t action`
- `uint8_t threat_level`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_collision.h`

5.34 `__mavlink_command_ack_t` Struct Reference

Public Attributes

- `uint16_t command`
- `uint8_t result`
- `uint8_t progress`
- `int32_t result_param2`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_command_ack.h`

5.35 `__mavlink_command_cancel_t` Struct Reference

Public Attributes

- `uint16_t command`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_command_cancel.h`

5.36 `__mavlink_command_int_t` Struct Reference

Public Attributes

- float **param1**
- float **param2**
- float **param3**
- float **param4**
- int32_t **x**
- int32_t **y**
- float **z**
- uint16_t **command**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **frame**
- uint8_t **current**
- uint8_t **autocontinue**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_command_int.h`

5.37 `__mavlink_command_long_t` Struct Reference

Public Attributes

- float **param1**
- float **param2**
- float **param3**
- float **param4**
- float **param5**
- float **param6**
- float **param7**
- uint16_t **command**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **confirmation**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_command_long.h`

5.38 `__mavlink_component_information_basic_t` Struct Reference

Public Attributes

- `uint64_t capabilities`
- `uint32_t time_boot_ms`
- `uint32_t time_manufacture_s`
- `char vendor_name [32]`
- `char model_name [32]`
- `char software_version [24]`
- `char hardware_version [24]`
- `char serial_number [32]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_component_information_basic.h`

5.39 `__mavlink_component_information_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `uint32_t general_metadata_file_crc`
- `uint32_t peripherals_metadata_file_crc`
- `char general_metadata_uri [100]`
- `char peripherals_metadata_uri [100]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_component_information.h`

5.40 `__mavlink_component_metadata_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `uint32_t file_crc`
- `char uri [100]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_component_metadata.h`

5.41 `__mavlink_control_system_state_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float x_acc`
- `float y_acc`
- `float z_acc`
- `float x_vel`
- `float y_vel`
- `float z_vel`
- `float x_pos`
- `float y_pos`
- `float z_pos`
- `float airspeed`
- `float vel_variance` [3]
- `float pos_variance` [3]
- `float q` [4]
- `float roll_rate`
- `float pitch_rate`
- `float yaw_rate`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_control_system_state.h`

5.42 `__mavlink_current_event_sequence_t` Struct Reference

Public Attributes

- `uint16_t sequence`
- `uint8_t flags`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_current_event_sequence.h`

5.43 `__mavlink_current_mode_t` Struct Reference

Public Attributes

- `uint32_t custom_mode`
- `uint32_t intended_custom_mode`
- `uint8_t standard_mode`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_current_mode.h`

5.44 __mavlink_data_stream_t Struct Reference

Public Attributes

- uint16_t **message_rate**
- uint8_t **stream_id**
- uint8_t **on_off**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_data_stream.h

5.45 __mavlink_data_transmission_handshake_t Struct Reference

Public Attributes

- uint32_t **size**
- uint16_t **width**
- uint16_t **height**
- uint16_t **packets**
- uint8_t **type**
- uint8_t **payload**
- uint8_t **jpg_quality**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_data_transmission_handshake.h

5.46 __mavlink_debug_float_array_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- uint16_t **array_id**
- char **name** [10]
- float **data** [58]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_debug_float_array.h

5.47 `__mavlink_debug_t` Struct Reference

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **value**
- `uint8_t` **ind**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_debug.h`

5.48 `__mavlink_debug_vect_t` Struct Reference

Public Attributes

- `uint64_t` **time_usec**
- `float` **x**
- `float` **y**
- `float` **z**
- `char` **name** [10]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_debug_vect.h`

5.49 `__mavlink_encapsulated_data_t` Struct Reference

Public Attributes

- `uint16_t` **seqnr**
- `uint8_t` **data** [253]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_encapsulated_data.h`

5.50 __mavlink_esc_info_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- uint32_t **error_count** [4]
- uint16_t **counter**
- uint16_t **failure_flags** [4]
- int16_t **temperature** [4]
- uint8_t **index**
- uint8_t **count**
- uint8_t **connection_type**
- uint8_t **info**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_esc_info.h

5.51 __mavlink_esc_status_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- int32_t **rpm** [4]
- float **voltage** [4]
- float **current** [4]
- uint8_t **index**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_esc_status.h

5.52 __mavlink_estimator_status_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **vel_ratio**
- float **pos_horiz_ratio**
- float **pos_vert_ratio**
- float **mag_ratio**
- float **hagl_ratio**
- float **tas_ratio**
- float **pos_horiz_accuracy**
- float **pos_vert_accuracy**
- uint16_t **flags**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_estimator_status.h

5.53 `__mavlink_event_t` Struct Reference

Public Attributes

- `uint32_t id`
- `uint32_t event_time_boot_ms`
- `uint16_t sequence`
- `uint8_t destination_component`
- `uint8_t destination_system`
- `uint8_t log_levels`
- `uint8_t arguments [40]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_event.h`

5.54 `__mavlink_extended_sys_state_t` Struct Reference

Public Attributes

- `uint8_t vtol_state`
- `uint8_t landed_state`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_extended_sys_state.h`

5.55 `__mavlink_fence_status_t` Struct Reference

Public Attributes

- `uint32_t breach_time`
- `uint16_t breach_count`
- `uint8_t breach_status`
- `uint8_t breach_type`
- `uint8_t breach_mitigation`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_fence_status.h`

5.56 __mavlink_field_info Struct Reference

Public Attributes

- const char * **name**
- const char * **print_format**
- mavlink_message_type_t **type**
- unsigned int **array_length**
- unsigned int **wire_offset**
- unsigned int **structure_offset**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/mavlink_types.h

5.57 __mavlink_figure_eight_execution_status_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **major_radius**
- float **minor_radius**
- float **orientation**
- int32_t **x**
- int32_t **y**
- float **z**
- uint8_t **frame**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_figure_eight_execution_status.h

5.58 __mavlink_file_transfer_protocol_t Struct Reference

Public Attributes

- uint8_t **target_network**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **payload** [251]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_file_transfer_protocol.h

5.59 `__mavlink_flight_information_t` Struct Reference

Public Attributes

- `uint64_t arming_time_utc`
- `uint64_t takeoff_time_utc`
- `uint64_t flight_uuid`
- `uint32_t time_boot_ms`
- `uint32_t landing_time`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_flight_information.h`

5.60 `__mavlink_follow_target_t` Struct Reference

Public Attributes

- `uint64_t timestamp`
- `uint64_t custom_state`
- `int32_t lat`
- `int32_t lon`
- `float alt`
- `float vel [3]`
- `float acc [3]`
- `float attitude_q [4]`
- `float rates [3]`
- `float position_cov [3]`
- `uint8_t est_capabilities`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_follow_target.h`

5.61 `__mavlink_fuel_status_t` Struct Reference

Public Attributes

- `float maximum_fuel`
- `float consumed_fuel`
- `float remaining_fuel`
- `float flow_rate`
- `float temperature`
- `uint32_t fuel_type`
- `uint8_t id`
- `uint8_t percent_remaining`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_fuel_status.h`

5.62 __mavlink_generator_status_t Struct Reference

Public Attributes

- uint64_t **status**
- float **battery_current**
- float **load_current**
- float **power_generated**
- float **bus_voltage**
- float **bat_current_setpoint**
- uint32_t **runtime**
- int32_t **time_until_maintenance**
- uint16_t **generator_speed**
- int16_t **rectifier_temperature**
- int16_t **generator_temperature**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_generator_status.h

5.63 __mavlink_gimbal_device_attitude_status_t Struct Reference

Public Attributes

- uint32_t **time_boot_ms**
- float **q** [4]
- float **angular_velocity_x**
- float **angular_velocity_y**
- float **angular_velocity_z**
- uint32_t **failure_flags**
- uint16_t **flags**
- uint8_t **target_system**
- uint8_t **target_component**
- float **delta_yaw**
- float **delta_yaw_velocity**
- uint8_t **gimbal_device_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_attitude_status.h

5.64 __mavlink_gimbal_device_information_t Struct Reference

Public Attributes

- uint64_t **uid**
- uint32_t **time_boot_ms**
- uint32_t **firmware_version**
- uint32_t **hardware_version**
- float **roll_min**
- float **roll_max**
- float **pitch_min**
- float **pitch_max**
- float **yaw_min**
- float **yaw_max**
- uint16_t **cap_flags**
- uint16_t **custom_cap_flags**
- char **vendor_name** [32]
- char **model_name** [32]
- char **custom_name** [32]
- uint8_t **gimbal_device_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_information.h

5.65 __mavlink_gimbal_device_set_attitude_t Struct Reference

Public Attributes

- float **q** [4]
- float **angular_velocity_x**
- float **angular_velocity_y**
- float **angular_velocity_z**
- uint16_t **flags**
- uint8_t **target_system**
- uint8_t **target_component**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_set_attitude.h

5.66 __mavlink_gimbal_manager_information_t Struct Reference

Public Attributes

- uint32_t **time_boot_ms**
- uint32_t **cap_flags**
- float **roll_min**
- float **roll_max**
- float **pitch_min**
- float **pitch_max**
- float **yaw_min**
- float **yaw_max**
- uint8_t **gimbal_device_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_information.h

5.67 __mavlink_gimbal_manager_set_attitude_t Struct Reference

Public Attributes

- uint32_t **flags**
- float **q** [4]
- float **angular_velocity_x**
- float **angular_velocity_y**
- float **angular_velocity_z**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **gimbal_device_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_attitude.h

5.68 __mavlink_gimbal_manager_set_manual_control_t Struct Reference

Public Attributes

- uint32_t **flags**
- float **pitch**
- float **yaw**
- float **pitch_rate**
- float **yaw_rate**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **gimbal_device_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_manual_control.h

5.69 `__mavlink_gimbal_manager_set_pitchyaw_t` Struct Reference

Public Attributes

- `uint32_t flags`
- `float pitch`
- `float yaw`
- `float pitch_rate`
- `float yaw_rate`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t gimbal_device_id`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_pitchyaw.h`

5.70 `__mavlink_gimbal_manager_status_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `uint32_t flags`
- `uint8_t gimbal_device_id`
- `uint8_t primary_control_sysid`
- `uint8_t primary_control_compid`
- `uint8_t secondary_control_sysid`
- `uint8_t secondary_control_compid`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_status.h`

5.71 `__mavlink_global_position_int_cov_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `int32_t lat`
- `int32_t lon`
- `int32_t alt`
- `int32_t relative_alt`
- `float vx`
- `float vy`
- `float vz`
- `float covariance` [36]
- `uint8_t estimator_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_global_position_int_cov.h`

5.72 `__mavlink_global_position_int_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `int32_t lat`
- `int32_t lon`
- `int32_t alt`
- `int32_t relative_alt`
- `int16_t vx`
- `int16_t vy`
- `int16_t vz`
- `uint16_t hdg`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/standard/mavlink_msg_global_position_int.h`

5.73 `__mavlink_global_vision_position_estimate_t` Struct Reference

Public Attributes

- `uint64_t usec`
- `float x`
- `float y`
- `float z`
- `float roll`
- `float pitch`
- `float yaw`
- `float covariance` [21]
- `uint8_t reset_counter`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_global_vision_position_estimate.h`

5.74 `__mavlink_gps2_rtk_t` Struct Reference

Public Attributes

- `uint32_t time_last_baseline_ms`
- `uint32_t tow`
- `int32_t baseline_a_mm`
- `int32_t baseline_b_mm`
- `int32_t baseline_c_mm`
- `uint32_t accuracy`
- `int32_t iar_num_hypotheses`
- `uint16_t wn`
- `uint8_t rtk_receiver_id`
- `uint8_t rtk_health`
- `uint8_t rtk_rate`
- `uint8_t nsats`
- `uint8_t baseline_coords_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_gps2_rtk.h`

5.75 `__mavlink_gps_inject_data_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t len`
- `uint8_t data` [110]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_inject_data.h`

5.76 `__mavlink_gps_rtcn_data_t` Struct Reference

Public Attributes

- `uint8_t flags`
- `uint8_t len`
- `uint8_t data` [180]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtcn_data.h`

5.77 `__mavlink_gps_rtk_t` Struct Reference

Public Attributes

- `uint32_t time_last_baseline_ms`
- `uint32_t tow`
- `int32_t baseline_a_mm`
- `int32_t baseline_b_mm`
- `int32_t baseline_c_mm`
- `uint32_t accuracy`
- `int32_t iar_num_hypotheses`
- `uint16_t wn`
- `uint8_t rtk_receiver_id`
- `uint8_t rtk_health`
- `uint8_t rtk_rate`
- `uint8_t nsats`
- `uint8_t baseline_coords_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtk.h`

5.78 `__mavlink_gps_status_t` Struct Reference

Public Attributes

- `uint8_t satellites_visible`
- `uint8_t satellite_prn` [20]
- `uint8_t satellite_used` [20]
- `uint8_t satellite_elevation` [20]
- `uint8_t satellite_azimuth` [20]
- `uint8_t satellite_snr` [20]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_status.h`

5.79 `__mavlink_heartbeat_t` Struct Reference

Public Attributes

- `uint32_t custom_mode`
- `uint8_t type`
- `uint8_t autopilot`
- `uint8_t base_mode`
- `uint8_t system_status`
- `uint8_t mavlink_version`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/minimal/mavlink_msg_heartbeat.h`

5.80 `__mavlink_high_latency2_t` Struct Reference

Public Attributes

- `uint32_t timestamp`
- `int32_t latitude`
- `int32_t longitude`
- `uint16_t custom_mode`
- `int16_t altitude`
- `int16_t target_altitude`
- `uint16_t target_distance`
- `uint16_t wp_num`
- `uint16_t failure_flags`
- `uint8_t type`
- `uint8_t autopilot`
- `uint8_t heading`
- `uint8_t target_heading`
- `uint8_t throttle`

- uint8_t **airspeed**
- uint8_t **airspeed_sp**
- uint8_t **groundspeed**
- uint8_t **windspeed**
- uint8_t **wind_heading**
- uint8_t **eph**
- uint8_t **epv**
- int8_t **temperature_air**
- int8_t **climb_rate**
- int8_t **battery**
- int8_t **custom0**
- int8_t **custom1**
- int8_t **custom2**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_high_latency2.h

5.81 __mavlink_high_latency_t Struct Reference

Public Attributes

- uint32_t **custom_mode**
- int32_t **latitude**
- int32_t **longitude**
- int16_t **roll**
- int16_t **pitch**
- uint16_t **heading**
- int16_t **heading_sp**
- int16_t **altitude_amsl**
- int16_t **altitude_sp**
- uint16_t **wp_distance**
- uint8_t **base_mode**
- uint8_t **landed_state**
- int8_t **throttle**
- uint8_t **airspeed**
- uint8_t **airspeed_sp**
- uint8_t **groundspeed**
- int8_t **climb_rate**
- uint8_t **gps_nsat**
- uint8_t **gps_fix_type**
- uint8_t **battery_remaining**
- int8_t **temperature**
- int8_t **temperature_air**
- uint8_t **failsafe**
- uint8_t **wp_num**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_high_latency.h

5.82 __mavlink_highres_imu_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **xacc**
- float **yacc**
- float **zacc**
- float **xgyro**
- float **ygyro**
- float **zgyro**
- float **xmag**
- float **ymag**
- float **zmag**
- float **abs_pressure**
- float **diff_pressure**
- float **pressure_alt**
- float **temperature**
- uint16_t **fields_updated**
- uint8_t **id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_highres_imu.h

5.83 __mavlink_hil_actuator_controls_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- uint64_t **flags**
- float **controls** [16]
- uint8_t **mode**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_hil_actuator_controls.h

5.84 __mavlink_hil_controls_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **roll_ailerons**
- float **pitch_elevator**
- float **yaw_rudder**
- float **throttle**
- float **aux1**
- float **aux2**
- float **aux3**
- float **aux4**
- uint8_t **mode**
- uint8_t **nav_mode**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_hil_controls.h

5.85 `__mavlink_hil_optical_flow_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `uint32_t integration_time_us`
- `float integrated_x`
- `float integrated_y`
- `float integrated_xgyro`
- `float integrated_ygyro`
- `float integrated_zgyro`
- `uint32_t time_delta_distance_us`
- `float distance`
- `int16_t temperature`
- `uint8_t sensor_id`
- `uint8_t quality`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_optical_flow.h`

5.86 `__mavlink_hil_rc_inputs_raw_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `uint16_t chan1_raw`
- `uint16_t chan2_raw`
- `uint16_t chan3_raw`
- `uint16_t chan4_raw`
- `uint16_t chan5_raw`
- `uint16_t chan6_raw`
- `uint16_t chan7_raw`
- `uint16_t chan8_raw`
- `uint16_t chan9_raw`
- `uint16_t chan10_raw`
- `uint16_t chan11_raw`
- `uint16_t chan12_raw`
- `uint8_t rssi`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_rc_inputs_raw.h`

5.87 __mavlink_hil_sensor_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **xacc**
- float **yacc**
- float **zacc**
- float **xgyro**
- float **ygyro**
- float **zgyro**
- float **xmag**
- float **ymag**
- float **zmag**
- float **abs_pressure**
- float **diff_pressure**
- float **pressure_alt**
- float **temperature**
- uint32_t **fields_updated**
- uint8_t **id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_hil_sensor.h

5.88 __mavlink_hil_state_quaternion_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **attitude_quaternion** [4]
- float **rollspeed**
- float **pitchspeed**
- float **yawspeed**
- int32_t **lat**
- int32_t **lon**
- int32_t **alt**
- int16_t **vx**
- int16_t **vy**
- int16_t **vz**
- uint16_t **ind_airspeed**
- uint16_t **true_airspeed**
- int16_t **xacc**
- int16_t **yacc**
- int16_t **zacc**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_hil_state_quaternion.h

5.89 `__mavlink_hil_state_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float roll`
- `float pitch`
- `float yaw`
- `float rollspeed`
- `float pitchspeed`
- `float yawspeed`
- `int32_t lat`
- `int32_t lon`
- `int32_t alt`
- `int16_t vx`
- `int16_t vy`
- `int16_t vz`
- `int16_t xacc`
- `int16_t yacc`
- `int16_t zacc`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_state.h`

5.90 `__mavlink_hygrometer_sensor_t` Struct Reference

Public Attributes

- `int16_t temperature`
- `uint16_t humidity`
- `uint8_t id`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_hygrometer_sensor.h`

5.91 `__mavlink_illuminator_status_t` Struct Reference

Public Attributes

- `uint32_t uptime_ms`
- `uint32_t error_status`
- `float brightness`
- `float strobe_period`
- `float strobe_duty_cycle`
- `float temp_c`
- `float min_strobe_period`
- `float max_strobe_period`
- `uint8_t enable`
- `uint8_t mode_bitmask`
- `uint8_t mode`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_illuminator_status.h`

5.92 __mavlink_isbd_link_status_t Struct Reference

Public Attributes

- uint64_t **timestamp**
- uint64_t **last_heartbeat**
- uint16_t **failed_sessions**
- uint16_t **successful_sessions**
- uint8_t **signal_quality**
- uint8_t **ring_pending**
- uint8_t **tx_session_pending**
- uint8_t **rx_session_pending**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_isbd_link_status.h

5.93 __mavlink_link_node_status_t Struct Reference

Public Attributes

- uint64_t **timestamp**
- uint32_t **tx_rate**
- uint32_t **rx_rate**
- uint32_t **messages_sent**
- uint32_t **messages_received**
- uint32_t **messages_lost**
- uint16_t **rx_parse_err**
- uint16_t **tx_overflows**
- uint16_t **rx_overflows**
- uint8_t **tx_buf**
- uint8_t **rx_buf**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_link_node_status.h

5.94 __mavlink_local_position_ned_cov_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **x**
- float **y**
- float **z**
- float **vx**
- float **vy**
- float **vz**
- float **ax**
- float **ay**
- float **az**
- float **covariance** [45]
- uint8_t **estimator_type**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_cov.h

5.95 `__mavlink_local_position_ned_system_global_offset_t` Struct Reference

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **x**
- `float` **y**
- `float` **z**
- `float` **roll**
- `float` **pitch**
- `float` **yaw**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_system_global_offset.h`

5.96 `__mavlink_local_position_ned_t` Struct Reference

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **x**
- `float` **y**
- `float` **z**
- `float` **vx**
- `float` **vy**
- `float` **vz**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned.h`

5.97 `__mavlink_log_data_t` Struct Reference

Public Attributes

- `uint32_t` **ofs**
- `uint16_t` **id**
- `uint8_t` **count**
- `uint8_t` **data** [90]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_log_data.h`

5.98 `__mavlink_log_entry_t` Struct Reference

Public Attributes

- `uint32_t time_utc`
- `uint32_t size`
- `uint16_t id`
- `uint16_t num_logs`
- `uint16_t last_log_num`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_log_entry.h`

5.99 `__mavlink_log_erase_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_log_erase.h`

5.100 `__mavlink_log_request_data_t` Struct Reference

Public Attributes

- `uint32_t ofs`
- `uint32_t count`
- `uint16_t id`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_log_request_data.h`

5.101 `__mavlink_log_request_end_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_log_request_end.h`

5.102 `__mavlink_log_request_list_t` Struct Reference

Public Attributes

- `uint16_t start`
- `uint16_t end`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_log_request_list.h`

5.103 `__mavlink_logging_ack_t` Struct Reference

Public Attributes

- `uint16_t sequence`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_logging_ack.h`

5.104 `__mavlink_logging_data_acked_t` Struct Reference

Public Attributes

- `uint16_t sequence`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t length`
- `uint8_t first_message_offset`
- `uint8_t data` [249]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_logging_data_acked.h`

5.105 `__mavlink_logging_data_t` Struct Reference

Public Attributes

- `uint16_t sequence`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t length`
- `uint8_t first_message_offset`
- `uint8_t data` [249]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_logging_data.h`

5.106 `__mavlink_manual_setpoint_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float roll`
- `float pitch`
- `float yaw`
- `float thrust`
- `uint8_t mode_switch`
- `uint8_t manual_override_switch`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_manual_setpoint.h`

5.107 `__mavlink_memory_vect_t` Struct Reference

Public Attributes

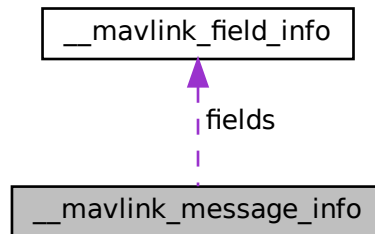
- `uint16_t address`
- `uint8_t ver`
- `uint8_t type`
- `int8_t value` [32]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_memory_vect.h`

5.108 __mavlink_message_info Struct Reference

Collaboration diagram for __mavlink_message_info:



Public Attributes

- uint32_t **msgid**
- const char * **name**
- unsigned **num_fields**
- [mavlink_field_info_t](#) **fields** [MAVLINK_MAX_FIELDS]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/mavlink_types.h

5.109 __mavlink_message_interval_t Struct Reference

Public Attributes

- int32_t **interval_us**
- uint16_t **message_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_message_interval.h

5.110 __mavlink_mission_ack_t Struct Reference

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **type**
- uint8_t **mission_type**
- uint32_t **opaque_id**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_mission_ack.h

5.111 __mavlink_mission_clear_all_t Struct Reference

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **mission_type**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_mission_clear_all.h

5.112 __mavlink_mission_item_int_t Struct Reference

Public Attributes

- float **param1**
- float **param2**
- float **param3**
- float **param4**
- int32_t **x**
- int32_t **y**
- float **z**
- uint16_t **seq**
- uint16_t **command**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **frame**
- uint8_t **current**
- uint8_t **autocontinue**
- uint8_t **mission_type**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_int.h

5.113 __mavlink_mission_item_reached_t Struct Reference

Public Attributes

- uint16_t **seq**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_reached.h

5.114 `__mavlink_mission_item_t` Struct Reference

Public Attributes

- float **param1**
- float **param2**
- float **param3**
- float **param4**
- float **x**
- float **y**
- float **z**
- uint16_t **seq**
- uint16_t **command**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **frame**
- uint8_t **current**
- uint8_t **autocontinue**
- uint8_t **mission_type**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_item.h`

5.115 `__mavlink_mission_request_int_t` Struct Reference

Public Attributes

- uint16_t **seq**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **mission_type**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_int.h`

5.116 `__mavlink_mission_request_list_t` Struct Reference

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **mission_type**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_list.h`

5.117 `__mavlink_mission_request_partial_list_t` Struct Reference

Public Attributes

- `int16_t start_index`
- `int16_t end_index`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t mission_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_partial_list.h`

5.118 `__mavlink_mission_request_t` Struct Reference

Public Attributes

- `uint16_t seq`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t mission_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_request.h`

5.119 `__mavlink_mission_set_current_t` Struct Reference

Public Attributes

- `uint16_t seq`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_set_current.h`

5.120 `__mavlink_mission_write_partial_list_t` Struct Reference

Public Attributes

- `int16_t start_index`
- `int16_t end_index`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t mission_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_write_partial_list.h`

5.121 `__mavlink_mount_orientation_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float roll`
- `float pitch`
- `float yaw`
- `float yaw_absolute`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_mount_orientation.h`

5.122 `__mavlink_msg_entry` Struct Reference

Public Attributes

- `uint32_t msgid`
- `uint8_t crc_extra`
- `uint8_t min_msg_len`
- `uint8_t max_msg_len`
- `uint8_t flags`
- `uint8_t target_system_ofs`
- `uint8_t target_component_ofs`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/mavlink_types.h`

5.123 `__mavlink_named_value_float_t` Struct Reference

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **value**
- `char` **name** [10]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_named_value_float.h`

5.124 `__mavlink_named_value_int_t` Struct Reference

Public Attributes

- `uint32_t` **time_boot_ms**
- `int32_t` **value**
- `char` **name** [10]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_named_value_int.h`

5.125 `__mavlink_nav_controller_output_t` Struct Reference

Public Attributes

- `float` **nav_roll**
- `float` **nav_pitch**
- `float` **alt_error**
- `float` **aspd_error**
- `float` **xtrack_error**
- `int16_t` **nav_bearing**
- `int16_t` **target_bearing**
- `uint16_t` **wp_dist**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_nav_controller_output.h`

5.126 `__mavlink_odometry_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float x`
- `float y`
- `float z`
- `float q [4]`
- `float vx`
- `float vy`
- `float vz`
- `float rollspeed`
- `float pitchspeed`
- `float yawspeed`
- `float pose_covariance [21]`
- `float velocity_covariance [21]`
- `uint8_t frame_id`
- `uint8_t child_frame_id`
- `uint8_t reset_counter`
- `uint8_t estimator_type`
- `int8_t quality`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_odometry.h`

5.127 `__mavlink_onboard_computer_status_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `uint32_t uptime`
- `uint32_t ram_usage`
- `uint32_t ram_total`
- `uint32_t storage_type [4]`
- `uint32_t storage_usage [4]`
- `uint32_t storage_total [4]`
- `uint32_t link_type [6]`
- `uint32_t link_tx_rate [6]`
- `uint32_t link_rx_rate [6]`
- `uint32_t link_tx_max [6]`
- `uint32_t link_rx_max [6]`
- `int16_t fan_speed [4]`
- `uint8_t type`
- `uint8_t cpu_cores [8]`
- `uint8_t cpu_combined [10]`
- `uint8_t gpu_cores [4]`
- `uint8_t gpu_combined [10]`
- `int8_t temperature_board`
- `int8_t temperature_core [8]`
- `uint16_t status_flags`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_onboard_computer_status.h`

5.128 `__mavlink_open_drone_id_arm_status_t` Struct Reference

Public Attributes

- `uint8_t status`
- `char error` [50]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_arm_status.h`

5.129 `__mavlink_open_drone_id_authentication_t` Struct Reference

Public Attributes

- `uint32_t timestamp`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t id_or_mac` [20]
- `uint8_t authentication_type`
- `uint8_t data_page`
- `uint8_t last_page_index`
- `uint8_t length`
- `uint8_t authentication_data` [23]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_authentication.h`

5.130 `__mavlink_open_drone_id_basic_id_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t id_or_mac` [20]
- `uint8_t id_type`
- `uint8_t ua_type`
- `uint8_t uas_id` [20]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_basic_id.h`

5.131 `__mavlink_open_drone_id_location_t` Struct Reference

Public Attributes

- `int32_t latitude`
- `int32_t longitude`
- `float altitude_barometric`
- `float altitude_geodetic`
- `float height`
- `float timestamp`
- `uint16_t direction`
- `uint16_t speed_horizontal`
- `int16_t speed_vertical`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t id_or_mac [20]`
- `uint8_t status`
- `uint8_t height_reference`
- `uint8_t horizontal_accuracy`
- `uint8_t vertical_accuracy`
- `uint8_t barometer_accuracy`
- `uint8_t speed_accuracy`
- `uint8_t timestamp_accuracy`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_location.h`

5.132 `__mavlink_open_drone_id_message_pack_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t id_or_mac [20]`
- `uint8_t single_message_size`
- `uint8_t msg_pack_size`
- `uint8_t messages [225]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_message_pack.h`

5.133 __mavlink_open_drone_id_operator_id_t Struct Reference

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **id_or_mac** [20]
- uint8_t **operator_id_type**
- char **operator_id** [20]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_operator_id.h

5.134 __mavlink_open_drone_id_self_id_t Struct Reference

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **id_or_mac** [20]
- uint8_t **description_type**
- char **description** [23]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_self_id.h

5.135 __mavlink_open_drone_id_system_t Struct Reference

Public Attributes

- int32_t **operator_latitude**
- int32_t **operator_longitude**
- float **area_ceiling**
- float **area_floor**
- float **operator_altitude_geo**
- uint32_t **timestamp**
- uint16_t **area_count**
- uint16_t **area_radius**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **id_or_mac** [20]
- uint8_t **operator_location_type**
- uint8_t **classification_type**
- uint8_t **category_eu**
- uint8_t **class_eu**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system.h

5.136 `__mavlink_open_drone_id_system_update_t` Struct Reference

Public Attributes

- `int32_t operator_latitude`
- `int32_t operator_longitude`
- `float operator_altitude_geo`
- `uint32_t timestamp`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system_update.h`

5.137 `__mavlink_optical_flow_rad_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `uint32_t integration_time_us`
- `float integrated_x`
- `float integrated_y`
- `float integrated_xgyro`
- `float integrated_ygyro`
- `float integrated_zgyro`
- `uint32_t time_delta_distance_us`
- `float distance`
- `int16_t temperature`
- `uint8_t sensor_id`
- `uint8_t quality`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_optical_flow_rad.h`

5.138 `__mavlink_orbit_execution_status_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float radius`
- `int32_t x`
- `int32_t y`
- `float z`
- `uint8_t frame`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_orbit_execution_status.h`

5.139 __mavlink_param_error_t Struct Reference

Public Attributes

- int16_t **param_index**
- uint8_t **target_system**
- uint8_t **target_component**
- char **param_id** [16]
- uint8_t **error**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_param_error.h

5.140 __mavlink_param_ext_ack_t Struct Reference

Public Attributes

- char **param_id** [16]
- char **param_value** [128]
- uint8_t **param_type**
- uint8_t **param_result**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_ack.h

5.141 __mavlink_param_ext_request_list_t Struct Reference

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_list.h

5.142 __mavlink_param_ext_request_read_t Struct Reference

Public Attributes

- int16_t **param_index**
- uint8_t **target_system**
- uint8_t **target_component**
- char **param_id** [16]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_read.h

5.143 `__mavlink_param_ext_set_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `char param_id [16]`
- `char param_value [128]`
- `uint8_t param_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_set.h`

5.144 `__mavlink_param_ext_value_t` Struct Reference

Public Attributes

- `uint16_t param_count`
- `uint16_t param_index`
- `char param_id [16]`
- `char param_value [128]`
- `uint8_t param_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_value.h`

5.145 `__mavlink_param_map_rc_t` Struct Reference

Public Attributes

- `float param_value0`
- `float scale`
- `float param_value_min`
- `float param_value_max`
- `int16_t param_index`
- `uint8_t target_system`
- `uint8_t target_component`
- `char param_id [16]`
- `uint8_t parameter_rc_channel_index`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_param_map_rc.h`

5.146 `__mavlink_param_request_list_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_param_request_list.h`

5.147 `__mavlink_param_request_read_t` Struct Reference

Public Attributes

- `int16_t param_index`
- `uint8_t target_system`
- `uint8_t target_component`
- `char param_id [16]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_param_request_read.h`

5.148 `__mavlink_param_set_t` Struct Reference

Public Attributes

- `float param_value`
- `uint8_t target_system`
- `uint8_t target_component`
- `char param_id [16]`
- `uint8_t param_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_param_set.h`

5.149 `__mavlink_param_value_t` Struct Reference

Public Attributes

- `float param_value`
- `uint16_t param_count`
- `uint16_t param_index`
- `char param_id [16]`
- `uint8_t param_type`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_param_value.h`

5.150 `__mavlink_ping_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `uint32_t seq`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_ping.h`

5.151 `__mavlink_play_tune_t` Struct Reference

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `char tune [30]`
- `char tune2 [200]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_play_tune.h`

5.152 `__mavlink_play_tune_v2_t` Struct Reference

Public Attributes

- `uint32_t format`
- `uint8_t target_system`
- `uint8_t target_component`
- `char tune [248]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_play_tune_v2.h`

5.153 `__mavlink_position_target_global_int_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `int32_t lat_int`
- `int32_t lon_int`
- `float alt`
- `float vx`
- `float vy`
- `float vz`
- `float afx`
- `float afy`
- `float afz`
- `float yaw`
- `float yaw_rate`
- `uint16_t type_mask`
- `uint8_t coordinate_frame`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_position_target_global_int.h`

5.154 `__mavlink_position_target_local_ned_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float x`
- `float y`
- `float z`
- `float vx`
- `float vy`
- `float vz`
- `float afx`
- `float afy`
- `float afz`
- `float yaw`
- `float yaw_rate`
- `uint16_t type_mask`
- `uint8_t coordinate_frame`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_position_target_local_ned.h`

5.155 `__mavlink_power_status_t` Struct Reference

Public Attributes

- `uint16_t Vcc`
- `uint16_t Vservo`
- `uint16_t flags`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_power_status.h`

5.156 `__mavlink_protocol_version_t` Struct Reference

Public Attributes

- `uint16_t version`
- `uint16_t min_version`
- `uint16_t max_version`
- `uint8_t spec_version_hash [8]`
- `uint8_t library_version_hash [8]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_protocol_version.h`

5.157 `__mavlink_radio_status_t` Struct Reference

Public Attributes

- `uint16_t rxerrors`
- `uint16_t fixed`
- `uint8_t rssi`
- `uint8_t remrssi`
- `uint8_t txbuf`
- `uint8_t noise`
- `uint8_t remnoise`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_radio_status.h`

5.158 `__mavlink_raw_pressure_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `int16_t press_abs`
- `int16_t press_diff1`
- `int16_t press_diff2`
- `int16_t temperature`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_raw_pressure.h`

5.159 `__mavlink_raw_rpm_t` Struct Reference

Public Attributes

- `float frequency`
- `uint8_t index`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_raw_rpm.h`

5.160 `__mavlink_rc_channels_override_t` Struct Reference

Public Attributes

- `uint16_t chan1_raw`
- `uint16_t chan2_raw`
- `uint16_t chan3_raw`
- `uint16_t chan4_raw`
- `uint16_t chan5_raw`
- `uint16_t chan6_raw`
- `uint16_t chan7_raw`
- `uint16_t chan8_raw`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint16_t chan9_raw`
- `uint16_t chan10_raw`
- `uint16_t chan11_raw`
- `uint16_t chan12_raw`
- `uint16_t chan13_raw`
- `uint16_t chan14_raw`
- `uint16_t chan15_raw`
- `uint16_t chan16_raw`
- `uint16_t chan17_raw`
- `uint16_t chan18_raw`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_override.h`

5.161 `__mavlink_rc_channels_raw_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `uint16_t chan1_raw`
- `uint16_t chan2_raw`
- `uint16_t chan3_raw`
- `uint16_t chan4_raw`
- `uint16_t chan5_raw`
- `uint16_t chan6_raw`
- `uint16_t chan7_raw`
- `uint16_t chan8_raw`
- `uint8_t port`
- `uint8_t rssi`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_raw.h`

5.162 `__mavlink_rc_channels_scaled_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `int16_t chan1_scaled`
- `int16_t chan2_scaled`
- `int16_t chan3_scaled`
- `int16_t chan4_scaled`
- `int16_t chan5_scaled`
- `int16_t chan6_scaled`
- `int16_t chan7_scaled`
- `int16_t chan8_scaled`
- `uint8_t port`
- `uint8_t rssi`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_scaled.h`

5.163 `__mavlink_rc_channels_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `uint16_t chan1_raw`
- `uint16_t chan2_raw`
- `uint16_t chan3_raw`
- `uint16_t chan4_raw`
- `uint16_t chan5_raw`
- `uint16_t chan6_raw`
- `uint16_t chan7_raw`
- `uint16_t chan8_raw`
- `uint16_t chan9_raw`
- `uint16_t chan10_raw`
- `uint16_t chan11_raw`
- `uint16_t chan12_raw`
- `uint16_t chan13_raw`
- `uint16_t chan14_raw`
- `uint16_t chan15_raw`
- `uint16_t chan16_raw`
- `uint16_t chan17_raw`
- `uint16_t chan18_raw`
- `uint8_t chancount`
- `uint8_t rssi`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels.h`

5.164 `__mavlink_relay_status_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `uint16_t on`
- `uint16_t present`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_relay_status.h`

5.165 `__mavlink_request_data_stream_t` Struct Reference

Public Attributes

- `uint16_t req_message_rate`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t req_stream_id`
- `uint8_t start_stop`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_request_data_stream.h`

5.166 `__mavlink_request_event_t` Struct Reference

Public Attributes

- `uint16_t first_sequence`
- `uint16_t last_sequence`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_request_event.h`

5.167 `__mavlink_resource_request_t` Struct Reference

Public Attributes

- `uint8_t request_id`
- `uint8_t uri_type`
- `uint8_t uri [120]`
- `uint8_t transfer_type`
- `uint8_t storage [120]`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_resource_request.h`

5.168 `__mavlink_response_event_error_t` Struct Reference

Public Attributes

- `uint16_t sequence`
- `uint16_t sequence_oldest_available`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t reason`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_response_event_error.h`

5.169 `__mavlink_safety_allowed_area_t` Struct Reference

Public Attributes

- `float p1x`
- `float p1y`
- `float p1z`
- `float p2x`
- `float p2y`
- `float p2z`
- `uint8_t frame`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_safety_allowed_area.h`

5.170 `__mavlink_safety_set_allowed_area_t` Struct Reference

Public Attributes

- `float p1x`
- `float p1y`
- `float p1z`
- `float p2x`
- `float p2y`
- `float p2z`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t frame`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_safety_set_allowed_area.h`

5.171 `__mavlink_scaled_imu2_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `int16_t xacc`
- `int16_t yacc`
- `int16_t zacc`
- `int16_t xgyro`
- `int16_t ygyro`
- `int16_t zgyro`
- `int16_t xmag`
- `int16_t ymag`
- `int16_t zmag`
- `int16_t temperature`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu2.h`

5.172 `__mavlink_scaled_imu3_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `int16_t xacc`
- `int16_t yacc`
- `int16_t zacc`
- `int16_t xgyro`
- `int16_t ygyro`
- `int16_t zgyro`
- `int16_t xmag`
- `int16_t ymag`
- `int16_t zmag`
- `int16_t temperature`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu3.h`

5.173 __mavlink_scaled_imu_t Struct Reference

Public Attributes

- uint32_t **time_boot_ms**
- int16_t **xacc**
- int16_t **yacc**
- int16_t **zacc**
- int16_t **xgyro**
- int16_t **ygyro**
- int16_t **zgyro**
- int16_t **xmag**
- int16_t **ymag**
- int16_t **zmag**
- int16_t **temperature**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu.h

5.174 __mavlink_scaled_pressure2_t Struct Reference

Public Attributes

- uint32_t **time_boot_ms**
- float **press_abs**
- float **press_diff**
- int16_t **temperature**
- int16_t **temperature_press_diff**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure2.h

5.175 __mavlink_scaled_pressure3_t Struct Reference

Public Attributes

- uint32_t **time_boot_ms**
- float **press_abs**
- float **press_diff**
- int16_t **temperature**
- int16_t **temperature_press_diff**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure3.h

5.176 `__mavlink_scaled_pressure_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float press_abs`
- `float press_diff`
- `int16_t temperature`
- `int16_t temperature_press_diff`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure.h`

5.177 `__mavlink_serial_control_t` Struct Reference

Public Attributes

- `uint32_t baudrate`
- `uint16_t timeout`
- `uint8_t device`
- `uint8_t flags`
- `uint8_t count`
- `uint8_t data [70]`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_serial_control.h`

5.178 `__mavlink_set_actuator_control_target_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float controls [8]`
- `uint8_t group_mlx`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_set_actuator_control_target.h`

5.179 `__mavlink_set_mode_t` Struct Reference

Public Attributes

- `uint32_t custom_mode`
- `uint8_t target_system`
- `uint8_t base_mode`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_set_mode.h`

5.180 `__mavlink_set_position_target_global_int_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `int32_t lat_int`
- `int32_t lon_int`
- `float alt`
- `float vx`
- `float vy`
- `float vz`
- `float afx`
- `float afy`
- `float afz`
- `float yaw`
- `float yaw_rate`
- `uint16_t type_mask`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t coordinate_frame`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_global_int.h`

5.181 `__mavlink_set_position_target_local_ned_t` Struct Reference

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **x**
- `float` **y**
- `float` **z**
- `float` **vx**
- `float` **vy**
- `float` **vz**
- `float` **afx**
- `float` **afy**
- `float` **afz**
- `float` **yaw**
- `float` **yaw_rate**
- `uint16_t` **type_mask**
- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **coordinate_frame**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_local_ned.h`

5.182 `__mavlink_setup_signing_t` Struct Reference

Public Attributes

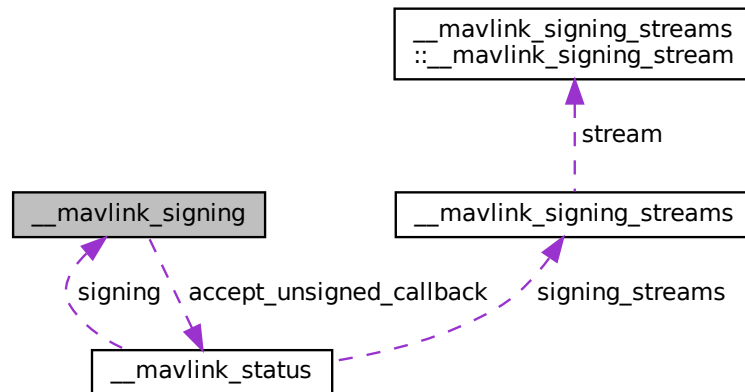
- `uint64_t` **initial_timestamp**
- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **secret_key** [32]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_setup_signing.h`

5.183 __mavlink_signing Struct Reference

Collaboration diagram for __mavlink_signing:



Public Attributes

- `uint8_t` [flags](#)
*MAVLINK_SIGNING_FLAG_**.
- `uint8_t` [link_id](#)
Same as MAVLINK_CHANNEL.
- `uint64_t` [timestamp](#)
Timestamp, in microseconds since UNIX epoch GMT.
- `uint8_t` **secret_key** [32]
- `mavlink_accept_unsigned_t` **accept_unsigned_callback**
- `mavlink_signing_status_t` **last_status**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/mavlink_types.h`

5.184 __mavlink_signing_streams::__mavlink_signing_stream Struct Reference

Public Attributes

- `uint8_t` [link_id](#)
ID of the link (MAVLINK_CHANNEL)
- `uint8_t` [sysid](#)
Remote system ID.
- `uint8_t` [compid](#)

Remote component ID.

- `uint8_t timestamp_bytes` [6]

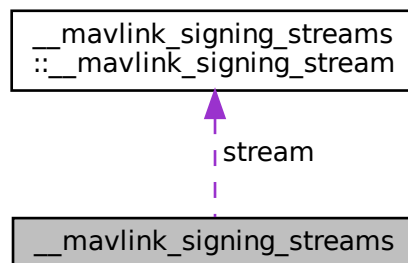
Timestamp, in microseconds since UNIX epoch GMT.

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/mavlink_types.h`

5.185 __mavlink_signing_streams Struct Reference

Collaboration diagram for __mavlink_signing_streams:



Classes

- struct `__mavlink_signing_stream`

Public Attributes

- `uint16_t num_signing_streams`
- struct `__mavlink_signing_streams::__mavlink_signing_stream stream` [MAVLINK_MAX_SIGNING_STREAMS]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/mavlink_types.h`

5.186 `__mavlink_sim_state_t` Struct Reference

Public Attributes

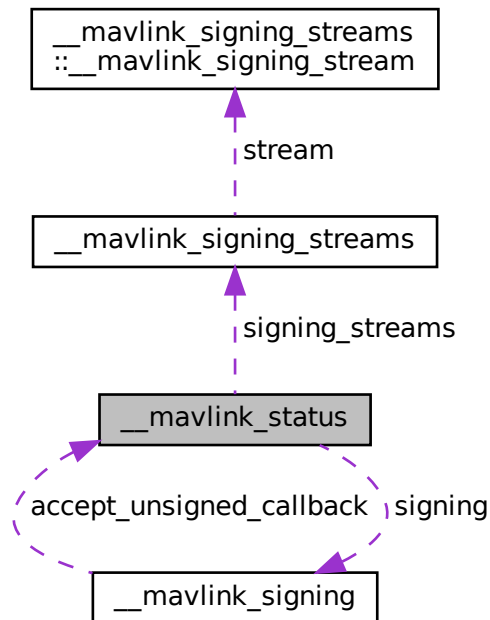
- float **q1**
- float **q2**
- float **q3**
- float **q4**
- float **roll**
- float **pitch**
- float **yaw**
- float **xacc**
- float **yacc**
- float **zacc**
- float **xgyro**
- float **ygyro**
- float **zgyro**
- float **lat**
- float **lon**
- float **alt**
- float **std_dev_horz**
- float **std_dev_vert**
- float **vn**
- float **ve**
- float **vd**
- int32_t **lat_int**
- int32_t **lon_int**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_sim_state.h`

5.187 __mavlink_status Struct Reference

Collaboration diagram for __mavlink_status:



Public Attributes

- `uint8_t msg_received`
Number of received messages.
- `uint8_t buffer_overrun`
Number of buffer overruns.
- `uint8_t parse_error`
Number of parse errors.
- `mavlink_parse_state_t parse_state`
Parsing state machine.
- `uint8_t packet_idx`
Index in current packet.
- `uint8_t current_rx_seq`
Sequence number of last packet received.
- `uint8_t current_tx_seq`
Sequence number of last packet sent.
- `uint16_t packet_rx_success_count`
Received packets.
- `uint16_t packet_rx_drop_count`
Number of packet drops.
- `uint8_t flags`

- `MAVLINK_STATUS_FLAG_*`
- `uint8_t signature_wait`
number of signature bytes left to receive
- `struct __mavlink_signing * signing`
optional signing state
- `struct __mavlink_signing_streams * signing_streams`
global record of stream timestamps

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/mavlink_types.h`

5.188 `__mavlink_storage_information_t` Struct Reference

Public Attributes

- `uint32_t time_boot_ms`
- `float total_capacity`
- `float used_capacity`
- `float available_capacity`
- `float read_speed`
- `float write_speed`
- `uint8_t storage_id`
- `uint8_t storage_count`
- `uint8_t status`
- `uint8_t type`
- `char name [32]`
- `uint8_t storage_usage`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_storage_information.h`

5.189 `__mavlink_supported_tunes_t` Struct Reference

Public Attributes

- `uint32_t format`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_supported_tunes.h`

5.190 `__mavlink_system_time_t` Struct Reference

Public Attributes

- `uint64_t time_unix_usec`
- `uint32_t time_boot_ms`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_system_time.h`

5.191 `__mavlink_terrain_check_t` Struct Reference

Public Attributes

- `int32_t lat`
- `int32_t lon`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_terrain_check.h`

5.192 `__mavlink_terrain_data_t` Struct Reference

Public Attributes

- `int32_t lat`
- `int32_t lon`
- `uint16_t grid_spacing`
- `int16_t data[16]`
- `uint8_t gridbit`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_terrain_data.h`

5.193 `__mavlink_terrain_report_t` Struct Reference

Public Attributes

- `int32_t lat`
- `int32_t lon`
- `float terrain_height`
- `float current_height`
- `uint16_t spacing`
- `uint16_t pending`
- `uint16_t loaded`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_terrain_report.h`

5.194 `__mavlink_terrain_request_t` Struct Reference

Public Attributes

- `uint64_t mask`
- `int32_t lat`
- `int32_t lon`
- `uint16_t grid_spacing`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_terrain_request.h`

5.195 `__mavlink_time_estimate_to_target_t` Struct Reference

Public Attributes

- `int32_t safe_return`
- `int32_t land`
- `int32_t mission_next_item`
- `int32_t mission_end`
- `int32_t commanded_action`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_time_estimate_to_target.h`

5.196 `__mavlink_timesync_t` Struct Reference

Public Attributes

- `int64_t tc1`
- `int64_t ts1`
- `uint8_t target_system`
- `uint8_t target_component`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_timesync.h`

5.197 `__mavlink_trajectory_representation_bezier_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float pos_x` [5]
- `float pos_y` [5]
- `float pos_z` [5]
- `float delta` [5]
- `float pos_yaw` [5]
- `uint8_t valid_points`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_bezier.h`

5.198 `__mavlink_trajectory_representation_waypoints_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float pos_x` [5]
- `float pos_y` [5]
- `float pos_z` [5]
- `float vel_x` [5]
- `float vel_y` [5]
- `float vel_z` [5]
- `float acc_x` [5]
- `float acc_y` [5]
- `float acc_z` [5]
- `float pos_yaw` [5]
- `float vel_yaw` [5]
- `uint16_t command` [5]
- `uint8_t valid_points`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_waypoints.h`

5.199 `__mavlink_tunnel_t` Struct Reference

Public Attributes

- `uint16_t payload_type`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t payload_length`
- `uint8_t payload` [128]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_tunnel.h`

5.200 `__mavlink_uavcan_node_info_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `uint32_t uptime_sec`
- `uint32_t sw_vcs_commit`
- `char name [80]`
- `uint8_t hw_version_major`
- `uint8_t hw_version_minor`
- `uint8_t hw_unique_id [16]`
- `uint8_t sw_version_major`
- `uint8_t sw_version_minor`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_info.h`

5.201 `__mavlink_uavcan_node_status_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `uint32_t uptime_sec`
- `uint16_t vendor_specific_status_code`
- `uint8_t health`
- `uint8_t mode`
- `uint8_t sub_mode`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_status.h`

5.202 `__mavlink_utm_global_position_t` Struct Reference

Public Attributes

- `uint64_t time`
- `int32_t lat`
- `int32_t lon`
- `int32_t alt`
- `int32_t relative_alt`
- `int32_t next_lat`
- `int32_t next_lon`
- `int32_t next_alt`
- `int16_t vx`
- `int16_t vy`
- `int16_t vz`

- uint16_t **h_acc**
- uint16_t **v_acc**
- uint16_t **vel_acc**
- uint16_t **update_rate**
- uint8_t **uas_id** [18]
- uint8_t **flight_state**
- uint8_t **flags**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_utm_global_position.h

5.203 **__mavlink_v2_extension_t** Struct Reference

Public Attributes

- uint16_t **message_type**
- uint8_t **target_network**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **payload** [249]

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_v2_extension.h

5.204 **__mavlink_vfr_hud_t** Struct Reference

Public Attributes

- float **airspeed**
- float **groundspeed**
- float **alt**
- float **climb**
- int16_t **heading**
- uint16_t **throttle**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_vfr_hud.h

5.205 `__mavlink_vibration_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float vibration_x`
- `float vibration_y`
- `float vibration_z`
- `uint32_t clipping_0`
- `uint32_t clipping_1`
- `uint32_t clipping_2`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_vibration.h`

5.206 `__mavlink_vicon_position_estimate_t` Struct Reference

Public Attributes

- `uint64_t usec`
- `float x`
- `float y`
- `float z`
- `float roll`
- `float pitch`
- `float yaw`
- `float covariance` [21]

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_vicon_position_estimate.h`

5.207 `__mavlink_video_stream_information_t` Struct Reference

Public Attributes

- `float framerate`
- `uint32_t bitrate`
- `uint16_t flags`
- `uint16_t resolution_h`
- `uint16_t resolution_v`
- `uint16_t rotation`
- `uint16_t hfov`
- `uint8_t stream_id`
- `uint8_t count`
- `uint8_t type`
- `char name` [32]
- `char uri` [160]
- `uint8_t encoding`
- `uint8_t camera_device_id`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_information.h`

5.208 `__mavlink_video_stream_status_t` Struct Reference

Public Attributes

- float **framerate**
- uint32_t **bitrate**
- uint16_t **flags**
- uint16_t **resolution_h**
- uint16_t **resolution_v**
- uint16_t **rotation**
- uint16_t **hfov**
- uint8_t **stream_id**
- uint8_t **camera_device_id**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_status.h`

5.209 `__mavlink_vision_position_estimate_t` Struct Reference

Public Attributes

- uint64_t **usec**
- float **x**
- float **y**
- float **z**
- float **roll**
- float **pitch**
- float **yaw**
- float **covariance** [21]
- uint8_t **reset_counter**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_vision_position_estimate.h`

5.210 `__mavlink_vision_speed_estimate_t` Struct Reference

Public Attributes

- uint64_t **usec**
- float **x**
- float **y**
- float **z**
- float **covariance** [9]
- uint8_t **reset_counter**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_vision_speed_estimate.h`

5.211 `__mavlink_wheel_distance_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `double distance` [16]
- `uint8_t count`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_wheel_distance.h`

5.212 `__mavlink_wifi_config_ap_t` Struct Reference

Public Attributes

- `char ssid` [32]
- `char password` [64]
- `int8_t mode`
- `int8_t response`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_wifi_config_ap.h`

5.213 `__mavlink_winch_status_t` Struct Reference

Public Attributes

- `uint64_t time_usec`
- `float line_length`
- `float speed`
- `float tension`
- `float voltage`
- `float current`
- `uint32_t status`
- `int16_t temperature`

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_winch_status.h`

5.214 __mavlink_wind_cov_t Struct Reference

Public Attributes

- uint64_t **time_usec**
- float **wind_x**
- float **wind_y**
- float **wind_z**
- float **var_horiz**
- float **var_vert**
- float **wind_alt**
- float **horiz_accuracy**
- float **vert_accuracy**

The documentation for this struct was generated from the following file:

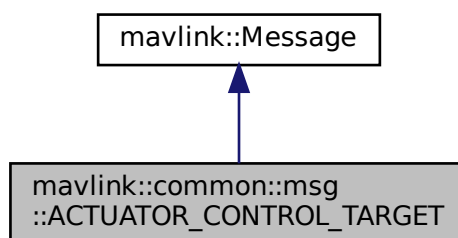
- single_hand_controller/include/mavlink/common/mavlink_msg_wind_cov.h

5.215 mavlink::common::msg::ACTUATOR_CONTROL_TARGET Struct Reference

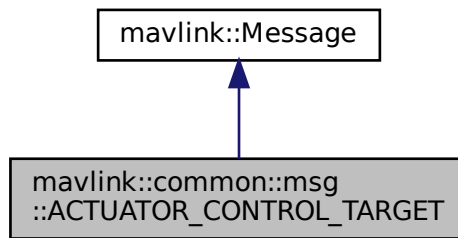
[ACTUATOR_CONTROL_TARGET](#) message.

```
#include <mavlink_msg_actuator_control_target.hpp>
```

Inheritance diagram for mavlink::common::msg::ACTUATOR_CONTROL_TARGET:



Collaboration diagram for mavlink::common::msg::ACTUATOR_CONTROL_TARGET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **group_mlx**
- `std::array< float, 8 >` **controls**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 140
- static constexpr `size_t` **LENGTH** = 41
- static constexpr `size_t` **MIN_LENGTH** = 41
- static constexpr `uint8_t` **CRC_EXTRA** = 181
- static constexpr auto **NAME** = "ACTUATOR_CONTROL_TARGET"

5.215.1 Detailed Description

[ACTUATOR_CONTROL_TARGET](#) message.

Set the vehicle attitude and body angular rates.

5.215.2 Member Function Documentation

5.215.2.1 deserialize()

```
void mavlink::common::msg::ACTUATOR_CONTROL_TARGET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.215.2.2 get_name()

```
std::string mavlink::common::msg::ACTUATOR_CONTROL_TARGET::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.215.2.3 serialize()

```
void mavlink::common::msg::ACTUATOR_CONTROL_TARGET::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

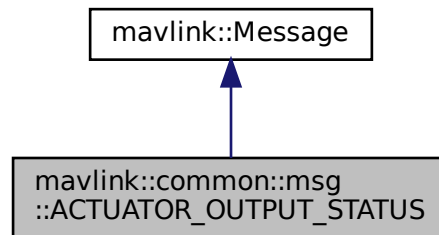
- `single_hand_controller/include/mavlink/common/mavlink_msg_actuator_control_target.hpp`

5.216 mavlink::common::msg::ACTUATOR_OUTPUT_STATUS Struct Reference

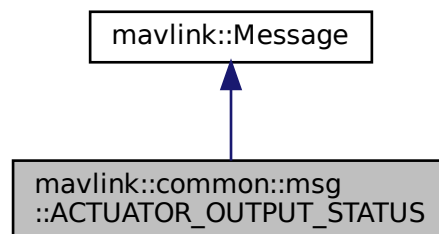
[ACTUATOR_OUTPUT_STATUS](#) message.

```
#include <mavlink_msg_actuator_output_status.hpp>
```

Inheritance diagram for mavlink::common::msg::ACTUATOR_OUTPUT_STATUS:



Collaboration diagram for mavlink::common::msg::ACTUATOR_OUTPUT_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- uint32_t **active**
- std::array< float, 32 > **actuator**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 375
- static constexpr size_t **LENGTH** = 140
- static constexpr size_t **MIN_LENGTH** = 140
- static constexpr uint8_t **CRC_EXTRA** = 251
- static constexpr auto **NAME** = "ACTUATOR_OUTPUT_STATUS"

5.216.1 Detailed Description

[ACTUATOR_OUTPUT_STATUS](#) message.

The raw values of the actuator outputs (e.g. on Pixhawk, from MAIN, AUX ports). This message supersedes [SERVO_OUTPUT_RAW](#).

5.216.2 Member Function Documentation

5.216.2.1 deserialize()

```
void mavlink::common::msg::ACTUATOR_OUTPUT_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.216.2.2 get_name()

```
std::string mavlink::common::msg::ACTUATOR_OUTPUT_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.216.2.3 serialize()

```
void mavlink::common::msg::ACTUATOR_OUTPUT_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

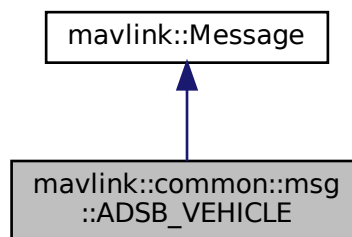
- single_hand_controller/include/mavlink/common/mavlink_msg_actuator_output_status.hpp

5.217 mavlink::common::msg::ADSB_VEHICLE Struct Reference

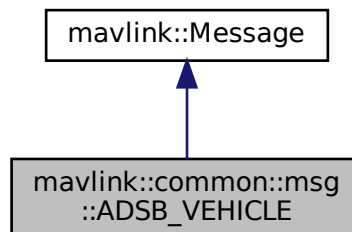
[ADSB_VEHICLE](#) message.

```
#include <mavlink_msg_adsb_vehicle.hpp>
```

Inheritance diagram for mavlink::common::msg::ADSB_VEHICLE:



Collaboration diagram for mavlink::common::msg::ADSB_VEHICLE:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- `void deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint32_t ICAO_address`
- `int32_t lat`
- `int32_t lon`
- `uint8_t altitude_type`
- `int32_t altitude`
- `uint16_t heading`
- `uint16_t hor_velocity`
- `int16_t ver_velocity`
- `std::array< char, 9 > callsign`
- `uint8_t emitter_type`
- `uint8_t tslc`
- `uint16_t flags`
- `uint16_t squawk`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 246`
- `static constexpr size_t LENGTH = 38`
- `static constexpr size_t MIN_LENGTH = 38`
- `static constexpr uint8_t CRC_EXTRA = 184`
- `static constexpr auto NAME = "ADSB_VEHICLE"`

5.217.1 Detailed Description

[ADSB_VEHICLE](#) message.

The location and information of an ADSB vehicle

5.217.2 Member Function Documentation

5.217.2.1 deserialize()

```
void mavlink::common::msg::ADSB_VEHICLE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.217.2.2 get_name()

```
std::string mavlink::common::msg::ADSB_VEHICLE::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.217.2.3 serialize()

```
void mavlink::common::msg::ADSB_VEHICLE::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

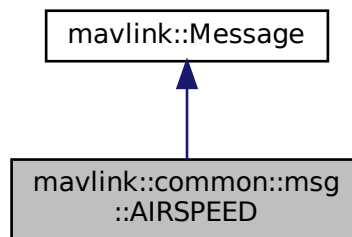
- single_hand_controller/include/mavlink/common/mavlink_msg_adsb_vehicle.hpp

5.218 mavlink::common::msg::AIRSPEED Struct Reference

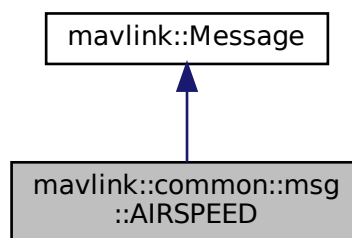
[AIRSPEED](#) message.

```
#include <mavlink_msg_airspeed.hpp>
```

Inheritance diagram for mavlink::common::msg::AIRSPEED:



Collaboration diagram for mavlink::common::msg::AIRSPEED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **id**
- `float` **airspeed**
- `int16_t` **temperature**
- `float` **raw_press**
- `uint8_t` **flags**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 295
- static constexpr size_t **LENGTH** = 12
- static constexpr size_t **MIN_LENGTH** = 12
- static constexpr uint8_t **CRC_EXTRA** = 234
- static constexpr auto **NAME** = "AIRSPEED"

5.218.1 Detailed Description

[AIRSPEED](#) message.

Airspeed information from a sensor.

5.218.2 Member Function Documentation

5.218.2.1 deserialize()

```
void mavlink::common::msg::AIRSPEED::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.218.2.2 get_name()

```
std::string mavlink::common::msg::AIRSPEED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.218.2.3 serialize()

```
void mavlink::common::msg::AIRSPEED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

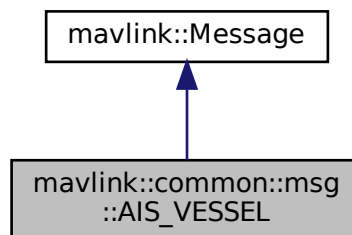
- `single_hand_controller/include/mavlink/common/mavlink_msg_airspeed.hpp`

5.219 mavlink::common::msg::AIS_VESSEL Struct Reference

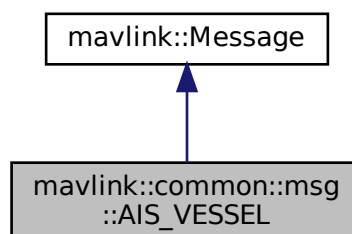
[AIS_VESSEL](#) message.

```
#include <mavlink_msg_ais_vessel.hpp>
```

Inheritance diagram for `mavlink::common::msg::AIS_VESSEL`:



Collaboration diagram for `mavlink::common::msg::AIS_VESSEL`:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint32_t` **MMSI**
- `int32_t` **lat**
- `int32_t` **lon**
- `uint16_t` **COG**
- `uint16_t` **heading**
- `uint16_t` **velocity**
- `int8_t` **turn_rate**
- `uint8_t` **navigational_status**
- `uint8_t` **type**
- `uint16_t` **dimension_bow**
- `uint16_t` **dimension_stern**
- `uint8_t` **dimension_port**
- `uint8_t` **dimension_starboard**
- `std::array< char, 7 >` **callsign**
- `std::array< char, 20 >` **name**
- `uint16_t` **tslc**
- `uint16_t` **flags**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 301
- static constexpr `size_t` **LENGTH** = 58
- static constexpr `size_t` **MIN_LENGTH** = 58
- static constexpr `uint8_t` **CRC_EXTRA** = 243
- static constexpr auto **NAME** = "AIS_VESSEL"

5.219.1 Detailed Description

[AIS_VESSEL](#) message.

The location and information of an AIS vessel

5.219.2 Member Function Documentation

5.219.2.1 deserialize()

```
void mavlink::common::msg::AIS_VESSEL::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.219.2.2 get_name()

```
std::string mavlink::common::msg::AIS_VESSEL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.219.2.3 serialize()

```
void mavlink::common::msg::AIS_VESSEL::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

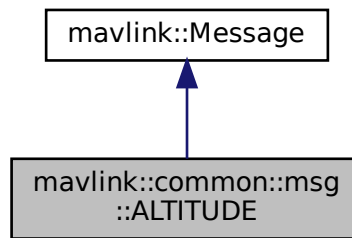
- single_hand_controller/include/mavlink/common/mavlink_msg_ais_vessel.hpp

5.220 mavlink::common::msg::ALTITUDE Struct Reference

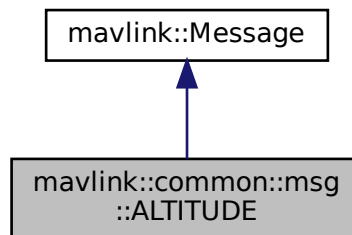
[ALTITUDE](#) message.

```
#include <mavlink_msg_altitude.hpp>
```

Inheritance diagram for mavlink::common::msg::ALTITUDE:



Collaboration diagram for mavlink::common::msg::ALTITUDE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` `time_usec`
- `float` `altitude_monotonic`

- float **altitude_amsl**
- float **altitude_local**
- float **altitude_relative**
- float **altitude_terrain**
- float **bottom_clearance**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 141
- static constexpr size_t **LENGTH** = 32
- static constexpr size_t **MIN_LENGTH** = 32
- static constexpr uint8_t **CRC_EXTRA** = 47
- static constexpr auto **NAME** = "ALTITUDE"

5.220.1 Detailed Description

[ALTITUDE](#) message.

The current system altitude.

5.220.2 Member Function Documentation

5.220.2.1 deserialize()

```
void mavlink::common::msg::ALTITUDE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.220.2.2 get_name()

```
std::string mavlink::common::msg::ALTITUDE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.220.2.3 serialize()

```
void mavlink::common::msg::ALTITUDE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/mavlink/common/mavlink_msg_altitude.hpp](#)

5.221 ATLAS_HC_ARM_DISARM_ACK Struct Reference

Public Attributes

- `uint8_t` **command**
- `uint8_t` **result**
- `uint8_t` **target_system** = HC_ID
- `uint8_t` **target_component** = HC_COMP_ID

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/message_structs.h](#)

5.222 ATLAS_HC_HEARTBEAT_BC Struct Reference

Public Attributes

- `uint8_t` **sys_id** = SCOUT_ID
- `uint8_t` **comp_id** = ATLAS_COMP_ID
- `uint8_t` **type** = MAV_TYPE_ONBOARD_CONTROLLER
- `uint8_t` **autopilot** = MAV_AUTOPILOT_GENERIC
- `uint8_t` **sys_status**
- `uint32_t` **custom_mode** = SCOUT_HEARTBEAT_IDENTIFIER

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/message_structs.h](#)

5.223 ATLAS_HC_SYS_STAT Struct Reference

Public Attributes

- uint16_t **voltage_battery**
- int8_t **battery_remaining**
- uint16_t **drop_rate_comm**

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/message_structs.h](#)

5.224 ATLAS_HC_TIMESYNC_RESP Struct Reference

Public Attributes

- uint64_t **tc1**
- uint64_t **ts1**
- uint8_t **target_sys** = HC_ID
- uint8_t **target_comp** = HC_COMP_ID

The documentation for this struct was generated from the following file:

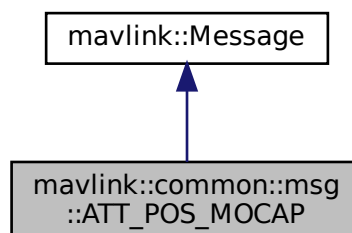
- [single_hand_controller/include/message_structs.h](#)

5.225 mavlink::common::msg::ATT_POS_MOCAP Struct Reference

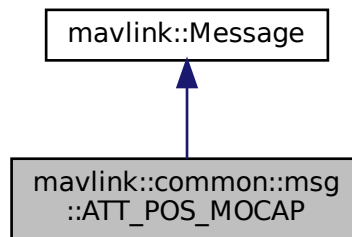
[ATT_POS_MOCAP](#) message.

```
#include <mavlink_msg_att_pos_mocap.hpp>
```

Inheritance diagram for mavlink::common::msg::ATT_POS_MOCAP:



Collaboration diagram for mavlink::common::msg::ATT_POS_MOCAP:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `std::array< float, 4 >` **q**
- `float` **x**
- `float` **y**
- `float` **z**
- `std::array< float, 21 >` **covariance**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 138
- static constexpr `size_t` **LENGTH** = 120
- static constexpr `size_t` **MIN_LENGTH** = 36
- static constexpr `uint8_t` **CRC_EXTRA** = 109
- static constexpr `auto` **NAME** = "ATT_POS_MOCAP"

5.225.1 Detailed Description

[ATT_POS_MOCAP](#) message.

Motion capture attitude and position

5.225.2 Member Function Documentation

5.225.2.1 deserialize()

```
void mavlink::common::msg::ATT_POS_MOCAP::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.225.2.2 get_name()

```
std::string mavlink::common::msg::ATT_POS_MOCAP::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.225.2.3 serialize()

```
void mavlink::common::msg::ATT_POS_MOCAP::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

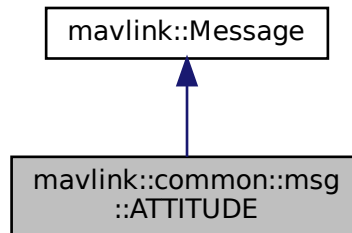
- single_hand_controller/include/mavlink/common/mavlink_msg_att_pos_mocap.hpp

5.226 mavlink::common::msg::ATTITUDE Struct Reference

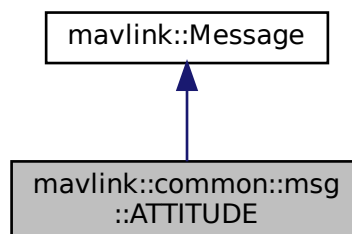
[ATTITUDE](#) message.

```
#include <mavlink_msg_attitude.hpp>
```

Inheritance diagram for mavlink::common::msg::ATTITUDE:



Collaboration diagram for mavlink::common::msg::ATTITUDE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- float **roll**
- float **pitch**
- float **yaw**
- float **rollspeed**
- float **pitchspeed**
- float **yawspeed**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 30
- static constexpr size_t **LENGTH** = 28
- static constexpr size_t **MIN_LENGTH** = 28
- static constexpr uint8_t **CRC_EXTRA** = 39
- static constexpr auto **NAME** = "ATTITUDE"

5.226.1 Detailed Description

[ATTITUDE](#) message.

The attitude in the aeronautical frame (right-handed, Z-down, Y-right, X-front, ZYX, intrinsic).

5.226.2 Member Function Documentation

5.226.2.1 deserialize()

```
void mavlink::common::msg::ATTITUDE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.226.2.2 get_name()

```
std::string mavlink::common::msg::ATTITUDE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.226.2.3 serialize()

```
void mavlink::common::msg::ATTITUDE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

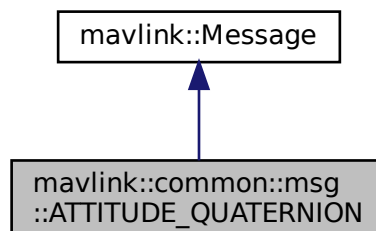
- single_hand_controller/include/mavlink/common/mavlink_msg_attitude.hpp

5.227 mavlink::common::msg::ATTITUDE_QUATERNION Struct Reference

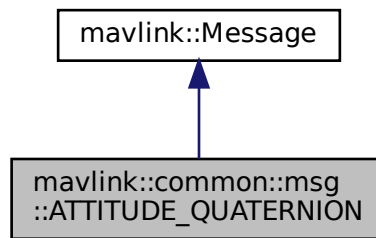
[ATTITUDE_QUATERNION](#) message.

```
#include <mavlink_msg_attitude_quaternion.hpp>
```

Inheritance diagram for mavlink::common::msg::ATTITUDE_QUATERNION:



Collaboration diagram for mavlink::common::msg::ATTITUDE_QUATERNION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **q1**
- `float` **q2**
- `float` **q3**
- `float` **q4**
- `float` **rollspeed**
- `float` **pitchspeed**
- `float` **yawspeed**
- `std::array< float, 4 >` **repr_offset_q**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 31
- static constexpr `size_t` **LENGTH** = 48
- static constexpr `size_t` **MIN_LENGTH** = 32
- static constexpr `uint8_t` **CRC_EXTRA** = 246
- static constexpr auto **NAME** = "ATTITUDE_QUATERNION"

5.227.1 Detailed Description

[ATTITUDE_QUATERNION](#) message.

The attitude in the aeronautical frame (right-handed, Z-down, X-front, Y-right), expressed as quaternion. Quaternion order is w, x, y, z and a zero rotation would be expressed as (1 0 0 0).

5.227.2 Member Function Documentation

5.227.2.1 deserialize()

```
void mavlink::common::msg::ATTITUDE_QUATERNION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.227.2.2 get_name()

```
std::string mavlink::common::msg::ATTITUDE_QUATERNION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.227.2.3 serialize()

```
void mavlink::common::msg::ATTITUDE_QUATERNION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

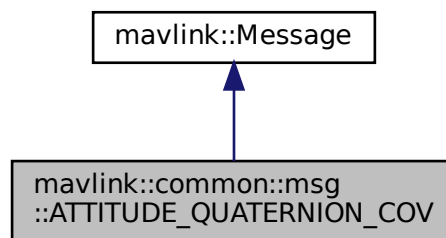
- `single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion.hpp`

5.228 mavlink::common::msg::ATTITUDE_QUATERNION_COV Struct Reference

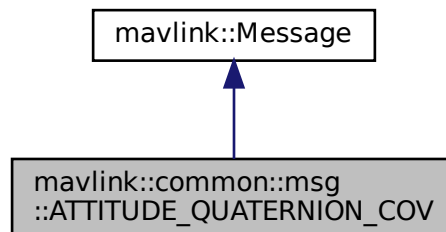
[ATTITUDE_QUATERNION_COV](#) message.

```
#include <mavlink_msg_attitude_quaternion_cov.hpp>
```

Inheritance diagram for `mavlink::common::msg::ATTITUDE_QUATERNION_COV`:



Collaboration diagram for `mavlink::common::msg::ATTITUDE_QUATERNION_COV`:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `std::array< float, 4 >` **q**
- `float` **rollspeed**
- `float` **pitchspeed**
- `float` **yawspeed**
- `std::array< float, 9 >` **covariance**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 61
- static constexpr `size_t` **LENGTH** = 72
- static constexpr `size_t` **MIN_LENGTH** = 72
- static constexpr `uint8_t` **CRC_EXTRA** = 167
- static constexpr auto **NAME** = "ATTITUDE_QUATERNION_COV"

5.228.1 Detailed Description

[ATTITUDE_QUATERNION_COV](#) message.

The attitude in the aeronautical frame (right-handed, Z-down, X-front, Y-right), expressed as quaternion. Quaternion order is w, x, y, z and a zero rotation would be expressed as (1 0 0 0).

5.228.2 Member Function Documentation

5.228.2.1 deserialize()

```
void mavlink::common::msg::ATTITUDE_QUATERNION_COV::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.228.2.2 get_name()

```
std::string mavlink::common::msg::ATTITUDE_QUATERNION_COV::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.228.2.3 serialize()

```
void mavlink::common::msg::ATTITUDE_QUATERNION_COV::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

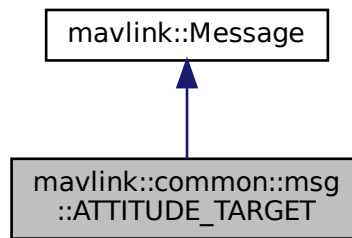
- single_hand_controller/include/mavlink/common/mavlink_msg_attitude_quaternion_cov.hpp

5.229 mavlink::common::msg::ATTITUDE_TARGET Struct Reference

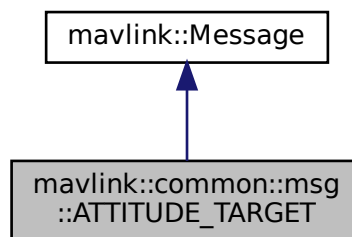
[ATTITUDE_TARGET](#) message.

```
#include <mavlink_msg_attitude_target.hpp>
```

Inheritance diagram for mavlink::common::msg::ATTITUDE_TARGET:



Collaboration diagram for mavlink::common::msg::ATTITUDE_TARGET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` [time_boot_ms](#)
- `uint8_t` [type_mask](#)

- `std::array< float, 4 > q`
- `float body_roll_rate`
- `float body_pitch_rate`
- `float body_yaw_rate`
- `float thrust`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 83`
- `static constexpr size_t LENGTH = 37`
- `static constexpr size_t MIN_LENGTH = 37`
- `static constexpr uint8_t CRC_EXTRA = 22`
- `static constexpr auto NAME = "ATTITUDE_TARGET"`

5.229.1 Detailed Description

[ATTITUDE_TARGET](#) message.

Reports the current commanded attitude of the vehicle as specified by the autopilot. This should match the commands sent in a [SET_ATTITUDE_TARGET](#) message if the vehicle is being controlled this way.

5.229.2 Member Function Documentation

5.229.2.1 deserialize()

```
void mavlink::common::msg::ATTITUDE_TARGET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.229.2.2 get_name()

```
std::string mavlink::common::msg::ATTITUDE_TARGET::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.229.2.3 serialize()

```
void mavlink::common::msg::ATTITUDE_TARGET::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

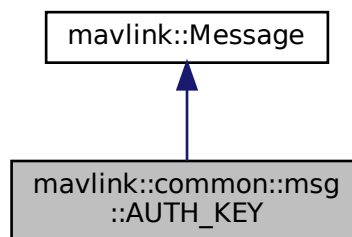
- single_hand_controller/include/mavlink/common/mavlink_msg_attitude_target.hpp

5.230 mavlink::common::msg::AUTH_KEY Struct Reference

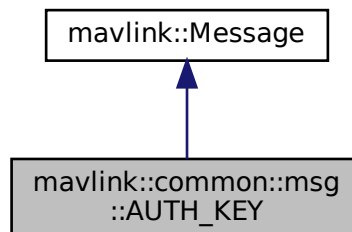
[AUTH_KEY](#) message.

```
#include <mavlink_msg_auth_key.hpp>
```

Inheritance diagram for mavlink::common::msg::AUTH_KEY:



Collaboration diagram for mavlink::common::msg::AUTH_KEY:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) ([mavlink::MsgMap](#) &map) const override
Serialize message.
- void [deserialize](#) ([mavlink::MsgMap](#) &map) override
Deserialize message.

Public Attributes

- `std::array< char, 32 >` **key**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 7
- static constexpr size_t **LENGTH** = 32
- static constexpr size_t **MIN_LENGTH** = 32
- static constexpr uint8_t **CRC_EXTRA** = 119
- static constexpr auto **NAME** = "AUTH_KEY"

5.230.1 Detailed Description

[AUTH_KEY](#) message.

Emit an encrypted signature / key identifying this system. PLEASE NOTE: This protocol has been kept simple, so transmitting the key requires an encrypted channel for true safety.

5.230.2 Member Function Documentation

5.230.2.1 deserialize()

```
void mavlink::common::msg::AUTH_KEY::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.230.2.2 `get_name()`

```
std::string mavlink::common::msg::AUTH_KEY::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.230.2.3 `serialize()`

```
void mavlink::common::msg::AUTH_KEY::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

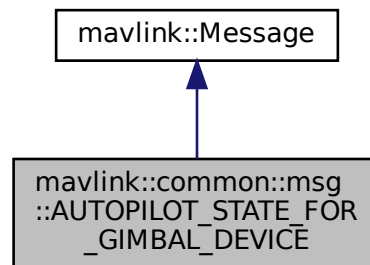
- `single_hand_controller/include/mavlink/common/mavlink_msg_auth_key.hpp`

5.231 `mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE` Struct Reference

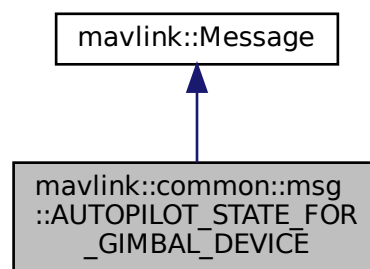
[AUTOPILOT_STATE_FOR_GIMBAL_DEVICE](#) message.

```
#include <mavlink_msg_autopilot_state_for_gimbal_device.hpp>
```

Inheritance diagram for mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE:



Collaboration diagram for mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint64_t **time_boot_us**
- std::array< float, 4 > **q**
- uint32_t **q_estimated_delay_us**
- float **vx**
- float **vy**
- float **vz**
- uint32_t **v_estimated_delay_us**
- float **feed_forward_angular_velocity_z**
- uint16_t **estimator_status**
- uint8_t **landed_state**
- float **angular_velocity_z**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 286
- static constexpr size_t **LENGTH** = 57
- static constexpr size_t **MIN_LENGTH** = 53
- static constexpr uint8_t **CRC_EXTRA** = 210
- static constexpr auto **NAME** = "AUTOPILOT_STATE_FOR_GIMBAL_DEVICE"

5.231.1 Detailed Description

[AUTOPILOT_STATE_FOR_GIMBAL_DEVICE](#) message.

Low level message containing autopilot state relevant for a gimbal device. This message is to be sent from the autopilot to the gimbal device component. The data of this message are for the gimbal device's estimator corrections, in particular horizon compensation, as well as indicates autopilot control intentions, e.g. feed forward angular control in the z-axis.

5.231.2 Member Function Documentation

5.231.2.1 deserialize()

```
void mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.231.2.2 get_name()

```
std::string mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.231.2.3 serialize()

```
void mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

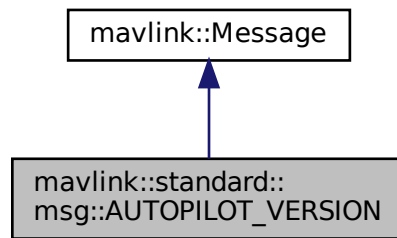
- single_hand_controller/include/mavlink/common/mavlink_msg_autopilot_state_for_gimbal_device.hpp

5.232 mavlink::standard::msg::AUTOPILOT_VERSION Struct Reference

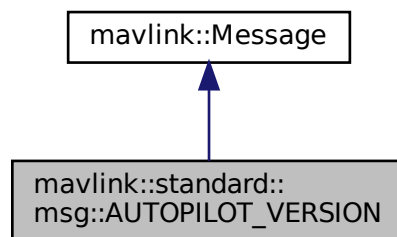
[AUTOPILOT_VERSION](#) message.

```
#include <mavlink_msg_autopilot_version.hpp>
```

Inheritance diagram for mavlink::standard::msg::AUTOPILOT_VERSION:



Collaboration diagram for mavlink::standard::msg::AUTOPILOT_VERSION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **capabilities**
- `uint32_t` **flight_sw_version**

- uint32_t **middleware_sw_version**
- uint32_t **os_sw_version**
- uint32_t **board_version**
- std::array< uint8_t, 8 > **flight_custom_version**
- std::array< uint8_t, 8 > **middleware_custom_version**
- std::array< uint8_t, 8 > **os_custom_version**
- uint16_t **vendor_id**
- uint16_t **product_id**
- uint64_t **uid**
- std::array< uint8_t, 18 > **uid2**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 148
- static constexpr size_t **LENGTH** = 78
- static constexpr size_t **MIN_LENGTH** = 60
- static constexpr uint8_t **CRC_EXTRA** = 178
- static constexpr auto **NAME** = "AUTOPILOT_VERSION"

5.232.1 Detailed Description

[AUTOPILOT_VERSION](#) message.

Version and capability of autopilot software. This should be emitted in response to a request with MAV_CMD_REQUEST_MESSAGE.

5.232.2 Member Function Documentation

5.232.2.1 deserialize()

```
void mavlink::standard::msg::AUTOPILOT_VERSION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.232.2.2 get_name()

```
std::string mavlink::standard::msg::AUTOPILOT_VERSION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.232.2.3 `serialize()`

```
void mavlink::standard::msg::AUTOPILOT_VERSION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

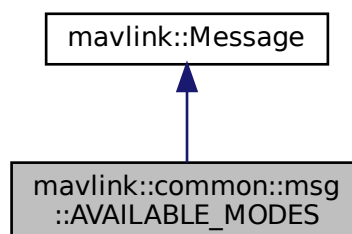
- `single_hand_controller/include/mavlink/standard/mavlink_msg_autopilot_version.hpp`

5.233 `mavlink::common::msg::AVAILABLE_MODES` Struct Reference

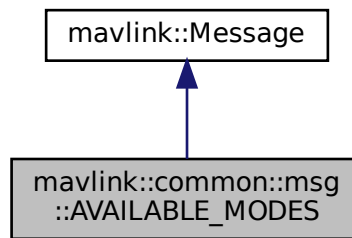
[AVAILABLE_MODES](#) message.

```
#include <mavlink_msg_available_modes.hpp>
```

Inheritance diagram for `mavlink::common::msg::AVAILABLE_MODES`:



Collaboration diagram for mavlink::common::msg::AVAILABLE_MODES:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **number_modes**
- uint8_t **mode_index**
- uint8_t **standard_mode**
- uint32_t **custom_mode**
- uint32_t **properties**
- std::array< char, 35 > **mode_name**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 435
- static constexpr size_t **LENGTH** = 46
- static constexpr size_t **MIN_LENGTH** = 46
- static constexpr uint8_t **CRC_EXTRA** = 134
- static constexpr auto **NAME** = "AVAILABLE_MODES"

5.233.1 Detailed Description

[AVAILABLE_MODES](#) message.

Information about a flight mode.

The message can be enumerated to get information for all modes, or requested for a particular mode, using MAV_SPECIFIC. Specify 0 in param2 to request that the message is emitted for all available modes or the specific index for a particular mode. The modes must be available/settable for the current vehicle/frame type. Each mode should only be emitted once (even if it is both standard and custom). Note that the current mode should be emitted in CURRENT_MODE, and that if the mode list can change then AVAILABLE_MODES should be emitted. See https://mavlink.io/en/services/standard_modes.html

5.233.2 Member Function Documentation

5.233.2.1 deserialize()

```
void mavlink::common::msg::AVAILABLE_MODES::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.233.2.2 get_name()

```
std::string mavlink::common::msg::AVAILABLE_MODES::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.233.2.3 serialize()

```
void mavlink::common::msg::AVAILABLE_MODES::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

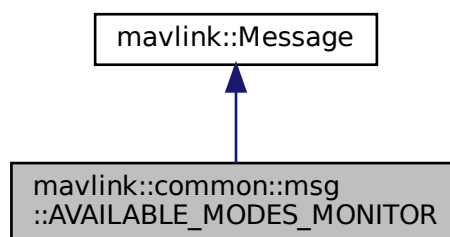
- single_hand_controller/include/mavlink/common/mavlink_msg_available_modes.hpp

5.234 mavlink::common::msg::AVAILABLE_MODES_MONITOR Struct Reference

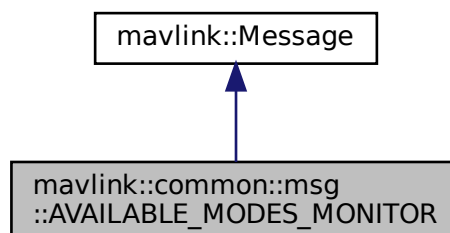
[AVAILABLE_MODES_MONITOR](#) message.

```
#include <mavlink_msg_available_modes_monitor.hpp>
```

Inheritance diagram for mavlink::common::msg::AVAILABLE_MODES_MONITOR:



Collaboration diagram for mavlink::common::msg::AVAILABLE_MODES_MONITOR:



Public Member Functions

- `std::string get_name (void) const` override
Get NAME constant.
- `Info get_message_info (void) const` override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map)` override
Deserialize message.

Public Attributes

- `uint8_t seq`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 437`
- `static constexpr size_t LENGTH = 1`
- `static constexpr size_t MIN_LENGTH = 1`
- `static constexpr uint8_t CRC_EXTRA = 30`
- `static constexpr auto NAME = "AVAILABLE_MODES_MONITOR"`

5.234.1 Detailed Description

[AVAILABLE_MODES_MONITOR](#) message.

A change to the sequence number indicates that the set of [AVAILABLE_MODES](#) has changed. A receiver must re-request all available modes whenever the sequence number changes. This is only emitted after the first change and should then be broadcast at low rate (nominally 0.3 Hz) and on change. See https://mavlink.io/en/services/standard_modes.html

5.234.2 Member Function Documentation

5.234.2.1 deserialize()

```
void mavlink::common::msg::AVAILABLE_MODES_MONITOR::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.234.2.2 get_name()

```
std::string mavlink::common::msg::AVAILABLE_MODES_MONITOR::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.234.2.3 serialize()

```
void mavlink::common::msg::AVAILABLE_MODES_MONITOR::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

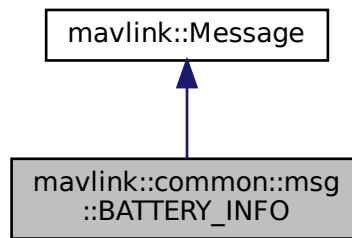
- single_hand_controller/include/mavlink/common/mavlink_msg_available_modes_monitor.hpp

5.235 mavlink::common::msg::BATTERY_INFO Struct Reference

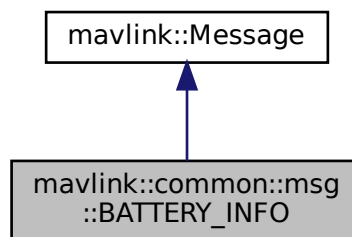
[BATTERY_INFO](#) message.

```
#include <mavlink_msg_battery_info.hpp>
```

Inheritance diagram for mavlink::common::msg::BATTERY_INFO:



Collaboration diagram for mavlink::common::msg::BATTERY_INFO:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **id**
- `uint8_t` **battery_function**

- uint8_t **type**
- uint8_t **state_of_health**
- uint8_t **cells_in_series**
- uint16_t **cycle_count**
- uint16_t **weight**
- float **discharge_minimum_voltage**
- float **charging_minimum_voltage**
- float **resting_minimum_voltage**
- float **charging_maximum_voltage**
- float **charging_maximum_current**
- float **nominal_voltage**
- float **discharge_maximum_current**
- float **discharge_maximum_burst_current**
- float **design_capacity**
- float **full_charge_capacity**
- std::array< char, 9 > **manufacture_date**
- std::array< char, 32 > **serial_number**
- std::array< char, 50 > **name**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 372
- static constexpr size_t **LENGTH** = 140
- static constexpr size_t **MIN_LENGTH** = 140
- static constexpr uint8_t **CRC_EXTRA** = 26
- static constexpr auto **NAME** = "BATTERY_INFO"

5.235.1 Detailed Description

[BATTERY_INFO](#) message.

Battery information that is static, or requires infrequent update.
This message should requested using MAV_CMD_REQUEST_MESSAGE and/or streamed at very low rate.
BATTERY_STATUS_V2 is used for higher-rate battery status information.

5.235.2 Member Function Documentation

5.235.2.1 deserialize()

```
void mavlink::common::msg::BATTERY_INFO::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.235.2.2 `get_name()`

```
std::string mavlink::common::msg::BATTERY_INFO::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.235.2.3 `serialize()`

```
void mavlink::common::msg::BATTERY_INFO::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

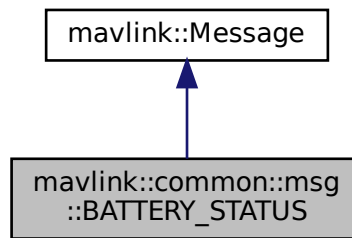
- `single_hand_controller/include/mavlink/common/mavlink_msg_battery_info.hpp`

5.236 `mavlink::common::msg::BATTERY_STATUS` Struct Reference

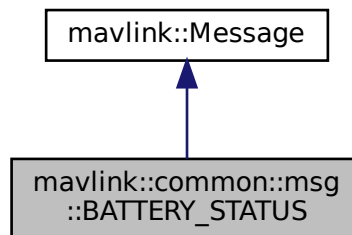
[BATTERY_STATUS](#) message.

```
#include <mavlink_msg_battery_status.hpp>
```

Inheritance diagram for mavlink::common::msg::BATTERY_STATUS:



Collaboration diagram for mavlink::common::msg::BATTERY_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **id**
- `uint8_t` **battery_function**

- uint8_t **type**
- int16_t **temperature**
- std::array< uint16_t, 10 > **voltages**
- int16_t **current_battery**
- int32_t **current_consumed**
- int32_t **energy_consumed**
- int8_t **battery_remaining**
- int32_t **time_remaining**
- uint8_t **charge_state**
- std::array< uint16_t, 4 > **voltages_ext**
- uint8_t **mode**
- uint32_t **fault_bitmask**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 147
- static constexpr size_t **LENGTH** = 54
- static constexpr size_t **MIN_LENGTH** = 36
- static constexpr uint8_t **CRC_EXTRA** = 154
- static constexpr auto **NAME** = "BATTERY_STATUS"

5.236.1 Detailed Description

[BATTERY_STATUS](#) message.

Battery information. Updates GCS with flight controller battery status. Smart batteries also use this message, but may additionally send [BATTERY_INFO](#).

5.236.2 Member Function Documentation

5.236.2.1 deserialize()

```
void mavlink::common::msg::BATTERY_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.236.2.2 get_name()

```
std::string mavlink::common::msg::BATTERY_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.236.2.3 serialize()

```
void mavlink::common::msg::BATTERY_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

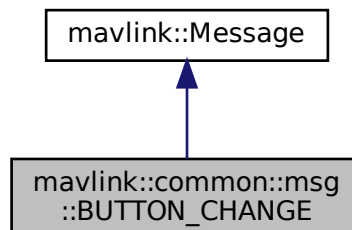
- single_hand_controller/include/mavlink/common/mavlink_msg_battery_status.hpp

5.237 mavlink::common::msg::BUTTON_CHANGE Struct Reference

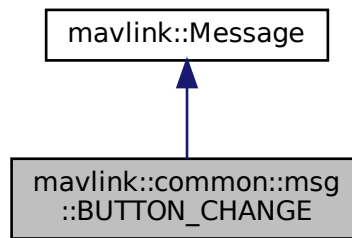
[BUTTON_CHANGE](#) message.

```
#include <mavlink_msg_button_change.hpp>
```

Inheritance diagram for mavlink::common::msg::BUTTON_CHANGE:



Collaboration diagram for mavlink::common::msg::BUTTON_CHANGE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint32_t` **last_change_ms**
- `uint8_t` **state**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 257
- static constexpr size_t **LENGTH** = 9
- static constexpr size_t **MIN_LENGTH** = 9
- static constexpr uint8_t **CRC_EXTRA** = 131
- static constexpr auto **NAME** = "BUTTON_CHANGE"

5.237.1 Detailed Description

[BUTTON_CHANGE](#) message.

Report button state change.

5.237.2 Member Function Documentation

5.237.2.1 deserialize()

```
void mavlink::common::msg::BUTTON_CHANGE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.237.2.2 get_name()

```
std::string mavlink::common::msg::BUTTON_CHANGE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.237.2.3 serialize()

```
void mavlink::common::msg::BUTTON_CHANGE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

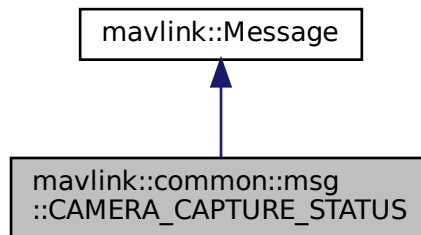
- `single_hand_controller/include/mavlink/common/mavlink_msg_button_change.hpp`

5.238 mavlink::common::msg::CAMERA_CAPTURE_STATUS Struct Reference

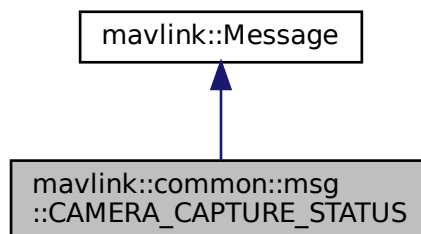
[CAMERA_CAPTURE_STATUS](#) message.

```
#include <mavlink_msg_camera_capture_status.hpp>
```

Inheritance diagram for mavlink::common::msg::CAMERA_CAPTURE_STATUS:



Collaboration diagram for mavlink::common::msg::CAMERA_CAPTURE_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- uint8_t **image_status**
- uint8_t **video_status**
- float **image_interval**
- uint32_t **recording_time_ms**
- float **available_capacity**
- int32_t **image_count**
- uint8_t **camera_device_id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 262
- static constexpr size_t **LENGTH** = 23
- static constexpr size_t **MIN_LENGTH** = 18
- static constexpr uint8_t **CRC_EXTRA** = 12
- static constexpr auto **NAME** = "CAMERA_CAPTURE_STATUS"

5.238.1 Detailed Description

[CAMERA_CAPTURE_STATUS](#) message.

Information about the status of a capture. Can be requested with a MAV_CMD_REQUEST_MESSAGE command.

5.238.2 Member Function Documentation

5.238.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_CAPTURE_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.238.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_CAPTURE_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.238.2.3 `serialize()`

```
void mavlink::common::msg::CAMERA_CAPTURE_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

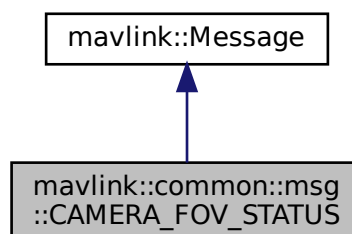
- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_capture_status.hpp`

5.239 `mavlink::common::msg::CAMERA_FOV_STATUS` Struct Reference

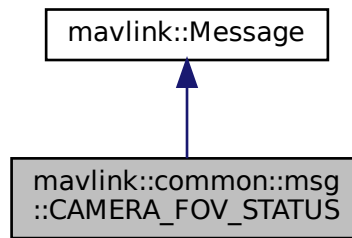
`CAMERA_FOV_STATUS` message.

```
#include <mavlink_msg_camera_fov_status.hpp>
```

Inheritance diagram for `mavlink::common::msg::CAMERA_FOV_STATUS`:



Collaboration diagram for mavlink::common::msg::CAMERA_FOV_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `int32_t` **lat_camera**
- `int32_t` **lon_camera**
- `int32_t` **alt_camera**
- `int32_t` **lat_image**
- `int32_t` **lon_image**
- `int32_t` **alt_image**
- `std::array< float, 4 >` **q**
- `float` **hfov**
- `float` **vfov**
- `uint8_t` **camera_device_id**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 271
- static constexpr `size_t` **LENGTH** = 53
- static constexpr `size_t` **MIN_LENGTH** = 52
- static constexpr `uint8_t` **CRC_EXTRA** = 22
- static constexpr auto **NAME** = "CAMERA_FOV_STATUS"

5.239.1 Detailed Description

[CAMERA_FOV_STATUS](#) message.

Information about the field of view of a camera. Can be requested with a MAV_CMD_REQUEST_MESSAGE command.

5.239.2 Member Function Documentation

5.239.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_FOV_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.239.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_FOV_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.239.2.3 serialize()

```
void mavlink::common::msg::CAMERA_FOV_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

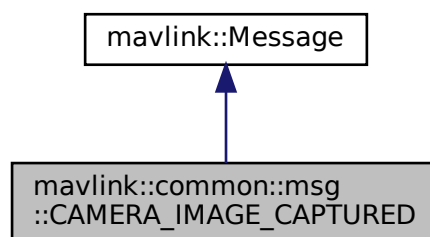
- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_fov_status.hpp`

5.240 mavlink::common::msg::CAMERA_IMAGE_CAPTURED Struct Reference

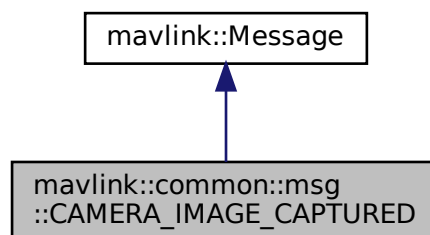
[CAMERA_IMAGE_CAPTURED](#) message.

```
#include <mavlink_msg_camera_image_captured.hpp>
```

Inheritance diagram for mavlink::common::msg::CAMERA_IMAGE_CAPTURED:



Collaboration diagram for mavlink::common::msg::CAMERA_IMAGE_CAPTURED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint64_t` **time_utc**
- `uint8_t` **camera_id**
- `int32_t` **lat**
- `int32_t` **lon**
- `int32_t` **alt**
- `int32_t` **relative_alt**
- `std::array< float, 4 >` **q**
- `int32_t` **image_index**
- `int8_t` **capture_result**
- `std::array< char, 205 >` **file_url**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 263
- static constexpr `size_t` **LENGTH** = 255
- static constexpr `size_t` **MIN_LENGTH** = 255
- static constexpr `uint8_t` **CRC_EXTRA** = 133
- static constexpr auto **NAME** = "CAMERA_IMAGE_CAPTURED"

5.240.1 Detailed Description

[CAMERA_IMAGE_CAPTURED](#) message.

Information about a captured image. This is emitted every time a message is captured. `MAV_CMD_REQUEST_MESSAGE` can be used to (re)request this message for a specific sequence number or range of sequence numbers: `MAV_CMD_REQUEST_MESSAGE.param2` indicates the sequence number the first image to send, or set to -1 to send the message for all sequence numbers. `MAV_CMD_REQUEST_MESSAGE.param3` is used to specify a range of messages to send: set to 0 (default) to send just the the message for the sequence number in param 2, set to -1 to send the message for the sequence number in param 2 and all the following sequence numbers, set to the sequence number of the final message in the range.

5.240.2 Member Function Documentation

5.240.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_IMAGE_CAPTURED::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.240.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_IMAGE_CAPTURED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.240.2.3 serialize()

```
void mavlink::common::msg::CAMERA_IMAGE_CAPTURED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

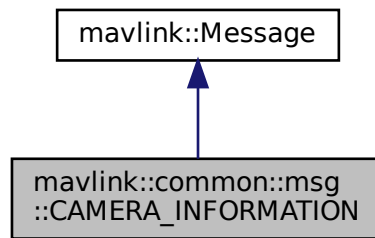
- single_hand_controller/include/mavlink/common/mavlink_msg_camera_image_captured.hpp

5.241 mavlink::common::msg::CAMERA_INFORMATION Struct Reference

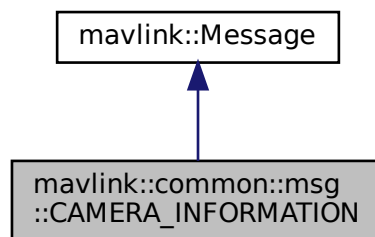
[CAMERA_INFORMATION](#) message.

```
#include <mavlink_msg_camera_information.hpp>
```


Inheritance diagram for mavlink::common::msg::CAMERA_INFORMATION:



Collaboration diagram for mavlink::common::msg::CAMERA_INFORMATION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `std::array< uint8_t, 32 >` **vendor_name**

- `std::array< uint8_t, 32 >` **model_name**
- `uint32_t` **firmware_version**
- `float` **focal_length**
- `float` **sensor_size_h**
- `float` **sensor_size_v**
- `uint16_t` **resolution_h**
- `uint16_t` **resolution_v**
- `uint8_t` **lens_id**
- `uint32_t` **flags**
- `uint16_t` **cam_definition_version**
- `std::array< char, 140 >` **cam_definition_uri**
- `uint8_t` **gimbal_device_id**
- `uint8_t` **camera_device_id**

Static Public Attributes

- `static constexpr msgid_t` **MSG_ID** = 259
- `static constexpr size_t` **LENGTH** = 237
- `static constexpr size_t` **MIN_LENGTH** = 235
- `static constexpr uint8_t` **CRC_EXTRA** = 92
- `static constexpr auto` **NAME** = "CAMERA_INFORMATION"

5.241.1 Detailed Description

[CAMERA_INFORMATION](#) message.

Information about a camera. Can be requested with a MAV_CMD_REQUEST_MESSAGE command.

5.241.2 Member Function Documentation

5.241.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_INFORMATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.241.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_INFORMATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.241.2.3 serialize()

```
void mavlink::common::msg::CAMERA_INFORMATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

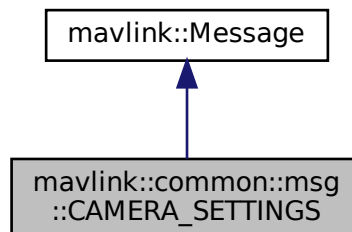
- single_hand_controller/include/mavlink/common/mavlink_msg_camera_information.hpp

5.242 mavlink::common::msg::CAMERA_SETTINGS Struct Reference

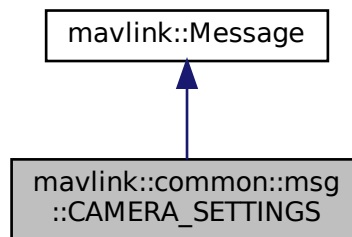
[CAMERA_SETTINGS](#) message.

```
#include <mavlink_msg_camera_settings.hpp>
```

Inheritance diagram for mavlink::common::msg::CAMERA_SETTINGS:



Collaboration diagram for mavlink::common::msg::CAMERA_SETTINGS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint8_t` **mode_id**
- `float` **zoomLevel**
- `float` **focusLevel**
- `uint8_t` **camera_device_id**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 260
- static constexpr `size_t` **LENGTH** = 14
- static constexpr `size_t` **MIN_LENGTH** = 5
- static constexpr `uint8_t` **CRC_EXTRA** = 146
- static constexpr auto **NAME** = "CAMERA_SETTINGS"

5.242.1 Detailed Description

[CAMERA_SETTINGS](#) message.

Settings of a camera. Can be requested with a MAV_CMD_REQUEST_MESSAGE command.

5.242.2 Member Function Documentation

5.242.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_SETTINGS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.242.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_SETTINGS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.242.2.3 serialize()

```
void mavlink::common::msg::CAMERA_SETTINGS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

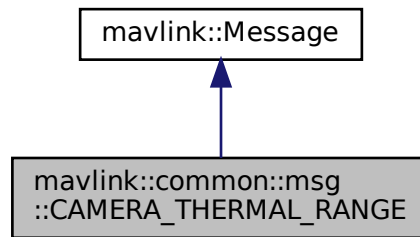
- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_settings.hpp`

5.243 mavlink::common::msg::CAMERA_THERMAL_RANGE Struct Reference

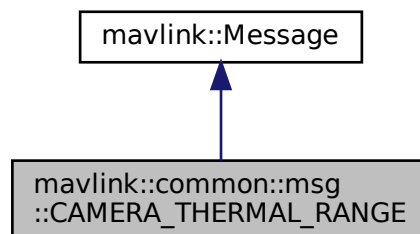
[CAMERA_THERMAL_RANGE](#) message.

```
#include <mavlink_msg_camera_thermal_range.hpp>
```

Inheritance diagram for mavlink::common::msg::CAMERA_THERMAL_RANGE:



Collaboration diagram for mavlink::common::msg::CAMERA_THERMAL_RANGE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- uint8_t **stream_id**
- uint8_t **camera_device_id**
- float **max**
- float **max_point_x**
- float **max_point_y**
- float **min**
- float **min_point_x**
- float **min_point_y**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 277
- static constexpr size_t **LENGTH** = 30
- static constexpr size_t **MIN_LENGTH** = 30
- static constexpr uint8_t **CRC_EXTRA** = 62
- static constexpr auto **NAME** = "CAMERA_THERMAL_RANGE"

5.243.1 Detailed Description

[CAMERA_THERMAL_RANGE](#) message.

Camera absolute thermal range. This can be streamed when the associated [VIDEO_STREAM_STATUS](#) flag field bit VIDEO_STREAM_STATUS_FLAGS_THERMAL_RANGE_ENABLED is set, but a GCS may choose to only request it for the current active stream. Use MAV_CMD_SET_MESSAGE_INTERVAL to define message interval (param3 indicates the stream id of the current camera, or 0 for all streams, param4 indicates the target camera_↔ device_id for autopilot-attached cameras or 0 for MAVLink cameras).

5.243.2 Member Function Documentation

5.243.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_THERMAL_RANGE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.243.2.2 `get_name()`

```
std::string mavlink::common::msg::CAMERA_THERMAL_RANGE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.243.2.3 `serialize()`

```
void mavlink::common::msg::CAMERA_THERMAL_RANGE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

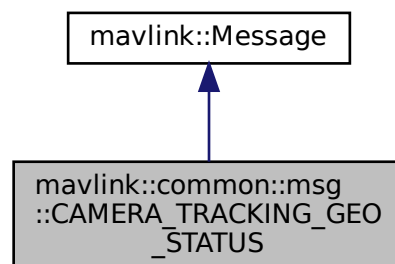
- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_thermal_range.hpp`

5.244 `mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS` Struct Reference

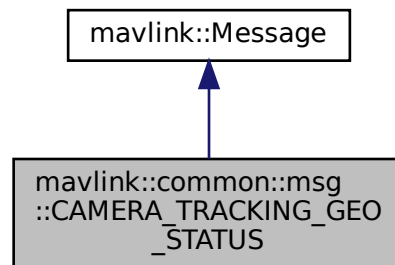
[CAMERA_TRACKING_GEO_STATUS](#) message.

```
#include <mavlink_msg_camera_tracking_geo_status.hpp>
```

Inheritance diagram for `mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS`:



Collaboration diagram for mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **tracking_status**
- int32_t **lat**
- int32_t **lon**
- float **alt**
- float **h_acc**
- float **v_acc**
- float **vel_n**
- float **vel_e**
- float **vel_d**
- float **vel_acc**
- float **dist**
- float **hdg**
- float **hdg_acc**
- uint8_t **camera_device_id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 276
- static constexpr size_t **LENGTH** = 50
- static constexpr size_t **MIN_LENGTH** = 49
- static constexpr uint8_t **CRC_EXTRA** = 18
- static constexpr auto **NAME** = "CAMERA_TRACKING_GEO_STATUS"

5.244.1 Detailed Description

[CAMERA_TRACKING_GEO_STATUS](#) message.

Camera tracking status, sent while in active tracking. Use MAV_CMD_SET_MESSAGE_INTERVAL to define message interval.

5.244.2 Member Function Documentation

5.244.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.244.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.244.2.3 serialize()

```
void mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

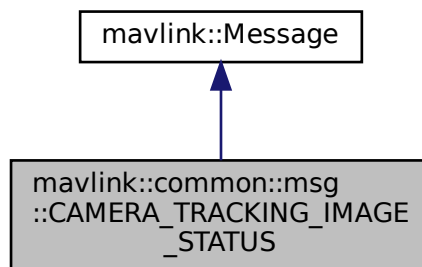
- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_geo_status.hpp`

5.245 mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS Struct Reference

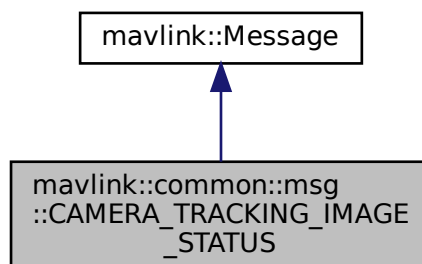
[CAMERA_TRACKING_IMAGE_STATUS](#) message.

```
#include <mavlink_msg_camera_tracking_image_status.hpp>
```

Inheritance diagram for mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS:



Collaboration diagram for mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint8_t` **tracking_status**
- `uint8_t` **tracking_mode**
- `uint8_t` **target_data**
- `float` **point_x**
- `float` **point_y**
- `float` **radius**
- `float` **rec_top_x**
- `float` **rec_top_y**
- `float` **rec_bottom_x**
- `float` **rec_bottom_y**
- `uint8_t` **camera_device_id**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 275
- static constexpr `size_t` **LENGTH** = 32
- static constexpr `size_t` **MIN_LENGTH** = 31
- static constexpr `uint8_t` **CRC_EXTRA** = 126
- static constexpr auto **NAME** = "CAMERA_TRACKING_IMAGE_STATUS"

5.245.1 Detailed Description

[CAMERA_TRACKING_IMAGE_STATUS](#) message.

Camera tracking status, sent while in active tracking. Use `MAV_CMD_SET_MESSAGE_INTERVAL` to define message interval.

5.245.2 Member Function Documentation

5.245.2.1 `deserialize()`

```
void mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.245.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.245.2.3 serialize()

```
void mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

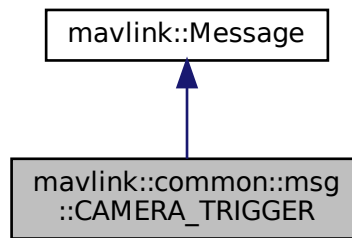
- single_hand_controller/include/mavlink/common/mavlink_msg_camera_tracking_image_status.hpp

5.246 mavlink::common::msg::CAMERA_TRIGGER Struct Reference

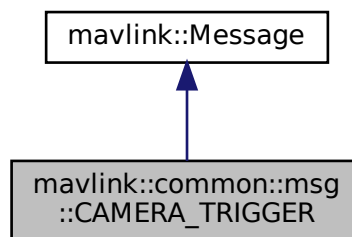
[CAMERA_TRIGGER](#) message.

```
#include <mavlink_msg_camera_trigger.hpp>
```

Inheritance diagram for mavlink::common::msg::CAMERA_TRIGGER:



Collaboration diagram for mavlink::common::msg::CAMERA_TRIGGER:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint32_t` **seq**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 112
- static constexpr size_t **LENGTH** = 12
- static constexpr size_t **MIN_LENGTH** = 12
- static constexpr uint8_t **CRC_EXTRA** = 174
- static constexpr auto **NAME** = "CAMERA_TRIGGER"

5.246.1 Detailed Description

[CAMERA_TRIGGER](#) message.

Camera-IMU triggering and synchronisation message.

5.246.2 Member Function Documentation

5.246.2.1 deserialize()

```
void mavlink::common::msg::CAMERA_TRIGGER::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.246.2.2 get_name()

```
std::string mavlink::common::msg::CAMERA_TRIGGER::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.246.2.3 serialize()

```
void mavlink::common::msg::CAMERA_TRIGGER::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

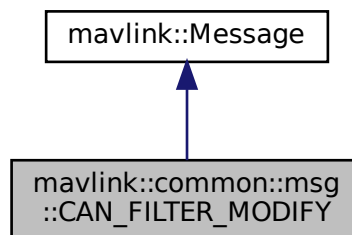
- `single_hand_controller/include/mavlink/common/mavlink_msg_camera_trigger.hpp`

5.247 mavlink::common::msg::CAN_FILTER_MODIFY Struct Reference

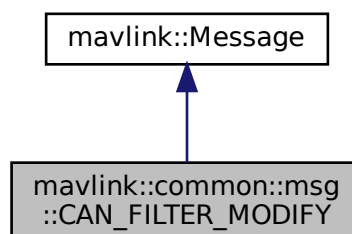
[CAN_FILTER_MODIFY](#) message.

```
#include <mavlink_msg_can_filter_modify.hpp>
```

Inheritance diagram for `mavlink::common::msg::CAN_FILTER_MODIFY`:



Collaboration diagram for `mavlink::common::msg::CAN_FILTER_MODIFY`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **bus**
- uint8_t **operation**
- uint8_t **num_ids**
- std::array< uint16_t, 16 > **ids**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 388
- static constexpr size_t **LENGTH** = 37
- static constexpr size_t **MIN_LENGTH** = 37
- static constexpr uint8_t **CRC_EXTRA** = 8
- static constexpr auto **NAME** = "CAN_FILTER_MODIFY"

5.247.1 Detailed Description

[CAN_FILTER_MODIFY](#) message.

Modify the filter of what CAN messages to forward over the mavlink. This can be used to make CAN forwarding work well on low bandwidth links. The filtering is applied on bits 8 to 24 of the CAN id (2nd and 3rd bytes) which corresponds to the DroneCAN message ID for DroneCAN. Filters with more than 16 IDs can be constructed by sending multiple [CAN_FILTER_MODIFY](#) messages.

5.247.2 Member Function Documentation

5.247.2.1 deserialize()

```
void mavlink::common::msg::CAN_FILTER_MODIFY::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.247.2.2 get_name()

```
std::string mavlink::common::msg::CAN_FILTER_MODIFY::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.247.2.3 serialize()

```
void mavlink::common::msg::CAN_FILTER_MODIFY::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

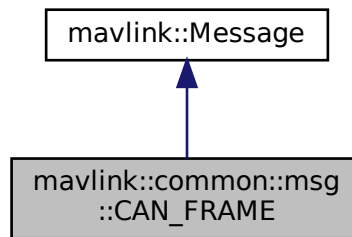
- single_hand_controller/include/mavlink/common/mavlink_msg_can_filter_modify.hpp

5.248 mavlink::common::msg::CAN_FRAME Struct Reference

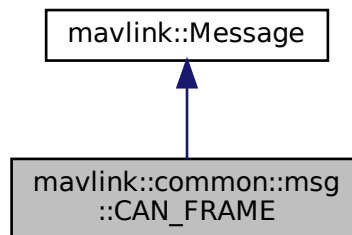
[CAN_FRAME](#) message.

```
#include <mavlink_msg_can_frame.hpp>
```

Inheritance diagram for mavlink::common::msg::CAN_FRAME:



Collaboration diagram for mavlink::common::msg::CAN_FRAME:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **bus**
- `uint8_t` **len**
- `uint32_t` **id**
- `std::array< uint8_t, 8 >` **data**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 386
- static constexpr size_t **LENGTH** = 16
- static constexpr size_t **MIN_LENGTH** = 16
- static constexpr uint8_t **CRC_EXTRA** = 132
- static constexpr auto **NAME** = "CAN_FRAME"

5.248.1 Detailed Description

[CAN_FRAME](#) message.

A forwarded CAN frame as requested by MAV_CMD_CAN_FORWARD.

5.248.2 Member Function Documentation

5.248.2.1 deserialize()

```
void mavlink::common::msg::CAN_FRAME::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.248.2.2 get_name()

```
std::string mavlink::common::msg::CAN_FRAME::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.248.2.3 serialize()

```
void mavlink::common::msg::CAN_FRAME::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

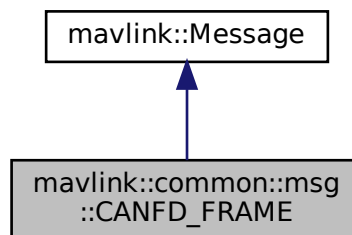
- `single_hand_controller/include/mavlink/common/mavlink_msg_can_frame.hpp`

5.249 mavlink::common::msg::CANFD_FRAME Struct Reference

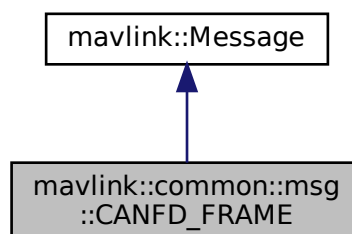
[CANFD_FRAME](#) message.

```
#include <mavlink_msg_canfd_frame.hpp>
```

Inheritance diagram for mavlink::common::msg::CANFD_FRAME:



Collaboration diagram for mavlink::common::msg::CANFD_FRAME:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **bus**
- `uint8_t` **len**
- `uint32_t` **id**
- `std::array< uint8_t, 64 >` **data**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 387
- static constexpr `size_t` **LENGTH** = 72
- static constexpr `size_t` **MIN_LENGTH** = 72
- static constexpr `uint8_t` **CRC_EXTRA** = 4
- static constexpr auto **NAME** = "CANFD_FRAME"

5.249.1 Detailed Description

[CANFD_FRAME](#) message.

A forwarded CANFD frame as requested by MAV_CMD_CAN_FORWARD. These are separated from [CAN_FRAME](#) as they need different handling (eg. TAO handling)

5.249.2 Member Function Documentation

5.249.2.1 deserialize()

```
void mavlink::common::msg::CANFD_FRAME::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.249.2.2 get_name()

```
std::string mavlink::common::msg::CANFD_FRAME::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.249.2.3 serialize()

```
void mavlink::common::msg::CANFD_FRAME::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

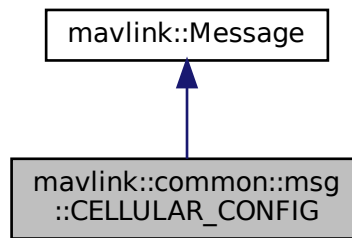
- single_hand_controller/include/mavlink/common/mavlink_msg_canfd_frame.hpp

5.250 mavlink::common::msg::CELLULAR_CONFIG Struct Reference

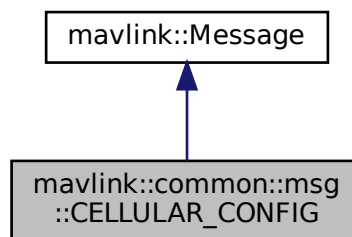
[CELLULAR_CONFIG](#) message.

```
#include <mavlink_msg_cellular_config.hpp>
```

Inheritance diagram for mavlink::common::msg::CELLULAR_CONFIG:



Collaboration diagram for mavlink::common::msg::CELLULAR_CONFIG:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` [enable_lte](#)
- `uint8_t` [enable_pin](#)

- std::array< char, 16 > **pin**
- std::array< char, 16 > **new_pin**
- std::array< char, 32 > **apn**
- std::array< char, 16 > **puk**
- uint8_t **roaming**
- uint8_t **response**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 336
- static constexpr size_t **LENGTH** = 84
- static constexpr size_t **MIN_LENGTH** = 84
- static constexpr uint8_t **CRC_EXTRA** = 245
- static constexpr auto **NAME** = "CELLULAR_CONFIG"

5.250.1 Detailed Description

[CELLULAR_CONFIG](#) message.

Configure cellular modems. This message is re-emitted as an acknowledgement by the modem. The message may also be explicitly requested using MAV_CMD_REQUEST_MESSAGE.

5.250.2 Member Function Documentation

5.250.2.1 deserialize()

```
void mavlink::common::msg::CELLULAR_CONFIG::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.250.2.2 get_name()

```
std::string mavlink::common::msg::CELLULAR_CONFIG::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.250.2.3 serialize()

```
void mavlink::common::msg::CELLULAR_CONFIG::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

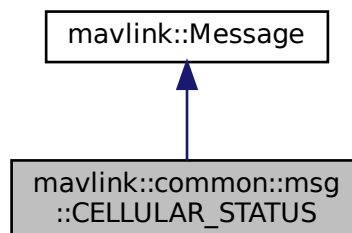
- single_hand_controller/include/mavlink/common/mavlink_msg_cellular_config.hpp

5.251 mavlink::common::msg::CELLULAR_STATUS Struct Reference

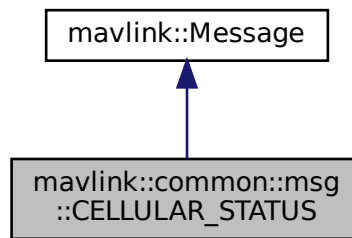
[CELLULAR_STATUS](#) message.

```
#include <mavlink_msg_cellular_status.hpp>
```

Inheritance diagram for mavlink::common::msg::CELLULAR_STATUS:



Collaboration diagram for mavlink::common::msg::CELLULAR_STATUS:



Public Member Functions

- `std::string` `get_name` (void) const override
Get NAME constant.
- `Info` `get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` `to_yaml` (void) const override
Make YAML-string from message content.
- void `serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- void `deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint8_t` `status`
- `uint8_t` `failure_reason`
- `uint8_t` `type`
- `uint8_t` `quality`
- `uint16_t` `mcc`
- `uint16_t` `mnc`
- `uint16_t` `lac`

Static Public Attributes

- static constexpr `msgid_t` `MSG_ID` = 334
- static constexpr `size_t` `LENGTH` = 10
- static constexpr `size_t` `MIN_LENGTH` = 10
- static constexpr `uint8_t` `CRC_EXTRA` = 72
- static constexpr auto `NAME` = "CELLULAR_STATUS"

5.251.1 Detailed Description

[CELLULAR_STATUS](#) message.

Report current used cellular network status

5.251.2 Member Function Documentation

5.251.2.1 deserialize()

```
void mavlink::common::msg::CELLULAR_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.251.2.2 get_name()

```
std::string mavlink::common::msg::CELLULAR_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.251.2.3 serialize()

```
void mavlink::common::msg::CELLULAR_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

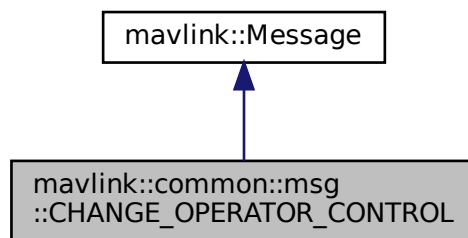
- `single_hand_controller/include/mavlink/common/mavlink_msg_cellular_status.hpp`

5.252 mavlink::common::msg::CHANGE_OPERATOR_CONTROL Struct Reference

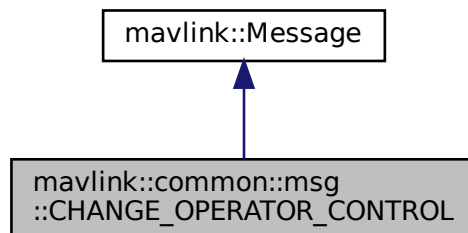
[CHANGE_OPERATOR_CONTROL](#) message.

```
#include <mavlink_msg_change_operator_control.hpp>
```

Inheritance diagram for mavlink::common::msg::CHANGE_OPERATOR_CONTROL:



Collaboration diagram for mavlink::common::msg::CHANGE_OPERATOR_CONTROL:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) ([mavlink::MsgMap](#) &map) const override
Serialize message.
- void [deserialize](#) ([mavlink::MsgMap](#) &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **control_request**
- `uint8_t` **version**
- `std::array< char, 25 >` **passkey**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 5
- static constexpr `size_t` **LENGTH** = 28
- static constexpr `size_t` **MIN_LENGTH** = 28
- static constexpr `uint8_t` **CRC_EXTRA** = 217
- static constexpr auto **NAME** = "CHANGE_OPERATOR_CONTROL"

5.252.1 Detailed Description

[CHANGE_OPERATOR_CONTROL](#) message.

Request to control this MAV

5.252.2 Member Function Documentation

5.252.2.1 `deserialize()`

```
void mavlink::common::msg::CHANGE_OPERATOR_CONTROL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.252.2.2 get_name()

```
std::string mavlink::common::msg::CHANGE_OPERATOR_CONTROL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.252.2.3 serialize()

```
void mavlink::common::msg::CHANGE_OPERATOR_CONTROL::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

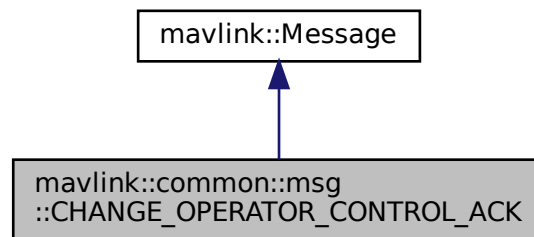
- single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control.hpp

5.253 mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK Struct Reference

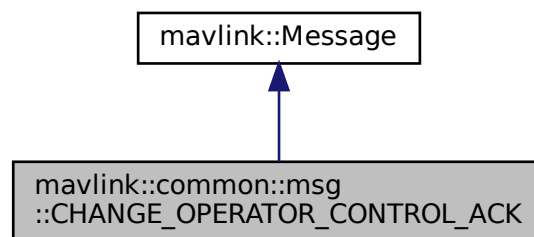
[CHANGE_OPERATOR_CONTROL_ACK](#) message.

```
#include <mavlink_msg_change_operator_control_ack.hpp>
```

Inheritance diagram for mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK:



Collaboration diagram for mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` [gcs_system_id](#)
- `uint8_t` [control_request](#)
- `uint8_t` [ack](#)

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 6
- static constexpr size_t **LENGTH** = 3
- static constexpr size_t **MIN_LENGTH** = 3
- static constexpr uint8_t **CRC_EXTRA** = 104
- static constexpr auto **NAME** = "CHANGE_OPERATOR_CONTROL_ACK"

5.253.1 Detailed Description

[CHANGE_OPERATOR_CONTROL_ACK](#) message.

Accept / deny control of this MAV

5.253.2 Member Function Documentation

5.253.2.1 deserialize()

```
void mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.253.2.2 get_name()

```
std::string mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.253.2.3 serialize()

```
void mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

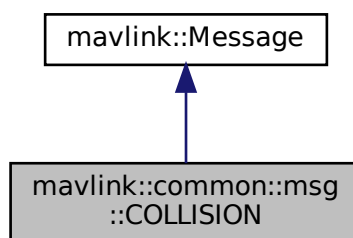
- `single_hand_controller/include/mavlink/common/mavlink_msg_change_operator_control_ack.hpp`

5.254 mavlink::common::msg::COLLISION Struct Reference

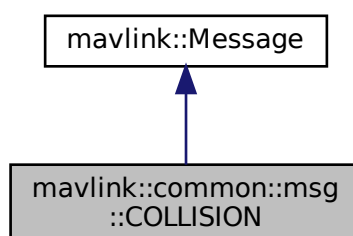
[COLLISION](#) message.

```
#include <mavlink_msg_collision.hpp>
```

Inheritance diagram for `mavlink::common::msg::COLLISION`:



Collaboration diagram for `mavlink::common::msg::COLLISION`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **src**
- uint32_t **id**
- uint8_t **action**
- uint8_t **threat_level**
- float **time_to_minimum_delta**
- float **altitude_minimum_delta**
- float **horizontal_minimum_delta**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 247
- static constexpr size_t **LENGTH** = 19
- static constexpr size_t **MIN_LENGTH** = 19
- static constexpr uint8_t **CRC_EXTRA** = 81
- static constexpr auto **NAME** = "COLLISION"

5.254.1 Detailed Description

[COLLISION](#) message.

Information about a potential collision

5.254.2 Member Function Documentation

5.254.2.1 deserialize()

```
void mavlink::common::msg::COLLISION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.254.2.2 get_name()

```
std::string mavlink::common::msg::COLLISION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.254.2.3 serialize()

```
void mavlink::common::msg::COLLISION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

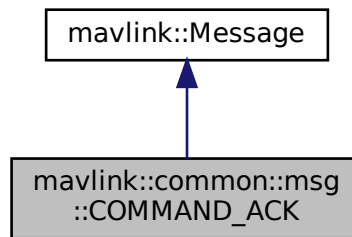
- single_hand_controller/include/mavlink/common/mavlink_msg_collision.hpp

5.255 mavlink::common::msg::COMMAND_ACK Struct Reference

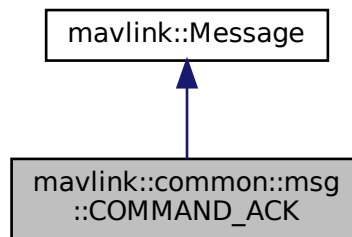
[COMMAND_ACK](#) message.

```
#include <mavlink_msg_command_ack.hpp>
```

Inheritance diagram for mavlink::common::msg::COMMAND_ACK:



Collaboration diagram for mavlink::common::msg::COMMAND_ACK:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint16_t command`
- `uint8_t result`
- `uint8_t progress`
- `int32_t result_param2`
- `uint8_t target_system`
- `uint8_t target_component`

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 77
- static constexpr size_t **LENGTH** = 10
- static constexpr size_t **MIN_LENGTH** = 3
- static constexpr uint8_t **CRC_EXTRA** = 143
- static constexpr auto **NAME** = "COMMAND_ACK"

5.255.1 Detailed Description

[COMMAND_ACK](#) message.

Report status of a command. Includes feedback whether the command was executed. The command microservice is documented at <https://mavlink.io/en/services/command.html>

5.255.2 Member Function Documentation

5.255.2.1 deserialize()

```
void mavlink::common::msg::COMMAND_ACK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.255.2.2 get_name()

```
std::string mavlink::common::msg::COMMAND_ACK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.255.2.3 serialize()

```
void mavlink::common::msg::COMMAND_ACK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

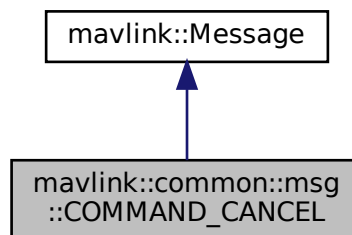
- `single_hand_controller/include/mavlink/common/mavlink_msg_command_ack.hpp`

5.256 mavlink::common::msg::COMMAND_CANCEL Struct Reference

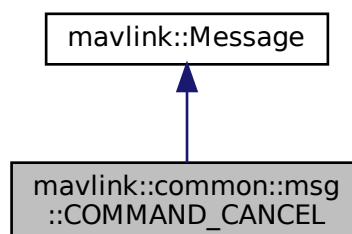
[COMMAND_CANCEL](#) message.

```
#include <mavlink_msg_command_cancel.hpp>
```

Inheritance diagram for mavlink::common::msg::COMMAND_CANCEL:



Collaboration diagram for mavlink::common::msg::COMMAND_CANCEL:



Public Member Functions

- `std::string` `get_name` (void) const override
Get NAME constant.
- `Info` `get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` `to_yaml` (void) const override
Make YAML-string from message content.
- void `serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- void `deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint8_t` `target_system`
- `uint8_t` `target_component`
- `uint16_t` `command`

Static Public Attributes

- static constexpr `msgid_t` `MSG_ID` = 80
- static constexpr `size_t` `LENGTH` = 4
- static constexpr `size_t` `MIN_LENGTH` = 4
- static constexpr `uint8_t` `CRC_EXTRA` = 14
- static constexpr auto `NAME` = "COMMAND_CANCEL"

5.256.1 Detailed Description

`COMMAND_CANCEL` message.

Cancel a long running command. The target system should respond with a `COMMAND_ACK` to the original command with result=`MAV_RESULT_CANCELLED` if the long running process was cancelled. If it has already completed, the cancel action can be ignored. The cancel action can be retried until some sort of acknowledgement to the original command has been received. The command microservice is documented at <https://mavlink.io/en/services/command.html>

5.256.2 Member Function Documentation

5.256.2.1 `deserialize()`

```
void mavlink::common::msg::COMMAND_CANCEL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.256.2.2 get_name()

```
std::string mavlink::common::msg::COMMAND_CANCEL::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.256.2.3 serialize()

```
void mavlink::common::msg::COMMAND_CANCEL::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

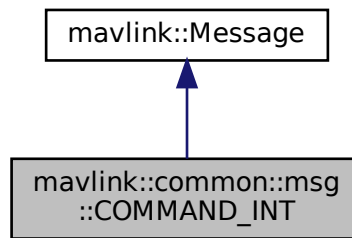
- single_hand_controller/include/mavlink/common/mavlink_msg_command_cancel.hpp

5.257 mavlink::common::msg::COMMAND_INT Struct Reference

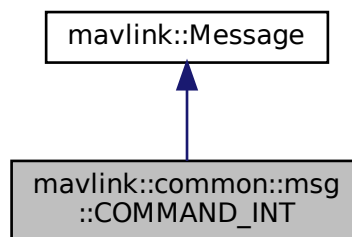
[COMMAND_INT](#) message.

```
#include <mavlink_msg_command_int.hpp>
```

Inheritance diagram for mavlink::common::msg::COMMAND_INT:



Collaboration diagram for mavlink::common::msg::COMMAND_INT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**

- uint8_t **frame**
- uint16_t **command**
- uint8_t **current**
- uint8_t **autocontinue**
- float **param1**
- float **param2**
- float **param3**
- float **param4**
- int32_t **x**
- int32_t **y**
- float **z**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 75
- static constexpr size_t **LENGTH** = 35
- static constexpr size_t **MIN_LENGTH** = 35
- static constexpr uint8_t **CRC_EXTRA** = 158
- static constexpr auto **NAME** = "COMMAND_INT"

5.257.1 Detailed Description

[COMMAND_INT](#) message.

Send a command with up to seven parameters to the MAV, where params 5 and 6 are integers and the other values are floats. This is preferred over [COMMAND_LONG](#) as it allows the MAV_FRAME to be specified for interpreting positional information, such as altitude. [COMMAND_INT](#) is also preferred when sending latitude and longitude data in params 5 and 6, as it allows for greater precision. Param 5 and 6 encode positional data as scaled integers, where the scaling depends on the actual command value. NaN or INT32_MAX may be used in float/integer params (respectively) to indicate optional/default values (e.g. to use the component's current latitude, yaw rather than a specific value). The command microservice is documented at <https://mavlink.io/en/services/command.html>

5.257.2 Member Function Documentation

5.257.2.1 deserialize()

```
void mavlink::common::msg::COMMAND_INT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.257.2.2 `get_name()`

```
std::string mavlink::common::msg::COMMAND_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.257.2.3 `serialize()`

```
void mavlink::common::msg::COMMAND_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

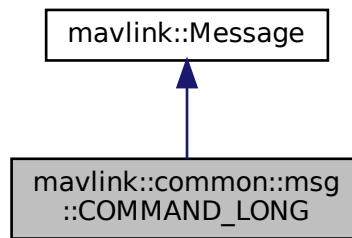
- `single_hand_controller/include/mavlink/common/mavlink_msg_command_int.hpp`

5.258 `mavlink::common::msg::COMMAND_LONG` Struct Reference

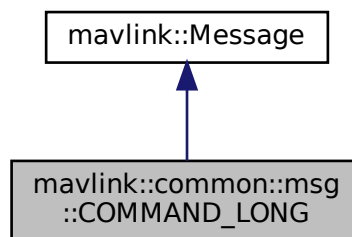
[COMMAND_LONG](#) message.

```
#include <mavlink_msg_command_long.hpp>
```

Inheritance diagram for mavlink::common::msg::COMMAND_LONG:



Collaboration diagram for mavlink::common::msg::COMMAND_LONG:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` [target_system](#)
- `uint8_t` [target_component](#)

- uint16_t **command**
- uint8_t **confirmation**
- float **param1**
- float **param2**
- float **param3**
- float **param4**
- float **param5**
- float **param6**
- float **param7**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 76
- static constexpr size_t **LENGTH** = 33
- static constexpr size_t **MIN_LENGTH** = 33
- static constexpr uint8_t **CRC_EXTRA** = 152
- static constexpr auto **NAME** = "COMMAND_LONG"

5.258.1 Detailed Description

[COMMAND_LONG](#) message.

Send a command with up to seven parameters to the MAV. [COMMAND_INT](#) is generally preferred when sending MAV_CMD commands that include positional information; it offers higher precision and allows the MAV_FRAME to be specified (which may otherwise be ambiguous, particularly for altitude). The command microservice is documented at <https://mavlink.io/en/services/command.html>

5.258.2 Member Function Documentation

5.258.2.1 deserialize()

```
void mavlink::common::msg::COMMAND_LONG::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.258.2.2 get_name()

```
std::string mavlink::common::msg::COMMAND_LONG::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.258.2.3 serialize()

```
void mavlink::common::msg::COMMAND_LONG::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

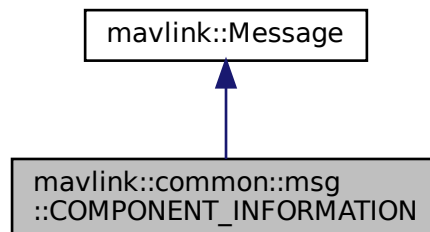
- single_hand_controller/include/mavlink/common/mavlink_msg_command_long.hpp

5.259 mavlink::common::msg::COMPONENT_INFORMATION Struct Reference

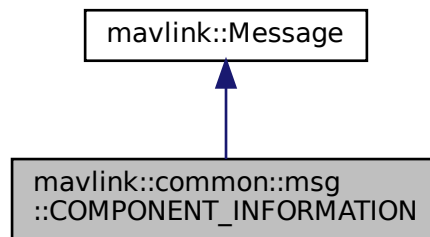
[COMPONENT_INFORMATION](#) message.

```
#include <mavlink_msg_component_information.hpp>
```

Inheritance diagram for mavlink::common::msg::COMPONENT_INFORMATION:



Collaboration diagram for mavlink::common::msg::COMPONENT_INFORMATION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint32_t` **general_metadata_file_crc**
- `std::array< char, 100 >` **general_metadata_uri**
- `uint32_t` **peripherals_metadata_file_crc**
- `std::array< char, 100 >` **peripherals_metadata_uri**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 395
- static constexpr size_t **LENGTH** = 212
- static constexpr size_t **MIN_LENGTH** = 212
- static constexpr uint8_t **CRC_EXTRA** = 0
- static constexpr auto **NAME** = "COMPONENT_INFORMATION"

5.259.1 Detailed Description

[COMPONENT_INFORMATION](#) message.

Component information message, which may be requested using `MAV_CMD_REQUEST_MESSAGE`.

5.259.2 Member Function Documentation

5.259.2.1 deserialize()

```
void mavlink::common::msg::COMPONENT_INFORMATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.259.2.2 get_name()

```
std::string mavlink::common::msg::COMPONENT_INFORMATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.259.2.3 serialize()

```
void mavlink::common::msg::COMPONENT_INFORMATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

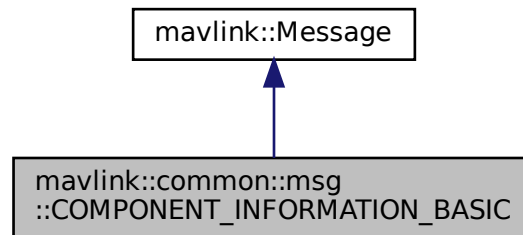
- `single_hand_controller/include/mavlink/common/mavlink_msg_component_information.hpp`

5.260 mavlink::common::msg::COMPONENT_INFORMATION_BASIC Struct Reference

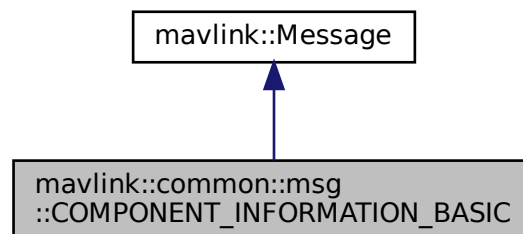
[COMPONENT_INFORMATION_BASIC](#) message.

```
#include <mavlink_msg_component_information_basic.hpp>
```

Inheritance diagram for mavlink::common::msg::COMPONENT_INFORMATION_BASIC:



Collaboration diagram for mavlink::common::msg::COMPONENT_INFORMATION_BASIC:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- uint64_t **capabilities**
- uint32_t **time_manufacture_s**
- std::array< char, 32 > **vendor_name**
- std::array< char, 32 > **model_name**
- std::array< char, 24 > **software_version**
- std::array< char, 24 > **hardware_version**
- std::array< char, 32 > **serial_number**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 396
- static constexpr size_t **LENGTH** = 160
- static constexpr size_t **MIN_LENGTH** = 160
- static constexpr uint8_t **CRC_EXTRA** = 50
- static constexpr auto **NAME** = "COMPONENT_INFORMATION_BASIC"

5.260.1 Detailed Description

[COMPONENT_INFORMATION_BASIC](#) message.

Basic component information data. Should be requested using MAV_CMD_REQUEST_MESSAGE on startup, or when required.

5.260.2 Member Function Documentation

5.260.2.1 deserialize()

```
void mavlink::common::msg::COMPONENT_INFORMATION_BASIC::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.260.2.2 get_name()

```
std::string mavlink::common::msg::COMPONENT_INFORMATION_BASIC::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.260.2.3 `serialize()`

```
void mavlink::common::msg::COMPONENT_INFORMATION_BASIC::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

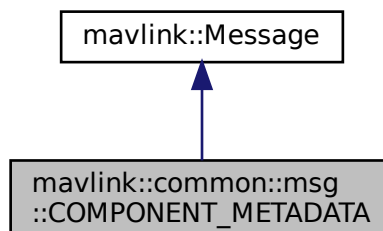
- `single_hand_controller/include/mavlink/common/mavlink_msg_component_information_basic.hpp`

5.261 `mavlink::common::msg::COMPONENT_METADATA` Struct Reference

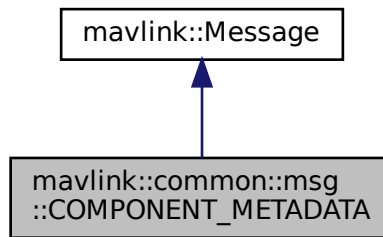
`COMPONENT_METADATA` message.

```
#include <mavlink_msg_component_metadata.hpp>
```

Inheritance diagram for `mavlink::common::msg::COMPONENT_METADATA`:



Collaboration diagram for mavlink::common::msg::COMPONENT_METADATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint32_t` **file_crc**
- `std::array< char, 100 >` **uri**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 397
- static constexpr `size_t` **LENGTH** = 108
- static constexpr `size_t` **MIN_LENGTH** = 108
- static constexpr `uint8_t` **CRC_EXTRA** = 182
- static constexpr auto **NAME** = "COMPONENT_METADATA"

5.261.1 Detailed Description

[COMPONENT_METADATA](#) message.

Component metadata message, which may be requested using `MAV_CMD_REQUEST_MESSAGE`.

This contains the MAVLink FTP URI and CRC for the component's general metadata file. The file must be hosted on the component, and may be xz compressed. The file CRC can be used for file caching.

The general metadata file can be read to get the locations of other metadata files (`COMP_METADATA_TYPE`) and t. For more information see: https://mavlink.io/en/services/component_information.html.

Note: Camera components should use `CAMERA_INFORMATION` instead, and autopilots may use both this message and A

5.261.2 Member Function Documentation

5.261.2.1 deserialize()

```
void mavlink::common::msg::COMPONENT_METADATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.261.2.2 get_name()

```
std::string mavlink::common::msg::COMPONENT_METADATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.261.2.3 serialize()

```
void mavlink::common::msg::COMPONENT_METADATA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

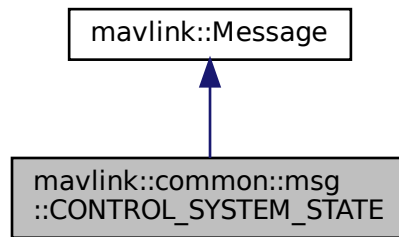
- `single_hand_controller/include/mavlink/common/mavlink_msg_component_metadata.hpp`

5.262 mavlink::common::msg::CONTROL_SYSTEM_STATE Struct Reference

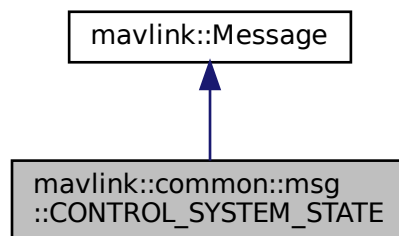
[CONTROL_SYSTEM_STATE](#) message.

```
#include <mavlink_msg_control_system_state.hpp>
```

Inheritance diagram for mavlink::common::msg::CONTROL_SYSTEM_STATE:



Collaboration diagram for mavlink::common::msg::CONTROL_SYSTEM_STATE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- float **x_acc**
- float **y_acc**
- float **z_acc**
- float **x_vel**
- float **y_vel**
- float **z_vel**
- float **x_pos**
- float **y_pos**
- float **z_pos**
- float **airspeed**
- std::array< float, 3 > **vel_variance**
- std::array< float, 3 > **pos_variance**
- std::array< float, 4 > **q**
- float **roll_rate**
- float **pitch_rate**
- float **yaw_rate**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 146
- static constexpr size_t **LENGTH** = 100
- static constexpr size_t **MIN_LENGTH** = 100
- static constexpr uint8_t **CRC_EXTRA** = 103
- static constexpr auto **NAME** = "CONTROL_SYSTEM_STATE"

5.262.1 Detailed Description

[CONTROL_SYSTEM_STATE](#) message.

The smoothed, monotonic system state used to feed the control loops of the system.

5.262.2 Member Function Documentation

5.262.2.1 deserialize()

```
void mavlink::common::msg::CONTROL_SYSTEM_STATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.262.2.2 get_name()

```
std::string mavlink::common::msg::CONTROL_SYSTEM_STATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.262.2.3 serialize()

```
void mavlink::common::msg::CONTROL_SYSTEM_STATE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

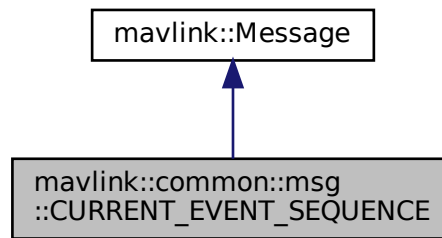
- single_hand_controller/include/mavlink/common/mavlink_msg_control_system_state.hpp

5.263 mavlink::common::msg::CURRENT_EVENT_SEQUENCE Struct Reference

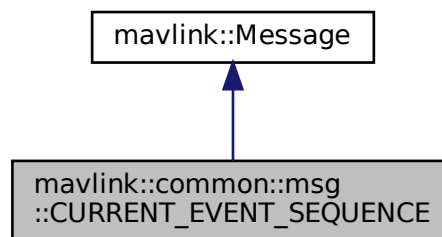
[CURRENT_EVENT_SEQUENCE](#) message.

```
#include <mavlink_msg_current_event_sequence.hpp>
```

Inheritance diagram for mavlink::common::msg::CURRENT_EVENT_SEQUENCE:



Collaboration diagram for mavlink::common::msg::CURRENT_EVENT_SEQUENCE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint16_t` **sequence**
- `uint8_t` **flags**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 411
- static constexpr size_t **LENGTH** = 3
- static constexpr size_t **MIN_LENGTH** = 3
- static constexpr uint8_t **CRC_EXTRA** = 106
- static constexpr auto **NAME** = "CURRENT_EVENT_SEQUENCE"

5.263.1 Detailed Description

[CURRENT_EVENT_SEQUENCE](#) message.

Regular broadcast for the current latest event sequence number for a component. This is used to check for dropped events.

5.263.2 Member Function Documentation

5.263.2.1 deserialize()

```
void mavlink::common::msg::CURRENT_EVENT_SEQUENCE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.263.2.2 get_name()

```
std::string mavlink::common::msg::CURRENT_EVENT_SEQUENCE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.263.2.3 serialize()

```
void mavlink::common::msg::CURRENT_EVENT_SEQUENCE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

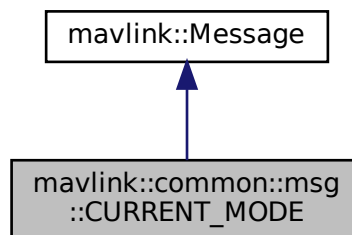
- `single_hand_controller/include/mavlink/common/mavlink_msg_current_event_sequence.hpp`

5.264 mavlink::common::msg::CURRENT_MODE Struct Reference

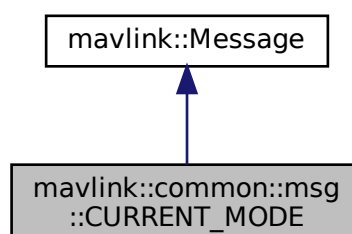
[CURRENT_MODE](#) message.

```
#include <mavlink_msg_current_mode.hpp>
```

Inheritance diagram for `mavlink::common::msg::CURRENT_MODE`:



Collaboration diagram for `mavlink::common::msg::CURRENT_MODE`:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t standard_mode`
- `uint32_t custom_mode`
- `uint32_t intended_custom_mode`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 436`
- `static constexpr size_t LENGTH = 9`
- `static constexpr size_t MIN_LENGTH = 9`
- `static constexpr uint8_t CRC_EXTRA = 193`
- `static constexpr auto NAME = "CURRENT_MODE"`

5.264.1 Detailed Description

`CURRENT_MODE` message.

Get the current mode. This should be emitted on any mode change, and broadcast at low rate (nominally 0.5 Hz). It may be requested using MAV_CMD_REQUEST_MESSAGE. See https://mavlink.io/en/services/standard_modes.html

5.264.2 Member Function Documentation

5.264.2.1 deserialize()

```
void mavlink::common::msg::CURRENT_MODE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.264.2.2 get_name()

```
std::string mavlink::common::msg::CURRENT_MODE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.264.2.3 serialize()

```
void mavlink::common::msg::CURRENT_MODE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

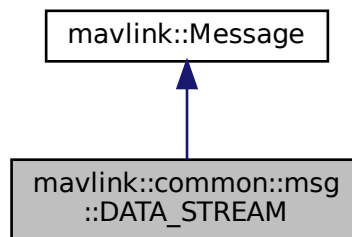
- single_hand_controller/include/mavlink/common/mavlink_msg_current_mode.hpp

5.265 mavlink::common::msg::DATA_STREAM Struct Reference

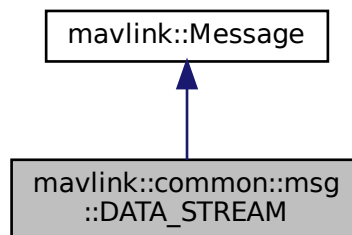
[DATA_STREAM](#) message.

```
#include <mavlink_msg_data_stream.hpp>
```

Inheritance diagram for mavlink::common::msg::DATA_STREAM:



Collaboration diagram for mavlink::common::msg::DATA_STREAM:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **stream_id**
- `uint16_t` **message_rate**
- `uint8_t` **on_off**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 67
- static constexpr size_t **LENGTH** = 4
- static constexpr size_t **MIN_LENGTH** = 4
- static constexpr uint8_t **CRC_EXTRA** = 21
- static constexpr auto **NAME** = "DATA_STREAM"

5.265.1 Detailed Description

[DATA_STREAM](#) message.

Data stream status information.

5.265.2 Member Function Documentation

5.265.2.1 deserialize()

```
void mavlink::common::msg::DATA_STREAM::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.265.2.2 get_name()

```
std::string mavlink::common::msg::DATA_STREAM::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.265.2.3 serialize()

```
void mavlink::common::msg::DATA_STREAM::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

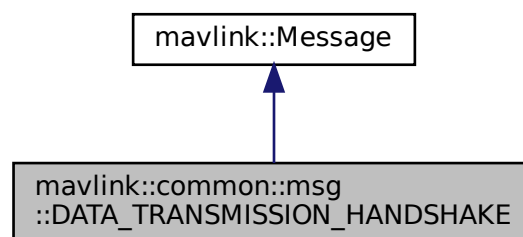
- single_hand_controller/include/mavlink/common/mavlink_msg_data_stream.hpp

5.266 mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE Struct Reference

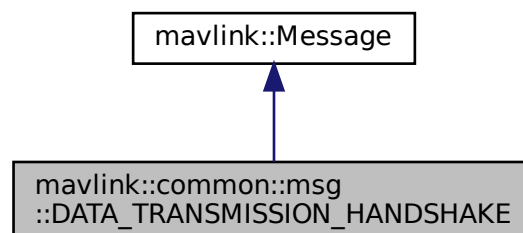
[DATA_TRANSMISSION_HANDSHAKE](#) message.

```
#include <mavlink_msg_data_transmission_handshake.hpp>
```

Inheritance diagram for mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE:



Collaboration diagram for mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **type**
- `uint32_t` **size**
- `uint16_t` **width**
- `uint16_t` **height**
- `uint16_t` **packets**
- `uint8_t` **payload**
- `uint8_t` **jpg_quality**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 130
- static constexpr `size_t` **LENGTH** = 13
- static constexpr `size_t` **MIN_LENGTH** = 13
- static constexpr `uint8_t` **CRC_EXTRA** = 29
- static constexpr auto **NAME** = "DATA_TRANSMISSION_HANDSHAKE"

5.266.1 Detailed Description

[DATA_TRANSMISSION_HANDSHAKE](#) message.

Handshake message to initiate, control and stop image streaming when using the Image Transmission Protocol:
https://mavlink.io/en/services/image_transmission.html.

5.266.2 Member Function Documentation

5.266.2.1 deserialize()

```
void mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.266.2.2 get_name()

```
std::string mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.266.2.3 serialize()

```
void mavlink::common::msg::DATA_TRANSMISSION_HANDSHAKE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

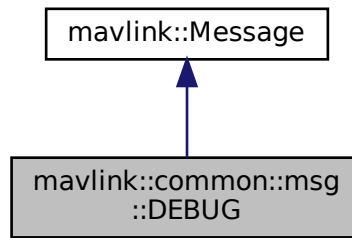
- single_hand_controller/include/mavlink/common/mavlink_msg_data_transmission_handshake.hpp

5.267 mavlink::common::msg::DEBUG Struct Reference

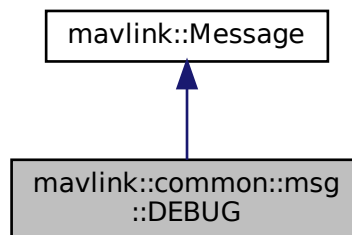
DEBUG message.

```
#include <mavlink_msg_debug.hpp>
```

Inheritance diagram for mavlink::common::msg::DEBUG:



Collaboration diagram for mavlink::common::msg::DEBUG:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` [time_boot_ms](#)
- `uint8_t` [ind](#)
- `float` [value](#)

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 254
- static constexpr size_t **LENGTH** = 9
- static constexpr size_t **MIN_LENGTH** = 9
- static constexpr uint8_t **CRC_EXTRA** = 46
- static constexpr auto **NAME** = "DEBUG"

5.267.1 Detailed Description

[DEBUG](#) message.

Send a debug value. The index is used to discriminate between values. These values show up in the plot of QGroundControl as [DEBUG](#) N.

5.267.2 Member Function Documentation

5.267.2.1 deserialize()

```
void mavlink::common::msg::DEBUG::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.267.2.2 get_name()

```
std::string mavlink::common::msg::DEBUG::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.267.2.3 serialize()

```
void mavlink::common::msg::DEBUG::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

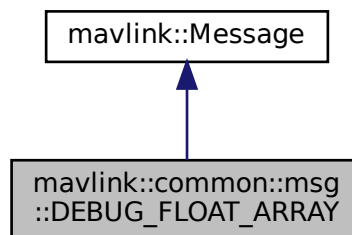
- `single_hand_controller/include/mavlink/common/mavlink_msg_debug.hpp`

5.268 `mavlink::common::msg::DEBUG_FLOAT_ARRAY` Struct Reference

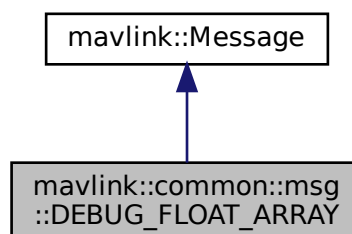
`DEBUG_FLOAT_ARRAY` message.

```
#include <mavlink_msg_debug_float_array.hpp>
```

Inheritance diagram for `mavlink::common::msg::DEBUG_FLOAT_ARRAY`:



Collaboration diagram for `mavlink::common::msg::DEBUG_FLOAT_ARRAY`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- std::array< char, 10 > **name**
- uint16_t **array_id**
- std::array< float, 58 > **data**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 350
- static constexpr size_t **LENGTH** = 252
- static constexpr size_t **MIN_LENGTH** = 20
- static constexpr uint8_t **CRC_EXTRA** = 232
- static constexpr auto **NAME** = "DEBUG_FLOAT_ARRAY"

5.268.1 Detailed Description

[DEBUG_FLOAT_ARRAY](#) message.

Large debug/prototyping array. The message uses the maximum available payload for data. The array_id and name fields are used to discriminate between messages in code and in user interfaces (respectively). Do not use in production code.

5.268.2 Member Function Documentation

5.268.2.1 deserialize()

```
void mavlink::common::msg::DEBUG_FLOAT_ARRAY::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.268.2.2 get_name()

```
std::string mavlink::common::msg::DEBUG_FLOAT_ARRAY::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.268.2.3 serialize()

```
void mavlink::common::msg::DEBUG_FLOAT_ARRAY::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

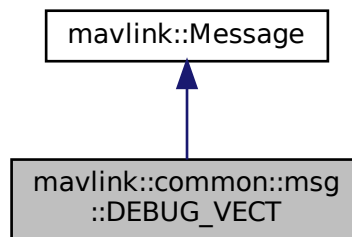
- single_hand_controller/include/mavlink/common/mavlink_msg_debug_float_array.hpp

5.269 mavlink::common::msg::DEBUG_VECT Struct Reference

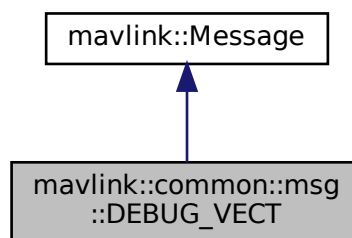
[DEBUG_VECT](#) message.

```
#include <mavlink_msg_debug_vect.hpp>
```


Inheritance diagram for mavlink::common::msg::DEBUG_VECT:



Collaboration diagram for mavlink::common::msg::DEBUG_VECT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `std::array< char, 10 >` **name**
- `uint64_t` **time_usec**
- `float` **x**
- `float` **y**
- `float` **z**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 250
- static constexpr size_t **LENGTH** = 30
- static constexpr size_t **MIN_LENGTH** = 30
- static constexpr uint8_t **CRC_EXTRA** = 49
- static constexpr auto **NAME** = "DEBUG_VECT"

5.269.1 Detailed Description

[DEBUG_VECT](#) message.

To debug something using a named 3D vector.

5.269.2 Member Function Documentation

5.269.2.1 deserialize()

```
void mavlink::common::msg::DEBUG_VECT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.269.2.2 get_name()

```
std::string mavlink::common::msg::DEBUG_VECT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.269.2.3 serialize()

```
void mavlink::common::msg::DEBUG_VECT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

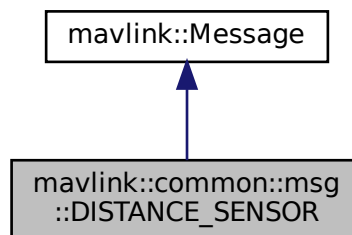
- single_hand_controller/include/mavlink/common/mavlink_msg_debug_vect.hpp

5.270 mavlink::common::msg::DISTANCE_SENSOR Struct Reference

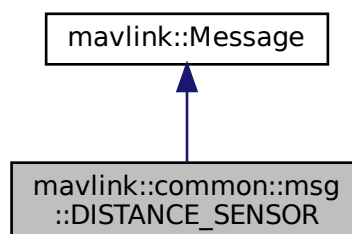
[DISTANCE_SENSOR](#) message.

```
#include <mavlink_msg_distance_sensor.hpp>
```

Inheritance diagram for mavlink::common::msg::DISTANCE_SENSOR:



Collaboration diagram for mavlink::common::msg::DISTANCE_SENSOR:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint16_t` **min_distance**
- `uint16_t` **max_distance**
- `uint16_t` **current_distance**
- `uint8_t` **type**
- `uint8_t` **id**
- `uint8_t` **orientation**
- `uint8_t` **covariance**
- `float` **horizontal_fov**
- `float` **vertical_fov**
- `std::array< float, 4 >` **quaternion**
- `uint8_t` **signal_quality**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 132
- static constexpr `size_t` **LENGTH** = 39
- static constexpr `size_t` **MIN_LENGTH** = 14
- static constexpr `uint8_t` **CRC_EXTRA** = 85
- static constexpr auto **NAME** = "DISTANCE_SENSOR"

5.270.1 Detailed Description

[DISTANCE_SENSOR](#) message.

Distance sensor information for an onboard rangefinder.

5.270.2 Member Function Documentation

5.270.2.1 `deserialize()`

```
void mavlink::common::msg::DISTANCE_SENSOR::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.270.2.2 get_name()

```
std::string mavlink::common::msg::DISTANCE_SENSOR::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.270.2.3 serialize()

```
void mavlink::common::msg::DISTANCE_SENSOR::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

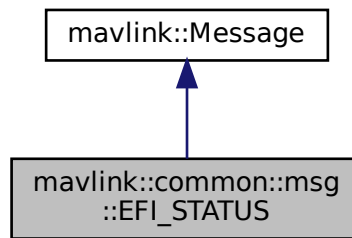
- single_hand_controller/include/mavlink/common/mavlink_msg_distance_sensor.hpp

5.271 mavlink::common::msg::EFI_STATUS Struct Reference

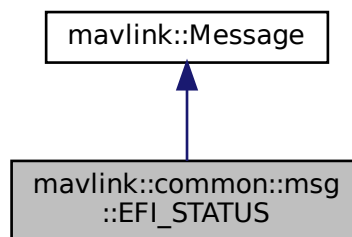
[EFI_STATUS](#) message.

```
#include <mavlink_msg_efi_status.hpp>
```

Inheritance diagram for mavlink::common::msg::EFI_STATUS:



Collaboration diagram for mavlink::common::msg::EFI_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **health**
- `float` **ecu_index**

- float **rpm**
- float **fuel_consumed**
- float **fuel_flow**
- float **engine_load**
- float **throttle_position**
- float **spark_dwell_time**
- float **barometric_pressure**
- float **intake_manifold_pressure**
- float **intake_manifold_temperature**
- float **cylinder_head_temperature**
- float **ignition_timing**
- float **injection_time**
- float **exhaust_gas_temperature**
- float **throttle_out**
- float **pt_compensation**
- float **ignition_voltage**
- float **fuel_pressure**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 225
- static constexpr size_t **LENGTH** = 73
- static constexpr size_t **MIN_LENGTH** = 65
- static constexpr uint8_t **CRC_EXTRA** = 208
- static constexpr auto **NAME** = "EFI_STATUS"

5.271.1 Detailed Description

[EFI_STATUS](#) message.

EFI status output

5.271.2 Member Function Documentation

5.271.2.1 deserialize()

```
void mavlink::common::msg::EFI_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.271.2.2 `get_name()`

```
std::string mavlink::common::msg::EFI_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.271.2.3 `serialize()`

```
void mavlink::common::msg::EFI_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

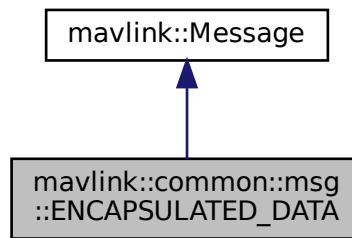
- `single_hand_controller/include/mavlink/common/mavlink_msg_efi_status.hpp`

5.272 `mavlink::common::msg::ENCAPSULATED_DATA` Struct Reference

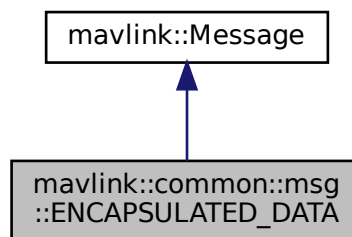
[ENCAPSULATED_DATA](#) message.

```
#include <mavlink_msg_encapsulated_data.hpp>
```


Inheritance diagram for mavlink::common::msg::ENCAPSULATED_DATA:



Collaboration diagram for mavlink::common::msg::ENCAPSULATED_DATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint16_t` **seqnr**
- `std::array< uint8_t, 253 >` **data**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 131
- static constexpr size_t **LENGTH** = 255
- static constexpr size_t **MIN_LENGTH** = 255
- static constexpr uint8_t **CRC_EXTRA** = 223
- static constexpr auto **NAME** = "ENCAPSULATED_DATA"

5.272.1 Detailed Description

[ENCAPSULATED_DATA](#) message.

Data packet for images sent using the Image Transmission Protocol: https://mavlink.io/en/services/image_transmission.html.

5.272.2 Member Function Documentation

5.272.2.1 deserialize()

```
void mavlink::common::msg::ENCAPSULATED_DATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.272.2.2 get_name()

```
std::string mavlink::common::msg::ENCAPSULATED_DATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.272.2.3 serialize()

```
void mavlink::common::msg::ENCAPSULATED_DATA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

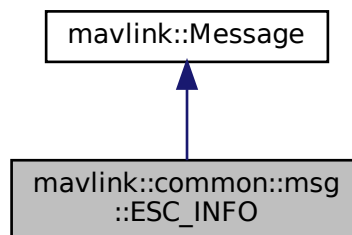
- single_hand_controller/include/mavlink/common/mavlink_msg_encapsulated_data.hpp

5.273 mavlink::common::msg::ESC_INFO Struct Reference

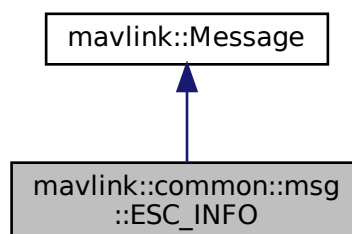
[ESC_INFO](#) message.

```
#include <mavlink_msg_esc_info.hpp>
```

Inheritance diagram for mavlink::common::msg::ESC_INFO:



Collaboration diagram for mavlink::common::msg::ESC_INFO:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t index`
- `uint64_t time_usec`
- `uint16_t counter`
- `uint8_t count`
- `uint8_t connection_type`
- `uint8_t info`
- `std::array< uint16_t, 4 > failure_flags`
- `std::array< uint32_t, 4 > error_count`
- `std::array< int16_t, 4 > temperature`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 290`
- `static constexpr size_t LENGTH = 46`
- `static constexpr size_t MIN_LENGTH = 46`
- `static constexpr uint8_t CRC_EXTRA = 251`
- `static constexpr auto NAME = "ESC_INFO"`

5.273.1 Detailed Description

[ESC_INFO](#) message.

ESC information for lower rate streaming. Recommended streaming rate 1Hz. See [ESC_STATUS](#) for higher-rate ESC data.

5.273.2 Member Function Documentation

5.273.2.1 deserialize()

```
void mavlink::common::msg::ESC_INFO::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.273.2.2 get_name()

```
std::string mavlink::common::msg::ESC_INFO::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.273.2.3 serialize()

```
void mavlink::common::msg::ESC_INFO::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

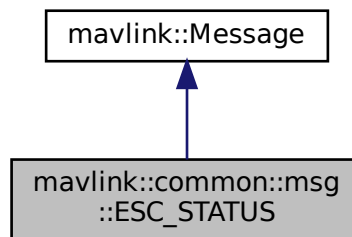
- single_hand_controller/include/mavlink/common/mavlink_msg_esc_info.hpp

5.274 mavlink::common::msg::ESC_STATUS Struct Reference

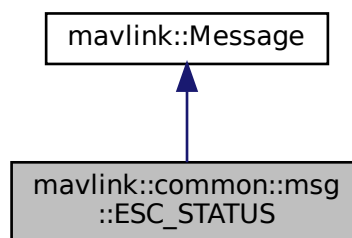
[ESC_STATUS](#) message.

```
#include <mavlink_msg_esc_status.hpp>
```

Inheritance diagram for mavlink::common::msg::ESC_STATUS:



Collaboration diagram for mavlink::common::msg::ESC_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **index**
- `uint64_t` **time_usec**
- `std::array< int32_t, 4 >` **rpm**
- `std::array< float, 4 >` **voltage**
- `std::array< float, 4 >` **current**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 291
- static constexpr size_t **LENGTH** = 57
- static constexpr size_t **MIN_LENGTH** = 57
- static constexpr uint8_t **CRC_EXTRA** = 10
- static constexpr auto **NAME** = "ESC_STATUS"

5.274.1 Detailed Description

[ESC_STATUS](#) message.

ESC information for higher rate streaming. Recommended streaming rate is ~ 10 Hz. Information that changes more slowly is sent in [ESC_INFO](#). It should typically only be streamed on high-bandwidth links (i.e. to a companion computer).

5.274.2 Member Function Documentation

5.274.2.1 deserialize()

```
void mavlink::common::msg::ESC_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.274.2.2 get_name()

```
std::string mavlink::common::msg::ESC_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.274.2.3 `serialize()`

```
void mavlink::common::msg::ESC_STATUS::serialize (  
    mavlink::MsgMap & map ) const    [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

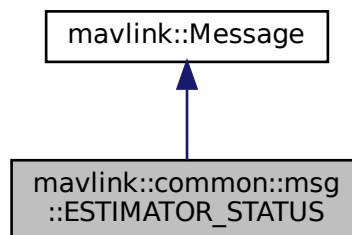
- `single_hand_controller/include/mavlink/common/mavlink_msg_esc_status.hpp`

5.275 mavlink::common::msg::ESTIMATOR_STATUS Struct Reference

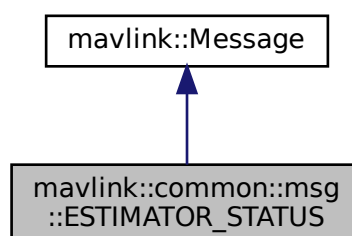
[ESTIMATOR_STATUS](#) message.

```
#include <mavlink_msg_estimator_status.hpp>
```

Inheritance diagram for `mavlink::common::msg::ESTIMATOR_STATUS`:



Collaboration diagram for `mavlink::common::msg::ESTIMATOR_STATUS`:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint16_t` **flags**
- `float` **vel_ratio**
- `float` **pos_horiz_ratio**
- `float` **pos_vert_ratio**
- `float` **mag_ratio**
- `float` **hagl_ratio**
- `float` **tas_ratio**
- `float` **pos_horiz_accuracy**
- `float` **pos_vert_accuracy**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 230
- static constexpr `size_t` **LENGTH** = 42
- static constexpr `size_t` **MIN_LENGTH** = 42
- static constexpr `uint8_t` **CRC_EXTRA** = 163
- static constexpr auto **NAME** = "ESTIMATOR_STATUS"

5.275.1 Detailed Description

[ESTIMATOR_STATUS](#) message.

Estimator status message including flags, innovation test ratios and estimated accuracies. The flags message is an integer bitmask containing information on which EKF outputs are valid. See the `ESTIMATOR_STATUS_FLAGS` enum definition for further information. The innovation test ratios show the magnitude of the sensor innovation divided by the innovation check threshold. Under normal operation the innovation test ratios should be below 0.5 with occasional values up to 1.0. Values greater than 1.0 should be rare under normal operation and indicate that a measurement has been rejected by the filter. The user should be notified if an innovation test ratio greater than 1.0 is recorded. Notifications for values in the range between 0.5 and 1.0 should be optional and controllable by the user.

5.275.2 Member Function Documentation

5.275.2.1 deserialize()

```
void mavlink::common::msg::ESTIMATOR_STATUS::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.275.2.2 get_name()

```
std::string mavlink::common::msg::ESTIMATOR_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.275.2.3 serialize()

```
void mavlink::common::msg::ESTIMATOR_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

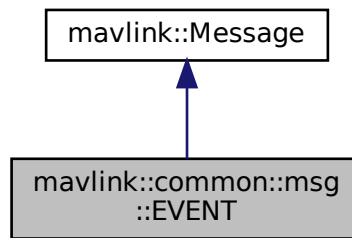
- single_hand_controller/include/mavlink/common/mavlink_msg_estimator_status.hpp

5.276 mavlink::common::msg::EVENT Struct Reference

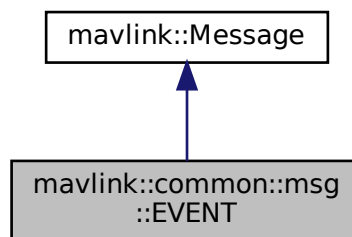
[EVENT](#) message.

```
#include <mavlink_msg_event.hpp>
```

Inheritance diagram for mavlink::common::msg::EVENT:



Collaboration diagram for mavlink::common::msg::EVENT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **destination_component**
- `uint8_t` **destination_system**

- uint32_t **id**
- uint32_t **event_time_boot_ms**
- uint16_t **sequence**
- uint8_t **log_levels**
- std::array< uint8_t, 40 > **arguments**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 410
- static constexpr size_t **LENGTH** = 53
- static constexpr size_t **MIN_LENGTH** = 53
- static constexpr uint8_t **CRC_EXTRA** = 160
- static constexpr auto **NAME** = "EVENT"

5.276.1 Detailed Description

[EVENT](#) message.

Event message. Each new event from a particular component gets a new sequence number. The same message might be sent multiple times if (re-)requested. Most events are broadcast, some can be specific to a target component (as receivers keep track of the sequence for missed events, all events need to be broadcast. Thus we use destination_component instead of target_component).

5.276.2 Member Function Documentation

5.276.2.1 deserialize()

```
void mavlink::common::msg::EVENT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.276.2.2 get_name()

```
std::string mavlink::common::msg::EVENT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.276.2.3 serialize()

```
void mavlink::common::msg::EVENT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

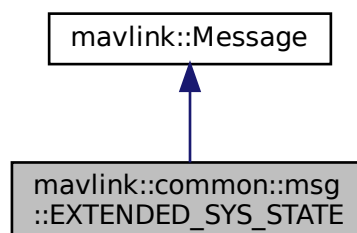
- single_hand_controller/include/mavlink/common/mavlink_msg_event.hpp

5.277 mavlink::common::msg::EXTENDED_SYS_STATE Struct Reference

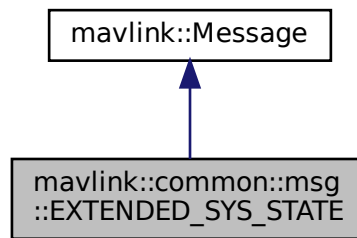
[EXTENDED_SYS_STATE](#) message.

```
#include <mavlink_msg_extended_sys_state.hpp>
```

Inheritance diagram for mavlink::common::msg::EXTENDED_SYS_STATE:



Collaboration diagram for mavlink::common::msg::EXTENDED_SYS_STATE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` `vtol_state`
- `uint8_t` `landed_state`

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 245
- static constexpr size_t **LENGTH** = 2
- static constexpr size_t **MIN_LENGTH** = 2
- static constexpr uint8_t **CRC_EXTRA** = 130
- static constexpr auto **NAME** = "EXTENDED_SYS_STATE"

5.277.1 Detailed Description

[EXTENDED_SYS_STATE](#) message.

Provides state for additional features

5.277.2 Member Function Documentation

5.277.2.1 deserialize()

```
void mavlink::common::msg::EXTENDED_SYS_STATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.277.2.2 get_name()

```
std::string mavlink::common::msg::EXTENDED_SYS_STATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.277.2.3 serialize()

```
void mavlink::common::msg::EXTENDED_SYS_STATE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

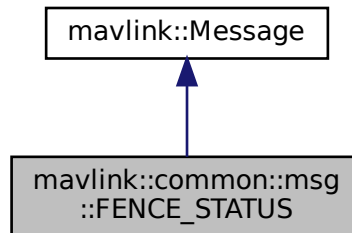
- single_hand_controller/include/mavlink/common/mavlink_msg_extended_sys_state.hpp

5.278 mavlink::common::msg::FENCE_STATUS Struct Reference

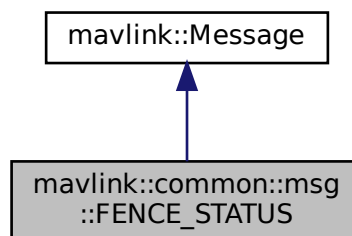
[FENCE_STATUS](#) message.

```
#include <mavlink_msg_fence_status.hpp>
```

Inheritance diagram for mavlink::common::msg::FENCE_STATUS:



Collaboration diagram for mavlink::common::msg::FENCE_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **breach_status**
- uint16_t **breach_count**
- uint8_t **breach_type**
- uint32_t **breach_time**
- uint8_t **breach_mitigation**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 162
- static constexpr size_t **LENGTH** = 9
- static constexpr size_t **MIN_LENGTH** = 8
- static constexpr uint8_t **CRC_EXTRA** = 189
- static constexpr auto **NAME** = "FENCE_STATUS"

5.278.1 Detailed Description

[FENCE_STATUS](#) message.

Status of geo-fencing. Sent in extended status stream when fencing enabled.

5.278.2 Member Function Documentation

5.278.2.1 deserialize()

```
void mavlink::common::msg::FENCE_STATUS::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.278.2.2 get_name()

```
std::string mavlink::common::msg::FENCE_STATUS::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.278.2.3 serialize()

```
void mavlink::common::msg::FENCE_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

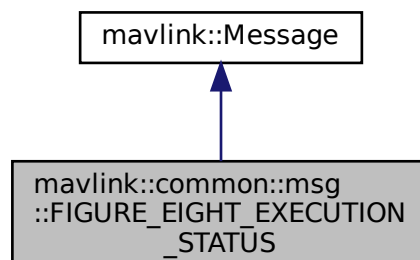
- single_hand_controller/include/mavlink/common/mavlink_msg_fence_status.hpp

5.279 mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS Struct Reference

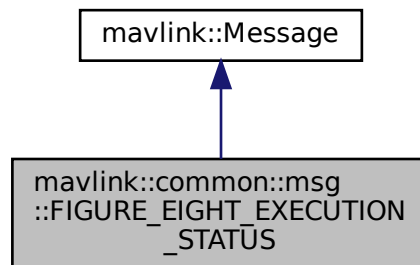
[FIGURE_EIGHT_EXECUTION_STATUS](#) message.

```
#include <mavlink_msg_figure_eight_execution_status.hpp>
```

Inheritance diagram for mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS:



Collaboration diagram for mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `float` **major_radius**
- `float` **minor_radius**
- `float` **orientation**
- `uint8_t` **frame**
- `int32_t` **x**
- `int32_t` **y**
- `float` **z**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 361
- static constexpr size_t **LENGTH** = 33
- static constexpr size_t **MIN_LENGTH** = 33
- static constexpr uint8_t **CRC_EXTRA** = 93
- static constexpr auto **NAME** = "FIGURE_EIGHT_EXECUTION_STATUS"

5.279.1 Detailed Description

[FIGURE_EIGHT_EXECUTION_STATUS](#) message.

Vehicle status report that is sent out while figure eight execution is in progress (see MAV_CMD_DO_FIGURE_EIGHT). This may typically send at low rates: of the order of 2Hz.

5.279.2 Member Function Documentation

5.279.2.1 deserialize()

```
void mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.279.2.2 get_name()

```
std::string mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.279.2.3 serialize()

```
void mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

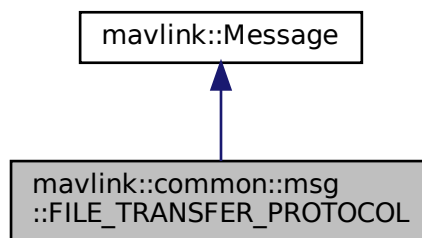
- single_hand_controller/include/mavlink/common/mavlink_msg_figure_eight_execution_status.hpp

5.280 mavlink::common::msg::FILE_TRANSFER_PROTOCOL Struct Reference

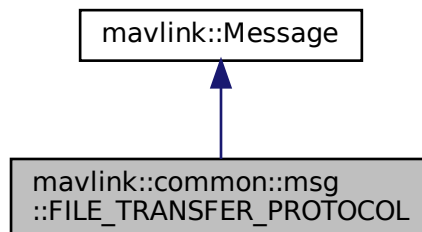
[FILE_TRANSFER_PROTOCOL](#) message.

```
#include <mavlink_msg_file_transfer_protocol.hpp>
```

Inheritance diagram for mavlink::common::msg::FILE_TRANSFER_PROTOCOL:



Collaboration diagram for mavlink::common::msg::FILE_TRANSFER_PROTOCOL:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t target_network`
- `uint8_t target_system`
- `uint8_t target_component`
- `std::array< uint8_t, 251 > payload`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 110`
- `static constexpr size_t LENGTH = 254`
- `static constexpr size_t MIN_LENGTH = 254`
- `static constexpr uint8_t CRC_EXTRA = 84`
- `static constexpr auto NAME = "FILE_TRANSFER_PROTOCOL"`

5.280.1 Detailed Description

`FILE_TRANSFER_PROTOCOL` message.

File transfer protocol message: <https://mavlink.io/en/services/ftp.html>.

5.280.2 Member Function Documentation

5.280.2.1 deserialize()

```
void mavlink::common::msg::FILE_TRANSFER_PROTOCOL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.280.2.2 get_name()

```
std::string mavlink::common::msg::FILE_TRANSFER_PROTOCOL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.280.2.3 serialize()

```
void mavlink::common::msg::FILE_TRANSFER_PROTOCOL::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

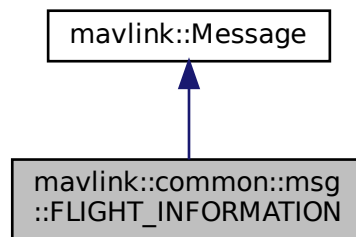
- single_hand_controller/include/mavlink/common/mavlink_msg_file_transfer_protocol.hpp

5.281 mavlink::common::msg::FLIGHT_INFORMATION Struct Reference

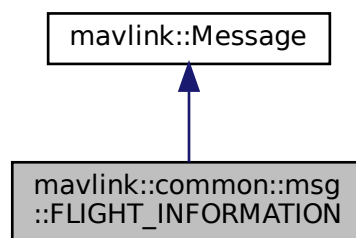
[FLIGHT_INFORMATION](#) message.

```
#include <mavlink_msg_flight_information.hpp>
```

Inheritance diagram for mavlink::common::msg::FLIGHT_INFORMATION:



Collaboration diagram for mavlink::common::msg::FLIGHT_INFORMATION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` [time_boot_ms](#)
- `uint64_t` [arming_time_utc](#)
- `uint64_t` [takeoff_time_utc](#)
- `uint64_t` [flight_uuid](#)
- `uint32_t` [landing_time](#)

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 264
- static constexpr size_t **LENGTH** = 32
- static constexpr size_t **MIN_LENGTH** = 28
- static constexpr uint8_t **CRC_EXTRA** = 49
- static constexpr auto **NAME** = "FLIGHT_INFORMATION"

5.281.1 Detailed Description

[FLIGHT_INFORMATION](#) message.

Flight information. This includes time since boot for arm, takeoff, and land, and a flight number. Takeoff and landing values reset to zero on arm. This can be requested using MAV_CMD_REQUEST_MESSAGE. Note, some fields are misnamed - timestamps are from boot (not UTC) and the flight_uuid is a sequence number.

5.281.2 Member Function Documentation

5.281.2.1 deserialize()

```
void mavlink::common::msg::FLIGHT_INFORMATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.281.2.2 get_name()

```
std::string mavlink::common::msg::FLIGHT_INFORMATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.281.2.3 `serialize()`

```
void mavlink::common::msg::FLIGHT_INFORMATION::serialize (  
    mavlink::MsgMap & map ) const    [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

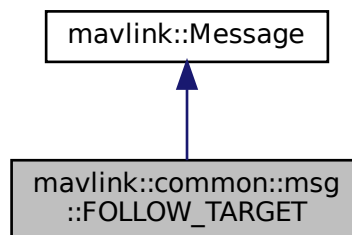
- single_hand_controller/include/mavlink/common/mavlink_msg_flight_information.hpp

5.282 mavlink::common::msg::FOLLOW_TARGET Struct Reference

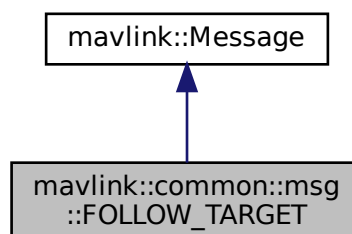
[FOLLOW_TARGET](#) message.

```
#include <mavlink_msg_follow_target.hpp>
```

Inheritance diagram for mavlink::common::msg::FOLLOW_TARGET:



Collaboration diagram for mavlink::common::msg::FOLLOW_TARGET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **timestamp**
- `uint8_t` **est_capabilities**
- `int32_t` **lat**
- `int32_t` **lon**
- `float` **alt**
- `std::array< float, 3 >` **vel**
- `std::array< float, 3 >` **acc**
- `std::array< float, 4 >` **attitude_q**
- `std::array< float, 3 >` **rates**
- `std::array< float, 3 >` **position_cov**
- `uint64_t` **custom_state**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 144
- static constexpr `size_t` **LENGTH** = 93
- static constexpr `size_t` **MIN_LENGTH** = 93
- static constexpr `uint8_t` **CRC_EXTRA** = 127
- static constexpr auto **NAME** = "FOLLOW_TARGET"

5.282.1 Detailed Description

[FOLLOW_TARGET](#) message.

Current motion information from a designated system

5.282.2 Member Function Documentation

5.282.2.1 [deserialize\(\)](#)

```
void mavlink::common::msg::FOLLOW_TARGET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.282.2.2 get_name()

```
std::string mavlink::common::msg::FOLLOW_TARGET::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.282.2.3 serialize()

```
void mavlink::common::msg::FOLLOW_TARGET::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

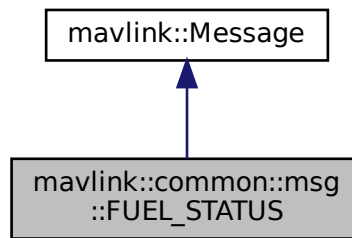
- single_hand_controller/include/mavlink/common/mavlink_msg_follow_target.hpp

5.283 mavlink::common::msg::FUEL_STATUS Struct Reference

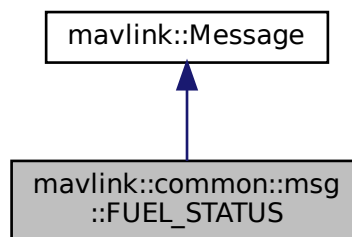
[FUEL_STATUS](#) message.

```
#include <mavlink_msg_fuel_status.hpp>
```

Inheritance diagram for mavlink::common::msg::FUEL_STATUS:



Collaboration diagram for mavlink::common::msg::FUEL_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` `id`
- `float` `maximum_fuel`

- float **consumed_fuel**
- float **remaining_fuel**
- uint8_t **percent_remaining**
- float **flow_rate**
- float **temperature**
- uint32_t **fuel_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 371
- static constexpr size_t **LENGTH** = 26
- static constexpr size_t **MIN_LENGTH** = 26
- static constexpr uint8_t **CRC_EXTRA** = 10
- static constexpr auto **NAME** = "FUEL_STATUS"

5.283.1 Detailed Description

[FUEL_STATUS](#) message.

Fuel status. This message provides "generic" fuel level information for in a GCS and for triggering failsafes in an autopilot. The fuel type and associated units for fields in this message are defined in the enum MAV_FUEL_TYPE.

The reported `consumed_fuel` and `remaining_fuel` must only be supplied if measured: they must not be inferred from the `maximum_fuel` and the other value. A recipient can assume that if these fields are supplied they are accurate. If not provided, the recipient can infer `remaining_fuel` from `maximum_fuel` and `consumed_fuel` on the assumption that the fuel was initially at its maximum (this is what battery monitors assume). Note however that this is an assumption, and the UI should prompt the user appropriately (i.e. notify user that they should fill the tank before boot).

This kind of information may also be sent in fuel-specific messages such as BATTERY_STATUS_V2. If both messages are sent for the same fuel system, the ids and corresponding information must match.

This should be streamed (nominally at 0.1 Hz).

5.283.2 Member Function Documentation

5.283.2.1 deserialize()

```
void mavlink::common::msg::FUEL_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.283.2.2 `get_name()`

```
std::string mavlink::common::msg::FUEL_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.283.2.3 `serialize()`

```
void mavlink::common::msg::FUEL_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

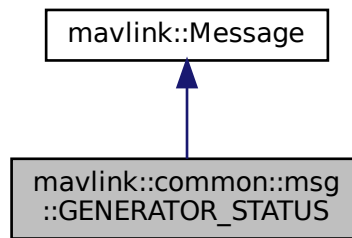
- `single_hand_controller/include/mavlink/common/mavlink_msg_fuel_status.hpp`

5.284 `mavlink::common::msg::GENERATOR_STATUS` Struct Reference

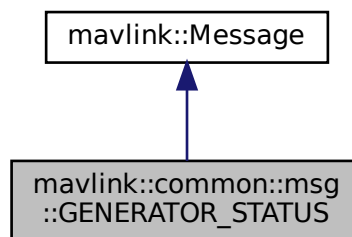
[GENERATOR_STATUS](#) message.

```
#include <mavlink_msg_generator_status.hpp>
```

Inheritance diagram for mavlink::common::msg::GENERATOR_STATUS:



Collaboration diagram for mavlink::common::msg::GENERATOR_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **status**
- `uint16_t` **generator_speed**

- float **battery_current**
- float **load_current**
- float **power_generated**
- float **bus_voltage**
- int16_t **rectifier_temperature**
- float **bat_current_setpoint**
- int16_t **generator_temperature**
- uint32_t **runtime**
- int32_t **time_until_maintenance**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 373
- static constexpr size_t **LENGTH** = 42
- static constexpr size_t **MIN_LENGTH** = 42
- static constexpr uint8_t **CRC_EXTRA** = 117
- static constexpr auto **NAME** = "GENERATOR_STATUS"

5.284.1 Detailed Description

[GENERATOR_STATUS](#) message.

Telemetry of power generation system. Alternator or mechanical generator.

5.284.2 Member Function Documentation

5.284.2.1 deserialize()

```
void mavlink::common::msg::GENERATOR_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.284.2.2 get_name()

```
std::string mavlink::common::msg::GENERATOR_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.284.2.3 serialize()

```
void mavlink::common::msg::GENERATOR_STATUS::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

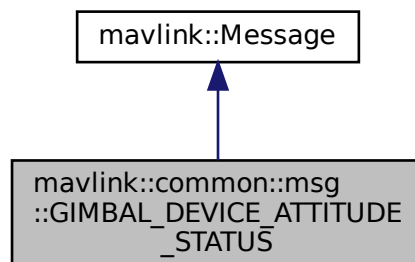
- single_hand_controller/include/mavlink/common/mavlink_msg_generator_status.hpp

5.285 mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS Struct Reference

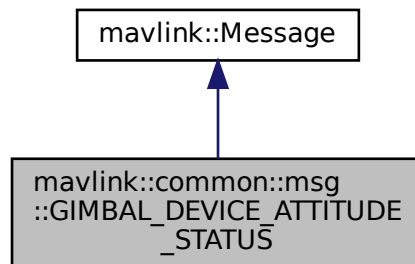
[GIMBAL_DEVICE_ATTITUDE_STATUS](#) message.

```
#include <mavlink_msg_gimbal_device_attitude_status.hpp>
```

Inheritance diagram for mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS:



Collaboration diagram for mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint32_t` **time_boot_ms**
- `uint16_t` **flags**
- `std::array< float, 4 >` **q**
- `float` **angular_velocity_x**
- `float` **angular_velocity_y**
- `float` **angular_velocity_z**
- `uint32_t` **failure_flags**
- `float` **delta_yaw**
- `float` **delta_yaw_velocity**
- `uint8_t` **gimbal_device_id**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 285
- static constexpr `size_t` **LENGTH** = 49
- static constexpr `size_t` **MIN_LENGTH** = 40
- static constexpr `uint8_t` **CRC_EXTRA** = 137
- static constexpr auto **NAME** = "GIMBAL_DEVICE_ATTITUDE_STATUS"

5.285.1 Detailed Description

[GIMBAL_DEVICE_ATTITUDE_STATUS](#) message.

Message reporting the status of a gimbal device. This message should be broadcast by a gimbal device component at a low regular rate (e.g. 5 Hz). For the angles encoded in the quaternion and the angular velocities holds: If the flag `GIMBAL_DEVICE_FLAGS_YAW_IN_VEHICLE_FRAME` is set, then they are relative to the vehicle heading (vehicle frame). If the flag `GIMBAL_DEVICE_FLAGS_YAW_IN_EARTH_FRAME` is set, then they are relative to absolute North (earth frame). If neither of these flags are set, then (for backwards compatibility) it holds: If the flag `GIMBAL_DEVICE_FLAGS_YAW_LOCK` is set, then they are relative to absolute North (earth frame), else they are relative to the vehicle heading (vehicle frame). Other conditions of the flags are not allowed. The quaternion and angular velocities in the other frame can be calculated from `delta_yaw` and `delta_yaw_velocity` as $q_earth = q_delta_yaw * q_vehicle$ and $w_earth = w_delta_yaw_velocity + w_vehicle$ (if not NaN). If neither the `GIMBAL_DEVICE_FLAGS_YAW_IN_VEHICLE_FRAME` nor the `GIMBAL_DEVICE_FLAGS_YAW_IN_EARTH_FRAME` flag is set, then (for backwards compatibility) the data in the `delta_yaw` and `delta_yaw_velocity` fields are to be ignored. New implementations should always set either `GIMBAL_DEVICE_FLAGS_YAW_IN_VEHICLE_FRAME` or `GIMBAL_DEVICE_FLAGS_YAW_IN_EARTH_FRAME`, and always should set `delta_yaw` and `delta_yaw_velocity` either to the proper value or NaN.

5.285.2 Member Function Documentation

5.285.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.285.2.2 get_name()

```
std::string mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.285.2.3 `serialize()`

```
void mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

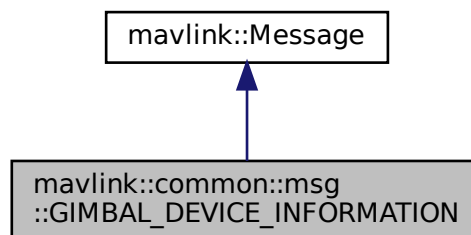
- `single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_attitude_status.hpp`

5.286 `mavlink::common::msg::GIMBAL_DEVICE_INFORMATION` Struct Reference

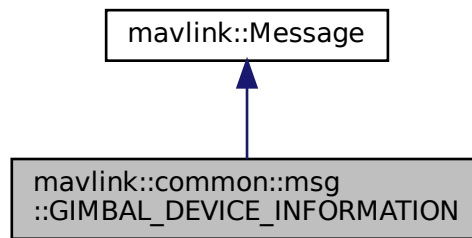
[GIMBAL_DEVICE_INFORMATION](#) message.

```
#include <mavlink_msg_gimbal_device_information.hpp>
```

Inheritance diagram for `mavlink::common::msg::GIMBAL_DEVICE_INFORMATION`:



Collaboration diagram for mavlink::common::msg::GIMBAL_DEVICE_INFORMATION:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t time_boot_ms`
- `std::array< char, 32 > vendor_name`
- `std::array< char, 32 > model_name`
- `std::array< char, 32 > custom_name`
- `uint32_t firmware_version`
- `uint32_t hardware_version`
- `uint64_t uid`
- `uint16_t cap_flags`
- `uint16_t custom_cap_flags`
- `float roll_min`
- `float roll_max`
- `float pitch_min`
- `float pitch_max`
- `float yaw_min`
- `float yaw_max`
- `uint8_t gimbal_device_id`

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 283
- static constexpr size_t **LENGTH** = 145
- static constexpr size_t **MIN_LENGTH** = 144
- static constexpr uint8_t **CRC_EXTRA** = 74
- static constexpr auto **NAME** = "GIMBAL_DEVICE_INFORMATION"

5.286.1 Detailed Description

[GIMBAL_DEVICE_INFORMATION](#) message.

Information about a low level gimbal. This message should be requested by the gimbal manager or a ground station using MAV_CMD_REQUEST_MESSAGE. The maximum angles and rates are the limits by hardware. However, the limits by software used are likely different/smaller and dependent on mode/settings/etc..

5.286.2 Member Function Documentation

5.286.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_DEVICE_INFORMATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.286.2.2 get_name()

```
std::string mavlink::common::msg::GIMBAL_DEVICE_INFORMATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.286.2.3 serialize()

```
void mavlink::common::msg::GIMBAL_DEVICE_INFORMATION::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

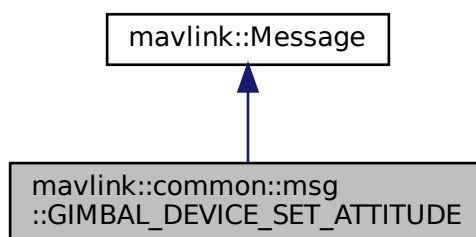
- `single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_information.hpp`

5.287 `mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE` Struct Reference

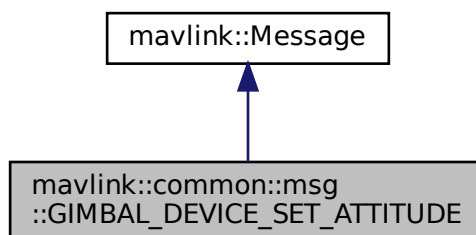
[GIMBAL_DEVICE_SET_ATTITUDE](#) message.

```
#include <mavlink_msg_gimbal_device_set_attitude.hpp>
```

Inheritance diagram for `mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE`:



Collaboration diagram for `mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint16_t **flags**
- std::array< float, 4 > **q**
- float **angular_velocity_x**
- float **angular_velocity_y**
- float **angular_velocity_z**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 284
- static constexpr size_t **LENGTH** = 32
- static constexpr size_t **MIN_LENGTH** = 32
- static constexpr uint8_t **CRC_EXTRA** = 99
- static constexpr auto **NAME** = "GIMBAL_DEVICE_SET_ATTITUDE"

5.287.1 Detailed Description

[GIMBAL_DEVICE_SET_ATTITUDE](#) message.

Low level message to control a gimbal device's attitude. This message is to be sent from the gimbal manager to the gimbal device component. The quaternion and angular velocities can be set to NaN according to use case. For the angles encoded in the quaternion and the angular velocities holds: If the flag `GIMBAL_DEVICE_FLAGS_YAW_IN_VEHICLE_FRAME` is set, then they are relative to the vehicle heading (vehicle frame). If the flag `GIMBAL_DEVICE_FLAGS_YAW_IN_EARTH_FRAME` is set, then they are relative to absolute North (earth frame). If neither of these flags are set, then (for backwards compatibility) it holds: If the flag `GIMBAL_DEVICE_FLAGS_YAW_LOCK` is set, then they are relative to absolute North (earth frame), else they are relative to the vehicle heading (vehicle frame). Setting both `GIMBAL_DEVICE_FLAGS_YAW_IN_VEHICLE_FRAME` and `GIMBAL_DEVICE_FLAGS_YAW_IN_EARTH_FRAME` is not allowed. These rules are to ensure backwards compatibility. New implementations should always set either `GIMBAL_DEVICE_FLAGS_YAW_IN_VEHICLE_FRAME` or `GIMBAL_DEVICE_FLAGS_YAW_IN_EARTH_FRAME`.

5.287.2 Member Function Documentation

5.287.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.287.2.2 get_name()

```
std::string mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.287.2.3 serialize()

```
void mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

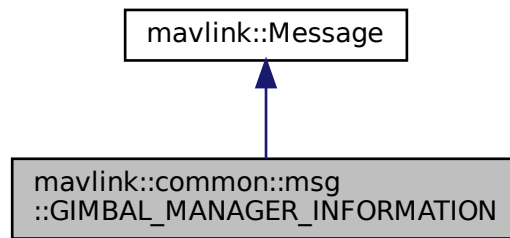
- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_device_set_attitude.hpp

5.288 mavlink::common::msg::GIMBAL_MANAGER_INFORMATION Struct Reference

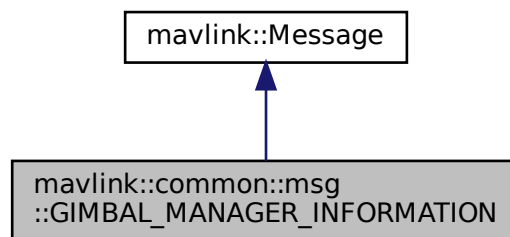
[GIMBAL_MANAGER_INFORMATION](#) message.

```
#include <mavlink_msg_gimbal_manager_information.hpp>
```

Inheritance diagram for mavlink::common::msg::GIMBAL_MANAGER_INFORMATION:



Collaboration diagram for mavlink::common::msg::GIMBAL_MANAGER_INFORMATION:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- uint32_t **cap_flags**

- uint8_t **gimbal_device_id**
- float **roll_min**
- float **roll_max**
- float **pitch_min**
- float **pitch_max**
- float **yaw_min**
- float **yaw_max**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 280
- static constexpr size_t **LENGTH** = 33
- static constexpr size_t **MIN_LENGTH** = 33
- static constexpr uint8_t **CRC_EXTRA** = 70
- static constexpr auto **NAME** = "GIMBAL_MANAGER_INFORMATION"

5.288.1 Detailed Description

[GIMBAL_MANAGER_INFORMATION](#) message.

Information about a high level gimbal manager. This message should be requested by a ground station using MAV_CMD_REQUEST_MESSAGE.

5.288.2 Member Function Documentation

5.288.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_INFORMATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.288.2.2 get_name()

```
std::string mavlink::common::msg::GIMBAL_MANAGER_INFORMATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.288.2.3 serialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_INFORMATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

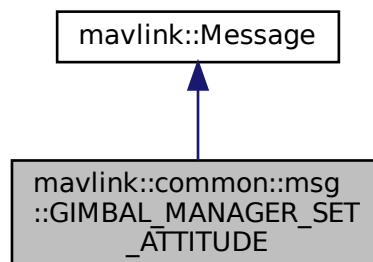
- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_information.hpp

5.289 mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE Struct Reference

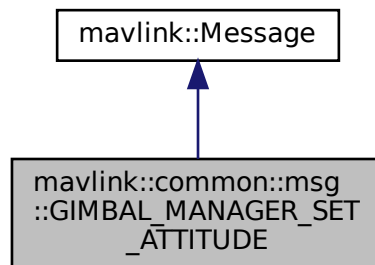
[GIMBAL_MANAGER_SET_ATTITUDE](#) message.

```
#include <mavlink_msg_gimbal_manager_set_attitude.hpp>
```

Inheritance diagram for mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE:



Collaboration diagram for mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint32_t` **flags**
- `uint8_t` **gimbal_device_id**
- `std::array< float, 4 >` **q**
- `float` **angular_velocity_x**
- `float` **angular_velocity_y**
- `float` **angular_velocity_z**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 282
- static constexpr `size_t` **LENGTH** = 35
- static constexpr `size_t` **MIN_LENGTH** = 35
- static constexpr `uint8_t` **CRC_EXTRA** = 123
- static constexpr auto **NAME** = "GIMBAL_MANAGER_SET_ATTITUDE"

5.289.1 Detailed Description

[GIMBAL_MANAGER_SET_ATTITUDE](#) message.

High level message to control a gimbal's attitude. This message is to be sent to the gimbal manager (e.g. from a ground station). Angles and rates can be set to NaN according to use case.

5.289.2 Member Function Documentation

5.289.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.289.2.2 get_name()

```
std::string mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.289.2.3 serialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

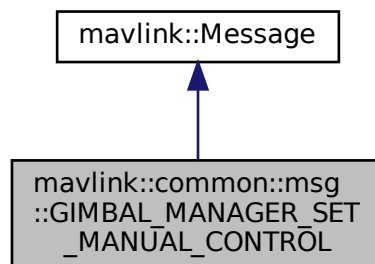
- `single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_attitude.hpp`

5.290 mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL Struct Reference

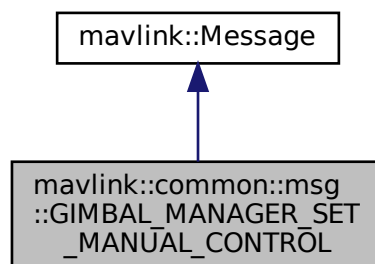
[GIMBAL_MANAGER_SET_MANUAL_CONTROL](#) message.

```
#include <mavlink_msg_gimbal_manager_set_manual_control.hpp>
```

Inheritance diagram for mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL:



Collaboration diagram for mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- Info [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint32_t **flags**
- uint8_t **gimbal_device_id**
- float **pitch**
- float **yaw**
- float **pitch_rate**
- float **yaw_rate**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 288
- static constexpr size_t **LENGTH** = 23
- static constexpr size_t **MIN_LENGTH** = 23
- static constexpr uint8_t **CRC_EXTRA** = 20
- static constexpr auto **NAME** = "GIMBAL_MANAGER_SET_MANUAL_CONTROL"

5.290.1 Detailed Description

[GIMBAL_MANAGER_SET_MANUAL_CONTROL](#) message.

High level message to control a gimbal manually. The angles or angular rates are unitless; the actual rates will depend on internal gimbal manager settings/configuration (e.g. set by parameters). This message is to be sent to the gimbal manager (e.g. from a ground station). Angles and rates can be set to NaN according to use case.

5.290.2 Member Function Documentation

5.290.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.290.2.2 get_name()

```
std::string mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.290.2.3 serialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

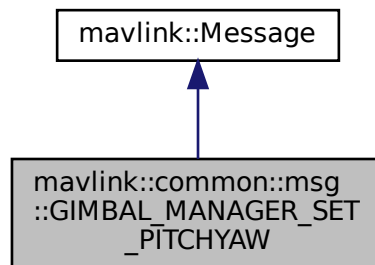
- single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_manual_control.hpp

5.291 mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW Struct Reference

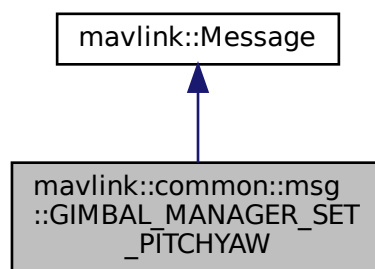
[GIMBAL_MANAGER_SET_PITCHYAW](#) message.

```
#include <mavlink_msg_gimbal_manager_set_pitchyaw.hpp>
```

Inheritance diagram for mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW:



Collaboration diagram for mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info](#) [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint32_t **flags**
- uint8_t **gimbal_device_id**
- float **pitch**
- float **yaw**
- float **pitch_rate**
- float **yaw_rate**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 287
- static constexpr size_t **LENGTH** = 23
- static constexpr size_t **MIN_LENGTH** = 23
- static constexpr uint8_t **CRC_EXTRA** = 1
- static constexpr auto **NAME** = "GIMBAL_MANAGER_SET_PITCHYAW"

5.291.1 Detailed Description

[GIMBAL_MANAGER_SET_PITCHYAW](#) message.

Set gimbal manager pitch and yaw angles (high rate message). This message is to be sent to the gimbal manager (e.g. from a ground station) and will be ignored by gimbal devices. Angles and rates can be set to NaN according to use case. Use MAV_CMD_DO_GIMBAL_MANAGER_PITCHYAW for low-rate adjustments that require confirmation.

5.291.2 Member Function Documentation

5.291.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.291.2.2 `get_name()`

```
std::string mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.291.2.3 `serialize()`

```
void mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

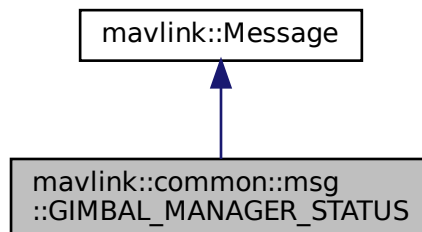
- `single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_set_pitchyaw.hpp`

5.292 `mavlink::common::msg::GIMBAL_MANAGER_STATUS` Struct Reference

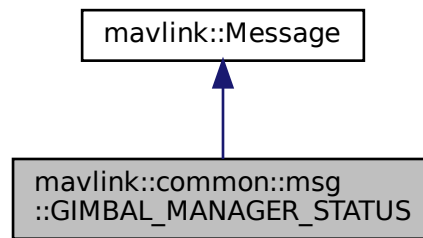
[GIMBAL_MANAGER_STATUS](#) message.

```
#include <mavlink_msg_gimbal_manager_status.hpp>
```

Inheritance diagram for `mavlink::common::msg::GIMBAL_MANAGER_STATUS`:



Collaboration diagram for mavlink::common::msg::GIMBAL_MANAGER_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint32_t` **flags**
- `uint8_t` **gimbal_device_id**
- `uint8_t` **primary_control_sysid**
- `uint8_t` **primary_control_compid**
- `uint8_t` **secondary_control_sysid**
- `uint8_t` **secondary_control_compid**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 281
- static constexpr `size_t` **LENGTH** = 13
- static constexpr `size_t` **MIN_LENGTH** = 13
- static constexpr `uint8_t` **CRC_EXTRA** = 48
- static constexpr auto **NAME** = "GIMBAL_MANAGER_STATUS"

5.292.1 Detailed Description

[GIMBAL_MANAGER_STATUS](#) message.

Current status about a high level gimbal manager. This message should be broadcast at a low regular rate (e.g. 5Hz).

5.292.2 Member Function Documentation

5.292.2.1 deserialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.292.2.2 get_name()

```
std::string mavlink::common::msg::GIMBAL_MANAGER_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.292.2.3 serialize()

```
void mavlink::common::msg::GIMBAL_MANAGER_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

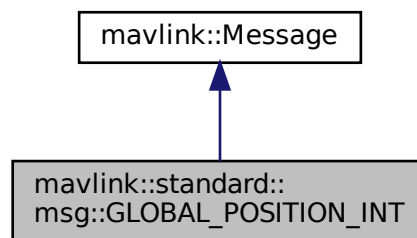
- `single_hand_controller/include/mavlink/common/mavlink_msg_gimbal_manager_status.hpp`

5.293 mavlink::standard::msg::GLOBAL_POSITION_INT Struct Reference

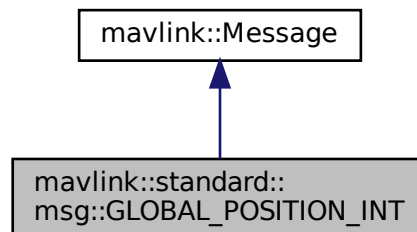
[GLOBAL_POSITION_INT](#) message.

```
#include <mavlink_msg_global_position_int.hpp>
```

Inheritance diagram for mavlink::standard::msg::GLOBAL_POSITION_INT:



Collaboration diagram for mavlink::standard::msg::GLOBAL_POSITION_INT:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- `void deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint32_t time_boot_ms`
- `int32_t lat`
- `int32_t lon`
- `int32_t alt`
- `int32_t relative_alt`
- `int16_t vx`
- `int16_t vy`
- `int16_t vz`
- `uint16_t hdg`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 33`
- `static constexpr size_t LENGTH = 28`
- `static constexpr size_t MIN_LENGTH = 28`
- `static constexpr uint8_t CRC_EXTRA = 104`
- `static constexpr auto NAME = "GLOBAL_POSITION_INT"`

5.293.1 Detailed Description

`GLOBAL_POSITION_INT` message.

The filtered global position (e.g. fused GPS and accelerometers). The position is in GPS-frame (right-handed, Z-up). It is designed as scaled integer message since the resolution of float is not sufficient.

5.293.2 Member Function Documentation

5.293.2.1 `deserialize()`

```
void mavlink::standard::msg::GLOBAL_POSITION_INT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.293.2.2 get_name()

```
std::string mavlink::standard::msg::GLOBAL_POSITION_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.293.2.3 serialize()

```
void mavlink::standard::msg::GLOBAL_POSITION_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

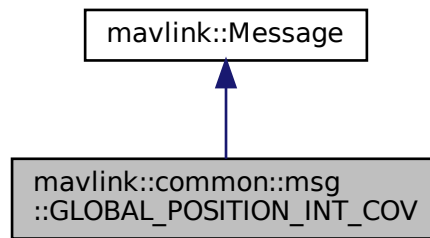
- single_hand_controller/include/mavlink/standard/mavlink_msg_global_position_int.hpp

5.294 mavlink::common::msg::GLOBAL_POSITION_INT_COV Struct Reference

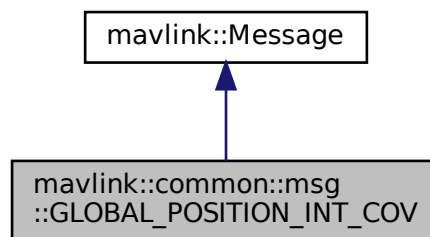
[GLOBAL_POSITION_INT_COV](#) message.

```
#include <mavlink_msg_global_position_int_cov.hpp>
```

Inheritance diagram for mavlink::common::msg::GLOBAL_POSITION_INT_COV:



Collaboration diagram for mavlink::common::msg::GLOBAL_POSITION_INT_COV:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **estimator_type**

- `int32_t lat`
- `int32_t lon`
- `int32_t alt`
- `int32_t relative_alt`
- `float vx`
- `float vy`
- `float vz`
- `std::array< float, 36 > covariance`

Static Public Attributes

- static constexpr `msgid_t MSG_ID` = 63
- static constexpr `size_t LENGTH` = 181
- static constexpr `size_t MIN_LENGTH` = 181
- static constexpr `uint8_t CRC_EXTRA` = 119
- static constexpr auto `NAME` = "GLOBAL_POSITION_INT_COV"

5.294.1 Detailed Description

[GLOBAL_POSITION_INT_COV](#) message.

The filtered global position (e.g. fused GPS and accelerometers). The position is in GPS-frame (right-handed, Z-up). It is designed as scaled integer message since the resolution of float is not sufficient. NOTE: This message is intended for onboard networks / companion computers and higher-bandwidth links and optimized for accuracy and completeness. Please use the GLOBAL_POSITION_INT message for a minimal subset.

5.294.2 Member Function Documentation

5.294.2.1 deserialize()

```
void mavlink::common::msg::GLOBAL_POSITION_INT_COV::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

<code>in</code>	<code>map</code>	
-----------------	------------------	--

Implements [mavlink::Message](#).

5.294.2.2 get_name()

```
std::string mavlink::common::msg::GLOBAL_POSITION_INT_COV::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.294.2.3 `serialize()`

```
void mavlink::common::msg::GLOBAL_POSITION_INT_COV::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

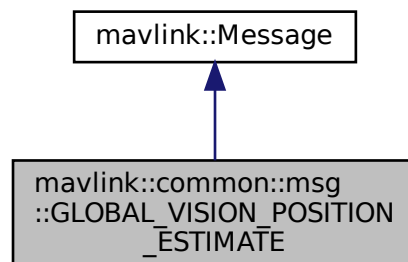
- `single_hand_controller/include/mavlink/common/mavlink_msg_global_position_int_cov.hpp`

5.295 `mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE` Struct Reference

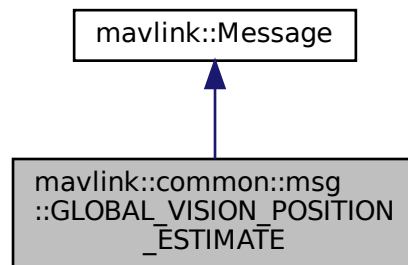
[GLOBAL_VISION_POSITION_ESTIMATE](#) message.

```
#include <mavlink_msg_global_vision_position_estimate.hpp>
```

Inheritance diagram for `mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE`:



Collaboration diagram for mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **usec**
- `float` **x**
- `float` **y**
- `float` **z**
- `float` **roll**
- `float` **pitch**
- `float` **yaw**
- `std::array< float, 21 >` **covariance**
- `uint8_t` **reset_counter**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 101
- static constexpr `size_t` **LENGTH** = 117
- static constexpr `size_t` **MIN_LENGTH** = 32
- static constexpr `uint8_t` **CRC_EXTRA** = 102
- static constexpr auto **NAME** = "GLOBAL_VISION_POSITION_ESTIMATE"

5.295.1 Detailed Description

[GLOBAL_VISION_POSITION_ESTIMATE](#) message.

Global position/attitude estimate from a vision source.

5.295.2 Member Function Documentation

5.295.2.1 deserialize()

```
void mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.295.2.2 get_name()

```
std::string mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.295.2.3 serialize()

```
void mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

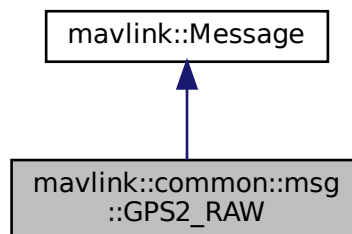
- `single_hand_controller/include/mavlink/common/mavlink_msg_global_vision_position_estimate.hpp`

5.296 mavlink::common::msg::GPS2_RAW Struct Reference

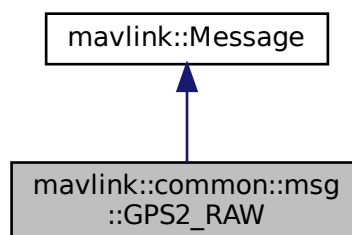
[GPS2_RAW](#) message.

```
#include <mavlink_msg_gps2_raw.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS2_RAW:



Collaboration diagram for mavlink::common::msg::GPS2_RAW:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **fix_type**
- `int32_t` **lat**
- `int32_t` **lon**
- `int32_t` **alt**
- `uint16_t` **eph**
- `uint16_t` **epv**
- `uint16_t` **vel**
- `uint16_t` **cog**
- `uint8_t` **satellites_visible**
- `uint8_t` **dgps_numch**
- `uint32_t` **dgps_age**
- `uint16_t` **yaw**
- `int32_t` **alt_ellipsoid**
- `uint32_t` **h_acc**
- `uint32_t` **v_acc**
- `uint32_t` **vel_acc**
- `uint32_t` **hdg_acc**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 124
- static constexpr `size_t` **LENGTH** = 57
- static constexpr `size_t` **MIN_LENGTH** = 35
- static constexpr `uint8_t` **CRC_EXTRA** = 87
- static constexpr auto **NAME** = "GPS2_RAW"

5.296.1 Detailed Description

[GPS2_RAW](#) message.

Second GPS data.

5.296.2 Member Function Documentation

5.296.2.1 deserialize()

```
void mavlink::common::msg::GPS2_RAW::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.296.2.2 get_name()

```
std::string mavlink::common::msg::GPS2_RAW::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.296.2.3 serialize()

```
void mavlink::common::msg::GPS2_RAW::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

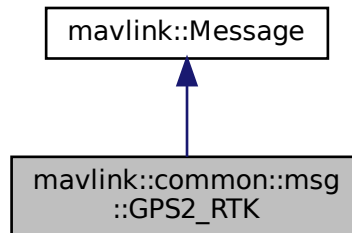
- single_hand_controller/include/mavlink/common/mavlink_msg_gps2_raw.hpp

5.297 mavlink::common::msg::GPS2_RTK Struct Reference

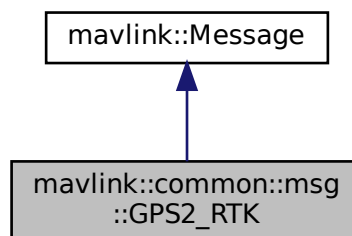
[GPS2_RTK](#) message.

```
#include <mavlink_msg_gps2_rtk.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS2_RTK:



Collaboration diagram for mavlink::common::msg::GPS2_RTK:



Public Member Functions

- `std::string get\_name (void) const` override
Get NAME constant.
- `Info get\_message\_info (void) const` override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to\_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map)` override
Deserialize message.

Public Attributes

- uint32_t **time_last_baseline_ms**
- uint8_t **rtk_receiver_id**
- uint16_t **wn**
- uint32_t **tow**
- uint8_t **rtk_health**
- uint8_t **rtk_rate**
- uint8_t **nsats**
- uint8_t **baseline_coords_type**
- int32_t **baseline_a_mm**
- int32_t **baseline_b_mm**
- int32_t **baseline_c_mm**
- uint32_t **accuracy**
- int32_t **iar_num_hypotheses**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 128
- static constexpr size_t **LENGTH** = 35
- static constexpr size_t **MIN_LENGTH** = 35
- static constexpr uint8_t **CRC_EXTRA** = 226
- static constexpr auto **NAME** = "GPS2_RTK"

5.297.1 Detailed Description

[GPS2_RTK](#) message.

RTK GPS data. Gives information on the relative baseline calculation the GPS is reporting

5.297.2 Member Function Documentation

5.297.2.1 deserialize()

```
void mavlink::common::msg::GPS2_RTK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.297.2.2 `get_name()`

```
std::string mavlink::common::msg::GPS2_RTK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.297.2.3 `serialize()`

```
void mavlink::common::msg::GPS2_RTK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

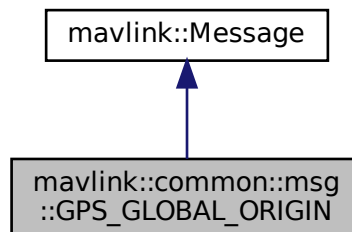
- `single_hand_controller/include/mavlink/common/mavlink_msg_gps2_rtk.hpp`

5.298 `mavlink::common::msg::GPS_GLOBAL_ORIGIN` Struct Reference

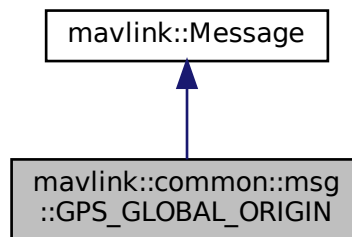
[GPS_GLOBAL_ORIGIN](#) message.

```
#include <mavlink_msg_gps_global_origin.hpp>
```

Inheritance diagram for `mavlink::common::msg::GPS_GLOBAL_ORIGIN`:



Collaboration diagram for mavlink::common::msg::GPS_GLOBAL_ORIGIN:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `int32_t` **latitude**
- `int32_t` **longitude**
- `int32_t` **altitude**
- `uint64_t` **time_usec**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 49
- static constexpr size_t **LENGTH** = 20
- static constexpr size_t **MIN_LENGTH** = 12
- static constexpr uint8_t **CRC_EXTRA** = 39
- static constexpr auto **NAME** = "GPS_GLOBAL_ORIGIN"

5.298.1 Detailed Description

[GPS_GLOBAL_ORIGIN](#) message.

Publishes the GPS coordinates of the vehicle local origin (0,0,0) position. Emitted whenever a new GPS-Local position mapping is requested or set - e.g. following [SET_GPS_GLOBAL_ORIGIN](#) message.

5.298.2 Member Function Documentation

5.298.2.1 deserialize()

```
void mavlink::common::msg::GPS_GLOBAL_ORIGIN::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.298.2.2 get_name()

```
std::string mavlink::common::msg::GPS_GLOBAL_ORIGIN::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.298.2.3 serialize()

```
void mavlink::common::msg::GPS_GLOBAL_ORIGIN::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

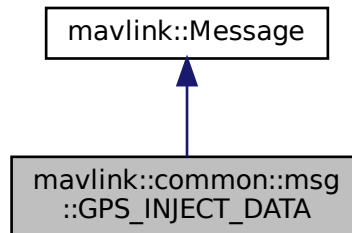
- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_global_origin.hpp`

5.299 mavlink::common::msg::GPS_INJECT_DATA Struct Reference

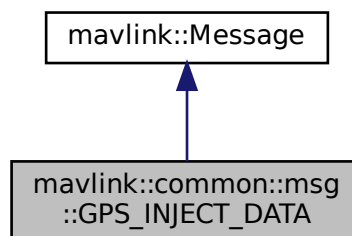
[GPS_INJECT_DATA](#) message.

```
#include <mavlink_msg_gps_inject_data.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS_INJECT_DATA:



Collaboration diagram for mavlink::common::msg::GPS_INJECT_DATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `uint8_t len`
- `std::array< uint8_t, 110 > data`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 123`
- `static constexpr size_t LENGTH = 113`
- `static constexpr size_t MIN_LENGTH = 113`
- `static constexpr uint8_t CRC_EXTRA = 250`
- `static constexpr auto NAME = "GPS_INJECT_DATA"`

5.299.1 Detailed Description

[GPS_INJECT_DATA](#) message.

Data for injecting into the onboard GPS (used for DGPS)

5.299.2 Member Function Documentation

5.299.2.1 deserialize()

```
void mavlink::common::msg::GPS_INJECT_DATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.299.2.2 get_name()

```
std::string mavlink::common::msg::GPS_INJECT_DATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.299.2.3 serialize()

```
void mavlink::common::msg::GPS_INJECT_DATA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

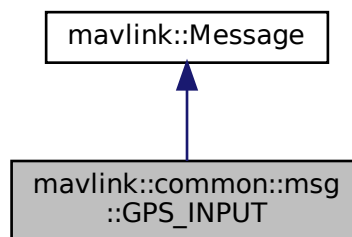
- single_hand_controller/include/mavlink/common/mavlink_msg_gps_inject_data.hpp

5.300 mavlink::common::msg::GPS_INPUT Struct Reference

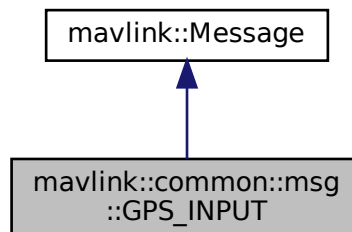
[GPS_INPUT](#) message.

```
#include <mavlink_msg_gps_input.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS_INPUT:



Collaboration diagram for mavlink::common::msg::GPS_INPUT:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t time_usec`
- `uint8_t gps_id`
- `uint16_t ignore_flags`
- `uint32_t time_week_ms`
- `uint16_t time_week`
- `uint8_t fix_type`
- `int32_t lat`
- `int32_t lon`
- `float alt`
- `float hdop`
- `float vdop`
- `float vn`
- `float ve`
- `float vd`
- `float speed_accuracy`
- `float horiz_accuracy`
- `float vert_accuracy`
- `uint8_t satellites_visible`
- `uint16_t yaw`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 232`
- `static constexpr size_t LENGTH = 65`
- `static constexpr size_t MIN_LENGTH = 63`
- `static constexpr uint8_t CRC_EXTRA = 151`
- `static constexpr auto NAME = "GPS_INPUT"`

5.300.1 Detailed Description

`GPS_INPUT` message.

GPS sensor input message. This is a raw sensor value sent by the GPS. This is NOT the global position estimate of the system.

5.300.2 Member Function Documentation

5.300.2.1 deserialize()

```
void mavlink::common::msg::GPS_INPUT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.300.2.2 get_name()

```
std::string mavlink::common::msg::GPS_INPUT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.300.2.3 serialize()

```
void mavlink::common::msg::GPS_INPUT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

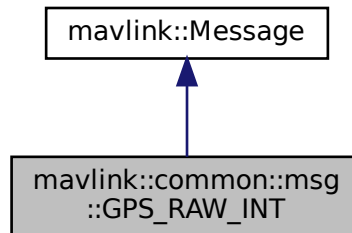
- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_input.hpp`

5.301 mavlink::common::msg::GPS_RAW_INT Struct Reference

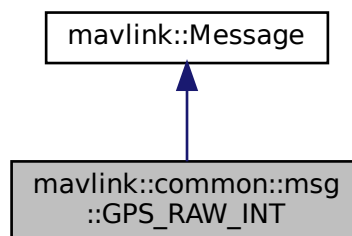
[GPS_RAW_INT](#) message.

```
#include <mavlink_msg_gps_raw_int.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS_RAW_INT:



Collaboration diagram for mavlink::common::msg::GPS_RAW_INT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- uint8_t **fix_type**
- int32_t **lat**
- int32_t **lon**
- int32_t **alt**
- uint16_t **eph**
- uint16_t **epv**
- uint16_t **vel**
- uint16_t **cog**
- uint8_t **satellites_visible**
- int32_t **alt_ellipsoid**
- uint32_t **h_acc**
- uint32_t **v_acc**
- uint32_t **vel_acc**
- uint32_t **hdg_acc**
- uint16_t **yaw**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 24
- static constexpr size_t **LENGTH** = 52
- static constexpr size_t **MIN_LENGTH** = 30
- static constexpr uint8_t **CRC_EXTRA** = 24
- static constexpr auto **NAME** = "GPS_RAW_INT"

5.301.1 Detailed Description

[GPS_RAW_INT](#) message.

The global position, as returned by the Global Positioning System (GPS). This is NOT the global position estimate of the system, but rather a RAW sensor value. See message GLOBAL_POSITION_INT for the global position estimate.

5.301.2 Member Function Documentation

5.301.2.1 deserialize()

```
void mavlink::common::msg::GPS_RAW_INT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.301.2.2 `get_name()`

```
std::string mavlink::common::msg::GPS_RAW_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.301.2.3 `serialize()`

```
void mavlink::common::msg::GPS_RAW_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

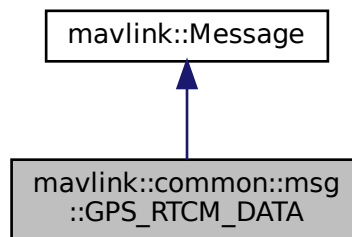
- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_raw_int.hpp`

5.302 `mavlink::common::msg::GPS_RTCM_DATA` Struct Reference

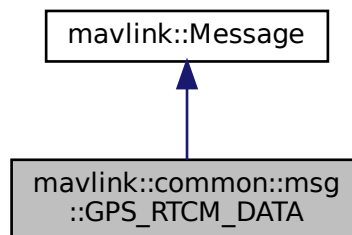
[GPS_RTCM_DATA](#) message.

```
#include <mavlink_msg_gps_rtc_data.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS_RTCM_DATA:



Collaboration diagram for mavlink::common::msg::GPS_RTCM_DATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **flags**
- `uint8_t` **len**
- `std::array< uint8_t, 180 >` **data**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 233
- static constexpr size_t **LENGTH** = 182
- static constexpr size_t **MIN_LENGTH** = 182
- static constexpr uint8_t **CRC_EXTRA** = 35
- static constexpr auto **NAME** = "GPS_RTCM_DATA"

5.302.1 Detailed Description

[GPS_RTCM_DATA](#) message.

RTCM message for injecting into the onboard GPS (used for DGPS)

5.302.2 Member Function Documentation

5.302.2.1 deserialize()

```
void mavlink::common::msg::GPS_RTCM_DATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.302.2.2 get_name()

```
std::string mavlink::common::msg::GPS_RTCM_DATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.302.2.3 serialize()

```
void mavlink::common::msg::GPS_RTCM_DATA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

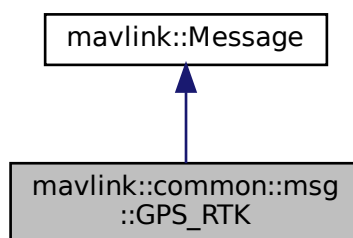
- `single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtc_data.hpp`

5.303 mavlink::common::msg::GPS_RTK Struct Reference

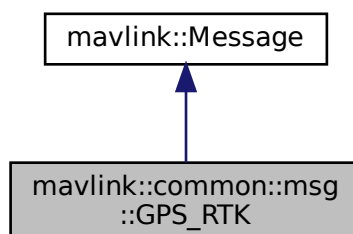
[GPS_RTK](#) message.

```
#include <mavlink_msg_gps_rtk.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS_RTK:



Collaboration diagram for mavlink::common::msg::GPS_RTK:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_last_baseline_ms**
- `uint8_t` **rtk_receiver_id**
- `uint16_t` **wn**
- `uint32_t` **tow**
- `uint8_t` **rtk_health**
- `uint8_t` **rtk_rate**
- `uint8_t` **nsats**
- `uint8_t` **baseline_coords_type**
- `int32_t` **baseline_a_mm**
- `int32_t` **baseline_b_mm**
- `int32_t` **baseline_c_mm**
- `uint32_t` **accuracy**
- `int32_t` **iar_num_hypotheses**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 127
- static constexpr `size_t` **LENGTH** = 35
- static constexpr `size_t` **MIN_LENGTH** = 35
- static constexpr `uint8_t` **CRC_EXTRA** = 25
- static constexpr auto **NAME** = "GPS_RTK"

5.303.1 Detailed Description

[GPS_RTK](#) message.

RTK GPS data. Gives information on the relative baseline calculation the GPS is reporting

5.303.2 Member Function Documentation

5.303.2.1 `deserialize()`

```
void mavlink::common::msg::GPS_RTK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.303.2.2 get_name()

```
std::string mavlink::common::msg::GPS_RTK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.303.2.3 serialize()

```
void mavlink::common::msg::GPS_RTK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

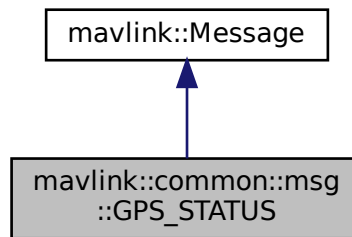
- single_hand_controller/include/mavlink/common/mavlink_msg_gps_rtk.hpp

5.304 mavlink::common::msg::GPS_STATUS Struct Reference

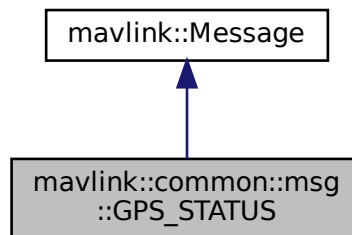
[GPS_STATUS](#) message.

```
#include <mavlink_msg_gps_status.hpp>
```

Inheritance diagram for mavlink::common::msg::GPS_STATUS:



Collaboration diagram for mavlink::common::msg::GPS_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **satellites_visible**
- `std::array< uint8_t, 20 >` **satellite_prn**
- `std::array< uint8_t, 20 >` **satellite_used**
- `std::array< uint8_t, 20 >` **satellite_elevation**
- `std::array< uint8_t, 20 >` **satellite_azimuth**
- `std::array< uint8_t, 20 >` **satellite_snr**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 25
- static constexpr size_t **LENGTH** = 101
- static constexpr size_t **MIN_LENGTH** = 101
- static constexpr uint8_t **CRC_EXTRA** = 23
- static constexpr auto **NAME** = "GPS_STATUS"

5.304.1 Detailed Description

[GPS_STATUS](#) message.

The positioning status, as reported by GPS. This message is intended to display status information about each satellite visible to the receiver. See message GLOBAL_POSITION_INT for the global position estimate. This message can contain information for up to 20 satellites.

5.304.2 Member Function Documentation

5.304.2.1 deserialize()

```
void mavlink::common::msg::GPS_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.304.2.2 get_name()

```
std::string mavlink::common::msg::GPS_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.304.2.3 `serialize()`

```
void mavlink::common::msg::GPS_STATUS::serialize (  
    mavlink::MsgMap & map ) const    [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/mavlink/common/mavlink_msg_gps_status.hpp](#)

5.305 HC_ATLAS_ARM_DISARM_CMD Struct Reference

Public Attributes

- `uint8_t target_system` = SCOUT_ID
- `uint8_t target_component` = ATLAS_COMP_ID
- `uint16_t command` = MAV_CMD_COMPONENT_ARM_DISARM
- `uint8_t confirmation` = 0
- `float param1`

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/message_structs.h](#)

5.306 HC_ATLAS_HEARTBEAT_BC Struct Reference

Public Attributes

- `uint8_t sys_id` = HC_ID
- `uint8_t comp_id` = HC_COMP_ID

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/message_structs.h](#)

5.307 HC_ATLAS_MANUAL_CONTROL_BC Struct Reference

Public Attributes

- `uint8_t target` = SCOUT_ID
- `int16_t x`
- `int16_t y`
- `uint16_t Push_buttons` = 0
- `uint16_t Tristate_Toggle_switches` = 0

The documentation for this struct was generated from the following file:

- [single_hand_controller/include/message_structs.h](#)

5.308 HC_ATLAS_TIMESYNC_REQ Struct Reference

Public Attributes

- int64_t **tc1** = 0
- int64_t **ts1**
- uint8_t **target_sys** = SCOUT_ID
- uint8_t **target_comp** = ATLAS_COMP_ID

The documentation for this struct was generated from the following file:

- single_hand_controller/include/[message_structs.h](#)

5.309 HC_RADIO_STATUS Struct Reference

Public Attributes

- uint8_t **rsi**
- uint8_t **remrsi**
- uint8_t **txbuf**
- uint8_t **noise**
- uint8_t **remnoise**
- uint16_t **rxerrors**
- uint16_t **fixed**

The documentation for this struct was generated from the following file:

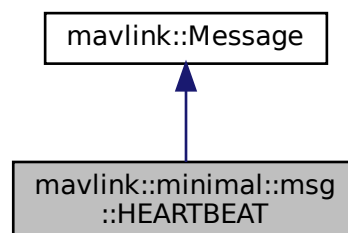
- single_hand_controller/include/[message_structs.h](#)

5.310 mavlink::minimal::msg::HEARTBEAT Struct Reference

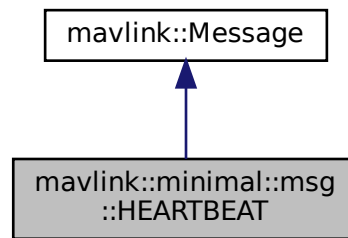
[HEARTBEAT](#) message.

```
#include <mavlink_msg_heartbeat.hpp>
```

Inheritance diagram for mavlink::minimal::msg::HEARTBEAT:



Collaboration diagram for mavlink::minimal::msg::HEARTBEAT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **type**
- `uint8_t` **autopilot**
- `uint8_t` **base_mode**
- `uint32_t` **custom_mode**
- `uint8_t` **system_status**
- `uint8_t` **mavlink_version**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 0
- static constexpr size_t **LENGTH** = 9
- static constexpr size_t **MIN_LENGTH** = 9
- static constexpr uint8_t **CRC_EXTRA** = 50
- static constexpr auto **NAME** = "HEARTBEAT"

5.310.1 Detailed Description

HEARTBEAT message.

The heartbeat message shows that a system or component is present and responding. The type and autopilot fields (along with the message component id), allow the receiving system to treat further messages from this system appropriately (e.g. by laying out the user interface based on the autopilot). This microservice is documented at <https://mavlink.io/en/services/heartbeat.html>

5.310.2 Member Function Documentation

5.310.2.1 deserialize()

```
void mavlink::minimal::msg::HEARTBEAT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.310.2.2 get_name()

```
std::string mavlink::minimal::msg::HEARTBEAT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.310.2.3 serialize()

```
void mavlink::minimal::msg::HEARTBEAT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

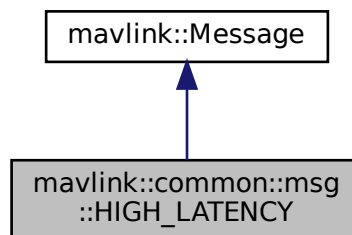
- `single_hand_controller/include/mavlink/minimal/mavlink_msg_heartbeat.hpp`

5.311 mavlink::common::msg::HIGH_LATENCY Struct Reference

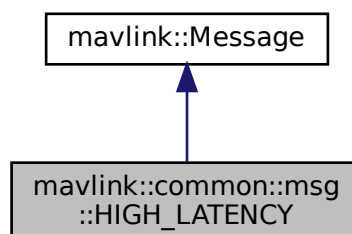
[HIGH_LATENCY](#) message.

```
#include <mavlink_msg_high_latency.hpp>
```

Inheritance diagram for mavlink::common::msg::HIGH_LATENCY:



Collaboration diagram for mavlink::common::msg::HIGH_LATENCY:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- `void deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint8_t base_mode`
- `uint32_t custom_mode`
- `uint8_t landed_state`
- `int16_t roll`
- `int16_t pitch`
- `uint16_t heading`
- `int8_t throttle`
- `int16_t heading_sp`
- `int32_t latitude`
- `int32_t longitude`
- `int16_t altitude_amsl`
- `int16_t altitude_sp`
- `uint8_t airspeed`
- `uint8_t airspeed_sp`
- `uint8_t groundspeed`
- `int8_t climb_rate`
- `uint8_t gps_nsat`
- `uint8_t gps_fix_type`
- `uint8_t battery_remaining`
- `int8_t temperature`
- `int8_t temperature_air`
- `uint8_t failsafe`
- `uint8_t wp_num`
- `uint16_t wp_distance`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 234`
- `static constexpr size_t LENGTH = 40`
- `static constexpr size_t MIN_LENGTH = 40`
- `static constexpr uint8_t CRC_EXTRA = 150`
- `static constexpr auto NAME = "HIGH_LATENCY"`

5.311.1 Detailed Description

`HIGH_LATENCY` message.

`Message` appropriate for high latency connections like Iridium

5.311.2 Member Function Documentation

5.311.2.1 deserialize()

```
void mavlink::common::msg::HIGH_LATENCY::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.311.2.2 get_name()

```
std::string mavlink::common::msg::HIGH_LATENCY::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.311.2.3 serialize()

```
void mavlink::common::msg::HIGH_LATENCY::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

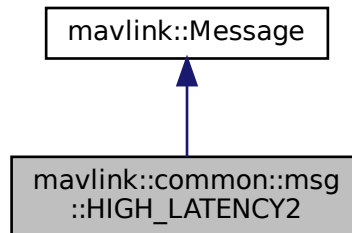
- `single_hand_controller/include/mavlink/common/mavlink_msg_high_latency.hpp`

5.312 mavlink::common::msg::HIGH_LATENCY2 Struct Reference

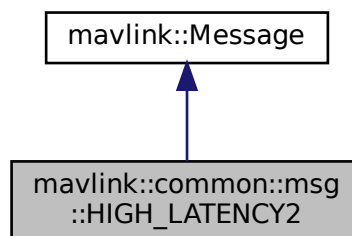
[HIGH_LATENCY2](#) message.

```
#include <mavlink_msg_high_latency2.hpp>
```

Inheritance diagram for mavlink::common::msg::HIGH_LATENCY2:



Collaboration diagram for mavlink::common::msg::HIGH_LATENCY2:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **timestamp**
- uint8_t **type**
- uint8_t **autopilot**
- uint16_t **custom_mode**
- int32_t **latitude**
- int32_t **longitude**
- int16_t **altitude**
- int16_t **target_altitude**
- uint8_t **heading**
- uint8_t **target_heading**
- uint16_t **target_distance**
- uint8_t **throttle**
- uint8_t **airspeed**
- uint8_t **airspeed_sp**
- uint8_t **groundspeed**
- uint8_t **windspeed**
- uint8_t **wind_heading**
- uint8_t **eph**
- uint8_t **epv**
- int8_t **temperature_air**
- int8_t **climb_rate**
- int8_t **battery**
- uint16_t **wp_num**
- uint16_t **failure_flags**
- int8_t **custom0**
- int8_t **custom1**
- int8_t **custom2**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 235
- static constexpr size_t **LENGTH** = 42
- static constexpr size_t **MIN_LENGTH** = 42
- static constexpr uint8_t **CRC_EXTRA** = 179
- static constexpr auto **NAME** = "HIGH_LATENCY2"

5.312.1 Detailed Description

[HIGH_LATENCY2](#) message.

[Message](#) appropriate for high latency connections like Iridium (version 2)

5.312.2 Member Function Documentation

5.312.2.1 deserialize()

```
void mavlink::common::msg::HIGH_LATENCY2::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.312.2.2 get_name()

```
std::string mavlink::common::msg::HIGH_LATENCY2::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.312.2.3 serialize()

```
void mavlink::common::msg::HIGH_LATENCY2::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

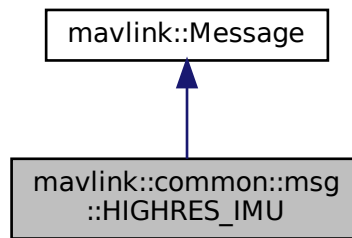
- single_hand_controller/include/mavlink/common/mavlink_msg_high_latency2.hpp

5.313 mavlink::common::msg::HIGHRES_IMU Struct Reference

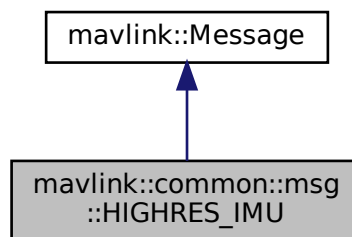
[HIGHRES_IMU](#) message.

```
#include <mavlink_msg_highres_imu.hpp>
```

Inheritance diagram for mavlink::common::msg::HIGHRES_IMU:



Collaboration diagram for mavlink::common::msg::HIGHRES_IMU:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `float` **xacc**

- float **yacc**
- float **zacc**
- float **xgyro**
- float **gyro**
- float **zgyro**
- float **xmag**
- float **ymag**
- float **zmag**
- float **abs_pressure**
- float **diff_pressure**
- float **pressure_alt**
- float **temperature**
- uint16_t **fields_updated**
- uint8_t **id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 105
- static constexpr size_t **LENGTH** = 63
- static constexpr size_t **MIN_LENGTH** = 62
- static constexpr uint8_t **CRC_EXTRA** = 93
- static constexpr auto **NAME** = "HIGHRES_IMU"

5.313.1 Detailed Description

[HIGHRES_IMU](#) message.

The IMU readings in SI units in NED body frame

5.313.2 Member Function Documentation

5.313.2.1 deserialize()

```
void mavlink::common::msg::HIGHRES_IMU::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.313.2.2 get_name()

```
std::string mavlink::common::msg::HIGHRES_IMU::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.313.2.3 serialize()

```
void mavlink::common::msg::HIGHRES_IMU::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

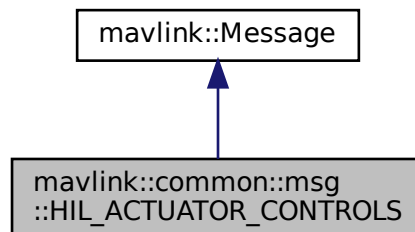
- single_hand_controller/include/mavlink/common/mavlink_msg_highres_imu.hpp

5.314 mavlink::common::msg::HIL_ACTUATOR_CONTROLS Struct Reference

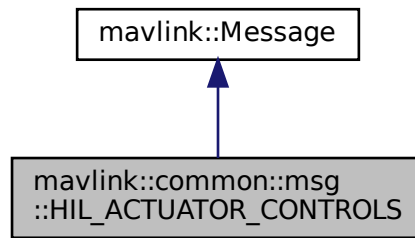
[HIL_ACTUATOR_CONTROLS](#) message.

```
#include <mavlink_msg_hil_actuator_controls.hpp>
```

Inheritance diagram for mavlink::common::msg::HIL_ACTUATOR_CONTROLS:



Collaboration diagram for mavlink::common::msg::HIL_ACTUATOR_CONTROLS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `std::array< float, 16 >` **controls**
- `uint8_t` **mode**
- `uint64_t` **flags**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 93
- static constexpr size_t **LENGTH** = 81
- static constexpr size_t **MIN_LENGTH** = 81
- static constexpr uint8_t **CRC_EXTRA** = 47
- static constexpr auto **NAME** = "HIL_ACTUATOR_CONTROLS"

5.314.1 Detailed Description

[HIL_ACTUATOR_CONTROLS](#) message.

Sent from autopilot to simulation. Hardware in the loop control outputs. Alternative to [HIL_CONTROLS](#).

5.314.2 Member Function Documentation

5.314.2.1 deserialize()

```
void mavlink::common::msg::HIL_ACTUATOR_CONTROLS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.314.2.2 get_name()

```
std::string mavlink::common::msg::HIL_ACTUATOR_CONTROLS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.314.2.3 serialize()

```
void mavlink::common::msg::HIL_ACTUATOR_CONTROLS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

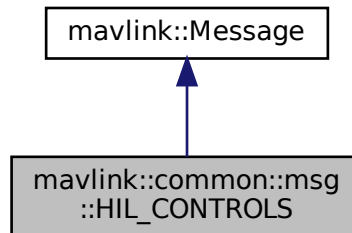
- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_actuator_controls.hpp`

5.315 mavlink::common::msg::HIL_CONTROLS Struct Reference

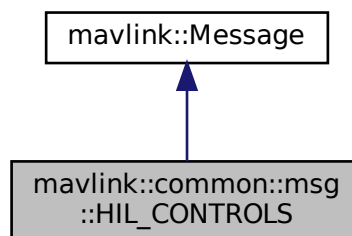
[HIL_CONTROLS](#) message.

```
#include <mavlink_msg_hil_controls.hpp>
```

Inheritance diagram for mavlink::common::msg::HIL_CONTROLS:



Collaboration diagram for mavlink::common::msg::HIL_CONTROLS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- float **roll_ailerons**
- float **pitch_elevator**
- float **yaw_rudder**
- float **throttle**
- float **aux1**
- float **aux2**
- float **aux3**
- float **aux4**
- uint8_t **mode**
- uint8_t **nav_mode**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 91
- static constexpr size_t **LENGTH** = 42
- static constexpr size_t **MIN_LENGTH** = 42
- static constexpr uint8_t **CRC_EXTRA** = 63
- static constexpr auto **NAME** = "HIL_CONTROLS"

5.315.1 Detailed Description

[HIL_CONTROLS](#) message.

Sent from autopilot to simulation. Hardware in the loop control outputs. Alternative to [HIL_ACTUATOR_CONTROLS](#).

5.315.2 Member Function Documentation

5.315.2.1 deserialize()

```
void mavlink::common::msg::HIL_CONTROLS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.315.2.2 `get_name()`

```
std::string mavlink::common::msg::HIL_CONTROLS::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.315.2.3 `serialize()`

```
void mavlink::common::msg::HIL_CONTROLS::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

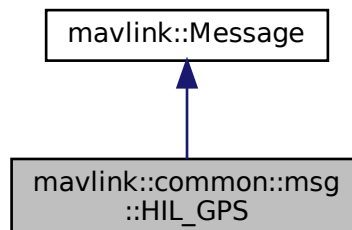
- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_controls.hpp`

5.316 `mavlink::common::msg::HIL_GPS` Struct Reference

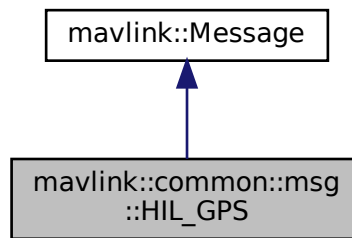
[HIL_GPS](#) message.

```
#include <mavlink_msg_hil_gps.hpp>
```

Inheritance diagram for `mavlink::common::msg::HIL_GPS`:



Collaboration diagram for mavlink::common::msg::HIL_GPS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- uint8_t **fix_type**
- int32_t **lat**
- int32_t **lon**
- int32_t **alt**
- uint16_t **eph**
- uint16_t **epv**
- uint16_t **vel**
- int16_t **vn**
- int16_t **ve**
- int16_t **vd**
- uint16_t **cog**
- uint8_t **satellites_visible**
- uint8_t **id**
- uint16_t **yaw**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 113
- static constexpr size_t **LENGTH** = 39
- static constexpr size_t **MIN_LENGTH** = 36
- static constexpr uint8_t **CRC_EXTRA** = 124
- static constexpr auto **NAME** = "HIL_GPS"

5.316.1 Detailed Description

[HIL_GPS](#) message.

The global position, as returned by the Global Positioning System (GPS). This is NOT the global position estimate of the system, but rather a RAW sensor value. See message GLOBAL_POSITION_INT for the global position estimate.

5.316.2 Member Function Documentation

5.316.2.1 deserialize()

```
void mavlink::common::msg::HIL_GPS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.316.2.2 get_name()

```
std::string mavlink::common::msg::HIL_GPS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.316.2.3 serialize()

```
void mavlink::common::msg::HIL_GPS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

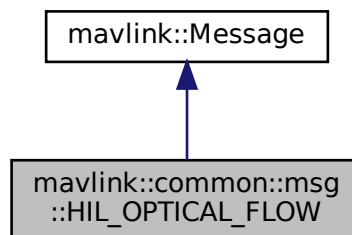
- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_gps.hpp`

5.317 `mavlink::common::msg::HIL_OPTICAL_FLOW` Struct Reference

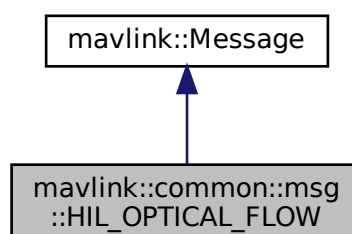
[HIL_OPTICAL_FLOW](#) message.

```
#include <mavlink_msg_hil_optical_flow.hpp>
```

Inheritance diagram for `mavlink::common::msg::HIL_OPTICAL_FLOW`:



Collaboration diagram for `mavlink::common::msg::HIL_OPTICAL_FLOW`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- uint8_t **sensor_id**
- uint32_t **integration_time_us**
- float **integrated_x**
- float **integrated_y**
- float **integrated_xgyro**
- float **integrated_ygyro**
- float **integrated_zgyro**
- int16_t **temperature**
- uint8_t **quality**
- uint32_t **time_delta_distance_us**
- float **distance**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 114
- static constexpr size_t **LENGTH** = 44
- static constexpr size_t **MIN_LENGTH** = 44
- static constexpr uint8_t **CRC_EXTRA** = 237
- static constexpr auto **NAME** = "HIL_OPTICAL_FLOW"

5.317.1 Detailed Description

[HIL_OPTICAL_FLOW](#) message.

Simulated optical flow from a flow sensor (e.g. PX4FLOW or optical mouse sensor)

5.317.2 Member Function Documentation

5.317.2.1 deserialize()

```
void mavlink::common::msg::HIL_OPTICAL_FLOW::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.317.2.2 get_name()

```
std::string mavlink::common::msg::HIL_OPTICAL_FLOW::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.317.2.3 serialize()

```
void mavlink::common::msg::HIL_OPTICAL_FLOW::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

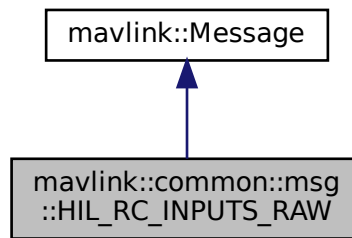
- single_hand_controller/include/mavlink/common/mavlink_msg_hil_optical_flow.hpp

5.318 mavlink::common::msg::HIL_RC_INPUTS_RAW Struct Reference

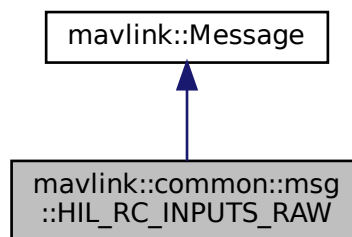
[HIL_RC_INPUTS_RAW](#) message.

```
#include <mavlink_msg_hil_rc_inputs_raw.hpp>
```

Inheritance diagram for mavlink::common::msg::HIL_RC_INPUTS_RAW:



Collaboration diagram for mavlink::common::msg::HIL_RC_INPUTS_RAW:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint16_t` **chan1_raw**

- uint16_t **chan2_raw**
- uint16_t **chan3_raw**
- uint16_t **chan4_raw**
- uint16_t **chan5_raw**
- uint16_t **chan6_raw**
- uint16_t **chan7_raw**
- uint16_t **chan8_raw**
- uint16_t **chan9_raw**
- uint16_t **chan10_raw**
- uint16_t **chan11_raw**
- uint16_t **chan12_raw**
- uint8_t **rsi**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 92
- static constexpr size_t **LENGTH** = 33
- static constexpr size_t **MIN_LENGTH** = 33
- static constexpr uint8_t **CRC_EXTRA** = 54
- static constexpr auto **NAME** = "HIL_RC_INPUTS_RAW"

5.318.1 Detailed Description

[HIL_RC_INPUTS_RAW](#) message.

Sent from simulation to autopilot. The RAW values of the RC channels received. The standard PPM modulation is as follows: 1000 microseconds: 0%, 2000 microseconds: 100%. Individual receivers/transmitters might violate this specification.

5.318.2 Member Function Documentation

5.318.2.1 deserialize()

```
void mavlink::common::msg::HIL_RC_INPUTS_RAW::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.318.2.2 get_name()

```
std::string mavlink::common::msg::HIL_RC_INPUTS_RAW::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.318.2.3 serialize()

```
void mavlink::common::msg::HIL_RC_INPUTS_RAW::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

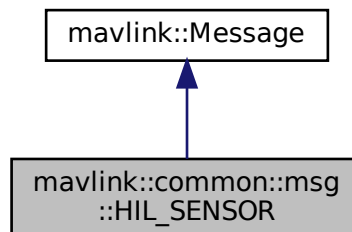
- single_hand_controller/include/mavlink/common/mavlink_msg_hil_rc_inputs_raw.hpp

5.319 mavlink::common::msg::HIL_SENSOR Struct Reference

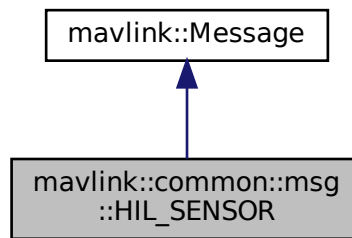
[HIL_SENSOR](#) message.

```
#include <mavlink_msg_hil_sensor.hpp>
```

Inheritance diagram for mavlink::common::msg::HIL_SENSOR:



Collaboration diagram for mavlink::common::msg::HIL_SENSOR:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) ([mavlink::MsgMap](#) &map) const override
Serialize message.
- void [deserialize](#) ([mavlink::MsgMap](#) &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `float` **xacc**
- `float` **yacc**
- `float` **zacc**
- `float` **xgyro**
- `float` **ygyro**
- `float` **zgyro**
- `float` **xmag**
- `float` **ymag**
- `float` **zmag**
- `float` **abs_pressure**
- `float` **diff_pressure**
- `float` **pressure_alt**
- `float` **temperature**
- `uint32_t` **fields_updated**
- `uint8_t` **id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 107
- static constexpr size_t **LENGTH** = 65
- static constexpr size_t **MIN_LENGTH** = 64
- static constexpr uint8_t **CRC_EXTRA** = 108
- static constexpr auto **NAME** = "HIL_SENSOR"

5.319.1 Detailed Description

[HIL_SENSOR](#) message.

The IMU readings in SI units in NED body frame

5.319.2 Member Function Documentation

5.319.2.1 deserialize()

```
void mavlink::common::msg::HIL_SENSOR::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.319.2.2 get_name()

```
std::string mavlink::common::msg::HIL_SENSOR::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.319.2.3 serialize()

```
void mavlink::common::msg::HIL_SENSOR::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

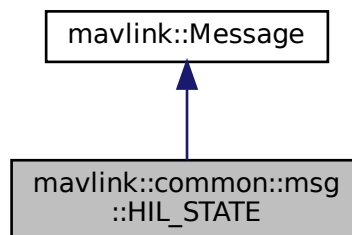
- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_sensor.hpp`

5.320 mavlink::common::msg::HIL_STATE Struct Reference

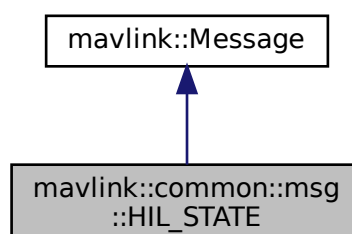
[HIL_STATE](#) message.

```
#include <mavlink_msg_hil_state.hpp>
```

Inheritance diagram for `mavlink::common::msg::HIL_STATE`:



Collaboration diagram for `mavlink::common::msg::HIL_STATE`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- Info [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- float **roll**
- float **pitch**
- float **yaw**
- float **rollspeed**
- float **pitchspeed**
- float **yawspeed**
- int32_t **lat**
- int32_t **lon**
- int32_t **alt**
- int16_t **vx**
- int16_t **vy**
- int16_t **vz**
- int16_t **xacc**
- int16_t **yacc**
- int16_t **zacc**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 90
- static constexpr size_t **LENGTH** = 56
- static constexpr size_t **MIN_LENGTH** = 56
- static constexpr uint8_t **CRC_EXTRA** = 183
- static constexpr auto **NAME** = "HIL_STATE"

5.320.1 Detailed Description

[HIL_STATE](#) message.

Sent from simulation to autopilot. This packet is useful for high throughput applications such as hardware in the loop simulations.

5.320.2 Member Function Documentation

5.320.2.1 deserialize()

```
void mavlink::common::msg::HIL_STATE::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.320.2.2 get_name()

```
std::string mavlink::common::msg::HIL_STATE::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.320.2.3 serialize()

```
void mavlink::common::msg::HIL_STATE::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

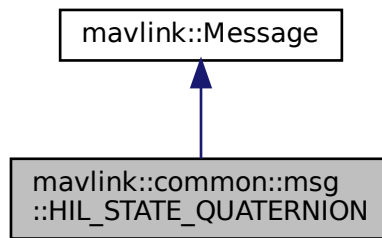
- single_hand_controller/include/mavlink/common/mavlink_msg_hil_state.hpp

5.321 mavlink::common::msg::HIL_STATE_QUATERNION Struct Reference

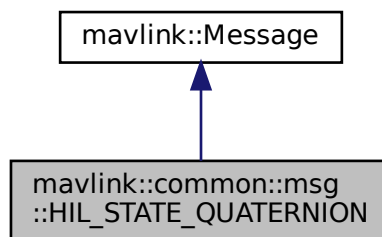
[HIL_STATE_QUATERNION](#) message.

```
#include <mavlink_msg_hil_state_quaternion.hpp>
```

Inheritance diagram for mavlink::common::msg::HIL_STATE_QUATERNION:



Collaboration diagram for mavlink::common::msg::HIL_STATE_QUATERNION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `std::array< float, 4 >` **attitude_quaternion**

- float **rollspeed**
- float **pitchspeed**
- float **yawspeed**
- int32_t **lat**
- int32_t **lon**
- int32_t **alt**
- int16_t **vx**
- int16_t **vy**
- int16_t **vz**
- uint16_t **ind_airspeed**
- uint16_t **true_airspeed**
- int16_t **xacc**
- int16_t **yacc**
- int16_t **zacc**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 115
- static constexpr size_t **LENGTH** = 64
- static constexpr size_t **MIN_LENGTH** = 64
- static constexpr uint8_t **CRC_EXTRA** = 4
- static constexpr auto **NAME** = "HIL_STATE_QUATERNION"

5.321.1 Detailed Description

[HIL_STATE_QUATERNION](#) message.

Sent from simulation to autopilot, avoids in contrast to [HIL_STATE](#) singularities. This packet is useful for high throughput applications such as hardware in the loop simulations.

5.321.2 Member Function Documentation

5.321.2.1 deserialize()

```
void mavlink::common::msg::HIL_STATE_QUATERNION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.321.2.2 `get_name()`

```
std::string mavlink::common::msg::HIL_STATE_QUATERNION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.321.2.3 `serialize()`

```
void mavlink::common::msg::HIL_STATE_QUATERNION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

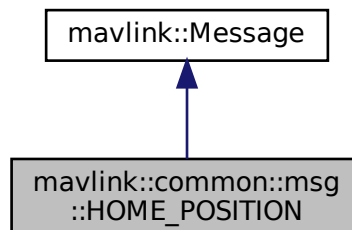
- `single_hand_controller/include/mavlink/common/mavlink_msg_hil_state_quaternion.hpp`

5.322 `mavlink::common::msg::HOME_POSITION` Struct Reference

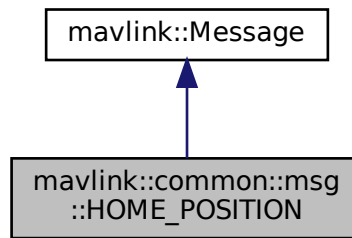
[HOME_POSITION](#) message.

```
#include <mavlink_msg_home_position.hpp>
```

Inheritance diagram for `mavlink::common::msg::HOME_POSITION`:



Collaboration diagram for mavlink::common::msg::HOME_POSITION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `int32_t` **latitude**
- `int32_t` **longitude**
- `int32_t` **altitude**
- `float` **x**
- `float` **y**
- `float` **z**
- `std::array< float, 4 >` **q**
- `float` **approach_x**
- `float` **approach_y**
- `float` **approach_z**
- `uint64_t` **time_usec**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 242
- static constexpr `size_t` **LENGTH** = 60
- static constexpr `size_t` **MIN_LENGTH** = 52
- static constexpr `uint8_t` **CRC_EXTRA** = 104
- static constexpr auto **NAME** = "HOME_POSITION"

5.322.1 Detailed Description

[HOME_POSITION](#) message.

Contains the home position. The home position is the default position that the system will return to and land on. The position must be set automatically by the system during the takeoff, and may also be explicitly set using `MAV_CMD_DO_SET_HOME`. The global and local positions encode the position in the respective coordinate frames, while the q parameter encodes the orientation of the surface. Under normal conditions it describes the heading and terrain slope, which can be used by the aircraft to adjust the approach. The approach 3D vector describes the point to which the system should fly in normal flight mode and then perform a landing sequence along the vector. Note: this message can be requested by sending the `MAV_CMD_REQUEST_MESSAGE` with param1=242 (or the deprecated `MAV_CMD_GET_HOME_POSITION` command).

5.322.2 Member Function Documentation

5.322.2.1 deserialize()

```
void mavlink::common::msg::HOME_POSITION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.322.2.2 get_name()

```
std::string mavlink::common::msg::HOME_POSITION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.322.2.3 serialize()

```
void mavlink::common::msg::HOME_POSITION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

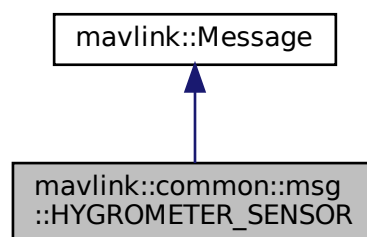
- `single_hand_controller/include/mavlink/common/mavlink_msg_home_position.hpp`

5.323 mavlink::common::msg::HYGROMETER_SENSOR Struct Reference

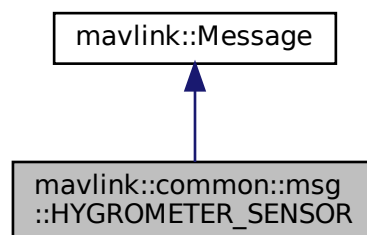
[HYGROMETER_SENSOR](#) message.

```
#include <mavlink_msg_hygrometer_sensor.hpp>
```

Inheritance diagram for mavlink::common::msg::HYGROMETER_SENSOR:



Collaboration diagram for mavlink::common::msg::HYGROMETER_SENSOR:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **id**
- `int16_t` **temperature**
- `uint16_t` **humidity**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 12920
- static constexpr `size_t` **LENGTH** = 5
- static constexpr `size_t` **MIN_LENGTH** = 5
- static constexpr `uint8_t` **CRC_EXTRA** = 20
- static constexpr auto **NAME** = "HYGROMETER_SENSOR"

5.323.1 Detailed Description

[HYGROMETER_SENSOR](#) message.

Temperature and humidity from hygrometer.

5.323.2 Member Function Documentation

5.323.2.1 deserialize()

```
void mavlink::common::msg::HYGROMETER_SENSOR::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.323.2.2 get_name()

```
std::string mavlink::common::msg::HYGROMETER_SENSOR::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.323.2.3 serialize()

```
void mavlink::common::msg::HYGROMETER_SENSOR::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

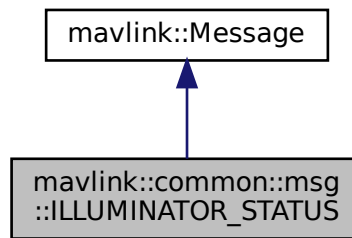
- single_hand_controller/include/mavlink/common/mavlink_msg_hygrometer_sensor.hpp

5.324 mavlink::common::msg::ILLUMINATOR_STATUS Struct Reference

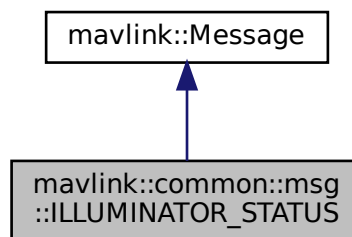
[ILLUMINATOR_STATUS](#) message.

```
#include <mavlink_msg_illuminator_status.hpp>
```

Inheritance diagram for mavlink::common::msg::ILLUMINATOR_STATUS:



Collaboration diagram for mavlink::common::msg::ILLUMINATOR_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **uptime_ms**
- `uint8_t` **enable**

- uint8_t **mode_bitmask**
- uint32_t **error_status**
- uint8_t **mode**
- float **brightness**
- float **strobe_period**
- float **strobe_duty_cycle**
- float **temp_c**
- float **min_strobe_period**
- float **max_strobe_period**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 440
- static constexpr size_t **LENGTH** = 35
- static constexpr size_t **MIN_LENGTH** = 35
- static constexpr uint8_t **CRC_EXTRA** = 66
- static constexpr auto **NAME** = "ILLUMINATOR_STATUS"

5.324.1 Detailed Description

[ILLUMINATOR_STATUS](#) message.

Illuminator status

5.324.2 Member Function Documentation

5.324.2.1 deserialize()

```
void mavlink::common::msg::ILLUMINATOR_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.324.2.2 get_name()

```
std::string mavlink::common::msg::ILLUMINATOR_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.324.2.3 `serialize()`

```
void mavlink::common::msg::ILLUMINATOR_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_illuminator_status.hpp`

5.325 `mavlink::Message::Info` Struct Reference

Public Attributes

- `msgid_t id`
- `size_t length`
- `size_t min_length`
- `uint8_t crc_extra`

The documentation for this struct was generated from the following file:

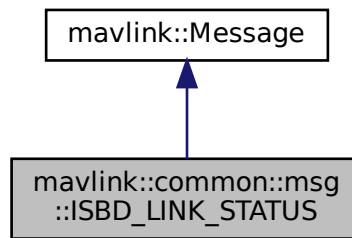
- `single_hand_controller/include/mavlink/message.hpp`

5.326 `mavlink::common::msg::ISBD_LINK_STATUS` Struct Reference

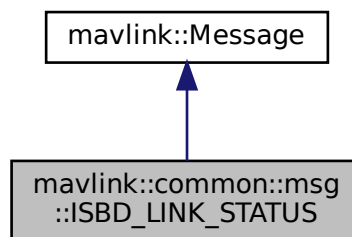
`ISBD_LINK_STATUS` message.

```
#include <mavlink_msg_isbd_link_status.hpp>
```


Inheritance diagram for mavlink::common::msg::ISBD_LINK_STATUS:



Collaboration diagram for mavlink::common::msg::ISBD_LINK_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **timestamp**
- `uint64_t` **last_heartbeat**

- uint16_t **failed_sessions**
- uint16_t **successful_sessions**
- uint8_t **signal_quality**
- uint8_t **ring_pending**
- uint8_t **tx_session_pending**
- uint8_t **rx_session_pending**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 335
- static constexpr size_t **LENGTH** = 24
- static constexpr size_t **MIN_LENGTH** = 24
- static constexpr uint8_t **CRC_EXTRA** = 225
- static constexpr auto **NAME** = "ISBD_LINK_STATUS"

5.326.1 Detailed Description

[ISBD_LINK_STATUS](#) message.

Status of the Iridium SBD link.

5.326.2 Member Function Documentation

5.326.2.1 deserialize()

```
void mavlink::common::msg::ISBD_LINK_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.326.2.2 get_name()

```
std::string mavlink::common::msg::ISBD_LINK_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.326.2.3 serialize()

```
void mavlink::common::msg::ISBD_LINK_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

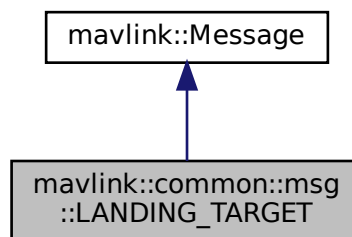
- single_hand_controller/include/mavlink/common/mavlink_msg_isbd_link_status.hpp

5.327 mavlink::common::msg::LANDING_TARGET Struct Reference

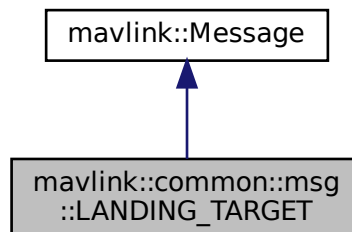
[LANDING_TARGET](#) message.

```
#include <mavlink_msg_landing_target.hpp>
```

Inheritance diagram for mavlink::common::msg::LANDING_TARGET:



Collaboration diagram for mavlink::common::msg::LANDING_TARGET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **target_num**
- `uint8_t` **frame**
- `float` **angle_x**
- `float` **angle_y**
- `float` **distance**
- `float` **size_x**
- `float` **size_y**
- `float` **x**
- `float` **y**
- `float` **z**
- `std::array< float, 4 >` **q**
- `uint8_t` **type**
- `uint8_t` **position_valid**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 149
- static constexpr `size_t` **LENGTH** = 60
- static constexpr `size_t` **MIN_LENGTH** = 30
- static constexpr `uint8_t` **CRC_EXTRA** = 200
- static constexpr auto **NAME** = "LANDING_TARGET"

5.327.1 Detailed Description

[LANDING_TARGET](#) message.

The location of a landing target. See: https://mavlink.io/en/services/landing_target.html

5.327.2 Member Function Documentation

5.327.2.1 `deserialize()`

```
void mavlink::common::msg::LANDING_TARGET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.327.2.2 get_name()

```
std::string mavlink::common::msg::LANDING_TARGET::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.327.2.3 serialize()

```
void mavlink::common::msg::LANDING_TARGET::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

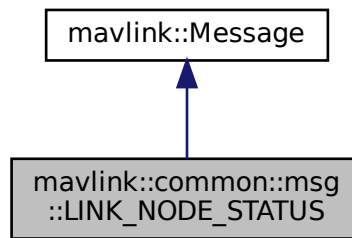
- single_hand_controller/include/mavlink/common/mavlink_msg_landing_target.hpp

5.328 mavlink::common::msg::LINK_NODE_STATUS Struct Reference

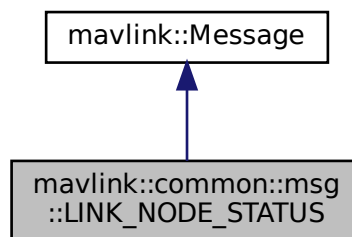
[LINK_NODE_STATUS](#) message.

```
#include <mavlink_msg_link_node_status.hpp>
```

Inheritance diagram for mavlink::common::msg::LINK_NODE_STATUS:



Collaboration diagram for mavlink::common::msg::LINK_NODE_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **timestamp**
- `uint8_t` **tx_buf**

- uint8_t **rx_buf**
- uint32_t **tx_rate**
- uint32_t **rx_rate**
- uint16_t **rx_parse_err**
- uint16_t **tx_overflows**
- uint16_t **rx_overflows**
- uint32_t **messages_sent**
- uint32_t **messages_received**
- uint32_t **messages_lost**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 8
- static constexpr size_t **LENGTH** = 36
- static constexpr size_t **MIN_LENGTH** = 36
- static constexpr uint8_t **CRC_EXTRA** = 117
- static constexpr auto **NAME** = "LINK_NODE_STATUS"

5.328.1 Detailed Description

[LINK_NODE_STATUS](#) message.

Status generated in each node in the communication chain and injected into MAVLink stream.

5.328.2 Member Function Documentation

5.328.2.1 deserialize()

```
void mavlink::common::msg::LINK_NODE_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.328.2.2 get_name()

```
std::string mavlink::common::msg::LINK_NODE_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.328.2.3 `serialize()`

```
void mavlink::common::msg::LINK_NODE_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

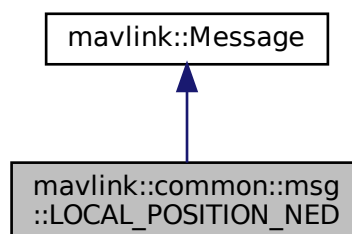
- `single_hand_controller/include/mavlink/common/mavlink_msg_link_node_status.hpp`

5.329 `mavlink::common::msg::LOCAL_POSITION_NED` Struct Reference

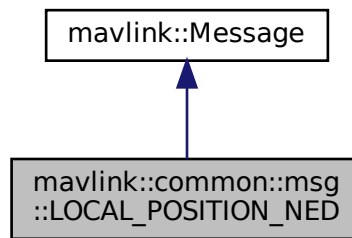
[LOCAL_POSITION_NED](#) message.

```
#include <mavlink_msg_local_position_ned.hpp>
```

Inheritance diagram for `mavlink::common::msg::LOCAL_POSITION_NED`:



Collaboration diagram for mavlink::common::msg::LOCAL_POSITION_NED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **x**
- `float` **y**
- `float` **z**
- `float` **vx**
- `float` **vy**
- `float` **vz**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 32
- static constexpr size_t **LENGTH** = 28
- static constexpr size_t **MIN_LENGTH** = 28
- static constexpr uint8_t **CRC_EXTRA** = 185
- static constexpr auto **NAME** = "LOCAL_POSITION_NED"

5.329.1 Detailed Description

[LOCAL_POSITION_NED](#) message.

The filtered local position (e.g. fused computer vision and accelerometers). Coordinate frame is right-handed, Z-axis down (aeronautical frame, NED / north-east-down convention)

5.329.2 Member Function Documentation

5.329.2.1 deserialize()

```
void mavlink::common::msg::LOCAL_POSITION_NED::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.329.2.2 get_name()

```
std::string mavlink::common::msg::LOCAL_POSITION_NED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.329.2.3 serialize()

```
void mavlink::common::msg::LOCAL_POSITION_NED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

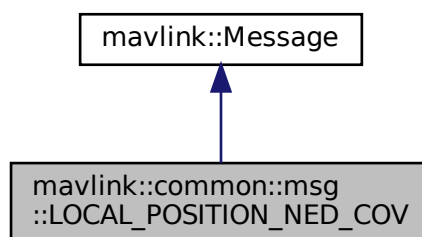
- single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned.hpp

5.330 mavlink::common::msg::LOCAL_POSITION_NED_COV Struct Reference

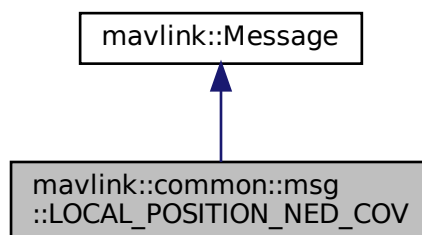
[LOCAL_POSITION_NED_COV](#) message.

```
#include <mavlink_msg_local_position_ned_cov.hpp>
```

Inheritance diagram for mavlink::common::msg::LOCAL_POSITION_NED_COV:



Collaboration diagram for mavlink::common::msg::LOCAL_POSITION_NED_COV:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **estimator_type**
- `float` **x**
- `float` **y**
- `float` **z**
- `float` **vx**
- `float` **vy**
- `float` **vz**
- `float` **ax**
- `float` **ay**
- `float` **az**
- `std::array< float, 45 >` **covariance**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 64
- static constexpr size_t **LENGTH** = 225
- static constexpr size_t **MIN_LENGTH** = 225
- static constexpr uint8_t **CRC_EXTRA** = 191
- static constexpr auto **NAME** = "LOCAL_POSITION_NED_COV"

5.330.1 Detailed Description

[LOCAL_POSITION_NED_COV](#) message.

The filtered local position (e.g. fused computer vision and accelerometers). Coordinate frame is right-handed, Z-axis down (aeronautical frame, NED / north-east-down convention)

5.330.2 Member Function Documentation

5.330.2.1 deserialize()

```
void mavlink::common::msg::LOCAL_POSITION_NED_COV::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.330.2.2 get_name()

```
std::string mavlink::common::msg::LOCAL_POSITION_NED_COV::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.330.2.3 serialize()

```
void mavlink::common::msg::LOCAL_POSITION_NED_COV::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

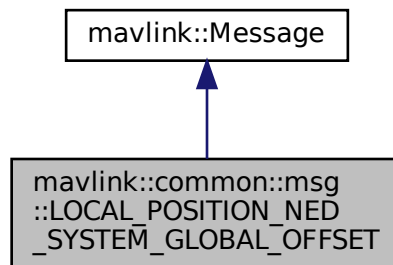
- single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_cov.hpp

**5.331 mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_↵
GLOBAL_OFFSET Struct Reference**

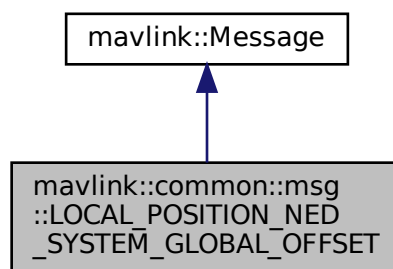
[LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET](#) message.

```
#include <mavlink_msg_local_position_ned_system_global_offset.hpp>
```

Inheritance diagram for mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET:



Collaboration diagram for mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- float **x**
- float **y**
- float **z**
- float **roll**
- float **pitch**
- float **yaw**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 89
- static constexpr size_t **LENGTH** = 28
- static constexpr size_t **MIN_LENGTH** = 28
- static constexpr uint8_t **CRC_EXTRA** = 231
- static constexpr auto **NAME** = "LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET"

5.331.1 Detailed Description

[LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET](#) message.

The offset in X, Y, Z and yaw between the [LOCAL_POSITION_NED](#) messages of MAV X and the global coordinate frame in NED coordinates. Coordinate frame is right-handed, Z-axis down (aeronautical frame, NED / north-east-down convention)

5.331.2 Member Function Documentation

5.331.2.1 deserialize()

```
void mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.331.2.2 get_name()

```
std::string mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.331.2.3 `serialize()`

```
void mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

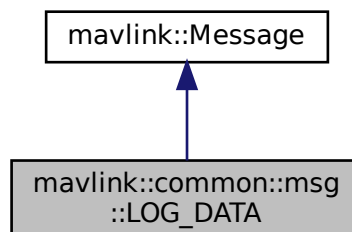
- `single_hand_controller/include/mavlink/common/mavlink_msg_local_position_ned_system_global_offset.↔hpp`

5.332 `mavlink::common::msg::LOG_DATA` Struct Reference

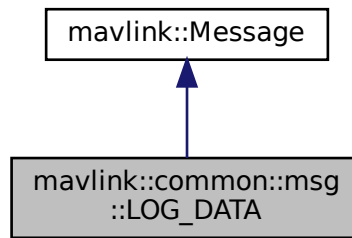
[LOG_DATA](#) message.

```
#include <mavlink_msg_log_data.hpp>
```

Inheritance diagram for `mavlink::common::msg::LOG_DATA`:



Collaboration diagram for mavlink::common::msg::LOG_DATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint16_t` **id**
- `uint32_t` **ofs**
- `uint8_t` **count**
- `std::array< uint8_t, 90 >` **data**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 120
- static constexpr `size_t` **LENGTH** = 97
- static constexpr `size_t` **MIN_LENGTH** = 97
- static constexpr `uint8_t` **CRC_EXTRA** = 134
- static constexpr auto **NAME** = "LOG_DATA"

5.332.1 Detailed Description

[LOG_DATA](#) message.

Reply to [LOG_REQUEST_DATA](#)

5.332.2 Member Function Documentation

5.332.2.1 deserialize()

```
void mavlink::common::msg::LOG_DATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.332.2.2 get_name()

```
std::string mavlink::common::msg::LOG_DATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.332.2.3 serialize()

```
void mavlink::common::msg::LOG_DATA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

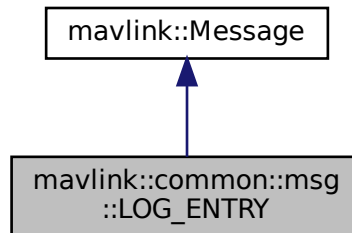
- `single_hand_controller/include/mavlink/common/mavlink_msg_log_data.hpp`

5.333 mavlink::common::msg::LOG_ENTRY Struct Reference

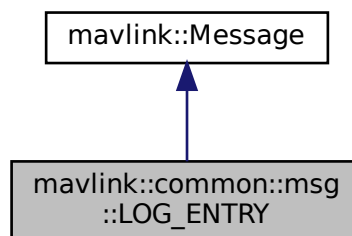
[LOG_ENTRY](#) message.

```
#include <mavlink_msg_log_entry.hpp>
```

Inheritance diagram for mavlink::common::msg::LOG_ENTRY:



Collaboration diagram for mavlink::common::msg::LOG_ENTRY:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint16_t **id**
- uint16_t **num_logs**
- uint16_t **last_log_num**
- uint32_t **time_utc**
- uint32_t **size**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 118
- static constexpr size_t **LENGTH** = 14
- static constexpr size_t **MIN_LENGTH** = 14
- static constexpr uint8_t **CRC_EXTRA** = 56
- static constexpr auto **NAME** = "LOG_ENTRY"

5.333.1 Detailed Description

[LOG_ENTRY](#) message.

Reply to [LOG_REQUEST_LIST](#)

5.333.2 Member Function Documentation

5.333.2.1 deserialize()

```
void mavlink::common::msg::LOG_ENTRY::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.333.2.2 get_name()

```
std::string mavlink::common::msg::LOG_ENTRY::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.333.2.3 serialize()

```
void mavlink::common::msg::LOG_ENTRY::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

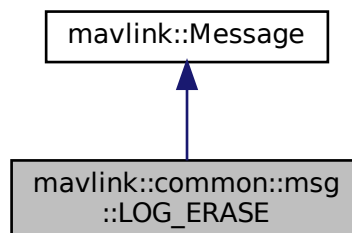
- single_hand_controller/include/mavlink/common/mavlink_msg_log_entry.hpp

5.334 mavlink::common::msg::LOG_ERASE Struct Reference

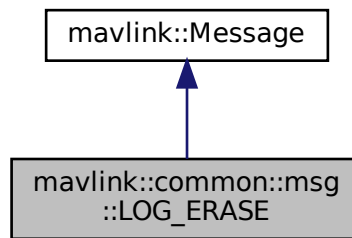
[LOG_ERASE](#) message.

```
#include <mavlink_msg_log_erase.hpp>
```

Inheritance diagram for mavlink::common::msg::LOG_ERASE:



Collaboration diagram for mavlink::common::msg::LOG_ERASE:



Public Member Functions

- `std::string get_name (void) const` override
Get NAME constant.
- `Info get_message_info (void) const` override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map) override`
Deserialize message.

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 121`
- `static constexpr size_t LENGTH = 2`
- `static constexpr size_t MIN_LENGTH = 2`
- `static constexpr uint8_t CRC_EXTRA = 237`
- `static constexpr auto NAME = "LOG_ERASE"`

5.334.1 Detailed Description

`LOG_ERASE` message.

Erase all logs

5.334.2 Member Function Documentation

5.334.2.1 deserialize()

```
void mavlink::common::msg::LOG_ERASE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.334.2.2 get_name()

```
std::string mavlink::common::msg::LOG_ERASE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.334.2.3 serialize()

```
void mavlink::common::msg::LOG_ERASE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

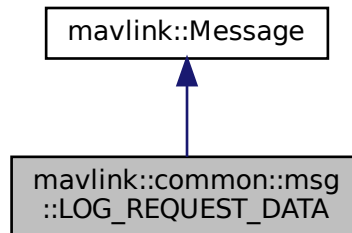
- single_hand_controller/include/mavlink/common/mavlink_msg_log_erase.hpp

5.335 mavlink::common::msg::LOG_REQUEST_DATA Struct Reference

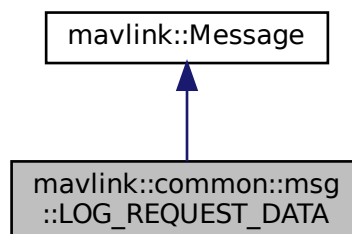
[LOG_REQUEST_DATA](#) message.

```
#include <mavlink_msg_log_request_data.hpp>
```

Inheritance diagram for mavlink::common::msg::LOG_REQUEST_DATA:



Collaboration diagram for mavlink::common::msg::LOG_REQUEST_DATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint16_t **id**
- uint32_t **ofs**
- uint32_t **count**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 119
- static constexpr size_t **LENGTH** = 12
- static constexpr size_t **MIN_LENGTH** = 12
- static constexpr uint8_t **CRC_EXTRA** = 116
- static constexpr auto **NAME** = "LOG_REQUEST_DATA"

5.335.1 Detailed Description

[LOG_REQUEST_DATA](#) message.

Request a chunk of a log

5.335.2 Member Function Documentation

5.335.2.1 deserialize()

```
void mavlink::common::msg::LOG_REQUEST_DATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.335.2.2 get_name()

```
std::string mavlink::common::msg::LOG_REQUEST_DATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.335.2.3 `serialize()`

```
void mavlink::common::msg::LOG_REQUEST_DATA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

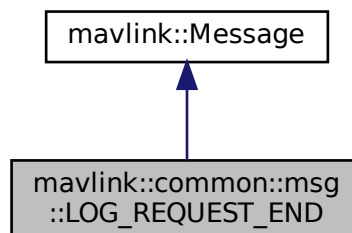
- `single_hand_controller/include/mavlink/common/mavlink_msg_log_request_data.hpp`

5.336 `mavlink::common::msg::LOG_REQUEST_END` Struct Reference

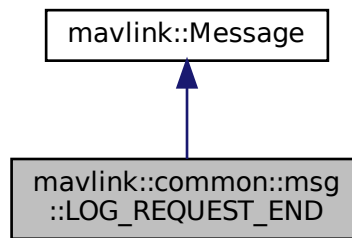
`LOG_REQUEST_END` message.

```
#include <mavlink_msg_log_request_end.hpp>
```

Inheritance diagram for `mavlink::common::msg::LOG_REQUEST_END`:



Collaboration diagram for mavlink::common::msg::LOG_REQUEST_END:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 122
- static constexpr size_t **LENGTH** = 2
- static constexpr size_t **MIN_LENGTH** = 2
- static constexpr uint8_t **CRC_EXTRA** = 203
- static constexpr auto **NAME** = "LOG_REQUEST_END"

5.336.1 Detailed Description

[LOG_REQUEST_END](#) message.

Stop log transfer and resume normal logging

5.336.2 Member Function Documentation

5.336.2.1 deserialize()

```
void mavlink::common::msg::LOG_REQUEST_END::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.336.2.2 get_name()

```
std::string mavlink::common::msg::LOG_REQUEST_END::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.336.2.3 serialize()

```
void mavlink::common::msg::LOG_REQUEST_END::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

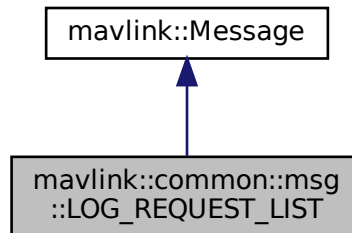
- `single_hand_controller/include/mavlink/common/mavlink_msg_log_request_end.hpp`

5.337 mavlink::common::msg::LOG_REQUEST_LIST Struct Reference

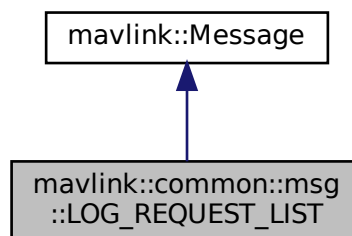
[LOG_REQUEST_LIST](#) message.

```
#include <mavlink_msg_log_request_list.hpp>
```

Inheritance diagram for mavlink::common::msg::LOG_REQUEST_LIST:



Collaboration diagram for mavlink::common::msg::LOG_REQUEST_LIST:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint16_t **start**
- uint16_t **end**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 117
- static constexpr size_t **LENGTH** = 6
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 128
- static constexpr auto **NAME** = "LOG_REQUEST_LIST"

5.337.1 Detailed Description

[LOG_REQUEST_LIST](#) message.

Request a list of available logs. On some systems calling this may stop on-board logging until [LOG_REQUEST_END](#) is called. If there are no log files available this request shall be answered with one [LOG_ENTRY](#) message with id = 0 and num_logs = 0. [LOG_ENTRY](#) messages can start with id 1 or 0. The ground station needs to be able to process either.

5.337.2 Member Function Documentation

5.337.2.1 deserialize()

```
void mavlink::common::msg::LOG_REQUEST_LIST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.337.2.2 get_name()

```
std::string mavlink::common::msg::LOG_REQUEST_LIST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.337.2.3 serialize()

```
void mavlink::common::msg::LOG_REQUEST_LIST::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

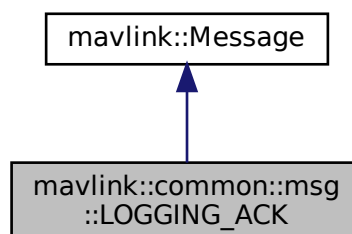
- single_hand_controller/include/mavlink/common/mavlink_msg_log_request_list.hpp

5.338 mavlink::common::msg::LOGGING_ACK Struct Reference

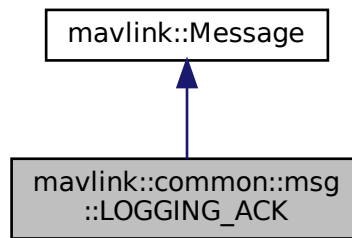
[LOGGING_ACK](#) message.

```
#include <mavlink_msg_logging_ack.hpp>
```

Inheritance diagram for mavlink::common::msg::LOGGING_ACK:



Collaboration diagram for mavlink::common::msg::LOGGING_ACK:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint16_t` **sequence**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 268
- static constexpr size_t **LENGTH** = 4
- static constexpr size_t **MIN_LENGTH** = 4
- static constexpr uint8_t **CRC_EXTRA** = 14
- static constexpr auto **NAME** = "LOGGING_ACK"

5.338.1 Detailed Description

[LOGGING_ACK](#) message.

An ack for a [LOGGING_DATA_ACKED](#) message

5.338.2 Member Function Documentation

5.338.2.1 deserialize()

```
void mavlink::common::msg::LOGGING_ACK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.338.2.2 get_name()

```
std::string mavlink::common::msg::LOGGING_ACK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.338.2.3 serialize()

```
void mavlink::common::msg::LOGGING_ACK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

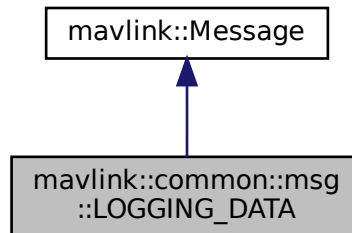
- `single_hand_controller/include/mavlink/common/mavlink_msg_logging_ack.hpp`

5.339 mavlink::common::msg::LOGGING_DATA Struct Reference

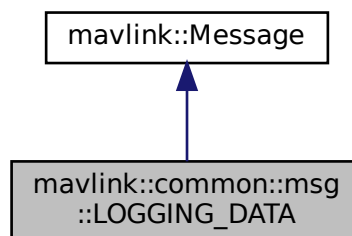
[LOGGING_DATA](#) message.

```
#include <mavlink_msg_logging_data.hpp>
```

Inheritance diagram for mavlink::common::msg::LOGGING_DATA:



Collaboration diagram for mavlink::common::msg::LOGGING_DATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint16_t **sequence**
- uint8_t **length**
- uint8_t **first_message_offset**
- std::array< uint8_t, 249 > **data**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 266
- static constexpr size_t **LENGTH** = 255
- static constexpr size_t **MIN_LENGTH** = 255
- static constexpr uint8_t **CRC_EXTRA** = 193
- static constexpr auto **NAME** = "LOGGING_DATA"

5.339.1 Detailed Description

[LOGGING_DATA](#) message.

A message containing logged data (see also MAV_CMD_LOGGING_START)

5.339.2 Member Function Documentation

5.339.2.1 deserialize()

```
void mavlink::common::msg::LOGGING_DATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.339.2.2 get_name()

```
std::string mavlink::common::msg::LOGGING_DATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.339.2.3 `serialize()`

```
void mavlink::common::msg::LOGGING_DATA::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

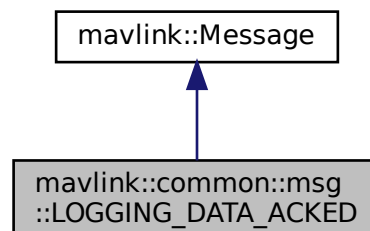
- `single_hand_controller/include/mavlink/common/mavlink_msg_logging_data.hpp`

5.340 `mavlink::common::msg::LOGGING_DATA_ACKED` Struct Reference

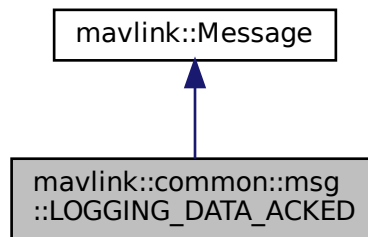
[LOGGING_DATA_ACKED](#) message.

```
#include <mavlink_msg_logging_data_acked.hpp>
```

Inheritance diagram for `mavlink::common::msg::LOGGING_DATA_ACKED`:



Collaboration diagram for mavlink::common::msg::LOGGING_DATA_ACKED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint16_t` **sequence**
- `uint8_t` **length**
- `uint8_t` **first_message_offset**
- `std::array< uint8_t, 249 >` **data**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 267
- static constexpr `size_t` **LENGTH** = 255
- static constexpr `size_t` **MIN_LENGTH** = 255
- static constexpr `uint8_t` **CRC_EXTRA** = 35
- static constexpr auto **NAME** = "LOGGING_DATA_ACKED"

5.340.1 Detailed Description

[LOGGING_DATA_ACKED](#) message.

A message containing logged data which requires a [LOGGING_ACK](#) to be sent back

5.340.2 Member Function Documentation

5.340.2.1 deserialize()

```
void mavlink::common::msg::LOGGING_DATA_ACKED::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.340.2.2 get_name()

```
std::string mavlink::common::msg::LOGGING_DATA_ACKED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.340.2.3 serialize()

```
void mavlink::common::msg::LOGGING_DATA_ACKED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

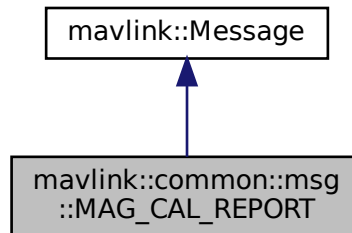
- `single_hand_controller/include/mavlink/common/mavlink_msg_logging_data_acked.hpp`

5.341 mavlink::common::msg::MAG_CAL_REPORT Struct Reference

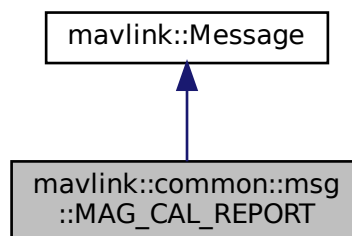
[MAG_CAL_REPORT](#) message.

```
#include <mavlink_msg_mag_cal_report.hpp>
```

Inheritance diagram for mavlink::common::msg::MAG_CAL_REPORT:



Collaboration diagram for mavlink::common::msg::MAG_CAL_REPORT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **compass_id**
- uint8_t **cal_mask**
- uint8_t **cal_status**
- uint8_t **autosaved**
- float **fitness**
- float **ofs_x**
- float **ofs_y**
- float **ofs_z**
- float **diag_x**
- float **diag_y**
- float **diag_z**
- float **offdiag_x**
- float **offdiag_y**
- float **offdiag_z**
- float **orientation_confidence**
- uint8_t **old_orientation**
- uint8_t **new_orientation**
- float **scale_factor**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 192
- static constexpr size_t **LENGTH** = 54
- static constexpr size_t **MIN_LENGTH** = 44
- static constexpr uint8_t **CRC_EXTRA** = 36
- static constexpr auto **NAME** = "MAG_CAL_REPORT"

5.341.1 Detailed Description

[MAG_CAL_REPORT](#) message.

Reports results of completed compass calibration. Sent until MAG_CAL_ACK received.

5.341.2 Member Function Documentation

5.341.2.1 deserialize()

```
void mavlink::common::msg::MAG_CAL_REPORT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.341.2.2 get_name()

```
std::string mavlink::common::msg::MAG_CAL_REPORT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.341.2.3 serialize()

```
void mavlink::common::msg::MAG_CAL_REPORT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

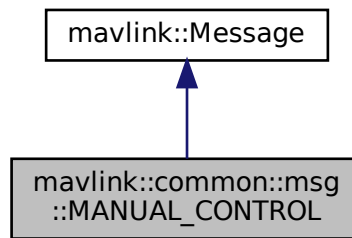
- single_hand_controller/include/mavlink/common/mavlink_msg_mag_cal_report.hpp

5.342 mavlink::common::msg::MANUAL_CONTROL Struct Reference

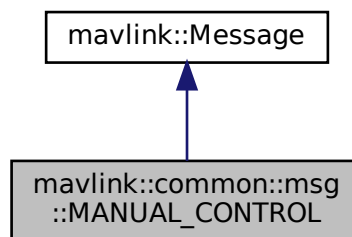
[MANUAL_CONTROL](#) message.

```
#include <mavlink_msg_manual_control.hpp>
```

Inheritance diagram for mavlink::common::msg::MANUAL_CONTROL:



Collaboration diagram for mavlink::common::msg::MANUAL_CONTROL:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` [target](#)
- `int16_t` [x](#)

- int16_t **y**
- int16_t **z**
- int16_t **r**
- uint16_t **buttons**
- uint16_t **buttons2**
- uint8_t **enabled_extensions**
- int16_t **s**
- int16_t **t**
- int16_t **aux1**
- int16_t **aux2**
- int16_t **aux3**
- int16_t **aux4**
- int16_t **aux5**
- int16_t **aux6**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 69
- static constexpr size_t **LENGTH** = 30
- static constexpr size_t **MIN_LENGTH** = 11
- static constexpr uint8_t **CRC_EXTRA** = 243
- static constexpr auto **NAME** = "MANUAL_CONTROL"

5.342.1 Detailed Description

[MANUAL_CONTROL](#) message.

Manual (joystick) control message. This message represents movement axes and button using standard joystick axes nomenclature. Unused axes can be disabled and buttons states are transmitted as individual on/off bits of a bitmask. For more information see https://mavlink.io/en/manual_control.html

5.342.2 Member Function Documentation

5.342.2.1 deserialize()

```
void mavlink::common::msg::MANUAL_CONTROL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.342.2.2 `get_name()`

```
std::string mavlink::common::msg::MANUAL_CONTROL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.342.2.3 `serialize()`

```
void mavlink::common::msg::MANUAL_CONTROL::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

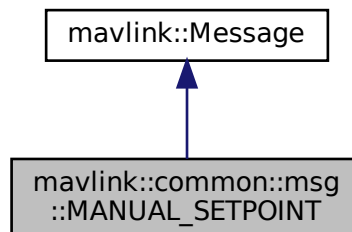
- `single_hand_controller/include/mavlink/common/mavlink_msg_manual_control.hpp`

5.343 `mavlink::common::msg::MANUAL_SETPOINT` Struct Reference

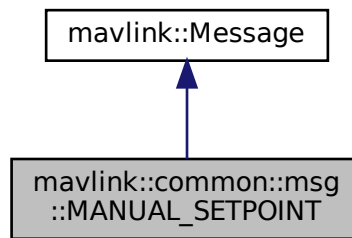
[MANUAL_SETPOINT](#) message.

```
#include <mavlink_msg_manual_setpoint.hpp>
```

Inheritance diagram for `mavlink::common::msg::MANUAL_SETPOINT`:



Collaboration diagram for mavlink::common::msg::MANUAL_SETPOINT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **roll**
- `float` **pitch**
- `float` **yaw**
- `float` **thrust**
- `uint8_t` **mode_switch**
- `uint8_t` **manual_override_switch**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 81
- static constexpr `size_t` **LENGTH** = 22
- static constexpr `size_t` **MIN_LENGTH** = 22
- static constexpr `uint8_t` **CRC_EXTRA** = 106
- static constexpr auto **NAME** = "MANUAL_SETPOINT"

5.343.1 Detailed Description

[MANUAL_SETPOINT](#) message.

Setpoint in roll, pitch, yaw and thrust from the operator

5.343.2 Member Function Documentation

5.343.2.1 deserialize()

```
void mavlink::common::msg::MANUAL_SETPOINT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.343.2.2 get_name()

```
std::string mavlink::common::msg::MANUAL_SETPOINT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.343.2.3 serialize()

```
void mavlink::common::msg::MANUAL_SETPOINT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_manual_setpoint.hpp`

5.344 mavlink_sha256_ctx Struct Reference

Public Attributes

- `uint32_t sz` [2]
- `uint32_t counter` [8]
- - union {
 - unsigned char **save_bytes** [64]
 - uint32_t **save_u32** [16]
- `u`

The documentation for this struct was generated from the following file:

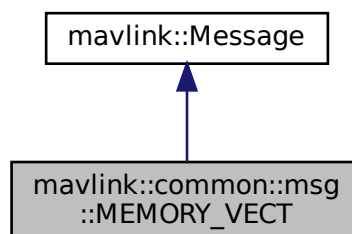
- `single_hand_controller/include/mavlink/mavlink_sha256.h`

5.345 mavlink::common::msg::MEMORY_VECT Struct Reference

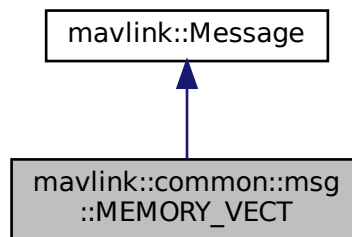
[MEMORY_VECT](#) message.

```
#include <mavlink_msg_memory_vect.hpp>
```

Inheritance diagram for `mavlink::common::msg::MEMORY_VECT`:



Collaboration diagram for mavlink::common::msg::MEMORY_VECT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint16_t` **address**
- `uint8_t` **ver**
- `uint8_t` **type**
- `std::array< int8_t, 32 >` **value**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 249
- static constexpr size_t **LENGTH** = 36
- static constexpr size_t **MIN_LENGTH** = 36
- static constexpr uint8_t **CRC_EXTRA** = 204
- static constexpr auto **NAME** = "MEMORY_VECT"

5.345.1 Detailed Description

[MEMORY_VECT](#) message.

Send raw controller memory. The use of this message is discouraged for normal packets, but a quite efficient way for testing new messages and getting experimental debug output.

5.345.2 Member Function Documentation

5.345.2.1 deserialize()

```
void mavlink::common::msg::MEMORY_VECT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.345.2.2 get_name()

```
std::string mavlink::common::msg::MEMORY_VECT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.345.2.3 serialize()

```
void mavlink::common::msg::MEMORY_VECT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_memory_vect.hpp`

5.346 mavlink::Message Struct Reference

MAVLink [Message](#) base class.

```
#include <message.hpp>
```

Inherited by [mavlink::common::msg::ACTUATOR_CONTROL_TARGET](#), [mavlink::common::msg::ACTUATOR_OUTPUT_STATUS](#), [mavlink::common::msg::ADSB_VEHICLE](#), [mavlink::common::msg::AIRSPEED](#), [mavlink::common::msg::AIS_VESSEL](#), [mavlink::common::msg::ALTITUDE](#), [mavlink::common::msg::ATTITUDE](#), [mavlink::common::msg::ATTITUDE_QUATERNION](#), [mavlink::common::msg::ATTITUDE_QUATERNION_COV](#), [mavlink::common::msg::ATTITUDE_TARGET](#), [mavlink::common::msg::ATTITUDE_TARGET_COV](#), [mavlink::common::msg::AUTH_KEY](#), [mavlink::common::msg::AUTOPILOT_STATE_FOR_GIMBAL_DEVICE](#), [mavlink::common::msg::AVAILABLE_MODES](#), [mavlink::common::msg::AVAILABLE_MODES_MONITOR](#), [mavlink::common::msg::BATTERY_STATUS](#), [mavlink::common::msg::BUTTON_CHANGE](#), [mavlink::common::msg::CAMERA_CAPTURED](#), [mavlink::common::msg::CAMERA_FOV_STATUS](#), [mavlink::common::msg::CAMERA_IMAGE_CAPTURED](#), [mavlink::common::msg::CAMERA_INFORMATION](#), [mavlink::common::msg::CAMERA_SETTINGS](#), [mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS](#), [mavlink::common::msg::CAMERA_TRACKING_IMAGE_STATUS](#), [mavlink::common::msg::CAMERA_TRIGGER](#), [mavlink::common::msg::CANFD_FRAME](#), [mavlink::common::msg::CAN_FILTER_MODE](#), [mavlink::common::msg::CAN_FRAME](#), [mavlink::common::msg::CELLULAR_CONFIG](#), [mavlink::common::msg::CELLULAR_STATUS](#), [mavlink::common::msg::CHANGE_OPERATOR_CONTROL](#), [mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK](#), [mavlink::common::msg::COLLISION](#), [mavlink::common::msg::COMMAND_ACK](#), [mavlink::common::msg::COMMAND_CANCEL](#), [mavlink::common::msg::COMMAND_INT](#), [mavlink::common::msg::COMMAND_LONG](#), [mavlink::common::msg::COMPONENT_INFORMATION_BASIC](#), [mavlink::common::msg::COMPONENT_METADATA](#), [mavlink::common::msg::CONTROL_SYSTEM_STATE](#), [mavlink::common::msg::CURRENT_EVENT_SEQUENCE](#), [mavlink::common::msg::CURRENT_MODE](#), [mavlink::common::msg::DATA_STREAM](#), [mavlink::common::msg::DATA_TRANSMISSION_INTERVAL](#), [mavlink::common::msg::DEBUG](#), [mavlink::common::msg::DEBUG_FLOAT_ARRAY](#), [mavlink::common::msg::DEBUG_VECT](#), [mavlink::common::msg::DISTANCE_SENSOR](#), [mavlink::common::msg::EFI_STATUS](#), [mavlink::common::msg::ENCAPSULATED_DATA](#), [mavlink::common::msg::ESC_INFO](#), [mavlink::common::msg::ESC_STATUS](#), [mavlink::common::msg::ESTIMATOR_STATUS](#), [mavlink::common::msg::EVENT](#), [mavlink::common::msg::EXTENDED_SYS_STATE](#), [mavlink::common::msg::FENCE_STATUS](#), [mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS](#), [mavlink::common::msg::FILE_TRANSFER_PROTOCOL](#), [mavlink::common::msg::FLIGHT_INFORMATION](#), [mavlink::common::msg::FOLLOW_TARGET](#), [mavlink::common::msg::FUEL_STATUS](#), [mavlink::common::msg::GENERATOR_STATUS](#), [mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS](#), [mavlink::common::msg::GIMBAL_DEVICE_INFORMATION](#), [mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE](#), [mavlink::common::msg::GIMBAL_MANAGER_INFORMATION](#), [mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE](#), [mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL](#), [mavlink::common::msg::GIMBAL_MANAGER_SET_PITCH_RATE](#), [mavlink::common::msg::GIMBAL_MANAGER_STATUS](#), [mavlink::common::msg::GLOBAL_POSITION_INT_COV](#), [mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE](#), [mavlink::common::msg::GPS2_RAW](#), [mavlink::common::msg::GPS_GLOBAL_ORIGIN](#), [mavlink::common::msg::GPS_INJECT_DATA](#), [mavlink::common::msg::GPS_INPUT_DATA](#), [mavlink::common::msg::GPS_RAW_INT](#), [mavlink::common::msg::GPS_RTCM_DATA](#), [mavlink::common::msg::GPS_RTK](#), [mavlink::common::msg::GPS_STATUS](#), [mavlink::common::msg::HIGHRES_IMU](#), [mavlink::common::msg::HIGH_LATENCY](#), [mavlink::common::msg::HIGH_LATENCY2](#), [mavlink::common::msg::HIL_ACTUATOR_CONTROLS](#), [mavlink::common::msg::HIL_CONFIG](#), [mavlink::common::msg::HIL_GPS](#), [mavlink::common::msg::HIL_OPTICAL_FLOW](#), [mavlink::common::msg::HIL_RC_INPUTS_RAW](#), [mavlink::common::msg::HIL_SENSOR](#), [mavlink::common::msg::HIL_STATE](#), [mavlink::common::msg::HIL_STATE_QUATERNION](#), [mavlink::common::msg::HOME_POSITION](#), [mavlink::common::msg::HYGROMETER_SENSOR](#), [mavlink::common::msg::ILLUMINATE](#), [mavlink::common::msg::ISBD_LINK_STATUS](#), [mavlink::common::msg::LANDING_TARGET](#), [mavlink::common::msg::LINK_NODE_STATUS](#), [mavlink::common::msg::LOCAL_POSITION_NED](#), [mavlink::common::msg::LOCAL_POSITION_NED_COV](#), [mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET](#), [mavlink::common::msg::LOGGING_ACK](#), [mavlink::common::msg::LOGGING_DATA](#), [mavlink::common::msg::LOGGING_DATA_ACKED](#), [mavlink::common::msg::LOG_DATA](#), [mavlink::common::msg::LOG_ENTRY](#), [mavlink::common::msg::LOG_ERASE](#), [mavlink::common::msg::LOG_REQUEST_DATA](#), [mavlink::common::msg::LOG_REQUEST_END](#), [mavlink::common::msg::LOG_REQUEST_LIST](#), [mavlink::common::msg::MAG_CALIBRATION](#), [mavlink::common::msg::MANUAL_CONTROL](#), [mavlink::common::msg::MANUAL_SETPOINT](#), [mavlink::common::msg::MEMORY_USAGE](#), [mavlink::common::msg::MESSAGE_INTERVAL](#), [mavlink::common::msg::MISSION_ACK](#), [mavlink::common::msg::MISSION_CLEAR](#), [mavlink::common::msg::MISSION_COUNT](#), [mavlink::common::msg::MISSION_CURRENT](#), [mavlink::common::msg::MISSION_ITEM](#), [mavlink::common::msg::MISSION_ITEM_INT](#), [mavlink::common::msg::MISSION_ITEM_REACHED](#), [mavlink::common::msg::MISSION_REQUEST](#), [mavlink::common::msg::MISSION_REQUEST_INT](#), [mavlink::common::msg::MISSION_REQUEST_LIST](#), [mavlink::common::msg::MISSION_REQUEST_LIST_ACK](#), [mavlink::common::msg::MISSION_SET_CURRENT](#), [mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST](#), [mavlink::common::msg::MOUNT_ORIENTATION](#), [mavlink::common::msg::NAMED_VALUE_FLOAT](#), [mavlink::common::msg::NAMED_VALUE_INT](#), [mavlink::common::msg::NAV_CONTROLLER_OUTPUT](#), [mavlink::common::msg::OBSTACLE_DISTANCE](#), [mavlink::common::msg::ODOMETRY](#), [mavlink::common::msg::ONBOARD_COMPUTER_STATUS](#), [mavlink::common::msg::OPEN_D](#)

mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION, mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID, mavlink::common::msg::OPEN_DRONE_ID_LOCATION, mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK, mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID, mavlink::common::msg::OPEN_DRONE_ID_SELF_ID, mavlink::common::msg::OPEN_DRONE_ID_SYSTEM, mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE, mavlink::common::msg::OPTICAL_FLOW, mavlink::common::msg::OPTICAL_FLOW_RAD, mavlink::common::msg::ORBIT_EXECUTION, mavlink::common::msg::PARAM_ERROR, mavlink::common::msg::PARAM_EXT_ACK, mavlink::common::msg::PARAM_EXT_REQUEST_READ, mavlink::common::msg::PARAM_EXT_REQUEST_WRITE, mavlink::common::msg::PARAM_EXT_SET, mavlink::common::msg::PARAM_MAP_RC, mavlink::common::msg::PARAM_REQUEST_LIST, mavlink::common::msg::PARAM_REQUEST_READ, mavlink::common::msg::PARAM_SET, mavlink::common::msg::PARAM_VALUE, mavlink::common::msg::PING, mavlink::common::msg::PLAY_TUNE, mavlink::common::msg::PLAY_TUNE_V2, mavlink::common::msg::POSITION_TARGET_GLOBAL, mavlink::common::msg::POSITION_TARGET_LOCAL_NED, mavlink::common::msg::POWER_STATUS, mavlink::common::msg::POWER_STATUS_REQUEST, mavlink::common::msg::RADIO_STATUS, mavlink::common::msg::RAW_IMU, mavlink::common::msg::RAW_PRESSURE, mavlink::common::msg::RAW_RPM, mavlink::common::msg::RC_CHANNELS, mavlink::common::msg::RC_CHANNELS_OVERRIDE, mavlink::common::msg::RC_CHANNELS_RAW, mavlink::common::msg::RC_CHANNELS_SCALED, mavlink::common::msg::RELEASE_MESSAGE, mavlink::common::msg::REQUEST_DATA_STREAM, mavlink::common::msg::REQUEST_EVENT, mavlink::common::msg::RESOURCE_REQUEST, mavlink::common::msg::RESPONSE_EVENT_ERROR, mavlink::common::msg::SAFETY_ALLOWED_AREA, mavlink::common::msg::SAFETY_SET_ALLOWED_AREA, mavlink::common::msg::SCALED_IMU, mavlink::common::msg::SCALED_IMU3, mavlink::common::msg::SCALED_PRESSURE, mavlink::common::msg::SCALED_PRESSURE3, mavlink::common::msg::SERIAL_CONTROL, mavlink::common::msg::SERVO_OUT, mavlink::common::msg::SETUP_SIGNING, mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET, mavlink::common::msg::SET_ATTITUDE_TARGET, mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN, mavlink::common::msg::SET_HOME_POSITION, mavlink::common::msg::SET_MODE, mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED, mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED, mavlink::common::msg::SIM_STATE, mavlink::common::msg::SIM_STATE, mavlink::common::msg::STATUSTEXT, mavlink::common::msg::STORAGE_INFORMATION, mavlink::common::msg::SUPPORTED_VERSION, mavlink::common::msg::SYSTEM_TIME, mavlink::common::msg::SYS_STATUS, mavlink::common::msg::TERRAIN_CHECK, mavlink::common::msg::TERRAIN_DATA, mavlink::common::msg::TERRAIN_REPORT, mavlink::common::msg::TERRAIN_REQUEST, mavlink::common::msg::TIMESYNC, mavlink::common::msg::TIME_ESTIMATE_TO_TARGET, mavlink::common::msg::TRAJECTORY, mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS, mavlink::common::msg::TUNNEL, mavlink::common::msg::UAVCAN_NODE_INFO, mavlink::common::msg::UAVCAN_NODE_STATUS, mavlink::common::msg::UTM_COORDINATE, mavlink::common::msg::V2_EXTENSION, mavlink::common::msg::VFR_HUD, mavlink::common::msg::VIBRATION, mavlink::common::msg::VICON_POSITION_ESTIMATE, mavlink::common::msg::VIDEO_STREAM_INFORMATION, mavlink::common::msg::VIDEO_STREAM_STATUS, mavlink::common::msg::VISION_POSITION_ESTIMATE, mavlink::common::msg::VISION_SPEED_ESTIMATE, mavlink::common::msg::WHEEL_DISTANCE, mavlink::common::msg::WIFI_CONFIG, mavlink::common::msg::WINCH_STATUS, mavlink::common::msg::WIND_COV, mavlink::minimal::msg::HEARTBEAT, mavlink::standard::msg::AUTOPILOT_VERSION, and mavlink::standard::msg::GLOBAL_POSITION_INT.

Classes

- struct [Info](#)

Public Member Functions

- virtual std::string [get_name](#) (void) const =0
Get NAME constant.
- virtual [Info](#) [get_message_info](#) (void) const =0
Get info needed for mavlink_finalize_message_xxx()
- virtual std::string [to_yaml](#) (void) const =0
Make YAML-string from message content.
- virtual void [serialize](#) ([MsgMap](#) &map) const =0
Serialize message.
- virtual void [deserialize](#) ([MsgMap](#) &msp)=0
Deserialize message.

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = UINT32_MAX
- static constexpr size_t **LENGTH** = 0
- static constexpr size_t **MIN_LENGTH** = 0
- static constexpr uint8_t **CRC_EXTRA** = 0
- static constexpr auto **NAME** = "BASE"

5.346.1 Detailed Description

MAVLink [Message](#) base class.

5.346.2 Member Function Documentation

5.346.2.1 deserialize()

```
virtual void mavlink::Message::deserialize (
    MsgMap & msp ) [pure virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implemented in [mavlink::standard::msg::GLOBAL_POSITION_INT](#), [mavlink::standard::msg::AUTOPILOT_VERSION](#), [mavlink::minimal::msg::HEARTBEAT](#), [mavlink::common::msg::WIND_COV](#), [mavlink::common::msg::WINCH_STATUS](#), [mavlink::common::msg::WIFI_CONFIG_AP](#), [mavlink::common::msg::WHEEL_DISTANCE](#), [mavlink::common::msg::VISION_SPEED](#), [mavlink::common::msg::VISION_POSITION_ESTIMATE](#), [mavlink::common::msg::VIDEO_STREAM_STATUS](#), [mavlink::common::msg::VIDEO_STREAM_INFORMATION](#), [mavlink::common::msg::VICON_POSITION_ESTIMATE](#), [mavlink::common::msg::VIBRATION](#), [mavlink::common::msg::VFR_HUD](#), [mavlink::common::msg::V2_EXTENSION](#), [mavlink::common::msg::UTM_GLOBAL_POSITION](#), [mavlink::common::msg::UAVCAN_NODE_STATUS](#), [mavlink::common::msg::UAVCAN_NODE_STATUS](#), [mavlink::common::msg::TUNNEL](#), [mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS](#), [mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER](#), [mavlink::common::msg::TIMESYNC](#), [mavlink::common::msg::TIME_ESTIMATE_TO_TARGET](#), [mavlink::common::msg::TERRAIN_REQUEST](#), [mavlink::common::msg::TERRAIN_DATA](#), [mavlink::common::msg::TERRAIN_CHECK](#), [mavlink::common::msg::SYSTEM_TIME](#), [mavlink::common::msg::SYS_STATUS](#), [mavlink::common::msg::SUPPORTED_TUNES](#), [mavlink::common::msg::STORAGE_INFORMATION](#), [mavlink::common::msg::STATUSTEXT](#), [mavlink::common::msg::SMART_BATTERY_INFO](#), [mavlink::common::msg::SIM_STATE](#), [mavlink::common::msg::SETUP_SIGNING](#), [mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED](#), [mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT](#), [mavlink::common::msg::SET_MODE](#), [mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN](#), [mavlink::common::msg::SET_ATTITUDE_TARGET](#), [mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET](#), [mavlink::common::msg::SERVO_OUTPUT_RAW](#), [mavlink::common::msg::SERIAL_CONTROL](#), [mavlink::common::msg::SCALED_PRESSURE3](#), [mavlink::common::msg::SCALED_PRESSURE2](#), [mavlink::common::msg::SCALED_PRESSURE](#), [mavlink::common::msg::SCALED_IMU3](#), [mavlink::common::msg::SCALED_IMU2](#), [mavlink::common::msg::SCALED_IMU](#), [mavlink::common::msg::SAFETY_SET_ALLOWED_AREA](#), [mavlink::common::msg::SAFETY_CHECK_REQUEST](#), [mavlink::common::msg::RESPONSE_EVENT_ERROR](#), [mavlink::common::msg::RESOURCE_REQUEST](#), [mavlink::common::msg::REQUEST_EVENT](#), [mavlink::common::msg::REQUEST_DATA_STREAM](#), [mavlink::common::msg::RELAY_STATUS](#), [mavlink::common::msg::RC_CHANNELS_SCALED](#), [mavlink::common::msg::RC_CHANNELS_RAW](#), [mavlink::common::msg::RC_CHANNELS_RAW](#).

mavlink::common::msg::RC_CHANNELS, mavlink::common::msg::RAW_RPM, mavlink::common::msg::RAW_PRESSURE, mavlink::common::msg::RAW_IMU, mavlink::common::msg::RADIO_STATUS, mavlink::common::msg::PROTOCOL_VERSION, mavlink::common::msg::POWER_STATUS, mavlink::common::msg::POSITION_TARGET_LOCAL_NED, mavlink::common::msg::PLAY_TUNE_V2, mavlink::common::msg::PLAY_TUNE, mavlink::common::msg::PING, mavlink::common::msg::PARAM_VALUE, mavlink::common::msg::PARAM_SET, mavlink::common::msg::PARAM_REQUEST_READ, mavlink::common::msg::PARAM_REQUEST_LIST, mavlink::common::msg::PARAM_MAP_RC, mavlink::common::msg::PARAM_EXT, mavlink::common::msg::PARAM_EXT_SET, mavlink::common::msg::PARAM_EXT_REQUEST_READ, mavlink::common::msg::PARAM_EXT_ACK, mavlink::common::msg::PARAM_ERROR, mavlink::common::msg::ORBIT_EXECUTION, mavlink::common::msg::OPTICAL_FLOW_RAD, mavlink::common::msg::OPTICAL_FLOW, mavlink::common::msg::OPEN_DRONE_ID, mavlink::common::msg::OPEN_DRONE_ID_SYSTEM, mavlink::common::msg::OPEN_DRONE_ID_SELF_ID, mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID, mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK, mavlink::common::msg::OPEN_DRONE_ID_LOCATION, mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID, mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION, mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS, mavlink::common::msg::ONBOARD_COMPUTER_STATUS, mavlink::common::msg::ODOMETRY, mavlink::common::msg::OBSTACLE_DISTANCE, mavlink::common::msg::NAV_CONTROLLER_OUTPUT, mavlink::common::msg::NAMED_VALUE_INT, mavlink::common::msg::NAMED_VALUE_FLOAT, mavlink::common::msg::MOUNT_ORIENTATION, mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST, mavlink::common::msg::MISSION_SET_CURRENT, mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST, mavlink::common::msg::MISSION_REQUEST_LIST, mavlink::common::msg::MISSION_REQUEST_INT, mavlink::common::msg::MISSION_ITEM_REACHED, mavlink::common::msg::MISSION_ITEM_INT, mavlink::common::msg::MISSION_CURRENT, mavlink::common::msg::MISSION_COUNT, mavlink::common::msg::MISSION_CLEAR, mavlink::common::msg::MISSION_ACK, mavlink::common::msg::MESSAGE_INTERVAL, mavlink::common::msg::MEMORY_VECTOR, mavlink::common::msg::MANUAL_SETPOINT, mavlink::common::msg::MANUAL_CONTROL, mavlink::common::msg::MAG_CAL_REPORT, mavlink::common::msg::LOGGING_DATA_ACKED, mavlink::common::msg::LOGGING_DATA, mavlink::common::msg::LOGGING_DATA_ACKED, mavlink::common::msg::LOG_REQUEST_LIST, mavlink::common::msg::LOG_REQUEST_END, mavlink::common::msg::LOG_REQUEST_ERASE, mavlink::common::msg::LOG_ERASE, mavlink::common::msg::LOG_ENTRY, mavlink::common::msg::LOG_DATA, mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET, mavlink::common::msg::LOCAL_POSITION_NED, mavlink::common::msg::LINK_NODE_STATUS, mavlink::common::msg::LANDING_GEAR_STATUS, mavlink::common::msg::ISBD_LINK_STATUS, mavlink::common::msg::ILLUMINATOR_STATUS, mavlink::common::msg::HYGROMETER, mavlink::common::msg::HOME_POSITION, mavlink::common::msg::HIL_STATE_QUATERNION, mavlink::common::msg::HIL_STATE_RAW, mavlink::common::msg::HIL_SENSOR, mavlink::common::msg::HIL_RC_INPUTS_RAW, mavlink::common::msg::HIL_OPTICAL_FLOW, mavlink::common::msg::HIL_GPS, mavlink::common::msg::HIL_CONTROLS, mavlink::common::msg::HIL_ACTUATOR_CONTROLS, mavlink::common::msg::HIGHRES_IMU, mavlink::common::msg::HIGH_LATENCY2, mavlink::common::msg::HIGH_LATENCY, mavlink::common::msg::GPS_STATUS, mavlink::common::msg::GPS_RTK, mavlink::common::msg::GPS_RTCM_DATA, mavlink::common::msg::GPS_RAW_INT, mavlink::common::msg::GPS_INPUT, mavlink::common::msg::GPS_INJECT_DATA, mavlink::common::msg::GPS_GLOBAL_ORIGIN, mavlink::common::msg::GPS2_RTK, mavlink::common::msg::GPS2_RAW, mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE, mavlink::common::msg::GLOBAL_POSITION_INT_COV, mavlink::common::msg::GIMBAL_MANAGER_STATUS, mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW, mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL, mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE, mavlink::common::msg::GIMBAL_MANAGER_INFORMATION, mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE, mavlink::common::msg::GIMBAL_DEVICE_INFORMATION, mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS, mavlink::common::msg::GENERATOR_STATUS, mavlink::common::msg::FUEL_STATUS, mavlink::common::msg::FOLLOW_TARGET, mavlink::common::msg::FLIGHT_INFORMATION, mavlink::common::msg::FILE_TRANSFER_PROTOCOL, mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS, mavlink::common::msg::FENCE_STATUS, mavlink::common::msg::EXTENDED_SYS_STATE, mavlink::common::msg::EVENT, mavlink::common::msg::ESTIMATOR_STATUS, mavlink::common::msg::ESC_STATUS, mavlink::common::msg::ESC_INFO, mavlink::common::msg::ENCAPSULATED_DATA, mavlink::common::msg::EFI_STATUS, mavlink::common::msg::DISTANCE_SENSOR, mavlink::common::msg::DEBUG_VECTOR, mavlink::common::msg::DEBUG_FLOAT_ARRAY, mavlink::common::msg::DEBUG, mavlink::common::msg::DATA_TRANSMISSION_INTERVAL, mavlink::common::msg::DATA_STREAM, mavlink::common::msg::CURRENT_MODE, mavlink::common::msg::CURRENT_EVENT_TIMESTAMP, mavlink::common::msg::CONTROL_SYSTEM_STATE, mavlink::common::msg::COMPONENT_METADATA, mavlink::common::msg::COMPONENT_INFORMATION_BASIC, mavlink::common::msg::COMPONENT_INFORMATION, mavlink::common::msg::COMMAND_LONG, mavlink::common::msg::COMMAND_INT, mavlink::common::msg::COMMAND_CANCEL, mavlink::common::msg::COMMAND_ACK, mavlink::common::msg::COLLISION, mavlink::common::msg::CHANGE_OPERATOR_CONTROL, mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK, mavlink::common::msg::CELLULAR_STATUS, mavlink::common::msg::CELLULAR_CONFIG, mavlink::common::msg::CANFD_FRAME, mavlink::common::msg::CAN_FRAME, mavlink::common::msg::CAN_FILTER_MODIFY, mavlink::common::msg::CAMERA_TRIGGER, mavlink::common::msg::CAMERA_TRIGGER_STATUS, mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS, mavlink::common::msg::CAMERA_THERMAL_RANGE, mavlink::common::msg::CAMERA_SETTINGS, mavlink::common::msg::CAMERA_INFORMATION, mavlink::common::msg::CAMERA_FOV_STATUS, mavlink::common::msg::CAMERA_FOV_STATUS, mavlink::common::msg::CAMERA_CAPTURE_STATUS,

mavlink::common::msg::BUTTON_CHANGE, mavlink::common::msg::BATTERY_STATUS, mavlink::common::msg::BATTERY_INFO, mavlink::common::msg::AVAILABLE_MODES_MONITOR, mavlink::common::msg::AVAILABLE_MODES, mavlink::common::msg::AUTH_KEY, mavlink::common::msg::ATTITUDE_TARGET, mavlink::common::msg::ATTITUDE_QUATERNION, mavlink::common::msg::ATTITUDE, mavlink::common::msg::ATT_POS_MOCAP, mavlink::common::msg::ALTITUDE, mavlink::common::msg::AIS_VESSEL, mavlink::common::msg::AIRSPEED, mavlink::common::msg::ADSB_VEHICLE, mavlink::common::msg::ACTUATOR_OUTPUT_STATUS, and mavlink::common::msg::AC

5.346.2.2 get_name()

```
virtual std::string mavlink::Message::get_name (
    void ) const [pure virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implemented in mavlink::standard::msg::GLOBAL_POSITION_INT, mavlink::standard::msg::AUTOPILOT_VERSION, mavlink::minimal::msg::HEARTBEAT, mavlink::common::msg::WIND_COV, mavlink::common::msg::WINCH_STATUS, mavlink::common::msg::WIFI_CONFIG_AP, mavlink::common::msg::WHEEL_DISTANCE, mavlink::common::msg::VISION_SPEED, mavlink::common::msg::VISION_POSITION_ESTIMATE, mavlink::common::msg::VIDEO_STREAM_STATUS, mavlink::common::msg::VIDEO_STREAM_INFORMATION, mavlink::common::msg::VICON_POSITION_ESTIMATE, mavlink::common::msg::VIBRATION, mavlink::common::msg::VFR_HUD, mavlink::common::msg::V2_EXTENSION, mavlink::common::msg::UTM_GLOBAL_POSITION, mavlink::common::msg::UAVCAN_NODE_STATUS, mavlink::common::msg::TUNNEL, mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS, mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER, mavlink::common::msg::TIMESYNC, mavlink::common::msg::TIME_ESTIMATE_TO_TARGET, mavlink::common::msg::TERRAIN_REQUEST, mavlink::common::msg::TERRAIN_DATA, mavlink::common::msg::TERRAIN_CHECK, mavlink::common::msg::SYSTEM_TIME, mavlink::common::msg::SYS_STATUS, mavlink::common::msg::SUPPORTED_TUNES, mavlink::common::msg::STORAGE_INFORMATION, mavlink::common::msg::STATUSTEXT, mavlink::common::msg::SMART_BATTERY_INFO, mavlink::common::msg::SIM_STATE, mavlink::common::msg::SETUP_SIGNING, mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED, mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT, mavlink::common::msg::SET_MODE, mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN, mavlink::common::msg::SET_ATTITUDE_TARGET, mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET, mavlink::common::msg::SERVO_OUTPUT_RAW, mavlink::common::msg::SERIAL_CONTROL, mavlink::common::msg::SCALED_PRESSURE3, mavlink::common::msg::SCALED_PRESSURE, mavlink::common::msg::SCALED_IMU3, mavlink::common::msg::SCALED_IMU2, mavlink::common::msg::SAFETY_SET_ALLOWED_AREA, mavlink::common::msg::SAFETY_RESPONSE_EVENT_ERROR, mavlink::common::msg::RESOURCE_REQUEST, mavlink::common::msg::REQUEST_EVENT, mavlink::common::msg::REQUEST_DATA_STREAM, mavlink::common::msg::RELAY_STATUS, mavlink::common::msg::RC_CHANNELS_SCALED, mavlink::common::msg::RC_CHANNELS_RAW, mavlink::common::msg::RC_CHANNELS, mavlink::common::msg::RAW_RPM, mavlink::common::msg::RAW_PRESSURE, mavlink::common::msg::RAW_IMU, mavlink::common::msg::RADIO_STATUS, mavlink::common::msg::PROTOCOL_VERSION, mavlink::common::msg::POWER_STATUS, mavlink::common::msg::POSITION_TARGET_LOCAL_NED, mavlink::common::msg::PLAY_TUNE_V2, mavlink::common::msg::PLAY_TUNE, mavlink::common::msg::PING, mavlink::common::msg::PARAM_VALUE, mavlink::common::msg::PARAM_SET, mavlink::common::msg::PARAM_REQUEST_READ, mavlink::common::msg::PARAM_REQUEST_LIST, mavlink::common::msg::PARAM_MAP_RC, mavlink::common::msg::PARAM_EXT_SET, mavlink::common::msg::PARAM_EXT_REQUEST_READ, mavlink::common::msg::PARAM_EXT_ACK, mavlink::common::msg::PARAM_ERROR, mavlink::common::msg::ORBIT_EXECUTION, mavlink::common::msg::OPTICAL_FLOW_RAD, mavlink::common::msg::OPTICAL_FLOW, mavlink::common::msg::OPEN_DRONE_ID_SYSTEM, mavlink::common::msg::OPEN_DRONE_ID_SELF_ID, mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID, mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK, mavlink::common::msg::OPEN_DRONE_ID_LOCATION, mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID, mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION, mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS, mavlink::common::msg::ONBOARD_COMPUTER_STATUS, mavlink::common::msg::ODOMETRY, mavlink::common::msg::OBSTACLE_AVOIDANCE, mavlink::common::msg::NAV_CONTROLLER_OUTPUT, mavlink::common::msg::NAMED_VALUE_INT, mavlink::common::msg::MOUNT_ORIENTATION, and mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST,

mavlink::common::msg::MISSION_SET_CURRENT, mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST,
 mavlink::common::msg::MISSION_REQUEST_LIST, mavlink::common::msg::MISSION_REQUEST_INT, mavlink::common::msg::MISSION_REQUEST_LIST,
 mavlink::common::msg::MISSION_ITEM_REACHED, mavlink::common::msg::MISSION_ITEM_INT, mavlink::common::msg::MISSION_ITEM_REACHED,
 mavlink::common::msg::MISSION_CURRENT, mavlink::common::msg::MISSION_COUNT, mavlink::common::msg::MISSION_CLEAR, mavlink::common::msg::MISSION_ACK,
 mavlink::common::msg::MESSAGE_INTERVAL, mavlink::common::msg::MEMORY_VECTOR, mavlink::common::msg::MANUAL_SETPOINT, mavlink::common::msg::MANUAL_CONTROL,
 mavlink::common::msg::MAG_CAL_REPORT, mavlink::common::msg::LOGGING_DATA_ACKED, mavlink::common::msg::LOGGING_DATA, mavlink::common::msg::LOGGING_DATA_ACKED,
 mavlink::common::msg::LOG_REQUEST_LIST, mavlink::common::msg::LOG_REQUEST_END, mavlink::common::msg::LOG_REQUEST_LIST, mavlink::common::msg::LOG_ERASE,
 mavlink::common::msg::LOG_ENTRY, mavlink::common::msg::LOG_DATA, mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET,
 mavlink::common::msg::LOCAL_POSITION_NED, mavlink::common::msg::LINK_NODE_STATUS, mavlink::common::msg::LANDING_GEAR_STATUS, mavlink::common::msg::ISBD_LINK_STATUS,
 mavlink::common::msg::ILLUMINATOR_STATUS, mavlink::common::msg::HYGROMETER_STATUS, mavlink::common::msg::HOME_POSITION, mavlink::common::msg::HIL_STATE_QUATERNION,
 mavlink::common::msg::HIL_STATE_RAW, mavlink::common::msg::HIL_SENSOR, mavlink::common::msg::HIL_RC_INPUTS_RAW, mavlink::common::msg::HIL_OPTICAL_FLOW,
 mavlink::common::msg::HIL_GPS, mavlink::common::msg::HIL_CONTROLS, mavlink::common::msg::HIL_ACTUATOR_CONTROLS, mavlink::common::msg::HIGHRES_IMU,
 mavlink::common::msg::HIGH_LATENCY2, mavlink::common::msg::HIGH_LATENCY, mavlink::common::msg::GPS_STATUS, mavlink::common::msg::GPS_RTK,
 mavlink::common::msg::GPS_RTCM_DATA, mavlink::common::msg::GPS_RAW_INT, mavlink::common::msg::GPS_INPUT, mavlink::common::msg::GPS_INJECT_DATA,
 mavlink::common::msg::GPS_GLOBAL_ORIGIN, mavlink::common::msg::GPS2_RTK, mavlink::common::msg::GPS2_RAW, mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE,
 mavlink::common::msg::GLOBAL_POSITION_INT_COV, mavlink::common::msg::GIMBAL_MANAGER_STATUS, mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW,
 mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL, mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE, mavlink::common::msg::GIMBAL_MANAGER_INFORMATION,
 mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE, mavlink::common::msg::GIMBAL_DEVICE_INFORMATION, mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS,
 mavlink::common::msg::GENERATOR_STATUS, mavlink::common::msg::FUEL_STATUS, mavlink::common::msg::FOLLOW_TARGET, mavlink::common::msg::FLIGHT_INFORMATION,
 mavlink::common::msg::FILE_TRANSFER_PROTOCOL, mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS, mavlink::common::msg::FENCE_STATUS,
 mavlink::common::msg::EXTENDED_SYS_STATE, mavlink::common::msg::EVENT, mavlink::common::msg::ESTIMATOR_STATUS, mavlink::common::msg::ESC_STATUS,
 mavlink::common::msg::ESC_INFO, mavlink::common::msg::ENCAPSULATED_DATA, mavlink::common::msg::EFI_STATUS, mavlink::common::msg::DISTANCE_SENSOR,
 mavlink::common::msg::DEBUG_VECTOR, mavlink::common::msg::DEBUG_FLOAT_ARRAY, mavlink::common::msg::DEBUG, mavlink::common::msg::DATA_TRANSMISSION_INTERVAL,
 mavlink::common::msg::DATA_STREAM, mavlink::common::msg::CURRENT_MODE, mavlink::common::msg::CURRENT_EVENT_TIMESTAMP, mavlink::common::msg::CONTROL_SYSTEM_STATE,
 mavlink::common::msg::COMPONENT_METADATA, mavlink::common::msg::COMPONENT_INFORMATION_BASIC, mavlink::common::msg::COMPONENT_INFORMATION,
 mavlink::common::msg::COMMAND_LONG, mavlink::common::msg::COMMAND_INT, mavlink::common::msg::COMMAND_CANCEL, mavlink::common::msg::COMMAND_ACK,
 mavlink::common::msg::COLLISION, mavlink::common::msg::CHANGE_OPERATOR_CONTROL, mavlink::common::msg::CELLULAR_STATUS, mavlink::common::msg::CELLULAR_CONFIG,
 mavlink::common::msg::CANFD_FRAME, mavlink::common::msg::CAN_FRAME, mavlink::common::msg::CAN_FILTER_MODIFY, mavlink::common::msg::CAMERA_TRIGGER,
 mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS, mavlink::common::msg::CAMERA_THERMAL_RANGE, mavlink::common::msg::CAMERA_SETTINGS,
 mavlink::common::msg::CAMERA_INFORMATION, mavlink::common::msg::CAMERA_FOV_STATUS, mavlink::common::msg::CAMERA_CAPTURE_STATUS,
 mavlink::common::msg::BUTTON_CHANGE, mavlink::common::msg::BATTERY_STATUS, mavlink::common::msg::BATTERY_INFO, mavlink::common::msg::AVAILABLE_MODES_MONITOR,
 mavlink::common::msg::AVAILABLE_MODES, mavlink::common::msg::AUTH_KEY, mavlink::common::msg::ATTITUDE_TARGET, mavlink::common::msg::ATTITUDE_QUATERNION,
 mavlink::common::msg::ATTITUDE, mavlink::common::msg::ATT_POS_MOCAP, mavlink::common::msg::ALTITUDE, mavlink::common::msg::AIS_VESSEL,
 mavlink::common::msg::AIRSPEED, mavlink::common::msg::ADSB_VEHICLE, mavlink::common::msg::ACTUATOR_OUTPUT_STATUS, and mavlink::common::msg::ACTUATOR_OUTPUT_STATUS

5.346.2.3 serialize()

```
virtual void mavlink::Message::serialize (
    MsgMap & map ) const [pure virtual]
```

Serialize message.

Parameters

out	map
-----	-----

Implemented in [mavlink::standard::msg::GLOBAL_POSITION_INT](#), [mavlink::standard::msg::AUTOPILOT_VERSION](#), [mavlink::minimal::msg::HEARTBEAT](#), [mavlink::common::msg::WIND_COV](#), [mavlink::common::msg::WINCH_STATUS](#), [mavlink::common::msg::WIFI_CONFIG_AP](#), [mavlink::common::msg::WHEEL_DISTANCE](#), [mavlink::common::msg::VISION_SPEED](#), [mavlink::common::msg::VISION_POSITION_ESTIMATE](#), [mavlink::common::msg::VIDEO_STREAM_STATUS](#), [mavlink::common::msg::VIDEO_STREAM_INFORMATION](#), [mavlink::common::msg::VICON_POSITION_ESTIMATE](#), [mavlink::common::msg::VIBRATION](#), [mavlink::common::msg::VFR_HUD](#), [mavlink::common::msg::V2_EXTENSION](#), [mavlink::common::msg::UTM_GLOBAL_POSITION](#), [mavlink::common::msg::UAVCAN_NODE_STATUS](#), [mavlink::common::msg::UA](#), [mavlink::common::msg::TUNNEL](#), [mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS](#), [mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER](#), [mavlink::common::msg::TIMESYNC](#), [mavlink::common::msg::TIME_ESTIMATE_TO_TARGET](#), [mavlink::common::msg::TERRAIN_REQUEST](#), [mavlink::common::msg::TE](#), [mavlink::common::msg::TERRAIN_DATA](#), [mavlink::common::msg::TERRAIN_CHECK](#), [mavlink::common::msg::SYSTEM_TIME](#), [mavlink::common::msg::SYS_STATUS](#), [mavlink::common::msg::SUPPORTED_TUNES](#), [mavlink::common::msg::STORAGE_INFORM](#), [mavlink::common::msg::STATUSTEXT](#), [mavlink::common::msg::SMART_BATTERY_INFO](#), [mavlink::common::msg::SIM_STATE](#), [mavlink::common::msg::SETUP_SIGNING](#), [mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED](#), [mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT](#), [mavlink::common::msg::SET_MODE](#), [mavlink::common::msg::SE](#), [mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN](#), [mavlink::common::msg::SET_ATTITUDE_TARGET](#), [mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET](#), [mavlink::common::msg::SERVO_OUTPUT_RAW](#), [mavlink::common::msg::SERIAL_CONTROL](#), [mavlink::common::msg::SCALED_PRESSURE3](#), [mavlink::common::msg::SCALED_PP](#), [mavlink::common::msg::SCALED_PRESSURE](#), [mavlink::common::msg::SCALED_IMU3](#), [mavlink::common::msg::SCALED_IMU2](#), [mavlink::common::msg::SCALED_IMU](#), [mavlink::common::msg::SAFETY_SET_ALLOWED_AREA](#), [mavlink::common::msg::SAFETY](#), [mavlink::common::msg::RESPONSE_EVENT_ERROR](#), [mavlink::common::msg::RESOURCE_REQUEST](#), [mavlink::common::msg::REQUEST_EVENT](#), [mavlink::common::msg::REQUEST_DATA_STREAM](#), [mavlink::common::msg::RELAY_S](#), [mavlink::common::msg::RC_CHANNELS_SCALED](#), [mavlink::common::msg::RC_CHANNELS_RAW](#), [mavlink::common::msg::RC_CH](#), [mavlink::common::msg::RC_CHANNELS](#), [mavlink::common::msg::RAW_RPM](#), [mavlink::common::msg::RAW_PRESSURE](#), [mavlink::common::msg::RAW_IMU](#), [mavlink::common::msg::RADIO_STATUS](#), [mavlink::common::msg::PROTOCOL_VERSION](#), [mavlink::common::msg::POWER_STATUS](#), [mavlink::common::msg::POSITION_TARGET_LOCAL_NED](#), [mavlink::common::msg::PO](#), [mavlink::common::msg::PLAY_TUNE_V2](#), [mavlink::common::msg::PLAY_TUNE](#), [mavlink::common::msg::PING](#), [mavlink::common::msg::PARAM_VALUE](#), [mavlink::common::msg::PARAM_SET](#), [mavlink::common::msg::PARAM_REQUEST_READ](#), [mavlink::common::msg::PARAM_REQUEST_LIST](#), [mavlink::common::msg::PARAM_MAP_RC](#), [mavlink::common::msg::PARAM_EX](#), [mavlink::common::msg::PARAM_EXT_SET](#), [mavlink::common::msg::PARAM_EXT_REQUEST_READ](#), [mavlink::common::msg::PAR](#), [mavlink::common::msg::PARAM_EXT_ACK](#), [mavlink::common::msg::PARAM_ERROR](#), [mavlink::common::msg::ORBIT_EXECUTION](#), [mavlink::common::msg::OPTICAL_FLOW_RAD](#), [mavlink::common::msg::OPTICAL_FLOW](#), [mavlink::common::msg::OPEN_DRONE](#), [mavlink::common::msg::OPEN_DRONE_ID_SYSTEM](#), [mavlink::common::msg::OPEN_DRONE_ID_SELF_ID](#), [mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID](#), [mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK](#), [mavlink::common::msg::OPEN_DRONE_ID_LOCATION](#), [mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID](#), [mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION](#), [mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS](#), [mavlink::common::msg::ONBOARD_COMPUTER_STATUS](#), [mavlink::common::msg::ODOMETRY](#), [mavlink::common::msg::OBSTAC](#), [mavlink::common::msg::NAV_CONTROLLER_OUTPUT](#), [mavlink::common::msg::NAMED_VALUE_INT](#), [mavlink::common::msg::NAMED](#), [mavlink::common::msg::MOUNT_ORIENTATION](#), [mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST](#), [mavlink::common::msg::MISSION_SET_CURRENT](#), [mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST](#), [mavlink::common::msg::MISSION_REQUEST_LIST](#), [mavlink::common::msg::MISSION_REQUEST_INT](#), [mavlink::common::msg::MI](#), [mavlink::common::msg::MISSION_ITEM_REACHED](#), [mavlink::common::msg::MISSION_ITEM_INT](#), [mavlink::common::msg::MISSIO](#), [mavlink::common::msg::MISSION_CURRENT](#), [mavlink::common::msg::MISSION_COUNT](#), [mavlink::common::msg::MISSION_CLEA](#), [mavlink::common::msg::MISSION_ACK](#), [mavlink::common::msg::MESSAGE_INTERVAL](#), [mavlink::common::msg::MEMORY_VECT](#), [mavlink::common::msg::MANUAL_SETPOINT](#), [mavlink::common::msg::MANUAL_CONTROL](#), [mavlink::common::msg::MAG_CAL_R](#), [mavlink::common::msg::LOGGING_DATA_ACKED](#), [mavlink::common::msg::LOGGING_DATA](#), [mavlink::common::msg::LOGGING_A](#), [mavlink::common::msg::LOG_REQUEST_LIST](#), [mavlink::common::msg::LOG_REQUEST_END](#), [mavlink::common::msg::LOG_REQ](#), [mavlink::common::msg::LOG_ERASE](#), [mavlink::common::msg::LOG_ENTRY](#), [mavlink::common::msg::LOG_DATA](#), [mavlink::common::msg::LOCAL_POSITION_NED_SYSTEM_GLOBAL_OFFSET](#), [mavlink::common::msg::LOCAL_POSITION_NED](#), [mavlink::common::msg::LOCAL_POSITION_NED](#), [mavlink::common::msg::LINK_NODE_STATUS](#), [mavlink::common::msg::LANDING](#), [mavlink::common::msg::ISBD_LINK_STATUS](#), [mavlink::common::msg::ILLUMINATOR_STATUS](#), [mavlink::common::msg::HYGROME](#), [mavlink::common::msg::HOME_POSITION](#), [mavlink::common::msg::HIL_STATE_QUATERNION](#), [mavlink::common::msg::HIL_STATE](#), [mavlink::common::msg::HIL_SENSOR](#), [mavlink::common::msg::HIL_RC_INPUTS_RAW](#), [mavlink::common::msg::HIL_OPTICAL_FL](#)

mavlink::common::msg::HIL_GPS, mavlink::common::msg::HIL_CONTROLS, mavlink::common::msg::HIL_ACTUATOR_CONTROLS, mavlink::common::msg::HIGHRES_IMU, mavlink::common::msg::HIGH_LATENCY2, mavlink::common::msg::HIGH_LATENCY, mavlink::common::msg::GPS_STATUS, mavlink::common::msg::GPS_RTK, mavlink::common::msg::GPS_RTCM_DATA, mavlink::common::msg::GPS_RAW_INT, mavlink::common::msg::GPS_INPUT, mavlink::common::msg::GPS_INJECT_DATA, mavlink::common::msg::GPS_GLOBAL_ORIGIN, mavlink::common::msg::GPS2_RTK, mavlink::common::msg::GPS2_RAW, mavlink::common::msg::GLOBAL_VISION_POSITION_ESTIMATE, mavlink::common::msg::GLOBAL_POSITION_INT_COV, mavlink::common::msg::GIMBAL_MANAGER_STATUS, mavlink::common::msg::GIMBAL_MANAGER_SET_PITCHYAW, mavlink::common::msg::GIMBAL_MANAGER_SET_MANUAL_CONTROL, mavlink::common::msg::GIMBAL_MANAGER_SET_ATTITUDE, mavlink::common::msg::GIMBAL_MANAGER_INFORMATION, mavlink::common::msg::GIMBAL_DEVICE_SET_ATTITUDE, mavlink::common::msg::GIMBAL_DEVICE_INFORMATION, mavlink::common::msg::GIMBAL_DEVICE_ATTITUDE_STATUS, mavlink::common::msg::GENERATOR_STATUS, mavlink::common::msg::FUEL_STATUS, mavlink::common::msg::FOLLOW_TARGET, mavlink::common::msg::FLIGHT_INFORMATION, mavlink::common::msg::FILE_TRANSFER_PROTOCOL, mavlink::common::msg::FIGURE_EIGHT_EXECUTION_STATUS, mavlink::common::msg::FENCE_STATUS, mavlink::common::msg::EXTENDED_SYS_STATE, mavlink::common::msg::EVENT, mavlink::common::msg::ESTIMATOR_STATUS, mavlink::common::msg::ESC_STATUS, mavlink::common::msg::ESC_INFO, mavlink::common::msg::ENCAPSULATED_DATA, mavlink::common::msg::EFI_STATUS, mavlink::common::msg::DISTANCE_SENSOR, mavlink::common::msg::DEBUG_VECT, mavlink::common::msg::DEBUG_FLOAT_ARRAY, mavlink::common::msg::DEBUG, mavlink::common::msg::DATA_TRANSMISSION, mavlink::common::msg::DATA_STREAM, mavlink::common::msg::CURRENT_MODE, mavlink::common::msg::CURRENT_EVENT, mavlink::common::msg::CONTROL_SYSTEM_STATE, mavlink::common::msg::COMPONENT_METADATA, mavlink::common::msg::COMPONENT_INFORMATION_BASIC, mavlink::common::msg::COMPONENT_INFORMATION, mavlink::common::msg::COMMAND_LONG, mavlink::common::msg::COMMAND_INT, mavlink::common::msg::COMMAND_CANCEL, mavlink::common::msg::COMMAND_ACK, mavlink::common::msg::COLLISION, mavlink::common::msg::CHANGE_OPERATOR_CONTROL, mavlink::common::msg::CHANGE_OPERATOR_CONTROL_ACK, mavlink::common::msg::CELLULAR_STATUS, mavlink::common::msg::CELLULAR_CONFIG, mavlink::common::msg::CANFD_FRAME, mavlink::common::msg::CAN_FRAME, mavlink::common::msg::CAN_FILTER_MODIFY, mavlink::common::msg::CAMERA_TRIGGER, mavlink::common::msg::CAMERA_TRIGGER_STATUS, mavlink::common::msg::CAMERA_TRACKING_GEO_STATUS, mavlink::common::msg::CAMERA_THERMAL_RANGE, mavlink::common::msg::CAMERA_SETTINGS, mavlink::common::msg::CAMERA_INFORMATION, mavlink::common::msg::CAMERA_INFORMATION2, mavlink::common::msg::CAMERA_FOV_STATUS, mavlink::common::msg::CAMERA_CAPTURE_STATUS, mavlink::common::msg::BUTTON_CHANGE, mavlink::common::msg::BATTERY_STATUS, mavlink::common::msg::BATTERY_INFO, mavlink::common::msg::AVAILABLE_MODES_MONITOR, mavlink::common::msg::AVAILABLE_MODES, mavlink::common::msg::AUTH_KEY, mavlink::common::msg::ATTITUDE_TARGET, mavlink::common::msg::ATTITUDE_QUATERNION, mavlink::common::msg::ATTITUDE_QUATERNION, mavlink::common::msg::ATTITUDE, mavlink::common::msg::ATT_POS_MOCAP, mavlink::common::msg::ALTITUDE, mavlink::common::msg::AIS_VESSEL, mavlink::common::msg::AIRSPEED, mavlink::common::msg::ADSB_VEHICLE, mavlink::common::msg::ACTUATOR_OUTPUT_STATUS, and mavlink::common::msg::AC

The documentation for this struct was generated from the following file:

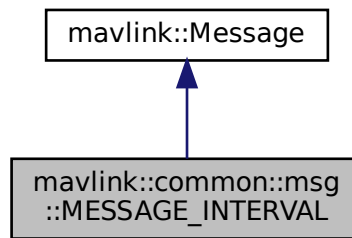
- single_hand_controller/include/mavlink/message.hpp

5.347 mavlink::common::msg::MESSAGE_INTERVAL Struct Reference

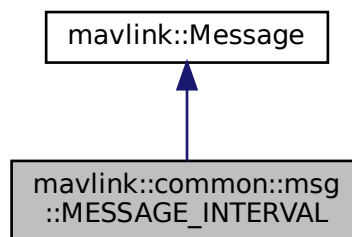
[MESSAGE_INTERVAL](#) message.

```
#include <mavlink_msg_message_interval.hpp>
```

Inheritance diagram for mavlink::common::msg::MESSAGE_INTERVAL:



Collaboration diagram for mavlink::common::msg::MESSAGE_INTERVAL:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint16_t` **message_id**
- `int32_t` **interval_us**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 244
- static constexpr size_t **LENGTH** = 6
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 95
- static constexpr auto **NAME** = "MESSAGE_INTERVAL"

5.347.1 Detailed Description

[MESSAGE_INTERVAL](#) message.

The interval between messages for a particular MAVLink message ID.

This message is sent in response to the MAV_CMD_REQUEST_MESSAGE command with param1=244 (this message) and pa

It may also be sent in response to MAV_CMD_GET_MESSAGE_INTERVAL. This interface replaces [DATA_STREAM](#).

5.347.2 Member Function Documentation

5.347.2.1 deserialize()

```
void mavlink::common::msg::MESSAGE_INTERVAL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.347.2.2 get_name()

```
std::string mavlink::common::msg::MESSAGE_INTERVAL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.347.2.3 serialize()

```
void mavlink::common::msg::MESSAGE_INTERVAL::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_message_interval.hpp`

5.348 message_sender Class Reference

pack data and write MAVLink packets to buffer.

```
#include <message_sender.hpp>
```

Public Member Functions

- [message_sender](#) (uint8_t *buffer, mavlink_message_t *message)
constructor for message sender holds message and buffer pointers.
- int [buffer_arm_disarm_cmd](#) (bool state)
pack an arm/disarm command into buffer
- int [buffer_heartbeat](#) ()
send heartbeat to buffer; heartbeat from hand controller is used to invoke ATLAS' radio to send RADIO_STATUS packets
- int [buffer_manual_control](#) (int x, int y, bool extra_feature_1_press, bool extra_feature_1_long_press, bool extra_feature_2_press, bool extra_feature_2_long_press, [directionToggle](#) dirTog, [speedToggle](#) spdTog)
send manual control packet to buffer
- int [buffer_timesync](#) ()

5.348.1 Detailed Description

pack data and write MAVLink packets to buffer.

all functions return the length of the packet, which is meant to be used when writing buffer to serial channel

5.348.2 Constructor & Destructor Documentation

5.348.2.1 message_sender()

```
message_sender::message_sender (
    uint8_t * buffer,
    mavlink_message_t * message ) [inline]
```

constructor for message sender holds message and buffer pointers.

Parameters

<i>buffer</i>	required to be sent by packet handler over UART
<i>message</i>	struct contains packed MAVLink message

5.348.3 Member Function Documentation

5.348.3.1 buffer_arm_disarm_cmd()

```
int message_sender::buffer_arm_disarm_cmd (
    bool state )
```

pack an arm/disarm command into buffer

Parameters

<i>state</i>	can be disarm(0), or arm(1)
--------------	-----------------------------

Returns

length of buffer

5.348.3.2 buffer_heartbeat()

```
int message_sender::buffer_heartbeat ( )
```

send heartbeat to buffer; heartbeat from hand controller is used to invoke ATLAS' radio to send RADIO_STATUS packets

Returns

length of buffer

5.348.3.3 buffer_manual_control()

```
int message_sender::buffer_manual_control (
    int x,
    int y,
    bool extra_feature_1_press,
    bool extra_feature_1_long_press,
    bool extra_feature_2_press,
    bool extra_feature_2_long_press,
    directionToggle dirTog,
    speedToggle spdTog )
```

send manual control packet to buffer

Parameters

<i>x</i>	normalized input for thumbstick along X direction (forward - backward)
<i>y</i>	normalized input for thumbstick along Y direction (left - right)
<i>extra_feature_1_press</i>	whether button A is pressed or not
<i>extra_feature_1_long_press</i>	whether button A is long pressed
<i>extra_feature_2_press</i>	whether button B is pressed or not
<i>extra_feature_2_long_press</i>	whether button B is long pressed
<i>dirTog</i>	enum value for the current position of direction toggle switch
<i>spdTog</i>	enum value for the current position of speed toggle switch

Returns

length of buffer

The documentation for this class was generated from the following files:

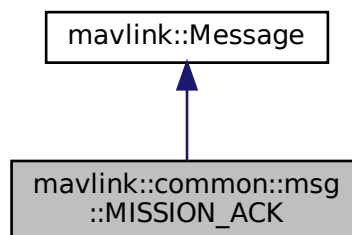
- [single_hand_controller/include/message_sender.hpp](#)
- [single_hand_controller/message_sender.cpp](#)

5.349 mavlink::common::msg::MISSION_ACK Struct Reference

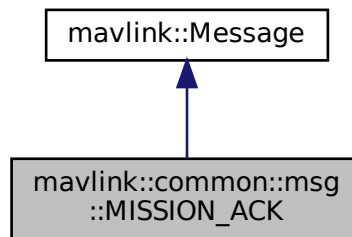
[MISSION_ACK](#) message.

```
#include <mavlink_msg_mission_ack.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_ACK:



Collaboration diagram for mavlink::common::msg::MISSION_ACK:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **type**
- `uint8_t` **mission_type**
- `uint32_t` **opaque_id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 47
- static constexpr size_t **LENGTH** = 8
- static constexpr size_t **MIN_LENGTH** = 3
- static constexpr uint8_t **CRC_EXTRA** = 153
- static constexpr auto **NAME** = "MISSION_ACK"

5.349.1 Detailed Description

[MISSION_ACK](#) message.

Acknowledgment message during waypoint handling. The type field states if this message is a positive ack (type=0) or if an error happened (type=non-zero).

5.349.2 Member Function Documentation

5.349.2.1 deserialize()

```
void mavlink::common::msg::MISSION_ACK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.349.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_ACK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.349.2.3 serialize()

```
void mavlink::common::msg::MISSION_ACK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

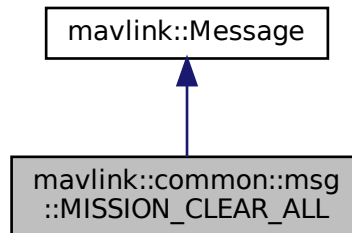
- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_ack.hpp`

5.350 mavlink::common::msg::MISSION_CLEAR_ALL Struct Reference

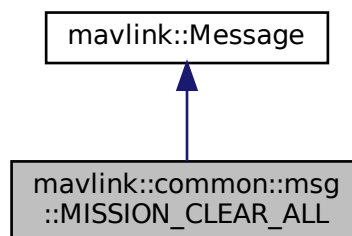
[MISSION_CLEAR_ALL](#) message.

```
#include <mavlink_msg_mission_clear_all.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_CLEAR_ALL:



Collaboration diagram for mavlink::common::msg::MISSION_CLEAR_ALL:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **mission_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 45
- static constexpr size_t **LENGTH** = 3
- static constexpr size_t **MIN_LENGTH** = 2
- static constexpr uint8_t **CRC_EXTRA** = 232
- static constexpr auto **NAME** = "MISSION_CLEAR_ALL"

5.350.1 Detailed Description

[MISSION_CLEAR_ALL](#) message.

Delete all mission items at once.

5.350.2 Member Function Documentation

5.350.2.1 deserialize()

```
void mavlink::common::msg::MISSION_CLEAR_ALL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.350.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_CLEAR_ALL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.350.2.3 serialize()

```
void mavlink::common::msg::MISSION_CLEAR_ALL::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

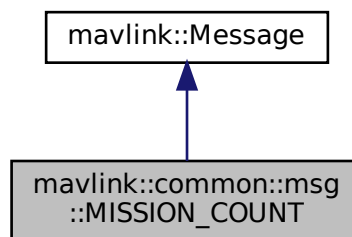
- single_hand_controller/include/mavlink/common/mavlink_msg_mission_clear_all.hpp

5.351 mavlink::common::msg::MISSION_COUNT Struct Reference

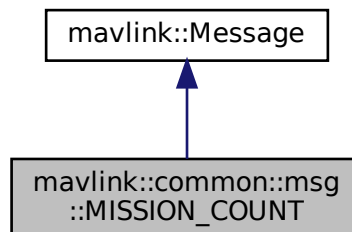
[MISSION_COUNT](#) message.

```
#include <mavlink_msg_mission_count.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_COUNT:



Collaboration diagram for mavlink::common::msg::MISSION_COUNT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint16_t` **count**
- `uint8_t` **mission_type**
- `uint32_t` **opaque_id**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 44
- static constexpr `size_t` **LENGTH** = 9
- static constexpr `size_t` **MIN_LENGTH** = 4
- static constexpr `uint8_t` **CRC_EXTRA** = 221
- static constexpr auto **NAME** = "MISSION_COUNT"

5.351.1 Detailed Description

[MISSION_COUNT](#) message.

This message is emitted as response to [MISSION_REQUEST_LIST](#) by the MAV and to initiate a write transaction. The GCS can then request the individual mission item based on the knowledge of the total number of waypoints.

5.351.2 Member Function Documentation

5.351.2.1 `deserialize()`

```
void mavlink::common::msg::MISSION_COUNT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.351.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_COUNT::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.351.2.3 serialize()

```
void mavlink::common::msg::MISSION_COUNT::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

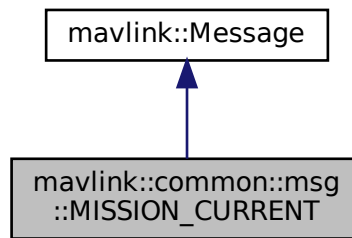
- single_hand_controller/include/mavlink/common/mavlink_msg_mission_count.hpp

5.352 mavlink::common::msg::MISSION_CURRENT Struct Reference

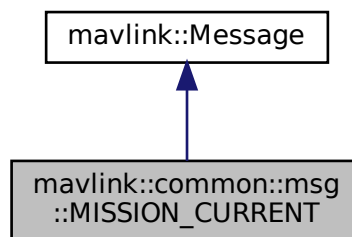
[MISSION_CURRENT](#) message.

```
#include <mavlink_msg_mission_current.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_CURRENT:



Collaboration diagram for mavlink::common::msg::MISSION_CURRENT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint16_t` **seq**
- `uint16_t` **total**

- uint8_t **mission_state**
- uint8_t **mission_mode**
- uint32_t **mission_id**
- uint32_t **fence_id**
- uint32_t **rally_points_id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 42
- static constexpr size_t **LENGTH** = 18
- static constexpr size_t **MIN_LENGTH** = 2
- static constexpr uint8_t **CRC_EXTRA** = 28
- static constexpr auto **NAME** = "MISSION_CURRENT"

5.352.1 Detailed Description

[MISSION_CURRENT](#) message.

Message that announces the sequence number of the current target mission item (that the system will fly toward). This message should be streamed all the time (nominally at 1Hz). This message should be emitted following a call to MAV_CMD_DO_SET_MISSION_CURRENT or MISSION_SET_CURRENT.

5.352.2 Member Function Documentation

5.352.2.1 deserialize()

```
void mavlink::common::msg::MISSION_CURRENT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.352.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_CURRENT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.352.2.3 serialize()

```
void mavlink::common::msg::MISSION_CURRENT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

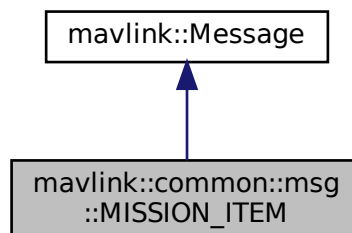
- single_hand_controller/include/mavlink/common/mavlink_msg_mission_current.hpp

5.353 mavlink::common::msg::MISSION_ITEM Struct Reference

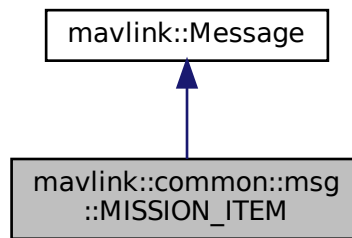
[MISSION_ITEM](#) message.

```
#include <mavlink_msg_mission_item.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_ITEM:



Collaboration diagram for mavlink::common::msg::MISSION_ITEM:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint16_t` **seq**
- `uint8_t` **frame**
- `uint16_t` **command**
- `uint8_t` **current**
- `uint8_t` **autocontinue**
- `float` **param1**
- `float` **param2**
- `float` **param3**
- `float` **param4**
- `float` **x**
- `float` **y**
- `float` **z**
- `uint8_t` **mission_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 39
- static constexpr size_t **LENGTH** = 38
- static constexpr size_t **MIN_LENGTH** = 37
- static constexpr uint8_t **CRC_EXTRA** = 254
- static constexpr auto **NAME** = "MISSION_ITEM"

5.353.1 Detailed Description

[MISSION_ITEM](#) message.

[Message](#) encoding a mission item. This message is emitted to announce the presence of a mission item and to set a mission item on the system. The mission item can be either in x, y, z meters (type: LOCAL) or x:lat, y:lon, z:altitude. Local frame is Z-down, right handed (NED), global frame is Z-up, right handed (ENU). NaN may be used to indicate an optional/default value (e.g. to use the system's current latitude or yaw rather than a specific value). See also <https://mavlink.io/en/services/mission.html>.

5.353.2 Member Function Documentation

5.353.2.1 deserialize()

```
void mavlink::common::msg::MISSION_ITEM::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.353.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_ITEM::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.353.2.3 serialize()

```
void mavlink::common::msg::MISSION_ITEM_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

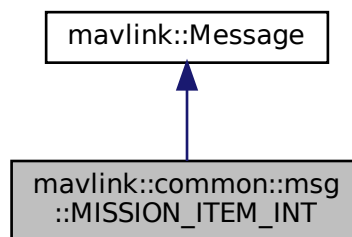
- single_hand_controller/include/mavlink/common/mavlink_msg_mission_item.hpp

5.354 mavlink::common::msg::MISSION_ITEM_INT Struct Reference

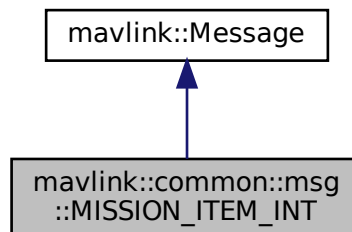
[MISSION_ITEM_INT](#) message.

```
#include <mavlink_msg_mission_item_int.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_ITEM_INT:



Collaboration diagram for mavlink::common::msg::MISSION_ITEM_INT:



Public Member Functions

- `std::string` `get_name` (void) const override
Get NAME constant.
- `Info` `get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` `to_yaml` (void) const override
Make YAML-string from message content.
- void `serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- void `deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint8_t` `target_system`
- `uint8_t` `target_component`
- `uint16_t` `seq`
- `uint8_t` `frame`
- `uint16_t` `command`
- `uint8_t` `current`
- `uint8_t` `autocontinue`
- `float` `param1`
- `float` `param2`
- `float` `param3`
- `float` `param4`
- `int32_t` `x`
- `int32_t` `y`
- `float` `z`
- `uint8_t` `mission_type`

Static Public Attributes

- static constexpr `msgid_t` `MSG_ID` = 73
- static constexpr `size_t` `LENGTH` = 38
- static constexpr `size_t` `MIN_LENGTH` = 37
- static constexpr `uint8_t` `CRC_EXTRA` = 38
- static constexpr auto `NAME` = "MISSION_ITEM_INT"

5.354.1 Detailed Description

`MISSION_ITEM_INT` message.

`Message` encoding a mission item. This message is emitted to announce the presence of a mission item and to set a mission item on the system. The mission item can be either in x, y, z meters (type: LOCAL) or x:lat, y:lon, z:altitude. Local frame is Z-down, right handed (NED), global frame is Z-up, right handed (ENU). NaN or INT32_MAX may be used in float/integer params (respectively) to indicate optional/default values (e.g. to use the component's current latitude, yaw rather than a specific value). See also <https://mavlink.io/en/services/mission.html>.

5.354.2 Member Function Documentation

5.354.2.1 deserialize()

```
void mavlink::common::msg::MISSION_ITEM_INT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.354.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_ITEM_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.354.2.3 serialize()

```
void mavlink::common::msg::MISSION_ITEM_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

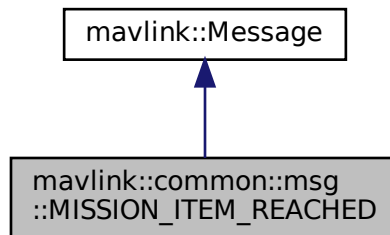
- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_int.hpp`

5.355 mavlink::common::msg::MISSION_ITEM_REACHED Struct Reference

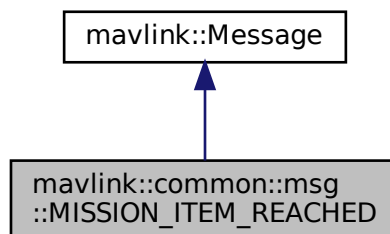
[MISSION_ITEM_REACHED](#) message.

```
#include <mavlink_msg_mission_item_reached.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_ITEM_REACHED:



Collaboration diagram for mavlink::common::msg::MISSION_ITEM_REACHED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint16_t **seq**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 46
- static constexpr size_t **LENGTH** = 2
- static constexpr size_t **MIN_LENGTH** = 2
- static constexpr uint8_t **CRC_EXTRA** = 11
- static constexpr auto **NAME** = "MISSION_ITEM_REACHED"

5.355.1 Detailed Description

[MISSION_ITEM_REACHED](#) message.

A certain mission item has been reached. The system will either hold this position (or circle on the orbit) or (if the autocontinue on the WP was set) continue to the next waypoint.

5.355.2 Member Function Documentation

5.355.2.1 deserialize()

```
void mavlink::common::msg::MISSION_ITEM_REACHED::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.355.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_ITEM_REACHED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.355.2.3 `serialize()`

```
void mavlink::common::msg::MISSION_ITEM_REACHED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

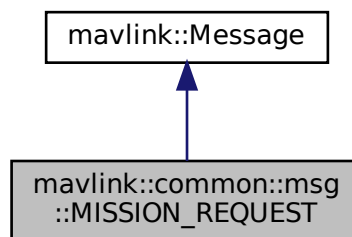
- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_item_reached.hpp`

5.356 `mavlink::common::msg::MISSION_REQUEST` Struct Reference

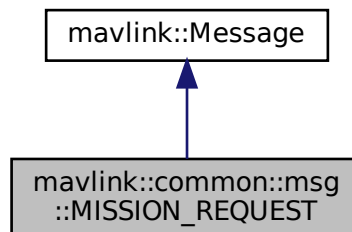
[MISSION_REQUEST](#) message.

```
#include <mavlink_msg_mission_request.hpp>
```

Inheritance diagram for `mavlink::common::msg::MISSION_REQUEST`:



Collaboration diagram for `mavlink::common::msg::MISSION_REQUEST`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- uint16_t **seq**
- uint8_t **mission_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 40
- static constexpr size_t **LENGTH** = 5
- static constexpr size_t **MIN_LENGTH** = 4
- static constexpr uint8_t **CRC_EXTRA** = 230
- static constexpr auto **NAME** = "MISSION_REQUEST"

5.356.1 Detailed Description

[MISSION_REQUEST](#) message.

Request the information of the mission item with the sequence number seq. The response of the system to this message should be a [MISSION_ITEM](#) message. <https://mavlink.io/en/services/mission.html>

5.356.2 Member Function Documentation

5.356.2.1 deserialize()

```
void mavlink::common::msg::MISSION_REQUEST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.356.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_REQUEST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.356.2.3 serialize()

```
void mavlink::common::msg::MISSION_REQUEST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

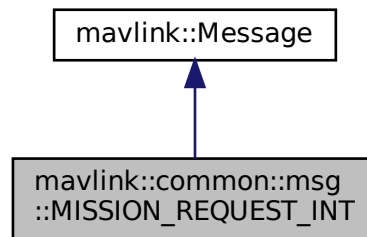
- single_hand_controller/include/mavlink/common/mavlink_msg_mission_request.hpp

5.357 mavlink::common::msg::MISSION_REQUEST_INT Struct Reference

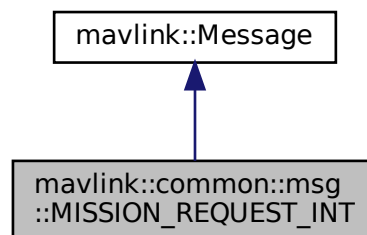
[MISSION_REQUEST_INT](#) message.

```
#include <mavlink_msg_mission_request_int.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_REQUEST_INT:



Collaboration diagram for mavlink::common::msg::MISSION_REQUEST_INT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint16_t` **seq**
- `uint8_t` **mission_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 51
- static constexpr size_t **LENGTH** = 5
- static constexpr size_t **MIN_LENGTH** = 4
- static constexpr uint8_t **CRC_EXTRA** = 196
- static constexpr auto **NAME** = "MISSION_REQUEST_INT"

5.357.1 Detailed Description

[MISSION_REQUEST_INT](#) message.

Request the information of the mission item with the sequence number seq. The response of the system to this message should be a [MISSION_ITEM_INT](#) message. <https://mavlink.io/en/services/mission.html>

5.357.2 Member Function Documentation

5.357.2.1 deserialize()

```
void mavlink::common::msg::MISSION_REQUEST_INT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.357.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_REQUEST_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.357.2.3 serialize()

```
void mavlink::common::msg::MISSION_REQUEST_INT::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

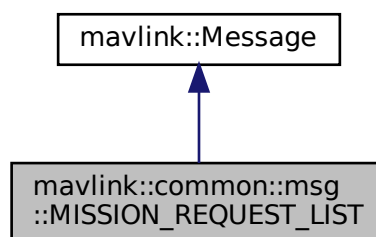
- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_int.hpp`

5.358 `mavlink::common::msg::MISSION_REQUEST_LIST` Struct Reference

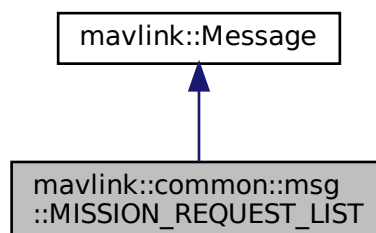
[MISSION_REQUEST_LIST](#) message.

```
#include <mavlink_msg_mission_request_list.hpp>
```

Inheritance diagram for `mavlink::common::msg::MISSION_REQUEST_LIST`:



Collaboration diagram for `mavlink::common::msg::MISSION_REQUEST_LIST`:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **mission_type**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 43
- static constexpr `size_t` **LENGTH** = 3
- static constexpr `size_t` **MIN_LENGTH** = 2
- static constexpr `uint8_t` **CRC_EXTRA** = 132
- static constexpr auto **NAME** = "MISSION_REQUEST_LIST"

5.358.1 Detailed Description

[MISSION_REQUEST_LIST](#) message.

Request the overall list of mission items from the system/component.

5.358.2 Member Function Documentation

5.358.2.1 deserialize()

```
void mavlink::common::msg::MISSION_REQUEST_LIST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.358.2.2 `get_name()`

```
std::string mavlink::common::msg::MISSION_REQUEST_LIST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.358.2.3 `serialize()`

```
void mavlink::common::msg::MISSION_REQUEST_LIST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

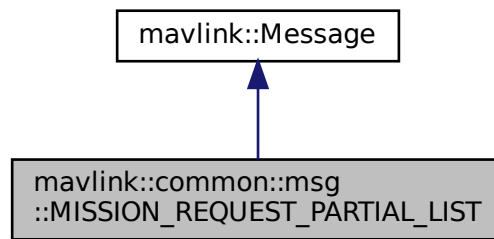
- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_list.hpp`

5.359 `mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST` Struct Reference

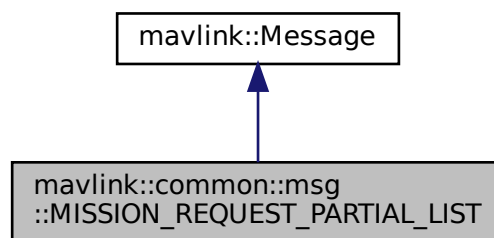
[MISSION_REQUEST_PARTIAL_LIST](#) message.

```
#include <mavlink_msg_mission_request_partial_list.hpp>
```


Inheritance diagram for mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST:



Collaboration diagram for mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `int16_t` **start_index**
- `int16_t` **end_index**
- `uint8_t` **mission_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 37
- static constexpr size_t **LENGTH** = 7
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 212
- static constexpr auto **NAME** = "MISSION_REQUEST_PARTIAL_LIST"

5.359.1 Detailed Description

[MISSION_REQUEST_PARTIAL_LIST](#) message.

Request a partial list of mission items from the system/component. <https://mavlink.io/en/services/mission.html>. If start and end index are the same, just send one waypoint.

5.359.2 Member Function Documentation

5.359.2.1 deserialize()

```
void mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.359.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.359.2.3 serialize()

```
void mavlink::common::msg::MISSION_REQUEST_PARTIAL_LIST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

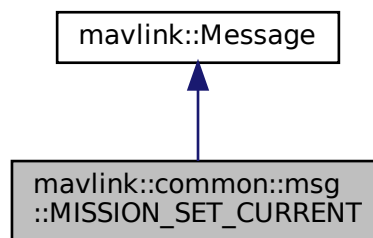
- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_request_partial_list.hpp`

5.360 mavlink::common::msg::MISSION_SET_CURRENT Struct Reference

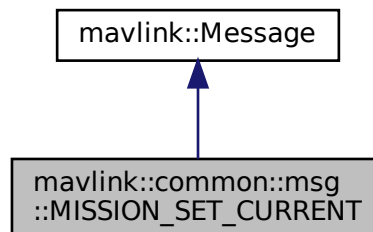
[MISSION_SET_CURRENT](#) message.

```
#include <mavlink_msg_mission_set_current.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_SET_CURRENT:



Collaboration diagram for mavlink::common::msg::MISSION_SET_CURRENT:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `uint16_t seq`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 41`
- `static constexpr size_t LENGTH = 4`
- `static constexpr size_t MIN_LENGTH = 4`
- `static constexpr uint8_t CRC_EXTRA = 28`
- `static constexpr auto NAME = "MISSION_SET_CURRENT"`

5.360.1 Detailed Description

`MISSION_SET_CURRENT` message.

Set the mission item with sequence number `seq` as the current item and emit `MISSION_CURRENT` (whether or not the mission is currently being executed, the system will continue to this new mission item on the shortest possible time). Note that mission jump repeat counters are not reset (see `MAV_CMD_DO_JUMP` param2).

This message may trigger a mission state-machine change on some systems: for example from `MISSION_STATE_NOT_STARTED` to `MISSION_STATE_RUNNING`. If the system is in mission mode, on those systems this command might therefore start, restart or resume the mission. If the system is not in mission mode this message must not trigger a switch to mission mode.

5.360.2 Member Function Documentation

5.360.2.1 deserialize()

```
void mavlink::common::msg::MISSION_SET_CURRENT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.360.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_SET_CURRENT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.360.2.3 serialize()

```
void mavlink::common::msg::MISSION_SET_CURRENT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

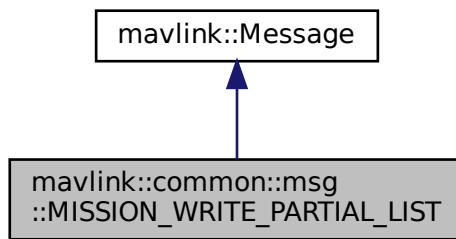
- single_hand_controller/include/mavlink/common/mavlink_msg_mission_set_current.hpp

5.361 mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST Struct Reference

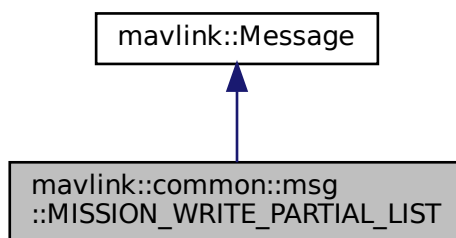
[MISSION_WRITE_PARTIAL_LIST](#) message.

```
#include <mavlink_msg_mission_write_partial_list.hpp>
```

Inheritance diagram for mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST:



Collaboration diagram for mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `int16_t` **start_index**
- `int16_t` **end_index**
- `uint8_t` **mission_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 38
- static constexpr size_t **LENGTH** = 7
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 9
- static constexpr auto **NAME** = "MISSION_WRITE_PARTIAL_LIST"

5.361.1 Detailed Description

[MISSION_WRITE_PARTIAL_LIST](#) message.

This message is sent to the MAV to write a partial list. If start index == end index, only one item will be transmitted / updated. If the start index is NOT 0 and above the current list size, this request should be REJECTED!

5.361.2 Member Function Documentation

5.361.2.1 deserialize()

```
void mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.361.2.2 get_name()

```
std::string mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.361.2.3 serialize()

```
void mavlink::common::msg::MISSION_WRITE_PARTIAL_LIST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

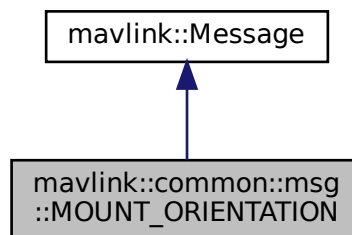
- `single_hand_controller/include/mavlink/common/mavlink_msg_mission_write_partial_list.hpp`

5.362 `mavlink::common::msg::MOUNT_ORIENTATION` Struct Reference

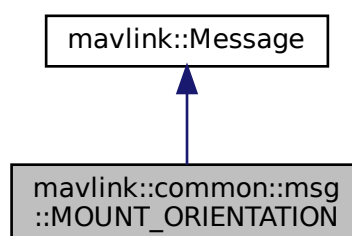
`MOUNT_ORIENTATION` message.

```
#include <mavlink_msg_mount_orientation.hpp>
```

Inheritance diagram for `mavlink::common::msg::MOUNT_ORIENTATION`:



Collaboration diagram for `mavlink::common::msg::MOUNT_ORIENTATION`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- float **roll**
- float **pitch**
- float **yaw**
- float **yaw_absolute**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 265
- static constexpr size_t **LENGTH** = 20
- static constexpr size_t **MIN_LENGTH** = 16
- static constexpr uint8_t **CRC_EXTRA** = 26
- static constexpr auto **NAME** = "MOUNT_ORIENTATION"

5.362.1 Detailed Description

[MOUNT_ORIENTATION](#) message.

Orientation of a mount

5.362.2 Member Function Documentation

5.362.2.1 deserialize()

```
void mavlink::common::msg::MOUNT_ORIENTATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.362.2.2 get_name()

```
std::string mavlink::common::msg::MOUNT_ORIENTATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.362.2.3 serialize()

```
void mavlink::common::msg::MOUNT_ORIENTATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_mount_orientation.hpp

5.363 mavlink::MsgMap Class Reference

Serialization helper wrapper for mavlink_message_t.

```
#include <msgmap.hpp>
```

Public Member Functions

- **MsgMap** (mavlink_message_t *p)
- **MsgMap** (mavlink_message_t &p)
- **MsgMap** (const mavlink_message_t *p)
- void **reset** ()
- void **reset** (uint32_t msgid, uint8_t len)
- template<typename _T >
void **operator**<< (const _T data)
- template<class _T, size_t _Size>
void **operator**<< (const std::array< _T, _Size > &data)
- template<typename _T >
void **operator**>> (_T &data)
- template<class _T, size_t _Size>
void **operator**>> (std::array< _T, _Size > &data)

5.363.1 Detailed Description

Serialization helper wrapper for mavlink_message_t.

The documentation for this class was generated from the following file:

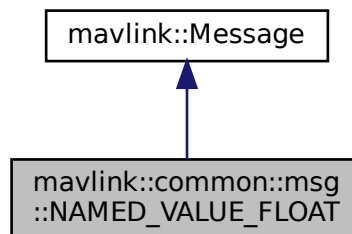
- single_hand_controller/include/mavlink/msgmap.hpp

5.364 mavlink::common::msg::NAMED_VALUE_FLOAT Struct Reference

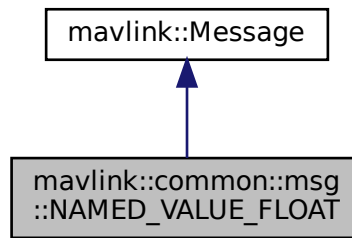
[NAMED_VALUE_FLOAT](#) message.

```
#include <mavlink_msg_named_value_float.hpp>
```

Inheritance diagram for mavlink::common::msg::NAMED_VALUE_FLOAT:



Collaboration diagram for mavlink::common::msg::NAMED_VALUE_FLOAT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `std::array< char, 10 >` **name**
- `float` **value**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 251
- static constexpr `size_t` **LENGTH** = 18
- static constexpr `size_t` **MIN_LENGTH** = 18
- static constexpr `uint8_t` **CRC_EXTRA** = 170
- static constexpr auto **NAME** = "NAMED_VALUE_FLOAT"

5.364.1 Detailed Description

[NAMED_VALUE_FLOAT](#) message.

Send a key-value pair as float. The use of this message is discouraged for normal packets, but a quite efficient way for testing new messages and getting experimental debug output.

5.364.2 Member Function Documentation

5.364.2.1 deserialize()

```
void mavlink::common::msg::NAMED_VALUE_FLOAT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.364.2.2 get_name()

```
std::string mavlink::common::msg::NAMED_VALUE_FLOAT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.364.2.3 serialize()

```
void mavlink::common::msg::NAMED_VALUE_FLOAT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

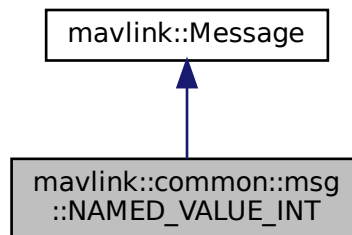
- single_hand_controller/include/mavlink/common/mavlink_msg_named_value_float.hpp

5.365 mavlink::common::msg::NAMED_VALUE_INT Struct Reference

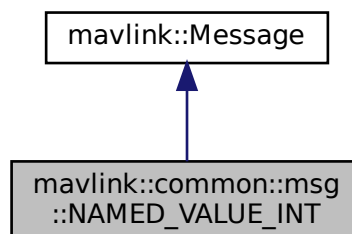
[NAMED_VALUE_INT](#) message.

```
#include <mavlink_msg_named_value_int.hpp>
```

Inheritance diagram for mavlink::common::msg::NAMED_VALUE_INT:



Collaboration diagram for mavlink::common::msg::NAMED_VALUE_INT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- std::array< char, 10 > **name**
- int32_t **value**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 252
- static constexpr size_t **LENGTH** = 18
- static constexpr size_t **MIN_LENGTH** = 18
- static constexpr uint8_t **CRC_EXTRA** = 44
- static constexpr auto **NAME** = "NAMED_VALUE_INT"

5.365.1 Detailed Description

[NAMED_VALUE_INT](#) message.

Send a key-value pair as integer. The use of this message is discouraged for normal packets, but a quite efficient way for testing new messages and getting experimental debug output.

5.365.2 Member Function Documentation

5.365.2.1 deserialize()

```
void mavlink::common::msg::NAMED_VALUE_INT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.365.2.2 get_name()

```
std::string mavlink::common::msg::NAMED_VALUE_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.365.2.3 `serialize()`

```
void mavlink::common::msg::NAMED_VALUE_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

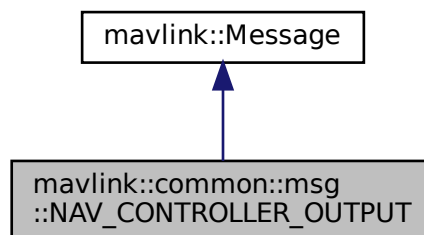
- `single_hand_controller/include/mavlink/common/mavlink_msg_named_value_int.hpp`

5.366 `mavlink::common::msg::NAV_CONTROLLER_OUTPUT` Struct Reference

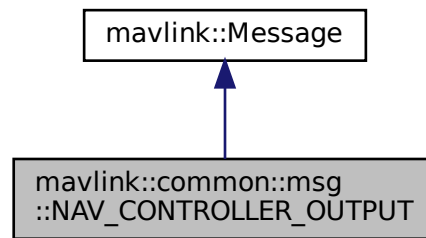
[NAV_CONTROLLER_OUTPUT](#) message.

```
#include <mavlink_msg_nav_controller_output.hpp>
```

Inheritance diagram for `mavlink::common::msg::NAV_CONTROLLER_OUTPUT`:



Collaboration diagram for mavlink::common::msg::NAV_CONTROLLER_OUTPUT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- float **nav_roll**
- float **nav_pitch**
- int16_t **nav_bearing**
- int16_t **target_bearing**
- uint16_t **wp_dist**
- float **alt_error**
- float **aspd_error**
- float **xtrack_error**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 62
- static constexpr size_t **LENGTH** = 26
- static constexpr size_t **MIN_LENGTH** = 26
- static constexpr uint8_t **CRC_EXTRA** = 183
- static constexpr auto **NAME** = "NAV_CONTROLLER_OUTPUT"

5.366.1 Detailed Description

[NAV_CONTROLLER_OUTPUT](#) message.

The state of the navigation and position controller.

5.366.2 Member Function Documentation

5.366.2.1 deserialize()

```
void mavlink::common::msg::NAV_CONTROLLER_OUTPUT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.366.2.2 get_name()

```
std::string mavlink::common::msg::NAV_CONTROLLER_OUTPUT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.366.2.3 serialize()

```
void mavlink::common::msg::NAV_CONTROLLER_OUTPUT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

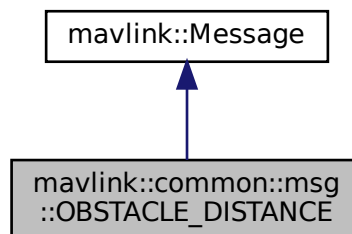
- `single_hand_controller/include/mavlink/common/mavlink_msg_nav_controller_output.hpp`

5.367 mavlink::common::msg::OBSTACLE_DISTANCE Struct Reference

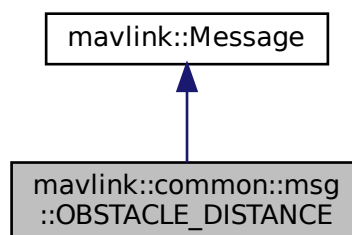
[OBSTACLE_DISTANCE](#) message.

```
#include <mavlink_msg_obstacle_distance.hpp>
```

Inheritance diagram for mavlink::common::msg::OBSTACLE_DISTANCE:



Collaboration diagram for mavlink::common::msg::OBSTACLE_DISTANCE:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- `void deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint64_t time_usec`
- `uint8_t sensor_type`
- `std::array< uint16_t, 72 > distances`
- `uint8_t increment`
- `uint16_t min_distance`
- `uint16_t max_distance`
- `float increment_f`
- `float angle_offset`
- `uint8_t frame`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 330`
- `static constexpr size_t LENGTH = 167`
- `static constexpr size_t MIN_LENGTH = 158`
- `static constexpr uint8_t CRC_EXTRA = 23`
- `static constexpr auto NAME = "OBSTACLE_DISTANCE"`

5.367.1 Detailed Description

[OBSTACLE_DISTANCE](#) message.

Obstacle distances in front of the sensor, starting from the left in increment degrees to the right

5.367.2 Member Function Documentation

5.367.2.1 deserialize()

```
void mavlink::common::msg::OBSTACLE_DISTANCE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.367.2.2 get_name()

```
std::string mavlink::common::msg::OBSTACLE_DISTANCE::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.367.2.3 serialize()

```
void mavlink::common::msg::OBSTACLE_DISTANCE::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

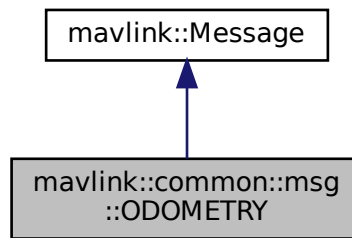
- single_hand_controller/include/mavlink/common/mavlink_msg_obstacle_distance.hpp

5.368 mavlink::common::msg::ODOMETRY Struct Reference

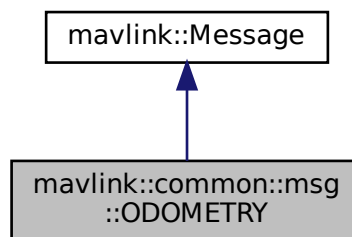
[ODOMETRY](#) message.

```
#include <mavlink_msg_odometry.hpp>
```

Inheritance diagram for mavlink::common::msg::ODOMETRY:



Collaboration diagram for mavlink::common::msg::ODOMETRY:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` [time_usec](#)
- `uint8_t` [frame_id](#)

- uint8_t **child_frame_id**
- float **x**
- float **y**
- float **z**
- std::array< float, 4 > **q**
- float **vx**
- float **vy**
- float **vz**
- float **rollspeed**
- float **pitchspeed**
- float **yawspeed**
- std::array< float, 21 > **pose_covariance**
- std::array< float, 21 > **velocity_covariance**
- uint8_t **reset_counter**
- uint8_t **estimator_type**
- int8_t **quality**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 331
- static constexpr size_t **LENGTH** = 233
- static constexpr size_t **MIN_LENGTH** = 230
- static constexpr uint8_t **CRC_EXTRA** = 91
- static constexpr auto **NAME** = "ODOMETRY"

5.368.1 Detailed Description

[ODOMETRY](#) message.

Odometry message to communicate odometry information with an external interface. Fits ROS REP 147 standard for aerial vehicles (<http://www.ros.org/reps/rep-0147.html>).

5.368.2 Member Function Documentation

5.368.2.1 deserialize()

```
void mavlink::common::msg::ODOMETRY::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.368.2.2 `get_name()`

```
std::string mavlink::common::msg::ODOMETRY::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.368.2.3 `serialize()`

```
void mavlink::common::msg::ODOMETRY::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

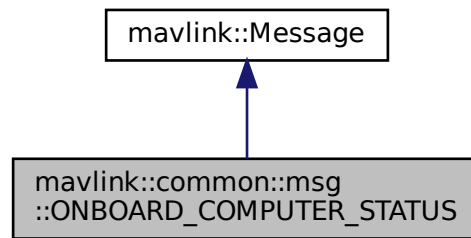
- `single_hand_controller/include/mavlink/common/mavlink_msg_odometry.hpp`

5.369 `mavlink::common::msg::ONBOARD_COMPUTER_STATUS` Struct Reference

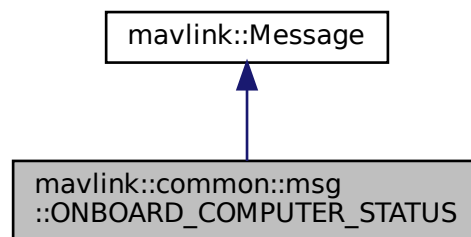
[ONBOARD_COMPUTER_STATUS](#) message.

```
#include <mavlink_msg_onboard_computer_status.hpp>
```


Inheritance diagram for mavlink::common::msg::ONBOARD_COMPUTER_STATUS:



Collaboration diagram for mavlink::common::msg::ONBOARD_COMPUTER_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint32_t` **uptime**

- `uint8_t` **type**
- `std::array< uint8_t, 8 >` **cpu_cores**
- `std::array< uint8_t, 10 >` **cpu_combined**
- `std::array< uint8_t, 4 >` **gpu_cores**
- `std::array< uint8_t, 10 >` **gpu_combined**
- `int8_t` **temperature_board**
- `std::array< int8_t, 8 >` **temperature_core**
- `std::array< int16_t, 4 >` **fan_speed**
- `uint32_t` **ram_usage**
- `uint32_t` **ram_total**
- `std::array< uint32_t, 4 >` **storage_type**
- `std::array< uint32_t, 4 >` **storage_usage**
- `std::array< uint32_t, 4 >` **storage_total**
- `std::array< uint32_t, 6 >` **link_type**
- `std::array< uint32_t, 6 >` **link_tx_rate**
- `std::array< uint32_t, 6 >` **link_rx_rate**
- `std::array< uint32_t, 6 >` **link_tx_max**
- `std::array< uint32_t, 6 >` **link_rx_max**
- `uint16_t` **status_flags**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 390
- static constexpr `size_t` **LENGTH** = 240
- static constexpr `size_t` **MIN_LENGTH** = 238
- static constexpr `uint8_t` **CRC_EXTRA** = 156
- static constexpr auto **NAME** = "ONBOARD_COMPUTER_STATUS"

5.369.1 Detailed Description

[ONBOARD_COMPUTER_STATUS](#) message.

Hardware status sent by an onboard computer.

5.369.2 Member Function Documentation

5.369.2.1 deserialize()

```
void mavlink::common::msg::ONBOARD_COMPUTER_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.369.2.2 get_name()

```
std::string mavlink::common::msg::ONBOARD_COMPUTER_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.369.2.3 serialize()

```
void mavlink::common::msg::ONBOARD_COMPUTER_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

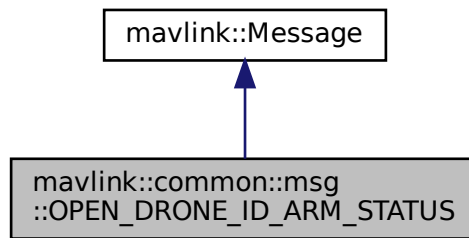
- single_hand_controller/include/mavlink/common/mavlink_msg_onboard_computer_status.hpp

5.370 mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS Struct Reference

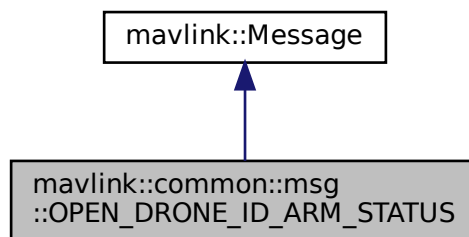
[OPEN_DRONE_ID_ARM_STATUS](#) message.

```
#include <mavlink_msg_open_drone_id_arm_status.hpp>
```

Inheritance diagram for mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS:



Collaboration diagram for mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **status**
- `std::array< char, 50 >` **error**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 12918
- static constexpr size_t **LENGTH** = 51
- static constexpr size_t **MIN_LENGTH** = 51
- static constexpr uint8_t **CRC_EXTRA** = 139
- static constexpr auto **NAME** = "OPEN_DRONE_ID_ARM_STATUS"

5.370.1 Detailed Description

[OPEN_DRONE_ID_ARM_STATUS](#) message.

Transmitter (remote ID system) is enabled and ready to start sending location and other required information. This is streamed by transmitter. A flight controller uses it as a condition to arm.

5.370.2 Member Function Documentation

5.370.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.370.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.370.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_ARM_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

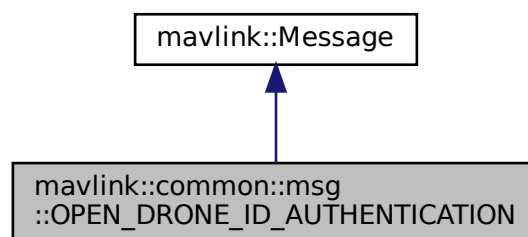
- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_arm_status.hpp`

5.371 `mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION` Struct Reference

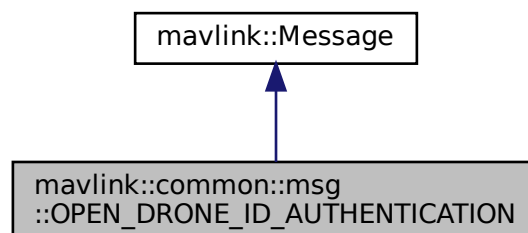
[OPEN_DRONE_ID_AUTHENTICATION](#) message.

```
#include <mavlink_msg_open_drone_id_authentication.hpp>
```

Inheritance diagram for `mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION`:



Collaboration diagram for `mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION`:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< uint8_t, 20 >` **id_or_mac**
- `uint8_t` **authentication_type**
- `uint8_t` **data_page**
- `uint8_t` **last_page_index**
- `uint8_t` **length**
- `uint32_t` **timestamp**
- `std::array< uint8_t, 23 >` **authentication_data**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 12902
- static constexpr `size_t` **LENGTH** = 53
- static constexpr `size_t` **MIN_LENGTH** = 53
- static constexpr `uint8_t` **CRC_EXTRA** = 140
- static constexpr auto **NAME** = "OPEN_DRONE_ID_AUTHENTICATION"

5.371.1 Detailed Description

[OPEN_DRONE_ID_AUTHENTICATION](#) message.

Data for filling the OpenDroneID Authentication message. The Authentication [Message](#) defines a field that can provide a means of authenticity for the identity of the UAS (Unmanned Aircraft System). The Authentication message can have two different formats. For data page 0, the fields PageCount, Length and TimeStamp are present and AuthData is only 17 bytes. For data page 1 through 15, PageCount, Length and TimeStamp are not present and the size of AuthData is 23 bytes.

5.371.2 Member Function Documentation

5.371.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.371.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.371.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_AUTHENTICATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

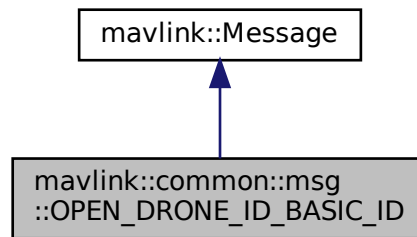
- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_authentication.hpp

5.372 mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID Struct Reference

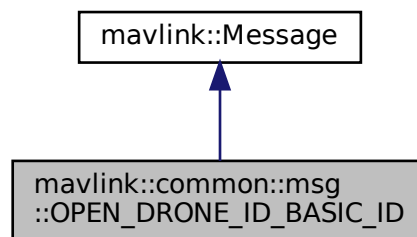
[OPEN_DRONE_ID_BASIC_ID](#) message.

```
#include <mavlink_msg_open_drone_id_basic_id.hpp>
```


Inheritance diagram for mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID:



Collaboration diagram for mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< uint8_t, 20 >` **id_or_mac**
- `uint8_t` **id_type**
- `uint8_t` **ua_type**
- `std::array< uint8_t, 20 >` **uas_id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 12900
- static constexpr size_t **LENGTH** = 44
- static constexpr size_t **MIN_LENGTH** = 44
- static constexpr uint8_t **CRC_EXTRA** = 114
- static constexpr auto **NAME** = "OPEN_DRONE_ID_BASIC_ID"

5.372.1 Detailed Description

[OPEN_DRONE_ID_BASIC_ID](#) message.

Data for filling the OpenDroneID Basic ID message. This and the below messages are primarily meant for feeding data to/from an OpenDroneID implementation. E.g. <https://github.com/opendroneid/opendroneid-core-c>. These messages are compatible with the ASTM F3411 Remote ID standard and the ASD-STAN prEN 4709-002 Direct Remote ID standard. Additional information and usage of these messages is documented at <https://mavlink.io/en/services/opendroneid.html>.

5.372.2 Member Function Documentation

5.372.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.372.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.372.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_BASIC_ID::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

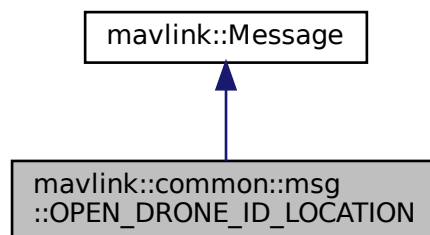
- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_basic_id.hpp

5.373 mavlink::common::msg::OPEN_DRONE_ID_LOCATION Struct Reference

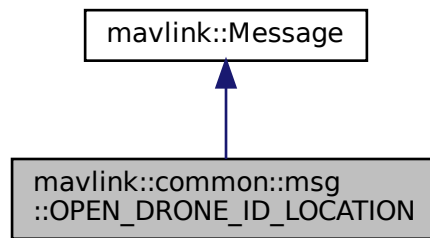
[OPEN_DRONE_ID_LOCATION](#) message.

```
#include <mavlink_msg_open_drone_id_location.hpp>
```

Inheritance diagram for mavlink::common::msg::OPEN_DRONE_ID_LOCATION:



Collaboration diagram for mavlink::common::msg::OPEN_DRONE_ID_LOCATION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< uint8_t, 20 >` **id_or_mac**
- `uint8_t` **status**
- `uint16_t` **direction**
- `uint16_t` **speed_horizontal**
- `int16_t` **speed_vertical**
- `int32_t` **latitude**
- `int32_t` **longitude**
- `float` **altitude_barometric**
- `float` **altitude_geodetic**
- `uint8_t` **height_reference**
- `float` **height**
- `uint8_t` **horizontal_accuracy**
- `uint8_t` **vertical_accuracy**
- `uint8_t` **barometer_accuracy**
- `uint8_t` **speed_accuracy**
- `float` **timestamp**
- `uint8_t` **timestamp_accuracy**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 12901
- static constexpr size_t **LENGTH** = 59
- static constexpr size_t **MIN_LENGTH** = 59
- static constexpr uint8_t **CRC_EXTRA** = 254
- static constexpr auto **NAME** = "OPEN_DRONE_ID_LOCATION"

5.373.1 Detailed Description

[OPEN_DRONE_ID_LOCATION](#) message.

Data for filling the OpenDroneID Location message. The float data types are 32-bit IEEE 754. The Location message provides the location, altitude, direction and speed of the aircraft.

5.373.2 Member Function Documentation

5.373.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_LOCATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.373.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_LOCATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.373.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_LOCATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

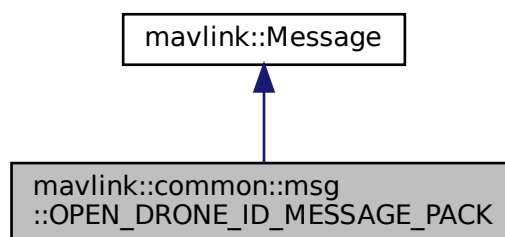
- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_location.hpp`

5.374 `mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK` Struct Reference

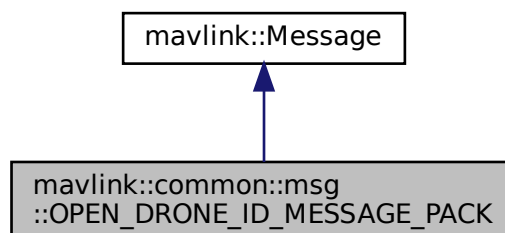
[OPEN_DRONE_ID_MESSAGE_PACK](#) message.

```
#include <mavlink_msg_open_drone_id_message_pack.hpp>
```

Inheritance diagram for `mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK`:



Collaboration diagram for `mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- std::array< uint8_t, 20 > **id_or_mac**
- uint8_t **single_message_size**
- uint8_t **msg_pack_size**
- std::array< uint8_t, 225 > **messages**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 12915
- static constexpr size_t **LENGTH** = 249
- static constexpr size_t **MIN_LENGTH** = 249
- static constexpr uint8_t **CRC_EXTRA** = 94
- static constexpr auto **NAME** = "OPEN_DRONE_ID_MESSAGE_PACK"

5.374.1 Detailed Description

[OPEN_DRONE_ID_MESSAGE_PACK](#) message.

An OpenDroneID message pack is a container for multiple encoded OpenDroneID messages (i.e. not in the format given for the above message descriptions but after encoding into the compressed OpenDroneID byte format). Used e.g. when transmitting on Bluetooth 5.0 Long Range/Extended Advertising or on WiFi Neighbor Aware Networking or on WiFi Beacon.

5.374.2 Member Function Documentation

5.374.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.374.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.374.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_MESSAGE_PACK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

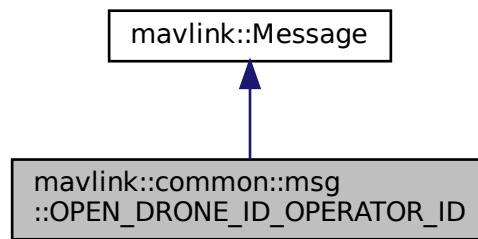
- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_message_pack.hpp

5.375 mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID Struct Reference

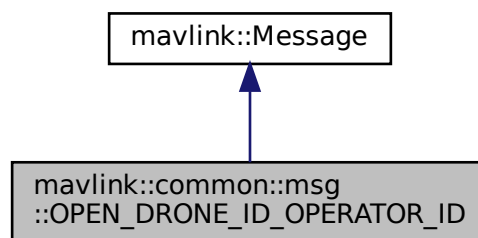
[OPEN_DRONE_ID_OPERATOR_ID](#) message.

```
#include <mavlink_msg_open_drone_id_operator_id.hpp>
```


Inheritance diagram for mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID:



Collaboration diagram for mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< uint8_t, 20 >` **id_or_mac**
- `uint8_t` **operator_id_type**
- `std::array< char, 20 >` **operator_id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 12905
- static constexpr size_t **LENGTH** = 43
- static constexpr size_t **MIN_LENGTH** = 43
- static constexpr uint8_t **CRC_EXTRA** = 49
- static constexpr auto **NAME** = "OPEN_DRONE_ID_OPERATOR_ID"

5.375.1 Detailed Description

[OPEN_DRONE_ID_OPERATOR_ID](#) message.

Data for filling the OpenDroneID Operator ID message, which contains the CAA (Civil Aviation Authority) issued operator ID.

5.375.2 Member Function Documentation

5.375.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.375.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.375.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_OPERATOR_ID::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

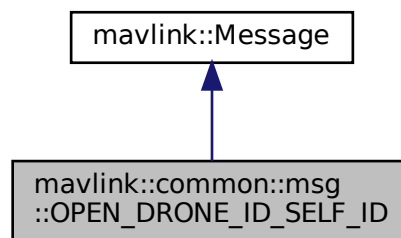
- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_operator_id.hpp

5.376 mavlink::common::msg::OPEN_DRONE_ID_SELF_ID Struct Reference

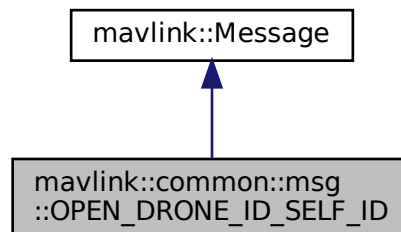
[OPEN_DRONE_ID_SELF_ID](#) message.

```
#include <mavlink_msg_open_drone_id_self_id.hpp>
```

Inheritance diagram for mavlink::common::msg::OPEN_DRONE_ID_SELF_ID:



Collaboration diagram for mavlink::common::msg::OPEN_DRONE_ID_SELF_ID:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< uint8_t, 20 >` **id_or_mac**
- `uint8_t` **description_type**
- `std::array< char, 23 >` **description**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 12903
- static constexpr `size_t` **LENGTH** = 46
- static constexpr `size_t` **MIN_LENGTH** = 46
- static constexpr `uint8_t` **CRC_EXTRA** = 249
- static constexpr auto **NAME** = "OPEN_DRONE_ID_SELF_ID"

5.376.1 Detailed Description

[OPEN_DRONE_ID_SELF_ID](#) message.

Data for filling the OpenDroneID Self ID message. The Self ID [Message](#) is an opportunity for the operator to (optionally) declare their identity and purpose of the flight. This message can provide additional information that could reduce the threat profile of a UA (Unmanned Aircraft) flying in a particular area or manner. This message can also be used to provide optional additional clarification in an emergency/remote ID system failure situation.

5.376.2 Member Function Documentation

5.376.2.1 `deserialize()`

```
void mavlink::common::msg::OPEN_DRONE_ID_SELF_ID::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.376.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_SELF_ID::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.376.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_SELF_ID::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

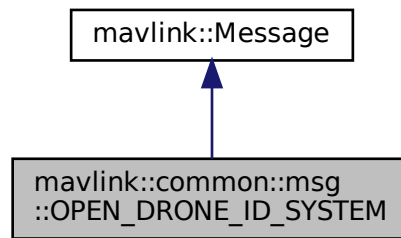
- single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_self_id.hpp

5.377 mavlink::common::msg::OPEN_DRONE_ID_SYSTEM Struct Reference

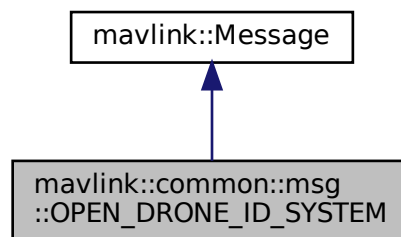
[OPEN_DRONE_ID_SYSTEM](#) message.

```
#include <mavlink_msg_open_drone_id_system.hpp>
```

Inheritance diagram for mavlink::common::msg::OPEN_DRONE_ID_SYSTEM:



Collaboration diagram for mavlink::common::msg::OPEN_DRONE_ID_SYSTEM:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**

- `std::array< uint8_t, 20 > id_or_mac`
- `uint8_t operator_location_type`
- `uint8_t classification_type`
- `int32_t operator_latitude`
- `int32_t operator_longitude`
- `uint16_t area_count`
- `uint16_t area_radius`
- `float area_ceiling`
- `float area_floor`
- `uint8_t category_eu`
- `uint8_t class_eu`
- `float operator_altitude_geo`
- `uint32_t timestamp`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 12904`
- `static constexpr size_t LENGTH = 54`
- `static constexpr size_t MIN_LENGTH = 54`
- `static constexpr uint8_t CRC_EXTRA = 77`
- `static constexpr auto NAME = "OPEN_DRONE_ID_SYSTEM"`

5.377.1 Detailed Description

[OPEN_DRONE_ID_SYSTEM](#) message.

Data for filling the OpenDroneID System message. The System [Message](#) contains general system information including the operator location/altitude and possible aircraft group and/or category/class information.

5.377.2 Member Function Documentation

5.377.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_SYSTEM::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

<code>in</code>	<code>map</code>	
-----------------	------------------	--

Implements [mavlink::Message](#).

5.377.2.2 `get_name()`

```
std::string mavlink::common::msg::OPEN_DRONE_ID_SYSTEM::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.377.2.3 `serialize()`

```
void mavlink::common::msg::OPEN_DRONE_ID_SYSTEM::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

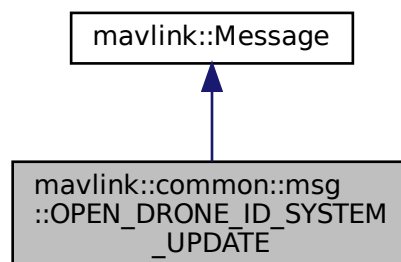
- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system.hpp`

5.378 `mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE` Struct Reference

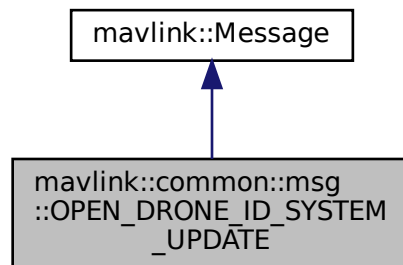
[OPEN_DRONE_ID_SYSTEM_UPDATE](#) message.

```
#include <mavlink_msg_open_drone_id_system_update.hpp>
```

Inheritance diagram for `mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE`:



Collaboration diagram for mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `int32_t` **operator_latitude**
- `int32_t` **operator_longitude**
- `float` **operator_altitude_geo**
- `uint32_t` **timestamp**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 12919
- static constexpr size_t **LENGTH** = 18
- static constexpr size_t **MIN_LENGTH** = 18
- static constexpr uint8_t **CRC_EXTRA** = 7
- static constexpr auto **NAME** = "OPEN_DRONE_ID_SYSTEM_UPDATE"

5.378.1 Detailed Description

[OPEN_DRONE_ID_SYSTEM_UPDATE](#) message.

Update the data in the [OPEN_DRONE_ID_SYSTEM](#) message with new location information. This can be sent to update the location information for the operator when no other information in the SYSTEM message has changed. This message allows for efficient operation on radio links which have limited uplink bandwidth while meeting requirements for update frequency of the operator location.

5.378.2 Member Function Documentation

5.378.2.1 deserialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.378.2.2 get_name()

```
std::string mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.378.2.3 serialize()

```
void mavlink::common::msg::OPEN_DRONE_ID_SYSTEM_UPDATE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

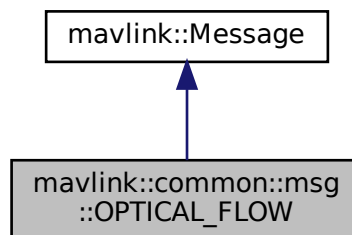
- `single_hand_controller/include/mavlink/common/mavlink_msg_open_drone_id_system_update.hpp`

5.379 mavlink::common::msg::OPTICAL_FLOW Struct Reference

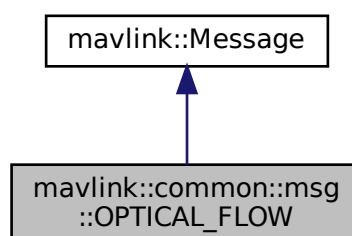
[OPTICAL_FLOW](#) message.

```
#include <mavlink_msg_optical_flow.hpp>
```

Inheritance diagram for mavlink::common::msg::OPTICAL_FLOW:



Collaboration diagram for mavlink::common::msg::OPTICAL_FLOW:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (`mavlink::MsgMap` &map) const override
Serialize message.
- `void deserialize` (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint64_t time_usec`
- `uint8_t sensor_id`
- `int16_t flow_x`
- `int16_t flow_y`
- `float flow_comp_m_x`
- `float flow_comp_m_y`
- `uint8_t quality`
- `float ground_distance`
- `float flow_rate_x`
- `float flow_rate_y`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 100`
- `static constexpr size_t LENGTH = 34`
- `static constexpr size_t MIN_LENGTH = 26`
- `static constexpr uint8_t CRC_EXTRA = 175`
- `static constexpr auto NAME = "OPTICAL_FLOW"`

5.379.1 Detailed Description

`OPTICAL_FLOW` message.

Optical flow from a flow sensor (e.g. optical mouse sensor)

5.379.2 Member Function Documentation

5.379.2.1 `deserialize()`

```
void mavlink::common::msg::OPTICAL_FLOW::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.379.2.2 get_name()

```
std::string mavlink::common::msg::OPTICAL_FLOW::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.379.2.3 serialize()

```
void mavlink::common::msg::OPTICAL_FLOW::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

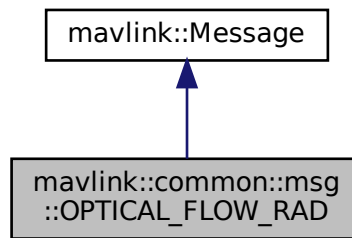
- single_hand_controller/include/mavlink/common/mavlink_msg_optical_flow.hpp

5.380 mavlink::common::msg::OPTICAL_FLOW_RAD Struct Reference

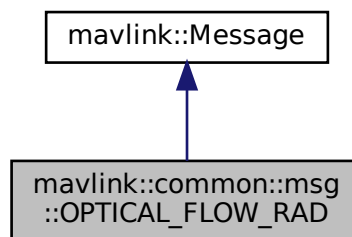
[OPTICAL_FLOW_RAD](#) message.

```
#include <mavlink_msg_optical_flow_rad.hpp>
```

Inheritance diagram for mavlink::common::msg::OPTICAL_FLOW_RAD:



Collaboration diagram for mavlink::common::msg::OPTICAL_FLOW_RAD:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` `time_usec`
- `uint8_t` `sensor_id`

- uint32_t **integration_time_us**
- float **integrated_x**
- float **integrated_y**
- float **integrated_xgyro**
- float **integrated_ygyro**
- float **integrated_zgyro**
- int16_t **temperature**
- uint8_t **quality**
- uint32_t **time_delta_distance_us**
- float **distance**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 106
- static constexpr size_t **LENGTH** = 44
- static constexpr size_t **MIN_LENGTH** = 44
- static constexpr uint8_t **CRC_EXTRA** = 138
- static constexpr auto **NAME** = "OPTICAL_FLOW_RAD"

5.380.1 Detailed Description

[OPTICAL_FLOW_RAD](#) message.

Optical flow from an angular rate flow sensor (e.g. PX4FLOW or mouse sensor)

5.380.2 Member Function Documentation

5.380.2.1 deserialize()

```
void mavlink::common::msg::OPTICAL_FLOW_RAD::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.380.2.2 get_name()

```
std::string mavlink::common::msg::OPTICAL_FLOW_RAD::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.380.2.3 `serialize()`

```
void mavlink::common::msg::OPTICAL_FLOW_RAD::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

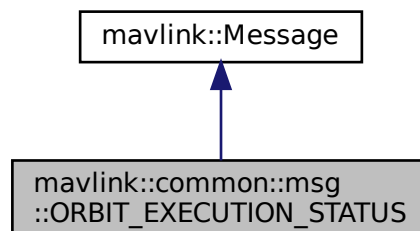
- `single_hand_controller/include/mavlink/common/mavlink_msg_optical_flow_rad.hpp`

5.381 `mavlink::common::msg::ORBIT_EXECUTION_STATUS` Struct Reference

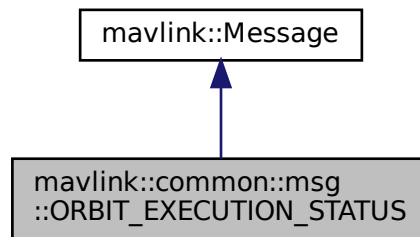
[ORBIT_EXECUTION_STATUS](#) message.

```
#include <mavlink_msg_orbit_execution_status.hpp>
```

Inheritance diagram for `mavlink::common::msg::ORBIT_EXECUTION_STATUS`:



Collaboration diagram for mavlink::common::msg::ORBIT_EXECUTION_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `float` **radius**
- `uint8_t` **frame**
- `int32_t` **x**
- `int32_t` **y**
- `float` **z**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 360
- static constexpr `size_t` **LENGTH** = 25
- static constexpr `size_t` **MIN_LENGTH** = 25
- static constexpr `uint8_t` **CRC_EXTRA** = 11
- static constexpr auto **NAME** = "ORBIT_EXECUTION_STATUS"

5.381.1 Detailed Description

[ORBIT_EXECUTION_STATUS](#) message.

Vehicle status report that is sent out while orbit execution is in progress (see MAV_CMD_DO_ORBIT).

5.381.2 Member Function Documentation

5.381.2.1 deserialize()

```
void mavlink::common::msg::ORBIT_EXECUTION_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.381.2.2 get_name()

```
std::string mavlink::common::msg::ORBIT_EXECUTION_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.381.2.3 serialize()

```
void mavlink::common::msg::ORBIT_EXECUTION_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_orbit_execution_status.hpp`

5.382 packet_receiver Class Reference

Methods within this class is meant to be invoked by the packet handler class, in order to process all received packets.

```
#include <packet_receiver.hpp>
```

Static Public Member Functions

- static void [receive_timesync](#) (mavlink_message_t *msg)
process timesync message; allows for accurate time synchronization on sending/receiving further signed packets
- static void [receive_radio_status](#) (mavlink_message_t *msg)
process radio status message to display signal strength on user display
- static void [receive_sys_status](#) (mavlink_message_t *msg)
process sys_status message to display battery percentage on user display
- static void [receive_ack](#) (mavlink_message_t *msg)
receive acknowledgement for all command long messages sent from device; includes accepted/rejected status using which the hand controller knows whether the packet is accepted or rejected
- static void [receive_heartbeat](#) (mavlink_message_t *msg)
receives and validates heartbeat messages received from scout, based on which the last_heartbeat_received_at variable is updated

5.382.1 Detailed Description

Methods within this class is meant to be invoked by the packet handler class, in order to process all received packets.

5.382.2 Member Function Documentation

5.382.2.1 [receive_ack\(\)](#)

```
void packet_receiver::receive_ack (
    mavlink_message_t * msg ) [static]
```

receive acknowledgement for all command long messages sent from device; includes accepted/rejected status using which the hand controller knows whether the packet is accepted or rejected

Parameters

<i>msg</i>	received message
------------	------------------

5.382.2.2 [receive_heartbeat\(\)](#)

```
void packet_receiver::receive_heartbeat (
    mavlink_message_t * msg ) [static]
```

receives and validates heartbeat messages received from scout, based on which the `last_heartbeat_received_at` variable is updated

Parameters

<i>msg</i>	received message
------------	------------------

5.382.2.3 receive_radio_status()

```
void packet_receiver::receive_radio_status (
    mavlink_message_t * msg ) [static]
```

process radio status message to display signal strength on user display

Parameters

<i>msg</i>	received message
------------	------------------

5.382.2.4 receive_sys_status()

```
void packet_receiver::receive_sys_status (
    mavlink_message_t * msg ) [static]
```

process `sys_status` message to display battery percentage on user display

Parameters

<i>msg</i>	received message
------------	------------------

5.382.2.5 receive_timesync()

```
void packet_receiver::receive_timesync (
    mavlink_message_t * msg ) [static]
```

process `timesync` message; allows for accurate time synchronization on sending/receiving further signed packets

Parameters

<i>msg</i>	received message
------------	------------------

The documentation for this class was generated from the following files:

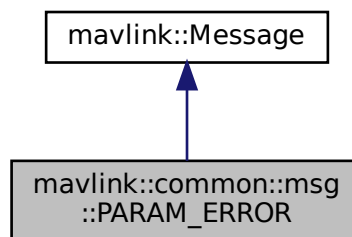
- [single_hand_controller/include/packet_receiver.hpp](#)
- [single_hand_controller/packet_receiver.cpp](#)

5.383 mavlink::common::msg::PARAM_ERROR Struct Reference

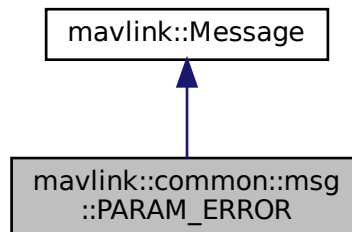
[PARAM_ERROR](#) message.

```
#include <mavlink_msg_param_error.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_ERROR:



Collaboration diagram for mavlink::common::msg::PARAM_ERROR:



Public Member Functions

- `std::string get\_name (void) const` override
Get NAME constant.
- `Info get\_message\_info (void) const` override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to\_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map)` override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- std::array< char, 16 > **param_id**
- int16_t **param_index**
- uint8_t **error**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 345
- static constexpr size_t **LENGTH** = 21
- static constexpr size_t **MIN_LENGTH** = 21
- static constexpr uint8_t **CRC_EXTRA** = 209
- static constexpr auto **NAME** = "PARAM_ERROR"

5.383.1 Detailed Description

[PARAM_ERROR](#) message.

Parameter set/get error. Returned from a MAVLink node in response to an error in the parameter protocol, for example failing to set a parameter because it does not exist.

5.383.2 Member Function Documentation

5.383.2.1 deserialize()

```
void mavlink::common::msg::PARAM_ERROR::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.383.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_ERROR::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.383.2.3 serialize()

```
void mavlink::common::msg::PARAM_ERROR::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

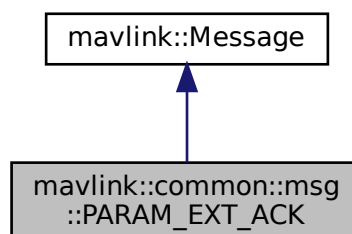
- single_hand_controller/include/mavlink/common/mavlink_msg_param_error.hpp

5.384 mavlink::common::msg::PARAM_EXT_ACK Struct Reference

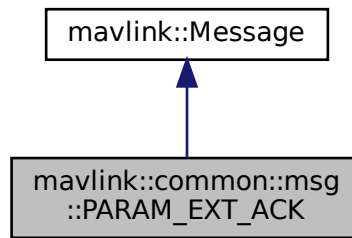
[PARAM_EXT_ACK](#) message.

```
#include <mavlink_msg_param_ext_ack.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_EXT_ACK:



Collaboration diagram for mavlink::common::msg::PARAM_EXT_ACK:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `std::array< char, 16 >` **param_id**
- `std::array< char, 128 >` **param_value**
- `uint8_t` **param_type**
- `uint8_t` **param_result**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 324
- static constexpr `size_t` **LENGTH** = 146
- static constexpr `size_t` **MIN_LENGTH** = 146
- static constexpr `uint8_t` **CRC_EXTRA** = 132
- static constexpr auto **NAME** = "PARAM_EXT_ACK"

5.384.1 Detailed Description

[PARAM_EXT_ACK](#) message.

Response from a [PARAM_EXT_SET](#) message.

5.384.2 Member Function Documentation

5.384.2.1 deserialize()

```
void mavlink::common::msg::PARAM_EXT_ACK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.384.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_EXT_ACK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.384.2.3 serialize()

```
void mavlink::common::msg::PARAM_EXT_ACK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

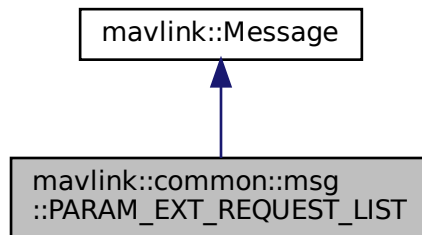
- single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_ack.hpp

5.385 mavlink::common::msg::PARAM_EXT_REQUEST_LIST Struct Reference

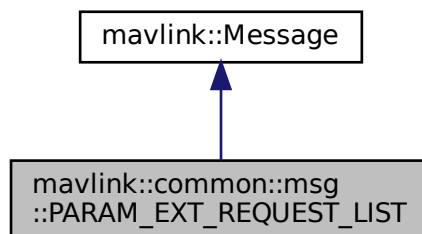
[PARAM_EXT_REQUEST_LIST](#) message.

```
#include <mavlink_msg_param_ext_request_list.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_EXT_REQUEST_LIST:



Collaboration diagram for mavlink::common::msg::PARAM_EXT_REQUEST_LIST:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 321
- static constexpr size_t **LENGTH** = 2
- static constexpr size_t **MIN_LENGTH** = 2
- static constexpr uint8_t **CRC_EXTRA** = 88
- static constexpr auto **NAME** = "PARAM_EXT_REQUEST_LIST"

5.385.1 Detailed Description

[PARAM_EXT_REQUEST_LIST](#) message.

Request all parameters of this component. All parameters should be emitted in response as [PARAM_EXT_VALUE](#).

5.385.2 Member Function Documentation

5.385.2.1 deserialize()

```
void mavlink::common::msg::PARAM_EXT_REQUEST_LIST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.385.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_EXT_REQUEST_LIST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.385.2.3 `serialize()`

```
void mavlink::common::msg::PARAM_EXT_REQUEST_LIST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

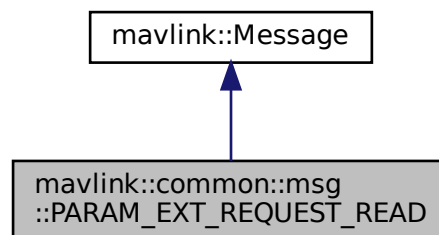
- `single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_list.hpp`

5.386 `mavlink::common::msg::PARAM_EXT_REQUEST_READ` Struct Reference

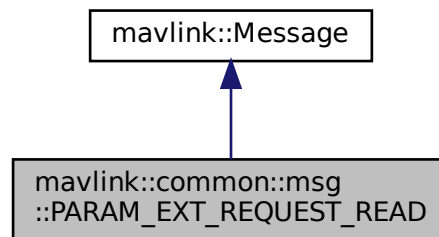
[PARAM_EXT_REQUEST_READ](#) message.

```
#include <mavlink_msg_param_ext_request_read.hpp>
```

Inheritance diagram for `mavlink::common::msg::PARAM_EXT_REQUEST_READ`:



Collaboration diagram for mavlink::common::msg::PARAM_EXT_REQUEST_READ:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< char, 16 >` **param_id**
- `int16_t` **param_index**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 320
- static constexpr `size_t` **LENGTH** = 20
- static constexpr `size_t` **MIN_LENGTH** = 20
- static constexpr `uint8_t` **CRC_EXTRA** = 243
- static constexpr auto **NAME** = "PARAM_EXT_REQUEST_READ"

5.386.1 Detailed Description

[PARAM_EXT_REQUEST_READ](#) message.

Request to read the value of a parameter with either the `param_id` string id or `param_index`. [PARAM_EXT_VALUE](#) should be emitted in response.

5.386.2 Member Function Documentation

5.386.2.1 deserialize()

```
void mavlink::common::msg::PARAM_EXT_REQUEST_READ::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.386.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_EXT_REQUEST_READ::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.386.2.3 serialize()

```
void mavlink::common::msg::PARAM_EXT_REQUEST_READ::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

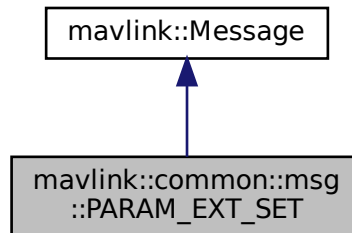
- single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_request_read.hpp

5.387 mavlink::common::msg::PARAM_EXT_SET Struct Reference

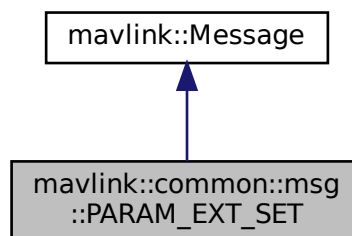
[PARAM_EXT_SET](#) message.

```
#include <mavlink_msg_param_ext_set.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_EXT_SET:



Collaboration diagram for mavlink::common::msg::PARAM_EXT_SET:



Public Member Functions

- `std::string get\_name (void) const` override
Get NAME constant.
- `Info get\_message\_info (void) const` override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string to\_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map)` override
Deserialize message.

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `std::array< char, 16 > param_id`
- `std::array< char, 128 > param_value`
- `uint8_t param_type`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 323`
- `static constexpr size_t LENGTH = 147`
- `static constexpr size_t MIN_LENGTH = 147`
- `static constexpr uint8_t CRC_EXTRA = 78`
- `static constexpr auto NAME = "PARAM_EXT_SET"`

5.387.1 Detailed Description

[PARAM_EXT_SET](#) message.

Set a parameter value. In order to deal with message loss (and retransmission of [PARAM_EXT_SET](#)), when setting a parameter value and the new value is the same as the current value, you will immediately get a `PARAM_ACK_ACCEPTED` response. If the current state is `PARAM_ACK_IN_PROGRESS`, you will accordingly receive a `PARAM_ACK_IN_PROGRESS` in response.

5.387.2 Member Function Documentation

5.387.2.1 deserialize()

```
void mavlink::common::msg::PARAM_EXT_SET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.387.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_EXT_SET::get_name (
    void ) const [inline], [override], [virtual]
```


Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.387.2.3 serialize()

```
void mavlink::common::msg::PARAM_EXT_SET::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

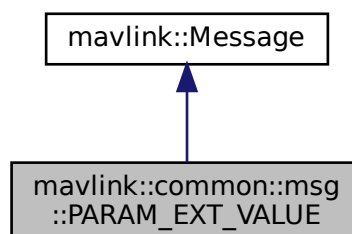
- single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_set.hpp

5.388 mavlink::common::msg::PARAM_EXT_VALUE Struct Reference

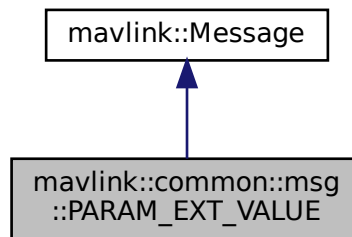
[PARAM_EXT_VALUE](#) message.

```
#include <mavlink_msg_param_ext_value.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_EXT_VALUE:



Collaboration diagram for mavlink::common::msg::PARAM_EXT_VALUE:



Public Member Functions

- `std::string get_name (void) const` override
Get NAME constant.
- `Info get_message_info (void) const` override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map) override`
Deserialize message.

Public Attributes

- `std::array< char, 16 > param_id`
- `std::array< char, 128 > param_value`
- `uint8_t param_type`
- `uint16_t param_count`
- `uint16_t param_index`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 322`
- `static constexpr size_t LENGTH = 149`
- `static constexpr size_t MIN_LENGTH = 149`
- `static constexpr uint8_t CRC_EXTRA = 243`
- `static constexpr auto NAME = "PARAM_EXT_VALUE"`

5.388.1 Detailed Description

[PARAM_EXT_VALUE](#) message.

Emit the value of a parameter. The inclusion of `param_count` and `param_index` in the message allows the recipient to keep track of received parameters and allows them to re-request missing parameters after a loss or timeout.

5.388.2 Member Function Documentation

5.388.2.1 deserialize()

```
void mavlink::common::msg::PARAM_EXT_VALUE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.388.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_EXT_VALUE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.388.2.3 serialize()

```
void mavlink::common::msg::PARAM_EXT_VALUE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

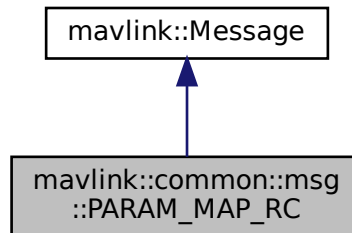
- `single_hand_controller/include/mavlink/common/mavlink_msg_param_ext_value.hpp`

5.389 mavlink::common::msg::PARAM_MAP_RC Struct Reference

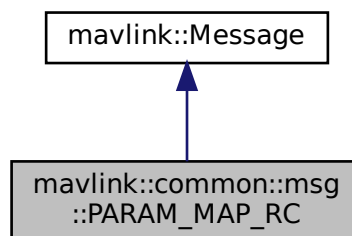
[PARAM_MAP_RC](#) message.

```
#include <mavlink_msg_param_map_rc.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_MAP_RC:



Collaboration diagram for mavlink::common::msg::PARAM_MAP_RC:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- std::array< char, 16 > **param_id**
- int16_t **param_index**
- uint8_t **parameter_rc_channel_index**
- float **param_value0**
- float **scale**
- float **param_value_min**
- float **param_value_max**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 50
- static constexpr size_t **LENGTH** = 37
- static constexpr size_t **MIN_LENGTH** = 37
- static constexpr uint8_t **CRC_EXTRA** = 78
- static constexpr auto **NAME** = "PARAM_MAP_RC"

5.389.1 Detailed Description

[PARAM_MAP_RC](#) message.

Bind a RC channel to a parameter. The parameter should change according to the RC channel value.

5.389.2 Member Function Documentation

5.389.2.1 deserialize()

```
void mavlink::common::msg::PARAM_MAP_RC::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.389.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_MAP_RC::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.389.2.3 `serialize()`

```
void mavlink::common::msg::PARAM_MAP_RC::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

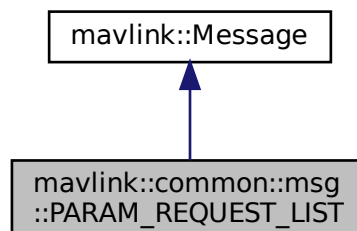
- `single_hand_controller/include/mavlink/common/mavlink_msg_param_map_rc.hpp`

5.390 `mavlink::common::msg::PARAM_REQUEST_LIST` Struct Reference

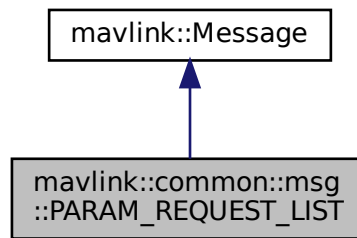
[PARAM_REQUEST_LIST](#) message.

```
#include <mavlink_msg_param_request_list.hpp>
```

Inheritance diagram for `mavlink::common::msg::PARAM_REQUEST_LIST`:



Collaboration diagram for mavlink::common::msg::PARAM_REQUEST_LIST:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 21`
- `static constexpr size_t LENGTH = 2`
- `static constexpr size_t MIN_LENGTH = 2`
- `static constexpr uint8_t CRC_EXTRA = 159`
- `static constexpr auto NAME = "PARAM_REQUEST_LIST"`

5.390.1 Detailed Description

`PARAM_REQUEST_LIST` message.

Request all parameters of this component. After this request, all parameters are emitted. The parameter microservice is documented at <https://mavlink.io/en/services/parameter.html>

5.390.2 Member Function Documentation

5.390.2.1 deserialize()

```
void mavlink::common::msg::PARAM_REQUEST_LIST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.390.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_REQUEST_LIST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.390.2.3 serialize()

```
void mavlink::common::msg::PARAM_REQUEST_LIST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

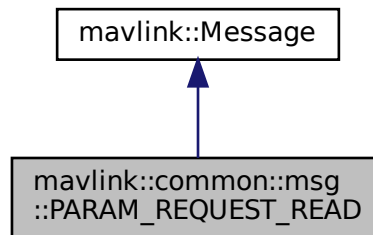
- `single_hand_controller/include/mavlink/common/mavlink_msg_param_request_list.hpp`

5.391 mavlink::common::msg::PARAM_REQUEST_READ Struct Reference

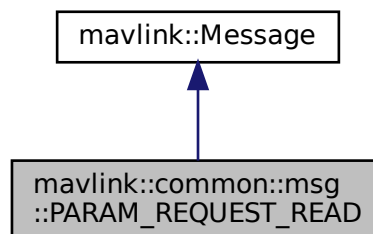
[PARAM_REQUEST_READ](#) message.

```
#include <mavlink_msg_param_request_read.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_REQUEST_READ:



Collaboration diagram for mavlink::common::msg::PARAM_REQUEST_READ:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- std::array< char, 16 > **param_id**
- int16_t **param_index**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 20
- static constexpr size_t **LENGTH** = 20
- static constexpr size_t **MIN_LENGTH** = 20
- static constexpr uint8_t **CRC_EXTRA** = 214
- static constexpr auto **NAME** = "PARAM_REQUEST_READ"

5.391.1 Detailed Description

[PARAM_REQUEST_READ](#) message.

Request to read the onboard parameter with the param_id string id. Onboard parameters are stored as key[const char*] -> value[float]. This allows to send a parameter to any other component (such as the GCS) without the need of previous knowledge of possible parameter names. Thus the same GCS can store different parameters for different autopilots. See also <https://mavlink.io/en/services/parameter.html> for a full documentation of QGroundControl and IMU code.

5.391.2 Member Function Documentation

5.391.2.1 deserialize()

```
void mavlink::common::msg::PARAM_REQUEST_READ::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.391.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_REQUEST_READ::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.391.2.3 serialize()

```
void mavlink::common::msg::PARAM_REQUEST_READ::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

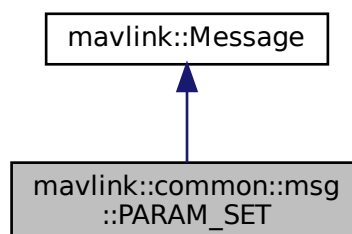
- single_hand_controller/include/mavlink/common/mavlink_msg_param_request_read.hpp

5.392 mavlink::common::msg::PARAM_SET Struct Reference

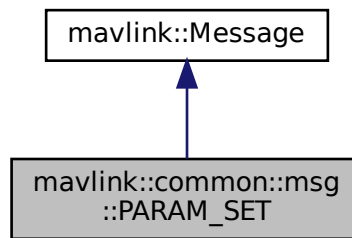
[PARAM_SET](#) message.

```
#include <mavlink_msg_param_set.hpp>
```

Inheritance diagram for mavlink::common::msg::PARAM_SET:



Collaboration diagram for mavlink::common::msg::PARAM_SET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< char, 16 >` **param_id**
- `float` **param_value**
- `uint8_t` **param_type**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 23
- static constexpr size_t **LENGTH** = 23
- static constexpr size_t **MIN_LENGTH** = 23
- static constexpr uint8_t **CRC_EXTRA** = 168
- static constexpr auto **NAME** = "PARAM_SET"

5.392.1 Detailed Description

[PARAM_SET](#) message.

Set a parameter value (write new value to permanent storage). The receiving component should acknowledge the new parameter value by broadcasting a [PARAM_VALUE](#) message (broadcasting ensures that multiple GCS all have an up-to-date list of all parameters). If the sending GCS did not receive a [PARAM_VALUE](#) within its timeout time, it should re-send the [PARAM_SET](#) message. The parameter microservice is documented at <https://mavlink.io/en/services/parameter.html>.

5.392.2 Member Function Documentation

5.392.2.1 deserialize()

```
void mavlink::common::msg::PARAM_SET::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.392.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_SET::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.392.2.3 serialize()

```
void mavlink::common::msg::PARAM_SET::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

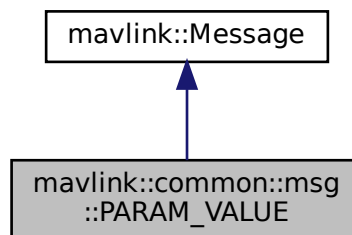
- `single_hand_controller/include/mavlink/common/mavlink_msg_param_set.hpp`

5.393 `mavlink::common::msg::PARAM_VALUE` Struct Reference

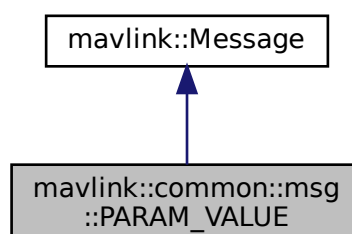
`PARAM_VALUE` message.

```
#include <mavlink_msg_param_value.hpp>
```

Inheritance diagram for `mavlink::common::msg::PARAM_VALUE`:



Collaboration diagram for `mavlink::common::msg::PARAM_VALUE`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- std::array< char, 16 > **param_id**
- float **param_value**
- uint8_t **param_type**
- uint16_t **param_count**
- uint16_t **param_index**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 22
- static constexpr size_t **LENGTH** = 25
- static constexpr size_t **MIN_LENGTH** = 25
- static constexpr uint8_t **CRC_EXTRA** = 220
- static constexpr auto **NAME** = "PARAM_VALUE"

5.393.1 Detailed Description

[PARAM_VALUE](#) message.

Emit the value of a onboard parameter. The inclusion of param_count and param_index in the message allows the recipient to keep track of received parameters and allows him to re-request missing parameters after a loss or timeout. The parameter microservice is documented at <https://mavlink.io/en/services/parameter.html>

5.393.2 Member Function Documentation

5.393.2.1 deserialize()

```
void mavlink::common::msg::PARAM_VALUE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.393.2.2 get_name()

```
std::string mavlink::common::msg::PARAM_VALUE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.393.2.3 serialize()

```
void mavlink::common::msg::PARAM_VALUE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

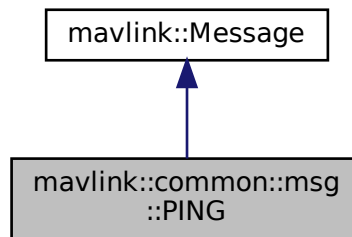
- single_hand_controller/include/mavlink/common/mavlink_msg_param_value.hpp

5.394 mavlink::common::msg::PING Struct Reference

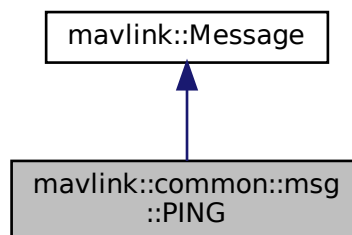
[PING](#) message.

```
#include <mavlink_msg_ping.hpp>
```


Inheritance diagram for mavlink::common::msg::PING:



Collaboration diagram for mavlink::common::msg::PING:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint32_t` **seq**
- `uint8_t` **target_system**
- `uint8_t` **target_component**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 4
- static constexpr size_t **LENGTH** = 14
- static constexpr size_t **MIN_LENGTH** = 14
- static constexpr uint8_t **CRC_EXTRA** = 237
- static constexpr auto **NAME** = "PING"

5.394.1 Detailed Description

PING message.

A ping message either requesting or responding to a ping. This allows to measure the system latencies, including serial port, radio modem and UDP connections. The ping microservice is documented at <https://mavlink.io/en/services/ping.html>

5.394.2 Member Function Documentation

5.394.2.1 deserialize()

```
void mavlink::common::msg::PING::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.394.2.2 get_name()

```
std::string mavlink::common::msg::PING::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.394.2.3 serialize()

```
void mavlink::common::msg::PING::serialize (  
    mavlink::MsgMap & map ) const    [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

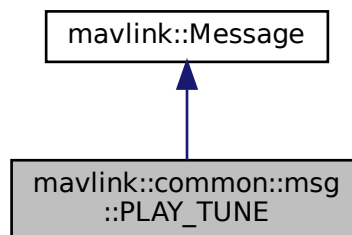
- `single_hand_controller/include/mavlink/common/mavlink_msg_ping.hpp`

5.395 mavlink::common::msg::PLAY_TUNE Struct Reference

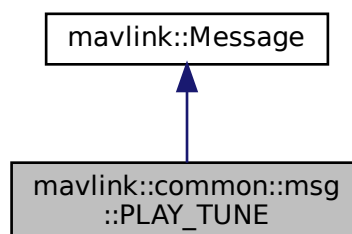
[PLAY_TUNE](#) message.

```
#include <mavlink_msg_play_tune.hpp>
```

Inheritance diagram for `mavlink::common::msg::PLAY_TUNE`:



Collaboration diagram for `mavlink::common::msg::PLAY_TUNE`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- uint8_t **target_component**
- std::array< char, 30 > **tune**
- std::array< char, 200 > **tune2**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 258
- static constexpr size_t **LENGTH** = 232
- static constexpr size_t **MIN_LENGTH** = 32
- static constexpr uint8_t **CRC_EXTRA** = 187
- static constexpr auto **NAME** = "PLAY_TUNE"

5.395.1 Detailed Description

[PLAY_TUNE](#) message.

Control vehicle tone generation (buzzer).

5.395.2 Member Function Documentation

5.395.2.1 deserialize()

```
void mavlink::common::msg::PLAY_TUNE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.395.2.2 get_name()

```
std::string mavlink::common::msg::PLAY_TUNE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.395.2.3 serialize()

```
void mavlink::common::msg::PLAY_TUNE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

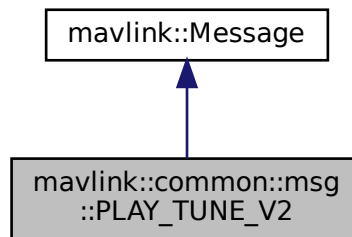
- single_hand_controller/include/mavlink/common/mavlink_msg_play_tune.hpp

5.396 mavlink::common::msg::PLAY_TUNE_V2 Struct Reference

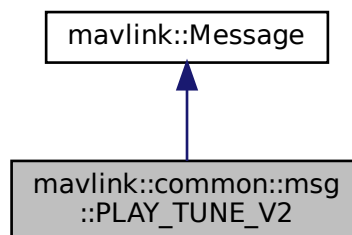
[PLAY_TUNE_V2](#) message.

```
#include <mavlink_msg_play_tune_v2.hpp>
```

Inheritance diagram for mavlink::common::msg::PLAY_TUNE_V2:



Collaboration diagram for mavlink::common::msg::PLAY_TUNE_V2:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint32_t` **format**
- `std::array< char, 248 >` **tune**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 400
- static constexpr size_t **LENGTH** = 254
- static constexpr size_t **MIN_LENGTH** = 254
- static constexpr uint8_t **CRC_EXTRA** = 110
- static constexpr auto **NAME** = "PLAY_TUNE_V2"

5.396.1 Detailed Description

[PLAY_TUNE_V2](#) message.

Play vehicle tone/tune (buzzer). Supersedes message [PLAY_TUNE](#).

5.396.2 Member Function Documentation

5.396.2.1 deserialize()

```
void mavlink::common::msg::PLAY_TUNE_V2::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.396.2.2 get_name()

```
std::string mavlink::common::msg::PLAY_TUNE_V2::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.396.2.3 serialize()

```
void mavlink::common::msg::PLAY_TUNE_V2::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

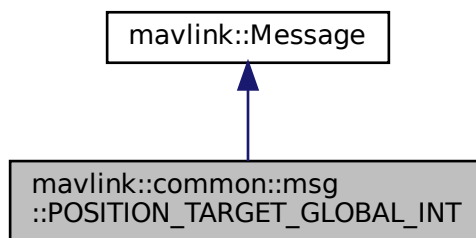
- single_hand_controller/include/mavlink/common/mavlink_msg_play_tune_v2.hpp

5.397 mavlink::common::msg::POSITION_TARGET_GLOBAL_INT Struct Reference

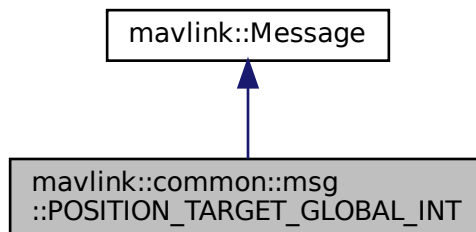
[POSITION_TARGET_GLOBAL_INT](#) message.

```
#include <mavlink_msg_position_target_global_int.hpp>
```

Inheritance diagram for mavlink::common::msg::POSITION_TARGET_GLOBAL_INT:



Collaboration diagram for mavlink::common::msg::POSITION_TARGET_GLOBAL_INT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint8_t` **coordinate_frame**
- `uint16_t` **type_mask**
- `int32_t` **lat_int**
- `int32_t` **lon_int**
- `float` **alt**
- `float` **vx**
- `float` **vy**
- `float` **vz**
- `float` **afx**
- `float` **afy**
- `float` **afz**
- `float` **yaw**
- `float` **yaw_rate**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 87
- static constexpr `size_t` **LENGTH** = 51
- static constexpr `size_t` **MIN_LENGTH** = 51
- static constexpr `uint8_t` **CRC_EXTRA** = 150
- static constexpr auto **NAME** = "POSITION_TARGET_GLOBAL_INT"

5.397.1 Detailed Description

[POSITION_TARGET_GLOBAL_INT](#) message.

Reports the current commanded vehicle position, velocity, and acceleration as specified by the autopilot. This should match the commands sent in [SET_POSITION_TARGET_GLOBAL_INT](#) if the vehicle is being controlled this way.

5.397.2 Member Function Documentation

5.397.2.1 `deserialize()`

```
void mavlink::common::msg::POSITION_TARGET_GLOBAL_INT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.397.2.2 get_name()

```
std::string mavlink::common::msg::POSITION_TARGET_GLOBAL_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.397.2.3 serialize()

```
void mavlink::common::msg::POSITION_TARGET_GLOBAL_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

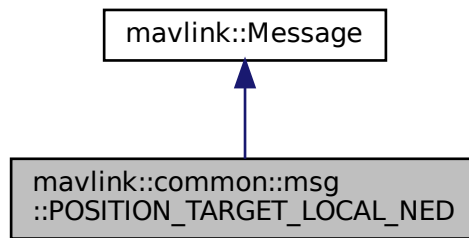
- single_hand_controller/include/mavlink/common/mavlink_msg_position_target_global_int.hpp

5.398 mavlink::common::msg::POSITION_TARGET_LOCAL_NED Struct Reference

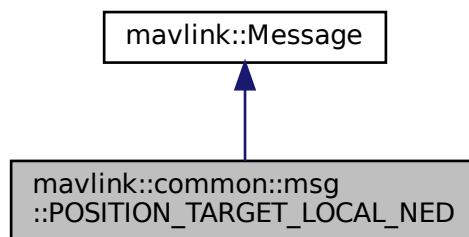
[POSITION_TARGET_LOCAL_NED](#) message.

```
#include <mavlink_msg_position_target_local_ned.hpp>
```

Inheritance diagram for mavlink::common::msg::POSITION_TARGET_LOCAL_NED:



Collaboration diagram for mavlink::common::msg::POSITION_TARGET_LOCAL_NED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint8_t` **coordinate_frame**

- uint16_t **type_mask**
- float **x**
- float **y**
- float **z**
- float **vx**
- float **vy**
- float **vz**
- float **afx**
- float **afy**
- float **afz**
- float **yaw**
- float **yaw_rate**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 85
- static constexpr size_t **LENGTH** = 51
- static constexpr size_t **MIN_LENGTH** = 51
- static constexpr uint8_t **CRC_EXTRA** = 140
- static constexpr auto **NAME** = "POSITION_TARGET_LOCAL_NED"

5.398.1 Detailed Description

[POSITION_TARGET_LOCAL_NED](#) message.

Reports the current commanded vehicle position, velocity, and acceleration as specified by the autopilot. This should match the commands sent in [SET_POSITION_TARGET_LOCAL_NED](#) if the vehicle is being controlled this way.

5.398.2 Member Function Documentation

5.398.2.1 deserialize()

```
void mavlink::common::msg::POSITION_TARGET_LOCAL_NED::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.398.2.2 `get_name()`

```
std::string mavlink::common::msg::POSITION_TARGET_LOCAL_NED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.398.2.3 `serialize()`

```
void mavlink::common::msg::POSITION_TARGET_LOCAL_NED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

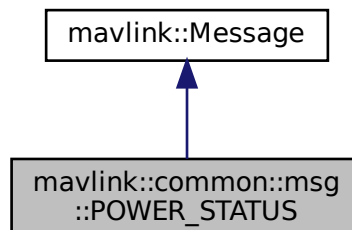
- `single_hand_controller/include/mavlink/common/mavlink_msg_position_target_local_ned.hpp`

5.399 `mavlink::common::msg::POWER_STATUS` Struct Reference

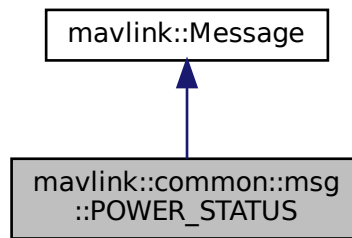
[POWER_STATUS](#) message.

```
#include <mavlink_msg_power_status.hpp>
```

Inheritance diagram for `mavlink::common::msg::POWER_STATUS`:



Collaboration diagram for mavlink::common::msg::POWER_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint16_t` **Vcc**
- `uint16_t` **Vservo**
- `uint16_t` **flags**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 125
- static constexpr size_t **LENGTH** = 6
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 203
- static constexpr auto **NAME** = "POWER_STATUS"

5.399.1 Detailed Description

[POWER_STATUS](#) message.

Power supply status

5.399.2 Member Function Documentation

5.399.2.1 deserialize()

```
void mavlink::common::msg::POWER_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.399.2.2 get_name()

```
std::string mavlink::common::msg::POWER_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.399.2.3 serialize()

```
void mavlink::common::msg::POWER_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

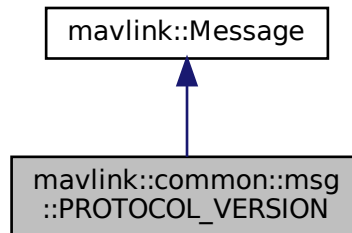
- `single_hand_controller/include/mavlink/common/mavlink_msg_power_status.hpp`

5.400 mavlink::common::msg::PROTOCOL_VERSION Struct Reference

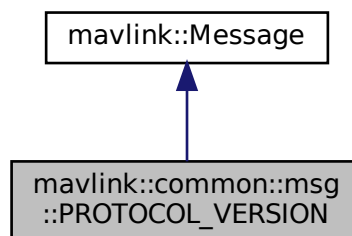
[PROTOCOL_VERSION](#) message.

```
#include <mavlink_msg_protocol_version.hpp>
```

Inheritance diagram for mavlink::common::msg::PROTOCOL_VERSION:



Collaboration diagram for mavlink::common::msg::PROTOCOL_VERSION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint16_t **version**
- uint16_t **min_version**
- uint16_t **max_version**
- std::array< uint8_t, 8 > **spec_version_hash**
- std::array< uint8_t, 8 > **library_version_hash**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 300
- static constexpr size_t **LENGTH** = 22
- static constexpr size_t **MIN_LENGTH** = 22
- static constexpr uint8_t **CRC_EXTRA** = 217
- static constexpr auto **NAME** = "PROTOCOL_VERSION"

5.400.1 Detailed Description

[PROTOCOL_VERSION](#) message.

Version and capability of protocol version. This message can be requested with MAV_CMD_REQUEST_MESSAGE and is used as part of the handshaking to establish which MAVLink version should be used on the network. Every node should respond to a request for [PROTOCOL_VERSION](#) to enable the handshaking. Library implementers should consider adding this into the default decoding state machine to allow the protocol core to respond directly.

5.400.2 Member Function Documentation

5.400.2.1 deserialize()

```
void mavlink::common::msg::PROTOCOL_VERSION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.400.2.2 get_name()

```
std::string mavlink::common::msg::PROTOCOL_VERSION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.400.2.3 serialize()

```
void mavlink::common::msg::PROTOCOL_VERSION::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

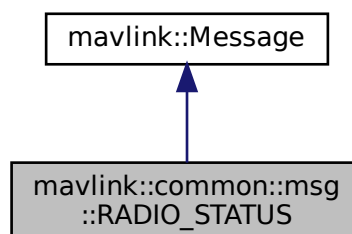
- single_hand_controller/include/mavlink/common/mavlink_msg_protocol_version.hpp

5.401 mavlink::common::msg::RADIO_STATUS Struct Reference

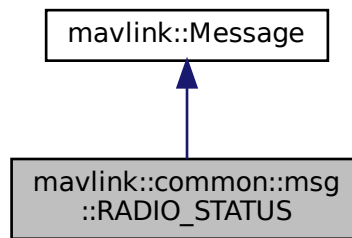
[RADIO_STATUS](#) message.

```
#include <mavlink_msg_radio_status.hpp>
```

Inheritance diagram for mavlink::common::msg::RADIO_STATUS:



Collaboration diagram for mavlink::common::msg::RADIO_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **rsi**
- `uint8_t` **remrsi**
- `uint8_t` **txbuf**
- `uint8_t` **noise**
- `uint8_t` **remnoise**
- `uint16_t` **rxerrors**
- `uint16_t` **fixed**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 109
- static constexpr size_t **LENGTH** = 9
- static constexpr size_t **MIN_LENGTH** = 9
- static constexpr uint8_t **CRC_EXTRA** = 185
- static constexpr auto **NAME** = "RADIO_STATUS"

5.401.1 Detailed Description

[RADIO_STATUS](#) message.

Status generated by radio and injected into MAVLink stream.

5.401.2 Member Function Documentation

5.401.2.1 deserialize()

```
void mavlink::common::msg::RADIO_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.401.2.2 get_name()

```
std::string mavlink::common::msg::RADIO_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.401.2.3 serialize()

```
void mavlink::common::msg::RADIO_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

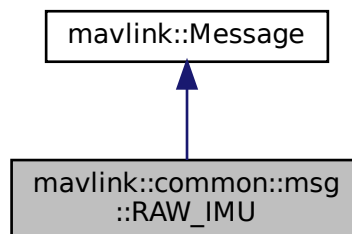
- `single_hand_controller/include/mavlink/common/mavlink_msg_radio_status.hpp`

5.402 `mavlink::common::msg::RAW_IMU` Struct Reference

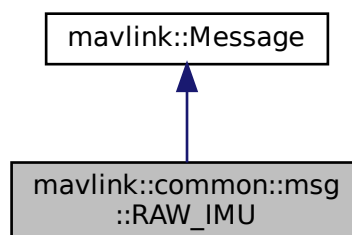
[RAW_IMU](#) message.

```
#include <mavlink_msg_raw_imu.hpp>
```

Inheritance diagram for `mavlink::common::msg::RAW_IMU`:



Collaboration diagram for `mavlink::common::msg::RAW_IMU`:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `int16_t` **xacc**
- `int16_t` **yacc**
- `int16_t` **zacc**
- `int16_t` **xgyro**
- `int16_t` **ygyro**
- `int16_t` **zgyro**
- `int16_t` **xmag**
- `int16_t` **ymag**
- `int16_t` **zmag**
- `uint8_t` **id**
- `int16_t` **temperature**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 27
- static constexpr size_t **LENGTH** = 29
- static constexpr size_t **MIN_LENGTH** = 26
- static constexpr uint8_t **CRC_EXTRA** = 144
- static constexpr auto **NAME** = "RAW_IMU"

5.402.1 Detailed Description

[RAW_IMU](#) message.

The RAW IMU readings for a 9DOF sensor, which is identified by the id (default IMU1). This message should always contain the true raw values without any scaling to allow data capture and system debugging.

5.402.2 Member Function Documentation

5.402.2.1 deserialize()

```
void mavlink::common::msg::RAW_IMU::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.402.2.2 get_name()

```
std::string mavlink::common::msg::RAW_IMU::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.402.2.3 serialize()

```
void mavlink::common::msg::RAW_IMU::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

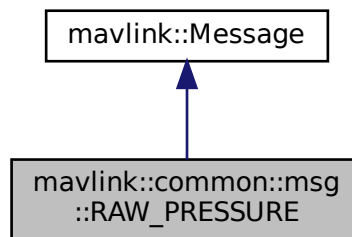
- single_hand_controller/include/mavlink/common/mavlink_msg_raw_imu.hpp

5.403 mavlink::common::msg::RAW_PRESSURE Struct Reference

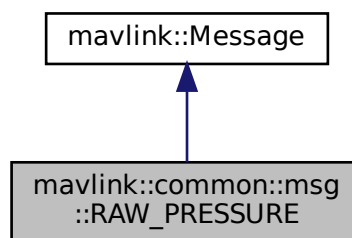
[RAW_PRESSURE](#) message.

```
#include <mavlink_msg_raw_pressure.hpp>
```


Inheritance diagram for mavlink::common::msg::RAW_PRESSURE:



Collaboration diagram for mavlink::common::msg::RAW_PRESSURE:



Public Member Functions

- `std::string get_name (void) const` override
Get NAME constant.
- `Info get_message_info (void) const` override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map)` override
Deserialize message.

Public Attributes

- `uint64_t time_usec`
- `int16_t press_abs`
- `int16_t press_diff1`
- `int16_t press_diff2`
- `int16_t temperature`

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 28
- static constexpr size_t **LENGTH** = 16
- static constexpr size_t **MIN_LENGTH** = 16
- static constexpr uint8_t **CRC_EXTRA** = 67
- static constexpr auto **NAME** = "RAW_PRESSURE"

5.403.1 Detailed Description

[RAW_PRESSURE](#) message.

The RAW pressure readings for the typical setup of one absolute pressure and one differential pressure sensor. The sensor values should be the raw, UNSCALED ADC values.

5.403.2 Member Function Documentation

5.403.2.1 deserialize()

```
void mavlink::common::msg::RAW_PRESSURE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.403.2.2 get_name()

```
std::string mavlink::common::msg::RAW_PRESSURE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.403.2.3 serialize()

```
void mavlink::common::msg::RAW_PRESSURE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

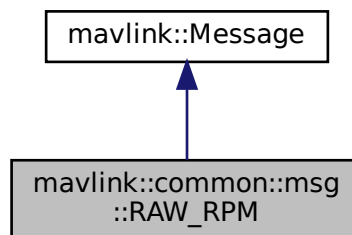
- `single_hand_controller/include/mavlink/common/mavlink_msg_raw_pressure.hpp`

5.404 mavlink::common::msg::RAW_RPM Struct Reference

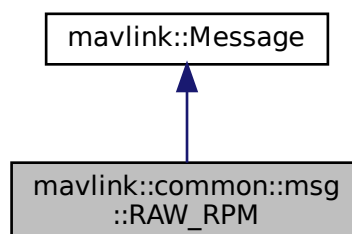
[RAW_RPM](#) message.

```
#include <mavlink_msg_raw_rpm.hpp>
```

Inheritance diagram for mavlink::common::msg::RAW_RPM:



Collaboration diagram for mavlink::common::msg::RAW_RPM:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **index**
- `float` **frequency**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 339
- static constexpr size_t **LENGTH** = 5
- static constexpr size_t **MIN_LENGTH** = 5
- static constexpr uint8_t **CRC_EXTRA** = 199
- static constexpr auto **NAME** = "RAW_RPM"

5.404.1 Detailed Description

[RAW_RPM](#) message.

RPM sensor data message.

5.404.2 Member Function Documentation

5.404.2.1 deserialize()

```
void mavlink::common::msg::RAW_RPM::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.404.2.2 get_name()

```
std::string mavlink::common::msg::RAW_RPM::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.404.2.3 serialize()

```
void mavlink::common::msg::RAW_RPM::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

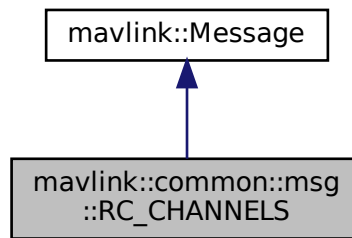
- single_hand_controller/include/mavlink/common/mavlink_msg_raw_rpm.hpp

5.405 mavlink::common::msg::RC_CHANNELS Struct Reference

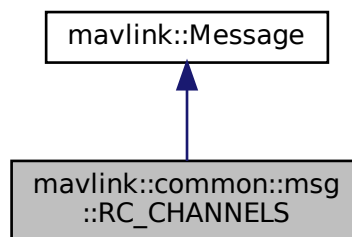
[RC_CHANNELS](#) message.

```
#include <mavlink_msg_rc_channels.hpp>
```

Inheritance diagram for mavlink::common::msg::RC_CHANNELS:



Collaboration diagram for mavlink::common::msg::RC_CHANNELS:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t time_boot_ms`
- `uint8_t chancount`

- uint16_t **chan1_raw**
- uint16_t **chan2_raw**
- uint16_t **chan3_raw**
- uint16_t **chan4_raw**
- uint16_t **chan5_raw**
- uint16_t **chan6_raw**
- uint16_t **chan7_raw**
- uint16_t **chan8_raw**
- uint16_t **chan9_raw**
- uint16_t **chan10_raw**
- uint16_t **chan11_raw**
- uint16_t **chan12_raw**
- uint16_t **chan13_raw**
- uint16_t **chan14_raw**
- uint16_t **chan15_raw**
- uint16_t **chan16_raw**
- uint16_t **chan17_raw**
- uint16_t **chan18_raw**
- uint8_t **rsi**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 65
- static constexpr size_t **LENGTH** = 42
- static constexpr size_t **MIN_LENGTH** = 42
- static constexpr uint8_t **CRC_EXTRA** = 118
- static constexpr auto **NAME** = "RC_CHANNELS"

5.405.1 Detailed Description

[RC_CHANNELS](#) message.

The PPM values of the RC channels received. The standard PPM modulation is as follows: 1000 microseconds↵ : 0%, 2000 microseconds: 100%. A value of UINT16_MAX implies the channel is unused. Individual receivers/transmitters might violate this specification.

5.405.2 Member Function Documentation

5.405.2.1 deserialize()

```
void mavlink::common::msg::RC_CHANNELS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.405.2.2 `get_name()`

```
std::string mavlink::common::msg::RC_CHANNELS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.405.2.3 `serialize()`

```
void mavlink::common::msg::RC_CHANNELS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

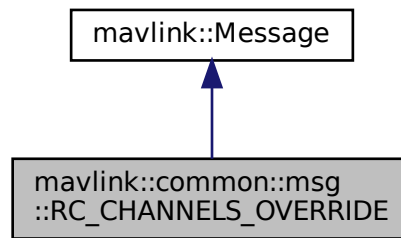
- `single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels.hpp`

5.406 `mavlink::common::msg::RC_CHANNELS_OVERRIDE` Struct Reference

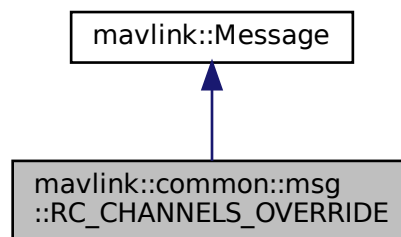
[RC_CHANNELS_OVERRIDE](#) message.

```
#include <mavlink_msg_rc_channels_override.hpp>
```


Inheritance diagram for mavlink::common::msg::RC_CHANNELS_OVERRIDE:



Collaboration diagram for mavlink::common::msg::RC_CHANNELS_OVERRIDE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` [target_system](#)
- `uint8_t` [target_component](#)

- uint16_t **chan1_raw**
- uint16_t **chan2_raw**
- uint16_t **chan3_raw**
- uint16_t **chan4_raw**
- uint16_t **chan5_raw**
- uint16_t **chan6_raw**
- uint16_t **chan7_raw**
- uint16_t **chan8_raw**
- uint16_t **chan9_raw**
- uint16_t **chan10_raw**
- uint16_t **chan11_raw**
- uint16_t **chan12_raw**
- uint16_t **chan13_raw**
- uint16_t **chan14_raw**
- uint16_t **chan15_raw**
- uint16_t **chan16_raw**
- uint16_t **chan17_raw**
- uint16_t **chan18_raw**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 70
- static constexpr size_t **LENGTH** = 38
- static constexpr size_t **MIN_LENGTH** = 18
- static constexpr uint8_t **CRC_EXTRA** = 124
- static constexpr auto **NAME** = "RC_CHANNELS_OVERRIDE"

5.406.1 Detailed Description

[RC_CHANNELS_OVERRIDE](#) message.

The RAW values of the RC channels sent to the MAV to override info received from the RC radio. The standard PPM modulation is as follows: 1000 microseconds: 0%, 2000 microseconds: 100%. Individual receivers/transmitters might violate this specification. Note carefully the semantic differences between the first 8 channels and the subsequent channels

5.406.2 Member Function Documentation

5.406.2.1 deserialize()

```
void mavlink::common::msg::RC_CHANNELS_OVERRIDE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.406.2.2 get_name()

```
std::string mavlink::common::msg::RC_CHANNELS_OVERRIDE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.406.2.3 serialize()

```
void mavlink::common::msg::RC_CHANNELS_OVERRIDE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

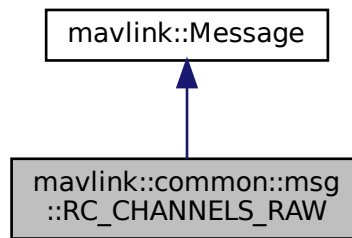
- single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_override.hpp

5.407 mavlink::common::msg::RC_CHANNELS_RAW Struct Reference

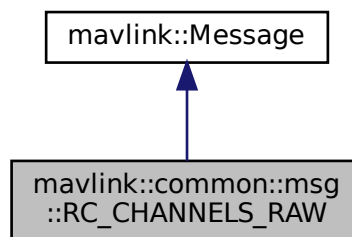
[RC_CHANNELS_RAW](#) message.

```
#include <mavlink_msg_rc_channels_raw.hpp>
```

Inheritance diagram for mavlink::common::msg::RC_CHANNELS_RAW:



Collaboration diagram for mavlink::common::msg::RC_CHANNELS_RAW:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint8_t` **port**

- uint16_t **chan1_raw**
- uint16_t **chan2_raw**
- uint16_t **chan3_raw**
- uint16_t **chan4_raw**
- uint16_t **chan5_raw**
- uint16_t **chan6_raw**
- uint16_t **chan7_raw**
- uint16_t **chan8_raw**
- uint8_t **rsi**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 35
- static constexpr size_t **LENGTH** = 22
- static constexpr size_t **MIN_LENGTH** = 22
- static constexpr uint8_t **CRC_EXTRA** = 244
- static constexpr auto **NAME** = "RC_CHANNELS_RAW"

5.407.1 Detailed Description

[RC_CHANNELS_RAW](#) message.

The RAW values of the RC channels received. The standard PPM modulation is as follows: 1000 microseconds↔ : 0%, 2000 microseconds: 100%. A value of UINT16_MAX implies the channel is unused. Individual receivers/transmitters might violate this specification.

5.407.2 Member Function Documentation

5.407.2.1 deserialize()

```
void mavlink::common::msg::RC_CHANNELS_RAW::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.407.2.2 get_name()

```
std::string mavlink::common::msg::RC_CHANNELS_RAW::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.407.2.3 `serialize()`

```
void mavlink::common::msg::RC_CHANNELS_RAW::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

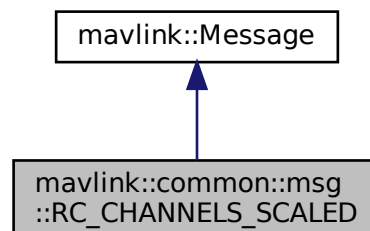
- `single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_raw.hpp`

5.408 `mavlink::common::msg::RC_CHANNELS_SCALED` Struct Reference

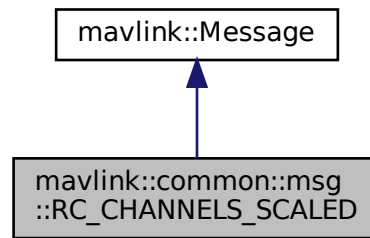
[RC_CHANNELS_SCALED](#) message.

```
#include <mavlink_msg_rc_channels_scaled.hpp>
```

Inheritance diagram for `mavlink::common::msg::RC_CHANNELS_SCALED`:



Collaboration diagram for mavlink::common::msg::RC_CHANNELS_SCALED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint8_t` **port**
- `int16_t` **chan1_scaled**
- `int16_t` **chan2_scaled**
- `int16_t` **chan3_scaled**
- `int16_t` **chan4_scaled**
- `int16_t` **chan5_scaled**
- `int16_t` **chan6_scaled**
- `int16_t` **chan7_scaled**
- `int16_t` **chan8_scaled**
- `uint8_t` **rsqi**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 34
- static constexpr `size_t` **LENGTH** = 22
- static constexpr `size_t` **MIN_LENGTH** = 22
- static constexpr `uint8_t` **CRC_EXTRA** = 237
- static constexpr auto **NAME** = "RC_CHANNELS_SCALED"

5.408.1 Detailed Description

[RC_CHANNELS_SCALED](#) message.

The scaled values of the RC channels received: (-100%) -10000, (0%) 0, (100%) 10000. Channels that are inactive should be set to INT16_MAX.

5.408.2 Member Function Documentation

5.408.2.1 deserialize()

```
void mavlink::common::msg::RC_CHANNELS_SCALED::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.408.2.2 get_name()

```
std::string mavlink::common::msg::RC_CHANNELS_SCALED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.408.2.3 serialize()

```
void mavlink::common::msg::RC_CHANNELS_SCALED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

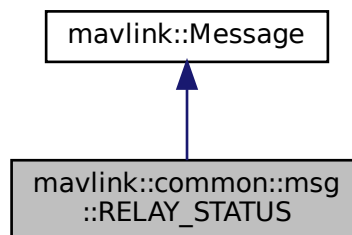
- single_hand_controller/include/mavlink/common/mavlink_msg_rc_channels_scaled.hpp

5.409 mavlink::common::msg::RELAY_STATUS Struct Reference

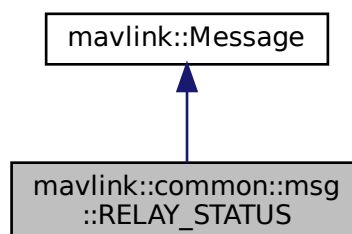
[RELAY_STATUS](#) message.

```
#include <mavlink_msg_relay_status.hpp>
```

Inheritance diagram for mavlink::common::msg::RELAY_STATUS:



Collaboration diagram for mavlink::common::msg::RELAY_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) ([mavlink::MsgMap](#) &map) const override
Serialize message.
- void [deserialize](#) ([mavlink::MsgMap](#) &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint16_t` **on**
- `uint16_t` **present**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 376
- static constexpr `size_t` **LENGTH** = 8
- static constexpr `size_t` **MIN_LENGTH** = 8
- static constexpr `uint8_t` **CRC_EXTRA** = 199
- static constexpr auto **NAME** = "RELAY_STATUS"

5.409.1 Detailed Description

[RELAY_STATUS](#) message.

Reports the on/off state of relays, as controlled by MAV_CMD_DO_SET_RELAY. [Message](#) streaming should be requested using MAV_CMD_SET_MESSAGE_INTERVAL. Note that it should not be sent on every relay state change to avoid flooding the link.

5.409.2 Member Function Documentation

5.409.2.1 deserialize()

```
void mavlink::common::msg::RELAY_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.409.2.2 get_name()

```
std::string mavlink::common::msg::RELAY_STATUS::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.409.2.3 serialize()

```
void mavlink::common::msg::RELAY_STATUS::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

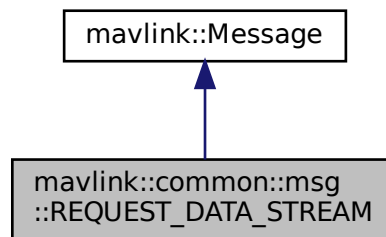
- single_hand_controller/include/mavlink/common/mavlink_msg_relay_status.hpp

5.410 mavlink::common::msg::REQUEST_DATA_STREAM Struct Reference

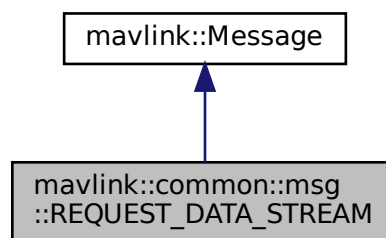
[REQUEST_DATA_STREAM](#) message.

```
#include <mavlink_msg_request_data_stream.hpp>
```

Inheritance diagram for mavlink::common::msg::REQUEST_DATA_STREAM:



Collaboration diagram for mavlink::common::msg::REQUEST_DATA_STREAM:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **req_stream_id**
- `uint16_t` **req_message_rate**
- `uint8_t` **start_stop**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 66
- static constexpr size_t **LENGTH** = 6
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 148
- static constexpr auto **NAME** = "REQUEST_DATA_STREAM"

5.410.1 Detailed Description

[REQUEST_DATA_STREAM](#) message.

Request a data stream.

5.410.2 Member Function Documentation

5.410.2.1 deserialize()

```
void mavlink::common::msg::REQUEST_DATA_STREAM::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.410.2.2 get_name()

```
std::string mavlink::common::msg::REQUEST_DATA_STREAM::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.410.2.3 serialize()

```
void mavlink::common::msg::REQUEST_DATA_STREAM::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

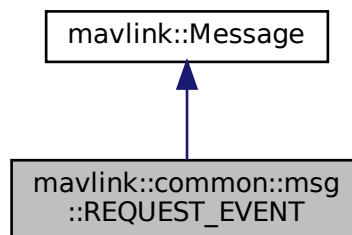
- `single_hand_controller/include/mavlink/common/mavlink_msg_request_data_stream.hpp`

5.411 mavlink::common::msg::REQUEST_EVENT Struct Reference

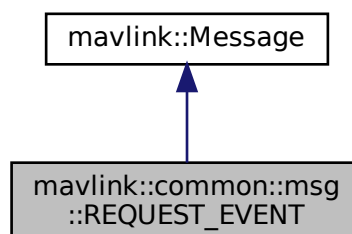
[REQUEST_EVENT](#) message.

```
#include <mavlink_msg_request_event.hpp>
```

Inheritance diagram for mavlink::common::msg::REQUEST_EVENT:



Collaboration diagram for mavlink::common::msg::REQUEST_EVENT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint16_t` **first_sequence**
- `uint16_t` **last_sequence**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 412
- static constexpr `size_t` **LENGTH** = 6
- static constexpr `size_t` **MIN_LENGTH** = 6
- static constexpr `uint8_t` **CRC_EXTRA** = 33
- static constexpr auto **NAME** = "REQUEST_EVENT"

5.411.1 Detailed Description

[REQUEST_EVENT](#) message.

Request one or more events to be (re-)sent. If `first_sequence==last_sequence`, only a single event is requested. Note that `first_sequence` can be larger than `last_sequence` (because the sequence number can wrap). Each sequence will trigger an [EVENT](#) or `EVENT_ERROR` response.

5.411.2 Member Function Documentation

5.411.2.1 `deserialize()`

```
void mavlink::common::msg::REQUEST_EVENT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.411.2.2 get_name()

```
std::string mavlink::common::msg::REQUEST_EVENT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.411.2.3 serialize()

```
void mavlink::common::msg::REQUEST_EVENT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

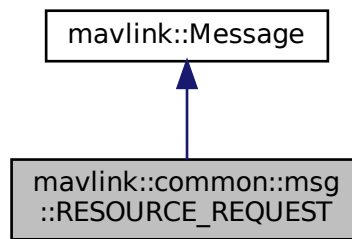
- single_hand_controller/include/mavlink/common/mavlink_msg_request_event.hpp

5.412 mavlink::common::msg::RESOURCE_REQUEST Struct Reference

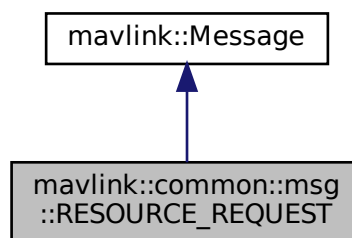
[RESOURCE_REQUEST](#) message.

```
#include <mavlink_msg_resource_request.hpp>
```


Inheritance diagram for mavlink::common::msg::RESOURCE_REQUEST:



Collaboration diagram for mavlink::common::msg::RESOURCE_REQUEST:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **request_id**
- `uint8_t` **uri_type**
- `std::array< uint8_t, 120 >` **uri**
- `uint8_t` **transfer_type**
- `std::array< uint8_t, 120 >` **storage**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 142
- static constexpr size_t **LENGTH** = 243
- static constexpr size_t **MIN_LENGTH** = 243
- static constexpr uint8_t **CRC_EXTRA** = 72
- static constexpr auto **NAME** = "RESOURCE_REQUEST"

5.412.1 Detailed Description

[RESOURCE_REQUEST](#) message.

The autopilot is requesting a resource (file, binary, other type of data)

5.412.2 Member Function Documentation

5.412.2.1 deserialize()

```
void mavlink::common::msg::RESOURCE_REQUEST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.412.2.2 get_name()

```
std::string mavlink::common::msg::RESOURCE_REQUEST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.412.2.3 serialize()

```
void mavlink::common::msg::RESOURCE_REQUEST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

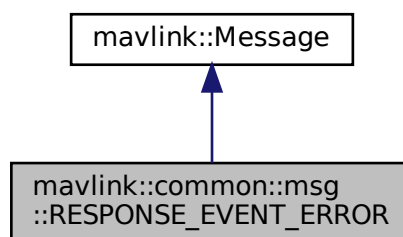
- `single_hand_controller/include/mavlink/common/mavlink_msg_resource_request.hpp`

5.413 mavlink::common::msg::RESPONSE_EVENT_ERROR Struct Reference

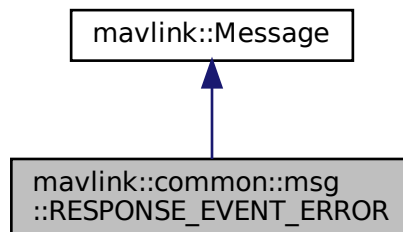
[RESPONSE_EVENT_ERROR](#) message.

```
#include <mavlink_msg_response_event_error.hpp>
```

Inheritance diagram for mavlink::common::msg::RESPONSE_EVENT_ERROR:



Collaboration diagram for mavlink::common::msg::RESPONSE_EVENT_ERROR:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t target_system`
- `uint8_t target_component`
- `uint16_t sequence`
- `uint16_t sequence_oldest_available`
- `uint8_t reason`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 413`
- `static constexpr size_t LENGTH = 7`
- `static constexpr size_t MIN_LENGTH = 7`
- `static constexpr uint8_t CRC_EXTRA = 77`
- `static constexpr auto NAME = "RESPONSE_EVENT_ERROR"`

5.413.1 Detailed Description

[RESPONSE_EVENT_ERROR](#) message.

Response to a [REQUEST_EVENT](#) in case of an error (e.g. the event is not available anymore).

5.413.2 Member Function Documentation

5.413.2.1 deserialize()

```
void mavlink::common::msg::RESPONSE_EVENT_ERROR::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.413.2.2 get_name()

```
std::string mavlink::common::msg::RESPONSE_EVENT_ERROR::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.413.2.3 serialize()

```
void mavlink::common::msg::RESPONSE_EVENT_ERROR::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

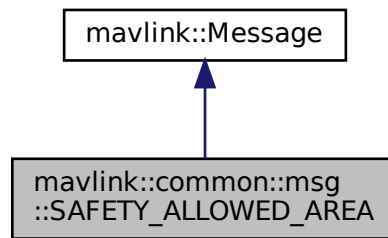
- single_hand_controller/include/mavlink/common/mavlink_msg_response_event_error.hpp

5.414 mavlink::common::msg::SAFETY_ALLOWED_AREA Struct Reference

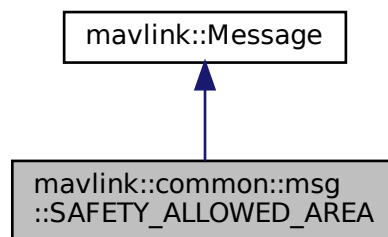
[SAFETY_ALLOWED_AREA](#) message.

```
#include <mavlink_msg_safety_allowed_area.hpp>
```

Inheritance diagram for mavlink::common::msg::SAFETY_ALLOWED_AREA:



Collaboration diagram for mavlink::common::msg::SAFETY_ALLOWED_AREA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **frame**
- `float` **p1x**

- float **p1y**
- float **p1z**
- float **p2x**
- float **p2y**
- float **p2z**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 55
- static constexpr size_t **LENGTH** = 25
- static constexpr size_t **MIN_LENGTH** = 25
- static constexpr uint8_t **CRC_EXTRA** = 3
- static constexpr auto **NAME** = "SAFETY_ALLOWED_AREA"

5.414.1 Detailed Description

[SAFETY_ALLOWED_AREA](#) message.

Read out the safety zone the MAV currently assumes.

5.414.2 Member Function Documentation

5.414.2.1 deserialize()

```
void mavlink::common::msg::SAFETY_ALLOWED_AREA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.414.2.2 get_name()

```
std::string mavlink::common::msg::SAFETY_ALLOWED_AREA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.414.2.3 `serialize()`

```
void mavlink::common::msg::SAFETY_ALLOWED_AREA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

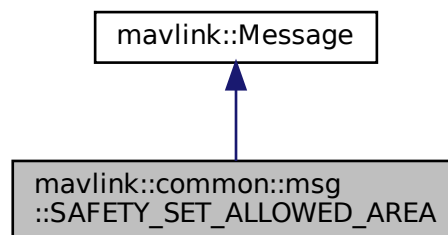
- `single_hand_controller/include/mavlink/common/mavlink_msg_safety_allowed_area.hpp`

5.415 `mavlink::common::msg::SAFETY_SET_ALLOWED_AREA` Struct Reference

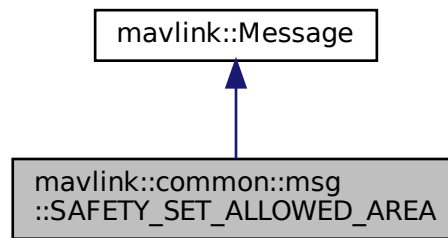
[SAFETY_SET_ALLOWED_AREA](#) message.

```
#include <mavlink_msg_safety_set_allowed_area.hpp>
```

Inheritance diagram for `mavlink::common::msg::SAFETY_SET_ALLOWED_AREA`:



Collaboration diagram for mavlink::common::msg::SAFETY_SET_ALLOWED_AREA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint8_t` **frame**
- `float` **p1x**
- `float` **p1y**
- `float` **p1z**
- `float` **p2x**
- `float` **p2y**
- `float` **p2z**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 54
- static constexpr `size_t` **LENGTH** = 27
- static constexpr `size_t` **MIN_LENGTH** = 27
- static constexpr `uint8_t` **CRC_EXTRA** = 15
- static constexpr auto **NAME** = "SAFETY_SET_ALLOWED_AREA"

5.415.1 Detailed Description

[SAFETY_SET_ALLOWED_AREA](#) message.

Set a safety zone (volume), which is defined by two corners of a cube. This message can be used to tell the MAV which setpoints/waypoints to accept and which to reject. Safety areas are often enforced by national or competition regulations.

5.415.2 Member Function Documentation

5.415.2.1 deserialize()

```
void mavlink::common::msg::SAFETY_SET_ALLOWED_AREA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.415.2.2 get_name()

```
std::string mavlink::common::msg::SAFETY_SET_ALLOWED_AREA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.415.2.3 serialize()

```
void mavlink::common::msg::SAFETY_SET_ALLOWED_AREA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

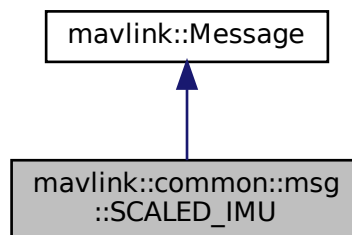
- single_hand_controller/include/mavlink/common/mavlink_msg_safety_set_allowed_area.hpp

5.416 mavlink::common::msg::SCALED_IMU Struct Reference

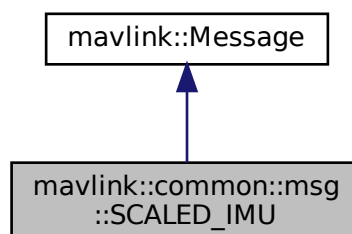
[SCALED_IMU](#) message.

```
#include <mavlink_msg_scaled_imu.hpp>
```

Inheritance diagram for mavlink::common::msg::SCALED_IMU:



Collaboration diagram for mavlink::common::msg::SCALED_IMU:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint32_t` [time_boot_ms](#)
- `int16_t` [xacc](#)
- `int16_t` [yacc](#)
- `int16_t` [zacc](#)
- `int16_t` [xgyro](#)
- `int16_t` [ygyro](#)
- `int16_t` [zgyro](#)
- `int16_t` [xmag](#)
- `int16_t` [ymag](#)
- `int16_t` [zmag](#)
- `int16_t` [temperature](#)

Static Public Attributes

- static constexpr `msgid_t` [MSG_ID](#) = 26
- static constexpr `size_t` [LENGTH](#) = 24
- static constexpr `size_t` [MIN_LENGTH](#) = 22
- static constexpr `uint8_t` [CRC_EXTRA](#) = 170
- static constexpr auto [NAME](#) = "SCALED_IMU"

5.416.1 Detailed Description

[SCALED_IMU](#) message.

The RAW IMU readings for the usual 9DOF sensor setup. This message should contain the scaled values to the described units

5.416.2 Member Function Documentation

5.416.2.1 [deserialize\(\)](#)

```
void mavlink::common::msg::SCALED_IMU::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.416.2.2 get_name()

```
std::string mavlink::common::msg::SCALED_IMU::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.416.2.3 serialize()

```
void mavlink::common::msg::SCALED_IMU::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

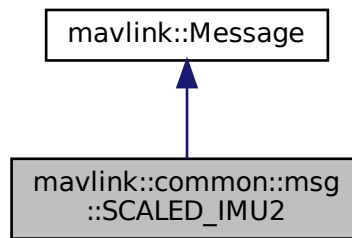
- single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu.hpp

5.417 mavlink::common::msg::SCALED_IMU2 Struct Reference

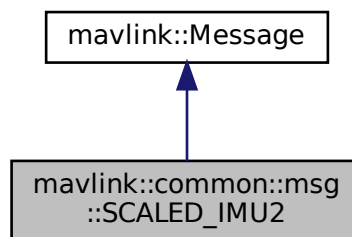
[SCALED_IMU2](#) message.

```
#include <mavlink_msg_scaled_imu2.hpp>
```

Inheritance diagram for mavlink::common::msg::SCALED_IMU2:



Collaboration diagram for mavlink::common::msg::SCALED_IMU2:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `int16_t` **xacc**

- `int16_t yacc`
- `int16_t zacc`
- `int16_t xgyro`
- `int16_t ygyro`
- `int16_t zgyro`
- `int16_t xmag`
- `int16_t ymag`
- `int16_t zmag`
- `int16_t temperature`

Static Public Attributes

- static constexpr `msgid_t MSG_ID` = 116
- static constexpr `size_t LENGTH` = 24
- static constexpr `size_t MIN_LENGTH` = 22
- static constexpr `uint8_t CRC_EXTRA` = 76
- static constexpr auto `NAME` = "SCALED_IMU2"

5.417.1 Detailed Description

[SCALED_IMU2](#) message.

The RAW IMU readings for secondary 9DOF sensor setup. This message should contain the scaled values to the described units

5.417.2 Member Function Documentation

5.417.2.1 deserialize()

```
void mavlink::common::msg::SCALED_IMU2::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.417.2.2 get_name()

```
std::string mavlink::common::msg::SCALED_IMU2::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.417.2.3 `serialize()`

```
void mavlink::common::msg::SCALED_IMU2::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

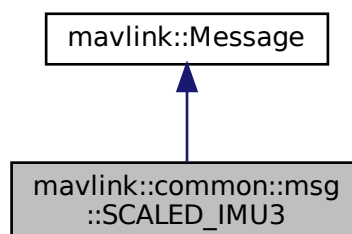
- `single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu2.hpp`

5.418 `mavlink::common::msg::SCALED_IMU3` Struct Reference

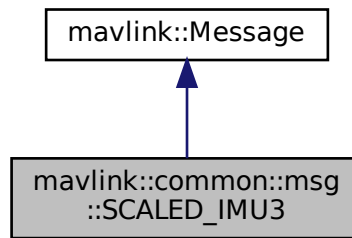
[SCALED_IMU3](#) message.

```
#include <mavlink_msg_scaled_imu3.hpp>
```

Inheritance diagram for `mavlink::common::msg::SCALED_IMU3`:



Collaboration diagram for mavlink::common::msg::SCALED_IMU3:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- int16_t **xacc**
- int16_t **yacc**
- int16_t **zacc**
- int16_t **xgyro**
- int16_t **ygyro**
- int16_t **zgyro**
- int16_t **xmag**
- int16_t **ymag**
- int16_t **zmag**
- int16_t **temperature**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 129
- static constexpr size_t **LENGTH** = 24
- static constexpr size_t **MIN_LENGTH** = 22
- static constexpr uint8_t **CRC_EXTRA** = 46
- static constexpr auto **NAME** = "SCALED_IMU3"

5.418.1 Detailed Description

[SCALED_IMU3](#) message.

The RAW IMU readings for 3rd 9DOF sensor setup. This message should contain the scaled values to the described units

5.418.2 Member Function Documentation

5.418.2.1 deserialize()

```
void mavlink::common::msg::SCALED_IMU3::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.418.2.2 get_name()

```
std::string mavlink::common::msg::SCALED_IMU3::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.418.2.3 serialize()

```
void mavlink::common::msg::SCALED_IMU3::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

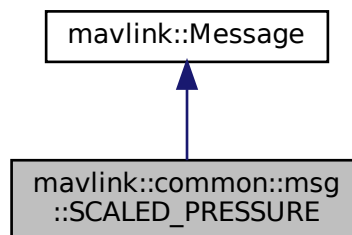
- `single_hand_controller/include/mavlink/common/mavlink_msg_scaled_imu3.hpp`

5.419 mavlink::common::msg::SCALED_PRESSURE Struct Reference

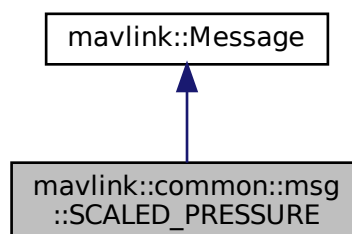
[SCALED_PRESSURE](#) message.

```
#include <mavlink_msg_scaled_pressure.hpp>
```

Inheritance diagram for mavlink::common::msg::SCALED_PRESSURE:



Collaboration diagram for mavlink::common::msg::SCALED_PRESSURE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **press_abs**
- `float` **press_diff**
- `int16_t` **temperature**
- `int16_t` **temperature_press_diff**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 29
- static constexpr `size_t` **LENGTH** = 16
- static constexpr `size_t` **MIN_LENGTH** = 14
- static constexpr `uint8_t` **CRC_EXTRA** = 115
- static constexpr auto **NAME** = "SCALED_PRESSURE"

5.419.1 Detailed Description

[SCALED_PRESSURE](#) message.

The pressure readings for the typical setup of one absolute and differential pressure sensor. The units are as specified in each field.

5.419.2 Member Function Documentation

5.419.2.1 deserialize()

```
void mavlink::common::msg::SCALED_PRESSURE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.419.2.2 get_name()

```
std::string mavlink::common::msg::SCALED_PRESSURE::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.419.2.3 serialize()

```
void mavlink::common::msg::SCALED_PRESSURE::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

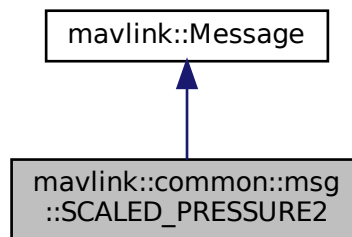
- single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure.hpp

5.420 mavlink::common::msg::SCALED_PRESSURE2 Struct Reference

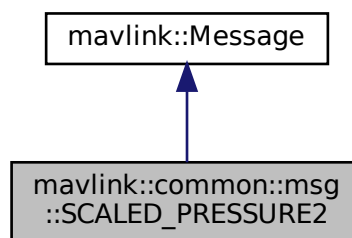
[SCALED_PRESSURE2](#) message.

```
#include <mavlink_msg_scaled_pressure2.hpp>
```

Inheritance diagram for mavlink::common::msg::SCALED_PRESSURE2:



Collaboration diagram for mavlink::common::msg::SCALED_PRESSURE2:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `float` **press_abs**
- `float` **press_diff**
- `int16_t` **temperature**
- `int16_t` **temperature_press_diff**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 137
- static constexpr size_t **LENGTH** = 16
- static constexpr size_t **MIN_LENGTH** = 14
- static constexpr uint8_t **CRC_EXTRA** = 195
- static constexpr auto **NAME** = "SCALED_PRESSURE2"

5.420.1 Detailed Description

[SCALED_PRESSURE2](#) message.

Barometer readings for 2nd barometer

5.420.2 Member Function Documentation

5.420.2.1 deserialize()

```
void mavlink::common::msg::SCALED_PRESSURE2::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.420.2.2 get_name()

```
std::string mavlink::common::msg::SCALED_PRESSURE2::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.420.2.3 serialize()

```
void mavlink::common::msg::SCALED_PRESSURE2::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

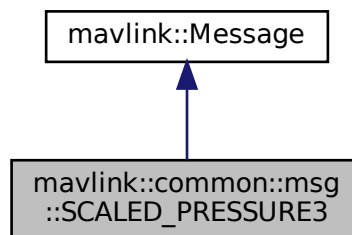
- `single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure2.hpp`

5.421 `mavlink::common::msg::SCALED_PRESSURE3` Struct Reference

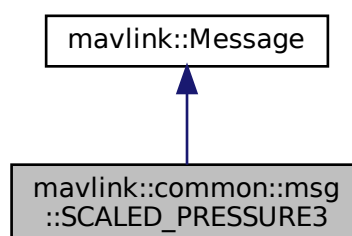
`SCALED_PRESSURE3` message.

```
#include <mavlink_msg_scaled_pressure3.hpp>
```

Inheritance diagram for `mavlink::common::msg::SCALED_PRESSURE3`:



Collaboration diagram for `mavlink::common::msg::SCALED_PRESSURE3`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- float **press_abs**
- float **press_diff**
- int16_t **temperature**
- int16_t **temperature_press_diff**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 143
- static constexpr size_t **LENGTH** = 16
- static constexpr size_t **MIN_LENGTH** = 14
- static constexpr uint8_t **CRC_EXTRA** = 131
- static constexpr auto **NAME** = "SCALED_PRESSURE3"

5.421.1 Detailed Description

[SCALED_PRESSURE3](#) message.

Barometer readings for 3rd barometer

5.421.2 Member Function Documentation

5.421.2.1 deserialize()

```
void mavlink::common::msg::SCALED_PRESSURE3::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.421.2.2 get_name()

```
std::string mavlink::common::msg::SCALED_PRESSURE3::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.421.2.3 serialize()

```
void mavlink::common::msg::SCALED_PRESSURE3::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

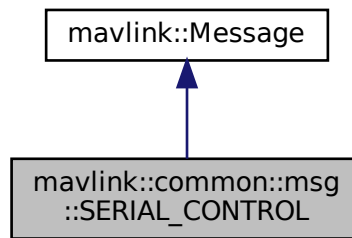
- single_hand_controller/include/mavlink/common/mavlink_msg_scaled_pressure3.hpp

5.422 mavlink::common::msg::SERIAL_CONTROL Struct Reference

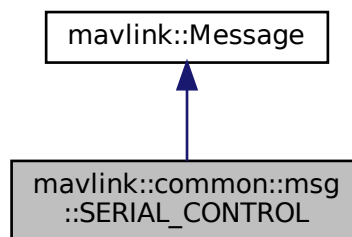
[SERIAL_CONTROL](#) message.

```
#include <mavlink_msg_serial_control.hpp>
```

Inheritance diagram for mavlink::common::msg::SERIAL_CONTROL:



Collaboration diagram for mavlink::common::msg::SERIAL_CONTROL:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **device**
- `uint8_t` **flags**

- uint16_t **timeout**
- uint32_t **baudrate**
- uint8_t **count**
- std::array< uint8_t, 70 > **data**
- uint8_t **target_system**
- uint8_t **target_component**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 126
- static constexpr size_t **LENGTH** = 81
- static constexpr size_t **MIN_LENGTH** = 79
- static constexpr uint8_t **CRC_EXTRA** = 220
- static constexpr auto **NAME** = "SERIAL_CONTROL"

5.422.1 Detailed Description

[SERIAL_CONTROL](#) message.

Control a serial port. This can be used for raw access to an onboard serial peripheral such as a GPS or telemetry radio. It is designed to make it possible to update the devices firmware via MAVLink messages or change the devices settings. A message with zero bytes can be used to change just the baudrate.

5.422.2 Member Function Documentation

5.422.2.1 deserialize()

```
void mavlink::common::msg::SERIAL_CONTROL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.422.2.2 get_name()

```
std::string mavlink::common::msg::SERIAL_CONTROL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.422.2.3 serialize()

```
void mavlink::common::msg::SERIAL_CONTROL::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

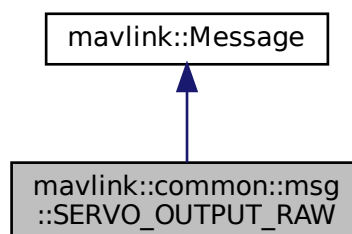
- single_hand_controller/include/mavlink/common/mavlink_msg_serial_control.hpp

5.423 mavlink::common::msg::SERVO_OUTPUT_RAW Struct Reference

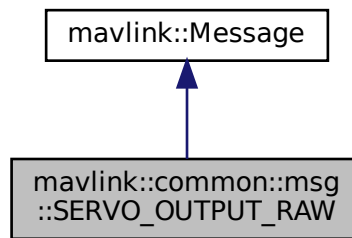
[SERVO_OUTPUT_RAW](#) message.

```
#include <mavlink_msg_servo_output_raw.hpp>
```

Inheritance diagram for mavlink::common::msg::SERVO_OUTPUT_RAW:



Collaboration diagram for mavlink::common::msg::SERVO_OUTPUT_RAW:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_usec**
- `uint8_t` **port**
- `uint16_t` **servo1_raw**
- `uint16_t` **servo2_raw**
- `uint16_t` **servo3_raw**
- `uint16_t` **servo4_raw**
- `uint16_t` **servo5_raw**
- `uint16_t` **servo6_raw**
- `uint16_t` **servo7_raw**
- `uint16_t` **servo8_raw**
- `uint16_t` **servo9_raw**
- `uint16_t` **servo10_raw**
- `uint16_t` **servo11_raw**
- `uint16_t` **servo12_raw**
- `uint16_t` **servo13_raw**
- `uint16_t` **servo14_raw**
- `uint16_t` **servo15_raw**
- `uint16_t` **servo16_raw**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 36
- static constexpr size_t **LENGTH** = 37
- static constexpr size_t **MIN_LENGTH** = 21
- static constexpr uint8_t **CRC_EXTRA** = 222
- static constexpr auto **NAME** = "SERVO_OUTPUT_RAW"

5.423.1 Detailed Description

[SERVO_OUTPUT_RAW](#) message.

Superseded by [ACTUATOR_OUTPUT_STATUS](#). The RAW values of the servo outputs (for RC input from the remote, use the [RC_CHANNELS](#) messages). The standard PPM modulation is as follows: 1000 microseconds: 0%, 2000 microseconds: 100%.

5.423.2 Member Function Documentation

5.423.2.1 deserialize()

```
void mavlink::common::msg::SERVO_OUTPUT_RAW::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.423.2.2 get_name()

```
std::string mavlink::common::msg::SERVO_OUTPUT_RAW::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.423.2.3 `serialize()`

```
void mavlink::common::msg::SERVO_OUTPUT_RAW::serialize (  
    mavlink::MsgMap & map ) const    [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

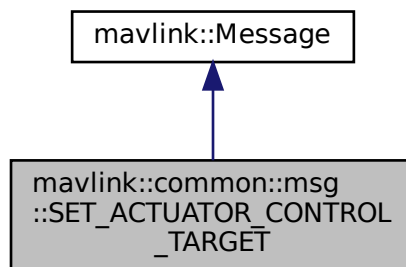
- single_hand_controller/include/mavlink/common/mavlink_msg_servo_output_raw.hpp

5.424 mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET Struct Reference

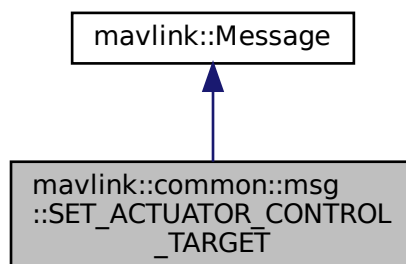
[SET_ACTUATOR_CONTROL_TARGET](#) message.

```
#include <mavlink_msg_set_actuator_control_target.hpp>
```

Inheritance diagram for mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET:



Collaboration diagram for mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **group_mlx**
- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< float, 8 >` **controls**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 139
- static constexpr `size_t` **LENGTH** = 43
- static constexpr `size_t` **MIN_LENGTH** = 43
- static constexpr `uint8_t` **CRC_EXTRA** = 168
- static constexpr auto **NAME** = "SET_ACTUATOR_CONTROL_TARGET"

5.424.1 Detailed Description

[SET_ACTUATOR_CONTROL_TARGET](#) message.

Set the vehicle attitude and body angular rates.

5.424.2 Member Function Documentation

5.424.2.1 deserialize()

```
void mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.424.2.2 get_name()

```
std::string mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.424.2.3 serialize()

```
void mavlink::common::msg::SET_ACTUATOR_CONTROL_TARGET::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

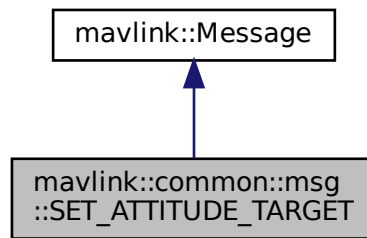
- single_hand_controller/include/mavlink/common/mavlink_msg_set_actuator_control_target.hpp

5.425 mavlink::common::msg::SET_ATTITUDE_TARGET Struct Reference

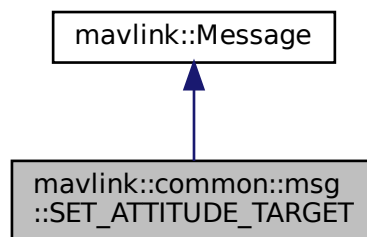
[SET_ATTITUDE_TARGET](#) message.

```
#include <mavlink_msg_set_attitude_target.hpp>
```

Inheritance diagram for mavlink::common::msg::SET_ATTITUDE_TARGET:



Collaboration diagram for mavlink::common::msg::SET_ATTITUDE_TARGET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint8_t` **target_system**

- uint8_t **target_component**
- uint8_t **type_mask**
- std::array< float, 4 > **q**
- float **body_roll_rate**
- float **body_pitch_rate**
- float **body_yaw_rate**
- float **thrust**
- std::array< float, 3 > **thrust_body**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 82
- static constexpr size_t **LENGTH** = 51
- static constexpr size_t **MIN_LENGTH** = 39
- static constexpr uint8_t **CRC_EXTRA** = 49
- static constexpr auto **NAME** = "SET_ATTITUDE_TARGET"

5.425.1 Detailed Description

[SET_ATTITUDE_TARGET](#) message.

Sets a desired vehicle attitude. Used by an external controller to command the vehicle (manual controller or other system).

5.425.2 Member Function Documentation

5.425.2.1 deserialize()

```
void mavlink::common::msg::SET_ATTITUDE_TARGET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.425.2.2 get_name()

```
std::string mavlink::common::msg::SET_ATTITUDE_TARGET::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.425.2.3 serialize()

```
void mavlink::common::msg::SET_ATTITUDE_TARGET::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

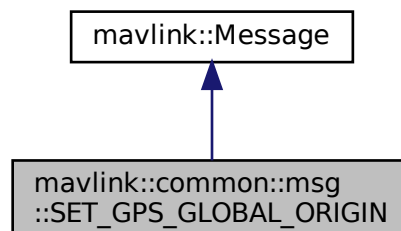
- single_hand_controller/include/mavlink/common/mavlink_msg_set_attitude_target.hpp

5.426 mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN Struct Reference

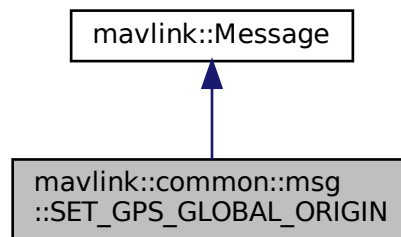
[SET_GPS_GLOBAL_ORIGIN](#) message.

```
#include <mavlink_msg_set_gps_global_origin.hpp>
```

Inheritance diagram for mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN:



Collaboration diagram for mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- Info [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- int32_t **latitude**
- int32_t **longitude**
- int32_t **altitude**
- uint64_t **time_usec**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 48
- static constexpr size_t **LENGTH** = 21
- static constexpr size_t **MIN_LENGTH** = 13
- static constexpr uint8_t **CRC_EXTRA** = 41
- static constexpr auto **NAME** = "SET_GPS_GLOBAL_ORIGIN"

5.426.1 Detailed Description

[SET_GPS_GLOBAL_ORIGIN](#) message.

Sets the GPS coordinates of the vehicle local origin (0,0,0) position. Vehicle should emit [GPS_GLOBAL_ORIGIN](#) irrespective of whether the origin is changed. This enables transform between the local coordinate frame and the global (GPS) coordinate frame, which may be necessary when (for example) indoor and outdoor settings are connected and the MAV should move from in- to outdoor.

5.426.2 Member Function Documentation

5.426.2.1 deserialize()

```
void mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.426.2.2 get_name()

```
std::string mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.426.2.3 serialize()

```
void mavlink::common::msg::SET_GPS_GLOBAL_ORIGIN::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

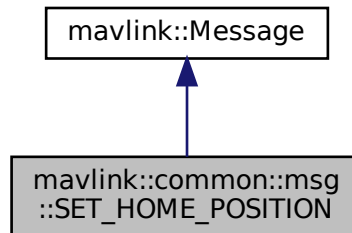
- `single_hand_controller/include/mavlink/common/mavlink_msg_set_gps_global_origin.hpp`

5.427 mavlink::common::msg::SET_HOME_POSITION Struct Reference

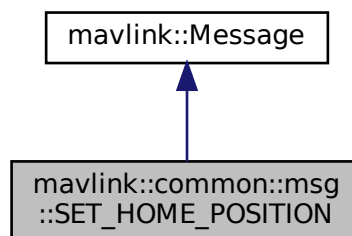
[SET_HOME_POSITION](#) message.

```
#include <mavlink_msg_set_home_position.hpp>
```

Inheritance diagram for mavlink::common::msg::SET_HOME_POSITION:



Collaboration diagram for mavlink::common::msg::SET_HOME_POSITION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **target_system**
- int32_t **latitude**
- int32_t **longitude**
- int32_t **altitude**
- float **x**
- float **y**
- float **z**
- std::array< float, 4 > **q**
- float **approach_x**
- float **approach_y**
- float **approach_z**
- uint64_t **time_usec**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 243
- static constexpr size_t **LENGTH** = 61
- static constexpr size_t **MIN_LENGTH** = 53
- static constexpr uint8_t **CRC_EXTRA** = 85
- static constexpr auto **NAME** = "SET_HOME_POSITION"

5.427.1 Detailed Description

[SET_HOME_POSITION](#) message.

Sets the home position.

The home position is the default position that the system will return to and land on.

The position is set automatically by the system during the takeoff (and may also be set using this message).

The global and local positions encode the position in the respective coordinate frames, while the q parameter

Under normal conditions it describes the heading and terrain slope, which can be used by the aircraft to adju

The approach 3D vector describes the point to which the system should fly in normal flight mode and then perf

Note: the current home position may be emitted in a HOME_POSITION message on request (using MAV_CMD_REQUEST_M

5.427.2 Member Function Documentation

5.427.2.1 deserialize()

```
void mavlink::common::msg::SET_HOME_POSITION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.427.2.2 get_name()

```
std::string mavlink::common::msg::SET_HOME_POSITION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.427.2.3 serialize()

```
void mavlink::common::msg::SET_HOME_POSITION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

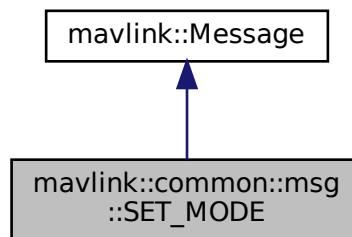
- single_hand_controller/include/mavlink/common/mavlink_msg_set_home_position.hpp

5.428 mavlink::common::msg::SET_MODE Struct Reference

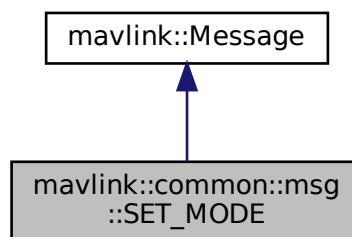
[SET_MODE](#) message.

```
#include <mavlink_msg_set_mode.hpp>
```

Inheritance diagram for mavlink::common::msg::SET_MODE:



Collaboration diagram for mavlink::common::msg::SET_MODE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **base_mode**
- `uint32_t` **custom_mode**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 11
- static constexpr size_t **LENGTH** = 6
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 89
- static constexpr auto **NAME** = "SET_MODE"

5.428.1 Detailed Description

[SET_MODE](#) message.

Set the system mode, as defined by enum MAV_MODE. There is no target component id as the mode is by definition for the overall aircraft, not only for one component.

5.428.2 Member Function Documentation

5.428.2.1 deserialize()

```
void mavlink::common::msg::SET_MODE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.428.2.2 get_name()

```
std::string mavlink::common::msg::SET_MODE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.428.2.3 serialize()

```
void mavlink::common::msg::SET_MODE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

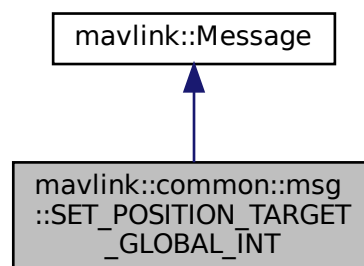
- `single_hand_controller/include/mavlink/common/mavlink_msg_set_mode.hpp`

5.429 mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT Struct Reference

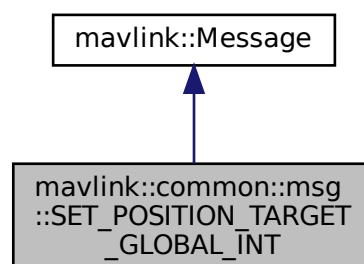
[SET_POSITION_TARGET_GLOBAL_INT](#) message.

```
#include <mavlink_msg_set_position_target_global_int.hpp>
```

Inheritance diagram for `mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT`:



Collaboration diagram for `mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **coordinate_frame**
- uint16_t **type_mask**
- int32_t **lat_int**
- int32_t **lon_int**
- float **alt**
- float **vx**
- float **vy**
- float **vz**
- float **afx**
- float **afy**
- float **afz**
- float **yaw**
- float **yaw_rate**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 86
- static constexpr size_t **LENGTH** = 53
- static constexpr size_t **MIN_LENGTH** = 53
- static constexpr uint8_t **CRC_EXTRA** = 5
- static constexpr auto **NAME** = "SET_POSITION_TARGET_GLOBAL_INT"

5.429.1 Detailed Description

[SET_POSITION_TARGET_GLOBAL_INT](#) message.

Sets a desired vehicle position, velocity, and/or acceleration in a global coordinate system (WGS84). Used by an external controller to command the vehicle (manual controller or other system).

5.429.2 Member Function Documentation

5.429.2.1 deserialize()

```
void mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.429.2.2 get_name()

```
std::string mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.429.2.3 serialize()

```
void mavlink::common::msg::SET_POSITION_TARGET_GLOBAL_INT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

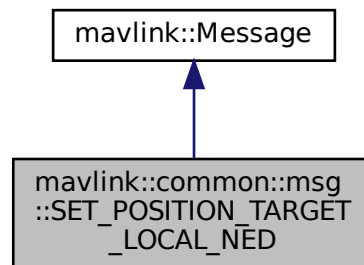
- single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_global_int.hpp

5.430 mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED Struct Reference

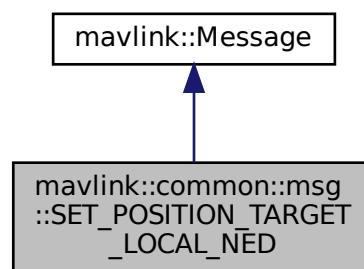
[SET_POSITION_TARGET_LOCAL_NED](#) message.

```
#include <mavlink_msg_set_position_target_local_ned.hpp>
```

Inheritance diagram for mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED:



Collaboration diagram for mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint32_t **time_boot_ms**
- uint8_t **target_system**
- uint8_t **target_component**
- uint8_t **coordinate_frame**
- uint16_t **type_mask**
- float **x**
- float **y**
- float **z**
- float **vx**
- float **vy**
- float **vz**
- float **afx**
- float **afy**
- float **afz**
- float **yaw**
- float **yaw_rate**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 84
- static constexpr size_t **LENGTH** = 53
- static constexpr size_t **MIN_LENGTH** = 53
- static constexpr uint8_t **CRC_EXTRA** = 143
- static constexpr auto **NAME** = "SET_POSITION_TARGET_LOCAL_NED"

5.430.1 Detailed Description

[SET_POSITION_TARGET_LOCAL_NED](#) message.

Sets a desired vehicle position in a local north-east-down coordinate frame. Used by an external controller to command the vehicle (manual controller or other system).

5.430.2 Member Function Documentation

5.430.2.1 deserialize()

```
void mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.430.2.2 `get_name()`

```
std::string mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.430.2.3 `serialize()`

```
void mavlink::common::msg::SET_POSITION_TARGET_LOCAL_NED::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

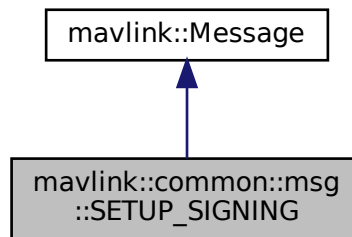
- `single_hand_controller/include/mavlink/common/mavlink_msg_set_position_target_local_ned.hpp`

5.431 `mavlink::common::msg::SETUP_SIGNING` Struct Reference

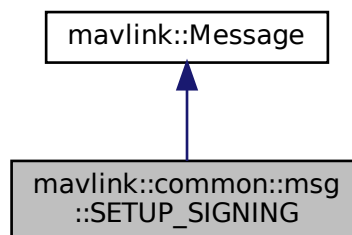
[SETUP_SIGNING](#) message.

```
#include <mavlink_msg_setup_signing.hpp>
```

Inheritance diagram for mavlink::common::msg::SETUP_SIGNING:



Collaboration diagram for mavlink::common::msg::SETUP_SIGNING:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `std::array< uint8_t, 32 >` **secret_key**
- `uint64_t` **initial_timestamp**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 256
- static constexpr size_t **LENGTH** = 42
- static constexpr size_t **MIN_LENGTH** = 42
- static constexpr uint8_t **CRC_EXTRA** = 71
- static constexpr auto **NAME** = "SETUP_SIGNING"

5.431.1 Detailed Description

[SETUP_SIGNING](#) message.

Setup a MAVLink2 signing key. If called with secret_key of all zero and zero initial_timestamp will disable signing

5.431.2 Member Function Documentation

5.431.2.1 deserialize()

```
void mavlink::common::msg::SETUP_SIGNING::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.431.2.2 get_name()

```
std::string mavlink::common::msg::SETUP_SIGNING::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.431.2.3 serialize()

```
void mavlink::common::msg::SETUP_SIGNING::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

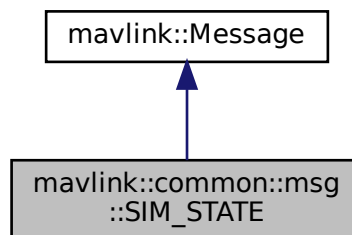
- `single_hand_controller/include/mavlink/common/mavlink_msg_setup_signing.hpp`

5.432 mavlink::common::msg::SIM_STATE Struct Reference

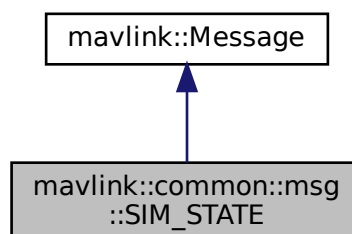
[SIM_STATE](#) message.

```
#include <mavlink_msg_sim_state.hpp>
```

Inheritance diagram for mavlink::common::msg::SIM_STATE:



Collaboration diagram for mavlink::common::msg::SIM_STATE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- float **q1**
- float **q2**
- float **q3**
- float **q4**
- float **roll**
- float **pitch**
- float **yaw**
- float **xacc**
- float **yacc**
- float **zacc**
- float **xgyro**
- float **ygyro**
- float **zgyro**
- float **lat**
- float **lon**
- float **alt**
- float **std_dev_horz**
- float **std_dev_vert**
- float **vn**
- float **ve**
- float **vd**
- `int32_t` **lat_int**
- `int32_t` **lon_int**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 108
- static constexpr `size_t` **LENGTH** = 92
- static constexpr `size_t` **MIN_LENGTH** = 84
- static constexpr `uint8_t` **CRC_EXTRA** = 32
- static constexpr auto **NAME** = "SIM_STATE"

5.432.1 Detailed Description

[SIM_STATE](#) message.

Status of simulation environment, if used

5.432.2 Member Function Documentation

5.432.2.1 deserialize()

```
void mavlink::common::msg::SIM_STATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.432.2.2 get_name()

```
std::string mavlink::common::msg::SIM_STATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.432.2.3 serialize()

```
void mavlink::common::msg::SIM_STATE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

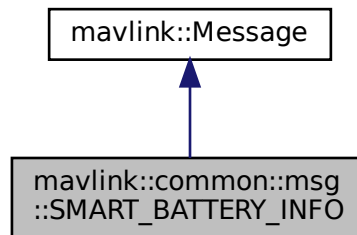
- `single_hand_controller/include/mavlink/common/mavlink_msg_sim_state.hpp`

5.433 mavlink::common::msg::SMART_BATTERY_INFO Struct Reference

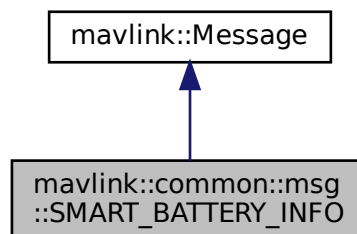
[SMART_BATTERY_INFO](#) message.

```
#include <mavlink_msg_smart_battery_info.hpp>
```

Inheritance diagram for mavlink::common::msg::SMART_BATTERY_INFO:



Collaboration diagram for mavlink::common::msg::SMART_BATTERY_INFO:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **id**
- uint8_t **battery_function**
- uint8_t **type**
- int32_t **capacity_full_specification**
- int32_t **capacity_full**
- uint16_t **cycle_count**
- std::array< char, 16 > **serial_number**
- std::array< char, 50 > **device_name**
- uint16_t **weight**
- uint16_t **discharge_minimum_voltage**
- uint16_t **charging_minimum_voltage**
- uint16_t **resting_minimum_voltage**
- uint16_t **charging_maximum_voltage**
- uint8_t **cells_in_series**
- uint32_t **discharge_maximum_current**
- uint32_t **discharge_maximum_burst_current**
- std::array< char, 11 > **manufacture_date**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 370
- static constexpr size_t **LENGTH** = 109
- static constexpr size_t **MIN_LENGTH** = 87
- static constexpr uint8_t **CRC_EXTRA** = 75
- static constexpr auto **NAME** = "SMART_BATTERY_INFO"

5.433.1 Detailed Description

[SMART_BATTERY_INFO](#) message.

Smart Battery information (static/infrequent update). Use for updates from: smart battery to flight stack, flight stack to GCS. Use [BATTERY_STATUS](#) for the frequent battery updates.

5.433.2 Member Function Documentation

5.433.2.1 deserialize()

```
void mavlink::common::msg::SMART_BATTERY_INFO::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.433.2.2 `get_name()`

```
std::string mavlink::common::msg::SMART_BATTERY_INFO::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.433.2.3 `serialize()`

```
void mavlink::common::msg::SMART_BATTERY_INFO::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

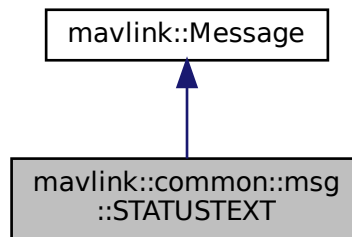
- `single_hand_controller/include/mavlink/common/mavlink_msg_smart_battery_info.hpp`

5.434 `mavlink::common::msg::STATUSTEXT` Struct Reference

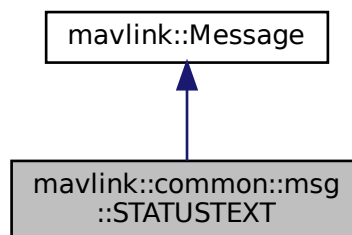
[STATUSTEXT](#) message.

```
#include <mavlink_msg_statustext.hpp>
```

Inheritance diagram for mavlink::common::msg::STATUSTEXT:



Collaboration diagram for mavlink::common::msg::STATUSTEXT:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **severity**
- `std::array< char, 50 >` **text**
- `uint16_t` **id**
- `uint8_t` **chunk_seq**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 253
- static constexpr size_t **LENGTH** = 54
- static constexpr size_t **MIN_LENGTH** = 51
- static constexpr uint8_t **CRC_EXTRA** = 83
- static constexpr auto **NAME** = "STATUSTEXT"

5.434.1 Detailed Description

[STATUSTEXT](#) message.

Status text message. These messages are printed in yellow in the COMM console of QGroundControl. WARNING: They consume quite some bandwidth, so use only for important status and error messages. If implemented wisely, these messages are buffered on the MCU and sent only at a limited rate (e.g. 10 Hz).

5.434.2 Member Function Documentation

5.434.2.1 deserialize()

```
void mavlink::common::msg::STATUSTEXT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.434.2.2 get_name()

```
std::string mavlink::common::msg::STATUSTEXT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.434.2.3 serialize()

```
void mavlink::common::msg::STATUSTEXT::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

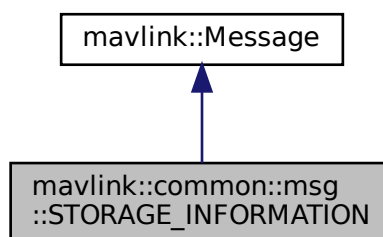
- `single_hand_controller/include/mavlink/common/mavlink_msg_statustext.hpp`

5.435 `mavlink::common::msg::STORAGE_INFORMATION` Struct Reference

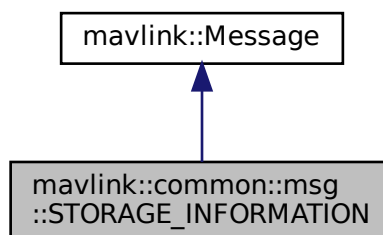
[STORAGE_INFORMATION](#) message.

```
#include <mavlink_msg_storage_information.hpp>
```

Inheritance diagram for `mavlink::common::msg::STORAGE_INFORMATION`:



Collaboration diagram for `mavlink::common::msg::STORAGE_INFORMATION`:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **time_boot_ms**
- `uint8_t` **storage_id**
- `uint8_t` **storage_count**
- `uint8_t` **status**
- `float` **total_capacity**
- `float` **used_capacity**
- `float` **available_capacity**
- `float` **read_speed**
- `float` **write_speed**
- `uint8_t` **type**
- `std::array< char, 32 >` **name**
- `uint8_t` **storage_usage**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 261
- static constexpr `size_t` **LENGTH** = 61
- static constexpr `size_t` **MIN_LENGTH** = 27
- static constexpr `uint8_t` **CRC_EXTRA** = 179
- static constexpr auto **NAME** = "STORAGE_INFORMATION"

5.435.1 Detailed Description

[STORAGE_INFORMATION](#) message.

Information about a storage medium. This message is sent in response to a request with MAV_CMD_REQUEST_MESSAGE and whenever the status of the storage changes (STORAGE_STATUS). Use MAV_CMD_REQUEST_MESSAGE.param2 to indicate the index/id of requested storage: 0 for all, 1 for first, 2 for second, etc.

5.435.2 Member Function Documentation

5.435.2.1 deserialize()

```
void mavlink::common::msg::STORAGE_INFORMATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.435.2.2 get_name()

```
std::string mavlink::common::msg::STORAGE_INFORMATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.435.2.3 serialize()

```
void mavlink::common::msg::STORAGE_INFORMATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

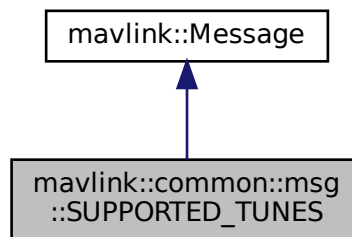
- single_hand_controller/include/mavlink/common/mavlink_msg_storage_information.hpp

5.436 mavlink::common::msg::SUPPORTED_TUNES Struct Reference

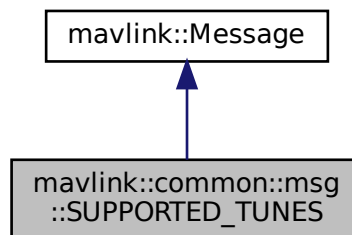
[SUPPORTED_TUNES](#) message.

```
#include <mavlink_msg_supported_tunes.hpp>
```

Inheritance diagram for mavlink::common::msg::SUPPORTED_TUNES:



Collaboration diagram for mavlink::common::msg::SUPPORTED_TUNES:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint32_t` **format**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 401
- static constexpr size_t **LENGTH** = 6
- static constexpr size_t **MIN_LENGTH** = 6
- static constexpr uint8_t **CRC_EXTRA** = 183
- static constexpr auto **NAME** = "SUPPORTED_TUNES"

5.436.1 Detailed Description

[SUPPORTED_TUNES](#) message.

Tune formats supported by vehicle. This should be emitted as response to MAV_CMD_REQUEST_MESSAGE.

5.436.2 Member Function Documentation

5.436.2.1 deserialize()

```
void mavlink::common::msg::SUPPORTED_TUNES::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.436.2.2 get_name()

```
std::string mavlink::common::msg::SUPPORTED_TUNES::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.436.2.3 serialize()

```
void mavlink::common::msg::SUPPORTED_TUNES::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

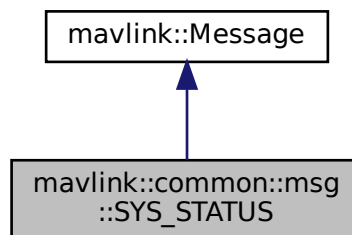
- single_hand_controller/include/mavlink/common/mavlink_msg_supported_tunes.hpp

5.437 mavlink::common::msg::SYS_STATUS Struct Reference

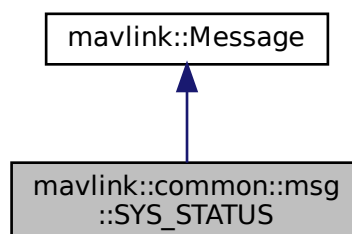
[SYS_STATUS](#) message.

```
#include <mavlink_msg_sys_status.hpp>
```

Inheritance diagram for mavlink::common::msg::SYS_STATUS:



Collaboration diagram for mavlink::common::msg::SYS_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint32_t` **onboard_control_sensors_present**
- `uint32_t` **onboard_control_sensors_enabled**
- `uint32_t` **onboard_control_sensors_health**
- `uint16_t` **load**
- `uint16_t` **voltage_battery**
- `int16_t` **current_battery**
- `int8_t` **battery_remaining**
- `uint16_t` **drop_rate_comm**
- `uint16_t` **errors_comm**
- `uint16_t` **errors_count1**
- `uint16_t` **errors_count2**
- `uint16_t` **errors_count3**
- `uint16_t` **errors_count4**
- `uint32_t` **onboard_control_sensors_present_extended**
- `uint32_t` **onboard_control_sensors_enabled_extended**
- `uint32_t` **onboard_control_sensors_health_extended**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 1
- static constexpr `size_t` **LENGTH** = 43
- static constexpr `size_t` **MIN_LENGTH** = 31
- static constexpr `uint8_t` **CRC_EXTRA** = 124
- static constexpr auto **NAME** = "SYS_STATUS"

5.437.1 Detailed Description

[SYS_STATUS](#) message.

Sensor and subsystem status information. Provides a compact representation of sensor/subsystem status and a few other basic statistics.

5.437.2 Member Function Documentation

5.437.2.1 deserialize()

```
void mavlink::common::msg::SYS_STATUS::deserialize (  
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.437.2.2 get_name()

```
std::string mavlink::common::msg::SYS_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.437.2.3 serialize()

```
void mavlink::common::msg::SYS_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

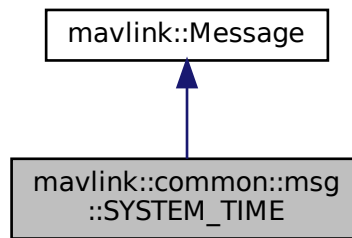
- single_hand_controller/include/mavlink/common/mavlink_msg_sys_status.hpp

5.438 mavlink::common::msg::SYSTEM_TIME Struct Reference

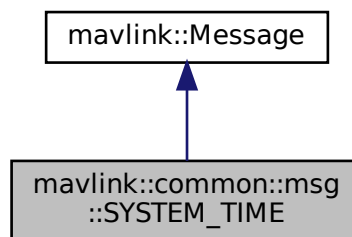
[SYSTEM_TIME](#) message.

```
#include <mavlink_msg_system_time.hpp>
```


Inheritance diagram for mavlink::common::msg::SYSTEM_TIME:



Collaboration diagram for mavlink::common::msg::SYSTEM_TIME:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` [time_unix_usec](#)
- `uint32_t` [time_boot_ms](#)

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 2
- static constexpr size_t **LENGTH** = 12
- static constexpr size_t **MIN_LENGTH** = 12
- static constexpr uint8_t **CRC_EXTRA** = 137
- static constexpr auto **NAME** = "SYSTEM_TIME"

5.438.1 Detailed Description

[SYSTEM_TIME](#) message.

The system time is the time of the sender's master clock. This can be emitted by flight controllers, onboard computers, or other components in the MAVLink network. Components that are using a less reliable time source, such as a battery-backed real time clock, can choose to match their system clock to that of a system that indicates a more recent time. This allows more broadly accurate date stamping of logs, and so on. If precise time synchronization is needed then use [TIMESYNC](#) instead.

5.438.2 Member Function Documentation

5.438.2.1 deserialize()

```
void mavlink::common::msg::SYSTEM_TIME::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.438.2.2 get_name()

```
std::string mavlink::common::msg::SYSTEM_TIME::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.438.2.3 serialize()

```
void mavlink::common::msg::SYSTEM_TIME::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

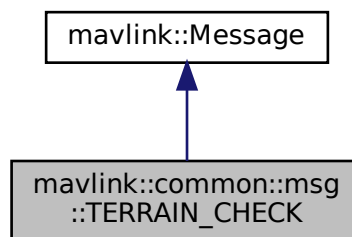
- single_hand_controller/include/mavlink/common/mavlink_msg_system_time.hpp

5.439 mavlink::common::msg::TERRAIN_CHECK Struct Reference

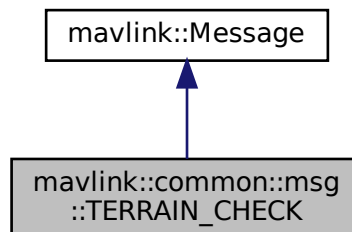
[TERRAIN_CHECK](#) message.

```
#include <mavlink_msg_terrain_check.hpp>
```

Inheritance diagram for mavlink::common::msg::TERRAIN_CHECK:



Collaboration diagram for mavlink::common::msg::TERRAIN_CHECK:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `int32_t` [lat](#)
- `int32_t` [lon](#)

Static Public Attributes

- static constexpr `msgid_t` [MSG_ID](#) = 135
- static constexpr `size_t` [LENGTH](#) = 8
- static constexpr `size_t` [MIN_LENGTH](#) = 8
- static constexpr `uint8_t` [CRC_EXTRA](#) = 203
- static constexpr auto [NAME](#) = "TERRAIN_CHECK"

5.439.1 Detailed Description

[TERRAIN_CHECK](#) message.

Request that the vehicle report terrain height at the given location (expected response is a [TERRAIN_REPORT](#)).
Used by GCS to check if vehicle has all terrain data needed for a mission.

5.439.2 Member Function Documentation

5.439.2.1 [deserialize\(\)](#)

```
void mavlink::common::msg::TERRAIN_CHECK::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.439.2.2 get_name()

```
std::string mavlink::common::msg::TERRAIN_CHECK::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.439.2.3 serialize()

```
void mavlink::common::msg::TERRAIN_CHECK::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

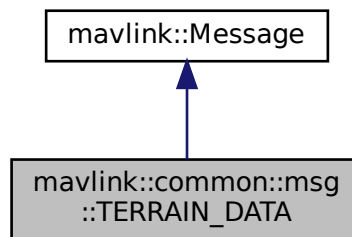
- single_hand_controller/include/mavlink/common/mavlink_msg_terrain_check.hpp

5.440 mavlink::common::msg::TERRAIN_DATA Struct Reference

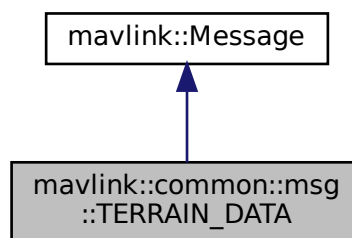
[TERRAIN_DATA](#) message.

```
#include <mavlink_msg_terrain_data.hpp>
```

Inheritance diagram for mavlink::common::msg::TERRAIN_DATA:



Collaboration diagram for mavlink::common::msg::TERRAIN_DATA:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `int32_t` **lat**
- `int32_t` **lon**
- `uint16_t` **grid_spacing**
- `uint8_t` **gridbit**
- `std::array< int16_t, 16 >` **data**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 134
- static constexpr size_t **LENGTH** = 43
- static constexpr size_t **MIN_LENGTH** = 43
- static constexpr uint8_t **CRC_EXTRA** = 229
- static constexpr auto **NAME** = "TERRAIN_DATA"

5.440.1 Detailed Description

[TERRAIN_DATA](#) message.

Terrain data sent from GCS. The lat/lon and grid_spacing must be the same as a lat/lon from a [TERRAIN_REQUEST](#). See terrain protocol docs: <https://mavlink.io/en/services/terrain.html>

5.440.2 Member Function Documentation

5.440.2.1 deserialize()

```
void mavlink::common::msg::TERRAIN_DATA::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.440.2.2 get_name()

```
std::string mavlink::common::msg::TERRAIN_DATA::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.440.2.3 serialize()

```
void mavlink::common::msg::TERRAIN_DATA::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

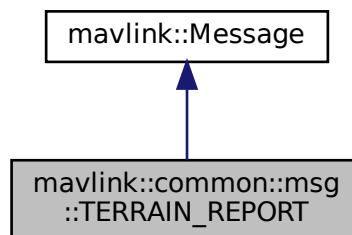
- `single_hand_controller/include/mavlink/common/mavlink_msg_terrain_data.hpp`

5.441 `mavlink::common::msg::TERRAIN_REPORT` Struct Reference

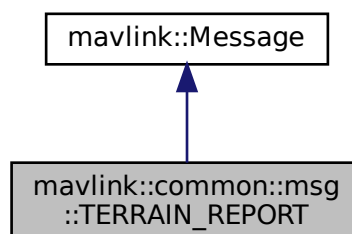
`TERRAIN_REPORT` message.

```
#include <mavlink_msg_terrain_report.hpp>
```

Inheritance diagram for `mavlink::common::msg::TERRAIN_REPORT`:



Collaboration diagram for `mavlink::common::msg::TERRAIN_REPORT`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- int32_t **lat**
- int32_t **lon**
- uint16_t **spacing**
- float **terrain_height**
- float **current_height**
- uint16_t **pending**
- uint16_t **loaded**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 136
- static constexpr size_t **LENGTH** = 22
- static constexpr size_t **MIN_LENGTH** = 22
- static constexpr uint8_t **CRC_EXTRA** = 1
- static constexpr auto **NAME** = "TERRAIN_REPORT"

5.441.1 Detailed Description

[TERRAIN_REPORT](#) message.

Streamed from drone to report progress of terrain map download (initiated by [TERRAIN_REQUEST](#)), or sent as a response to a [TERRAIN_CHECK](#) request. See terrain protocol docs: <https://mavlink.io/en/services/terrain.html>

5.441.2 Member Function Documentation

5.441.2.1 deserialize()

```
void mavlink::common::msg::TERRAIN_REPORT::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.441.2.2 get_name()

```
std::string mavlink::common::msg::TERRAIN_REPORT::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.441.2.3 serialize()

```
void mavlink::common::msg::TERRAIN_REPORT::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

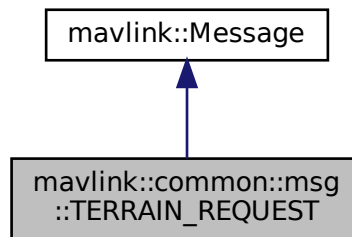
- single_hand_controller/include/mavlink/common/mavlink_msg_terrain_report.hpp

5.442 mavlink::common::msg::TERRAIN_REQUEST Struct Reference

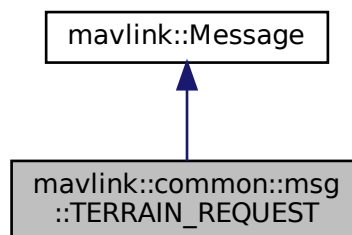
[TERRAIN_REQUEST](#) message.

```
#include <mavlink_msg_terrain_request.hpp>
```

Inheritance diagram for mavlink::common::msg::TERRAIN_REQUEST:



Collaboration diagram for mavlink::common::msg::TERRAIN_REQUEST:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `int32_t` **lat**
- `int32_t` **lon**
- `uint16_t` **grid_spacing**
- `uint64_t` **mask**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 133
- static constexpr size_t **LENGTH** = 18
- static constexpr size_t **MIN_LENGTH** = 18
- static constexpr uint8_t **CRC_EXTRA** = 6
- static constexpr auto **NAME** = "TERRAIN_REQUEST"

5.442.1 Detailed Description

[TERRAIN_REQUEST](#) message.

Request for terrain data and terrain status. See terrain protocol docs: <https://mavlink.io/en/services/terrain.html>

5.442.2 Member Function Documentation

5.442.2.1 deserialize()

```
void mavlink::common::msg::TERRAIN_REQUEST::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.442.2.2 get_name()

```
std::string mavlink::common::msg::TERRAIN_REQUEST::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.442.2.3 serialize()

```
void mavlink::common::msg::TERRAIN_REQUEST::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_terrain_request.hpp`

5.443 thumbstickControl Struct Reference

thumbstick control is used within this library

```
#include <IOhandler.hpp>
```

Public Attributes

- `int X`
- `int Y`

5.443.1 Detailed Description

thumbstick control is used within this library

The documentation for this struct was generated from the following file:

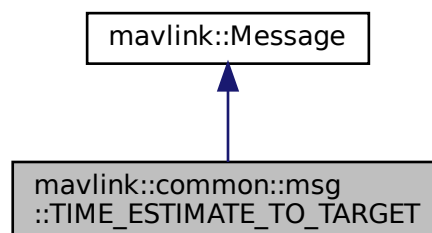
- `single_hand_controller/include/IOhandler.hpp`

5.444 mavlink::common::msg::TIME_ESTIMATE_TO_TARGET Struct Reference

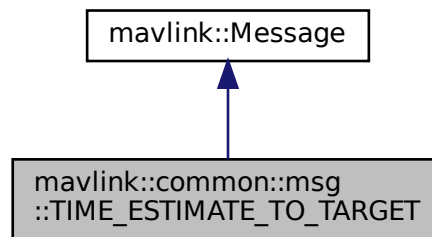
[TIME_ESTIMATE_TO_TARGET](#) message.

```
#include <mavlink_msg_time_estimate_to_target.hpp>
```

Inheritance diagram for `mavlink::common::msg::TIME_ESTIMATE_TO_TARGET`:



Collaboration diagram for mavlink::common::msg::TIME_ESTIMATE_TO_TARGET:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `int32_t` [safe_return](#)
- `int32_t` [land](#)
- `int32_t` [mission_next_item](#)
- `int32_t` [mission_end](#)
- `int32_t` [commanded_action](#)

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 380
- static constexpr size_t **LENGTH** = 20
- static constexpr size_t **MIN_LENGTH** = 20
- static constexpr uint8_t **CRC_EXTRA** = 232
- static constexpr auto **NAME** = "TIME_ESTIMATE_TO_TARGET"

5.444.1 Detailed Description

[TIME_ESTIMATE_TO_TARGET](#) message.

Time/duration estimates for various events and actions given the current vehicle state and position.

5.444.2 Member Function Documentation

5.444.2.1 deserialize()

```
void mavlink::common::msg::TIME_ESTIMATE_TO_TARGET::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.444.2.2 get_name()

```
std::string mavlink::common::msg::TIME_ESTIMATE_TO_TARGET::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.444.2.3 serialize()

```
void mavlink::common::msg::TIME_ESTIMATE_TO_TARGET::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

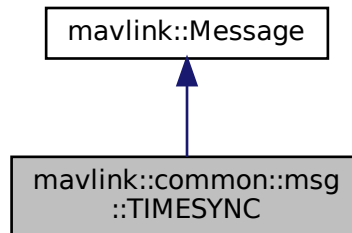
- `single_hand_controller/include/mavlink/common/mavlink_msg_time_estimate_to_target.hpp`

5.445 mavlink::common::msg::TIMESYNC Struct Reference

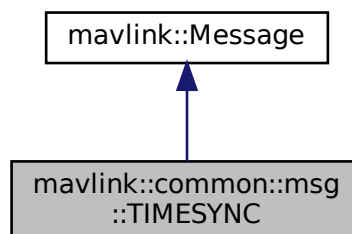
[TIMESYNC](#) message.

```
#include <mavlink_msg_timesync.hpp>
```

Inheritance diagram for mavlink::common::msg::TIMESYNC:



Collaboration diagram for mavlink::common::msg::TIMESYNC:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- int64_t **tc1**
- int64_t **ts1**
- uint8_t **target_system**
- uint8_t **target_component**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 111
- static constexpr size_t **LENGTH** = 18
- static constexpr size_t **MIN_LENGTH** = 16
- static constexpr uint8_t **CRC_EXTRA** = 34
- static constexpr auto **NAME** = "TIMESYNC"

5.445.1 Detailed Description

TIMESYNC message.

Time synchronization message.

The message is used for both timesync requests and responses.

The request is sent with 'ts1=syncing component timestamp' and 'tcl=0', and may be broadcast or targeted to a system.

The response is sent with 'ts1=syncing component timestamp' (mirror back unchanged), and 'tcl=responding component timestamp'.

Systems can determine if they are receiving a request or response based on the value of 'tc'.

If the response has 'target_system==target_component==0' the remote system has not been updated to use the correct timestamp.

Timestamps are UNIX Epoch time or time since system boot in nanoseconds (the timestamp format can be inferred from the 'ts1' field).

The message sequence is repeated numerous times with results being filtered/averaged to estimate the offset.

See also: <https://mavlink.io/en/services/timesync.html>.

5.445.2 Member Function Documentation

5.445.2.1 deserialize()

```
void mavlink::common::msg::TIMESYNC::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.445.2.2 `get_name()`

```
std::string mavlink::common::msg::TIMESYNC::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.445.2.3 `serialize()`

```
void mavlink::common::msg::TIMESYNC::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

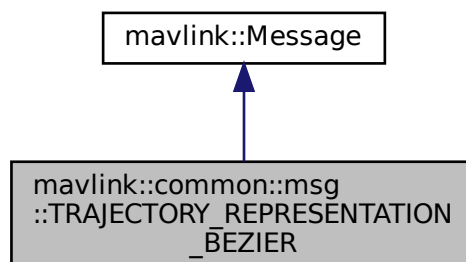
- `single_hand_controller/include/mavlink/common/mavlink_msg_timesync.hpp`

5.446 `mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER` Struct Reference

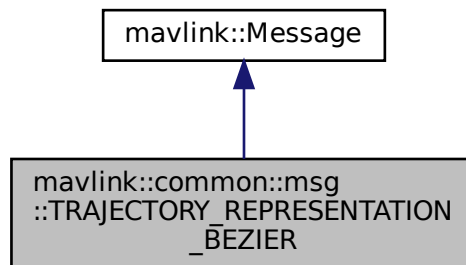
[TRAJECTORY_REPRESENTATION_BEZIER](#) message.

```
#include <mavlink_msg_trajectory_representation_bezier.hpp>
```

Inheritance diagram for `mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER`:



Collaboration diagram for mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **valid_points**
- `std::array< float, 5 >` **pos_x**
- `std::array< float, 5 >` **pos_y**
- `std::array< float, 5 >` **pos_z**
- `std::array< float, 5 >` **delta**
- `std::array< float, 5 >` **pos_yaw**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 333
- static constexpr size_t **LENGTH** = 109
- static constexpr size_t **MIN_LENGTH** = 109
- static constexpr uint8_t **CRC_EXTRA** = 231
- static constexpr auto **NAME** = "TRAJECTORY_REPRESENTATION_BEZIER"

5.446.1 Detailed Description

[TRAJECTORY_REPRESENTATION_BEZIER](#) message.

Describe a trajectory using an array of up-to 5 bezier control points in the local frame (MAV_FRAME_LOCAL_NED).

5.446.2 Member Function Documentation

5.446.2.1 deserialize()

```
void mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.446.2.2 get_name()

```
std::string mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.446.2.3 serialize()

```
void mavlink::common::msg::TRAJECTORY_REPRESENTATION_BEZIER::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

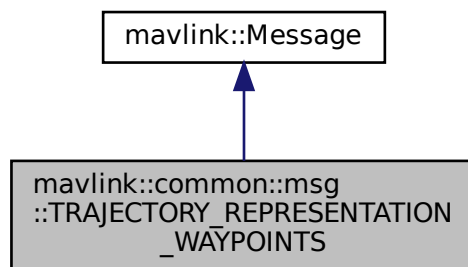
- `single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_bezier.hpp`

5.447 mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS Struct Reference

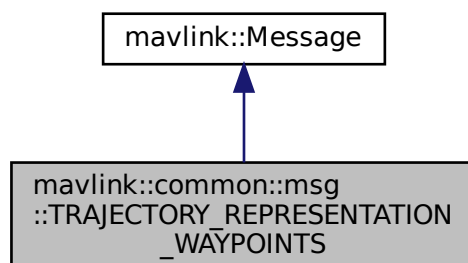
[TRAJECTORY_REPRESENTATION_WAYPOINTS](#) message.

```
#include <mavlink_msg_trajectory_representation_waypoints.hpp>
```

Inheritance diagram for mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS:



Collaboration diagram for mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `uint8_t` **valid_points**
- `std::array< float, 5 >` **pos_x**
- `std::array< float, 5 >` **pos_y**
- `std::array< float, 5 >` **pos_z**
- `std::array< float, 5 >` **vel_x**
- `std::array< float, 5 >` **vel_y**
- `std::array< float, 5 >` **vel_z**
- `std::array< float, 5 >` **acc_x**
- `std::array< float, 5 >` **acc_y**
- `std::array< float, 5 >` **acc_z**
- `std::array< float, 5 >` **pos_yaw**
- `std::array< float, 5 >` **vel_yaw**
- `std::array< uint16_t, 5 >` **command**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 332
- static constexpr `size_t` **LENGTH** = 239
- static constexpr `size_t` **MIN_LENGTH** = 239
- static constexpr `uint8_t` **CRC_EXTRA** = 236
- static constexpr auto **NAME** = "TRAJECTORY_REPRESENTATION_WAYPOINTS"

5.447.1 Detailed Description

[TRAJECTORY_REPRESENTATION_WAYPOINTS](#) message.

Describe a trajectory using an array of up-to 5 waypoints in the local frame (MAV_FRAME_LOCAL_NED).

5.447.2 Member Function Documentation

5.447.2.1 deserialize()

```
void mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.447.2.2 get_name()

```
std::string mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.447.2.3 serialize()

```
void mavlink::common::msg::TRAJECTORY_REPRESENTATION_WAYPOINTS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

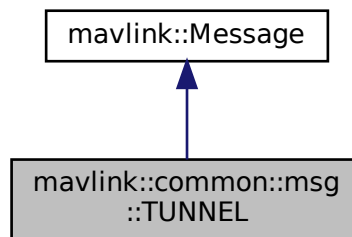
- single_hand_controller/include/mavlink/common/mavlink_msg_trajectory_representation_waypoints.hpp

5.448 mavlink::common::msg::TUNNEL Struct Reference

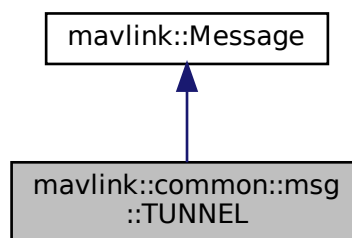
[TUNNEL](#) message.

```
#include <mavlink_msg_tunnel.hpp>
```

Inheritance diagram for mavlink::common::msg::TUNNEL:



Collaboration diagram for mavlink::common::msg::TUNNEL:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **target_system**
- `uint8_t` **target_component**
- `uint16_t` **payload_type**
- `uint8_t` **payload_length**
- `std::array< uint8_t, 128 >` **payload**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 385
- static constexpr size_t **LENGTH** = 133
- static constexpr size_t **MIN_LENGTH** = 133
- static constexpr uint8_t **CRC_EXTRA** = 147
- static constexpr auto **NAME** = "TUNNEL"

5.448.1 Detailed Description

[TUNNEL](#) message.

[Message](#) for transporting "arbitrary" variable-length data from one component to another (broadcast is not forbidden, but discouraged). The encoding of the data is usually extension specific, i.e. determined by the source, and is usually not documented as part of the MAVLink specification.

5.448.2 Member Function Documentation

5.448.2.1 deserialize()

```
void mavlink::common::msg::TUNNEL::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.448.2.2 get_name()

```
std::string mavlink::common::msg::TUNNEL::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.448.2.3 `serialize()`

```
void mavlink::common::msg::TUNNEL::serialize (  
    mavlink::MsgMap & map ) const    [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

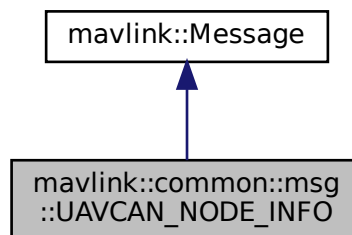
- `single_hand_controller/include/mavlink/common/mavlink_msg_tunnel.hpp`

5.449 mavlink::common::msg::UAVCAN_NODE_INFO Struct Reference

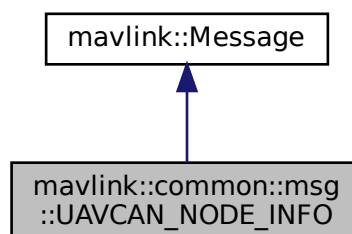
[UAVCAN_NODE_INFO](#) message.

```
#include <mavlink_msg_uavcan_node_info.hpp>
```

Inheritance diagram for `mavlink::common::msg::UAVCAN_NODE_INFO`:



Collaboration diagram for `mavlink::common::msg::UAVCAN_NODE_INFO`:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t time_usec`
- `uint32_t uptime_sec`
- `std::array< char, 80 > name`
- `uint8_t hw_version_major`
- `uint8_t hw_version_minor`
- `std::array< uint8_t, 16 > hw_unique_id`
- `uint8_t sw_version_major`
- `uint8_t sw_version_minor`
- `uint32_t sw_vcs_commit`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 311`
- `static constexpr size_t LENGTH = 116`
- `static constexpr size_t MIN_LENGTH = 116`
- `static constexpr uint8_t CRC_EXTRA = 95`
- `static constexpr auto NAME = "UAVCAN_NODE_INFO"`

5.449.1 Detailed Description

[UAVCAN_NODE_INFO](#) message.

General information describing a particular UAVCAN node. Please refer to the definition of the UAVCAN service "uavcan.protocol.GetNodeInfo" for the background information. This message should be emitted by the system whenever a new node appears online, or an existing node reboots. Additionally, it can be emitted upon request from the other end of the MAVLink channel (see MAV_CMD_UAVCAN_GET_NODE_INFO). It is also not prohibited to emit this message unconditionally at a low frequency. The UAVCAN specification is available at <http://uavcan.org>.

5.449.2 Member Function Documentation

5.449.2.1 deserialize()

```
void mavlink::common::msg::UAVCAN_NODE_INFO::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.449.2.2 get_name()

```
std::string mavlink::common::msg::UAVCAN_NODE_INFO::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.449.2.3 serialize()

```
void mavlink::common::msg::UAVCAN_NODE_INFO::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

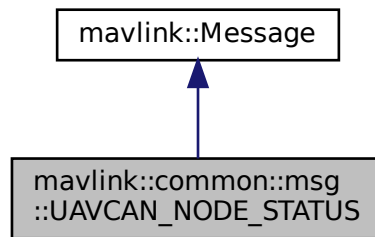
- single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_info.hpp

5.450 mavlink::common::msg::UAVCAN_NODE_STATUS Struct Reference

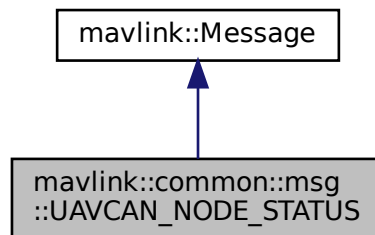
[UAVCAN_NODE_STATUS](#) message.

```
#include <mavlink_msg_uavcan_node_status.hpp>
```

Inheritance diagram for mavlink::common::msg::UAVCAN_NODE_STATUS:



Collaboration diagram for mavlink::common::msg::UAVCAN_NODE_STATUS:



Public Member Functions

- `std::string get_name (void) const` override
Get NAME constant.
- `Info get_message_info (void) const` override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map)` override
Deserialize message.

Public Attributes

- `uint64_t time_usec`
- `uint32_t uptime_sec`
- `uint8_t health`
- `uint8_t mode`
- `uint8_t sub_mode`
- `uint16_t vendor_specific_status_code`

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 310
- static constexpr size_t **LENGTH** = 17
- static constexpr size_t **MIN_LENGTH** = 17
- static constexpr uint8_t **CRC_EXTRA** = 28
- static constexpr auto **NAME** = "UAVCAN_NODE_STATUS"

5.450.1 Detailed Description

[UAVCAN_NODE_STATUS](#) message.

General status information of an UAVCAN node. Please refer to the definition of the UAVCAN message "uavcan.protocol.NodeStatus" for the background information. The UAVCAN specification is available at <http://uavcan.org>.

5.450.2 Member Function Documentation

5.450.2.1 deserialize()

```
void mavlink::common::msg::UAVCAN_NODE_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.450.2.2 get_name()

```
std::string mavlink::common::msg::UAVCAN_NODE_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.450.2.3 `serialize()`

```
void mavlink::common::msg::UAVCAN_NODE_STATUS::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/common/mavlink_msg_uavcan_node_status.hpp`

5.451 mavlink::impl::UIntBuffer< _T > Struct Template Reference

Public Types

- typedef [UIntBufferHelper](#)< sizeof(_T)>::Type **Type**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/msgmap.hpp`

5.452 mavlink::impl::UIntBufferHelper< _N > Struct Template Reference

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/msgmap.hpp`

5.453 mavlink::impl::UIntBufferHelper< 1 > Struct Reference

Public Types

- typedef uint8_t **Type**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/msgmap.hpp`

5.454 mavlink::impl::UIntBufferHelper< 2 > Struct Reference

Public Types

- typedef uint16_t **Type**

The documentation for this struct was generated from the following file:

- `single_hand_controller/include/mavlink/msgmap.hpp`

5.455 mavlink::impl::UIntBufferHelper< 4 > Struct Reference

Public Types

- typedef uint32_t **Type**

The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/msgmap.hpp

5.456 mavlink::impl::UIntBufferHelper< 8 > Struct Reference

Public Types

- typedef uint64_t **Type**

The documentation for this struct was generated from the following file:

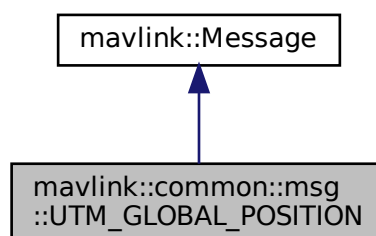
- single_hand_controller/include/mavlink/msgmap.hpp

5.457 mavlink::common::msg::UTM_GLOBAL_POSITION Struct Reference

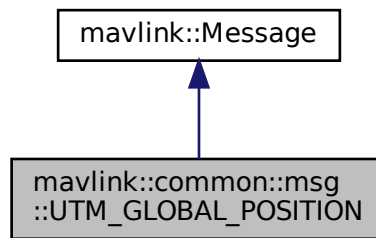
[UTM_GLOBAL_POSITION](#) message.

```
#include <mavlink_msg_utm_global_position.hpp>
```

Inheritance diagram for mavlink::common::msg::UTM_GLOBAL_POSITION:



Collaboration diagram for mavlink::common::msg::UTM_GLOBAL_POSITION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time**
- `std::array< uint8_t, 18 >` **uas_id**
- `int32_t` **lat**
- `int32_t` **lon**
- `int32_t` **alt**
- `int32_t` **relative_alt**
- `int16_t` **vx**
- `int16_t` **vy**
- `int16_t` **vz**
- `uint16_t` **h_acc**
- `uint16_t` **v_acc**
- `uint16_t` **vel_acc**
- `int32_t` **next_lat**
- `int32_t` **next_lon**
- `int32_t` **next_alt**
- `uint16_t` **update_rate**
- `uint8_t` **flight_state**
- `uint8_t` **flags**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 340
- static constexpr size_t **LENGTH** = 70
- static constexpr size_t **MIN_LENGTH** = 70
- static constexpr uint8_t **CRC_EXTRA** = 99
- static constexpr auto **NAME** = "UTM_GLOBAL_POSITION"

5.457.1 Detailed Description

[UTM_GLOBAL_POSITION](#) message.

The global position resulting from GPS and sensor fusion.

5.457.2 Member Function Documentation

5.457.2.1 deserialize()

```
void mavlink::common::msg::UTM_GLOBAL_POSITION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.457.2.2 get_name()

```
std::string mavlink::common::msg::UTM_GLOBAL_POSITION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.457.2.3 serialize()

```
void mavlink::common::msg::UTM_GLOBAL_POSITION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

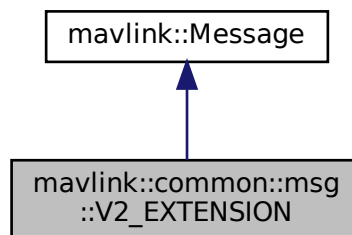
- single_hand_controller/include/mavlink/common/mavlink_msg_utm_global_position.hpp

5.458 mavlink::common::msg::V2_EXTENSION Struct Reference

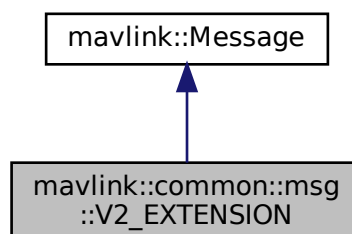
[V2_EXTENSION](#) message.

```
#include <mavlink_msg_v2_extension.hpp>
```

Inheritance diagram for mavlink::common::msg::V2_EXTENSION:



Collaboration diagram for mavlink::common::msg::V2_EXTENSION:



Public Member Functions

- `std::string get_name` (void) const override
Get NAME constant.
- `Info get_message_info` (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml` (void) const override
Make YAML-string from message content.
- `void serialize` (mavlink::MsgMap &map) const override
Serialize message.
- `void deserialize` (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t target_network`
- `uint8_t target_system`
- `uint8_t target_component`
- `uint16_t message_type`
- `std::array< uint8_t, 249 > payload`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 248`
- `static constexpr size_t LENGTH = 254`
- `static constexpr size_t MIN_LENGTH = 254`
- `static constexpr uint8_t CRC_EXTRA = 8`
- `static constexpr auto NAME = "V2_EXTENSION"`

5.458.1 Detailed Description

[V2_EXTENSION](#) message.

[Message](#) implementing parts of the V2 payload specs in V1 frames for transitional support.

5.458.2 Member Function Documentation

5.458.2.1 deserialize()

```
void mavlink::common::msg::V2_EXTENSION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.458.2.2 get_name()

```
std::string mavlink::common::msg::V2_EXTENSION::get_name (  
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.458.2.3 serialize()

```
void mavlink::common::msg::V2_EXTENSION::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

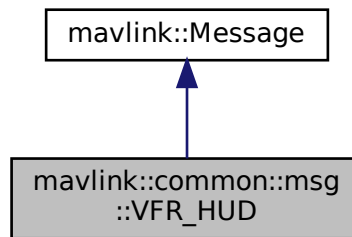
- single_hand_controller/include/mavlink/common/mavlink_msg_v2_extension.hpp

5.459 mavlink::common::msg::VFR_HUD Struct Reference

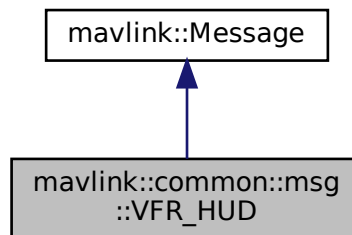
[VFR_HUD](#) message.

```
#include <mavlink_msg_vfr_hud.hpp>
```

Inheritance diagram for mavlink::common::msg::VFR_HUD:



Collaboration diagram for mavlink::common::msg::VFR_HUD:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- float **airspeed**
- float **groundspeed**
- int16_t **heading**
- uint16_t **throttle**
- float **alt**
- float **climb**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 74
- static constexpr size_t **LENGTH** = 20
- static constexpr size_t **MIN_LENGTH** = 20
- static constexpr uint8_t **CRC_EXTRA** = 20
- static constexpr auto **NAME** = "VFR_HUD"

5.459.1 Detailed Description

[VFR_HUD](#) message.

Metrics typically displayed on a HUD for fixed wing aircraft.

5.459.2 Member Function Documentation

5.459.2.1 deserialize()

```
void mavlink::common::msg::VFR_HUD::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.459.2.2 get_name()

```
std::string mavlink::common::msg::VFR_HUD::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.459.2.3 serialize()

```
void mavlink::common::msg::VFR_HUD::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

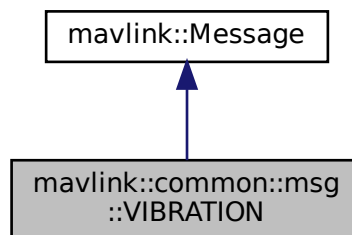
- `single_hand_controller/include/mavlink/common/mavlink_msg_vfr_hud.hpp`

5.460 `mavlink::common::msg::VIBRATION` Struct Reference

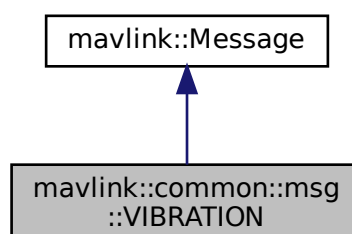
[VIBRATION](#) message.

```
#include <mavlink_msg_vibration.hpp>
```

Inheritance diagram for `mavlink::common::msg::VIBRATION`:



Collaboration diagram for `mavlink::common::msg::VIBRATION`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- float **vibration_x**
- float **vibration_y**
- float **vibration_z**
- uint32_t **clipping_0**
- uint32_t **clipping_1**
- uint32_t **clipping_2**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 241
- static constexpr size_t **LENGTH** = 32
- static constexpr size_t **MIN_LENGTH** = 32
- static constexpr uint8_t **CRC_EXTRA** = 90
- static constexpr auto **NAME** = "VIBRATION"

5.460.1 Detailed Description

[VIBRATION](#) message.

Vibration levels and accelerometer clipping

5.460.2 Member Function Documentation

5.460.2.1 deserialize()

```
void mavlink::common::msg::VIBRATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.460.2.2 get_name()

```
std::string mavlink::common::msg::VIBRATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.460.2.3 serialize()

```
void mavlink::common::msg::VIBRATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

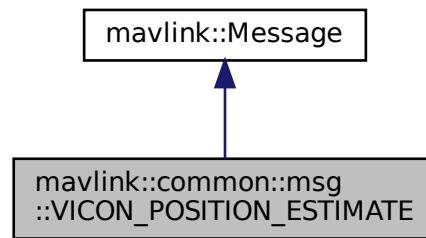
- single_hand_controller/include/mavlink/common/mavlink_msg_vibration.hpp

5.461 mavlink::common::msg::VICON_POSITION_ESTIMATE Struct Reference

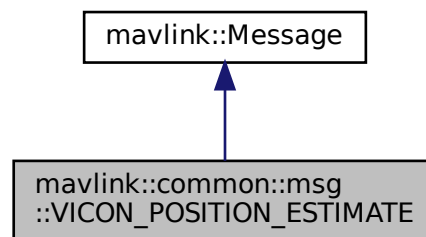
[VICON_POSITION_ESTIMATE](#) message.

```
#include <mavlink_msg_vicon_position_estimate.hpp>
```

Inheritance diagram for mavlink::common::msg::VICON_POSITION_ESTIMATE:



Collaboration diagram for mavlink::common::msg::VICON_POSITION_ESTIMATE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` [usec](#)
- `float` [x](#)

- float **y**
- float **z**
- float **roll**
- float **pitch**
- float **yaw**
- std::array< float, 21 > **covariance**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 104
- static constexpr size_t **LENGTH** = 116
- static constexpr size_t **MIN_LENGTH** = 32
- static constexpr uint8_t **CRC_EXTRA** = 56
- static constexpr auto **NAME** = "VICON_POSITION_ESTIMATE"

5.461.1 Detailed Description

[VICON_POSITION_ESTIMATE](#) message.

Global position estimate from a Vicon motion system source.

5.461.2 Member Function Documentation

5.461.2.1 deserialize()

```
void mavlink::common::msg::VICON_POSITION_ESTIMATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.461.2.2 get_name()

```
std::string mavlink::common::msg::VICON_POSITION_ESTIMATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.461.2.3 serialize()

```
void mavlink::common::msg::VICON_POSITION_ESTIMATE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

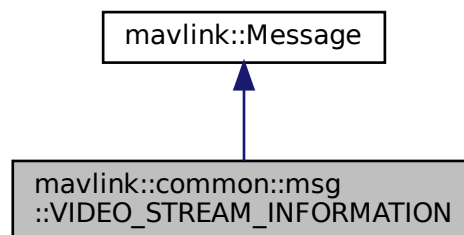
- single_hand_controller/include/mavlink/common/mavlink_msg_vicon_position_estimate.hpp

5.462 mavlink::common::msg::VIDEO_STREAM_INFORMATION Struct Reference

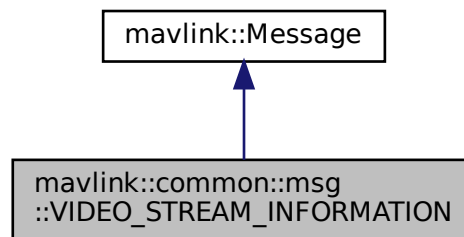
[VIDEO_STREAM_INFORMATION](#) message.

```
#include <mavlink_msg_video_stream_information.hpp>
```

Inheritance diagram for mavlink::common::msg::VIDEO_STREAM_INFORMATION:



Collaboration diagram for mavlink::common::msg::VIDEO_STREAM_INFORMATION:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint8_t` **stream_id**
- `uint8_t` **count**
- `uint8_t` **type**
- `uint16_t` **flags**
- `float` **framerate**
- `uint16_t` **resolution_h**
- `uint16_t` **resolution_v**
- `uint32_t` **bitrate**
- `uint16_t` **rotation**
- `uint16_t` **hfov**
- `std::array< char, 32 >` **name**
- `std::array< char, 160 >` **uri**
- `uint8_t` **encoding**
- `uint8_t` **camera_device_id**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 269
- static constexpr `size_t` **LENGTH** = 215
- static constexpr `size_t` **MIN_LENGTH** = 213
- static constexpr `uint8_t` **CRC_EXTRA** = 109
- static constexpr auto **NAME** = "VIDEO_STREAM_INFORMATION"

5.462.1 Detailed Description

[VIDEO_STREAM_INFORMATION](#) message.

Information about video stream. It may be requested using MAV_CMD_REQUEST_MESSAGE, where param2 indicates the video stream id: 0 for all streams, 1 for first, 2 for second, etc.

5.462.2 Member Function Documentation

5.462.2.1 deserialize()

```
void mavlink::common::msg::VIDEO_STREAM_INFORMATION::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.462.2.2 get_name()

```
std::string mavlink::common::msg::VIDEO_STREAM_INFORMATION::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.462.2.3 serialize()

```
void mavlink::common::msg::VIDEO_STREAM_INFORMATION::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

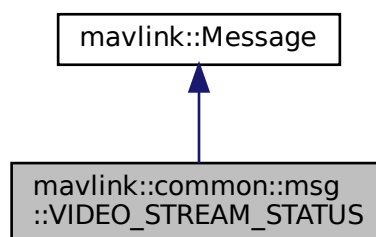
- `single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_information.hpp`

5.463 `mavlink::common::msg::VIDEO_STREAM_STATUS` Struct Reference

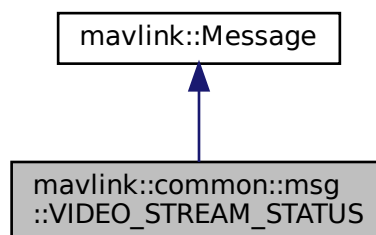
[VIDEO_STREAM_STATUS](#) message.

```
#include <mavlink_msg_video_stream_status.hpp>
```

Inheritance diagram for `mavlink::common::msg::VIDEO_STREAM_STATUS`:



Collaboration diagram for `mavlink::common::msg::VIDEO_STREAM_STATUS`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint8_t **stream_id**
- uint16_t **flags**
- float **framerate**
- uint16_t **resolution_h**
- uint16_t **resolution_v**
- uint32_t **bitrate**
- uint16_t **rotation**
- uint16_t **hfov**
- uint8_t **camera_device_id**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 270
- static constexpr size_t **LENGTH** = 20
- static constexpr size_t **MIN_LENGTH** = 19
- static constexpr uint8_t **CRC_EXTRA** = 59
- static constexpr auto **NAME** = "VIDEO_STREAM_STATUS"

5.463.1 Detailed Description

[VIDEO_STREAM_STATUS](#) message.

Information about the status of a video stream. It may be requested using MAV_CMD_REQUEST_MESSAGE.

5.463.2 Member Function Documentation

5.463.2.1 deserialize()

```
void mavlink::common::msg::VIDEO_STREAM_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.463.2.2 get_name()

```
std::string mavlink::common::msg::VIDEO_STREAM_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.463.2.3 serialize()

```
void mavlink::common::msg::VIDEO_STREAM_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

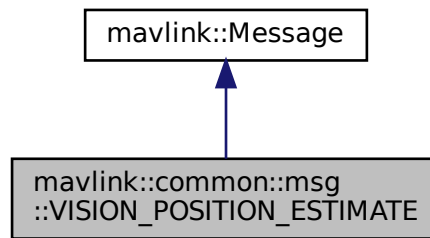
- single_hand_controller/include/mavlink/common/mavlink_msg_video_stream_status.hpp

5.464 mavlink::common::msg::VISION_POSITION_ESTIMATE Struct Reference

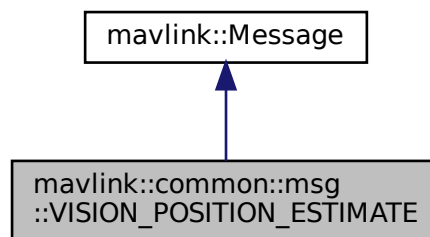
[VISION_POSITION_ESTIMATE](#) message.

```
#include <mavlink_msg_vision_position_estimate.hpp>
```

Inheritance diagram for mavlink::common::msg::VISION_POSITION_ESTIMATE:



Collaboration diagram for mavlink::common::msg::VISION_POSITION_ESTIMATE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` `usec`
- `float` `x`

- float **y**
- float **z**
- float **roll**
- float **pitch**
- float **yaw**
- std::array< float, 21 > **covariance**
- uint8_t **reset_counter**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 102
- static constexpr size_t **LENGTH** = 117
- static constexpr size_t **MIN_LENGTH** = 32
- static constexpr uint8_t **CRC_EXTRA** = 158
- static constexpr auto **NAME** = "VISION_POSITION_ESTIMATE"

5.464.1 Detailed Description

[VISION_POSITION_ESTIMATE](#) message.

Local position/attitude estimate from a vision source.

5.464.2 Member Function Documentation

5.464.2.1 deserialize()

```
void mavlink::common::msg::VISION_POSITION_ESTIMATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.464.2.2 get_name()

```
std::string mavlink::common::msg::VISION_POSITION_ESTIMATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.464.2.3 serialize()

```
void mavlink::common::msg::VISION_POSITION_ESTIMATE::serialize (  
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

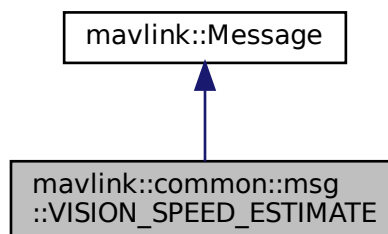
- single_hand_controller/include/mavlink/common/mavlink_msg_vision_position_estimate.hpp

5.465 mavlink::common::msg::VISION_SPEED_ESTIMATE Struct Reference

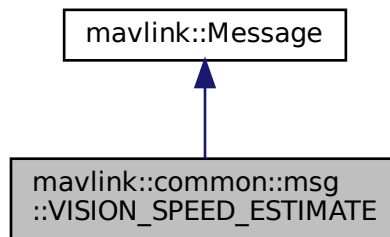
[VISION_SPEED_ESTIMATE](#) message.

```
#include <mavlink_msg_vision_speed_estimate.hpp>
```

Inheritance diagram for mavlink::common::msg::VISION_SPEED_ESTIMATE:



Collaboration diagram for mavlink::common::msg::VISION_SPEED_ESTIMATE:



Public Member Functions

- `std::string get_name (void) const` override
Get NAME constant.
- `Info get_message_info (void) const` override
Get info needed for mavlink_finalize_message_xxx()
- `std::string to_yaml (void) const` override
Make YAML-string from message content.
- `void serialize (mavlink::MsgMap &map) const` override
Serialize message.
- `void deserialize (mavlink::MsgMap &map) override`
Deserialize message.

Public Attributes

- `uint64_t usec`
- `float x`
- `float y`
- `float z`
- `std::array< float, 9 > covariance`
- `uint8_t reset_counter`

Static Public Attributes

- `static constexpr msgid_t MSG_ID = 103`
- `static constexpr size_t LENGTH = 57`
- `static constexpr size_t MIN_LENGTH = 20`
- `static constexpr uint8_t CRC_EXTRA = 208`
- `static constexpr auto NAME = "VISION_SPEED_ESTIMATE"`

5.465.1 Detailed Description

`VISION_SPEED_ESTIMATE` message.

Speed estimate from a vision source.

5.465.2 Member Function Documentation

5.465.2.1 deserialize()

```
void mavlink::common::msg::VISION_SPEED_ESTIMATE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.465.2.2 get_name()

```
std::string mavlink::common::msg::VISION_SPEED_ESTIMATE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.465.2.3 serialize()

```
void mavlink::common::msg::VISION_SPEED_ESTIMATE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

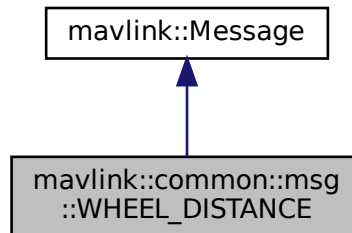
- `single_hand_controller/include/mavlink/common/mavlink_msg_vision_speed_estimate.hpp`

5.466 mavlink::common::msg::WHEEL_DISTANCE Struct Reference

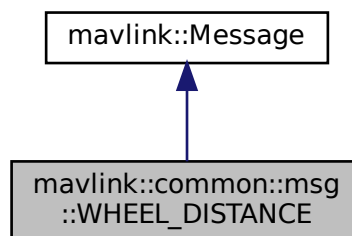
[WHEEL_DISTANCE](#) message.

```
#include <mavlink_msg_wheel_distance.hpp>
```

Inheritance diagram for mavlink::common::msg::WHEEL_DISTANCE:



Collaboration diagram for mavlink::common::msg::WHEEL_DISTANCE:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- uint64_t **time_usec**
- uint8_t **count**
- std::array< double, 16 > **distance**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 9000
- static constexpr size_t **LENGTH** = 137
- static constexpr size_t **MIN_LENGTH** = 137
- static constexpr uint8_t **CRC_EXTRA** = 113
- static constexpr auto **NAME** = "WHEEL_DISTANCE"

5.466.1 Detailed Description

[WHEEL_DISTANCE](#) message.

Cumulative distance traveled for each reported wheel.

5.466.2 Member Function Documentation

5.466.2.1 deserialize()

```
void mavlink::common::msg::WHEEL_DISTANCE::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.466.2.2 get_name()

```
std::string mavlink::common::msg::WHEEL_DISTANCE::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.466.2.3 `serialize()`

```
void mavlink::common::msg::WHEEL_DISTANCE::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

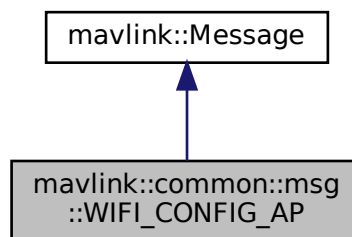
- `single_hand_controller/include/mavlink/common/mavlink_msg_wheel_distance.hpp`

5.467 `mavlink::common::msg::WIFI_CONFIG_AP` Struct Reference

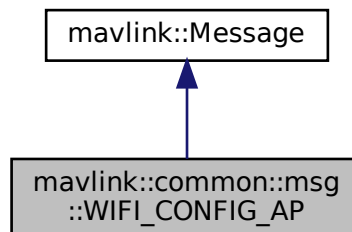
[WIFI_CONFIG_AP](#) message.

```
#include <mavlink_msg_wifi_config_ap.hpp>
```

Inheritance diagram for `mavlink::common::msg::WIFI_CONFIG_AP`:



Collaboration diagram for `mavlink::common::msg::WIFI_CONFIG_AP`:



Public Member Functions

- std::string [get_name](#) (void) const override
Get NAME constant.
- [Info get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- std::string [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- std::array< char, 32 > **ssid**
- std::array< char, 64 > **password**
- int8_t **mode**
- int8_t **response**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 299
- static constexpr size_t **LENGTH** = 98
- static constexpr size_t **MIN_LENGTH** = 96
- static constexpr uint8_t **CRC_EXTRA** = 19
- static constexpr auto **NAME** = "WIFI_CONFIG_AP"

5.467.1 Detailed Description

[WIFI_CONFIG_AP](#) message.

Configure WiFi AP SSID, password, and mode. This message is re-emitted as an acknowledgement by the AP. The message may also be explicitly requested using MAV_CMD_REQUEST_MESSAGE

5.467.2 Member Function Documentation

5.467.2.1 deserialize()

```
void mavlink::common::msg::WIFI_CONFIG_AP::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.467.2.2 get_name()

```
std::string mavlink::common::msg::WIFI_CONFIG_AP::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.467.2.3 serialize()

```
void mavlink::common::msg::WIFI_CONFIG_AP::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

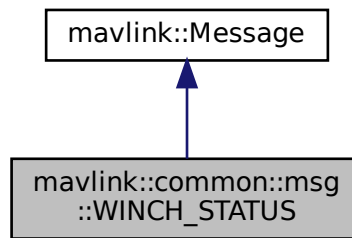
- single_hand_controller/include/mavlink/common/mavlink_msg_wifi_config_ap.hpp

5.468 mavlink::common::msg::WINCH_STATUS Struct Reference

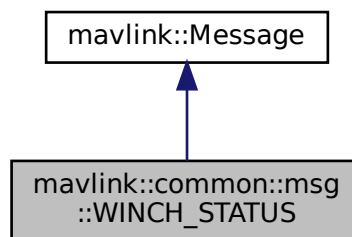
[WINCH_STATUS](#) message.

```
#include <mavlink_msg_winch_status.hpp>
```

Inheritance diagram for mavlink::common::msg::WINCH_STATUS:



Collaboration diagram for mavlink::common::msg::WINCH_STATUS:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for mavlink_finalize_message_xxx()
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (mavlink::MsgMap &map) const override
Serialize message.
- void [deserialize](#) (mavlink::MsgMap &map) override
Deserialize message.

Public Attributes

- `uint64_t` [time_usec](#)
- `float` [line_length](#)

- float **speed**
- float **tension**
- float **voltage**
- float **current**
- int16_t **temperature**
- uint32_t **status**

Static Public Attributes

- static constexpr msgid_t **MSG_ID** = 9005
- static constexpr size_t **LENGTH** = 34
- static constexpr size_t **MIN_LENGTH** = 34
- static constexpr uint8_t **CRC_EXTRA** = 117
- static constexpr auto **NAME** = "WINCH_STATUS"

5.468.1 Detailed Description

[WINCH_STATUS](#) message.

Winch status.

5.468.2 Member Function Documentation

5.468.2.1 deserialize()

```
void mavlink::common::msg::WINCH_STATUS::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	map	
----	-----	--

Implements [mavlink::Message](#).

5.468.2.2 get_name()

```
std::string mavlink::common::msg::WINCH_STATUS::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.468.2.3 serialize()

```
void mavlink::common::msg::WINCH_STATUS::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	map	
-----	-----	--

Implements [mavlink::Message](#).

The documentation for this struct was generated from the following file:

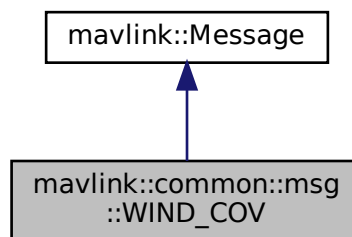
- single_hand_controller/include/mavlink/common/mavlink_msg_winch_status.hpp

5.469 mavlink::common::msg::WIND_COV Struct Reference

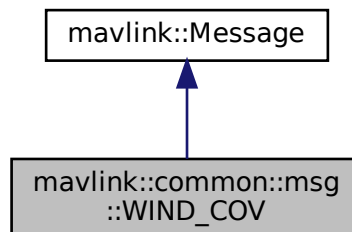
[WIND_COV](#) message.

```
#include <mavlink_msg_wind_cov.hpp>
```

Inheritance diagram for mavlink::common::msg::WIND_COV:



Collaboration diagram for mavlink::common::msg::WIND_COV:



Public Member Functions

- `std::string` [get_name](#) (void) const override
Get NAME constant.
- `Info` [get_message_info](#) (void) const override
Get info needed for `mavlink_finalize_message_xxx()`
- `std::string` [to_yaml](#) (void) const override
Make YAML-string from message content.
- void [serialize](#) (`mavlink::MsgMap` &map) const override
Serialize message.
- void [deserialize](#) (`mavlink::MsgMap` &map) override
Deserialize message.

Public Attributes

- `uint64_t` **time_usec**
- `float` **wind_x**
- `float` **wind_y**
- `float` **wind_z**
- `float` **var_horiz**
- `float` **var_vert**
- `float` **wind_alt**
- `float` **horiz_accuracy**
- `float` **vert_accuracy**

Static Public Attributes

- static constexpr `msgid_t` **MSG_ID** = 231
- static constexpr `size_t` **LENGTH** = 40
- static constexpr `size_t` **MIN_LENGTH** = 40
- static constexpr `uint8_t` **CRC_EXTRA** = 105
- static constexpr auto **NAME** = "WIND_COV"

5.469.1 Detailed Description

[WIND_COV](#) message.

Wind estimate from vehicle. Note that despite the name, this message does not actually contain any covariances but instead variability and accuracy fields in terms of standard deviation (1-STD).

5.469.2 Member Function Documentation

5.469.2.1 `deserialize()`

```
void mavlink::common::msg::WIND_COV::deserialize (
    mavlink::MsgMap & msp ) [inline], [override], [virtual]
```

Deserialize message.

Parameters

in	<i>map</i>	
----	------------	--

Implements [mavlink::Message](#).

5.469.2.2 get_name()

```
std::string mavlink::common::msg::WIND_COV::get_name (
    void ) const [inline], [override], [virtual]
```

Get NAME constant.

Helper for overloaded class access.

Implements [mavlink::Message](#).

5.469.2.3 serialize()

```
void mavlink::common::msg::WIND_COV::serialize (
    mavlink::MsgMap & map ) const [inline], [override], [virtual]
```

Serialize message.

Parameters

out	<i>map</i>	
-----	------------	--

Implements [mavlink::Message](#).

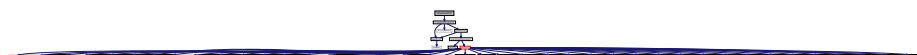
The documentation for this struct was generated from the following file:

- single_hand_controller/include/mavlink/common/mavlink_msg_wind_cov.hpp

File Documentation

```
#include "include/displayHandler.hpp"
```

Include dependency graph for displayHandler.cpp:



- void `displayLogo` ()
display newspace logo in white background and blue foreground colors
- void `displayInvertedLogo` ()
display newspace logo in grey background and light blue foreground (to look like an translucent grey film filter)
- void `setFont1` ()
set font style to 1
- void `initDisplayComm` ()
initialize display communications (serial baud rate)
- void `clear_display` ()
clear LCD display
- void `displayUGV_status` (ugv_status s)
update UGV status display
- void `displaySpeed` (int speed)
update speed display
- void `displayRSSI` ()
update RSSI stat display
- void `displayBattery` ()
update battery stat display
- void `updateDisplay` ()
this function is meant to be called once every 1 second, and only updates RSSI, and battery stats.
- void `displayInfo` (String message)
display info message and wait for any user input to continue execution
- bool **`displayError`** (String message)
- void `displayBasic` ()
set the screen with symbols/text that is required to understand the output of the display updates

Variables

- struct buttons **user_input**
- ugv_status **current_state**
- ugv_status **prev_state**
- TFT_22_ILI9225 **tft** = TFT_22_ILI9225(TFT_RST, TFT_RS, TFT_CS, TFT_SDI, TFT_CLK, TFT_LED, TFT↵_BRIGHTNESS)

6.1.1 Detailed Description

Version

0.1

Author

Nikhil Tom Jose

Date

22/04/2026

Part of displayHandler library. Defines variables and functions used for displayHandler.h

6.1.2 Function Documentation

6.1.2.1 updateDisplay()

```
void updateDisplay ( )
```

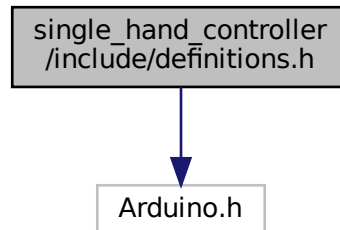
this function is meant to be called once every 1 second, and only updates RSSI, and battery stats.

call [displaySpeed\(\)](#) and [displayUGV_status\(\)](#) separately when the values are updated (must be event based, and not periodic)

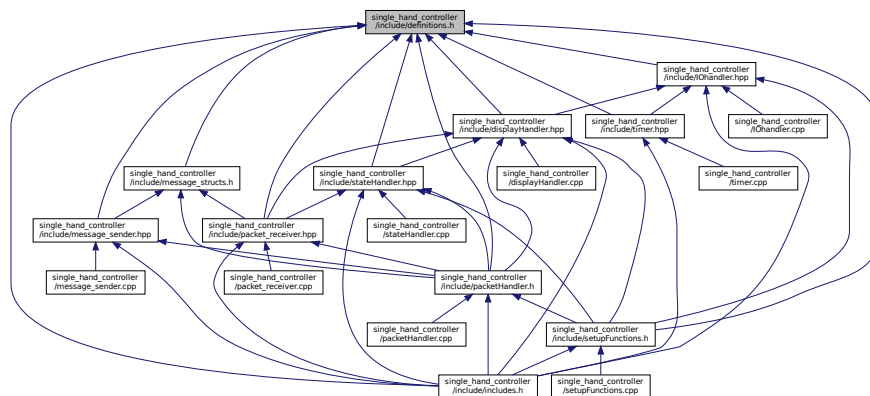
6.2 single_hand_controller/include/definitions.h File Reference

```
#include "Arduino.h"
```

Include dependency graph for definitions.h:



This graph shows which files directly or indirectly include this file:



Macros

- `#define SCOUT_ID 1`
- `#define ATLAS_COMP_ID 191`
- `#define HC_ID 2`
- `#define HC_COMP_ID 1`
- `#define SCOUT_HEARTBEAT_IDENTIFIER 1191`
- `#define SECONDS_10 10000`
- `#define SECONDS_3 3000`
- `#define SECONDS_2 2000`
- `#define MS_20 20`
- `#define MS_10 10`
- `#define HEARTBEAT_TIMEOUT SECONDS_3`
- `#define LONG_PRESS_DURATION SECONDS_2 * 1000`
- `#define TIMESYNC_MSG_WAIT SECONDS_10`

- #define **MANUAL_CONTROL_MSG_WAIT** MS_10
- #define **OFP_LOOP_TIME** MS_20
- #define **XY_NORMALIZED_MAX** 100
- #define **ANALOG_OUTPUT_MAX** 1023
- #define **XY_UPPER_LIMIT** 462
- *512 is the midpoint of 0 to 1023 hence, 50 units towards either side is allowed*
- #define **XY_LOWER_LIMIT** 562
- #define **FILTER_SAMPLES** 10
- #define **STATE_DISARM** 0
- #define **STATE_ARM** 1
- #define **BAUD_RATE** 115200
- #define **LCD_DISPLAY_BAUD_RATE** 9600
- #define **PROTOTYPE**
- #define **_DEBUG_**
- #define **IF_PROTOTYPE**(CODE) CODE
- #define **DEBUG**(CODE) CODE
- #define **IF_DEBUG**(CODE) CODE
- #define **TFT_RST** A4
- #define **TFT_RS** A3
- #define **TFT_CS** A5
- #define **TFT_SDI** A2
- #define **TFT_CLK** A7
- #define **TFT_LED** A9
- #define **TFT_BRIGHTNESS** 200
- #define **XPIN** A0
- #define **YPIN** A1
- #define **BUTTON_NEUTRAL** 2
- #define **BUTTON_INC_SPEED** 3
- #define **BUTTON_DEC_SPEED** 5
- #define **BUTTON_ARM_DISARM** 4
- #define **TOGGLE_FORWARD** 6
- #define **TOGGLE_REVERSE** 8

Enumerations

- enum **ugv_status** { **disconnected** , **active** , **standby** , **unknown** }

Variables

- unsigned char const **signing_key** [32] = {0x2d,0x3d,0x67,0xb6,0xa9,0x92,0x1b,0x1a,0xb0,0x93,0x11,0x73,0xe2,0xda,0x1c,0xe2,0x1c,0xe2,0x1c,0xe2,0x1c,0xe2,0x1c,0xe2,0x1c,0xe2,0x1c,0xe2,0x1c,0xe2,0x1c,0xe2}
- *signing key used to sign packets [meant to be changed on deployment]*

6.2.1 Detailed Description

Version

0.1

Author

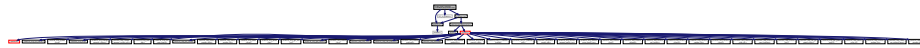
Nikhil Tom Jose

Date

22/04/2026

All definitions, used within every header file is included here.


```
#include "definitions.h"
#include "TFT_22_ILI9225.h"
#include "IOhandler.hpp"
Include dependency graph for displayHandler.hpp:
```



```

graph TD
    A["single_hand_controller  
/include/displayHandler.hpp"]
    B["single_hand_controller  
/displayHandler.cpp"]
    C["single_hand_controller  
/include/stateHandler.hpp"]
    D["single_hand_controller  
/stateHandler.cpp"]
    E["single_hand_controller  
/include/packet_receiver.hpp"]
    F["single_hand_controller  
/include/packetHandler.h"]
    G["single_hand_controller  
/packetHandler.cpp"]
    H["single_hand_controller  
/packet_receiver.cpp"]
    I["single_hand_controller  
/include/includes.h"]
    J["single_hand_controller  
/include/setupFunctions.h"]

    A --> B
    A --> C
    A --> D
    A --> E
    A --> F
    A --> G
    A --> H
    A --> I
    A --> J
    B --> A
    B --> C
    B --> D
    B --> E
    B --> F
    B --> G
    B --> H
    B --> I
    B --> J
    C --> A
    C --> B
    C --> D
    C --> E
    C --> F
    C --> G
    C --> H
    C --> I
    C --> J
    D --> A
    D --> B
    D --> C
    D --> E
    D --> F
    D --> G
    D --> H
    D --> I
    D --> J
    E --> A
    E --> B
    E --> C
    E --> D
    E --> F
    E --> G
    E --> H
    E --> I
    E --> J
    F --> A
    F --> B
    F --> C
    F --> D
    F --> E
    F --> G
    F --> H
    F --> I
    F --> J
    G --> A
    G --> B
    G --> C
    G --> D
    G --> E
    G --> F
    G --> H
    G --> I
    G --> J
    H --> A
    H --> B
    H --> C
    H --> D
    H --> E
    H --> F
    H --> G
    H --> I
    H --> J
    I --> A
    I --> B
    I --> C
    I --> D
    I --> E
    I --> F
    I --> G
    I --> H
    I --> J
    J --> A
    J --> B
    J --> C
    J --> D
    J --> E
    J --> F
    J --> G
    J --> H
    J --> I
  
```

- void **setupDisplay** ()
clear LCD display
- **IF_PROTOTYPE** (void **displayLogo**();void **displayInvertedLogo**();) bool **displayError**(String message)
display error message and confirm with user whether to proceed or wait
- void **displayInfo** (String message)
display info message and wait for any user input to continue execution
- void **initDisplayComm** ()
initialize display communications (serial baud rate)
- void **displayUGV_status** (ugv_status)
update UGV status display
- void **displaySpeed** (int speed)
update speed display
- void **displayRSSI** ()
update RSSI stat display
- void **displayBattery** ()
update battery stat display

- void `displayBasic()`
set the screen with symbols/text that is required to understand the output of the display updates
- void `displayLogo()`
display newspace logo in white background and blue foreground colors
- void `displayInvertedLogo()`
display newspace logo in grey background and light blue foreground (to look like an translucent grey film filter)
- void `updateDisplay()`
this function is meant to be called once every 1 second, and only updates RSSI, and battery stats.
- void `setFont1()`
set font style to 1

6.3.1 Detailed Description

Version

0.1

Author

Nikhil Tom Jose

Date

22/04/2026 Takes care of all display-related functionality

6.3.2 Function Documentation

6.3.2.1 `setupDisplay()`

```
void setupDisplay ( )
```

1. setup display baud rate
2. clear display
3. display logo/initial messages
4. display text/symbols for status indication

6.3.2.2 `updateDisplay()`

```
void updateDisplay ( )
```

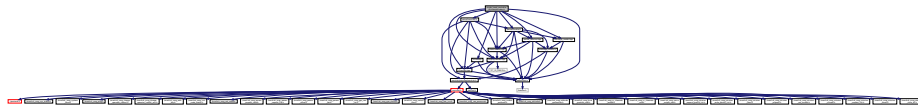
this function is meant to be called once every 1 second, and only updates RSSI, and battery stats.

call `displaySpeed()` and `displayUGV_status()` separately when the values are updated (must be event based, and not periodic)

6.4 single_hand_controller/include/includes.h File Reference

```
#include "definitions.h"
#include "setupFunctions.h"
#include "packetHandler.h"
#include "packet_receiver.hpp"
#include "message_sender.hpp"
#include "timer.hpp"
#include "IOhandler.hpp"
#include "displayHandler.hpp"
#include "stateHandler.hpp"
```

Include dependency graph for includes.h:



6.4.1 Detailed Description

Version

0.1

Author

Nikhil Tom Jose

Date

22/04/2026 all the header files that are included within the main file

6.5 single_hand_controller/include/IOhandler.hpp File Reference

```
#include "definitions.h"
#include "mavlink/common/mavlink.h"
```

Include dependency graph for IOhandler.hpp:



6.5.2.1 getXY()

```
void getXY (
    struct thumbstickControl * control )
```

get thumbstick's normalized position (-100 to 100)

Parameters

<i>control</i>	-> the current position of the thumbstick is written into this struct
----------------	---

6.5.2.2 getXY_raw()

```
void getXY_raw (
    struct thumbstickControl * control )
```

get thumbstick's raw position (0 to 1023)

Parameters

<i>control</i>	-> the current position of the thumbstick is written into this struct
----------------	---

6.6 single_hand_controller/include/mavlink/common/common.h File Reference

MAVLink comm protocol generated from common.xml.

```
#include "../protocol.h"
#include "../mavlink_msg_sys_status.h"
#include "../mavlink_msg_system_time.h"
#include "../mavlink_msg_ping.h"
#include "../mavlink_msg_change_operator_control.h"
#include "../mavlink_msg_change_operator_control_ack.h"
#include "../mavlink_msg_auth_key.h"
#include "../mavlink_msg_link_node_status.h"
#include "../mavlink_msg_set_mode.h"
#include "../mavlink_msg_param_request_read.h"
#include "../mavlink_msg_param_request_list.h"
#include "../mavlink_msg_param_value.h"
#include "../mavlink_msg_param_set.h"
#include "../mavlink_msg_gps_raw_int.h"
#include "../mavlink_msg_gps_status.h"
#include "../mavlink_msg_scaled_imu.h"
#include "../mavlink_msg_raw_imu.h"
#include "../mavlink_msg_raw_pressure.h"
#include "../mavlink_msg_scaled_pressure.h"
#include "../mavlink_msg_attitude.h"
#include "../mavlink_msg_attitude_quaternion.h"
#include "../mavlink_msg_local_position_ned.h"
#include "../mavlink_msg_rc_channels_scaled.h"
#include "../mavlink_msg_rc_channels_raw.h"
#include "../mavlink_msg_servo_output_raw.h"
#include "../mavlink_msg_mission_request_partial_list.h"
#include "../mavlink_msg_mission_write_partial_list.h"
#include "../mavlink_msg_mission_item.h"
```

```
#include "../mavlink_msg_mission_request.h"
#include "../mavlink_msg_mission_set_current.h"
#include "../mavlink_msg_mission_current.h"
#include "../mavlink_msg_mission_request_list.h"
#include "../mavlink_msg_mission_count.h"
#include "../mavlink_msg_mission_clear_all.h"
#include "../mavlink_msg_mission_item_reached.h"
#include "../mavlink_msg_mission_ack.h"
#include "../mavlink_msg_set_gps_global_origin.h"
#include "../mavlink_msg_gps_global_origin.h"
#include "../mavlink_msg_param_map_rc.h"
#include "../mavlink_msg_mission_request_int.h"
#include "../mavlink_msg_safety_set_allowed_area.h"
#include "../mavlink_msg_safety_allowed_area.h"
#include "../mavlink_msg_attitude_quaternion_cov.h"
#include "../mavlink_msg_nav_controller_output.h"
#include "../mavlink_msg_global_position_int_cov.h"
#include "../mavlink_msg_local_position_ned_cov.h"
#include "../mavlink_msg_rc_channels.h"
#include "../mavlink_msg_request_data_stream.h"
#include "../mavlink_msg_data_stream.h"
#include "../mavlink_msg_manual_control.h"
#include "../mavlink_msg_rc_channels_override.h"
#include "../mavlink_msg_mission_item_int.h"
#include "../mavlink_msg_vfr_hud.h"
#include "../mavlink_msg_command_int.h"
#include "../mavlink_msg_command_long.h"
#include "../mavlink_msg_command_ack.h"
#include "../mavlink_msg_command_cancel.h"
#include "../mavlink_msg_manual_setpoint.h"
#include "../mavlink_msg_set_attitude_target.h"
#include "../mavlink_msg_attitude_target.h"
#include "../mavlink_msg_set_position_target_local_ned.h"
#include "../mavlink_msg_position_target_local_ned.h"
#include "../mavlink_msg_set_position_target_global_int.h"
#include "../mavlink_msg_position_target_global_int.h"
#include "../mavlink_msg_local_position_ned_system_global_offset.h"
#include "../mavlink_msg_hil_state.h"
#include "../mavlink_msg_hil_controls.h"
#include "../mavlink_msg_hil_rc_inputs_raw.h"
#include "../mavlink_msg_hil_actuator_controls.h"
#include "../mavlink_msg_optical_flow.h"
#include "../mavlink_msg_global_vision_position_estimate.h"
#include "../mavlink_msg_vision_position_estimate.h"
#include "../mavlink_msg_vision_speed_estimate.h"
#include "../mavlink_msg_vicon_position_estimate.h"
#include "../mavlink_msg_highres_imu.h"
#include "../mavlink_msg_optical_flow_rad.h"
#include "../mavlink_msg_hil_sensor.h"
#include "../mavlink_msg_sim_state.h"
#include "../mavlink_msg_radio_status.h"
#include "../mavlink_msg_file_transfer_protocol.h"
#include "../mavlink_msg_timesync.h"
#include "../mavlink_msg_camera_trigger.h"
#include "../mavlink_msg_hil_gps.h"
#include "../mavlink_msg_hil_optical_flow.h"
#include "../mavlink_msg_hil_state_quaternion.h"
#include "../mavlink_msg_scaled_imu2.h"
```

```
#include "../mavlink_msg_log_request_list.h"
#include "../mavlink_msg_log_entry.h"
#include "../mavlink_msg_log_request_data.h"
#include "../mavlink_msg_log_data.h"
#include "../mavlink_msg_log_erase.h"
#include "../mavlink_msg_log_request_end.h"
#include "../mavlink_msg_gps_inject_data.h"
#include "../mavlink_msg_gps2_raw.h"
#include "../mavlink_msg_power_status.h"
#include "../mavlink_msg_serial_control.h"
#include "../mavlink_msg_gps_rtk.h"
#include "../mavlink_msg_gps2_rtk.h"
#include "../mavlink_msg_scaled_imu3.h"
#include "../mavlink_msg_data_transmission_handshake.h"
#include "../mavlink_msg_encapsulated_data.h"
#include "../mavlink_msg_distance_sensor.h"
#include "../mavlink_msg_terrain_request.h"
#include "../mavlink_msg_terrain_data.h"
#include "../mavlink_msg_terrain_check.h"
#include "../mavlink_msg_terrain_report.h"
#include "../mavlink_msg_scaled_pressure2.h"
#include "../mavlink_msg_att_pos_mocap.h"
#include "../mavlink_msg_set_actuator_control_target.h"
#include "../mavlink_msg_actuator_control_target.h"
#include "../mavlink_msg_altitude.h"
#include "../mavlink_msg_resource_request.h"
#include "../mavlink_msg_scaled_pressure3.h"
#include "../mavlink_msg_follow_target.h"
#include "../mavlink_msg_control_system_state.h"
#include "../mavlink_msg_battery_status.h"
#include "../mavlink_msg_landing_target.h"
#include "../mavlink_msg_fence_status.h"
#include "../mavlink_msg_mag_cal_report.h"
#include "../mavlink_msg_efi_status.h"
#include "../mavlink_msg_estimator_status.h"
#include "../mavlink_msg_wind_cov.h"
#include "../mavlink_msg_gps_input.h"
#include "../mavlink_msg_gps_rtcmm_data.h"
#include "../mavlink_msg_high_latency.h"
#include "../mavlink_msg_high_latency2.h"
#include "../mavlink_msg_vibration.h"
#include "../mavlink_msg_home_position.h"
#include "../mavlink_msg_set_home_position.h"
#include "../mavlink_msg_message_interval.h"
#include "../mavlink_msg_extended_sys_state.h"
#include "../mavlink_msg_adsb_vehicle.h"
#include "../mavlink_msg_collision.h"
#include "../mavlink_msg_v2_extension.h"
#include "../mavlink_msg_memory_vect.h"
#include "../mavlink_msg_debug_vect.h"
#include "../mavlink_msg_named_value_float.h"
#include "../mavlink_msg_named_value_int.h"
#include "../mavlink_msg_statustext.h"
#include "../mavlink_msg_debug.h"
#include "../mavlink_msg_setup_signing.h"
#include "../mavlink_msg_button_change.h"
#include "../mavlink_msg_play_tune.h"
#include "../mavlink_msg_camera_information.h"
```



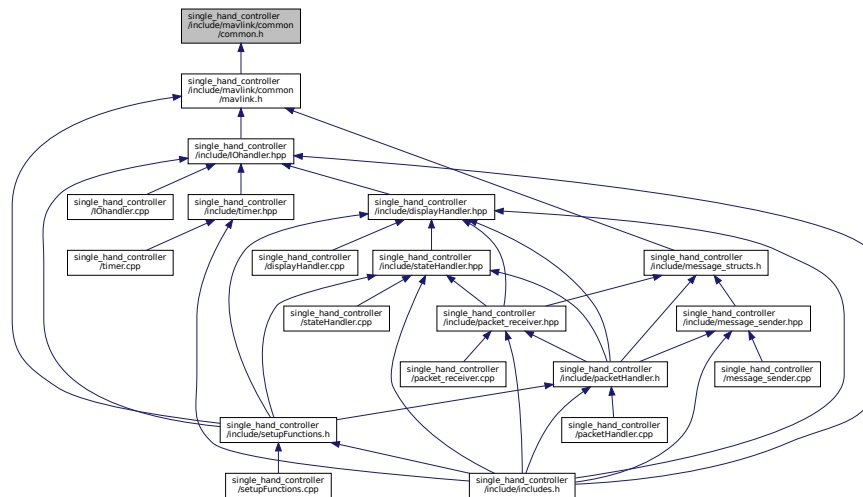
```
#include "../mavlink_msg_camera_settings.h"
#include "../mavlink_msg_storage_information.h"
#include "../mavlink_msg_camera_capture_status.h"
#include "../mavlink_msg_camera_image_captured.h"
#include "../mavlink_msg_flight_information.h"
#include "../mavlink_msg_mount_orientation.h"
#include "../mavlink_msg_logging_data.h"
#include "../mavlink_msg_logging_data_acked.h"
#include "../mavlink_msg_logging_ack.h"
#include "../mavlink_msg_video_stream_information.h"
#include "../mavlink_msg_video_stream_status.h"
#include "../mavlink_msg_camera_fov_status.h"
#include "../mavlink_msg_camera_tracking_image_status.h"
#include "../mavlink_msg_camera_tracking_geo_status.h"
#include "../mavlink_msg_camera_thermal_range.h"
#include "../mavlink_msg_gimbal_manager_information.h"
#include "../mavlink_msg_gimbal_manager_status.h"
#include "../mavlink_msg_gimbal_manager_set_attitude.h"
#include "../mavlink_msg_gimbal_device_information.h"
#include "../mavlink_msg_gimbal_device_set_attitude.h"
#include "../mavlink_msg_gimbal_device_attitude_status.h"
#include "../mavlink_msg_autopilot_state_for_gimbal_device.h"
#include "../mavlink_msg_gimbal_manager_set_pitchyaw.h"
#include "../mavlink_msg_gimbal_manager_set_manual_control.h"
#include "../mavlink_msg_esc_info.h"
#include "../mavlink_msg_esc_status.h"
#include "../mavlink_msg_airspeed.h"
#include "../mavlink_msg_wifi_config_ap.h"
#include "../mavlink_msg_protocol_version.h"
#include "../mavlink_msg_ais_vessel.h"
#include "../mavlink_msg_uavcan_node_status.h"
#include "../mavlink_msg_uavcan_node_info.h"
#include "../mavlink_msg_param_ext_request_read.h"
#include "../mavlink_msg_param_ext_request_list.h"
#include "../mavlink_msg_param_ext_value.h"
#include "../mavlink_msg_param_ext_set.h"
#include "../mavlink_msg_param_ext_ack.h"
#include "../mavlink_msg_obstacle_distance.h"
#include "../mavlink_msg_odometry.h"
#include "../mavlink_msg_trajectory_representation_waypoints.h"
#include "../mavlink_msg_trajectory_representation_bezier.h"
#include "../mavlink_msg_cellular_status.h"
#include "../mavlink_msg_isbd_link_status.h"
#include "../mavlink_msg_cellular_config.h"
#include "../mavlink_msg_raw_rpm.h"
#include "../mavlink_msg_utm_global_position.h"
#include "../mavlink_msg_param_error.h"
#include "../mavlink_msg_debug_float_array.h"
#include "../mavlink_msg_orbit_execution_status.h"
#include "../mavlink_msg_smart_battery_info.h"
#include "../mavlink_msg_figure_eight_execution_status.h"
#include "../mavlink_msg_fuel_status.h"
#include "../mavlink_msg_battery_info.h"
#include "../mavlink_msg_generator_status.h"
#include "../mavlink_msg_actuator_output_status.h"
#include "../mavlink_msg_relay_status.h"
#include "../mavlink_msg_time_estimate_to_target.h"
#include "../mavlink_msg_tunnel.h"
```

```

#include "../mavlink_msg_can_frame.h"
#include "../mavlink_msg_onboard_computer_status.h"
#include "../mavlink_msg_component_information.h"
#include "../mavlink_msg_component_information_basic.h"
#include "../mavlink_msg_component_metadata.h"
#include "../mavlink_msg_play_tune_v2.h"
#include "../mavlink_msg_supported_tunes.h"
#include "../mavlink_msg_event.h"
#include "../mavlink_msg_current_event_sequence.h"
#include "../mavlink_msg_request_event.h"
#include "../mavlink_msg_response_event_error.h"
#include "../mavlink_msg_available_modes.h"
#include "../mavlink_msg_current_mode.h"
#include "../mavlink_msg_available_modes_monitor.h"
#include "../mavlink_msg_illuminator_status.h"
#include "../mavlink_msg_canfd_frame.h"
#include "../mavlink_msg_can_filter_modify.h"
#include "../mavlink_msg_wheel_distance.h"
#include "../mavlink_msg_winch_status.h"
#include "../mavlink_msg_open_drone_id_basic_id.h"
#include "../mavlink_msg_open_drone_id_location.h"
#include "../mavlink_msg_open_drone_id_authentication.h"
#include "../mavlink_msg_open_drone_id_self_id.h"
#include "../mavlink_msg_open_drone_id_system.h"
#include "../mavlink_msg_open_drone_id_operator_id.h"
#include "../mavlink_msg_open_drone_id_message_pack.h"
#include "../mavlink_msg_open_drone_id_arm_status.h"
#include "../mavlink_msg_open_drone_id_system_update.h"
#include "../mavlink_msg_hygrometer_sensor.h"
#include "../standard/standard.h"

```

This graph shows which files directly or indirectly include this file:



Macros

- `#define MAVLINK_COMMON_H`
- `#define MAVLINK_COMMON_XML_HASH -6952494756206127088`
- `#define MAVLINK_MESSAGE_LENGTHS {}`

- #define **MAVLINK_MESSAGE_CRCS** {{0, 50, 9, 9, 0, 0, 0}, {1, 124, 31, 43, 0, 0, 0}, {2, 137, 12, 12, 0, 0, 0}, {4, 237, 14, 14, 3, 12, 13}, {5, 217, 28, 28, 1, 0, 0}, {6, 104, 3, 3, 0, 0, 0}, {7, 119, 32, 32, 0, 0, 0}, {8, 117, 36, 36, 0, 0, 0}, {11, 89, 6, 6, 1, 4, 0}, {20, 214, 20, 20, 3, 2, 3}, {21, 159, 2, 2, 3, 0, 1}, {22, 220, 25, 25, 0, 0, 0}, {23, 168, 23, 23, 3, 4, 5}, {24, 24, 30, 52, 0, 0, 0}, {25, 23, 101, 101, 0, 0, 0}, {26, 170, 22, 24, 0, 0, 0}, {27, 144, 26, 29, 0, 0, 0}, {28, 67, 16, 16, 0, 0, 0}, {29, 115, 14, 16, 0, 0, 0}, {30, 39, 28, 28, 0, 0, 0}, {31, 246, 32, 48, 0, 0, 0}, {32, 185, 28, 28, 0, 0, 0}, {33, 104, 28, 28, 0, 0, 0}, {34, 237, 22, 22, 0, 0, 0}, {35, 244, 22, 22, 0, 0, 0}, {36, 222, 21, 37, 0, 0, 0}, {37, 212, 6, 7, 3, 4, 5}, {38, 9, 6, 7, 3, 4, 5}, {39, 254, 37, 38, 3, 32, 33}, {40, 230, 4, 5, 3, 2, 3}, {41, 28, 4, 4, 3, 2, 3}, {42, 28, 2, 18, 0, 0, 0}, {43, 132, 2, 3, 3, 0, 1}, {44, 221, 4, 9, 3, 2, 3}, {45, 232, 2, 3, 3, 0, 1}, {46, 11, 2, 2, 0, 0, 0}, {47, 153, 3, 8, 3, 0, 1}, {48, 41, 13, 21, 1, 12, 0}, {49, 39, 12, 20, 0, 0, 0}, {50, 78, 37, 37, 3, 18, 19}, {51, 196, 4, 5, 3, 2, 3}, {54, 15, 27, 27, 3, 24, 25}, {55, 3, 25, 25, 0, 0, 0}, {61, 167, 72, 72, 0, 0, 0}, {62, 183, 26, 26, 0, 0, 0}, {63, 119, 181, 181, 0, 0, 0}, {64, 191, 225, 225, 0, 0, 0}, {65, 118, 42, 42, 0, 0, 0}, {66, 148, 6, 6, 3, 2, 3}, {67, 21, 4, 4, 0, 0, 0}, {69, 243, 11, 30, 1, 10, 0}, {70, 124, 18, 38, 3, 16, 17}, {73, 38, 37, 38, 3, 32, 33}, {74, 20, 20, 20, 0, 0, 0}, {75, 158, 35, 35, 3, 30, 31}, {76, 152, 33, 33, 3, 30, 31}, {77, 143, 3, 10, 3, 8, 9}, {80, 14, 4, 4, 3, 2, 3}, {81, 106, 22, 22, 0, 0, 0}, {82, 49, 39, 51, 3, 36, 37}, {83, 22, 37, 37, 0, 0, 0}, {84, 143, 53, 53, 3, 50, 51}, {85, 140, 51, 51, 0, 0, 0}, {86, 5, 53, 53, 3, 50, 51}, {87, 150, 51, 51, 0, 0, 0}, {89, 231, 28, 28, 0, 0, 0}, {90, 183, 56, 56, 0, 0, 0}, {91, 63, 42, 42, 0, 0, 0}, {92, 54, 33, 33, 0, 0, 0}, {93, 47, 81, 81, 0, 0, 0}, {100, 175, 26, 34, 0, 0, 0}, {101, 102, 32, 117, 0, 0, 0}, {102, 158, 32, 117, 0, 0, 0}, {103, 208, 20, 57, 0, 0, 0}, {104, 56, 32, 116, 0, 0, 0}, {105, 93, 62, 63, 0, 0, 0}, {106, 138, 44, 44, 0, 0, 0}, {107, 108, 64, 65, 0, 0, 0}, {108, 32, 84, 92, 0, 0, 0}, {109, 185, 9, 9, 0, 0, 0}, {110, 84, 254, 254, 3, 1, 2}, {111, 34, 16, 18, 3, 16, 17}, {112, 174, 12, 12, 0, 0, 0}, {113, 124, 36, 39, 0, 0, 0}, {114, 237, 44, 44, 0, 0, 0}, {115, 4, 64, 64, 0, 0, 0}, {116, 76, 22, 24, 0, 0, 0}, {117, 128, 6, 6, 3, 4, 5}, {118, 56, 14, 14, 0, 0, 0}, {119, 116, 12, 12, 3, 10, 11}, {120, 134, 97, 97, 0, 0, 0}, {121, 237, 2, 2, 3, 0, 1}, {122, 203, 2, 2, 3, 0, 1}, {123, 250, 113, 113, 3, 0, 1}, {124, 87, 35, 57, 0, 0, 0}, {125, 203, 6, 6, 0, 0, 0}, {126, 220, 79, 81, 3, 79, 80}, {127, 25, 35, 35, 0, 0, 0}, {128, 226, 35, 35, 0, 0, 0}, {129, 46, 22, 24, 0, 0, 0}, {130, 29, 13, 13, 0, 0, 0}, {131, 223, 255, 255, 0, 0, 0}, {132, 85, 14, 39, 0, 0, 0}, {133, 6, 18, 18, 0, 0, 0}, {134, 229, 43, 43, 0, 0, 0}, {135, 203, 8, 8, 0, 0, 0}, {136, 1, 22, 22, 0, 0, 0}, {137, 195, 14, 16, 0, 0, 0}, {138, 109, 36, 120, 0, 0, 0}, {139, 168, 43, 43, 3, 41, 42}, {140, 181, 41, 41, 0, 0, 0}, {141, 47, 32, 32, 0, 0, 0}, {142, 72, 243, 243, 0, 0, 0}, {143, 131, 14, 16, 0, 0, 0}, {144, 127, 93, 93, 0, 0, 0}, {146, 103, 100, 100, 0, 0, 0}, {147, 154, 36, 54, 0, 0, 0}, {148, 178, 60, 78, 0, 0, 0}, {149, 200, 30, 60, 0, 0, 0}, {162, 189, 8, 9, 0, 0, 0}, {192, 36, 44, 54, 0, 0, 0}, {225, 208, 65, 73, 0, 0, 0}, {230, 163, 42, 42, 0, 0, 0}, {231, 105, 40, 40, 0, 0, 0}, {232, 151, 63, 65, 0, 0, 0}, {233, 35, 182, 182, 0, 0, 0}, {234, 150, 40, 40, 0, 0, 0}, {235, 179, 42, 42, 0, 0, 0}, {241, 90, 32, 32, 0, 0, 0}, {242, 104, 52, 60, 0, 0, 0}, {243, 85, 53, 61, 1, 52, 0}, {244, 95, 6, 6, 0, 0, 0}, {245, 130, 2, 2, 0, 0, 0}, {246, 184, 38, 38, 0, 0, 0}, {247, 81, 19, 19, 0, 0, 0}, {248, 8, 254, 254, 3, 3, 4}, {249, 204, 36, 36, 0, 0, 0}, {250, 49, 30, 30, 0, 0, 0}, {251, 170, 18, 18, 0, 0, 0}, {252, 44, 18, 18, 0, 0, 0}, {253, 83, 51, 54, 0, 0, 0}, {254, 46, 9, 9, 0, 0, 0}, {256, 71, 42, 42, 3, 8, 9}, {257, 131, 9, 9, 0, 0, 0}, {258, 187, 32, 232, 3, 0, 1}, {259, 92, 235, 237, 0, 0, 0}, {260, 146, 5, 14, 0, 0, 0}, {261, 179, 27, 61, 0, 0, 0}, {262, 12, 18, 23, 0, 0, 0}, {263, 133, 255, 255, 0, 0, 0}, {264, 49, 28, 32, 0, 0, 0}, {265, 26, 16, 20, 0, 0, 0}, {266, 193, 255, 255, 3, 2, 3}, {267, 35, 255, 255, 3, 2, 3}, {268, 14, 4, 4, 3, 2, 3}, {269, 109, 213, 215, 0, 0, 0}, {270, 59, 19, 20, 0, 0, 0}, {271, 22, 52, 53, 0, 0, 0}, {275, 126, 31, 32, 0, 0, 0}, {276, 18, 49, 50, 0, 0, 0}, {277, 62, 30, 30, 0, 0, 0}, {280, 70, 33, 33, 0, 0, 0}, {281, 48, 13, 13, 0, 0, 0}, {282, 123, 35, 35, 3, 32, 33}, {283, 74, 144, 145, 0, 0, 0}, {284, 99, 32, 32, 3, 30, 31}, {285, 137, 40, 49, 3, 38, 39}, {286, 210, 53, 57, 3, 50, 51}, {287, 1, 23, 23, 3, 20, 21}, {288, 20, 23, 23, 3, 20, 21}, {290, 251, 46, 46, 0, 0, 0}, {291, 10, 57, 57, 0, 0, 0}, {295, 234, 12, 12, 0, 0, 0}, {299, 19, 96, 98, 0, 0, 0}, {300, 217, 22, 22, 0, 0, 0}, {301, 243, 58, 58, 0, 0, 0}, {310, 28, 17, 17, 0, 0, 0}, {311, 95, 116, 116, 0, 0, 0}, {320, 243, 20, 20, 3, 2, 3}, {321, 88, 2, 2, 3, 0, 1}, {322, 243, 149, 149, 0, 0, 0}, {323, 78, 147, 147, 3, 0, 1}, {324, 132, 146, 146, 0, 0, 0}, {330, 23, 158, 167, 0, 0, 0}, {331, 91, 230, 233, 0, 0, 0}, {332, 236, 239, 239, 0, 0, 0}, {333, 231, 109, 109, 0, 0, 0}, {334, 72, 10, 10, 0, 0, 0}, {335, 225, 24, 24, 0, 0, 0}, {336, 245, 84, 84, 0, 0, 0}, {339, 199, 5, 5, 0, 0, 0}, {340, 99, 70, 70, 0, 0, 0}, {345, 209, 21, 21, 3, 2, 3}, {350, 232, 20, 252, 0, 0, 0}, {360, 11, 25, 25, 0, 0, 0}, {361, 93, 33, 33, 0, 0, 0}, {370, 75, 87, 109, 0, 0, 0}, {371, 10, 26, 26, 0, 0, 0}, {372, 26, 140, 140, 0, 0, 0}, {373, 117, 42, 42, 0, 0, 0}, {375, 251, 140, 140, 0, 0, 0}, {376, 199, 8, 8, 0, 0, 0}, {380, 232, 20, 20, 0, 0, 0}, {385, 147, 133, 133, 3, 2, 3}, {386, 132, 16, 16, 3, 4, 5}, {387, 4, 72, 72, 3, 4, 5}, {388, 8, 37, 37, 3, 32, 33}, {390, 156, 238, 240, 0, 0, 0}, {395, 0, 212, 212, 0, 0, 0}, {396, 50, 160, 160, 0, 0, 0}, {397, 182, 108, 108, 0, 0, 0}, {400, 110, 254, 254, 3, 4, 5}, {401, 183, 6, 6, 3, 4, 5}, {410, 160, 53, 53, 0, 0, 0}, {411, 106, 3, 3, 0, 0, 0}, {412, 33, 6, 6, 3, 4, 5}, {413, 77, 7, 7, 3, 4, 5}, {435, 134, 46, 46, 0, 0, 0}, {436, 193, 9, 9, 0, 0, 0}, {437, 30, 1, 1, 0, 0, 0}, {440, 66, 35, 35, 0, 0, 0}, {9000, 113, 137, 137, 0, 0, 0}, {9005, 117, 34, 34, 0, 0, 0}, {12900, 114, 44, 44, 3, 0, 1}, {12901, 254, 59, 59, 3, 30, 31}, {12902, 140, 53, 53, 3, 4, 5}, {12903, 249, 46, 46, 3, 0, 1}, {12904, 77, 54, 54, 3, 28, 29}, {12905, 49, 43, 43, 3, 0, 1}, {12915, 94, 249, 249, 3, 0, 1}, {12918, 139, 51, 51, 0, 0, 0}, {12919, 7, 18, 18, 3, 16, 17}, {12920, 20, 5, 5, 0, 0, 0}}
- #define **MAVLINK_ENABLED_COMMON**

- #define [HAVE_ENUM_HL_FAILURE_FLAG](#)
Flags to report failure cases over the high latency telemetry.
- #define [HAVE_ENUM_MAV_GOTO](#)
Actions that may be specified in MAV_CMD_OVERRIDE_GOTO to override mission execution.
- #define [HAVE_ENUM_MAV_MODE](#)
Predefined OR-combined MAV_MODE_FLAG values.
- #define [HAVE_ENUM_MAV_SYS_STATUS_SENSOR](#)
These encode the sensors whose status is sent as part of the SYS_STATUS message.
- #define [HAVE_ENUM_MAV_SYS_STATUS_SENSOR_EXTENDED](#)
These encode the sensors whose status is sent as part of the SYS_STATUS message in the extended fields.
- #define [HAVE_ENUM_MAV_FRAME](#)
Coordinate frames used by MAVLink.
- #define [HAVE_ENUM_MAVLINK_DATA_STREAM_TYPE](#)

- #define [HAVE_ENUM_FENCE_BREACH](#)

- #define [HAVE_ENUM_FENCE_MITIGATE](#)
Actions being taken to mitigate/prevent fence breach.
- #define [HAVE_ENUM_FENCE_TYPE](#)
Fence types to enable or disable when using MAV_CMD_DO_FENCE_ENABLE.
- #define [HAVE_ENUM_MAV_MOUNT_MODE](#)
Enumeration of possible mount operation modes.
- #define [HAVE_ENUM_GIMBAL_DEVICE_CAP_FLAGS](#)
Gimbal device (low level) capability flags (bitmap).
- #define [HAVE_ENUM_GIMBAL_MANAGER_CAP_FLAGS](#)
Gimbal manager high level capability flags (bitmap).
- #define [HAVE_ENUM_GIMBAL_DEVICE_FLAGS](#)
Flags for gimbal device (lower level) operation.
- #define [HAVE_ENUM_GIMBAL_MANAGER_FLAGS](#)
Flags for high level gimbal manager operation The first 16 bits are identical to the GIMBAL_DEVICE_FLAGS.
- #define [HAVE_ENUM_GIMBAL_DEVICE_ERROR_FLAGS](#)
Gimbal device (low level) error flags (bitmap, 0 means no error)
- #define [HAVE_ENUM_GRIPPER_ACTIONS](#)
Gripper actions.
- #define [HAVE_ENUM_WINCH_ACTIONS](#)
Winch actions.
- #define [HAVE_ENUM_UAVCAN_NODE_HEALTH](#)
Generalized UAVCAN node health.
- #define [HAVE_ENUM_UAVCAN_NODE_MODE](#)
Generalized UAVCAN node mode.
- #define [HAVE_ENUM_ESC_CONNECTION_TYPE](#)
Indicates the ESC connection type.
- #define [HAVE_ENUM_ESC_FAILURE_FLAGS](#)
Flags to report ESC failures.
- #define [HAVE_ENUM_STORAGE_STATUS](#)
Flags to indicate the status of camera storage.
- #define [HAVE_ENUM_STORAGE_TYPE](#)
Flags to indicate the type of storage.
- #define [HAVE_ENUM_STORAGE_USAGE_FLAG](#)

Flags to indicate usage for a particular storage (see `STORAGE_INFORMATION.storage_usage` and `MAV_CMD_SET_STORAGE_USAGE`).

- #define `HAVE_ENUM_ORBIT_YAW_BEHAVIOUR`
Yaw behaviour during orbit flight.
- #define `HAVE_ENUM_WIFI_CONFIG_AP_RESPONSE`
Possible responses from a `WIFI_CONFIG_AP` message.
- #define `HAVE_ENUM_CELLULAR_CONFIG_RESPONSE`
Possible responses from a `CELLULAR_CONFIG` message.
- #define `HAVE_ENUM_WIFI_CONFIG_AP_MODE`
WiFi Mode.
- #define `HAVE_ENUM_COMP_METADATA_TYPE`
Supported component metadata types.
- #define `HAVE_ENUM_ACTUATOR_CONFIGURATION`
Actuator configuration, used to change a setting on an actuator.
- #define `HAVE_ENUM_ACTUATOR_OUTPUT_FUNCTION`
Actuator output function.
- #define `HAVE_ENUM_AUTOTUNE_AXIS`
Axes that will be autotuned by `MAV_CMD_DO_AUTOTUNE_ENABLE`.
- #define `HAVE_ENUM_PREFLIGHT_STORAGE_PARAMETER_ACTION`
Actions for reading/writing parameters between persistent and volatile storage when using `MAV_CMD_PREFLIGHT_STORAGE`.
- #define `HAVE_ENUM_PREFLIGHT_STORAGE_MISSION_ACTION`
Actions for reading and writing plan information (mission, rally points, geofence) between persistent and volatile storage when using `MAV_CMD_PREFLIGHT_STORAGE`.
- #define `HAVE_ENUM_REBOOT_SHUTDOWN_ACTION`
Reboot/shutdown action for selected component in `MAV_CMD_PREFLIGHT_REBOOT_SHUTDOWN`.
- #define `HAVE_ENUM_REBOOT_SHUTDOWN_CONDITIONS`
Specifies the conditions under which the `MAV_CMD_PREFLIGHT_REBOOT_SHUTDOWN` command should be accepted.
- #define `HAVE_ENUM_MAV_CMD`
Commands to be executed by the MAV.
- #define `HAVE_ENUM_MAV_DATA_STREAM`
A data stream is not a fixed set of messages, but rather a recommendation to the autopilot software.
- #define `HAVE_ENUM_MAV_ROI`
The ROI (region of interest) for the vehicle.
- #define `HAVE_ENUM_MAV_PARAM_TYPE`
Specifies the datatype of a MAVLink parameter.
- #define `HAVE_ENUM_MAV_PARAM_ERROR`
Parameter protocol error types (see `PARAM_ERROR`).
- #define `HAVE_ENUM_MAV_PARAM_EXT_TYPE`
Specifies the datatype of a MAVLink extended parameter.
- #define `HAVE_ENUM_MAV_RESULT`
Result from a MAVLink command (`MAV_CMD`)
- #define `HAVE_ENUM_MAV_MISSION_RESULT`
Result of mission operation (in a `MISSION_ACK` message).
- #define `HAVE_ENUM_MAV_SEVERITY`
Indicates the severity level, generally used for status messages to indicate their relative urgency.
- #define `HAVE_ENUM_MAV_POWER_STATUS`
Power supply status flags (bitmask)
- #define `HAVE_ENUM_SERIAL_CONTROL_DEV`
`SERIAL_CONTROL` device types.

- #define [HAVE_ENUM_SERIAL_CONTROL_FLAG](#)
SERIAL_CONTROL flags (bitmask)
- #define [HAVE_ENUM_MAV_DISTANCE_SENSOR](#)
Enumeration of distance sensor types.
- #define [HAVE_ENUM_MAV_SENSOR_ORIENTATION](#)
Enumeration of sensor orientation, according to its rotations.
- #define [HAVE_ENUM_MAV_MISSION_TYPE](#)
Type of mission items being requested/sent in mission protocol.
- #define [HAVE_ENUM_MAV_ESTIMATOR_TYPE](#)
Enumeration of estimator types.
- #define [HAVE_ENUM_MAV_BATTERY_TYPE](#)
Enumeration of battery types.
- #define [HAVE_ENUM_MAV_BATTERY_FUNCTION](#)
Enumeration of battery functions.
- #define [HAVE_ENUM_MAV_BATTERY_CHARGE_STATE](#)
Enumeration for battery charge states.
- #define [HAVE_ENUM_MAV_BATTERY_MODE](#)
Battery mode.
- #define [HAVE_ENUM_MAV_BATTERY_FAULT](#)
Smart battery supply status/fault flags (bitmask) for health indication.
- #define [HAVE_ENUM_MAV_FUEL_TYPE](#)
Fuel types for use in FUEL_TYPE.
- #define [HAVE_ENUM_MAV_GENERATOR_STATUS_FLAG](#)
Flags to report status/failure cases for a power generator (used in GENERATOR_STATUS).
- #define [HAVE_ENUM_MAV_VTOL_STATE](#)
Enumeration of VTOL states.
- #define [HAVE_ENUM_MAV_LANDED_STATE](#)
Enumeration of landed detector states.
- #define [HAVE_ENUM_ADSB_ALTITUDE_TYPE](#)
Enumeration of the ADSB altimeter types.
- #define [HAVE_ENUM_ADSB_EMITTER_TYPE](#)
ADSB classification for the type of vehicle emitting the transponder signal.
- #define [HAVE_ENUM_ADSB_FLAGS](#)
These flags indicate status such as data validity of each data source.
- #define [HAVE_ENUM_MAV_DO_REPOSITION_FLAGS](#)
Bitmap of options for the MAV_CMD_DO_REPOSITION.
- #define [HAVE_ENUM_SPEED_TYPE](#)
Speed setpoint types used in MAV_CMD_DO_CHANGE_SPEED.
- #define [HAVE_ENUM_ESTIMATOR_STATUS_FLAGS](#)
Flags in ESTIMATOR_STATUS message.
- #define [HAVE_ENUM_MOTOR_TEST_ORDER](#)
Sequence that motors are tested when using MAV_CMD_DO_MOTOR_TEST.
- #define [HAVE_ENUM_MOTOR_TEST_THROTTLE_TYPE](#)
Defines how throttle value is represented in MAV_CMD_DO_MOTOR_TEST.
- #define [HAVE_ENUM_GPS_INPUT_IGNORE_FLAGS](#)

- #define [HAVE_ENUM_MAV_COLLISION_ACTION](#)
Possible actions an aircraft can take to avoid a collision.
- #define [HAVE_ENUM_MAV_COLLISION_THREAT_LEVEL](#)
Aircraft-rated danger from this threat.

- #define [HAVE_ENUM_MAV_COLLISION_SRC](#)
Source of information about this collision.
 - #define [HAVE_ENUM_GPS_FIX_TYPE](#)
Type of GPS fix.
 - #define [HAVE_ENUM_RTK_BASELINE_COORDINATE_SYSTEM](#)
RTK GPS baseline coordinate system, used for RTK corrections.
 - #define [HAVE_ENUM_LANDING_TARGET_TYPE](#)
Type of landing target.
 - #define [HAVE_ENUM_VTOL_TRANSITION_HEADING](#)
Direction of VTOL transition.
 - #define [HAVE_ENUM_CAMERA_CAP_FLAGS](#)
Camera capability flags (Bitmap)
 - #define [HAVE_ENUM_VIDEO_STREAM_STATUS_FLAGS](#)
Stream status flags (Bitmap)
 - #define [HAVE_ENUM_VIDEO_STREAM_TYPE](#)
Video stream types.
 - #define [HAVE_ENUM_VIDEO_STREAM_ENCODING](#)
Video stream encodings.
 - #define [HAVE_ENUM_CAMERA_TRACKING_STATUS_FLAGS](#)
Camera tracking status flags.
 - #define [HAVE_ENUM_CAMERA_TRACKING_MODE](#)
Camera tracking modes.
 - #define [HAVE_ENUM_CAMERA_TRACKING_TARGET_DATA](#)
Camera tracking target data (shows where tracked target is within image)
 - #define [HAVE_ENUM_CAMERA_ZOOM_TYPE](#)
Zoom types for MAV_CMD_SET_CAMERA_ZOOM.
 - #define [HAVE_ENUM_SET_FOCUS_TYPE](#)
Focus types for MAV_CMD_SET_CAMERA_FOCUS.
 - #define [HAVE_ENUM_CAMERA_SOURCE](#)
Camera sources for MAV_CMD_SET_CAMERA_SOURCE.
 - #define [HAVE_ENUM_PARAM_ACK](#)
Result from PARAM_EXT_SET message.
 - #define [HAVE_ENUM_CAMERA_MODE](#)
Camera Modes.
 - #define [HAVE_ENUM_MAV_ARM_AUTH_DENIED_REASON](#)
-
- #define [HAVE_ENUM_RC_TYPE](#)
RC type.
 - #define [HAVE_ENUM_RC_SUB_TYPE](#)
RC sub-type of types defined in RC_TYPE.
 - #define [HAVE_ENUM_ENGINE_CONTROL_OPTIONS](#)
Engine control options.
 - #define [HAVE_ENUM_POSITION_TARGET_TYPERMASK](#)
Bitmap to indicate which dimensions should be ignored by the vehicle: a value of 0b0000000000000000 or 0b0000001000000000 indicates that none of the setpoint dimensions should be ignored.
 - #define [HAVE_ENUM_ATTITUDE_TARGET_TYPERMASK](#)
Bitmap to indicate which dimensions should be ignored by the vehicle: a value of 0b00000000 indicates that none of the setpoint dimensions should be ignored.
 - #define [HAVE_ENUM_UTM_FLIGHT_STATE](#)
Airborne status of UAS.

- #define [HAVE_ENUM_UTM_DATA_AVAIL_FLAGS](#)
Flags for the global position report.
- #define [HAVE_ENUM_CELLULAR_STATUS_FLAG](#)
These flags encode the cellular network status.
- #define [HAVE_ENUM_CELLULAR_NETWORK_FAILED_REASON](#)
These flags are used to diagnose the failure state of `CELLULAR_STATUS`.
- #define [HAVE_ENUM_CELLULAR_NETWORK_RADIO_TYPE](#)
Cellular network radio type.
- #define [HAVE_ENUM_PRECISION_LAND_MODE](#)
Precision land modes (used in `MAV_CMD_NAV_LAND`).
- #define [HAVE_ENUM_PARACHUTE_ACTION](#)
Parachute actions.
- #define [HAVE_ENUM_MAV_TUNNEL_PAYLOAD_TYPE](#)
- #define [HAVE_ENUM_MAV_ODID_ID_TYPE](#)
- #define [HAVE_ENUM_MAV_ODID_UA_TYPE](#)
- #define [HAVE_ENUM_MAV_ODID_STATUS](#)
- #define [HAVE_ENUM_MAV_ODID_HEIGHT_REF](#)
- #define [HAVE_ENUM_MAV_ODID_HOR_ACC](#)
- #define [HAVE_ENUM_MAV_ODID_VER_ACC](#)
- #define [HAVE_ENUM_MAV_ODID_SPEED_ACC](#)
- #define [HAVE_ENUM_MAV_ODID_TIME_ACC](#)
- #define [HAVE_ENUM_MAV_ODID_AUTH_TYPE](#)
- #define [HAVE_ENUM_MAV_ODID_DESC_TYPE](#)
- #define [HAVE_ENUM_MAV_ODID_OPERATOR_LOCATION_TYPE](#)
- #define [HAVE_ENUM_MAV_ODID_CLASSIFICATION_TYPE](#)
- #define [HAVE_ENUM_MAV_ODID_CATEGORY_EU](#)

- #define `HAVE_ENUM_MAV_ODID_CLASS_EU`
- #define `HAVE_ENUM_MAV_ODID_OPERATOR_ID_TYPE`
- #define `HAVE_ENUM_MAV_ODID_ARM_STATUS`
- #define `HAVE_ENUM_TUNE_FORMAT`
Tune formats (used for vehicle buzzer/tone generation).
- #define `HAVE_ENUM_AIS_TYPE`
Type of AIS vessel, enum duplicated from AIS standard, <https://gpsd.gitlab.io/gpsd/AIVDM.html>.
- #define `HAVE_ENUM_AIS_NAV_STATUS`
Navigational status of AIS vessel, enum duplicated from AIS standard, <https://gpsd.gitlab.io/gpsd/AIVDM.html>.
- #define `HAVE_ENUM_AIS_FLAGS`
These flags are used in the AIS_VESSEL.fields bitmask to indicate validity of data in the other message fields.
- #define `HAVE_ENUM_FAILURE_UNIT`
List of possible units where failures can be injected.
- #define `HAVE_ENUM_FAILURE_TYPE`
List of possible failure type to inject.
- #define `HAVE_ENUM_NAV_VTOL_LAND_OPTIONS`
- #define `HAVE_ENUM_MAV_WINCH_STATUS_FLAG`
Winch status flags used in WINCH_STATUS.
- #define `HAVE_ENUM_MAG_CAL_STATUS`
- #define `HAVE_ENUM_MAV_EVENT_ERROR_REASON`
Reason for an event error response.
- #define `HAVE_ENUM_MAV_EVENT_CURRENT_SEQUENCE_FLAGS`
Flags for CURRENT_EVENT_SEQUENCE.
- #define `HAVE_ENUM_HIL_SENSOR_UPDATED_FLAGS`
Flags in the HIL_SENSOR message indicate which fields have updated since the last message.
- #define `HAVE_ENUM_HIGHRES_IMU_UPDATED_FLAGS`
Flags in the HIGHRES_IMU message indicate which fields have updated since the last message.
- #define `HAVE_ENUM_CAN_FILTER_OP`
- #define `HAVE_ENUM_MAV_FTP_ERR`
MAV FTP error codes (may be used in FILE_TRANSFER_PROTOCOL).
- #define `HAVE_ENUM_MAV_FTP_OPCODE`
MAV FTP opcodes (may be used in FILE_TRANSFER_PROTOCOL).
- #define `HAVE_ENUM_MISSION_STATE`
States of the mission state machine.
- #define `HAVE_ENUM_SAFETY_SWITCH_STATE`
Possible safety switch states.
- #define `HAVE_ENUM_ILLUMINATOR_MODE`
Modes of illuminator.
- #define `HAVE_ENUM_ILLUMINATOR_ERROR_FLAGS`

- *Illuminator module error flags (bitmap, 0 means no error)*
- `#define HAVE_ENUM_MAV_STANDARD_MODE`
Standard modes with a well understood meaning across flight stacks and vehicle types.
- `#define HAVE_ENUM_MAV_MODE_PROPERTY`
Mode properties.
- `#define HAVE_ENUM_HIL_ACTUATOR_CONTROLS_FLAGS`
Flags used in HIL_ACTUATOR_CONTROLS message.
- `#define HAVE_ENUM_COMPUTER_STATUS_FLAGS`
Flags used to report computer status.
- `#define HAVE_ENUM_AIRSPPEED_SENSOR_FLAGS`
Airspeed sensor flags.
- `#define MAVLINK_VERSION 3`

Typedefs

- `typedef enum HL_FAILURE_FLAG HL_FAILURE_FLAG`
- `typedef enum MAV_GOTO MAV_GOTO`
- `typedef enum MAV_MODE MAV_MODE`
- `typedef enum MAV_SYS_STATUS_SENSOR MAV_SYS_STATUS_SENSOR`
- `typedef enum MAV_SYS_STATUS_SENSOR_EXTENDED MAV_SYS_STATUS_SENSOR_EXTENDED`
- `typedef enum MAV_FRAME MAV_FRAME`
- `typedef enum MAVLINK_DATA_STREAM_TYPE MAVLINK_DATA_STREAM_TYPE`
- `typedef enum FENCE_BREACH FENCE_BREACH`
- `typedef enum FENCE_MITIGATE FENCE_MITIGATE`
- `typedef enum FENCE_TYPE FENCE_TYPE`
- `typedef enum MAV_MOUNT_MODE MAV_MOUNT_MODE`
- `typedef enum GIMBAL_DEVICE_CAP_FLAGS GIMBAL_DEVICE_CAP_FLAGS`
- `typedef enum GIMBAL_MANAGER_CAP_FLAGS GIMBAL_MANAGER_CAP_FLAGS`
- `typedef enum GIMBAL_DEVICE_FLAGS GIMBAL_DEVICE_FLAGS`
- `typedef enum GIMBAL_MANAGER_FLAGS GIMBAL_MANAGER_FLAGS`
- `typedef enum GIMBAL_DEVICE_ERROR_FLAGS GIMBAL_DEVICE_ERROR_FLAGS`
- `typedef enum GRIPPER_ACTIONS GRIPPER_ACTIONS`
- `typedef enum WINCH_ACTIONS WINCH_ACTIONS`
- `typedef enum UAVCAN_NODE_HEALTH UAVCAN_NODE_HEALTH`
- `typedef enum UAVCAN_NODE_MODE UAVCAN_NODE_MODE`
- `typedef enum ESC_CONNECTION_TYPE ESC_CONNECTION_TYPE`
- `typedef enum ESC_FAILURE_FLAGS ESC_FAILURE_FLAGS`
- `typedef enum STORAGE_STATUS STORAGE_STATUS`
- `typedef enum STORAGE_TYPE STORAGE_TYPE`
- `typedef enum STORAGE_USAGE_FLAG STORAGE_USAGE_FLAG`
- `typedef enum ORBIT_YAW_BEHAVIOUR ORBIT_YAW_BEHAVIOUR`
- `typedef enum WIFI_CONFIG_AP_RESPONSE WIFI_CONFIG_AP_RESPONSE`
- `typedef enum CELLULAR_CONFIG_RESPONSE CELLULAR_CONFIG_RESPONSE`
- `typedef enum WIFI_CONFIG_AP_MODE WIFI_CONFIG_AP_MODE`
- `typedef enum COMP_METADATA_TYPE COMP_METADATA_TYPE`
- `typedef enum ACTUATOR_CONFIGURATION ACTUATOR_CONFIGURATION`
- `typedef enum ACTUATOR_OUTPUT_FUNCTION ACTUATOR_OUTPUT_FUNCTION`
- `typedef enum AUTOTUNE_AXIS AUTOTUNE_AXIS`
- `typedef enum PREFLIGHT_STORAGE_PARAMETER_ACTION PREFLIGHT_STORAGE_PARAMETER_↵
_ACTION`
- `typedef enum PREFLIGHT_STORAGE_MISSION_ACTION PREFLIGHT_STORAGE_MISSION_ACTION`
- `typedef enum REBOOT_SHUTDOWN_ACTION REBOOT_SHUTDOWN_ACTION`
- `typedef enum REBOOT_SHUTDOWN_CONDITIONS REBOOT_SHUTDOWN_CONDITIONS`

- typedef enum MAV_CMD **MAV_CMD**
- typedef enum MAV_DATA_STREAM **MAV_DATA_STREAM**
- typedef enum MAV_ROI **MAV_ROI**
- typedef enum MAV_PARAM_TYPE **MAV_PARAM_TYPE**
- typedef enum MAV_PARAM_ERROR **MAV_PARAM_ERROR**
- typedef enum MAV_PARAM_EXT_TYPE **MAV_PARAM_EXT_TYPE**
- typedef enum MAV_RESULT **MAV_RESULT**
- typedef enum MAV_MISSION_RESULT **MAV_MISSION_RESULT**
- typedef enum MAV_SEVERITY **MAV_SEVERITY**
- typedef enum MAV_POWER_STATUS **MAV_POWER_STATUS**
- typedef enum SERIAL_CONTROL_DEV **SERIAL_CONTROL_DEV**
- typedef enum SERIAL_CONTROL_FLAG **SERIAL_CONTROL_FLAG**
- typedef enum MAV_DISTANCE_SENSOR **MAV_DISTANCE_SENSOR**
- typedef enum MAV_SENSOR_ORIENTATION **MAV_SENSOR_ORIENTATION**
- typedef enum MAV_MISSION_TYPE **MAV_MISSION_TYPE**
- typedef enum MAV_ESTIMATOR_TYPE **MAV_ESTIMATOR_TYPE**
- typedef enum MAV_BATTERY_TYPE **MAV_BATTERY_TYPE**
- typedef enum MAV_BATTERY_FUNCTION **MAV_BATTERY_FUNCTION**
- typedef enum MAV_BATTERY_CHARGE_STATE **MAV_BATTERY_CHARGE_STATE**
- typedef enum MAV_BATTERY_MODE **MAV_BATTERY_MODE**
- typedef enum MAV_BATTERY_FAULT **MAV_BATTERY_FAULT**
- typedef enum MAV_FUEL_TYPE **MAV_FUEL_TYPE**
- typedef enum MAV_GENERATOR_STATUS_FLAG **MAV_GENERATOR_STATUS_FLAG**
- typedef enum MAV_VTOL_STATE **MAV_VTOL_STATE**
- typedef enum MAV_LANDED_STATE **MAV_LANDED_STATE**
- typedef enum ADSB_ALTITUDE_TYPE **ADSB_ALTITUDE_TYPE**
- typedef enum ADSB_EMITTER_TYPE **ADSB_EMITTER_TYPE**
- typedef enum ADSB_FLAGS **ADSB_FLAGS**
- typedef enum MAV_DO_REPOSITION_FLAGS **MAV_DO_REPOSITION_FLAGS**
- typedef enum SPEED_TYPE **SPEED_TYPE**
- typedef enum ESTIMATOR_STATUS_FLAGS **ESTIMATOR_STATUS_FLAGS**
- typedef enum MOTOR_TEST_ORDER **MOTOR_TEST_ORDER**
- typedef enum MOTOR_TEST_THROTTLE_TYPE **MOTOR_TEST_THROTTLE_TYPE**
- typedef enum GPS_INPUT_IGNORE_FLAGS **GPS_INPUT_IGNORE_FLAGS**
- typedef enum MAV_COLLISION_ACTION **MAV_COLLISION_ACTION**
- typedef enum MAV_COLLISION_THREAT_LEVEL **MAV_COLLISION_THREAT_LEVEL**
- typedef enum MAV_COLLISION_SRC **MAV_COLLISION_SRC**
- typedef enum GPS_FIX_TYPE **GPS_FIX_TYPE**
- typedef enum RTK_BASELINE_COORDINATE_SYSTEM **RTK_BASELINE_COORDINATE_SYSTEM**
- typedef enum LANDING_TARGET_TYPE **LANDING_TARGET_TYPE**
- typedef enum VTOL_TRANSITION_HEADING **VTOL_TRANSITION_HEADING**
- typedef enum CAMERA_CAP_FLAGS **CAMERA_CAP_FLAGS**
- typedef enum VIDEO_STREAM_STATUS_FLAGS **VIDEO_STREAM_STATUS_FLAGS**
- typedef enum VIDEO_STREAM_TYPE **VIDEO_STREAM_TYPE**
- typedef enum VIDEO_STREAM_ENCODING **VIDEO_STREAM_ENCODING**
- typedef enum CAMERA_TRACKING_STATUS_FLAGS **CAMERA_TRACKING_STATUS_FLAGS**
- typedef enum CAMERA_TRACKING_MODE **CAMERA_TRACKING_MODE**
- typedef enum CAMERA_TRACKING_TARGET_DATA **CAMERA_TRACKING_TARGET_DATA**
- typedef enum CAMERA_ZOOM_TYPE **CAMERA_ZOOM_TYPE**
- typedef enum SET_FOCUS_TYPE **SET_FOCUS_TYPE**
- typedef enum CAMERA_SOURCE **CAMERA_SOURCE**
- typedef enum PARAM_ACK **PARAM_ACK**
- typedef enum CAMERA_MODE **CAMERA_MODE**
- typedef enum MAV_ARM_AUTH_DENIED_REASON **MAV_ARM_AUTH_DENIED_REASON**
- typedef enum RC_TYPE **RC_TYPE**

- typedef enum RC_SUB_TYPE **RC_SUB_TYPE**
- typedef enum ENGINE_CONTROL_OPTIONS **ENGINE_CONTROL_OPTIONS**
- typedef enum POSITION_TARGET_TYEMASK **POSITION_TARGET_TYEMASK**
- typedef enum ATTITUDE_TARGET_TYEMASK **ATTITUDE_TARGET_TYEMASK**
- typedef enum UTM_FLIGHT_STATE **UTM_FLIGHT_STATE**
- typedef enum UTM_DATA_AVAIL_FLAGS **UTM_DATA_AVAIL_FLAGS**
- typedef enum CELLULAR_STATUS_FLAG **CELLULAR_STATUS_FLAG**
- typedef enum CELLULAR_NETWORK_FAILED_REASON **CELLULAR_NETWORK_FAILED_REASON**
- typedef enum CELLULAR_NETWORK_RADIO_TYPE **CELLULAR_NETWORK_RADIO_TYPE**
- typedef enum PRECISION_LAND_MODE **PRECISION_LAND_MODE**
- typedef enum PARACHUTE_ACTION **PARACHUTE_ACTION**
- typedef enum MAV_TUNNEL_PAYLOAD_TYPE **MAV_TUNNEL_PAYLOAD_TYPE**
- typedef enum MAV_ODID_ID_TYPE **MAV_ODID_ID_TYPE**
- typedef enum MAV_ODID_UA_TYPE **MAV_ODID_UA_TYPE**
- typedef enum MAV_ODID_STATUS **MAV_ODID_STATUS**
- typedef enum MAV_ODID_HEIGHT_REF **MAV_ODID_HEIGHT_REF**
- typedef enum MAV_ODID_HOR_ACC **MAV_ODID_HOR_ACC**
- typedef enum MAV_ODID_VER_ACC **MAV_ODID_VER_ACC**
- typedef enum MAV_ODID_SPEED_ACC **MAV_ODID_SPEED_ACC**
- typedef enum MAV_ODID_TIME_ACC **MAV_ODID_TIME_ACC**
- typedef enum MAV_ODID_AUTH_TYPE **MAV_ODID_AUTH_TYPE**
- typedef enum MAV_ODID_DESC_TYPE **MAV_ODID_DESC_TYPE**
- typedef enum MAV_ODID_OPERATOR_LOCATION_TYPE **MAV_ODID_OPERATOR_LOCATION_TYPE**
- typedef enum MAV_ODID_CLASSIFICATION_TYPE **MAV_ODID_CLASSIFICATION_TYPE**
- typedef enum MAV_ODID_CATEGORY_EU **MAV_ODID_CATEGORY_EU**
- typedef enum MAV_ODID_CLASS_EU **MAV_ODID_CLASS_EU**
- typedef enum MAV_ODID_OPERATOR_ID_TYPE **MAV_ODID_OPERATOR_ID_TYPE**
- typedef enum MAV_ODID_ARM_STATUS **MAV_ODID_ARM_STATUS**
- typedef enum TUNE_FORMAT **TUNE_FORMAT**
- typedef enum AIS_TYPE **AIS_TYPE**
- typedef enum AIS_NAV_STATUS **AIS_NAV_STATUS**
- typedef enum AIS_FLAGS **AIS_FLAGS**
- typedef enum FAILURE_UNIT **FAILURE_UNIT**
- typedef enum FAILURE_TYPE **FAILURE_TYPE**
- typedef enum NAV_VTOL_LAND_OPTIONS **NAV_VTOL_LAND_OPTIONS**
- typedef enum MAV_WINCH_STATUS_FLAG **MAV_WINCH_STATUS_FLAG**
- typedef enum MAG_CAL_STATUS **MAG_CAL_STATUS**
- typedef enum MAV_EVENT_ERROR_REASON **MAV_EVENT_ERROR_REASON**
- typedef enum MAV_EVENT_CURRENT_SEQUENCE_FLAGS **MAV_EVENT_CURRENT_SEQUENCE_FLAGS**
- typedef enum HIL_SENSOR_UPDATED_FLAGS **HIL_SENSOR_UPDATED_FLAGS**
- typedef enum HIGHRES_IMU_UPDATED_FLAGS **HIGHRES_IMU_UPDATED_FLAGS**
- typedef enum CAN_FILTER_OP **CAN_FILTER_OP**
- typedef enum MAV_FTP_ERR **MAV_FTP_ERR**
- typedef enum MAV_FTP_OPCODE **MAV_FTP_OPCODE**
- typedef enum MISSION_STATE **MISSION_STATE**
- typedef enum SAFETY_SWITCH_STATE **SAFETY_SWITCH_STATE**
- typedef enum ILLUMINATOR_MODE **ILLUMINATOR_MODE**
- typedef enum ILLUMINATOR_ERROR_FLAGS **ILLUMINATOR_ERROR_FLAGS**
- typedef enum MAV_STANDARD_MODE **MAV_STANDARD_MODE**
- typedef enum MAV_MODE_PROPERTY **MAV_MODE_PROPERTY**
- typedef enum HIL_ACTUATOR_CONTROLS_FLAGS **HIL_ACTUATOR_CONTROLS_FLAGS**
- typedef enum COMPUTER_STATUS_FLAGS **COMPUTER_STATUS_FLAGS**
- typedef enum AIRSPEED_SENSOR_FLAGS **AIRSPEED_SENSOR_FLAGS**

Enumerations

- enum HL_FAILURE_FLAG {
 HL_FAILURE_FLAG_GPS =1 , HL_FAILURE_FLAG_DIFFERENTIAL_PRESSURE =2 , HL_FAILURE_FLAG_ABSOLUTE_PRESSURE =4 , HL_FAILURE_FLAG_3D_ACCEL =8 ,
 HL_FAILURE_FLAG_3D_GYRO =16 , HL_FAILURE_FLAG_3D_MAG =32 , HL_FAILURE_FLAG_TERRAIN =64 , HL_FAILURE_FLAG_BATTERY =128 ,
 HL_FAILURE_FLAG_RC_RECEIVER =256 , HL_FAILURE_FLAG_OFFBOARD_LINK =512 , HL_FAILURE_FLAG_ENGINE =1024 , HL_FAILURE_FLAG_GEOFENCE =2048 ,
 HL_FAILURE_FLAG_ESTIMATOR =4096 , HL_FAILURE_FLAG_MISSION =8192 , HL_FAILURE_FLAG_ENUM_END =8193 }
- enum MAV_GOTO {
 MAV_GOTO_DO_HOLD =0 , MAV_GOTO_DO_CONTINUE =1 , MAV_GOTO_HOLD_AT_CURRENT_POSITION =2 , MAV_GOTO_HOLD_AT_SPECIFIED_POSITION =3 ,
 MAV_GOTO_ENUM_END =4 }
- enum MAV_MODE {
 MAV_MODE_PREFLIGHT =0 , MAV_MODE_MANUAL_DISARMED =64 , MAV_MODE_TEST_DISARMED =66 , MAV_MODE_STABILIZE_DISARMED =80 ,
 MAV_MODE_GUIDED_DISARMED =88 , MAV_MODE_AUTO_DISARMED =92 , MAV_MODE_MANUAL_ARMED =192 , MAV_MODE_TEST_ARMED =194 ,
 MAV_MODE_STABILIZE_ARMED =208 , MAV_MODE_GUIDED_ARMED =216 , MAV_MODE_AUTO_ARMED =220 , MAV_MODE_ENUM_END =221 }
- enum MAV_SYS_STATUS_SENSOR {
 MAV_SYS_STATUS_SENSOR_3D_GYRO =1 , MAV_SYS_STATUS_SENSOR_3D_ACCEL =2 , MAV_SYS_STATUS_SENSOR_3D_MAG =4 , MAV_SYS_STATUS_SENSOR_ABSOLUTE_PRESSURE =8 ,
 MAV_SYS_STATUS_SENSOR_DIFFERENTIAL_PRESSURE =16 , MAV_SYS_STATUS_SENSOR_GPS =32 , MAV_SYS_STATUS_SENSOR_OPTICAL_FLOW =64 , MAV_SYS_STATUS_SENSOR_VISION_POSITION =128 ,
 MAV_SYS_STATUS_SENSOR_LASER_POSITION =256 , MAV_SYS_STATUS_SENSOR_EXTERNAL_GROUND_TRUTH =512 , MAV_SYS_STATUS_SENSOR_ANGULAR_RATE_CONTROL =1024 , MAV_SYS_STATUS_SENSOR_ATTITUDE_STABILIZATION =2048 ,
 MAV_SYS_STATUS_SENSOR_YAW_POSITION =4096 , MAV_SYS_STATUS_SENSOR_Z_ALTITUDE_CONTROL =8192 , MAV_SYS_STATUS_SENSOR_XY_POSITION_CONTROL =16384 , MAV_SYS_STATUS_SENSOR_MOTOR_OUTPUTS =32768 ,
 MAV_SYS_STATUS_SENSOR_RC_RECEIVER =65536 , MAV_SYS_STATUS_SENSOR_3D_GYRO2 =131072 , MAV_SYS_STATUS_SENSOR_3D_ACCEL2 =262144 , MAV_SYS_STATUS_SENSOR_3D_MAG2 =524288 ,
 MAV_SYS_STATUS_GEOFENCE =1048576 , MAV_SYS_STATUS_AHRS =2097152 , MAV_SYS_STATUS_TERRAIN =4194304 , MAV_SYS_STATUS_REVERSE_MOTOR =8388608 ,
 MAV_SYS_STATUS_LOGGING =16777216 , MAV_SYS_STATUS_SENSOR_BATTERY =33554432 , MAV_SYS_STATUS_SENSOR_PROXIMITY =67108864 , MAV_SYS_STATUS_SENSOR_SATCOM =134217728 ,
 MAV_SYS_STATUS_PREARM_CHECK =268435456 , MAV_SYS_STATUS_OBSTACLE_AVOIDANCE =536870912 , MAV_SYS_STATUS_SENSOR_PROPULSION =1073741824 , MAV_SYS_STATUS_EXTENSION_USED =2147483648 ,
 MAV_SYS_STATUS_SENSOR_ENUM_END =2147483649 }
- enum MAV_SYS_STATUS_SENSOR_EXTENDED { MAV_SYS_STATUS_RECOVERY_SYSTEM =1 , MAV_SYS_STATUS_SENSOR_EXTENDED_ENUM_END =2 }
- enum MAV_FRAME {
 MAV_FRAME_GLOBAL =0 , MAV_FRAME_LOCAL_NED =1 , MAV_FRAME_MISSION =2 , MAV_FRAME_GLOBAL_RELATIVE_ALT =3 ,
 MAV_FRAME_LOCAL_ENU =4 , MAV_FRAME_GLOBAL_INT =5 , MAV_FRAME_GLOBAL_RELATIVE_ALT_INT =6 , MAV_FRAME_LOCAL_OFFSET_NED =7 ,
 MAV_FRAME_BODY_NED =8 , MAV_FRAME_BODY_OFFSET_NED =9 , MAV_FRAME_GLOBAL_TERRAIN_ALT =10 , MAV_FRAME_GLOBAL_TERRAIN_ALT_INT =11 ,
 MAV_FRAME_BODY_FRD =12 , MAV_FRAME_RESERVED_13 =13 , MAV_FRAME_RESERVED_14 =14 , MAV_FRAME_RESERVED_15 =15 ,
 MAV_FRAME_RESERVED_16 =16 , MAV_FRAME_RESERVED_17 =17 , MAV_FRAME_RESERVED_18 =18

```

=18 , MAV_FRAME_RESERVED_19 =19 ,
MAV_FRAME_LOCAL_FRD =20 , MAV_FRAME_LOCAL_FLU =21 , MAV_FRAME_ENUM_END =22 }
• enum MAVLINK_DATA_STREAM_TYPE {
  MAVLINK_DATA_STREAM_IMG_JPEG =0 , MAVLINK_DATA_STREAM_IMG_BMP =1 , MAVLINK_↵
  DATA_STREAM_IMG_RAW8U =2 , MAVLINK_DATA_STREAM_IMG_RAW32U =3 ,
  MAVLINK_DATA_STREAM_IMG_PGM =4 , MAVLINK_DATA_STREAM_IMG_PNG =5 , MAVLINK_↵
  DATA_STREAM_TYPE_ENUM_END =6 }
• enum FENCE_BREACH {
  FENCE_BREACH_NONE =0 , FENCE_BREACH_MINALT =1 , FENCE_BREACH_MAXALT =2 , FENCE_↵
  _BREACH_BOUNDARY =3 ,
  FENCE_BREACH_ENUM_END =4 }
• enum FENCE_MITIGATE { FENCE_MITIGATE_UNKNOWN =0 , FENCE_MITIGATE_NONE =1 , FENCE_↵
  _MITIGATE_VEL_LIMIT =2 , FENCE_MITIGATE_ENUM_END =3 }
• enum FENCE_TYPE {
  FENCE_TYPE_ALT_MAX =1 , FENCE_TYPE_CIRCLE =2 , FENCE_TYPE_POLYGON =4 , FENCE_↵
  TYPE_ALT_MIN =8 ,
  FENCE_TYPE_ENUM_END =9 }
• enum MAV_MOUNT_MODE {
  MAV_MOUNT_MODE_RETRACT =0 , MAV_MOUNT_MODE_NEUTRAL =1 , MAV_MOUNT_MODE_↵
  MAVLINK_TARGETING =2 , MAV_MOUNT_MODE_RC_TARGETING =3 ,
  MAV_MOUNT_MODE_GPS_POINT =4 , MAV_MOUNT_MODE_SYSID_TARGET =5 , MAV_MOUNT_↵
  MODE_HOME_LOCATION =6 , MAV_MOUNT_MODE_ENUM_END =7 }
• enum GIMBAL_DEVICE_CAP_FLAGS {
  GIMBAL_DEVICE_CAP_FLAGS_HAS_RETRACT =1 , GIMBAL_DEVICE_CAP_FLAGS_HAS_NEUTRAL
  =2 , GIMBAL_DEVICE_CAP_FLAGS_HAS_ROLL_AXIS =4 , GIMBAL_DEVICE_CAP_FLAGS_HAS_↵
  ROLL_FOLLOW =8 ,
  GIMBAL_DEVICE_CAP_FLAGS_HAS_ROLL_LOCK =16 , GIMBAL_DEVICE_CAP_FLAGS_HAS_↵
  PITCH_AXIS =32 , GIMBAL_DEVICE_CAP_FLAGS_HAS_PITCH_FOLLOW =64 , GIMBAL_DEVICE_↵
  CAP_FLAGS_HAS_PITCH_LOCK =128 ,
  GIMBAL_DEVICE_CAP_FLAGS_HAS_YAW_AXIS =256 , GIMBAL_DEVICE_CAP_FLAGS_HAS_YAW_↵
  _FOLLOW =512 , GIMBAL_DEVICE_CAP_FLAGS_HAS_YAW_LOCK =1024 , GIMBAL_DEVICE_CAP_↵
  _FLAGS_SUPPORTS_INFINITE_YAW =2048 ,
  GIMBAL_DEVICE_CAP_FLAGS_SUPPORTS_YAW_IN_EARTH_FRAME =4096 , GIMBAL_DEVICE_↵
  CAP_FLAGS_HAS_RC_INPUTS =8192 , GIMBAL_DEVICE_CAP_FLAGS_ENUM_END =8193 }
• enum GIMBAL_MANAGER_CAP_FLAGS {
  GIMBAL_MANAGER_CAP_FLAGS_HAS_RETRACT =1 , GIMBAL_MANAGER_CAP_FLAGS_HAS_↵
  NEUTRAL =2 , GIMBAL_MANAGER_CAP_FLAGS_HAS_ROLL_AXIS =4 , GIMBAL_MANAGER_CAP_↵
  _FLAGS_HAS_ROLL_FOLLOW =8 ,
  GIMBAL_MANAGER_CAP_FLAGS_HAS_ROLL_LOCK =16 , GIMBAL_MANAGER_CAP_FLAGS_↵
  HAS_PITCH_AXIS =32 , GIMBAL_MANAGER_CAP_FLAGS_HAS_PITCH_FOLLOW =64 , GIMBAL_↵
  MANAGER_CAP_FLAGS_HAS_PITCH_LOCK =128 ,
  GIMBAL_MANAGER_CAP_FLAGS_HAS_YAW_AXIS =256 , GIMBAL_MANAGER_CAP_FLAGS_HAS_↵
  _YAW_FOLLOW =512 , GIMBAL_MANAGER_CAP_FLAGS_HAS_YAW_LOCK =1024 , GIMBAL_↵
  MANAGER_CAP_FLAGS_SUPPORTS_INFINITE_YAW =2048 ,
  GIMBAL_MANAGER_CAP_FLAGS_SUPPORTS_YAW_IN_EARTH_FRAME =4096 , GIMBAL_↵
  MANAGER_CAP_FLAGS_HAS_RC_INPUTS =8192 , GIMBAL_MANAGER_CAP_FLAGS_CAN_POINT_↵
  _LOCATION_LOCAL =65536 , GIMBAL_MANAGER_CAP_FLAGS_CAN_POINT_LOCATION_GLOBAL
  =131072 ,
  GIMBAL_MANAGER_CAP_FLAGS_ENUM_END =131073 }
• enum GIMBAL_DEVICE_FLAGS {
  GIMBAL_DEVICE_FLAGS_RETRACT =1 , GIMBAL_DEVICE_FLAGS_NEUTRAL =2 , GIMBAL_↵
  DEVICE_FLAGS_ROLL_LOCK =4 , GIMBAL_DEVICE_FLAGS_PITCH_LOCK =8 ,
  GIMBAL_DEVICE_FLAGS_YAW_LOCK =16 , GIMBAL_DEVICE_FLAGS_YAW_IN_VEHICLE_FRAME
  =32 , GIMBAL_DEVICE_FLAGS_YAW_IN_EARTH_FRAME =64 , GIMBAL_DEVICE_FLAGS_↵
  ACCEPTS_YAW_IN_EARTH_FRAME =128 ,
  GIMBAL_DEVICE_FLAGS_RC_EXCLUSIVE =256 , GIMBAL_DEVICE_FLAGS_RC_MIXED =512 ,
  GIMBAL_DEVICE_FLAGS_ENUM_END =513 }

```

- enum GIMBAL_MANAGER_FLAGS {
GIMBAL_MANAGER_FLAGS_RETRACT =1 , GIMBAL_MANAGER_FLAGS_NEUTRAL =2 , GIMBAL_MANAGER_FLAGS_ROLL_LOCK =4 , GIMBAL_MANAGER_FLAGS_PITCH_LOCK =8 ,
GIMBAL_MANAGER_FLAGS_YAW_LOCK =16 , GIMBAL_MANAGER_FLAGS_YAW_IN_VEHICLE_FRAME =32 , GIMBAL_MANAGER_FLAGS_YAW_IN_EARTH_FRAME =64 , GIMBAL_MANAGER_FLAGS_ACCEPTS_YAW_IN_EARTH_FRAME =128 ,
GIMBAL_MANAGER_FLAGS_RC_EXCLUSIVE =256 , GIMBAL_MANAGER_FLAGS_RC_MIXED =512 , GIMBAL_MANAGER_FLAGS_ENUM_END =513 }
- enum GIMBAL_DEVICE_ERROR_FLAGS {
GIMBAL_DEVICE_ERROR_FLAGS_AT_ROLL_LIMIT =1 , GIMBAL_DEVICE_ERROR_FLAGS_AT_PITCH_LIMIT =2 , GIMBAL_DEVICE_ERROR_FLAGS_AT_YAW_LIMIT =4 , GIMBAL_DEVICE_ERROR_FLAGS_ENCODER_ERROR =8 ,
GIMBAL_DEVICE_ERROR_FLAGS_POWER_ERROR =16 , GIMBAL_DEVICE_ERROR_FLAGS_MOTOR_ERROR =32 , GIMBAL_DEVICE_ERROR_FLAGS_SOFTWARE_ERROR =64 , GIMBAL_DEVICE_ERROR_FLAGS_COMMS_ERROR =128 ,
GIMBAL_DEVICE_ERROR_FLAGS_CALIBRATION_RUNNING =256 , GIMBAL_DEVICE_ERROR_FLAGS_NO_MANAGER =512 , GIMBAL_DEVICE_ERROR_FLAGS_ENUM_END =513 }
- enum GRIPPER_ACTIONS { GRIPPER_ACTION_RELEASE =0 , GRIPPER_ACTION_GRAB =1 , GRIPPER_ACTION_HOLD =2 , GRIPPER_ACTIONS_ENUM_END =3 }
- enum WINCH_ACTIONS {
WINCH_RELAXED =0 , WINCH_RELATIVE_LENGTH_CONTROL =1 , WINCH_RATE_CONTROL =2 , WINCH_LOCK =3 ,
WINCH_DELIVER =4 , WINCH_HOLD =5 , WINCH_RETRACT =6 , WINCH_LOAD_LINE =7 , WINCH_ABANDON_LINE =8 , WINCH_LOAD_PAYLOAD =9 , WINCH_ACTIONS_ENUM_END =10 }
- enum UAVCAN_NODE_HEALTH {
UAVCAN_NODE_HEALTH_OK =0 , UAVCAN_NODE_HEALTH_WARNING =1 , UAVCAN_NODE_HEALTH_ERROR =2 , UAVCAN_NODE_HEALTH_CRITICAL =3 ,
UAVCAN_NODE_HEALTH_ENUM_END =4 }
- enum UAVCAN_NODE_MODE {
UAVCAN_NODE_MODE_OPERATIONAL =0 , UAVCAN_NODE_MODE_INITIALIZATION =1 , UAVCAN_NODE_MODE_MAINTENANCE =2 , UAVCAN_NODE_MODE_SOFTWARE_UPDATE =3 ,
UAVCAN_NODE_MODE_OFFLINE =7 , UAVCAN_NODE_MODE_ENUM_END =8 }
- enum ESC_CONNECTION_TYPE {
ESC_CONNECTION_TYPE_PPM =0 , ESC_CONNECTION_TYPE_SERIAL =1 , ESC_CONNECTION_TYPE_ONESHOT =2 , ESC_CONNECTION_TYPE_I2C =3 ,
ESC_CONNECTION_TYPE_CAN =4 , ESC_CONNECTION_TYPE_DSHOT =5 , ESC_CONNECTION_TYPE_ENUM_END =6 }
- enum ESC_FAILURE_FLAGS {
ESC_FAILURE_OVER_CURRENT =1 , ESC_FAILURE_OVER_VOLTAGE =2 , ESC_FAILURE_OVER_TEMPERATURE =4 , ESC_FAILURE_OVER_RPM =8 ,
ESC_FAILURE_INCONSISTENT_CMD =16 , ESC_FAILURE_MOTOR_STUCK =32 , ESC_FAILURE_GENERIC =64 , ESC_FAILURE_FLAGS_ENUM_END =65 }
- enum STORAGE_STATUS {
STORAGE_STATUS_EMPTY =0 , STORAGE_STATUS_UNFORMATTED =1 , STORAGE_STATUS_READY =2 , STORAGE_STATUS_NOT_SUPPORTED =3 ,
STORAGE_STATUS_ENUM_END =4 }
- enum STORAGE_TYPE {
STORAGE_TYPE_UNKNOWN =0 , STORAGE_TYPE_USB_STICK =1 , STORAGE_TYPE_SD =2 , STORAGE_TYPE_MICROSD =3 ,
STORAGE_TYPE_CF =4 , STORAGE_TYPE_CFE =5 , STORAGE_TYPE_XQD =6 , STORAGE_TYPE_HD =7 ,
STORAGE_TYPE_OTHER =254 , STORAGE_TYPE_ENUM_END =255 }
- enum STORAGE_USAGE_FLAG {
STORAGE_USAGE_FLAG_SET =1 , STORAGE_USAGE_FLAG_PHOTO =2 , STORAGE_USAGE_FLAG_VIDEO =4 , STORAGE_USAGE_FLAG_LOGS =8 ,
STORAGE_USAGE_FLAG_ENUM_END =9 }

- enum ORBIT_YAW_BEHAVIOUR {
ORBIT_YAW_BEHAVIOUR_HOLD_FRONT_TO_CIRCLE_CENTER =0 , ORBIT_YAW_BEHAVIOUR_HOLD_INITIAL_HEADING =1 , ORBIT_YAW_BEHAVIOUR_UNCONTROLLED =2 , ORBIT_YAW_BEHAVIOUR_HOLD_FRONT_TANGENT_TO_CIRCLE =3 ,
ORBIT_YAW_BEHAVIOUR_RC_CONTROLLED =4 , ORBIT_YAW_BEHAVIOUR_UNCHANGED =5 ,
ORBIT_YAW_BEHAVIOUR_ENUM_END =6 }
- enum WIFI_CONFIG_AP_RESPONSE {
WIFI_CONFIG_AP_RESPONSE_UNDEFINED =0 , WIFI_CONFIG_AP_RESPONSE_ACCEPTED =1 ,
WIFI_CONFIG_AP_RESPONSE_REJECTED =2 , WIFI_CONFIG_AP_RESPONSE_MODE_ERROR =3 ,
WIFI_CONFIG_AP_RESPONSE_SSID_ERROR =4 , WIFI_CONFIG_AP_RESPONSE_PASSWORD_ERROR =5 , WIFI_CONFIG_AP_RESPONSE_ENUM_END =6 }
- enum CELLULAR_CONFIG_RESPONSE {
CELLULAR_CONFIG_RESPONSE_ACCEPTED =0 , CELLULAR_CONFIG_RESPONSE_APN_ERROR =1 ,
CELLULAR_CONFIG_RESPONSE_PIN_ERROR =2 , CELLULAR_CONFIG_RESPONSE_REJECTED =3 ,
CELLULAR_CONFIG_BLOCKED_PUK_REQUIRED =4 , CELLULAR_CONFIG_RESPONSE_ENUM_END =5 }
- enum WIFI_CONFIG_AP_MODE {
WIFI_CONFIG_AP_MODE_UNDEFINED =0 , WIFI_CONFIG_AP_MODE_AP =1 , WIFI_CONFIG_AP_MODE_STATION =2 , WIFI_CONFIG_AP_MODE_DISABLED =3 ,
WIFI_CONFIG_AP_MODE_ENUM_END =4 }
- enum COMP_METADATA_TYPE {
COMP_METADATA_TYPE_GENERAL =0 , COMP_METADATA_TYPE_PARAMETER =1 , COMP_METADATA_TYPE_COMMANDS =2 , COMP_METADATA_TYPE_PERIPHERALS =3 ,
COMP_METADATA_TYPE_EVENTS =4 , COMP_METADATA_TYPE_ACTUATORS =5 , COMP_METADATA_TYPE_ENUM_END =6 }
- enum ACTUATOR_CONFIGURATION {
ACTUATOR_CONFIGURATION_NONE =0 , ACTUATOR_CONFIGURATION_BEEP =1 , ACTUATOR_CONFIGURATION_3D_MODE_ON =2 , ACTUATOR_CONFIGURATION_3D_MODE_OFF =3 ,
ACTUATOR_CONFIGURATION_SPIN_DIRECTION1 =4 , ACTUATOR_CONFIGURATION_SPIN_DIRECTION2 =5 , ACTUATOR_CONFIGURATION_ENUM_END =6 }
- enum ACTUATOR_OUTPUT_FUNCTION {
ACTUATOR_OUTPUT_FUNCTION_NONE =0 , ACTUATOR_OUTPUT_FUNCTION_MOTOR1 =1 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR2 =2 , ACTUATOR_OUTPUT_FUNCTION_MOTOR3 =3 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR4 =4 , ACTUATOR_OUTPUT_FUNCTION_MOTOR5 =5 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR6 =6 , ACTUATOR_OUTPUT_FUNCTION_MOTOR7 =7 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR8 =8 , ACTUATOR_OUTPUT_FUNCTION_MOTOR9 =9 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR10 =10 , ACTUATOR_OUTPUT_FUNCTION_MOTOR11 =11 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR12 =12 , ACTUATOR_OUTPUT_FUNCTION_MOTOR13 =13 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR14 =14 , ACTUATOR_OUTPUT_FUNCTION_MOTOR15 =15 ,
ACTUATOR_OUTPUT_FUNCTION_MOTOR16 =16 , ACTUATOR_OUTPUT_FUNCTION_SERVO1 =33 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO2 =34 , ACTUATOR_OUTPUT_FUNCTION_SERVO3 =35 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO4 =36 , ACTUATOR_OUTPUT_FUNCTION_SERVO5 =37 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO6 =38 , ACTUATOR_OUTPUT_FUNCTION_SERVO7 =39 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO8 =40 , ACTUATOR_OUTPUT_FUNCTION_SERVO9 =41 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO10 =42 , ACTUATOR_OUTPUT_FUNCTION_SERVO11 =43 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO12 =44 , ACTUATOR_OUTPUT_FUNCTION_SERVO13 =45 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO14 =46 , ACTUATOR_OUTPUT_FUNCTION_SERVO15 =47 ,
ACTUATOR_OUTPUT_FUNCTION_SERVO16 =48 , ACTUATOR_OUTPUT_FUNCTION_ENUM_END =49
}
- enum AUTOTUNE_AXIS { AUTOTUNE_AXIS_ROLL =1 , AUTOTUNE_AXIS_PITCH =2 , AUTOTUNE_AXIS_YAW =4 , AUTOTUNE_AXIS_ENUM_END =5 }
- enum PREFLIGHT_STORAGE_PARAMETER_ACTION {
PARAM_READ_PERSISTENT =0 , PARAM_WRITE_PERSISTENT =1 , PARAM_RESET_CONFIG_DEFAULT =2 , PARAM_RESET_SENSOR_DEFAULT =3 ,
PARAM_RESET_ALL_DEFAULT =4 , PREFLIGHT_STORAGE_PARAMETER_ACTION_ENUM_END =5 }
- enum PREFLIGHT_STORAGE_MISSION_ACTION { MISSION_READ_PERSISTENT =0 , MISSION_WRITE_PERSISTENT =1 , MISSION_RESET_DEFAULT =2 , PREFLIGHT_STORAGE_MISSION_ENUM_END =3 }


```

ACTION_ENUM_END =3 }
• enum REBOOT_SHUTDOWN_ACTION {
    REBOOT_SHUTDOWN_ACTION_NONE =0 , REBOOT_SHUTDOWN_ACTION_REBOOT =1 , REBOOT_↵
    _SHUTDOWN_ACTION_SHUTDOWN =2 , REBOOT_SHUTDOWN_ACTION_REBOOT_TO_BOOTLOADER
    =3 ,
    REBOOT_SHUTDOWN_ACTION_POWER_ON =4 , REBOOT_SHUTDOWN_ACTION_ENUM_END =5 }
• enum REBOOT_SHUTDOWN_CONDITIONS { REBOOT_SHUTDOWN_CONDITIONS_SAFETY_↵
    _INTERLOCKED =0 , REBOOT_SHUTDOWN_CONDITIONS_FORCE =20190226 , REBOOT_↵
    SHUTDOWN_CONDITIONS_ENUM_END =20190227 }
• enum MAV_CMD {
    MAV_CMD_NAV_WAYPOINT =16 , MAV_CMD_NAV_LOITER_UNLIM =17 , MAV_CMD_NAV_LOITER_↵
    TURNS =18 , MAV_CMD_NAV_LOITER_TIME =19 ,
    MAV_CMD_NAV_RETURN_TO_LAUNCH =20 , MAV_CMD_NAV_LAND =21 , MAV_CMD_NAV_↵
    TAKEOFF =22 , MAV_CMD_NAV_LAND_LOCAL =23 ,
    MAV_CMD_NAV_TAKEOFF_LOCAL =24 , MAV_CMD_NAV_FOLLOW =25 , MAV_CMD_NAV_↵
    CONTINUE_AND_CHANGE_ALT =30 , MAV_CMD_NAV_LOITER_TO_ALT =31 ,
    MAV_CMD_DO_FOLLOW =32 , MAV_CMD_DO_FOLLOW_REPOSITION =33 , MAV_CMD_DO_ORBIT
    =34 , MAV_CMD_DO_FIGURE_EIGHT =35 ,
    MAV_CMD_NAV_ROI =80 , MAV_CMD_NAV_PATHPLANNING =81 , MAV_CMD_NAV_SPLINE_↵
    WAYPOINT =82 , MAV_CMD_NAV_VTOL_TAKEOFF =84 ,
    MAV_CMD_NAV_VTOL_LAND =85 , MAV_CMD_NAV_GUIDED_ENABLE =92 , MAV_CMD_NAV_DELAY
    =93 , MAV_CMD_NAV_PAYLOAD_PLACE =94 ,
    MAV_CMD_NAV_LAST =95 , MAV_CMD_CONDITION_DELAY =112 , MAV_CMD_CONDITION_↵
    CHANGE_ALT =113 , MAV_CMD_CONDITION_DISTANCE =114 ,
    MAV_CMD_CONDITION_YAW =115 , MAV_CMD_CONDITION_LAST =159 , MAV_CMD_DO_SET_MODE
    =176 , MAV_CMD_DO_JUMP =177 ,
    MAV_CMD_DO_CHANGE_SPEED =178 , MAV_CMD_DO_SET_HOME =179 , MAV_CMD_DO_SET_↵
    PARAMETER =180 , MAV_CMD_DO_SET_RELAY =181 ,
    MAV_CMD_DO_REPEAT_RELAY =182 , MAV_CMD_DO_SET_SERVO =183 , MAV_CMD_DO_REPEAT_↵
    _SERVO =184 , MAV_CMD_DO_FLIGHTTERMINATION =185 ,
    MAV_CMD_DO_CHANGE_ALTITUDE =186 , MAV_CMD_DO_SET_ACTUATOR =187 , MAV_CMD_DO_↵
    _RETURN_PATH_START =188 , MAV_CMD_DO_LAND_START =189 ,
    MAV_CMD_DO_RALLY_LAND =190 , MAV_CMD_DO_GO_AROUND =191 , MAV_CMD_DO_↵
    REPOSITION =192 , MAV_CMD_DO_PAUSE_CONTINUE =193 ,
    MAV_CMD_DO_SET_REVERSE =194 , MAV_CMD_DO_SET_ROI_LOCATION =195 , MAV_CMD_DO_↵
    SET_ROI_WPNEXT_OFFSET =196 , MAV_CMD_DO_SET_ROI_NONE =197 ,
    MAV_CMD_DO_SET_ROI_SYSID =198 , MAV_CMD_DO_CONTROL_VIDEO =200 , MAV_CMD_DO_↵
    SET_ROI =201 , MAV_CMD_DO_DIGICAM_CONFIGURE =202 ,
    MAV_CMD_DO_DIGICAM_CONTROL =203 , MAV_CMD_DO_MOUNT_CONFIGURE =204 , MAV_CMD_↵
    _DO_MOUNT_CONTROL =205 , MAV_CMD_DO_SET_CAM_TRIGG_DIST =206 ,
    MAV_CMD_DO_FENCE_ENABLE =207 , MAV_CMD_DO_PARACHUTE =208 , MAV_CMD_DO_↵
    MOTOR_TEST =209 , MAV_CMD_DO_INVERTED_FLIGHT =210 ,
    MAV_CMD_DO_GRIPPER =211 , MAV_CMD_DO_AUTOTUNE_ENABLE =212 , MAV_CMD_NAV_SET_↵
    _YAW_SPEED =213 , MAV_CMD_DO_SET_CAM_TRIGG_INTERVAL =214 ,
    MAV_CMD_DO_MOUNT_CONTROL_QUAT =220 , MAV_CMD_DO_GUIDED_MASTER =221 , MAV_↵
    CMD_DO_GUIDED_LIMITS =222 , MAV_CMD_DO_ENGINE_CONTROL =223 ,
    MAV_CMD_DO_SET_MISSION_CURRENT =224 , MAV_CMD_DO_LAST =240 , MAV_CMD_↵
    PREFLIGHT_CALIBRATION =241 , MAV_CMD_PREFLIGHT_SET_SENSOR_OFFSETS =242 ,
    MAV_CMD_PREFLIGHT_UAVCAN =243 , MAV_CMD_PREFLIGHT_STORAGE =245 , MAV_CMD_↵
    PREFLIGHT_REBOOT_SHUTDOWN =246 , MAV_CMD_OVERRIDE_GOTO =252 ,
    MAV_CMD_OBLIQUE_SURVEY =260 , MAV_CMD_DO_SET_STANDARD_MODE =262 , MAV_CMD_↵
    MISSION_START =300 , MAV_CMD_ACTUATOR_TEST =310 ,
    MAV_CMD_CONFIGURE_ACTUATOR =311 , MAV_CMD_COMPONENT_ARM_DISARM =400 , MAV_↵
    CMD_RUN_PREARM_CHECKS =401 , MAV_CMD_ILLUMINATOR_ON_OFF =405 ,
    MAV_CMD_DO_ILLUMINATOR_CONFIGURE =406 , MAV_CMD_GET_HOME_POSITION =410 , MAV_↵
    CMD_INJECT_FAILURE =420 , MAV_CMD_START_RX_PAIR =500 ,
    MAV_CMD_GET_MESSAGE_INTERVAL =510 , MAV_CMD_SET_MESSAGE_INTERVAL =511 , MAV_↵
    CMD_REQUEST_MESSAGE =512 , MAV_CMD_REQUEST_PROTOCOL_VERSION =519 ,

```

```

MAV_CMD_REQUEST_AUTOPILOT_CAPABILITIES =520 , MAV_CMD_REQUEST_CAMERA_↵
INFORMATION =521 , MAV_CMD_REQUEST_CAMERA_SETTINGS =522 , MAV_CMD_REQUEST_↵
_STORAGE_INFORMATION =525 ,
MAV_CMD_STORAGE_FORMAT =526 , MAV_CMD_REQUEST_CAMERA_CAPTURE_STATUS =527 ,
MAV_CMD_REQUEST_FLIGHT_INFORMATION =528 , MAV_CMD_RESET_CAMERA_SETTINGS =529 ,
MAV_CMD_SET_CAMERA_MODE =530 , MAV_CMD_SET_CAMERA_ZOOM =531 , MAV_CMD_SET_↵
CAMERA_FOCUS =532 , MAV_CMD_SET_STORAGE_USAGE =533 ,
MAV_CMD_SET_CAMERA_SOURCE =534 , MAV_CMD_JUMP_TAG =600 , MAV_CMD_DO_JUMP_TAG
=601 , MAV_CMD_DO_GIMBAL_MANAGER_PITCHYAW =1000 ,
MAV_CMD_DO_GIMBAL_MANAGER_CONFIGURE =1001 , MAV_CMD_IMAGE_START_CAPTURE
=2000 , MAV_CMD_IMAGE_STOP_CAPTURE =2001 , MAV_CMD_REQUEST_CAMERA_IMAGE_↵
CAPTURE =2002 ,
MAV_CMD_DO_TRIGGER_CONTROL =2003 , MAV_CMD_CAMERA_TRACK_POINT =2004 , MAV_↵
CMD_CAMERA_TRACK_RECTANGLE =2005 , MAV_CMD_CAMERA_STOP_TRACKING =2010 ,
MAV_CMD_VIDEO_START_CAPTURE =2500 , MAV_CMD_VIDEO_STOP_CAPTURE =2501 , MAV_↵
CMD_VIDEO_START_STREAMING =2502 , MAV_CMD_VIDEO_STOP_STREAMING =2503 ,
MAV_CMD_REQUEST_VIDEO_STREAM_INFORMATION =2504 , MAV_CMD_REQUEST_VIDEO_↵
STREAM_STATUS =2505 , MAV_CMD_LOGGING_START =2510 , MAV_CMD_LOGGING_STOP =2511 ,
MAV_CMD_AIRFRAME_CONFIGURATION =2520 , MAV_CMD_CONTROL_HIGH_LATENCY =2600 ,
MAV_CMD_PANORAMA_CREATE =2800 , MAV_CMD_DO_VTOL_TRANSITION =3000 ,
MAV_CMD_ARM_AUTHORIZATION_REQUEST =3001 , MAV_CMD_SET_GUIDED_SUBMODE_↵
STANDARD =4000 , MAV_CMD_SET_GUIDED_SUBMODE_CIRCLE =4001 , MAV_CMD_CONDITION_↵
GATE =4501 ,
MAV_CMD_NAV_FENCE_RETURN_POINT =5000 , MAV_CMD_NAV_FENCE_POLYGON_VERTEX_↵
INCLUSION =5001 , MAV_CMD_NAV_FENCE_POLYGON_VERTEX_EXCLUSION =5002 , MAV_CMD_↵
NAV_FENCE_CIRCLE_INCLUSION =5003 ,
MAV_CMD_NAV_FENCE_CIRCLE_EXCLUSION =5004 , MAV_CMD_NAV_RALLY_POINT =5100 , MAV_↵
_CMD_UAVCAN_GET_NODE_INFO =5200 , MAV_CMD_DO_SET_SAFETY_SWITCH_STATE =5300 ,
MAV_CMD_DO_ADSB_OUT_IDENT =10001 , MAV_CMD_PAYLOAD_PREPARE_DEPLOY =30001 ,
MAV_CMD_PAYLOAD_CONTROL_DEPLOY =30002 , MAV_CMD_WAYPOINT_USER_1 =31000 ,
MAV_CMD_WAYPOINT_USER_2 =31001 , MAV_CMD_WAYPOINT_USER_3 =31002 , MAV_CMD_↵
WAYPOINT_USER_4 =31003 , MAV_CMD_WAYPOINT_USER_5 =31004 ,
MAV_CMD_SPATIAL_USER_1 =31005 , MAV_CMD_SPATIAL_USER_2 =31006 , MAV_CMD_SPATIAL_↵
_USER_3 =31007 , MAV_CMD_SPATIAL_USER_4 =31008 ,
MAV_CMD_SPATIAL_USER_5 =31009 , MAV_CMD_USER_1 =31010 , MAV_CMD_USER_2 =31011 ,
MAV_CMD_USER_3 =31012 ,
MAV_CMD_USER_4 =31013 , MAV_CMD_USER_5 =31014 , MAV_CMD_CAN_FORWARD =32000 ,
MAV_CMD_FIXED_MAG_CAL_YAW =42006 ,
MAV_CMD_DO_WINCH =42600 , MAV_CMD_EXTERNAL_POSITION_ESTIMATE =43003 , MAV_CMD_↵
_ENUM_END =43004 }
• enum MAV_DATA_STREAM {
MAV_DATA_STREAM_ALL =0 , MAV_DATA_STREAM_RAW_SENSORS =1 , MAV_DATA_STREAM_↵
EXTENDED_STATUS =2 , MAV_DATA_STREAM_RC_CHANNELS =3 ,
MAV_DATA_STREAM_RAW_CONTROLLER =4 , MAV_DATA_STREAM_POSITION =6 , MAV_DATA_↵
STREAM_EXTRA1 =10 , MAV_DATA_STREAM_EXTRA2 =11 ,
MAV_DATA_STREAM_EXTRA3 =12 , MAV_DATA_STREAM_ENUM_END =13 }
• enum MAV_ROI {
MAV_ROI_NONE =0 , MAV_ROI_WPNEXT =1 , MAV_ROI_WPINDEX =2 , MAV_ROI_LOCATION =3 ,
MAV_ROI_TARGET =4 , MAV_ROI_ENUM_END =5 }
• enum MAV_PARAM_TYPE {
MAV_PARAM_TYPE_UINT8 =1 , MAV_PARAM_TYPE_INT8 =2 , MAV_PARAM_TYPE_UINT16 =3 ,
MAV_PARAM_TYPE_INT16 =4 ,
MAV_PARAM_TYPE_UINT32 =5 , MAV_PARAM_TYPE_INT32 =6 , MAV_PARAM_TYPE_UINT64 =7 ,
MAV_PARAM_TYPE_INT64 =8 ,
MAV_PARAM_TYPE_REAL32 =9 , MAV_PARAM_TYPE_REAL64 =10 , MAV_PARAM_TYPE_ENUM_↵
END =11 }
• enum MAV_PARAM_ERROR {
MAV_PARAM_ERROR_NO_ERROR =0 , MAV_PARAM_ERROR_DOES_NOT_EXIST =1 , MAV_↵

```

```

PARAM_ERROR_VALUE_OUT_OF_RANGE =2 , MAV_PARAM_ERROR_PERMISSION_DENIED =3
,
MAV_PARAM_ERROR_COMPONENT_NOT_FOUND =4 , MAV_PARAM_ERROR_READ_ONLY =5 ,
MAV_PARAM_ERROR_TYPE_UNSUPPORTED =6 , MAV_PARAM_ERROR_TYPE_MISMATCH =7 ,
MAV_PARAM_ERROR_READ_FAIL =8 , MAV_PARAM_ERROR_ENUM_END =9 }
• enum MAV_PARAM_EXT_TYPE {
MAV_PARAM_EXT_TYPE_UINT8 =1 , MAV_PARAM_EXT_TYPE_INT8 =2 , MAV_PARAM_EXT_TYPE_↵
_UINT16 =3 , MAV_PARAM_EXT_TYPE_INT16 =4 ,
MAV_PARAM_EXT_TYPE_UINT32 =5 , MAV_PARAM_EXT_TYPE_INT32 =6 , MAV_PARAM_EXT_↵
_TYPE_UINT64 =7 , MAV_PARAM_EXT_TYPE_INT64 =8 ,
MAV_PARAM_EXT_TYPE_REAL32 =9 , MAV_PARAM_EXT_TYPE_REAL64 =10 , MAV_PARAM_EXT_↵
_TYPE_CUSTOM =11 , MAV_PARAM_EXT_TYPE_ENUM_END =12 }
• enum MAV_RESULT {
MAV_RESULT_ACCEPTED =0 , MAV_RESULT_TEMPORARILY_REJECTED =1 , MAV_RESULT_↵
DENIED =2 , MAV_RESULT_UNSUPPORTED =3 ,
MAV_RESULT_FAILED =4 , MAV_RESULT_IN_PROGRESS =5 , MAV_RESULT_CANCELLED =6 , MAV_↵
_RESULT_COMMAND_LONG_ONLY =7 ,
MAV_RESULT_COMMAND_INT_ONLY =8 , MAV_RESULT_COMMAND_UNSUPPORTED_MAV_FRAME
=9 , MAV_RESULT_NOT_IN_CONTROL =10 , MAV_RESULT_ENUM_END =11 }
• enum MAV_MISSION_RESULT {
MAV_MISSION_ACCEPTED =0 , MAV_MISSION_ERROR =1 , MAV_MISSION_UNSUPPORTED_FRAME
=2 , MAV_MISSION_UNSUPPORTED =3 ,
MAV_MISSION_NO_SPACE =4 , MAV_MISSION_INVALID =5 , MAV_MISSION_INVALID_PARAM1 =6 ,
MAV_MISSION_INVALID_PARAM2 =7 ,
MAV_MISSION_INVALID_PARAM3 =8 , MAV_MISSION_INVALID_PARAM4 =9 , MAV_MISSION_↵
INVALID_PARAM5_X =10 , MAV_MISSION_INVALID_PARAM6_Y =11 ,
MAV_MISSION_INVALID_PARAM7 =12 , MAV_MISSION_INVALID_SEQUENCE =13 , MAV_MISSION_↵
DENIED =14 , MAV_MISSION_OPERATION_CANCELLED =15 ,
MAV_MISSION_RESULT_ENUM_END =16 }
• enum MAV_SEVERITY {
MAV_SEVERITY_EMERGENCY =0 , MAV_SEVERITY_ALERT =1 , MAV_SEVERITY_CRITICAL =2 ,
MAV_SEVERITY_ERROR =3 ,
MAV_SEVERITY_WARNING =4 , MAV_SEVERITY_NOTICE =5 , MAV_SEVERITY_INFO =6 , MAV_↵
_SEVERITY_DEBUG =7 ,
MAV_SEVERITY_ENUM_END =8 }
• enum MAV_POWER_STATUS {
MAV_POWER_STATUS_BRICK_VALID =1 , MAV_POWER_STATUS_SERVO_VALID =2 , MAV_POWER_↵
_STATUS_USB_CONNECTED =4 , MAV_POWER_STATUS_PERIPH_OVERCURRENT =8 ,
MAV_POWER_STATUS_PERIPH_HIPOWER_OVERCURRENT =16 , MAV_POWER_STATUS_CHANGED
=32 , MAV_POWER_STATUS_ENUM_END =33 }
• enum SERIAL_CONTROL_DEV {
SERIAL_CONTROL_DEV_TELEM1 =0 , SERIAL_CONTROL_DEV_TELEM2 =1 , SERIAL_CONTROL_↵
_DEV_GPS1 =2 , SERIAL_CONTROL_DEV_GPS2 =3 ,
SERIAL_CONTROL_DEV_SHELL =10 , SERIAL_CONTROL_SERIAL0 =100 , SERIAL_CONTROL_↵
_SERIAL1 =101 , SERIAL_CONTROL_SERIAL2 =102 ,
SERIAL_CONTROL_SERIAL3 =103 , SERIAL_CONTROL_SERIAL4 =104 , SERIAL_CONTROL_↵
_SERIAL5 =105 , SERIAL_CONTROL_SERIAL6 =106 ,
SERIAL_CONTROL_SERIAL7 =107 , SERIAL_CONTROL_SERIAL8 =108 , SERIAL_CONTROL_↵
_SERIAL9 =109 , SERIAL_CONTROL_DEV_ENUM_END =110 }
• enum SERIAL_CONTROL_FLAG {
SERIAL_CONTROL_FLAG_REPLY =1 , SERIAL_CONTROL_FLAG_RESPOND =2 , SERIAL_↵
_CONTROL_FLAG_EXCLUSIVE =4 , SERIAL_CONTROL_FLAG_BLOCKING =8 ,
SERIAL_CONTROL_FLAG_MULTI =16 , SERIAL_CONTROL_FLAG_ENUM_END =17 }
• enum MAV_DISTANCE_SENSOR {
MAV_DISTANCE_SENSOR_LASER =0 , MAV_DISTANCE_SENSOR_ULTRASOUND =1 , MAV_↵
_DISTANCE_SENSOR_INFRARED =2 , MAV_DISTANCE_SENSOR_RADAR =3 ,
MAV_DISTANCE_SENSOR_UNKNOWN =4 , MAV_DISTANCE_SENSOR_ENUM_END =5 }

```

- enum MAV_SENSOR_ORIENTATION {
 MAV_SENSOR_ROTATION_NONE =0 , MAV_SENSOR_ROTATION_YAW_45 =1 , MAV_SENSOR_↵
 ROTATION_YAW_90 =2 , MAV_SENSOR_ROTATION_YAW_135 =3 ,
 MAV_SENSOR_ROTATION_YAW_180 =4 , MAV_SENSOR_ROTATION_YAW_225 =5 , MAV_SENSOR_↵
 ROTATION_YAW_270 =6 , MAV_SENSOR_ROTATION_YAW_315 =7 ,
 MAV_SENSOR_ROTATION_ROLL_180 =8 , MAV_SENSOR_ROTATION_ROLL_180_YAW_45 =9 , MAV_↵
 _SENSOR_ROTATION_ROLL_180_YAW_90 =10 , MAV_SENSOR_ROTATION_ROLL_180_YAW_135 =11
 ,
 MAV_SENSOR_ROTATION_PITCH_180 =12 , MAV_SENSOR_ROTATION_ROLL_180_YAW_225 =13 ,
 MAV_SENSOR_ROTATION_ROLL_180_YAW_270 =14 , MAV_SENSOR_ROTATION_ROLL_180_YAW_↵
 _315 =15 ,
 MAV_SENSOR_ROTATION_ROLL_90 =16 , MAV_SENSOR_ROTATION_ROLL_90_YAW_45 =17 , MAV_↵
 _SENSOR_ROTATION_ROLL_90_YAW_90 =18 , MAV_SENSOR_ROTATION_ROLL_90_YAW_135 =19 ,
 MAV_SENSOR_ROTATION_ROLL_270 =20 , MAV_SENSOR_ROTATION_ROLL_270_YAW_45 =21 ,
 MAV_SENSOR_ROTATION_ROLL_270_YAW_90 =22 , MAV_SENSOR_ROTATION_ROLL_270_YAW_↵
 135 =23 ,
 MAV_SENSOR_ROTATION_PITCH_90 =24 , MAV_SENSOR_ROTATION_PITCH_270 =25 , MAV_↵
 SENSOR_ROTATION_PITCH_180_YAW_90 =26 , MAV_SENSOR_ROTATION_PITCH_180_YAW_270 =27
 ,
 MAV_SENSOR_ROTATION_ROLL_90_PITCH_90 =28 , MAV_SENSOR_ROTATION_ROLL_180_PITCH_↵
 _90 =29 , MAV_SENSOR_ROTATION_ROLL_270_PITCH_90 =30 , MAV_SENSOR_ROTATION_ROLL_↵
 90_PITCH_180 =31 ,
 MAV_SENSOR_ROTATION_ROLL_270_PITCH_180 =32 , MAV_SENSOR_ROTATION_ROLL_90_↵
 PITCH_270 =33 , MAV_SENSOR_ROTATION_ROLL_180_PITCH_270 =34 , MAV_SENSOR_ROTATION_↵
 _ROLL_270_PITCH_270 =35 ,
 MAV_SENSOR_ROTATION_ROLL_90_PITCH_180_YAW_90 =36 , MAV_SENSOR_ROTATION_ROLL_↵
 90_YAW_270 =37 , MAV_SENSOR_ROTATION_ROLL_90_PITCH_68_YAW_293 =38 , MAV_SENSOR_↵
 ROTATION_PITCH_315 =39 ,
 MAV_SENSOR_ROTATION_ROLL_90_PITCH_315 =40 , MAV_SENSOR_ROTATION_CUSTOM =100 ,
 MAV_SENSOR_ORIENTATION_ENUM_END =101 }
- enum MAV_MISSION_TYPE {
 MAV_MISSION_TYPE_MISSION =0 , MAV_MISSION_TYPE_FENCE =1 , MAV_MISSION_TYPE_RALLY
 =2 , MAV_MISSION_TYPE_ALL =255 ,
 MAV_MISSION_TYPE_ENUM_END =256 }
- enum MAV_ESTIMATOR_TYPE {
 MAV_ESTIMATOR_TYPE_UNKNOWN =0 , MAV_ESTIMATOR_TYPE_NAIVE =1 , MAV_ESTIMATOR_↵
 TYPE_VISION =2 , MAV_ESTIMATOR_TYPE_VIO =3 ,
 MAV_ESTIMATOR_TYPE_GPS =4 , MAV_ESTIMATOR_TYPE_GPS_INS =5 , MAV_ESTIMATOR_TYPE_↵
 _MOCAP =6 , MAV_ESTIMATOR_TYPE_LIDAR =7 ,
 MAV_ESTIMATOR_TYPE_AUTOPILOT =8 , MAV_ESTIMATOR_TYPE_ENUM_END =9 }
- enum MAV_BATTERY_TYPE {
 MAV_BATTERY_TYPE_UNKNOWN =0 , MAV_BATTERY_TYPE_LIPO =1 , MAV_BATTERY_TYPE_LIFE
 =2 , MAV_BATTERY_TYPE_LION =3 ,
 MAV_BATTERY_TYPE_NIMH =4 , MAV_BATTERY_TYPE_ENUM_END =5 }
- enum MAV_BATTERY_FUNCTION {
 MAV_BATTERY_FUNCTION_UNKNOWN =0 , MAV_BATTERY_FUNCTION_ALL =1 , MAV_BATTERY_↵
 FUNCTION_PROPULSION =2 , MAV_BATTERY_FUNCTION_AVIONICS =3 ,
 MAV_BATTERY_FUNCTION_PAYLOAD =4 , MAV_BATTERY_FUNCTION_ENUM_END =5 }
- enum MAV_BATTERY_CHARGE_STATE {
 MAV_BATTERY_CHARGE_STATE_UNDEFINED =0 , MAV_BATTERY_CHARGE_STATE_OK =1 , MAV_↵
 _BATTERY_CHARGE_STATE_LOW =2 , MAV_BATTERY_CHARGE_STATE_CRITICAL =3 ,
 MAV_BATTERY_CHARGE_STATE_EMERGENCY =4 , MAV_BATTERY_CHARGE_STATE_FAILED =5 ,
 MAV_BATTERY_CHARGE_STATE_UNHEALTHY =6 , MAV_BATTERY_CHARGE_STATE_CHARGING =7
 ,
 MAV_BATTERY_CHARGE_STATE_ENUM_END =8 }
- enum MAV_BATTERY_MODE { MAV_BATTERY_MODE_UNKNOWN =0 , MAV_BATTERY_MODE_↵
 AUTO_DISCHARGING =1 , MAV_BATTERY_MODE_HOT_SWAP =2 , MAV_BATTERY_MODE_ENUM_↵
 _END =3 }

- enum MAV_BATTERY_FAULT {
 MAV_BATTERY_FAULT_DEEP_DISCHARGE =1 , MAV_BATTERY_FAULT_SPIKES =2 , MAV_BATTERY_FAULT_CELL_FAIL =4 , MAV_BATTERY_FAULT_OVER_CURRENT =8 ,
 MAV_BATTERY_FAULT_OVER_TEMPERATURE =16 , MAV_BATTERY_FAULT_UNDER_TEMPERATURE =32 , MAV_BATTERY_FAULT_INCOMPATIBLE_VOLTAGE =64 , MAV_BATTERY_FAULT_INCOMPATIBLE_FIRMWARE =128 ,
 BATTERY_FAULT_INCOMPATIBLE_CELLS_CONFIGURATION =256 , MAV_BATTERY_FAULT_ENUM_END =257 }
- enum MAV_FUEL_TYPE { MAV_FUEL_TYPE_UNKNOWN =0 , MAV_FUEL_TYPE_LIQUID =1 , MAV_FUEL_TYPE_GAS =2 , MAV_FUEL_TYPE_ENUM_END =3 }
- enum MAV_GENERATOR_STATUS_FLAG {
 MAV_GENERATOR_STATUS_FLAG_OFF =1 , MAV_GENERATOR_STATUS_FLAG_READY =2 , MAV_GENERATOR_STATUS_FLAG_GENERATING =4 , MAV_GENERATOR_STATUS_FLAG_CHARGING =8 ,
 MAV_GENERATOR_STATUS_FLAG_REDUCED_POWER =16 , MAV_GENERATOR_STATUS_FLAG_MAXPOWER =32 , MAV_GENERATOR_STATUS_FLAG_OVERTEMP_WARNING =64 , MAV_GENERATOR_STATUS_FLAG_OVERTEMP_FAULT =128 ,
 MAV_GENERATOR_STATUS_FLAG_ELECTRONICS_OVERTEMP_WARNING =256 , MAV_GENERATOR_STATUS_FLAG_ELECTRONICS_OVERTEMP_FAULT =512 , MAV_GENERATOR_STATUS_FLAG_ELECTRONICS_FAULT =1024 , MAV_GENERATOR_STATUS_FLAG_POWERSOURCE_FAULT =2048 ,
 MAV_GENERATOR_STATUS_FLAG_COMMUNICATION_WARNING =4096 , MAV_GENERATOR_STATUS_FLAG_COOLING_WARNING =8192 , MAV_GENERATOR_STATUS_FLAG_POWER_RAIL_FAULT =16384 ,
 MAV_GENERATOR_STATUS_FLAG_OVERCURRENT_FAULT =32768 , MAV_GENERATOR_STATUS_FLAG_BATTERY_OVERCHARGE_CURRENT_FAULT =65536 , MAV_GENERATOR_STATUS_FLAG_OVERVOLTAGE_FAULT =131072 ,
 MAV_GENERATOR_STATUS_FLAG_BATTERY_UNDERVOLT_FAULT =262144 , MAV_GENERATOR_STATUS_FLAG_START_INHIBITED =524288 ,
 MAV_GENERATOR_STATUS_FLAG_MAINTENANCE_REQUIRED =1048576 , MAV_GENERATOR_STATUS_FLAG_WARMING_UP =2097152 , MAV_GENERATOR_STATUS_FLAG_IDLE =4194304 ,
 MAV_GENERATOR_STATUS_FLAG_ENUM_END =4194305 }
- enum MAV_VTOL_STATE {
 MAV_VTOL_STATE_UNDEFINED =0 , MAV_VTOL_STATE_TRANSITION_TO_FW =1 , MAV_VTOL_STATE_TRANSITION_TO_MC =2 , MAV_VTOL_STATE_MC =3 ,
 MAV_VTOL_STATE_FW =4 , MAV_VTOL_STATE_ENUM_END =5 }
- enum MAV_LANDED_STATE {
 MAV_LANDED_STATE_UNDEFINED =0 , MAV_LANDED_STATE_ON_GROUND =1 , MAV_LANDED_STATE_IN_AIR =2 , MAV_LANDED_STATE_TAKEOFF =3 ,
 MAV_LANDED_STATE_LANDING =4 , MAV_LANDED_STATE_ENUM_END =5 }
- enum ADSB_ALTITUDE_TYPE { ADSB_ALTITUDE_TYPE_PRESSURE_QNH =0 , ADSB_ALTITUDE_TYPE_GEOMETRIC =1 , ADSB_ALTITUDE_TYPE_ENUM_END =2 }
- enum ADSB_EMITTER_TYPE {
 ADSB_EMITTER_TYPE_NO_INFO =0 , ADSB_EMITTER_TYPE_LIGHT =1 , ADSB_EMITTER_TYPE_SMALL =2 , ADSB_EMITTER_TYPE_LARGE =3 ,
 ADSB_EMITTER_TYPE_HIGH_VORTEX_LARGE =4 , ADSB_EMITTER_TYPE_HEAVY =5 , ADSB_EMITTER_TYPE_HIGHLY_MANUV =6 , ADSB_EMITTER_TYPE_ROTOCRAFT =7 ,
 ADSB_EMITTER_TYPE_UNASSIGNED =8 , ADSB_EMITTER_TYPE_GLIDER =9 , ADSB_EMITTER_TYPE_LIGHTER_AIR =10 , ADSB_EMITTER_TYPE_PARACHUTE =11 ,
 ADSB_EMITTER_TYPE_ULTRA_LIGHT =12 , ADSB_EMITTER_TYPE_UNASSIGNED2 =13 , ADSB_EMITTER_TYPE_UAV =14 , ADSB_EMITTER_TYPE_SPACE =15 ,
 ADSB_EMITTER_TYPE_UNASSIGNED3 =16 , ADSB_EMITTER_TYPE_EMERGENCY_SURFACE =17 , ADSB_EMITTER_TYPE_SERVICE_SURFACE =18 , ADSB_EMITTER_TYPE_POINT_OBSTACLE =19 ,
 ADSB_EMITTER_TYPE_ENUM_END =20 }
- enum ADSB_FLAGS {
 ADSB_FLAGS_VALID_COORDS =1 , ADSB_FLAGS_VALID_ALTITUDE =2 , ADSB_FLAGS_VALID_HEADING =4 , ADSB_FLAGS_VALID_VELOCITY =8 ,
 ADSB_FLAGS_VALID_CALLSIGN =16 , ADSB_FLAGS_VALID_SQUAWK =32 , ADSB_FLAGS_SIMULATED =64 , ADSB_FLAGS_VERTICAL_VELOCITY_VALID =128 ,
 ADSB_FLAGS_BARO_VALID =256 , ADSB_FLAGS_SOURCE_UAT =32768 , ADSB_FLAGS_ENUM_END =32769 }

- enum **MAV_DO_REPOSITION_FLAGS** { **MAV_DO_REPOSITION_FLAGS_CHANGE_MODE** =1 , **MAV_DO_REPOSITION_FLAGS_RELATIVE_YAW** =2 , **MAV_DO_REPOSITION_FLAGS_ENUM_END** =3 }
- enum **SPEED_TYPE** { **SPEED_TYPE_AIRSPPEED** =0 , **SPEED_TYPE_GROUNDSPEED** =1 , **SPEED_TYPE_CLIMB_SPEED** =2 , **SPEED_TYPE_DESCENT_SPEED** =3 , **SPEED_TYPE_ENUM_END** =4 }
- enum **ESTIMATOR_STATUS_FLAGS** { **ESTIMATOR_ATTITUDE** =1 , **ESTIMATOR_VELOCITY_HORIZ** =2 , **ESTIMATOR_VELOCITY_VERT** =4 , **ESTIMATOR_POS_HORIZ_REL** =8 , **ESTIMATOR_POS_HORIZ_ABS** =16 , **ESTIMATOR_POS_VERT_ABS** =32 , **ESTIMATOR_POS_VERT_AGL** =64 , **ESTIMATOR_CONST_POS_MODE** =128 , **ESTIMATOR_PRED_POS_HORIZ_REL** =256 , **ESTIMATOR_PRED_POS_HORIZ_ABS** =512 , **ESTIMATOR_GPS_GLITCH** =1024 , **ESTIMATOR_ACCEL_ERROR** =2048 , **ESTIMATOR_STATUS_FLAGS_ENUM_END** =2049 }
- enum **MOTOR_TEST_ORDER** { **MOTOR_TEST_ORDER_DEFAULT** =0 , **MOTOR_TEST_ORDER_SEQUENCE** =1 , **MOTOR_TEST_ORDER_BOARD** =2 , **MOTOR_TEST_ORDER_ENUM_END** =3 }
- enum **MOTOR_TEST_THROTTLE_TYPE** { **MOTOR_TEST_THROTTLE_PERCENT** =0 , **MOTOR_TEST_THROTTLE_PWM** =1 , **MOTOR_TEST_THROTTLE_PILOT** =2 , **MOTOR_TEST_COMPASS_CAL** =3 , **MOTOR_TEST_THROTTLE_TYPE_ENUM_END** =4 }
- enum **GPS_INPUT_IGNORE_FLAGS** { **GPS_INPUT_IGNORE_FLAG_ALT** =1 , **GPS_INPUT_IGNORE_FLAG_HDOP** =2 , **GPS_INPUT_IGNORE_FLAG_VDOP** =4 , **GPS_INPUT_IGNORE_FLAG_VEL_HORIZ** =8 , **GPS_INPUT_IGNORE_FLAG_VEL_VERT** =16 , **GPS_INPUT_IGNORE_FLAG_SPEED_ACCURACY** =32 , **GPS_INPUT_IGNORE_FLAG_HORIZONTAL_ACCURACY** =64 , **GPS_INPUT_IGNORE_FLAG_VERTICAL_ACCURACY** =128 , **GPS_INPUT_IGNORE_FLAGS_ENUM_END** =129 }
- enum **MAV_COLLISION_ACTION** { **MAV_COLLISION_ACTION_NONE** =0 , **MAV_COLLISION_ACTION_REPORT** =1 , **MAV_COLLISION_ACTION_ASCEND_OR_DESCEND** =2 , **MAV_COLLISION_ACTION_MOVE_HORIZONTALLY** =3 , **MAV_COLLISION_ACTION_MOVE_PERPENDICULAR** =4 , **MAV_COLLISION_ACTION_RTL** =5 , **MAV_COLLISION_ACTION_HOVER** =6 , **MAV_COLLISION_ACTION_ENUM_END** =7 }
- enum **MAV_COLLISION_THREAT_LEVEL** { **MAV_COLLISION_THREAT_LEVEL_NONE** =0 , **MAV_COLLISION_THREAT_LEVEL_LOW** =1 , **MAV_COLLISION_THREAT_LEVEL_HIGH** =2 , **MAV_COLLISION_THREAT_LEVEL_ENUM_END** =3 }
- enum **MAV_COLLISION_SRC** { **MAV_COLLISION_SRC_ADSB** =0 , **MAV_COLLISION_SRC_MAVLINK_GPS_GLOBAL_INT** =1 , **MAV_COLLISION_SRC_ENUM_END** =2 }
- enum **GPS_FIX_TYPE** { **GPS_FIX_TYPE_NO_GPS** =0 , **GPS_FIX_TYPE_NO_FIX** =1 , **GPS_FIX_TYPE_2D_FIX** =2 , **GPS_FIX_TYPE_3D_FIX** =3 , **GPS_FIX_TYPE_DGPS** =4 , **GPS_FIX_TYPE_RTK_FLOAT** =5 , **GPS_FIX_TYPE_RTK_FIXED** =6 , **GPS_FIX_TYPE_STATIC** =7 , **GPS_FIX_TYPE_PPP** =8 , **GPS_FIX_TYPE_ENUM_END** =9 }
- enum **RTK_BASELINE_COORDINATE_SYSTEM** { **RTK_BASELINE_COORDINATE_SYSTEM_ECEF** =0 , **RTK_BASELINE_COORDINATE_SYSTEM_NED** =1 , **RTK_BASELINE_COORDINATE_SYSTEM_ENUM_END** =2 }
- enum **LANDING_TARGET_TYPE** { **LANDING_TARGET_TYPE_LIGHT_BEACON** =0 , **LANDING_TARGET_TYPE_RADIO_BEACON** =1 , **LANDING_TARGET_TYPE_VISION_FIDUCIAL** =2 , **LANDING_TARGET_TYPE_VISION_OTHER** =3 , **LANDING_TARGET_TYPE_ENUM_END** =4 }
- enum **VTOL_TRANSITION_HEADING** { **VTOL_TRANSITION_HEADING_VEHICLE_DEFAULT** =0 , **VTOL_TRANSITION_HEADING_NEXT_WAYPOINT** =1 , **VTOL_TRANSITION_HEADING_TAKEOFF** =2 , **VTOL_TRANSITION_HEADING_SPECIFIED** =3 , **VTOL_TRANSITION_HEADING_ANY** =4 , **VTOL_TRANSITION_HEADING_ENUM_END** =5 }
- enum **CAMERA_CAP_FLAGS** { **CAMERA_CAP_FLAGS_CAPTURE_VIDEO** =1 , **CAMERA_CAP_FLAGS_CAPTURE_IMAGE** =2 ,

```

CAMERA_CAP_FLAGS_HAS_MODES =4 , CAMERA_CAP_FLAGS_CAN_CAPTURE_IMAGE_IN_↵
VIDEO_MODE =8 ,
CAMERA_CAP_FLAGS_CAN_CAPTURE_VIDEO_IN_IMAGE_MODE =16 , CAMERA_CAP_FLAGS_↵
HAS_IMAGE_SURVEY_MODE =32 , CAMERA_CAP_FLAGS_HAS_BASIC_ZOOM =64 , CAMERA_↵
CAP_FLAGS_HAS_BASIC_FOCUS =128 ,
CAMERA_CAP_FLAGS_HAS_VIDEO_STREAM =256 , CAMERA_CAP_FLAGS_HAS_TRACKING_↵
_POINT =512 , CAMERA_CAP_FLAGS_HAS_TRACKING_RECTANGLE =1024 , CAMERA_CAP_↵
FLAGS_HAS_TRACKING_GEO_STATUS =2048 ,
CAMERA_CAP_FLAGS_HAS_THERMAL_RANGE =4096 , CAMERA_CAP_FLAGS_HAS_MTI =8192 ,
CAMERA_CAP_FLAGS_ENUM_END =8193 }
• enum VIDEO_STREAM_STATUS_FLAGS { VIDEO_STREAM_STATUS_FLAGS_RUNNING =1 , VIDEO_↵
_STREAM_STATUS_FLAGS_THERMAL =2 , VIDEO_STREAM_STATUS_FLAGS_THERMAL_RANGE_↵
_ENABLED =4 , VIDEO_STREAM_STATUS_FLAGS_ENUM_END =5 }
• enum VIDEO_STREAM_TYPE {
VIDEO_STREAM_TYPE_RTSP =0 , VIDEO_STREAM_TYPE RTPUDP =1 , VIDEO_STREAM_TYPE_↵
TCP_MPEG =2 , VIDEO_STREAM_TYPE_MPEG_TS =3 ,
VIDEO_STREAM_TYPE_ENUM_END =4 }
• enum VIDEO_STREAM_ENCODING { VIDEO_STREAM_ENCODING_UNKNOWN =0 , VIDEO_↵
_STREAM_ENCODING_H264 =1 , VIDEO_STREAM_ENCODING_H265 =2 , VIDEO_STREAM_↵
ENCODING_ENUM_END =3 }
• enum CAMERA_TRACKING_STATUS_FLAGS {
CAMERA_TRACKING_STATUS_FLAGS_IDLE =0 , CAMERA_TRACKING_STATUS_FLAGS_ACTIVE =1
, CAMERA_TRACKING_STATUS_FLAGS_ERROR =2 , CAMERA_TRACKING_STATUS_FLAGS_MTI =4
,
CAMERA_TRACKING_STATUS_FLAGS_COASTING =8 , CAMERA_TRACKING_STATUS_FLAGS_↵
ENUM_END =9 }
• enum CAMERA_TRACKING_MODE { CAMERA_TRACKING_MODE_NONE =0 , CAMERA_↵
TRACKING_MODE_POINT =1 , CAMERA_TRACKING_MODE_RECTANGLE =2 , CAMERA_↵
TRACKING_MODE_ENUM_END =3 }
• enum CAMERA_TRACKING_TARGET_DATA { CAMERA_TRACKING_TARGET_DATA_EMBEDDED =1
, CAMERA_TRACKING_TARGET_DATA_RENDERED =2 , CAMERA_TRACKING_TARGET_DATA_IN_↵
STATUS =4 , CAMERA_TRACKING_TARGET_DATA_ENUM_END =5 }
• enum CAMERA_ZOOM_TYPE {
ZOOM_TYPE_STEP =0 , ZOOM_TYPE_CONTINUOUS =1 , ZOOM_TYPE_RANGE =2 , ZOOM_TYPE_↵
FOCAL_LENGTH =3 ,
ZOOM_TYPE_HORIZONTAL_FOV =4 , CAMERA_ZOOM_TYPE_ENUM_END =5 }
• enum SET_FOCUS_TYPE {
FOCUS_TYPE_STEP =0 , FOCUS_TYPE_CONTINUOUS =1 , FOCUS_TYPE_RANGE =2 , FOCUS_↵
TYPE_METERS =3 ,
FOCUS_TYPE_AUTO =4 , FOCUS_TYPE_AUTO_SINGLE =5 , FOCUS_TYPE_AUTO_CONTINUOUS =6 ,
SET_FOCUS_TYPE_ENUM_END =7 }
• enum CAMERA_SOURCE {
CAMERA_SOURCE_DEFAULT =0 , CAMERA_SOURCE_RGB =1 , CAMERA_SOURCE_IR =2 ,
CAMERA_SOURCE_NDVI =3 ,
CAMERA_SOURCE_ENUM_END =4 }
• enum PARAM_ACK {
PARAM_ACK_ACCEPTED =0 , PARAM_ACK_VALUE_UNSUPPORTED =1 , PARAM_ACK_FAILED =2 ,
PARAM_ACK_IN_PROGRESS =3 ,
PARAM_ACK_ENUM_END =4 }
• enum CAMERA_MODE { CAMERA_MODE_IMAGE =0 , CAMERA_MODE_VIDEO =1 , CAMERA_↵
MODE_IMAGE_SURVEY =2 , CAMERA_MODE_ENUM_END =3 }
• enum MAV_ARM_AUTH_DENIED_REASON {
MAV_ARM_AUTH_DENIED_REASON_GENERIC =0 , MAV_ARM_AUTH_DENIED_REASON_NONE
=1 , MAV_ARM_AUTH_DENIED_REASON_INVALID_WAYPOINT =2 , MAV_ARM_AUTH_DENIED_↵
REASON_TIMEOUT =3 ,
MAV_ARM_AUTH_DENIED_REASON_AIRSPACE_IN_USE =4 , MAV_ARM_AUTH_DENIED_REASON_↵
_BAD_WEATHER =5 , MAV_ARM_AUTH_DENIED_REASON_ENUM_END =6 }
• enum RC_TYPE { RC_TYPE_SPEKTRUM =0 , RC_TYPE_CRSF =1 , RC_TYPE_ENUM_END =2 }

```

- enum RC_SUB_TYPE { RC_SUB_TYPE_SPEKTRUM_DSM2 =0 , RC_SUB_TYPE_SPEKTRUM_DSMX =1 , RC_SUB_TYPE_SPEKTRUM_DSMX8 =2 , RC_SUB_TYPE_ENUM_END =3 }
- enum ENGINE_CONTROL_OPTIONS { ENGINE_CONTROL_OPTIONS_ALLOW_START_WHILE_↵
DISARMED =1 , ENGINE_CONTROL_OPTIONS_ENUM_END =2 }
- enum POSITION_TARGET_TPEMASK {
POSITION_TARGET_TPEMASK_X_IGNORE =1 , POSITION_TARGET_TPEMASK_Y_IGNORE =2 ,
POSITION_TARGET_TPEMASK_Z_IGNORE =4 , POSITION_TARGET_TPEMASK_VX_IGNORE =8 ,
POSITION_TARGET_TPEMASK_VY_IGNORE =16 , POSITION_TARGET_TPEMASK_VZ_IGNORE
=32 , POSITION_TARGET_TPEMASK_AX_IGNORE =64 , POSITION_TARGET_TPEMASK_AY_↵
IGNORE =128 ,
POSITION_TARGET_TPEMASK_AZ_IGNORE =256 , POSITION_TARGET_TPEMASK_FORCE_SET
=512 , POSITION_TARGET_TPEMASK_YAW_IGNORE =1024 , POSITION_TARGET_TPEMASK_↵
YAW_RATE_IGNORE =2048 ,
POSITION_TARGET_TPEMASK_ENUM_END =2049 }
- enum ATTITUDE_TARGET_TPEMASK {
ATTITUDE_TARGET_TPEMASK_BODY_ROLL_RATE_IGNORE =1 , ATTITUDE_TARGET_TPEMASK_↵
_BODY_PITCH_RATE_IGNORE =2 , ATTITUDE_TARGET_TPEMASK_BODY_YAW_RATE_IGNORE =4
 , ATTITUDE_TARGET_TPEMASK_THRUST_BODY_SET =32 ,
ATTITUDE_TARGET_TPEMASK_THROTTLE_IGNORE =64 , ATTITUDE_TARGET_TPEMASK_↵
ATTITUDE_IGNORE =128 , ATTITUDE_TARGET_TPEMASK_ENUM_END =129 }
- enum UTM_FLIGHT_STATE {
UTM_FLIGHT_STATE_UNKNOWN =1 , UTM_FLIGHT_STATE_GROUND =2 , UTM_FLIGHT_STATE_↵
AIRBORNE =3 , UTM_FLIGHT_STATE_EMERGENCY =16 ,
UTM_FLIGHT_STATE_NOCTRL =32 , UTM_FLIGHT_STATE_ENUM_END =33 }
- enum UTM_DATA_AVAIL_FLAGS {
UTM_DATA_AVAIL_FLAGS_TIME_VALID =1 , UTM_DATA_AVAIL_FLAGS_UAS_ID_AVAILABLE =2 ,
UTM_DATA_AVAIL_FLAGS_POSITION_AVAILABLE =4 , UTM_DATA_AVAIL_FLAGS_ALTITUDE_↵
AVAILABLE =8 ,
UTM_DATA_AVAIL_FLAGS_RELATIVE_ALTITUDE_AVAILABLE =16 , UTM_DATA_AVAIL_FLAGS_↵
HORIZONTAL_VELO_AVAILABLE =32 , UTM_DATA_AVAIL_FLAGS_VERTICAL_VELO_AVAILABLE
=64 , UTM_DATA_AVAIL_FLAGS_NEXT_WAYPOINT_AVAILABLE =128 ,
UTM_DATA_AVAIL_FLAGS_ENUM_END =129 }
- enum CELLULAR_STATUS_FLAG {
CELLULAR_STATUS_FLAG_UNKNOWN =0 , CELLULAR_STATUS_FLAG_FAILED =1 , CELLULAR_↵
STATUS_FLAG_INITIALIZING =2 , CELLULAR_STATUS_FLAG_LOCKED =3 ,
CELLULAR_STATUS_FLAG_DISABLED =4 , CELLULAR_STATUS_FLAG_DISABLING =5 , CELLULAR_↵
_STATUS_FLAG_ENABLING =6 , CELLULAR_STATUS_FLAG_ENABLED =7 ,
CELLULAR_STATUS_FLAG_SEARCHING =8 , CELLULAR_STATUS_FLAG_REGISTERED =9 ,
CELLULAR_STATUS_FLAG_DISCONNECTING =10 , CELLULAR_STATUS_FLAG_CONNECTING =11 ,
CELLULAR_STATUS_FLAG_CONNECTED =12 , CELLULAR_STATUS_FLAG_ENUM_END =13 }
- enum CELLULAR_NETWORK_FAILED_REASON {
CELLULAR_NETWORK_FAILED_REASON_NONE =0 , CELLULAR_NETWORK_FAILED_REASON_↵
_UNKNOWN =1 , CELLULAR_NETWORK_FAILED_REASON_SIM_MISSING =2 , CELLULAR_↵
NETWORK_FAILED_REASON_SIM_ERROR =3 ,
CELLULAR_NETWORK_FAILED_REASON_ENUM_END =4 }
- enum CELLULAR_NETWORK_RADIO_TYPE {
CELLULAR_NETWORK_RADIO_TYPE_NONE =0 , CELLULAR_NETWORK_RADIO_TYPE_GSM =1 ,
CELLULAR_NETWORK_RADIO_TYPE_CDMA =2 , CELLULAR_NETWORK_RADIO_TYPE_WCDMA =3
 ,
CELLULAR_NETWORK_RADIO_TYPE_LTE =4 , CELLULAR_NETWORK_RADIO_TYPE_ENUM_END
=5 }
- enum PRECISION_LAND_MODE { PRECISION_LAND_MODE_DISABLED =0 , PRECISION_LAND_↵
MODE_OPPORTUNISTIC =1 , PRECISION_LAND_MODE_REQUIRED =2 , PRECISION_LAND_MODE_↵
_ENUM_END =3 }
- enum PARACHUTE_ACTION { PARACHUTE_DISABLE =0 , PARACHUTE_ENABLE =1 , PARACHUTE_↵
_RELEASE =2 , PARACHUTE_ACTION_ENUM_END =3 }

- enum MAV_TUNNEL_PAYLOAD_TYPE {
 MAV_TUNNEL_PAYLOAD_TYPE_UNKNOWN =0 , MAV_TUNNEL_PAYLOAD_TYPE_STORM32_↵
 RESERVED0 =200 , MAV_TUNNEL_PAYLOAD_TYPE_STORM32_RESERVED1 =201 , MAV_TUNNEL_↵
 _PAYLOAD_TYPE_STORM32_RESERVED2 =202 ,
 MAV_TUNNEL_PAYLOAD_TYPE_STORM32_RESERVED3 =203 , MAV_TUNNEL_PAYLOAD_TYPE_↵
 STORM32_RESERVED4 =204 , MAV_TUNNEL_PAYLOAD_TYPE_STORM32_RESERVED5 =205 , MAV_↵
 _TUNNEL_PAYLOAD_TYPE_STORM32_RESERVED6 =206 ,
 MAV_TUNNEL_PAYLOAD_TYPE_STORM32_RESERVED7 =207 , MAV_TUNNEL_PAYLOAD_TYPE_↵
 STORM32_RESERVED8 =208 , MAV_TUNNEL_PAYLOAD_TYPE_STORM32_RESERVED9 =209 , MAV_↵
 _TUNNEL_PAYLOAD_TYPE_MODALAI_REMOTE_OSD =210 ,
 MAV_TUNNEL_PAYLOAD_TYPE_MODALAI_ESC_UART_PASSTHRU =211 , MAV_TUNNEL_↵
 PAYLOAD_TYPE_MODALAI_IO_UART_PASSTHRU =212 , MAV_TUNNEL_PAYLOAD_TYPE_ENUM_↵
 END =213 }
- enum MAV_ODID_ID_TYPE {
 MAV_ODID_ID_TYPE_NONE =0 , MAV_ODID_ID_TYPE_SERIAL_NUMBER =1 , MAV_ODID_ID_TYPE_↵
 _CAA_REGISTRATION_ID =2 , MAV_ODID_ID_TYPE_UTM_ASSIGNED_UUID =3 ,
 MAV_ODID_ID_TYPE_SPECIFIC_SESSION_ID =4 , MAV_ODID_ID_TYPE_ENUM_END =5 }
- enum MAV_ODID_UA_TYPE {
 MAV_ODID_UA_TYPE_NONE =0 , MAV_ODID_UA_TYPE_AEROPLANE =1 , MAV_ODID_UA_TYPE_↵
 HELICOPTER_OR_MULTIROTOR =2 , MAV_ODID_UA_TYPE_GYROPLANE =3 ,
 MAV_ODID_UA_TYPE_HYBRID_LIFT =4 , MAV_ODID_UA_TYPE_ORNITHOPTER =5 , MAV_ODID_↵
 UA_TYPE_GLIDER =6 , MAV_ODID_UA_TYPE_KITE =7 ,
 MAV_ODID_UA_TYPE_FREE_BALLOON =8 , MAV_ODID_UA_TYPE_CAPTIVE_BALLOON =9 , MAV_↵
 ODID_UA_TYPE_AIRSHIP =10 , MAV_ODID_UA_TYPE_FREE_FALL_PARACHUTE =11 ,
 MAV_ODID_UA_TYPE_ROCKET =12 , MAV_ODID_UA_TYPE_TETHERED_POWERED_AIRCRAFT =13
 , MAV_ODID_UA_TYPE_GROUND_OBSTACLE =14 , MAV_ODID_UA_TYPE_OTHER =15 ,
 MAV_ODID_UA_TYPE_ENUM_END =16 }
- enum MAV_ODID_STATUS {
 MAV_ODID_STATUS_UNDECLARED =0 , MAV_ODID_STATUS_GROUND =1 , MAV_ODID_STATUS_↵
 AIRBORNE =2 , MAV_ODID_STATUS_EMERGENCY =3 ,
 MAV_ODID_STATUS_REMOTE_ID_SYSTEM_FAILURE =4 , MAV_ODID_STATUS_ENUM_END =5 }
- enum MAV_ODID_HEIGHT_REF { MAV_ODID_HEIGHT_REF_OVER_TAKEOFF =0 , MAV_ODID_↵
 HEIGHT_REF_OVER_GROUND =1 , MAV_ODID_HEIGHT_REF_ENUM_END =2 }
- enum MAV_ODID_HOR_ACC {
 MAV_ODID_HOR_ACC_UNKNOWN =0 , MAV_ODID_HOR_ACC_10NM =1 , MAV_ODID_HOR_ACC_4_↵
 NM =2 , MAV_ODID_HOR_ACC_2NM =3 ,
 MAV_ODID_HOR_ACC_1NM =4 , MAV_ODID_HOR_ACC_0_5NM =5 , MAV_ODID_HOR_ACC_0_3NM =6
 , MAV_ODID_HOR_ACC_0_1NM =7 ,
 MAV_ODID_HOR_ACC_0_05NM =8 , MAV_ODID_HOR_ACC_30_METER =9 , MAV_ODID_HOR_ACC_↵
 _10_METER =10 , MAV_ODID_HOR_ACC_3_METER =11 ,
 MAV_ODID_HOR_ACC_1_METER =12 , MAV_ODID_HOR_ACC_ENUM_END =13 }
- enum MAV_ODID_VER_ACC {
 MAV_ODID_VER_ACC_UNKNOWN =0 , MAV_ODID_VER_ACC_150_METER =1 , MAV_ODID_VER_↵
 ACC_45_METER =2 , MAV_ODID_VER_ACC_25_METER =3 ,
 MAV_ODID_VER_ACC_10_METER =4 , MAV_ODID_VER_ACC_3_METER =5 , MAV_ODID_VER_ACC_↵
 _1_METER =6 , MAV_ODID_VER_ACC_ENUM_END =7 }
- enum MAV_ODID_SPEED_ACC {
 MAV_ODID_SPEED_ACC_UNKNOWN =0 , MAV_ODID_SPEED_ACC_10_METERS_PER_SECOND =1
 , MAV_ODID_SPEED_ACC_3_METERS_PER_SECOND =2 , MAV_ODID_SPEED_ACC_1_METERS_↵
 PER_SECOND =3 ,
 MAV_ODID_SPEED_ACC_0_3_METERS_PER_SECOND =4 , MAV_ODID_SPEED_ACC_ENUM_END =5
 }
- enum MAV_ODID_TIME_ACC {
 MAV_ODID_TIME_ACC_UNKNOWN =0 , MAV_ODID_TIME_ACC_0_1_SECOND =1 , MAV_ODID_↵
 TIME_ACC_0_2_SECOND =2 , MAV_ODID_TIME_ACC_0_3_SECOND =3 ,
 MAV_ODID_TIME_ACC_0_4_SECOND =4 , MAV_ODID_TIME_ACC_0_5_SECOND =5 , MAV_ODID_↵
 TIME_ACC_0_6_SECOND =6 , MAV_ODID_TIME_ACC_0_7_SECOND =7 ,

```

MAV_ODID_TIME_ACC_0_8_SECOND =8 , MAV_ODID_TIME_ACC_0_9_SECOND =9 , MAV_ODID_↵
TIME_ACC_1_0_SECOND =10 , MAV_ODID_TIME_ACC_1_1_SECOND =11 ,
MAV_ODID_TIME_ACC_1_2_SECOND =12 , MAV_ODID_TIME_ACC_1_3_SECOND =13 , MAV_ODID_↵
TIME_ACC_1_4_SECOND =14 , MAV_ODID_TIME_ACC_1_5_SECOND =15 ,
MAV_ODID_TIME_ACC_ENUM_END =16 }
• enum MAV_ODID_AUTH_TYPE {
MAV_ODID_AUTH_TYPE_NONE =0 , MAV_ODID_AUTH_TYPE_UAS_ID_SIGNATURE =1 , MAV_↵
ODID_AUTH_TYPE_OPERATOR_ID_SIGNATURE =2 , MAV_ODID_AUTH_TYPE_MESSAGE_SET_↵
SIGNATURE =3 ,
MAV_ODID_AUTH_TYPE_NETWORK_REMOTE_ID =4 , MAV_ODID_AUTH_TYPE_SPECIFIC_↵
AUTHENTICATION =5 , MAV_ODID_AUTH_TYPE_ENUM_END =6 }
• enum MAV_ODID_DESC_TYPE { MAV_ODID_DESC_TYPE_TEXT =0 , MAV_ODID_DESC_TYPE_↵
EMERGENCY =1 , MAV_ODID_DESC_TYPE_EXTENDED_STATUS =2 , MAV_ODID_DESC_TYPE_↵
ENUM_END =3 }
• enum MAV_ODID_OPERATOR_LOCATION_TYPE { MAV_ODID_OPERATOR_LOCATION_TYPE_↵
TAKEOFF =0 , MAV_ODID_OPERATOR_LOCATION_TYPE_LIVE_GNSS =1 , MAV_ODID_OPERATOR_↵
LOCATION_TYPE_FIXED =2 , MAV_ODID_OPERATOR_LOCATION_TYPE_ENUM_END =3 }
• enum MAV_ODID_CLASSIFICATION_TYPE { MAV_ODID_CLASSIFICATION_TYPE_UNDECLARED =0 ,
MAV_ODID_CLASSIFICATION_TYPE_EU =1 , MAV_ODID_CLASSIFICATION_TYPE_ENUM_END =2 }
• enum MAV_ODID_CATEGORY_EU {
MAV_ODID_CATEGORY_EU_UNDECLARED =0 , MAV_ODID_CATEGORY_EU_OPEN =1 , MAV_ODID_↵
CATEGORY_EU_SPECIFIC =2 , MAV_ODID_CATEGORY_EU_CERTIFIED =3 ,
MAV_ODID_CATEGORY_EU_ENUM_END =4 }
• enum MAV_ODID_CLASS_EU {
MAV_ODID_CLASS_EU_UNDECLARED =0 , MAV_ODID_CLASS_EU_CLASS_0 =1 , MAV_ODID_↵
CLASS_EU_CLASS_1 =2 , MAV_ODID_CLASS_EU_CLASS_2 =3 ,
MAV_ODID_CLASS_EU_CLASS_3 =4 , MAV_ODID_CLASS_EU_CLASS_4 =5 , MAV_ODID_CLASS_↵
EU_CLASS_5 =6 , MAV_ODID_CLASS_EU_CLASS_6 =7 ,
MAV_ODID_CLASS_EU_ENUM_END =8 }
• enum MAV_ODID_OPERATOR_ID_TYPE { MAV_ODID_OPERATOR_ID_TYPE_CAA =0 , MAV_ODID_↵
OPERATOR_ID_TYPE_ENUM_END =1 }
• enum MAV_ODID_ARM_STATUS { MAV_ODID_ARM_STATUS_GOOD_TO_ARM =0 , MAV_ODID_↵
ARM_STATUS_PRE_ARM_FAIL_GENERIC =1 , MAV_ODID_ARM_STATUS_ENUM_END =2 }
• enum TUNE_FORMAT { TUNE_FORMAT_QBASIC1_1 =1 , TUNE_FORMAT_MML_MODERN =2 ,
TUNE_FORMAT_ENUM_END =3 }
• enum AIS_TYPE {
AIS_TYPE_UNKNOWN =0 , AIS_TYPE_RESERVED_1 =1 , AIS_TYPE_RESERVED_2 =2 , AIS_TYPE_↵
RESERVED_3 =3 ,
AIS_TYPE_RESERVED_4 =4 , AIS_TYPE_RESERVED_5 =5 , AIS_TYPE_RESERVED_6 =6 , AIS_↵
TYPE_RESERVED_7 =7 ,
AIS_TYPE_RESERVED_8 =8 , AIS_TYPE_RESERVED_9 =9 , AIS_TYPE_RESERVED_10 =10 , AIS_↵
TYPE_RESERVED_11 =11 ,
AIS_TYPE_RESERVED_12 =12 , AIS_TYPE_RESERVED_13 =13 , AIS_TYPE_RESERVED_14 =14 ,
AIS_TYPE_RESERVED_15 =15 ,
AIS_TYPE_RESERVED_16 =16 , AIS_TYPE_RESERVED_17 =17 , AIS_TYPE_RESERVED_18 =18 ,
AIS_TYPE_RESERVED_19 =19 ,
AIS_TYPE_WIG =20 , AIS_TYPE_WIG_HAZARDOUS_A =21 , AIS_TYPE_WIG_HAZARDOUS_B =22 ,
AIS_TYPE_WIG_HAZARDOUS_C =23 ,
AIS_TYPE_WIG_HAZARDOUS_D =24 , AIS_TYPE_WIG_RESERVED_1 =25 , AIS_TYPE_WIG_↵
RESERVED_2 =26 , AIS_TYPE_WIG_RESERVED_3 =27 ,
AIS_TYPE_WIG_RESERVED_4 =28 , AIS_TYPE_WIG_RESERVED_5 =29 , AIS_TYPE_FISHING =30 ,
AIS_TYPE_TOWING =31 ,
AIS_TYPE_TOWING_LARGE =32 , AIS_TYPE_DREDGING =33 , AIS_TYPE_DIVING =34 , AIS_TYPE_↵
MILITARY =35 ,
AIS_TYPE_SAILING =36 , AIS_TYPE_PLEASURE =37 , AIS_TYPE_RESERVED_20 =38 , AIS_TYPE_↵
RESERVED_21 =39 ,
AIS_TYPE_HSC =40 , AIS_TYPE_HSC_HAZARDOUS_A =41 , AIS_TYPE_HSC_HAZARDOUS_B =42 ,
AIS_TYPE_HSC_HAZARDOUS_C =43 ,

```

```

AIS_TYPE_HSC_HAZARDOUS_D =44 , AIS_TYPE_HSC_RESERVED_1 =45 , AIS_TYPE_HSC_↵
RESERVED_2 =46 , AIS_TYPE_HSC_RESERVED_3 =47 ,
AIS_TYPE_HSC_RESERVED_4 =48 , AIS_TYPE_HSC_UNKNOWN =49 , AIS_TYPE_PILOT =50 , AIS_↵
_TYPE_SAR =51 ,
AIS_TYPE_TUG =52 , AIS_TYPE_PORT_TENDER =53 , AIS_TYPE_ANTI_POLLUTION =54 , AIS_↵
TYPE_LAW_ENFORCEMENT =55 ,
AIS_TYPE_SPARE_LOCAL_1 =56 , AIS_TYPE_SPARE_LOCAL_2 =57 , AIS_TYPE_MEDICAL_↵
TRANSPORT =58 , AIS_TYPE_NONECOMBATANT =59 ,
AIS_TYPE_PASSENGER =60 , AIS_TYPE_PASSENGER_HAZARDOUS_A =61 , AIS_TYPE_↵
PASSENGER_HAZARDOUS_B =62 , AIS_TYPE_PASSENGER_HAZARDOUS_C =63 ,
AIS_TYPE_PASSENGER_HAZARDOUS_D =64 , AIS_TYPE_PASSENGER_RESERVED_1 =65 , AIS_↵
TYPE_PASSENGER_RESERVED_2 =66 , AIS_TYPE_PASSENGER_RESERVED_3 =67 ,
AIS_TYPE_PASSENGER_RESERVED_4 =68 , AIS_TYPE_PASSENGER_UNKNOWN =69 , AIS_TYPE_↵
_CARGO =70 , AIS_TYPE_CARGO_HAZARDOUS_A =71 ,
AIS_TYPE_CARGO_HAZARDOUS_B =72 , AIS_TYPE_CARGO_HAZARDOUS_C =73 , AIS_TYPE_↵
CARGO_HAZARDOUS_D =74 , AIS_TYPE_CARGO_RESERVED_1 =75 ,
AIS_TYPE_CARGO_RESERVED_2 =76 , AIS_TYPE_CARGO_RESERVED_3 =77 , AIS_TYPE_↵
CARGO_RESERVED_4 =78 , AIS_TYPE_CARGO_UNKNOWN =79 ,
AIS_TYPE_TANKER =80 , AIS_TYPE_TANKER_HAZARDOUS_A =81 , AIS_TYPE_TANKER_↵
HAZARDOUS_B =82 , AIS_TYPE_TANKER_HAZARDOUS_C =83 ,
AIS_TYPE_TANKER_HAZARDOUS_D =84 , AIS_TYPE_TANKER_RESERVED_1 =85 , AIS_TYPE_↵
TANKER_RESERVED_2 =86 , AIS_TYPE_TANKER_RESERVED_3 =87 ,
AIS_TYPE_TANKER_RESERVED_4 =88 , AIS_TYPE_TANKER_UNKNOWN =89 , AIS_TYPE_OTHER
=90 , AIS_TYPE_OTHER_HAZARDOUS_A =91 ,
AIS_TYPE_OTHER_HAZARDOUS_B =92 , AIS_TYPE_OTHER_HAZARDOUS_C =93 , AIS_TYPE_↵
OTHER_HAZARDOUS_D =94 , AIS_TYPE_OTHER_RESERVED_1 =95 ,
AIS_TYPE_OTHER_RESERVED_2 =96 , AIS_TYPE_OTHER_RESERVED_3 =97 , AIS_TYPE_OTHER_↵
_RESERVED_4 =98 , AIS_TYPE_OTHER_UNKNOWN =99 ,
AIS_TYPE_ENUM_END =100 }
• enum AIS_NAV_STATUS {
AIS_NAV_STATUS_UNDER_WAY =0 , AIS_NAV_STATUS_ANCHORED =1 , AIS_NAV_STATUS_UN_↵
COMMANDED =2 , AIS_NAV_STATUS_RESTRICTED_MANOEUVERABILITY =3 ,
AIS_NAV_STATUS_DRAUGHT_CONSTRAINED =4 , AIS_NAV_STATUS_MOORED =5 , AIS_NAV_↵
STATUS_AGROUND =6 , AIS_NAV_STATUS_FISHING =7 ,
AIS_NAV_STATUS_SAILING =8 , AIS_NAV_STATUS_RESERVED_HSC =9 , AIS_NAV_STATUS_↵
RESERVED_WIG =10 , AIS_NAV_STATUS_RESERVED_1 =11 ,
AIS_NAV_STATUS_RESERVED_2 =12 , AIS_NAV_STATUS_RESERVED_3 =13 , AIS_NAV_STATUS_↵
AIS_SART =14 , AIS_NAV_STATUS_UNKNOWN =15 ,
AIS_NAV_STATUS_ENUM_END =16 }
• enum AIS_FLAGS {
AIS_FLAGS_POSITION_ACCURACY =1 , AIS_FLAGS_VALID_COG =2 , AIS_FLAGS_VALID_VELOCITY
=4 , AIS_FLAGS_HIGH_VELOCITY =8 ,
AIS_FLAGS_VALID_TURN_RATE =16 , AIS_FLAGS_TURN_RATE_SIGN_ONLY =32 , AIS_FLAGS_↵
VALID_DIMENSIONS =64 , AIS_FLAGS_LARGE_BOW_DIMENSION =128 ,
AIS_FLAGS_LARGE_STERN_DIMENSION =256 , AIS_FLAGS_LARGE_PORT_DIMENSION =512 ,
AIS_FLAGS_LARGE_STARBOARD_DIMENSION =1024 , AIS_FLAGS_VALID_CALLSIGN =2048 ,
AIS_FLAGS_VALID_NAME =4096 , AIS_FLAGS_ENUM_END =4097 }
• enum FAILURE_UNIT {
FAILURE_UNIT_SENSOR_GYRO =0 , FAILURE_UNIT_SENSOR_ACCEL =1 , FAILURE_UNIT_↵
SENSOR_MAG =2 , FAILURE_UNIT_SENSOR_BARO =3 ,
FAILURE_UNIT_SENSOR_GPS =4 , FAILURE_UNIT_SENSOR_OPTICAL_FLOW =5 , FAILURE_UNIT_↵
_SENSOR_VIO =6 , FAILURE_UNIT_SENSOR_DISTANCE_SENSOR =7 ,
FAILURE_UNIT_SENSOR_AIRSPEED =8 , FAILURE_UNIT_SYSTEM_BATTERY =100 , FAILURE_↵
UNIT_SYSTEM_MOTOR =101 , FAILURE_UNIT_SYSTEM_SERVO =102 ,
FAILURE_UNIT_SYSTEM_AVOIDANCE =103 , FAILURE_UNIT_SYSTEM_RC_SIGNAL =104 , FAILURE_↵
_UNIT_SYSTEM_MAVLINK_SIGNAL =105 , FAILURE_UNIT_ENUM_END =106 }
• enum FAILURE_TYPE {
FAILURE_TYPE_OK =0 , FAILURE_TYPE_OFF =1 , FAILURE_TYPE_STUCK =2 , FAILURE_TYPE_↵

```

```

GARBAGE =3 ,
FAILURE_TYPE_WRONG =4 , FAILURE_TYPE_SLOW =5 , FAILURE_TYPE_DELAYED =6 , FAILURE_↵
_TYPE_INTERMITTENT =7 ,
FAILURE_TYPE_ENUM_END =8 }
• enum NAV_VTOL_LAND_OPTIONS { NAV_VTOL_LAND_OPTIONS_DEFAULT =0 , NAV_VTOL_LAND_↵
_OPTIONS_FW_DESCENT =1 , NAV_VTOL_LAND_OPTIONS_HOVER_DESCENT =2 , NAV_VTOL_↵
LAND_OPTIONS_ENUM_END =3 }
• enum MAV_WINCH_STATUS_FLAG {
MAV_WINCH_STATUS_HEALTHY =1 , MAV_WINCH_STATUS_FULLY_RETRACTED =2 , MAV_WINCH_↵
_STATUS_MOVING =4 , MAV_WINCH_STATUS_CLUTCH_ENGAGED =8 ,
MAV_WINCH_STATUS_LOCKED =16 , MAV_WINCH_STATUS_DROPPING =32 , MAV_WINCH_↵
STATUS_ARRESTING =64 , MAV_WINCH_STATUS_GROUND_SENSE =128 ,
MAV_WINCH_STATUS_RETRACTING =256 , MAV_WINCH_STATUS_REDELIVER =512 , MAV_WINCH_↵
_STATUS_ABANDON_LINE =1024 , MAV_WINCH_STATUS_LOCKING =2048 ,
MAV_WINCH_STATUS_LOAD_LINE =4096 , MAV_WINCH_STATUS_LOAD_PAYLOAD =8192 , MAV_↵
WINCH_STATUS_FLAG_ENUM_END =8193 }
• enum MAG_CAL_STATUS {
MAG_CAL_NOT_STARTED =0 , MAG_CAL_WAITING_TO_START =1 , MAG_CAL_RUNNING_STEP_↵
ONE =2 , MAG_CAL_RUNNING_STEP_TWO =3 ,
MAG_CAL_SUCCESS =4 , MAG_CAL_FAILED =5 , MAG_CAL_BAD_ORIENTATION =6 , MAG_CAL_↵
BAD_RADIUS =7 ,
MAG_CAL_STATUS_ENUM_END =8 }
• enum MAV_EVENT_ERROR_REASON { MAV_EVENT_ERROR_REASON_UNAVAILABLE =0 , MAV_↵
EVENT_ERROR_REASON_ENUM_END =1 }
• enum MAV_EVENT_CURRENT_SEQUENCE_FLAGS { MAV_EVENT_CURRENT_SEQUENCE_FLAGS_↵
_RESET =1 , MAV_EVENT_CURRENT_SEQUENCE_FLAGS_ENUM_END =2 }
• enum HIL_SENSOR_UPDATED_FLAGS {
HIL_SENSOR_UPDATED_XACC =1 , HIL_SENSOR_UPDATED_YACC =2 , HIL_SENSOR_UPDATED_↵
ZACC =4 , HIL_SENSOR_UPDATED_XGYRO =8 ,
HIL_SENSOR_UPDATED_YGYRO =16 , HIL_SENSOR_UPDATED_ZGYRO =32 , HIL_SENSOR_↵
UPDATED_XMAG =64 , HIL_SENSOR_UPDATED_YMAG =128 ,
HIL_SENSOR_UPDATED_ZMAG =256 , HIL_SENSOR_UPDATED_ABS_PRESSURE =512 , HIL_↵
SENSOR_UPDATED_DIFF_PRESSURE =1024 , HIL_SENSOR_UPDATED_PRESSURE_ALT =2048 ,
HIL_SENSOR_UPDATED_TEMPERATURE =4096 , HIL_SENSOR_UPDATED_RESET =2147483648 ,
HIL_SENSOR_UPDATED_FLAGS_ENUM_END =2147483649 }
• enum HIGHRES_IMU_UPDATED_FLAGS {
HIGHRES_IMU_UPDATED_XACC =1 , HIGHRES_IMU_UPDATED_YACC =2 , HIGHRES_IMU_↵
UPDATED_ZACC =4 , HIGHRES_IMU_UPDATED_XGYRO =8 ,
HIGHRES_IMU_UPDATED_YGYRO =16 , HIGHRES_IMU_UPDATED_ZGYRO =32 , HIGHRES_IMU_↵
UPDATED_XMAG =64 , HIGHRES_IMU_UPDATED_YMAG =128 ,
HIGHRES_IMU_UPDATED_ZMAG =256 , HIGHRES_IMU_UPDATED_ABS_PRESSURE =512 ,
HIGHRES_IMU_UPDATED_DIFF_PRESSURE =1024 , HIGHRES_IMU_UPDATED_PRESSURE_ALT
=2048 ,
HIGHRES_IMU_UPDATED_TEMPERATURE =4096 , HIGHRES_IMU_UPDATED_FLAGS_ENUM_END
=4097 }
• enum CAN_FILTER_OP { CAN_FILTER_REPLACE =0 , CAN_FILTER_ADD =1 , CAN_FILTER_REMOVE
=2 , CAN_FILTER_OP_ENUM_END =3 }
• enum MAV_FTP_ERR {
MAV_FTP_ERR_NONE =0 , MAV_FTP_ERR_FAIL =1 , MAV_FTP_ERR_FAILERRNO =2 , MAV_FTP_↵
ERR_INVALIDDATASIZE =3 ,
MAV_FTP_ERR_INVALIDSESSION =4 , MAV_FTP_ERR_NOSESSIONSAVAILABLE =5 , MAV_FTP_↵
ERR_EOF =6 , MAV_FTP_ERR_UNKNOWNCOMMAND =7 ,
MAV_FTP_ERR_FILEEXISTS =8 , MAV_FTP_ERR_FILEPROTECTED =9 , MAV_FTP_ERR_↵
FILENOTFOUND =10 , MAV_FTP_ERR_ENUM_END =11 }
• enum MAV_FTP_OPCODE {
MAV_FTP_OPCODE_NONE =0 , MAV_FTP_OPCODE_TERMINATESSESSION =1 , MAV_FTP_OPCODE_↵
_RESETSESSION =2 , MAV_FTP_OPCODE_LISTDIRECTORY =3 ,

```

- ```

MAV FTP_OPCODE_OPENFILERO =4 , MAV FTP_OPCODE_READFILE =5 , MAV FTP_OPCODE_↵
CREATEFILE =6 , MAV FTP_OPCODE_WRITEFILE =7 ,
MAV FTP_OPCODE_REMOVEFILE =8 , MAV FTP_OPCODE_CREATEDIRECTORY =9 , MAV FTP_↵
OPCODE_REMOVEDIRECTORY =10 , MAV FTP_OPCODE_OPENFILEWO =11 ,
MAV FTP_OPCODE_TRUNCATEFILE =12 , MAV FTP_OPCODE_RENAME =13 , MAV FTP_OPCODE_↵
_CALCFILECRC =14 , MAV FTP_OPCODE_BURSTREADFILE =15 ,
MAV FTP_OPCODE_ACK =128 , MAV FTP_OPCODE_NAK =129 , MAV FTP_OPCODE_ENUM_END
=130 }

```
- enum MISSION\_STATE {  
MISSION\_STATE\_UNKNOWN =0 , MISSION\_STATE\_NO\_MISSION =1 , MISSION\_STATE\_NOT\_↵  
STARTED =2 , MISSION\_STATE\_ACTIVE =3 ,  
MISSION\_STATE\_PAUSED =4 , MISSION\_STATE\_COMPLETE =5 , MISSION\_STATE\_ENUM\_END =6 }
  - enum SAFETY\_SWITCH\_STATE { SAFETY\_SWITCH\_STATE\_SAFE =0 , SAFETY\_SWITCH\_STATE\_↵  
DANGEROUS =1 , SAFETY\_SWITCH\_STATE\_ENUM\_END =2 }
  - enum ILLUMINATOR\_MODE { ILLUMINATOR\_MODE\_UNKNOWN =0 , ILLUMINATOR\_MODE\_↵  
INTERNAL\_CONTROL =1 , ILLUMINATOR\_MODE\_EXTERNAL\_SYNC =2 , ILLUMINATOR\_MODE\_↵  
\_ENUM\_END =3 }
  - enum ILLUMINATOR\_ERROR\_FLAGS { ILLUMINATOR\_ERROR\_FLAGS\_THERMAL\_THROTTLING =1  
, ILLUMINATOR\_ERROR\_FLAGS\_OVER\_TEMPERATURE\_SHUTDOWN =2 , ILLUMINATOR\_ERROR\_↵  
FLAGS\_THERMISTOR\_FAILURE =4 , ILLUMINATOR\_ERROR\_FLAGS\_ENUM\_END =5 }
  - enum MAV\_STANDARD\_MODE {  
MAV\_STANDARD\_MODE\_NON\_STANDARD =0 , MAV\_STANDARD\_MODE\_POSITION\_HOLD =1 ,  
MAV\_STANDARD\_MODE\_ORBIT =2 , MAV\_STANDARD\_MODE\_CRUISE =3 ,  
MAV\_STANDARD\_MODE\_ALTITUDE\_HOLD =4 , MAV\_STANDARD\_MODE\_SAFE\_RECOVERY =5 ,  
MAV\_STANDARD\_MODE\_MISSION =6 , MAV\_STANDARD\_MODE\_LAND =7 ,  
MAV\_STANDARD\_MODE\_TAKEOFF =8 , MAV\_STANDARD\_MODE\_ENUM\_END =9 }
  - enum MAV\_MODE\_PROPERTY { MAV\_MODE\_PROPERTY\_ADVANCED =1 , MAV\_MODE\_↵  
PROPERTY\_NOT\_USER\_SELECTABLE =2 , MAV\_MODE\_PROPERTY\_AUTO\_MODE =4 , MAV\_↵  
MODE\_PROPERTY\_ENUM\_END =5 }
  - enum HIL\_ACTUATOR\_CONTROLS\_FLAGS { HIL\_ACTUATOR\_CONTROLS\_FLAGS\_LOCKSTEP =1 ,  
HIL\_ACTUATOR\_CONTROLS\_FLAGS\_ENUM\_END =2 }
  - enum COMPUTER\_STATUS\_FLAGS {  
COMPUTER\_STATUS\_FLAGS\_UNDER\_VOLTAGE =1 , COMPUTER\_STATUS\_FLAGS\_CPU\_THROTTLE  
=2 , COMPUTER\_STATUS\_FLAGS\_THERMAL\_THROTTLE =4 , COMPUTER\_STATUS\_FLAGS\_DISK\_↵  
\_FULL =8 ,  
COMPUTER\_STATUS\_FLAGS\_ENUM\_END =9 }
  - enum AIRSPEED\_SENSOR\_FLAGS { AIRSPEED\_SENSOR\_UNHEALTHY =1 , AIRSPEED\_SENSOR\_↵  
USING =2 , AIRSPEED\_SENSOR\_FLAGS\_ENUM\_END =3 }

### 6.6.1 Detailed Description

MAVLink comm protocol generated from common.xml.

See also

<http://mavlink.org>

### 6.6.2 Macro Definition Documentation

#### 6.6.2.1 HAVE\_ENUM\_ACTUATOR\_CONFIGURATION

```
#define HAVE_ENUM_ACTUATOR_CONFIGURATION
```

Actuator configuration, used to change a setting on an actuator.

Component information metadata can be used to know which outputs support which commands.

#### 6.6.2.2 HAVE\_ENUM\_ACTUATOR\_OUTPUT\_FUNCTION

```
#define HAVE_ENUM_ACTUATOR_OUTPUT_FUNCTION
```

Actuator output function.

Values greater or equal to 1000 are autopilot-specific.

#### 6.6.2.3 HAVE\_ENUM\_ADSB\_FLAGS

```
#define HAVE_ENUM_ADSB_FLAGS
```

These flags indicate status such as data validity of each data source.

Set = data valid

#### 6.6.2.4 HAVE\_ENUM\_AIS\_FLAGS

```
#define HAVE_ENUM_AIS_FLAGS
```

These flags are used in the AIS\_VESSEL.fields bitmask to indicate validity of data in the other message fields.

When set, the data is valid.

#### 6.6.2.5 HAVE\_ENUM\_AUTOTUNE\_AXIS

```
#define HAVE_ENUM_AUTOTUNE_AXIS
```

Axes that will be autotuned by MAV\_CMD\_DO\_AUTOTUNE\_ENABLE.

Note that at least one flag must be set in MAV\_CMD\_DO\_AUTOTUNE\_ENABLE.param2: if none are set, the flight stack will tune its default set of axes.

#### 6.6.2.6 HAVE\_ENUM\_COMP\_METADATA\_TYPE

```
#define HAVE_ENUM_COMP_METADATA_TYPE
```

Supported component metadata types.

These are used in the "general" metadata file returned by COMPONENT\_METADATA to provide information about supported metadata types. The types are not used directly in MAVLink messages.

### 6.6.2.7 HAVE\_ENUM\_FENCE\_TYPE

```
#define HAVE_ENUM_FENCE_TYPE
```

Fence types to enable or disable when using MAV\_CMD\_DO\_FENCE\_ENABLE.

Note that at least one of these flags must be set in MAV\_CMD\_DO\_FENCE\_ENABLE.param2. If none are set, the flight stack will ignore the field and enable/disable its default set of fences (usually all of them).

### 6.6.2.8 HAVE\_ENUM\_GIMBAL\_MANAGER\_CAP\_FLAGS

```
#define HAVE_ENUM_GIMBAL_MANAGER_CAP_FLAGS
```

Gimbal manager high level capability flags (bitmap).

The first 16 bits are identical to the GIMBAL\_DEVICE\_CAP\_FLAGS. However, the gimbal manager does not need to copy the flags from the gimbal but can also enhance the capabilities and thus add flags.

### 6.6.2.9 HAVE\_ENUM\_MAV\_BATTERY\_FAULT

```
#define HAVE_ENUM_MAV_BATTERY_FAULT
```

Smart battery supply status/fault flags (bitmask) for health indication.

The battery must also report either MAV\_BATTERY\_CHARGE\_STATE\_FAILED or MAV\_BATTERY\_CHARGE\_STATE\_UNHEALTHY if any of these are set.

### 6.6.2.10 HAVE\_ENUM\_MAV\_BATTERY\_MODE

```
#define HAVE_ENUM_MAV_BATTERY_MODE
```

Battery mode.

Note, the normal operation mode (i.e. when flying) should be reported as MAV\_BATTERY\_MODE\_UNKNOWN to allow message trimming in normal flight.

### 6.6.2.11 HAVE\_ENUM\_MAV\_CMD

```
#define HAVE_ENUM_MAV_CMD
```

Commands to be executed by the MAV.

They can be executed on user request, or as part of a mission script. If the action is used in a mission, the parameter mapping to the waypoint/mission message is as follows: Param 1, Param 2, Param 3, Param 4, X↔: Param 5, Y:Param 6, Z:Param 7. This command list is similar what ARINC 424 is for commercial aircraft: A data format how to interpret waypoint/mission data. NaN and INT32\_MAX may be used in float/integer params (respectively) to indicate optional/default values (e.g. to use the component's current yaw or latitude rather than a specific value). See [https://mavlink.io/en/guide/xml\\_schema.html#MAV\\_CMD](https://mavlink.io/en/guide/xml_schema.html#MAV_CMD) for information about the structure of the MAV\_CMD entries

#### 6.6.2.12 HAVE\_ENUM\_MAV\_DATA\_STREAM

```
#define HAVE_ENUM_MAV_DATA_STREAM
```

A data stream is not a fixed set of messages, but rather a recommendation to the autopilot software.

Individual autopilots may or may not obey the recommended messages.

#### 6.6.2.13 HAVE\_ENUM\_MAV\_FRAME

```
#define HAVE_ENUM_MAV_FRAME
```

Coordinate frames used by MAVLink.

Not all frames are supported by all commands, messages, or vehicles.

Global frames use the following naming conventions:

- "GLOBAL": Global coordinate frame with WGS84 latitude/longitude and altitude positive over mean sea level (MSL) by default. The following modifiers may be used with "GLOBAL":
  - "RELATIVE\_ALT": Altitude is relative to the vehicle home position rather than MSL.
  - "TERRAIN\_ALT": Altitude is relative to ground level rather than MSL.
  - "INT": Latitude/longitude (in degrees) are scaled by multiplying by 1E7.

Local frames use the following naming conventions:

- "LOCAL": Origin of local frame is fixed relative to earth. Unless otherwise specified this origin is the origin of the vehicle position-estimator ("EKF").
- "BODY": Origin of local frame travels with the vehicle. NOTE, "BODY" does NOT indicate alignment of frame axis with vehicle attitude.
- "OFFSET": Deprecated synonym for "BODY" (origin travels with the vehicle). Not to be used for new frames.

Some deprecated frames do not follow these conventions (e.g. MAV\_FRAME\_BODY\_NED and MAV\_FRAME\_↔ BODY\_OFFSET\_NED).

#### 6.6.2.14 HAVE\_ENUM\_MAV\_FTP\_ERR

```
#define HAVE_ENUM_MAV_FTP_ERR
```

MAV FTP error codes (may be used in FILE\_TRANSFER\_PROTOCOL).

See <https://mavlink.io/en/services/ftp.html>.

#### 6.6.2.15 HAVE\_ENUM\_MAV\_FTP\_OPCODE

```
#define HAVE_ENUM_MAV_FTP_OPCODE
```

MAV FTP opcodes (may be used in FILE\_TRANSFER\_PROTOCOL).

See <https://mavlink.io/en/services/ftp.html>.



#### 6.6.2.16 HAVE\_ENUM\_MAV\_FUEL\_TYPE

```
#define HAVE_ENUM_MAV_FUEL_TYPE
```

Fuel types for use in FUEL\_TYPE.

Fuel types specify the units for the maximum, available and consumed fuel, and for the flow rates.

#### 6.6.2.17 HAVE\_ENUM\_MAV\_GENERATOR\_STATUS\_FLAG

```
#define HAVE_ENUM_MAV_GENERATOR_STATUS_FLAG
```

Flags to report status/failure cases for a power generator (used in GENERATOR\_STATUS).

Note that FAULTS are conditions that cause the generator to fail. Warnings are conditions that require attention before the next use (they indicate the system is not operating properly).

#### 6.6.2.18 HAVE\_ENUM\_MAV\_MODE

```
#define HAVE_ENUM_MAV_MODE
```

Predefined OR-combined MAV\_MODE\_FLAG values.

These can simplify using the flags when setting modes. Note that manual input is enabled in all modes as a safety override.

#### 6.6.2.19 HAVE\_ENUM\_MAV\_MOUNT\_MODE

```
#define HAVE_ENUM_MAV_MOUNT_MODE
```

Enumeration of possible mount operation modes.

This message is used by obsolete/deprecated gimbal messages.

#### 6.6.2.20 HAVE\_ENUM\_MAV\_ROI

```
#define HAVE_ENUM_MAV_ROI
```

The ROI (region of interest) for the vehicle.

This can be used by the vehicle for camera/vehicle attitude alignment (see MAV\_CMD\_NAV\_ROI).

#### 6.6.2.21 HAVE\_ENUM\_MAV\_SEVERITY

```
#define HAVE_ENUM_MAV_SEVERITY
```

Indicates the severity level, generally used for status messages to indicate their relative urgency.

Based on RFC-5424 using expanded definitions at: <http://www.kiwisyslog.com/kb/info←:-syslog-message-levels/>.

#### 6.6.2.22 HAVE\_ENUM\_MAV\_STANDARD\_MODE

```
#define HAVE_ENUM_MAV_STANDARD_MODE
```

Standard modes with a well understood meaning across flight stacks and vehicle types.

For example, most flight stack have the concept of a "return" or "RTL" mode that takes a vehicle to safety, even though the precise mechanics of this mode may differ. The modes supported by a flight stack can be queried using `AVAILABLE_MODES` and set using `MAV_CMD_DO_SET_STANDARD_MODE`. The current mode is streamed in `CURRENT_MODE`. See [https://mavlink.io/en/services/standard\\_modes.html](https://mavlink.io/en/services/standard_modes.html)

#### 6.6.2.23 HAVE\_ENUM\_MISSION\_STATE

```
#define HAVE_ENUM_MISSION_STATE
```

States of the mission state machine.

Note that these states are independent of whether the mission is in a mode that can execute mission items or not (is suspended). They may not all be relevant on all vehicles.

#### 6.6.2.24 HAVE\_ENUM\_PARACHUTE\_ACTION

```
#define HAVE_ENUM_PARACHUTE_ACTION
```

Parachute actions.

Trigger release and enable/disable auto-release.

#### 6.6.2.25 HAVE\_ENUM\_POSITION\_TARGET\_TYEMASK

```
#define HAVE_ENUM_POSITION_TARGET_TYEMASK
```

Bitmap to indicate which dimensions should be ignored by the vehicle: a value of 0b0000000000000000 or 0b0000000100000000 indicates that none of the setpoint dimensions should be ignored.

If bit 9 is set the floats `afx` `afy` `afz` should be interpreted as force instead of acceleration.

#### 6.6.2.26 HAVE\_ENUM\_PREFLIGHT\_STORAGE\_MISSION\_ACTION

```
#define HAVE_ENUM_PREFLIGHT_STORAGE_MISSION_ACTION
```

Actions for reading and writing plan information (mission, rally points, geofence) between persistent and volatile storage when using `MAV_CMD_PREFLIGHT_STORAGE`.

(Commonly missions are loaded from persistent storage (flash/EEPROM) into volatile storage (RAM) on startup and written back when they are changed.)

### 6.6.2.27 HAVE\_ENUM\_PREFLIGHT\_STORAGE\_PARAMETER\_ACTION

```
#define HAVE_ENUM_PREFLIGHT_STORAGE_PARAMETER_ACTION
```

Actions for reading/writing parameters between persistent and volatile storage when using MAV\_CMD\_↔ PREFLIGHT\_STORAGE.

(Commonly parameters are loaded from persistent storage (flash/EEPROM) into volatile storage (RAM) on startup and written back when they are changed.)

### 6.6.2.28 HAVE\_ENUM\_RC\_SUB\_TYPE

```
#define HAVE_ENUM_RC_SUB_TYPE
```

RC sub-type of types defined in RC\_TYPE.

Used in MAV\_CMD\_START\_RX\_PAIR. Ignored if value does not correspond to the set RC\_TYPE.

### 6.6.2.29 HAVE\_ENUM\_RC\_TYPE

```
#define HAVE_ENUM_RC_TYPE
```

RC type.

Used in MAV\_CMD\_START\_RX\_PAIR.

## 6.7 single\_hand\_controller/include/mavlink/common/common.hpp File Reference

MAVLink comm protocol generated from common.xml.

```
#include <array>
#include <stdint>
#include <sstream>
#include "../message.hpp"
#include "../mavlink_msg_sys_status.hpp"
#include "../mavlink_msg_system_time.hpp"
#include "../mavlink_msg_ping.hpp"
#include "../mavlink_msg_change_operator_control.hpp"
#include "../mavlink_msg_change_operator_control_ack.hpp"
#include "../mavlink_msg_auth_key.hpp"
#include "../mavlink_msg_link_node_status.hpp"
#include "../mavlink_msg_set_mode.hpp"
#include "../mavlink_msg_param_request_read.hpp"
#include "../mavlink_msg_param_request_list.hpp"
#include "../mavlink_msg_param_value.hpp"
#include "../mavlink_msg_param_set.hpp"
#include "../mavlink_msg_gps_raw_int.hpp"
#include "../mavlink_msg_gps_status.hpp"
#include "../mavlink_msg_scaled_imu.hpp"
#include "../mavlink_msg_raw_imu.hpp"
```

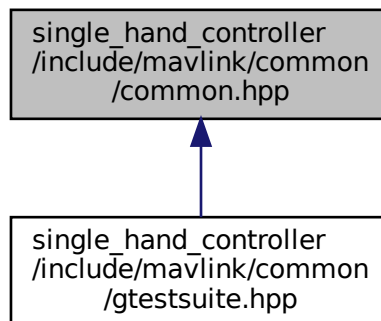
```
#include "../mavlink_msg_raw_pressure.hpp"
#include "../mavlink_msg_scaled_pressure.hpp"
#include "../mavlink_msg_attitude.hpp"
#include "../mavlink_msg_attitude_quaternion.hpp"
#include "../mavlink_msg_local_position_ned.hpp"
#include "../mavlink_msg_rc_channels_scaled.hpp"
#include "../mavlink_msg_rc_channels_raw.hpp"
#include "../mavlink_msg_servo_output_raw.hpp"
#include "../mavlink_msg_mission_request_partial_list.hpp"
#include "../mavlink_msg_mission_write_partial_list.hpp"
#include "../mavlink_msg_mission_item.hpp"
#include "../mavlink_msg_mission_request.hpp"
#include "../mavlink_msg_mission_set_current.hpp"
#include "../mavlink_msg_mission_current.hpp"
#include "../mavlink_msg_mission_request_list.hpp"
#include "../mavlink_msg_mission_count.hpp"
#include "../mavlink_msg_mission_clear_all.hpp"
#include "../mavlink_msg_mission_item_reached.hpp"
#include "../mavlink_msg_mission_ack.hpp"
#include "../mavlink_msg_set_gps_global_origin.hpp"
#include "../mavlink_msg_gps_global_origin.hpp"
#include "../mavlink_msg_param_map_rc.hpp"
#include "../mavlink_msg_mission_request_int.hpp"
#include "../mavlink_msg_safety_set_allowed_area.hpp"
#include "../mavlink_msg_safety_allowed_area.hpp"
#include "../mavlink_msg_attitude_quaternion_cov.hpp"
#include "../mavlink_msg_nav_controller_output.hpp"
#include "../mavlink_msg_global_position_int_cov.hpp"
#include "../mavlink_msg_local_position_ned_cov.hpp"
#include "../mavlink_msg_rc_channels.hpp"
#include "../mavlink_msg_request_data_stream.hpp"
#include "../mavlink_msg_data_stream.hpp"
#include "../mavlink_msg_manual_control.hpp"
#include "../mavlink_msg_rc_channels_override.hpp"
#include "../mavlink_msg_mission_item_int.hpp"
#include "../mavlink_msg_vfr_hud.hpp"
#include "../mavlink_msg_command_int.hpp"
#include "../mavlink_msg_command_long.hpp"
#include "../mavlink_msg_command_ack.hpp"
#include "../mavlink_msg_command_cancel.hpp"
#include "../mavlink_msg_manual_setpoint.hpp"
#include "../mavlink_msg_set_attitude_target.hpp"
#include "../mavlink_msg_attitude_target.hpp"
#include "../mavlink_msg_set_position_target_local_ned.hpp"
#include "../mavlink_msg_position_target_local_ned.hpp"
#include "../mavlink_msg_set_position_target_global_int.hpp"
#include "../mavlink_msg_position_target_global_int.hpp"
#include "../mavlink_msg_local_position_ned_system_global_offset.hpp"
#include "../mavlink_msg_hil_state.hpp"
#include "../mavlink_msg_hil_controls.hpp"
#include "../mavlink_msg_hil_rc_inputs_raw.hpp"
#include "../mavlink_msg_hil_actuator_controls.hpp"
#include "../mavlink_msg_optical_flow.hpp"
#include "../mavlink_msg_global_vision_position_estimate.hpp"
#include "../mavlink_msg_vision_position_estimate.hpp"
#include "../mavlink_msg_vision_speed_estimate.hpp"
#include "../mavlink_msg_vicon_position_estimate.hpp"
#include "../mavlink_msg_highres_imu.hpp"
```

```
#include "../mavlink_msg_optical_flow_rad.hpp"
#include "../mavlink_msg_hil_sensor.hpp"
#include "../mavlink_msg_sim_state.hpp"
#include "../mavlink_msg_radio_status.hpp"
#include "../mavlink_msg_file_transfer_protocol.hpp"
#include "../mavlink_msg_timesync.hpp"
#include "../mavlink_msg_camera_trigger.hpp"
#include "../mavlink_msg_hil_gps.hpp"
#include "../mavlink_msg_hil_optical_flow.hpp"
#include "../mavlink_msg_hil_state_quaternion.hpp"
#include "../mavlink_msg_scaled_imu2.hpp"
#include "../mavlink_msg_log_request_list.hpp"
#include "../mavlink_msg_log_entry.hpp"
#include "../mavlink_msg_log_request_data.hpp"
#include "../mavlink_msg_log_data.hpp"
#include "../mavlink_msg_log_erase.hpp"
#include "../mavlink_msg_log_request_end.hpp"
#include "../mavlink_msg_gps_inject_data.hpp"
#include "../mavlink_msg_gps2_raw.hpp"
#include "../mavlink_msg_power_status.hpp"
#include "../mavlink_msg_serial_control.hpp"
#include "../mavlink_msg_gps_rtk.hpp"
#include "../mavlink_msg_gps2_rtk.hpp"
#include "../mavlink_msg_scaled_imu3.hpp"
#include "../mavlink_msg_data_transmission_handshake.hpp"
#include "../mavlink_msg_encapsulated_data.hpp"
#include "../mavlink_msg_distance_sensor.hpp"
#include "../mavlink_msg_terrain_request.hpp"
#include "../mavlink_msg_terrain_data.hpp"
#include "../mavlink_msg_terrain_check.hpp"
#include "../mavlink_msg_terrain_report.hpp"
#include "../mavlink_msg_scaled_pressure2.hpp"
#include "../mavlink_msg_att_pos_mocap.hpp"
#include "../mavlink_msg_set_actuator_control_target.hpp"
#include "../mavlink_msg_actuator_control_target.hpp"
#include "../mavlink_msg_altitude.hpp"
#include "../mavlink_msg_resource_request.hpp"
#include "../mavlink_msg_scaled_pressure3.hpp"
#include "../mavlink_msg_follow_target.hpp"
#include "../mavlink_msg_control_system_state.hpp"
#include "../mavlink_msg_battery_status.hpp"
#include "../mavlink_msg_landing_target.hpp"
#include "../mavlink_msg_fence_status.hpp"
#include "../mavlink_msg_mag_cal_report.hpp"
#include "../mavlink_msg_efi_status.hpp"
#include "../mavlink_msg_estimator_status.hpp"
#include "../mavlink_msg_wind_cov.hpp"
#include "../mavlink_msg_gps_input.hpp"
#include "../mavlink_msg_gps_rtcmm_data.hpp"
#include "../mavlink_msg_high_latency.hpp"
#include "../mavlink_msg_high_latency2.hpp"
#include "../mavlink_msg_vibration.hpp"
#include "../mavlink_msg_home_position.hpp"
#include "../mavlink_msg_set_home_position.hpp"
#include "../mavlink_msg_message_interval.hpp"
#include "../mavlink_msg_extended_sys_state.hpp"
#include "../mavlink_msg_adsb_vehicle.hpp"
#include "../mavlink_msg_collision.hpp"
```

```
#include "../mavlink_msg_v2_extension.hpp"
#include "../mavlink_msg_memory_vect.hpp"
#include "../mavlink_msg_debug_vect.hpp"
#include "../mavlink_msg_named_value_float.hpp"
#include "../mavlink_msg_named_value_int.hpp"
#include "../mavlink_msg_statustext.hpp"
#include "../mavlink_msg_debug.hpp"
#include "../mavlink_msg_setup_signing.hpp"
#include "../mavlink_msg_button_change.hpp"
#include "../mavlink_msg_play_tune.hpp"
#include "../mavlink_msg_camera_information.hpp"
#include "../mavlink_msg_camera_settings.hpp"
#include "../mavlink_msg_storage_information.hpp"
#include "../mavlink_msg_camera_capture_status.hpp"
#include "../mavlink_msg_camera_image_captured.hpp"
#include "../mavlink_msg_flight_information.hpp"
#include "../mavlink_msg_mount_orientation.hpp"
#include "../mavlink_msg_logging_data.hpp"
#include "../mavlink_msg_logging_data_acked.hpp"
#include "../mavlink_msg_logging_ack.hpp"
#include "../mavlink_msg_video_stream_information.hpp"
#include "../mavlink_msg_video_stream_status.hpp"
#include "../mavlink_msg_camera_fov_status.hpp"
#include "../mavlink_msg_camera_tracking_image_status.hpp"
#include "../mavlink_msg_camera_tracking_geo_status.hpp"
#include "../mavlink_msg_camera_thermal_range.hpp"
#include "../mavlink_msg_gimbal_manager_information.hpp"
#include "../mavlink_msg_gimbal_manager_status.hpp"
#include "../mavlink_msg_gimbal_manager_set_attitude.hpp"
#include "../mavlink_msg_gimbal_device_information.hpp"
#include "../mavlink_msg_gimbal_device_set_attitude.hpp"
#include "../mavlink_msg_gimbal_device_attitude_status.hpp"
#include "../mavlink_msg_autopilot_state_for_gimbal_device.hpp"
#include "../mavlink_msg_gimbal_manager_set_pitchyaw.hpp"
#include "../mavlink_msg_gimbal_manager_set_manual_control.hpp"
#include "../mavlink_msg_esc_info.hpp"
#include "../mavlink_msg_esc_status.hpp"
#include "../mavlink_msg_airspeed.hpp"
#include "../mavlink_msg_wifi_config_ap.hpp"
#include "../mavlink_msg_protocol_version.hpp"
#include "../mavlink_msg_ais_vessel.hpp"
#include "../mavlink_msg_uavcan_node_status.hpp"
#include "../mavlink_msg_uavcan_node_info.hpp"
#include "../mavlink_msg_param_ext_request_read.hpp"
#include "../mavlink_msg_param_ext_request_list.hpp"
#include "../mavlink_msg_param_ext_value.hpp"
#include "../mavlink_msg_param_ext_set.hpp"
#include "../mavlink_msg_param_ext_ack.hpp"
#include "../mavlink_msg_obstacle_distance.hpp"
#include "../mavlink_msg_odometry.hpp"
#include "../mavlink_msg_trajectory_representation_waypoints.hpp"
#include "../mavlink_msg_trajectory_representation_bezier.hpp"
#include "../mavlink_msg_cellular_status.hpp"
#include "../mavlink_msg_isbd_link_status.hpp"
#include "../mavlink_msg_cellular_config.hpp"
#include "../mavlink_msg_raw_rpm.hpp"
#include "../mavlink_msg_utm_global_position.hpp"
#include "../mavlink_msg_param_error.hpp"
```

```
#include "../mavlink_msg_debug_float_array.hpp"
#include "../mavlink_msg_orbit_execution_status.hpp"
#include "../mavlink_msg_smart_battery_info.hpp"
#include "../mavlink_msg_figure_eight_execution_status.hpp"
#include "../mavlink_msg_fuel_status.hpp"
#include "../mavlink_msg_battery_info.hpp"
#include "../mavlink_msg_generator_status.hpp"
#include "../mavlink_msg_actuator_output_status.hpp"
#include "../mavlink_msg_relay_status.hpp"
#include "../mavlink_msg_time_estimate_to_target.hpp"
#include "../mavlink_msg_tunnel.hpp"
#include "../mavlink_msg_can_frame.hpp"
#include "../mavlink_msg_onboard_computer_status.hpp"
#include "../mavlink_msg_component_information.hpp"
#include "../mavlink_msg_component_information_basic.hpp"
#include "../mavlink_msg_component_metadata.hpp"
#include "../mavlink_msg_play_tune_v2.hpp"
#include "../mavlink_msg_supported_tunes.hpp"
#include "../mavlink_msg_event.hpp"
#include "../mavlink_msg_current_event_sequence.hpp"
#include "../mavlink_msg_request_event.hpp"
#include "../mavlink_msg_response_event_error.hpp"
#include "../mavlink_msg_available_modes.hpp"
#include "../mavlink_msg_current_mode.hpp"
#include "../mavlink_msg_available_modes_monitor.hpp"
#include "../mavlink_msg_illuminator_status.hpp"
#include "../mavlink_msg_canfd_frame.hpp"
#include "../mavlink_msg_can_filter_modify.hpp"
#include "../mavlink_msg_wheel_distance.hpp"
#include "../mavlink_msg_winch_status.hpp"
#include "../mavlink_msg_open_drone_id_basic_id.hpp"
#include "../mavlink_msg_open_drone_id_location.hpp"
#include "../mavlink_msg_open_drone_id_authentication.hpp"
#include "../mavlink_msg_open_drone_id_self_id.hpp"
#include "../mavlink_msg_open_drone_id_system.hpp"
#include "../mavlink_msg_open_drone_id_operator_id.hpp"
#include "../mavlink_msg_open_drone_id_message_pack.hpp"
#include "../mavlink_msg_open_drone_id_arm_status.hpp"
#include "../mavlink_msg_open_drone_id_system_update.hpp"
#include "../mavlink_msg_hygrometer_sensor.hpp"
#include "../standard/standard.hpp"
```

This graph shows which files directly or indirectly include this file:



## Macros

- `#define MAVLINK_STX 253`

## Enumerations

- enum class `mavlink::common::HL_FAILURE_FLAG` : `uint16_t` {  
**GPS** =1 , **DIFFERENTIAL\_PRESSURE** =2 , **ABSOLUTE\_PRESSURE** =4 , **FLAG\_3D\_ACCEL** =8 ,  
**FLAG\_3D\_GYRO** =16 , **FLAG\_3D\_MAG** =32 , **TERRAIN** =64 , **BATTERY** =128 ,  
**RC\_RECEIVER** =256 , **OFFBOARD\_LINK** =512 , **ENGINE** =1024 , **GEOFENCE** =2048 ,  
**ESTIMATOR** =4096 , **MISSION** =8192 }  
*Flags to report failure cases over the high latency telemetry.*
- enum class `mavlink::common::MAV_GOTO` { **DO\_HOLD** =0 , **DO\_CONTINUE** =1 , **HOLD\_AT\_CURRENT** ←  
**\_POSITION** =2 , **HOLD\_AT\_SPECIFIED\_POSITION** =3 }  
*Actions that may be specified in MAV\_CMD\_OVERRIDE\_GOTO to override mission execution.*
- enum class `mavlink::common::MAV_MODE` : `uint8_t` {  
**PREFLIGHT** =0 , **MANUAL\_DISARMED** =64 , **TEST\_DISARMED** =66 , **STABILIZE\_DISARMED** =80 ,  
**GUIDED\_DISARMED** =88 , **AUTO\_DISARMED** =92 , **MANUAL\_ARMED** =192 , **TEST\_ARMED** =194 ,  
**STABILIZE\_ARMED** =208 , **GUIDED\_ARMED** =216 , **AUTO\_ARMED** =220 }  
*Predefined OR-combined MAV\_MODE\_FLAG values.*
- enum class `mavlink::common::MAV_SYS_STATUS_SENSOR` : `uint32_t` {  
**SENSOR\_3D\_GYRO** =1 , **SENSOR\_3D\_ACCEL** =2 , **SENSOR\_3D\_MAG** =4 , **ABSOLUTE\_PRESSURE** =8  
, **DIFFERENTIAL\_PRESSURE** =16 , **GPS** =32 , **OPTICAL\_FLOW** =64 , **VISION\_POSITION** =128 ,  
**LASER\_POSITION** =256 , **EXTERNAL\_GROUND\_TRUTH** =512 , **ANGULAR\_RATE\_CONTROL** =1024 ,  
**ATTITUDE\_STABILIZATION** =2048 ,  
**YAW\_POSITION** =4096 , **Z\_ALTITUDE\_CONTROL** =8192 , **XY\_POSITION\_CONTROL** =16384 , **MOTOR** ←  
**\_OUTPUTS** =32768 ,  
**RC\_RECEIVER** =65536 , **SENSOR\_3D\_GYRO2** =131072 , **SENSOR\_3D\_ACCEL2** =262144 , **SENSOR** ←  
**\_3D\_MAG2** =524288 ,  
**GEOFENCE** =1048576 , **AHRS** =2097152 , **TERRAIN** =4194304 , **REVERSE\_MOTOR** =8388608 ,  
**LOGGING** =16777216 , **BATTERY** =33554432 , **PROXIMITY** =67108864 , **SATCOM** =134217728 ,  
**PREARM\_CHECK** =268435456 , **OBSTACLE\_AVOIDANCE** =536870912 , **PROPULSION** =1073741824 ,  
**EXTENSION\_USED** =2147483648 }



*These encode the sensors whose status is sent as part of the SYS\_STATUS message.*

- enum class `mavlink::common::MAV_SYS_STATUS_SENSOR_EXTENDED` : `uint32_t` { `RECOVERY_↵`  
`SYSTEM` =1 }

*These encode the sensors whose status is sent as part of the SYS\_STATUS message in the extended fields.*

- enum class `mavlink::common::MAV_FRAME` : `uint8_t` {  
`GLOBAL` =0 , `LOCAL_NED` =1 , `MISSION` =2 , `GLOBAL_RELATIVE_ALT` =3 ,  
`LOCAL_ENU` =4 , `GLOBAL_INT` =5 , `GLOBAL_RELATIVE_ALT_INT` =6 , `LOCAL_OFFSET_NED` =7 ,  
`BODY_NED` =8 , `BODY_OFFSET_NED` =9 , `GLOBAL_TERRAIN_ALT` =10 , `GLOBAL_TERRAIN_ALT_INT`  
=11 ,  
`BODY_FRD` =12 , `RESERVED_13` =13 , `RESERVED_14` =14 , `RESERVED_15` =15 ,  
`RESERVED_16` =16 , `RESERVED_17` =17 , `RESERVED_18` =18 , `RESERVED_19` =19 ,  
`LOCAL_FRD` =20 , `LOCAL_FLU` =21 }

*Coordinate frames used by MAVLink.*

- enum class `mavlink::common::MAVLINK_DATA_STREAM_TYPE` : `uint8_t` {  
`IMG_JPEG` =0 , `IMG_BMP` =1 , `IMG_RAW8U` =2 , `IMG_RAW32U` =3 ,  
`IMG_PGM` =4 , `IMG_PNG` =5 }

- enum class `mavlink::common::FENCE_BREACH` : `uint8_t` { `NONE` =0 , `MINALT` =1 , `MAXALT` =2 , `BOUND-  
ARY` =3 }

- enum class `mavlink::common::FENCE_MITIGATE` : `uint8_t` { `UNKNOWN` =0 , `NONE` =1 , `VEL_LIMIT` =2 }

*Actions being taken to mitigate/prevent fence breach.*

- enum class `mavlink::common::FENCE_TYPE` { `ALT_MAX` =1 , `CIRCLE` =2 , `POLYGON` =4 , `ALT_MIN` =8 }

*Fence types to enable or disable when using MAV\_CMD\_DO\_FENCE\_ENABLE.*

- enum class `mavlink::common::MAV_MOUNT_MODE` {  
`RETRACT` =0 , `NEUTRAL` =1 , `MAVLINK_TARGETING` =2 , `RC_TARGETING` =3 ,  
`GPS_POINT` =4 , `SYSID_TARGET` =5 , `HOME_LOCATION` =6 }

*Enumeration of possible mount operation modes.*

- enum class `mavlink::common::GIMBAL_DEVICE_CAP_FLAGS` : `uint16_t` {  
`HAS_RETRACT` =1 , `HAS_NEUTRAL` =2 , `HAS_ROLL_AXIS` =4 , `HAS_ROLL_FOLLOW` =8 ,  
`HAS_ROLL_LOCK` =16 , `HAS_PITCH_AXIS` =32 , `HAS_PITCH_FOLLOW` =64 , `HAS_PITCH_LOCK` =128  
,  
`HAS_YAW_AXIS` =256 , `HAS_YAW_FOLLOW` =512 , `HAS_YAW_LOCK` =1024 , `SUPPORTS_INFINITE_↵`  
`_YAW` =2048 ,  
`SUPPORTS_YAW_IN_EARTH_FRAME` =4096 , `HAS_RC_INPUTS` =8192 }

*Gimbal device (low level) capability flags (bitmap).*

- enum class `mavlink::common::GIMBAL_MANAGER_CAP_FLAGS` : `uint32_t` {  
`HAS_RETRACT` =1 , `HAS_NEUTRAL` =2 , `HAS_ROLL_AXIS` =4 , `HAS_ROLL_FOLLOW` =8 ,  
`HAS_ROLL_LOCK` =16 , `HAS_PITCH_AXIS` =32 , `HAS_PITCH_FOLLOW` =64 , `HAS_PITCH_LOCK` =128  
,  
`HAS_YAW_AXIS` =256 , `HAS_YAW_FOLLOW` =512 , `HAS_YAW_LOCK` =1024 , `SUPPORTS_INFINITE_↵`  
`_YAW` =2048 ,  
`SUPPORTS_YAW_IN_EARTH_FRAME` =4096 , `HAS_RC_INPUTS` =8192 , `CAN_POINT_LOCATION_↵`  
`LOCAL` =65536 , `CAN_POINT_LOCATION_GLOBAL` =131072 }

*Gimbal manager high level capability flags (bitmap).*

- enum class `mavlink::common::GIMBAL_DEVICE_FLAGS` : `uint16_t` {  
`RETRACT` =1 , `NEUTRAL` =2 , `ROLL_LOCK` =4 , `PITCH_LOCK` =8 ,  
`YAW_LOCK` =16 , `YAW_IN_VEHICLE_FRAME` =32 , `YAW_IN_EARTH_FRAME` =64 , `ACCEPTS_YAW_↵`  
`IN_EARTH_FRAME` =128 ,  
`RC_EXCLUSIVE` =256 , `RC_MIXED` =512 }

*Flags for gimbal device (lower level) operation.*

- enum class `mavlink::common::GIMBAL_MANAGER_FLAGS` : `uint32_t` {  
**RETRACT** =1 , **NEUTRAL** =2 , **ROLL\_LOCK** =4 , **PITCH\_LOCK** =8 ,  
**YAW\_LOCK** =16 , **YAW\_IN\_VEHICLE\_FRAME** =32 , **YAW\_IN\_EARTH\_FRAME** =64 , **ACCEPTS\_YAW\_↵**  
**IN\_EARTH\_FRAME** =128 ,  
**RC\_EXCLUSIVE** =256 , **RC\_MIXED** =512 }  
*Flags for high level gimbal manager operation The first 16 bits are identical to the GIMBAL\_DEVICE\_FLAGS.*
- enum class `mavlink::common::GIMBAL_DEVICE_ERROR_FLAGS` : `uint32_t` {  
**AT\_ROLL\_LIMIT** =1 , **AT\_PITCH\_LIMIT** =2 , **AT\_YAW\_LIMIT** =4 , **ENCODER\_ERROR** =8 ,  
**POWER\_ERROR** =16 , **MOTOR\_ERROR** =32 , **SOFTWARE\_ERROR** =64 , **COMMS\_ERROR** =128 ,  
**CALIBRATION\_RUNNING** =256 , **NO\_MANAGER** =512 }  
*Gimbal device (low level) error flags (bitmap, 0 means no error)*
- enum class `mavlink::common::GRIPPER_ACTIONS` { **ACTION\_RELEASE** =0 , **ACTION\_GRAB** =1 ,  
**ACTION\_HOLD** =2 }  
*Gripper actions.*
- enum class `mavlink::common::WINCH_ACTIONS` {  
**RELAXED** =0 , **RELATIVE\_LENGTH\_CONTROL** =1 , **RATE\_CONTROL** =2 , **LOCK** =3 ,  
**DELIVER** =4 , **HOLD** =5 , **RETRACT** =6 , **LOAD\_LINE** =7 ,  
**ABANDON\_LINE** =8 , **LOAD\_PAYLOAD** =9 }  
*Winch actions.*
- enum class `mavlink::common::UAVCAN_NODE_HEALTH` : `uint8_t` { **OK** =0 , **WARNING** =1 , **ERROR** =2 ,  
**CRITICAL** =3 }  
*Generalized UAVCAN node health.*
- enum class `mavlink::common::UAVCAN_NODE_MODE` : `uint8_t` {  
**OPERATIONAL** =0 , **INITIALIZATION** =1 , **MAINTENANCE** =2 , **SOFTWARE\_UPDATE** =3 ,  
**OFFLINE** =7 }  
*Generalized UAVCAN node mode.*
- enum class `mavlink::common::ESC_CONNECTION_TYPE` : `uint8_t` {  
**PPM** =0 , **SERIAL** =1 , **ONESHOT** =2 , **I2C** =3 ,  
**CAN** =4 , **DSHOT** =5 }  
*Indicates the ESC connection type.*
- enum class `mavlink::common::ESC_FAILURE_FLAGS` : `uint16_t` {  
**OVER\_CURRENT** =1 , **OVER\_VOLTAGE** =2 , **OVER\_TEMPERATURE** =4 , **OVER\_RPM** =8 ,  
**INCONSISTENT\_CMD** =16 , **MOTOR\_STUCK** =32 , **GENERIC** =64 }  
*Flags to report ESC failures.*
- enum class `mavlink::common::STORAGE_STATUS` : `uint8_t` { **EMPTY** =0 , **UNFORMATTED** =1 , **READY**  
=2 , **NOT\_SUPPORTED** =3 }  
*Flags to indicate the status of camera storage.*
- enum class `mavlink::common::STORAGE_TYPE` : `uint8_t` {  
**UNKNOWN** =0 , **USB\_STICK** =1 , **SD** =2 , **MICROSD** =3 ,  
**CF** =4 , **CFE** =5 , **XQD** =6 , **HD** =7 ,  
**OTHER** =254 }  
*Flags to indicate the type of storage.*
- enum class `mavlink::common::STORAGE_USAGE_FLAG` : `uint8_t` { **SET** =1 , **PHOTO** =2 , **VIDEO** =4 ,  
**LOGS** =8 }  
*Flags to indicate usage for a particular storage (see STORAGE\_INFORMATION.storage\_usage and MAV\_CMD\_↵*  
*SET\_STORAGE\_USAGE).*
- enum class `mavlink::common::ORBIT_YAW_BEHAVIOUR` {  
**HOLD\_FRONT\_TO\_CIRCLE\_CENTER** =0 , **HOLD\_INITIAL\_HEADING** =1 , **UNCONTROLLED** =2 ,  
**HOLD\_FRONT\_TANGENT\_TO\_CIRCLE** =3 ,  
**RC\_CONTROLLED** =4 , **UNCHANGED** =5 }  
*Yaw behaviour during orbit flight.*
- enum class `mavlink::common::WIFI_CONFIG_AP_RESPONSE` : `int8_t` {  
**UNDEFINED** =0 , **ACCEPTED** =1 , **REJECTED** =2 , **MODE\_ERROR** =3 ,  
**SSID\_ERROR** =4 , **PASSWORD\_ERROR** =5 }  
*Possible responses from a WIFI\_CONFIG\_AP message.*

- enum class `mavlink::common::CELLULAR_CONFIG_RESPONSE` : `uint8_t` {  
**ACCEPTED** =0 , **APN\_ERROR** =1 , **PIN\_ERROR** =2 , **REJECTED** =3 ,  
**BLOCKED\_PUK\_REQUIRED** =4 }  
*Possible responses from a CELLULAR\_CONFIG message.*
- enum class `mavlink::common::WIFI_CONFIG_AP_MODE` : `int8_t` { **UNDEFINED** =0 , **AP** =1 , **STATION** =2 ,  
**DISABLED** =3 }  
*WiFi Mode.*
- enum class `mavlink::common::COMP_METADATA_TYPE` {  
**GENERAL** =0 , **PARAMETER** =1 , **COMMANDS** =2 , **PERIPHERALS** =3 ,  
**EVENTS** =4 , **ACTUATORS** =5 }  
*Supported component metadata types.*
- enum class `mavlink::common::ACTUATOR_CONFIGURATION` {  
**NONE** =0 , **BEEP** =1 , **CONFIGURATION\_3D\_MODE\_ON** =2 , **CONFIGURATION\_3D\_MODE\_OFF** =3 ,  
**SPIN\_DIRECTION1** =4 , **SPIN\_DIRECTION2** =5 }  
*Actuator configuration, used to change a setting on an actuator.*
- enum class `mavlink::common::ACTUATOR_OUTPUT_FUNCTION` {  
**NONE** =0 , **MOTOR1** =1 , **MOTOR2** =2 , **MOTOR3** =3 ,  
**MOTOR4** =4 , **MOTOR5** =5 , **MOTOR6** =6 , **MOTOR7** =7 ,  
**MOTOR8** =8 , **MOTOR9** =9 , **MOTOR10** =10 , **MOTOR11** =11 ,  
**MOTOR12** =12 , **MOTOR13** =13 , **MOTOR14** =14 , **MOTOR15** =15 ,  
**MOTOR16** =16 , **SERVO1** =33 , **SERVO2** =34 , **SERVO3** =35 ,  
**SERVO4** =36 , **SERVO5** =37 , **SERVO6** =38 , **SERVO7** =39 ,  
**SERVO8** =40 , **SERVO9** =41 , **SERVO10** =42 , **SERVO11** =43 ,  
**SERVO12** =44 , **SERVO13** =45 , **SERVO14** =46 , **SERVO15** =47 ,  
**SERVO16** =48 }  
*Actuator output function.*
- enum class `mavlink::common::AUTOTUNE_AXIS` { **ROLL** =1 , **PITCH** =2 , **YAW** =4 }  
*Axes that will be autotuned by MAV\_CMD\_DO\_AUTOTUNE\_ENABLE.*
- enum class `mavlink::common::PREFLIGHT_STORAGE_PARAMETER_ACTION` {  
**PARAM\_READ\_PERSISTENT** =0 , **PARAM\_WRITE\_PERSISTENT** =1 , **PARAM\_RESET\_CONFIG** ←  
**DEFAULT** =2 , **PARAM\_RESET\_SENSOR\_DEFAULT** =3 ,  
**PARAM\_RESET\_ALL\_DEFAULT** =4 }  
*Actions for reading/writing parameters between persistent and volatile storage when using MAV\_CMD\_←  
PREFLIGHT\_STORAGE.*
- enum class `mavlink::common::PREFLIGHT_STORAGE_MISSION_ACTION` { **MISSION\_READ** ←  
**PERSISTENT** =0 , **MISSION\_WRITE\_PERSISTENT** =1 , **MISSION\_RESET\_DEFAULT** =2 }  
*Actions for reading and writing plan information (mission, rally points, geofence) between persistent and volatile  
storage when using MAV\_CMD\_PREFLIGHT\_STORAGE.*
- enum class `mavlink::common::REBOOT_SHUTDOWN_ACTION` {  
**NONE** =0 , **REBOOT** =1 , **SHUTDOWN** =2 , **REBOOT\_TO\_BOOTLOADER** =3 ,  
**POWER\_ON** =4 }  
*Reboot/shutdown action for selected component in MAV\_CMD\_PREFLIGHT\_REBOOT\_SHUTDOWN.*
- enum class `mavlink::common::REBOOT_SHUTDOWN_CONDITIONS` { **SAFETY\_INTERLOCKED** =0 ,  
**FORCE** =20190226 }  
*Specifies the conditions under which the MAV\_CMD\_PREFLIGHT\_REBOOT\_SHUTDOWN command should be ac-  
cepted.*
- enum class `mavlink::common::MAV_CMD` : `uint16_t` {  
**NAV\_WAYPOINT** =16 , **NAV\_LOITER\_UNLIM** =17 , **NAV\_LOITER\_TURNS** =18 , **NAV\_LOITER\_TIME** =19  
, **NAV\_RETURN\_TO\_LAUNCH** =20 , **NAV\_LAND** =21 , **NAV\_TAKEOFF** =22 , **NAV\_LAND\_LOCAL** =23 ,  
**NAV\_TAKEOFF\_LOCAL** =24 , **NAV\_FOLLOW** =25 , **NAV\_CONTINUE\_AND\_CHANGE\_ALT** =30 , **NAV\_←  
\_LOITER\_TO\_ALT** =31 ,  
**DO\_FOLLOW** =32 , **DO\_FOLLOW\_REPOSITION** =33 , **DO\_ORBIT** =34 , **DO\_FIGURE\_EIGHT** =35 ,  
**NAV\_ROI** =80 , **NAV\_PATHPLANNING** =81 , **NAV\_SPLINE\_WAYPOINT** =82 , **NAV\_VTOL\_TAKEOFF** =84 ,  
**NAV\_VTOL\_LAND** =85 , **NAV\_GUIDED\_ENABLE** =92 , **NAV\_DELAY** =93 , **NAV\_PAYLOAD\_PLACE** =94 ,  
**NAV\_LAST** =95 , **CONDITION\_DELAY** =112 , **CONDITION\_CHANGE\_ALT** =113 , **CONDITION\_DISTANCE**

```

=114 ,
CONDITION_YAW =115 , CONDITION_LAST =159 , DO_SET_MODE =176 , DO_JUMP =177 ,
DO_CHANGE_SPEED =178 , DO_SET_HOME =179 , DO_SET_PARAMETER =180 , DO_SET_RELAY
=181 ,
DO_REPEAT_RELAY =182 , DO_SET_SERVO =183 , DO_REPEAT_SERVO =184 , DO_FLIGHTTERMINATION
=185 ,
DO_CHANGE_ALTITUDE =186 , DO_SET_ACTUATOR =187 , DO_RETURN_PATH_START =188 , DO↵
_LAND_START =189 ,
DO_RALLY_LAND =190 , DO_GO_AROUND =191 , DO_REPOSITION =192 , DO_PAUSE_CONTINUE
=193 ,
DO_SET_REVERSE =194 , DO_SET_ROI_LOCATION =195 , DO_SET_ROI_WPNEXT_OFFSET =196 ,
DO_SET_ROI_NONE =197 ,
DO_SET_ROI_SYSID =198 , DO_CONTROL_VIDEO =200 , DO_SET_ROI =201 , DO_DIGICAM↵
CONFIGURE =202 ,
DO_DIGICAM_CONTROL =203 , DO_MOUNT_CONFIGURE =204 , DO_MOUNT_CONTROL =205 , DO↵
_SET_CAM_TRIGG_DIST =206 ,
DO_FENCE_ENABLE =207 , DO_PARACHUTE =208 , DO_MOTOR_TEST =209 , DO_INVERTED↵
FLIGHT =210 ,
DO_GRIPPER =211 , DO_AUTOTUNE_ENABLE =212 , NAV_SET_YAW_SPEED =213 , DO_SET_CAM↵
_TRIGG_INTERVAL =214 ,
DO_MOUNT_CONTROL_QUAT =220 , DO_GUIDED_MASTER =221 , DO_GUIDED_LIMITS =222 , DO↵
_ENGINE_CONTROL =223 ,
DO_SET_MISSION_CURRENT =224 , DO_LAST =240 , PREFLIGHT_CALIBRATION =241 , PREFLIGHT↵
_SET_SENSOR_OFFSETS =242 ,
PREFLIGHT_UAVCAN =243 , PREFLIGHT_STORAGE =245 , PREFLIGHT_REBOOT_SHUTDOWN =246
, OVERRIDE_GOTO =252 ,
OBLIQUE_SURVEY =260 , DO_SET_STANDARD_MODE =262 , MISSION_START =300 , ACTUATOR↵
TEST =310 ,
CONFIGURE_ACTUATOR =311 , COMPONENT_ARM_DISARM =400 , RUN_PREARM_CHECKS =401 ,
ILLUMINATOR_ON_OFF =405 ,
DO_ILLUMINATOR_CONFIGURE =406 , GET_HOME_POSITION =410 , INJECT_FAILURE =420 ,
START_RX_PAIR =500 ,
GET_MESSAGE_INTERVAL =510 , SET_MESSAGE_INTERVAL =511 , REQUEST_MESSAGE =512 ,
REQUEST_PROTOCOL_VERSION =519 ,
REQUEST_AUTOPILOT_CAPABILITIES =520 , REQUEST_CAMERA_INFORMATION =521 , REQUEST↵
_CAMERA_SETTINGS =522 , REQUEST_STORAGE_INFORMATION =525 ,
STORAGE_FORMAT =526 , REQUEST_CAMERA_CAPTURE_STATUS =527 , REQUEST_FLIGHT↵
INFORMATION =528 , RESET_CAMERA_SETTINGS =529 ,
SET_CAMERA_MODE =530 , SET_CAMERA_ZOOM =531 , SET_CAMERA_FOCUS =532 , SET↵
_STORAGE_USAGE =533 ,
SET_CAMERA_SOURCE =534 , JUMP_TAG =600 , DO_JUMP_TAG =601 , DO_GIMBAL_MANAGER↵
PITCHYAW =1000 ,
DO_GIMBAL_MANAGER_CONFIGURE =1001 , IMAGE_START_CAPTURE =2000 , IMAGE_STOP↵
CAPTURE =2001 , REQUEST_CAMERA_IMAGE_CAPTURE =2002 ,
DO_TRIGGER_CONTROL =2003 , CAMERA_TRACK_POINT =2004 , CAMERA_TRACK_RECTANGLE
=2005 , CAMERA_STOP_TRACKING =2010 ,
VIDEO_START_CAPTURE =2500 , VIDEO_STOP_CAPTURE =2501 , VIDEO_START_STREAMING
=2502 , VIDEO_STOP_STREAMING =2503 ,
REQUEST_VIDEO_STREAM_INFORMATION =2504 , REQUEST_VIDEO_STREAM_STATUS =2505 ,
LOGGING_START =2510 , LOGGING_STOP =2511 ,
AIRFRAME_CONFIGURATION =2520 , CONTROL_HIGH_LATENCY =2600 , PANORAMA_CREATE
=2800 , DO_VTOL_TRANSITION =3000 ,
ARM_AUTHORIZATION_REQUEST =3001 , SET_GUIDED_SUBMODE_STANDARD =4000 , SET↵
_GUIDED_SUBMODE_CIRCLE =4001 , CONDITION_GATE =4501 ,
NAV_FENCE_RETURN_POINT =5000 , NAV_FENCE_POLYGON_VERTEX_INCLUSION =5001 , NAV↵
_FENCE_POLYGON_VERTEX_EXCLUSION =5002 , NAV_FENCE_CIRCLE_INCLUSION =5003 ,
NAV_FENCE_CIRCLE_EXCLUSION =5004 , NAV_RALLY_POINT =5100 , UAVCAN_GET_NODE_INFO
=5200 , DO_SET_SAFETY_SWITCH_STATE =5300 ,

```

```
DO_ADSB_OUT_IDENT =10001 , PAYLOAD_PREPARE_DEPLOY =30001 , PAYLOAD_CONTROL_↵
DEPLOY =30002 , WAYPOINT_USER_1 =31000 ,
WAYPOINT_USER_2 =31001 , WAYPOINT_USER_3 =31002 , WAYPOINT_USER_4 =31003 ,
WAYPOINT_USER_5 =31004 ,
SPATIAL_USER_1 =31005 , SPATIAL_USER_2 =31006 , SPATIAL_USER_3 =31007 , SPATIAL_USER_4
=31008 ,
SPATIAL_USER_5 =31009 , USER_1 =31010 , USER_2 =31011 , USER_3 =31012 ,
USER_4 =31013 , USER_5 =31014 , CAN_FORWARD =32000 , FIXED_MAG_CAL_YAW =42006 ,
DO_WINCH =42600 , EXTERNAL_POSITION_ESTIMATE =43003 }
```

*Commands to be executed by the MAV.*

- enum class [mavlink::common::MAV\\_DATA\\_STREAM](#) : uint8\_t {  
**ALL** =0 , **RAW\_SENSORS** =1 , **EXTENDED\_STATUS** =2 , **RC\_CHANNELS** =3 ,  
**RAW\_CONTROLLER** =4 , **POSITION** =6 , **EXTRA1** =10 , **EXTRA2** =11 ,  
**EXTRA3** =12 }

*A data stream is not a fixed set of messages, but rather a recommendation to the autopilot software.*

- enum class [mavlink::common::MAV\\_ROI](#) {  
**NONE** =0 , **WPNEXT** =1 , **WPINDEX** =2 , **LOCATION** =3 ,  
**TARGET** =4 }

*The ROI (region of interest) for the vehicle.*

- enum class [mavlink::common::MAV\\_PARAM\\_TYPE](#) : uint8\_t {  
**UINT8** =1 , **INT8** =2 , **UINT16** =3 , **INT16** =4 ,  
**UINT32** =5 , **INT32** =6 , **UINT64** =7 , **INT64** =8 ,  
**REAL32** =9 , **REAL64** =10 }

*Specifies the datatype of a MAVLink parameter.*

- enum class [mavlink::common::MAV\\_PARAM\\_ERROR](#) : uint8\_t {  
**NO\_ERROR** =0 , **DOES\_NOT\_EXIST** =1 , **VALUE\_OUT\_OF\_RANGE** =2 , **PERMISSION\_DENIED** =3 ,  
**COMPONENT\_NOT\_FOUND** =4 , **READ\_ONLY** =5 , **TYPE\_UNSUPPORTED** =6 , **TYPE\_MISMATCH** =7 ,  
**READ\_FAIL** =8 }

*Parameter protocol error types (see PARAM\_ERROR).*

- enum class [mavlink::common::MAV\\_PARAM\\_EXT\\_TYPE](#) : uint8\_t {  
**UINT8** =1 , **INT8** =2 , **UINT16** =3 , **INT16** =4 ,  
**UINT32** =5 , **INT32** =6 , **UINT64** =7 , **INT64** =8 ,  
**REAL32** =9 , **REAL64** =10 , **CUSTOM** =11 }

*Specifies the datatype of a MAVLink extended parameter.*

- enum class [mavlink::common::MAV\\_RESULT](#) : uint8\_t {  
**ACCEPTED** =0 , **TEMPORARILY\_REJECTED** =1 , **DENIED** =2 , **UNSUPPORTED** =3 ,  
**FAILED** =4 , **IN\_PROGRESS** =5 , **CANCELLED** =6 , **COMMAND\_LONG\_ONLY** =7 ,  
**COMMAND\_INT\_ONLY** =8 , **COMMAND\_UNSUPPORTED\_MAV\_FRAME** =9 , **NOT\_IN\_CONTROL** =10 }

*Result from a MAVLink command (MAV\_CMD)*

- enum class [mavlink::common::MAV\\_MISSION\\_RESULT](#) : uint8\_t {  
**ACCEPTED** =0 , **ERROR** =1 , **UNSUPPORTED\_FRAME** =2 , **UNSUPPORTED** =3 ,  
**NO\_SPACE** =4 , **INVALID** =5 , **INVALID\_PARAM1** =6 , **INVALID\_PARAM2** =7 ,  
**INVALID\_PARAM3** =8 , **INVALID\_PARAM4** =9 , **INVALID\_PARAM5\_X** =10 , **INVALID\_PARAM6\_Y** =11 ,  
**INVALID\_PARAM7** =12 , **INVALID\_SEQUENCE** =13 , **DENIED** =14 , **OPERATION\_CANCELLED** =15 }

*Result of mission operation (in a MISSION\_ACK message).*

- enum class [mavlink::common::MAV\\_SEVERITY](#) : uint8\_t {  
**EMERGENCY** =0 , **ALERT** =1 , **CRITICAL** =2 , **ERROR** =3 ,  
**WARNING** =4 , **NOTICE** =5 , **INFO** =6 , **DEBUG** =7 }

*Indicates the severity level, generally used for status messages to indicate their relative urgency.*

- enum class [mavlink::common::MAV\\_POWER\\_STATUS](#) : uint16\_t {  
**BRICK\_VALID** =1 , **SERVO\_VALID** =2 , **USB\_CONNECTED** =4 , **PERIPH\_OVERCURRENT** =8 ,  
**PERIPH\_HIPOWER\_OVERCURRENT** =16 , **CHANGED** =32 }

*Power supply status flags (bitmask)*

- enum class [mavlink::common::SERIAL\\_CONTROL\\_DEV](#) : uint8\_t {  
**TELEM1** =0 , **TELEM2** =1 , **GPS1** =2 , **GPS2** =3 ,  
**SHELL** =10 , **SERIAL0** =100 , **SERIAL1** =101 , **SERIAL2** =102 ,  
**SERIAL3** =103 , **SERIAL4** =104 , **SERIAL5** =105 , **SERIAL6** =106 ,  
**SERIAL7** =107 , **SERIAL8** =108 , **SERIAL9** =109 }  
*SERIAL\_CONTROL device types.*
- enum class [mavlink::common::SERIAL\\_CONTROL\\_FLAG](#) : uint8\_t {  
**REPLY** =1 , **RESPOND** =2 , **EXCLUSIVE** =4 , **BLOCKING** =8 ,  
**MULTI** =16 }  
*SERIAL\_CONTROL flags (bitmask)*
- enum class [mavlink::common::MAV\\_DISTANCE\\_SENSOR](#) : uint8\_t {  
**LASER** =0 , **ULTRASOUND** =1 , **INFRARED** =2 , **RADAR** =3 ,  
**UNKNOWN** =4 }  
*Enumeration of distance sensor types.*
- enum class [mavlink::common::MAV\\_SENSOR\\_ORIENTATION](#) : uint8\_t {  
**ROTATION\_NONE** =0 , **ROTATION\_YAW\_45** =1 , **ROTATION\_YAW\_90** =2 , **ROTATION\_YAW\_135** =3 ,  
**ROTATION\_YAW\_180** =4 , **ROTATION\_YAW\_225** =5 , **ROTATION\_YAW\_270** =6 , **ROTATION\_YAW\_315** =7 ,  
**ROTATION\_ROLL\_180** =8 , **ROTATION\_ROLL\_180\_YAW\_45** =9 , **ROTATION\_ROLL\_180\_YAW\_90** =10 ,  
**ROTATION\_ROLL\_180\_YAW\_135** =11 ,  
**ROTATION\_PITCH\_180** =12 , **ROTATION\_ROLL\_180\_YAW\_225** =13 , **ROTATION\_ROLL\_180\_YAW\_270**  
=14 , **ROTATION\_ROLL\_180\_YAW\_315** =15 ,  
**ROTATION\_ROLL\_90** =16 , **ROTATION\_ROLL\_90\_YAW\_45** =17 , **ROTATION\_ROLL\_90\_YAW\_90** =18 ,  
**ROTATION\_ROLL\_90\_YAW\_135** =19 ,  
**ROTATION\_ROLL\_270** =20 , **ROTATION\_ROLL\_270\_YAW\_45** =21 , **ROTATION\_ROLL\_270\_YAW\_90** =22 ,  
**ROTATION\_ROLL\_270\_YAW\_135** =23 ,  
**ROTATION\_PITCH\_90** =24 , **ROTATION\_PITCH\_270** =25 , **ROTATION\_PITCH\_180\_YAW\_90** =26 ,  
**ROTATION\_PITCH\_180\_YAW\_270** =27 ,  
**ROTATION\_ROLL\_90\_PITCH\_90** =28 , **ROTATION\_ROLL\_180\_PITCH\_90** =29 , **ROTATION\_ROLL\_270\_↵**  
\_PITCH\_90 =30 , **ROTATION\_ROLL\_90\_PITCH\_180** =31 ,  
**ROTATION\_ROLL\_270\_PITCH\_180** =32 , **ROTATION\_ROLL\_90\_PITCH\_270** =33 , **ROTATION\_ROLL\_↵**  
180\_PITCH\_270 =34 , **ROTATION\_ROLL\_270\_PITCH\_270** =35 ,  
**ROTATION\_ROLL\_90\_PITCH\_180\_YAW\_90** =36 , **ROTATION\_ROLL\_90\_YAW\_270** =37 , **ROTATION\_↵**  
ROLL\_90\_PITCH\_68\_YAW\_293 =38 , **ROTATION\_PITCH\_315** =39 ,  
**ROTATION\_ROLL\_90\_PITCH\_315** =40 , **ROTATION\_CUSTOM** =100 }  
*Enumeration of sensor orientation, according to its rotations.*
- enum class [mavlink::common::MAV\\_MISSION\\_TYPE](#) : uint8\_t { **MISSION** =0 , **FENCE** =1 , **RALLY** =2 , **ALL**  
=255 }  
*Type of mission items being requested/sent in mission protocol.*
- enum class [mavlink::common::MAV\\_ESTIMATOR\\_TYPE](#) : uint8\_t {  
**UNKNOWN** =0 , **NAIVE** =1 , **VISION** =2 , **VIO** =3 ,  
**GPS** =4 , **GPS\_INS** =5 , **MOCAP** =6 , **LIDAR** =7 ,  
**AUTOPILOT** =8 }  
*Enumeration of estimator types.*
- enum class [mavlink::common::MAV\\_BATTERY\\_TYPE](#) : uint8\_t {  
**UNKNOWN** =0 , **LIPO** =1 , **LIFE** =2 , **LION** =3 ,  
**NIMH** =4 }  
*Enumeration of battery types.*
- enum class [mavlink::common::MAV\\_BATTERY\\_FUNCTION](#) : uint8\_t {  
**UNKNOWN** =0 , **ALL** =1 , **PROPULSION** =2 , **AVIONICS** =3 ,  
**PAYLOAD** =4 }  
*Enumeration of battery functions.*
- enum class [mavlink::common::MAV\\_BATTERY\\_CHARGE\\_STATE](#) : uint8\_t {  
**UNDEFINED** =0 , **OK** =1 , **LOW** =2 , **CRITICAL** =3 ,  
**EMERGENCY** =4 , **FAILED** =5 , **UNHEALTHY** =6 , **CHARGING** =7 }  
*Enumeration for battery charge states.*

- enum class `mavlink::common::MAV_BATTERY_MODE` : `uint8_t` { `UNKNOWN` =0 , `AUTO_DISCHARGING` =1 , `HOT_SWAP` =2 }  
*Battery mode.*
- enum class `mavlink::common::MAV_BATTERY_FAULT` : `uint32_t` { `DEEP_DISCHARGE` =1 , `SPIKES` =2 , `CELL_FAIL` =4 , `OVER_CURRENT` =8 , `OVER_TEMPERATURE` =16 , `UNDER_TEMPERATURE` =32 , `INCOMPATIBLE_VOLTAGE` =64 , `INCOMPATIBLE_FIRMWARE` =128 , `BATTERY_FAULT_INCOMPATIBLE_CELLS_CONFIGURATION` =256 }  
*Smart battery supply status/fault flags (bitmask) for health indication.*
- enum class `mavlink::common::MAV_FUEL_TYPE` : `uint32_t` { `UNKNOWN` =0 , `LIQUID` =1 , `GAS` =2 }  
*Fuel types for use in FUEL\_TYPE.*
- enum class `mavlink::common::MAV_GENERATOR_STATUS_FLAG` : `uint64_t` { `OFF` =1 , `READY` =2 , `GENERATING` =4 , `CHARGING` =8 , `REDUCED_POWER` =16 , `MAXPOWER` =32 , `OVERTEMP_WARNING` =64 , `OVERTEMP_FAULT` =128 , `ELECTRONICS_OVERTEMP_WARNING` =256 , `ELECTRONICS_OVERTEMP_FAULT` =512 , `ELECTRONICS_FAULT` =1024 , `POWERSOURCE_FAULT` =2048 , `COMMUNICATION_WARNING` =4096 , `COOLING_WARNING` =8192 , `POWER_RAIL_FAULT` =16384 , `OVERCURRENT_FAULT` =32768 , `BATTERY_OVERCHARGE_CURRENT_FAULT` =65536 , `OVERVOLTAGE_FAULT` =131072 , `BATTERY_UNDERVOLT_FAULT` =262144 , `START_INHIBITED` =524288 , `MAINTENANCE_REQUIRED` =1048576 , `WARMING_UP` =2097152 , `IDLE` =4194304 }  
*Flags to report status/failure cases for a power generator (used in GENERATOR\_STATUS).*
- enum class `mavlink::common::MAV_VTOL_STATE` : `uint8_t` { `UNDEFINED` =0 , `TRANSITION_TO_FW` =1 , `TRANSITION_TO_MC` =2 , `MC` =3 , `FW` =4 }  
*Enumeration of VTOL states.*
- enum class `mavlink::common::MAV_LANDED_STATE` : `uint8_t` { `UNDEFINED` =0 , `ON_GROUND` =1 , `IN_AIR` =2 , `TAKEOFF` =3 , `LANDING` =4 }  
*Enumeration of landed detector states.*
- enum class `mavlink::common::ADSB_ALTITUDE_TYPE` : `uint8_t` { `PRESSURE_QNH` =0 , `GEOMETRIC` =1 }  
*Enumeration of the ADSB altimeter types.*
- enum class `mavlink::common::ADSB_EMITTER_TYPE` : `uint8_t` { `NO_INFO` =0 , `LIGHT` =1 , `SMALL` =2 , `LARGE` =3 , `HIGH_VORTEX_LARGE` =4 , `HEAVY` =5 , `HIGHLY_MANUV` =6 , `ROTOCRAFT` =7 , `UNASSIGNED` =8 , `GLIDER` =9 , `LIGHTER_AIR` =10 , `PARACHUTE` =11 , `ULTRA_LIGHT` =12 , `UNASSIGNED2` =13 , `UAV` =14 , `SPACE` =15 , `UNASSIGNED3` =16 , `EMERGENCY_SURFACE` =17 , `SERVICE_SURFACE` =18 , `POINT_OBSTACLE` =19 }  
*ADSB classification for the type of vehicle emitting the transponder signal.*
- enum class `mavlink::common::ADSB_FLAGS` : `uint16_t` { `VALID_COORDS` =1 , `VALID_ALTITUDE` =2 , `VALID_HEADING` =4 , `VALID_VELOCITY` =8 , `VALID_CALLSIGN` =16 , `VALID_SQUAWK` =32 , `SIMULATED` =64 , `VERTICAL_VELOCITY_VALID` =128 , `BARO_VALID` =256 , `SOURCE_UAT` =32768 }  
*These flags indicate status such as data validity of each data source.*
- enum class `mavlink::common::MAV_DO_REPOSITION_FLAGS` { `CHANGE_MODE` =1 , `RELATIVE_YAW` =2 }  
*Bitmap of options for the MAV\_CMD\_DO\_REPOSITION.*
- enum class `mavlink::common::SPEED_TYPE` { `AIRSPEED` =0 , `GROUNDSPEED` =1 , `CLIMB_SPEED` =2 , `DESCENT_SPEED` =3 }  
*Speed setpoint types used in MAV\_CMD\_DO\_CHANGE\_SPEED.*
- enum class `mavlink::common::ESTIMATOR_STATUS_FLAGS` : `uint16_t` { `ATTITUDE` =1 , `VELOCITY_HORIZ` =2 , `VELOCITY_VERT` =4 , `POS_HORIZ_REL` =8 , `POS_HORIZ_ABS` =16 , `POS_VERT_ABS` =32 , `POS_VERT_AGL` =64 , `CONST_POS_MODE` =128 ,



**PRED\_POS\_HORIZ\_REL** =256 , **PRED\_POS\_HORIZ\_ABS** =512 , **GPS\_GLITCH** =1024 , **ACCEL\_ERROR** =2048 }

*Flags in ESTIMATOR\_STATUS message.*

- enum class **mavlink::common::MOTOR\_TEST\_ORDER** { **DEFAULT** =0 , **SEQUENCE** =1 , **BOARD** =2 }  
*Sequence that motors are tested when using MAV\_CMD\_DO\_MOTOR\_TEST.*
- enum class **mavlink::common::MOTOR\_TEST\_THROTTLE\_TYPE** { **PERCENT** =0 , **PWM** =1 , **PILOT** =2 , **COMPASS\_CAL** =3 }  
*Defines how throttle value is represented in MAV\_CMD\_DO\_MOTOR\_TEST.*
- enum class **mavlink::common::GPS\_INPUT\_IGNORE\_FLAGS** : uint16\_t { **FLAG\_ALT** =1 , **FLAG\_HDOP** =2 , **FLAG\_VDOP** =4 , **FLAG\_VEL\_HORIZ** =8 , **FLAG\_VEL\_VERT** =16 , **FLAG\_SPEED\_ACCURACY** =32 , **FLAG\_HORIZONTAL\_ACCURACY** =64 , **FLAG\_VERTICAL\_ACCURACY** =128 }
- enum class **mavlink::common::MAV\_COLLISION\_ACTION** : uint8\_t { **NONE** =0 , **REPORT** =1 , **ASCEND\_OR\_DESCEND** =2 , **MOVE\_HORIZONTALLY** =3 , **MOVE\_PERPENDICULAR** =4 , **RTL** =5 , **HOVER** =6 }  
*Possible actions an aircraft can take to avoid a collision.*
- enum class **mavlink::common::MAV\_COLLISION\_THREAT\_LEVEL** : uint8\_t { **NONE** =0 , **LOW** =1 , **HIGH** =2 }  
*Aircraft-rated danger from this threat.*
- enum class **mavlink::common::MAV\_COLLISION\_SRC** : uint8\_t { **ADSB** =0 , **MAVLINK\_GPS\_GLOBAL\_INT** =1 }  
*Source of information about this collision.*
- enum class **mavlink::common::GPS\_FIX\_TYPE** : uint8\_t { **NO\_GPS** =0 , **NO\_FIX** =1 , **TYPE\_2D\_FIX** =2 , **TYPE\_3D\_FIX** =3 , **DGPS** =4 , **RTK\_FLOAT** =5 , **RTK\_FIXED** =6 , **STATIC** =7 , **PPP** =8 }  
*Type of GPS fix.*
- enum class **mavlink::common::RTK\_BASELINE\_COORDINATE\_SYSTEM** : uint8\_t { **ECEF** =0 , **NED** =1 }  
*RTK GPS baseline coordinate system, used for RTK corrections.*
- enum class **mavlink::common::LANDING\_TARGET\_TYPE** : uint8\_t { **LIGHT\_BEACON** =0 , **RADIO\_BEACON** =1 , **VISION\_FIDUCIAL** =2 , **VISION\_OTHER** =3 }  
*Type of landing target.*
- enum class **mavlink::common::VTOL\_TRANSITION\_HEADING** { **VEHICLE\_DEFAULT** =0 , **NEXT\_WAYPOINT** =1 , **TAKEOFF** =2 , **SPECIFIED** =3 , **ANY** =4 }  
*Direction of VTOL transition.*
- enum class **mavlink::common::CAMERA\_CAP\_FLAGS** : uint32\_t { **CAPTURE\_VIDEO** =1 , **CAPTURE\_IMAGE** =2 , **HAS\_MODES** =4 , **CAN\_CAPTURE\_IMAGE\_IN\_VIDEO\_MODE** =8 , **CAN\_CAPTURE\_VIDEO\_IN\_IMAGE\_MODE** =16 , **HAS\_IMAGE\_SURVEY\_MODE** =32 , **HAS\_BASIC\_ZOOM** =64 , **HAS\_BASIC\_FOCUS** =128 , **HAS\_VIDEO\_STREAM** =256 , **HAS\_TRACKING\_POINT** =512 , **HAS\_TRACKING\_RECTANGLE** =1024 , **HAS\_TRACKING\_GEO\_STATUS** =2048 , **HAS\_THERMAL\_RANGE** =4096 , **HAS\_MTI** =8192 }  
*Camera capability flags (Bitmap)*
- enum class **mavlink::common::VIDEO\_STREAM\_STATUS\_FLAGS** : uint16\_t { **RUNNING** =1 , **THERMAL** =2 , **THERMAL\_RANGE\_ENABLED** =4 }  
*Stream status flags (Bitmap)*
- enum class **mavlink::common::VIDEO\_STREAM\_TYPE** : uint8\_t { **RTSP** =0 , **RTPUDP** =1 , **TCP\_MPEG** =2 , **MPEG\_TS** =3 }  
*Video stream types.*



- enum class [mavlink::common::VIDEO\\_STREAM\\_ENCODING](#) : uint8\_t { **UNKNOWN** =0 , **H264** =1 , **H265** =2 }  
*Video stream encodings.*
- enum class [mavlink::common::CAMERA\\_TRACKING\\_STATUS\\_FLAGS](#) : uint8\_t { **IDLE** =0 , **ACTIVE** =1 , **ERROR** =2 , **MTI** =4 , **COASTING** =8 }  
*Camera tracking status flags.*
- enum class [mavlink::common::CAMERA\\_TRACKING\\_MODE](#) : uint8\_t { **NONE** =0 , **POINT** =1 , **RECTANGLE** =2 }  
*Camera tracking modes.*
- enum class [mavlink::common::CAMERA\\_TRACKING\\_TARGET\\_DATA](#) : uint8\_t { **EMBEDDED** =1 , **RENDERED** =2 , **IN\_STATUS** =4 }  
*Camera tracking target data (shows where tracked target is within image)*
- enum class [mavlink::common::CAMERA\\_ZOOM\\_TYPE](#) { **ZOOM\_TYPE\_STEP** =0 , **ZOOM\_TYPE\_CONTINUOUS** =1 , **ZOOM\_TYPE\_RANGE** =2 , **ZOOM\_TYPE\_FOCAL\_LENGTH** =3 , **ZOOM\_TYPE\_HORIZONTAL\_FOV** =4 }  
*Zoom types for MAV\_CMD\_SET\_CAMERA\_ZOOM.*
- enum class [mavlink::common::SET\\_FOCUS\\_TYPE](#) { **FOCUS\_TYPE\_STEP** =0 , **FOCUS\_TYPE\_CONTINUOUS** =1 , **FOCUS\_TYPE\_RANGE** =2 , **FOCUS\_TYPE\_METERS** =3 , **FOCUS\_TYPE\_AUTO** =4 , **FOCUS\_TYPE\_AUTO\_SINGLE** =5 , **FOCUS\_TYPE\_AUTO\_CONTINUOUS** =6 }  
*Focus types for MAV\_CMD\_SET\_CAMERA\_FOCUS.*
- enum class [mavlink::common::CAMERA\\_SOURCE](#) { **DEFAULT** =0 , **RGB** =1 , **IR** =2 , **NDVI** =3 }  
*Camera sources for MAV\_CMD\_SET\_CAMERA\_SOURCE.*
- enum class [mavlink::common::PARAM\\_ACK](#) : uint8\_t { **ACCEPTED** =0 , **VALUE\_UNSUPPORTED** =1 , **FAILED** =2 , **IN\_PROGRESS** =3 }  
*Result from PARAM\_EXT\_SET message.*
- enum class [mavlink::common::CAMERA\\_MODE](#) : uint8\_t { **IMAGE** =0 , **VIDEO** =1 , **IMAGE\_SURVEY** =2 }  
*Camera Modes.*
- enum class [mavlink::common::MAV\\_ARM\\_AUTH\\_DENIED\\_REASON](#) { **GENERIC** =0 , **NONE** =1 , **INVALID\_WAYPOINT** =2 , **TIMEOUT** =3 , **AIRSPACE\_IN\_USE** =4 , **BAD\_WEATHER** =5 }
- enum class [mavlink::common::RC\\_TYPE](#) { **SPEKTRUM** =0 , **CRSF** =1 }  
*RC type.*
- enum class [mavlink::common::RC\\_SUB\\_TYPE](#) { **SPEKTRUM\_DSM2** =0 , **SPEKTRUM\_DSMX** =1 , **SPEKTRUM\_DSMX8** =2 }  
*RC sub-type of types defined in RC\_TYPE.*
- enum class [mavlink::common::ENGINE\\_CONTROL\\_OPTIONS](#) { **ALLOW\_START\_WHILE\_DISARMED** =1 }  
*Engine control options.*
- enum class [mavlink::common::POSITION\\_TARGET\\_TYEMASK](#) : uint16\_t { **X\_IGNORE** =1 , **Y\_IGNORE** =2 , **Z\_IGNORE** =4 , **VX\_IGNORE** =8 , **VY\_IGNORE** =16 , **VZ\_IGNORE** =32 , **AX\_IGNORE** =64 , **AY\_IGNORE** =128 , **AZ\_IGNORE** =256 , **FORCE\_SET** =512 , **YAW\_IGNORE** =1024 , **YAW\_RATE\_IGNORE** =2048 }  
*Bitmap to indicate which dimensions should be ignored by the vehicle: a value of 0b0000000000000000 or 0b0000001000000000 indicates that none of the setpoint dimensions should be ignored.*
- enum class [mavlink::common::ATTITUDE\\_TARGET\\_TYEMASK](#) : uint8\_t { **BODY\_ROLL\_RATE\_IGNORE** =1 , **BODY\_PITCH\_RATE\_IGNORE** =2 , **BODY\_YAW\_RATE\_IGNORE** =4 , **THRUST\_BODY\_SET** =32 , **THROTTLE\_IGNORE** =64 , **ATTITUDE\_IGNORE** =128 }

Bitmap to indicate which dimensions should be ignored by the vehicle: a value of 0b00000000 indicates that none of the setpoint dimensions should be ignored.

- enum class `mavlink::common::UTM_FLIGHT_STATE` : `uint8_t` {  
**UNKNOWN** =1 , **GROUND** =2 , **AIRBORNE** =3 , **EMERGENCY** =16 ,  
**NOCTRL** =32 }

Airborne status of UAS.

- enum class `mavlink::common::UTM_DATA_AVAIL_FLAGS` : `uint8_t` {  
**TIME\_VALID** =1 , **UAS\_ID\_AVAILABLE** =2 , **POSITION\_AVAILABLE** =4 , **ALTITUDE\_AVAILABLE** =8 ,  
**RELATIVE\_ALTITUDE\_AVAILABLE** =16 , **HORIZONTAL\_VELO\_AVAILABLE** =32 , **VERTICAL\_VELO\_AVAILABLE** =64 , **NEXT\_WAYPOINT\_AVAILABLE** =128 }

Flags for the global position report.

- enum class `mavlink::common::CELLULAR_STATUS_FLAG` : `uint8_t` {  
**UNKNOWN** =0 , **FAILED** =1 , **INITIALIZING** =2 , **LOCKED** =3 ,  
**DISABLED** =4 , **DISABLING** =5 , **ENABLING** =6 , **ENABLED** =7 ,  
**SEARCHING** =8 , **REGISTERED** =9 , **DISCONNECTING** =10 , **CONNECTING** =11 ,  
**CONNECTED** =12 }

These flags encode the cellular network status.

- enum class `mavlink::common::CELLULAR_NETWORK_FAILED_REASON` : `uint8_t` { **NONE** =0 , **UNKNOWN** =1 , **SIM\_MISSING** =2 , **SIM\_ERROR** =3 }

These flags are used to diagnose the failure state of `CELLULAR_STATUS`.

- enum class `mavlink::common::CELLULAR_NETWORK_RADIO_TYPE` : `uint8_t` {  
**NONE** =0 , **GSM** =1 , **CDMA** =2 , **WCDMA** =3 ,  
**LTE** =4 }

Cellular network radio type.

- enum class `mavlink::common::PRECISION_LAND_MODE` { **DISABLED** =0 , **OPPORTUNISTIC** =1 , **REQUIRED** =2 }

Precision land modes (used in `MAV_CMD_NAV_LAND`).

- enum class `mavlink::common::PARACHUTE_ACTION` { **DISABLE** =0 , **ENABLE** =1 , **RELEASE** =2 }

Parachute actions.

- enum class `mavlink::common::MAV_TUNNEL_PAYLOAD_TYPE` : `uint16_t` {  
**UNKNOWN** =0 , **STORM32\_RESERVED0** =200 , **STORM32\_RESERVED1** =201 , **STORM32\_RESERVED2** =202 ,  
**STORM32\_RESERVED3** =203 , **STORM32\_RESERVED4** =204 , **STORM32\_RESERVED5** =205 ,  
**STORM32\_RESERVED6** =206 ,  
**STORM32\_RESERVED7** =207 , **STORM32\_RESERVED8** =208 , **STORM32\_RESERVED9** =209 ,  
**MODALAI\_REMOTE\_OSD** =210 ,  
**MODALAI\_ESC\_UART\_PASSTHRU** =211 , **MODALAI\_IO\_UART\_PASSTHRU** =212 }

- enum class `mavlink::common::MAV_ODID_ID_TYPE` : `uint8_t` {  
**NONE** =0 , **SERIAL\_NUMBER** =1 , **CAA\_REGISTRATION\_ID** =2 , **UTM\_ASSIGNED\_UUID** =3 ,  
**SPECIFIC\_SESSION\_ID** =4 }

- enum class `mavlink::common::MAV_ODID_UA_TYPE` : `uint8_t` {  
**NONE** =0 , **AEROPLANE** =1 , **HELICOPTER\_OR\_MULTIROTOR** =2 , **GYROPLANE** =3 ,  
**HYBRID\_LIFT** =4 , **ORNITHOPTER** =5 , **GLIDER** =6 , **KITE** =7 ,  
**FREE\_BALLOON** =8 , **CAPTIVE\_BALLOON** =9 , **AIRSHIP** =10 , **FREE\_FALL\_PARACHUTE** =11 ,  
**ROCKET** =12 , **TETHERED\_POWERED\_AIRCRAFT** =13 , **GROUND\_OBSACLE** =14 , **OTHER** =15 }

- enum class `mavlink::common::MAV_ODID_STATUS` : `uint8_t` {  
**UNDECLARED** =0 , **GROUND** =1 , **AIRBORNE** =2 , **EMERGENCY** =3 ,  
**REMOTE\_ID\_SYSTEM\_FAILURE** =4 }

- enum class `mavlink::common::MAV_ODID_HEIGHT_REF` : `uint8_t` { `OVER_TAKEOFF` =0 , `OVER_GROUND` =1 }
- enum class `mavlink::common::MAV_ODID_HOR_ACC` : `uint8_t` { `UNKNOWN` =0 , `ACC_10NM` =1 , `ACC_4NM` =2 , `ACC_2NM` =3 , `ACC_1NM` =4 , `ACC_0_5NM` =5 , `ACC_0_3NM` =6 , `ACC_0_1NM` =7 , `ACC_0_05NM` =8 , `ACC_30_METER` =9 , `ACC_10_METER` =10 , `ACC_3_METER` =11 , `ACC_1_METER` =12 }
- enum class `mavlink::common::MAV_ODID_VER_ACC` : `uint8_t` { `UNKNOWN` =0 , `ACC_150_METER` =1 , `ACC_45_METER` =2 , `ACC_25_METER` =3 , `ACC_10_METER` =4 , `ACC_3_METER` =5 , `ACC_1_METER` =6 }
- enum class `mavlink::common::MAV_ODID_SPEED_ACC` : `uint8_t` { `UNKNOWN` =0 , `ACC_10_METERS_PER_SECOND` =1 , `ACC_3_METERS_PER_SECOND` =2 , `ACC_1_METERS_PER_SECOND` =3 , `ACC_0_3_METERS_PER_SECOND` =4 }
- enum class `mavlink::common::MAV_ODID_TIME_ACC` : `uint8_t` { `UNKNOWN` =0 , `ACC_0_1_SECOND` =1 , `ACC_0_2_SECOND` =2 , `ACC_0_3_SECOND` =3 , `ACC_0_4_SECOND` =4 , `ACC_0_5_SECOND` =5 , `ACC_0_6_SECOND` =6 , `ACC_0_7_SECOND` =7 , `ACC_0_8_SECOND` =8 , `ACC_0_9_SECOND` =9 , `ACC_1_0_SECOND` =10 , `ACC_1_1_SECOND` =11 , `ACC_1_2_SECOND` =12 , `ACC_1_3_SECOND` =13 , `ACC_1_4_SECOND` =14 , `ACC_1_5_SECOND` =15 }
- enum class `mavlink::common::MAV_ODID_AUTH_TYPE` : `uint8_t` { `NONE` =0 , `UAS_ID_SIGNATURE` =1 , `OPERATOR_ID_SIGNATURE` =2 , `MESSAGE_SET_SIGNATURE` =3 , `NETWORK_REMOTE_ID` =4 , `SPECIFIC_AUTHENTICATION` =5 }
- enum class `mavlink::common::MAV_ODID_DESC_TYPE` : `uint8_t` { `TEXT` =0 , `EMERGENCY` =1 , `EXTENDED_STATUS` =2 }
- enum class `mavlink::common::MAV_ODID_OPERATOR_LOCATION_TYPE` : `uint8_t` { `TAKEOFF` =0 , `LIVE_GNSS` =1 , `FIXED` =2 }
- enum class `mavlink::common::MAV_ODID_CLASSIFICATION_TYPE` : `uint8_t` { `UNDECLARED` =0 , `EU` =1 }
- enum class `mavlink::common::MAV_ODID_CATEGORY_EU` : `uint8_t` { `UNDECLARED` =0 , `OPEN` =1 , `SPECIFIC` =2 , `CERTIFIED` =3 }
- enum class `mavlink::common::MAV_ODID_CLASS_EU` : `uint8_t` { `UNDECLARED` =0 , `CLASS_0` =1 , `CLASS_1` =2 , `CLASS_2` =3 , `CLASS_3` =4 , `CLASS_4` =5 , `CLASS_5` =6 , `CLASS_6` =7 }

- enum class `mavlink::common::MAV_ODID_OPERATOR_ID_TYPE` : `uint8_t` { `CAA` =0 }
- enum class `mavlink::common::MAV_ODID_ARM_STATUS` : `uint8_t` { `GOOD_TO_ARM` =0 , `PRE_ARM` ← `FAIL_GENERIC` =1 }
- enum class `mavlink::common::TUNE_FORMAT` : `uint32_t` { `QBASIC1_1` =1 , `MML_MODERN` =2 }  
*Tune formats (used for vehicle buzzer/tone generation).*
- enum class `mavlink::common::AIS_TYPE` : `uint8_t` {  
`UNKNOWN` =0 , `RESERVED_1` =1 , `RESERVED_2` =2 , `RESERVED_3` =3 ,  
`RESERVED_4` =4 , `RESERVED_5` =5 , `RESERVED_6` =6 , `RESERVED_7` =7 ,  
`RESERVED_8` =8 , `RESERVED_9` =9 , `RESERVED_10` =10 , `RESERVED_11` =11 ,  
`RESERVED_12` =12 , `RESERVED_13` =13 , `RESERVED_14` =14 , `RESERVED_15` =15 ,  
`RESERVED_16` =16 , `RESERVED_17` =17 , `RESERVED_18` =18 , `RESERVED_19` =19 ,  
`WIG` =20 , `WIG_HAZARDOUS_A` =21 , `WIG_HAZARDOUS_B` =22 , `WIG_HAZARDOUS_C` =23 ,  
`WIG_HAZARDOUS_D` =24 , `WIG_RESERVED_1` =25 , `WIG_RESERVED_2` =26 , `WIG_RESERVED_3` =27 ,  
, `WIG_RESERVED_4` =28 , `WIG_RESERVED_5` =29 , `FISHING` =30 , `TOWING` =31 ,  
`TOWING_LARGE` =32 , `DREDGING` =33 , `DIVING` =34 , `MILITARY` =35 ,  
`SAILING` =36 , `PLEASURE` =37 , `RESERVED_20` =38 , `RESERVED_21` =39 ,  
`HSC` =40 , `HSC_HAZARDOUS_A` =41 , `HSC_HAZARDOUS_B` =42 , `HSC_HAZARDOUS_C` =43 ,  
`HSC_HAZARDOUS_D` =44 , `HSC_RESERVED_1` =45 , `HSC_RESERVED_2` =46 , `HSC_RESERVED_3` =47 ,  
`HSC_RESERVED_4` =48 , `HSC_UNKNOWN` =49 , `PILOT` =50 , `SAR` =51 ,  
`TUG` =52 , `PORT_TENDER` =53 , `ANTI_POLLUTION` =54 , `LAW_ENFORCEMENT` =55 ,  
`SPARE_LOCAL_1` =56 , `SPARE_LOCAL_2` =57 , `MEDICAL_TRANSPORT` =58 , `NONECOMBATANT` =59 ,  
, `PASSENGER` =60 , `PASSENGER_HAZARDOUS_A` =61 , `PASSENGER_HAZARDOUS_B` =62 ,  
`PASSENGER_HAZARDOUS_C` =63 ,  
`PASSENGER_HAZARDOUS_D` =64 , `PASSENGER_RESERVED_1` =65 , `PASSENGER_RESERVED_2` =66 ,  
`PASSENGER_RESERVED_3` =67 ,  
`PASSENGER_RESERVED_4` =68 , `PASSENGER_UNKNOWN` =69 , `CARGO` =70 , `CARGO` ← `HAZARDOUS_A` =71 ,  
`CARGO_HAZARDOUS_B` =72 , `CARGO_HAZARDOUS_C` =73 , `CARGO_HAZARDOUS_D` =74 ,  
`CARGO_RESERVED_1` =75 ,  
`CARGO_RESERVED_2` =76 , `CARGO_RESERVED_3` =77 , `CARGO_RESERVED_4` =78 , `CARGO` ← `UNKNOWN` =79 ,  
`TANKER` =80 , `TANKER_HAZARDOUS_A` =81 , `TANKER_HAZARDOUS_B` =82 , `TANKER` ← `HAZARDOUS_C` =83 ,  
`TANKER_HAZARDOUS_D` =84 , `TANKER_RESERVED_1` =85 , `TANKER_RESERVED_2` =86 , `TANKER` ← `RESERVED_3` =87 ,  
`TANKER_RESERVED_4` =88 , `TANKER_UNKNOWN` =89 , `OTHER` =90 , `OTHER_HAZARDOUS_A` =91 ,  
`OTHER_HAZARDOUS_B` =92 , `OTHER_HAZARDOUS_C` =93 , `OTHER_HAZARDOUS_D` =94 , `OTHER` ← `RESERVED_1` =95 ,  
`OTHER_RESERVED_2` =96 , `OTHER_RESERVED_3` =97 , `OTHER_RESERVED_4` =98 , `OTHER` ← `UNKNOWN` =99 }
- Type of AIS vessel, enum duplicated from AIS standard, <https://gpsd.gitlab.io/gpsd/AIVDM.html>.*
- enum class `mavlink::common::AIS_NAV_STATUS` : `uint8_t` {  
`UNDER_WAY` =0 , `ANCHORED` =1 , `UN_COMMANDED` =2 , `RESTRICTED_MANOEUVRABILITY` =3 ,  
`DRAUGHT_CONSTRAINED` =4 , `MOORED` =5 , `AGROUND` =6 , `FISHING` =7 ,  
`SAILING` =8 , `RESERVED_HSC` =9 , `RESERVED_WIG` =10 , `RESERVED_1` =11 ,  
`RESERVED_2` =12 , `RESERVED_3` =13 , `AIS_SART` =14 , `UNKNOWN` =15 }
- Navigational status of AIS vessel, enum duplicated from AIS standard, <https://gpsd.gitlab.io/gpsd/AIVDM.html>.*
- enum class `mavlink::common::AIS_FLAGS` : `uint16_t` {  
`POSITION_ACCURACY` =1 , `VALID_COG` =2 , `VALID_VELOCITY` =4 , `HIGH_VELOCITY` =8 ,

**VALID\_TURN\_RATE** =16 , **TURN\_RATE\_SIGN\_ONLY** =32 , **VALID\_DIMENSIONS** =64 , **LARGE\_BOW\_DIMENSION** =128 ,  
**LARGE\_STERN\_DIMENSION** =256 , **LARGE\_PORT\_DIMENSION** =512 , **LARGE\_STARBOARD\_DIMENSION** =1024 , **VALID\_CALLSIGN** =2048 ,  
**VALID\_NAME** =4096 }

*These flags are used in the AIS\_VESSEL.fields bitmask to indicate validity of data in the other message fields.*

- enum class **mavlink::common::FAILURE\_UNIT** {  
**SENSOR\_GYRO** =0 , **SENSOR\_ACCEL** =1 , **SENSOR\_MAG** =2 , **SENSOR\_BARO** =3 ,  
**SENSOR\_GPS** =4 , **SENSOR\_OPTICAL\_FLOW** =5 , **SENSOR\_VIO** =6 , **SENSOR\_DISTANCE\_SENSOR** =7 ,  
**SENSOR\_AIRSPEED** =8 , **SYSTEM\_BATTERY** =100 , **SYSTEM\_MOTOR** =101 , **SYSTEM\_SERVO** =102 ,  
**SYSTEM\_AVOIDANCE** =103 , **SYSTEM\_RC\_SIGNAL** =104 , **SYSTEM\_MAVLINK\_SIGNAL** =105 }

*List of possible units where failures can be injected.*

- enum class **mavlink::common::FAILURE\_TYPE** {  
**OK** =0 , **OFF** =1 , **STUCK** =2 , **GARBAGE** =3 ,  
**WRONG** =4 , **SLOW** =5 , **DELAYED** =6 , **INTERMITTENT** =7 }

*List of possible failure type to inject.*

- enum class **mavlink::common::NAV\_VTOL\_LAND\_OPTIONS** { **DEFAULT** =0 , **FW\_DESCENT** =1 ,  
**HOVER\_DESCENT** =2 }

- enum class **mavlink::common::MAV\_WINCH\_STATUS\_FLAG** : uint32\_t {  
**HEALTHY** =1 , **FULLY\_RETRACTED** =2 , **MOVING** =4 , **CLUTCH\_ENGAGED** =8 ,  
**LOCKED** =16 , **DROPPING** =32 , **ARRESTING** =64 , **GROUND\_SENSE** =128 ,  
**RETRACTING** =256 , **REDELIVER** =512 , **ABANDON\_LINE** =1024 , **LOCKING** =2048 ,  
**LOAD\_LINE** =4096 , **LOAD\_PAYLOAD** =8192 }

*Winch status flags used in WINCH\_STATUS.*

- enum class **mavlink::common::MAG\_CAL\_STATUS** : uint8\_t {  
**NOT\_STARTED** =0 , **WAITING\_TO\_START** =1 , **RUNNING\_STEP\_ONE** =2 , **RUNNING\_STEP\_TWO** =3 ,  
**SUCCESS** =4 , **FAILED** =5 , **BAD\_ORIENTATION** =6 , **BAD\_RADIUS** =7 }

- enum class **mavlink::common::MAV\_EVENT\_ERROR\_REASON** : uint8\_t { **UNAVAILABLE** =0 }

*Reason for an event error response.*

- enum class **mavlink::common::MAV\_EVENT\_CURRENT\_SEQUENCE\_FLAGS** : uint8\_t { **RESET** =1 }

*Flags for CURRENT\_EVENT\_SEQUENCE.*

- enum class **mavlink::common::HIL\_SENSOR\_UPDATED\_FLAGS** : uint32\_t {  
**XACC** =1 , **YACC** =2 , **ZACC** =4 , **XGYRO** =8 ,  
**YGYRO** =16 , **ZGYRO** =32 , **XMAG** =64 , **YMAG** =128 ,  
**ZMAG** =256 , **ABS\_PRESSURE** =512 , **DIFF\_PRESSURE** =1024 , **PRESSURE\_ALT** =2048 ,  
**TEMPERATURE** =4096 , **RESET** =2147483648 }

*Flags in the HIL\_SENSOR message indicate which fields have updated since the last message.*

- enum class **mavlink::common::HIGHRES\_IMU\_UPDATED\_FLAGS** : uint16\_t {  
**XACC** =1 , **YACC** =2 , **ZACC** =4 , **XGYRO** =8 ,  
**YGYRO** =16 , **ZGYRO** =32 , **XMAG** =64 , **YMAG** =128 ,  
**ZMAG** =256 , **ABS\_PRESSURE** =512 , **DIFF\_PRESSURE** =1024 , **PRESSURE\_ALT** =2048 ,  
**TEMPERATURE** =4096 }

*Flags in the HIGHRES\_IMU message indicate which fields have updated since the last message.*

- enum class **mavlink::common::CAN\_FILTER\_OP** : uint8\_t { **REPLACE** =0 , **ADD** =1 , **REMOVE** =2 }

- enum class **mavlink::common::MAV\_FTP\_ERR** {  
**NONE** =0 , **FAIL** =1 , **FAILERRNO** =2 , **INVALIDDATASIZE** =3 ,  
**INVALIDSESSION** =4 , **NOSESSIONSAVAILABLE** =5 , **EOF** =6 , **UNKNOWNCOMMAND** =7 ,  
**FILEEXISTS** =8 , **FILEPROTECTED** =9 , **FILENOTFOUND** =10 }

*MAV FTP error codes (may be used in FILE\_TRANSFER\_PROTOCOL).*

- enum class [mavlink::common::MAV\\_FTP\\_OPCODE](#) {  
**NONE** =0 , **TERMINATESSESSION** =1 , **RESETSESSION** =2 , **LISTDIRECTORY** =3 ,  
**OPENFILERO** =4 , **READFILE** =5 , **CREATEFILE** =6 , **WRITEFILE** =7 ,  
**REMOVEFILE** =8 , **CREATEDIRECTORY** =9 , **REMOVEDIRECTORY** =10 , **OPENFILEWO** =11 ,  
**TRUNCATEFILE** =12 , **RENAME** =13 , **CALCFILECRC** =14 , **BURSTREADFILE** =15 ,  
**ACK** =128 , **NAK** =129 }

*MAV FTP opcodes (may be used in FILE\_TRANSFER\_PROTOCOL).*

- enum class [mavlink::common::MISSION\\_STATE](#) : uint8\_t {  
**UNKNOWN** =0 , **NO\_MISSION** =1 , **NOT\_STARTED** =2 , **ACTIVE** =3 ,  
**PAUSED** =4 , **COMPLETE** =5 }

*States of the mission state machine.*

- enum class [mavlink::common::SAFETY\\_SWITCH\\_STATE](#) { **SAFE** =0 , **DANGEROUS** =1 }

*Possible safety switch states.*

- enum class [mavlink::common::ILLUMINATOR\\_MODE](#) : uint8\_t { **UNKNOWN** =0 , **INTERNAL\_CONTROL** =1 , **EXTERNAL\_SYNC** =2 }

*Modes of illuminator.*

- enum class [mavlink::common::ILLUMINATOR\\_ERROR\\_FLAGS](#) : uint32\_t { **THERMAL\_THROTTLING** =1 , **OVER\_TEMPERATURE\_SHUTDOWN** =2 , **THERMISTOR\_FAILURE** =4 }

*Illuminator module error flags (bitmap, 0 means no error)*

- enum class [mavlink::common::MAV\\_STANDARD\\_MODE](#) : uint8\_t {  
**NON\_STANDARD** =0 , **POSITION\_HOLD** =1 , **ORBIT** =2 , **CRUISE** =3 ,  
**ALTITUDE\_HOLD** =4 , **SAFE\_RECOVERY** =5 , **MISSION** =6 , **LAND** =7 ,  
**TAKEOFF** =8 }

*Standard modes with a well understood meaning across flight stacks and vehicle types.*

- enum class [mavlink::common::MAV\\_MODE\\_PROPERTY](#) : uint32\_t { **ADVANCED** =1 , **NOT\_USER\_SELECTABLE** =2 , **AUTO\_MODE** =4 }

*Mode properties.*

- enum class [mavlink::common::HIL\\_ACTUATOR\\_CONTROLS\\_FLAGS](#) : uint64\_t { **LOCKSTEP** =1 }

*Flags used in HIL\_ACTUATOR\_CONTROLS message.*

- enum class [mavlink::common::COMPUTER\\_STATUS\\_FLAGS](#) : uint16\_t { **UNDER\_VOLTAGE** =1 , **CPU\_THROTTLE** =2 , **THERMAL\_THROTTLE** =4 , **DISK\_FULL** =8 }

*Flags used to report computer status.*

- enum class [mavlink::common::AIRSPEED\\_SENSOR\\_FLAGS](#) : uint8\_t { **UNHEALTHY** =1 , **USING** =2 }

*Airspeed sensor flags.*

## Variables

- constexpr std::array< [mavlink\\_msg\\_entry\\_t](#), 233 > [mavlink::common::MESSAGE\\_ENTRIES](#) {{ {0, 50, 9, 9, 0, 0, 0}, {1, 124, 31, 43, 0, 0, 0}, {2, 137, 12, 12, 0, 0, 0}, {4, 237, 14, 14, 3, 12, 13}, {5, 217, 28, 28, 1, 0, 0}, {6, 104, 3, 3, 0, 0, 0}, {7, 119, 32, 32, 0, 0, 0}, {8, 117, 36, 36, 0, 0, 0}, {11, 89, 6, 6, 1, 4, 0}, {20, 214, 20, 20, 3, 2, 3}, {21, 159, 2, 2, 3, 0, 1}, {22, 220, 25, 25, 0, 0, 0}, {23, 168, 23, 23, 3, 4, 5}, {24, 24, 30, 52, 0, 0, 0}, {25, 23, 101, 101, 0, 0, 0}, {26, 170, 22, 24, 0, 0, 0}, {27, 144, 26, 29, 0, 0, 0}, {28, 67, 16, 16, 0, 0, 0}, {29, 115, 14, 16, 0, 0, 0}, {30, 39, 28, 28, 0, 0, 0}, {31, 246, 32, 48, 0, 0, 0}, {32, 185, 28, 28, 0, 0, 0}, {33, 104, 28, 28, 0, 0, 0}, {34, 237, 22, 22, 0, 0, 0}, {35, 244, 22, 22, 0, 0, 0}, {36, 222, 21, 37, 0, 0, 0}, {37, 212, 6, 7, 3, 4, 5}, {38, 9, 6, 7, 3, 4, 5}, {39, 254, 37, 38, 3, 32, 33}, {40, 230, 4, 5, 3, 2, 3}, {41, 28, 4, 4, 3, 2, 3}, {42, 28, 2, 18, 0, 0, 0}, {43, 132, 2, 3, 3, 0, 1}, {44, 221, 4, 9, 3, 2, 3}, {45, 232, 2, 3, 3, 0, 1}, {46, 11, 2, 2, 0, 0, 0}, {47, 153, 3, 8, 3, 0, 1}, {48, 41, 13, 21, 1, 12, 0}, {49, 39, 12, 20, 0, 0, 0}, {50, 78, 37, 37, 3, 18, 19}, {51, 196, 4, 5, 3, 2, 3}, {54, 15, 27, 27, 3, 24, 25}, {55, 3, 25, 25, 0, 0, 0}, {61, 167, 72, 72, 0, 0, 0}, {62, 183, 26, 26, 0, 0, 0}, {63, 119, 181, 181, 0, 0, 0}, {64, 191, 225, 225, 0, 0, 0}, {65, 118, 42, 42, 0, 0, 0}, {66, 148, 6, 6, 3, 2, 3}, {67, 21, 4, 4, 0, 0, 0}, {69, 243, 11, 30, 1, 10, 0}, {70, 124, 18, 38, 3, 16, 17}, {73, 38, 37, 38, 3, 32, 33}, {74, 20, 20, 20, 0, 0, 0}, {75, 158, 35, 35, 3, 30, 31}, {76, 152, 33, 33, 3, 30, 31}, {77, 143, 3, 10, 3, 8, 9}, {80, 14, 4, 4, 3, 2, 3}, {81, 106, 22, 22, 0, 0, 0}, {82, 49, 39, 51, 3, 36, 37}, {83, 22, 37, 37, 0, 0, 0}, {84, 143, 53, 53, 3, 50, 51}, {85, 140, 51, 51, 0, 0, 0}, {86, 5, 53, 53, 3, 50, 51}, {87, 150, 51, 51, 0, 0, 0}, {89, 231, 28, 28, 0, 0, 0},

0}, {90, 183, 56, 56, 0, 0, 0}, {91, 63, 42, 42, 0, 0, 0}, {92, 54, 33, 33, 0, 0, 0}, {93, 47, 81, 81, 0, 0, 0}, {100, 175, 26, 34, 0, 0, 0}, {101, 102, 32, 117, 0, 0, 0}, {102, 158, 32, 117, 0, 0, 0}, {103, 208, 20, 57, 0, 0, 0}, {104, 56, 32, 116, 0, 0, 0}, {105, 93, 62, 63, 0, 0, 0}, {106, 138, 44, 44, 0, 0, 0}, {107, 108, 64, 65, 0, 0, 0}, {108, 32, 84, 92, 0, 0, 0}, {109, 185, 9, 9, 0, 0, 0}, {110, 84, 254, 254, 3, 1, 2}, {111, 34, 16, 18, 3, 16, 17}, {112, 174, 12, 12, 0, 0, 0}, {113, 124, 36, 39, 0, 0, 0}, {114, 237, 44, 44, 0, 0, 0}, {115, 4, 64, 64, 0, 0, 0}, {116, 76, 22, 24, 0, 0, 0}, {117, 128, 6, 6, 3, 4, 5}, {118, 56, 14, 14, 0, 0, 0}, {119, 116, 12, 12, 3, 10, 11}, {120, 134, 97, 97, 0, 0, 0}, {121, 237, 2, 2, 3, 0, 1}, {122, 203, 2, 2, 3, 0, 1}, {123, 250, 113, 113, 3, 0, 1}, {124, 87, 35, 57, 0, 0, 0}, {125, 203, 6, 6, 0, 0, 0}, {126, 220, 79, 81, 3, 79, 80}, {127, 25, 35, 35, 0, 0, 0}, {128, 226, 35, 35, 0, 0, 0}, {129, 46, 22, 24, 0, 0, 0}, {130, 29, 13, 13, 0, 0, 0}, {131, 223, 255, 255, 0, 0, 0}, {132, 85, 14, 39, 0, 0, 0}, {133, 6, 18, 18, 0, 0, 0}, {134, 229, 43, 43, 0, 0, 0}, {135, 203, 8, 8, 0, 0, 0}, {136, 1, 22, 22, 0, 0, 0}, {137, 195, 14, 16, 0, 0, 0}, {138, 109, 36, 120, 0, 0, 0}, {139, 168, 43, 43, 3, 41, 42}, {140, 181, 41, 41, 0, 0, 0}, {141, 47, 32, 32, 0, 0, 0}, {142, 72, 243, 243, 0, 0, 0}, {143, 131, 14, 16, 0, 0, 0}, {144, 127, 93, 93, 0, 0, 0}, {146, 103, 100, 100, 0, 0, 0}, {147, 154, 36, 54, 0, 0, 0}, {148, 178, 60, 78, 0, 0, 0}, {149, 200, 30, 60, 0, 0, 0}, {162, 189, 8, 9, 0, 0, 0}, {192, 36, 44, 54, 0, 0, 0}, {225, 208, 65, 73, 0, 0, 0}, {230, 163, 42, 42, 0, 0, 0}, {231, 105, 40, 40, 0, 0, 0}, {232, 151, 63, 65, 0, 0, 0}, {233, 35, 182, 182, 0, 0, 0}, {234, 150, 40, 40, 0, 0, 0}, {235, 179, 42, 42, 0, 0, 0}, {241, 90, 32, 32, 0, 0, 0}, {242, 104, 52, 60, 0, 0, 0}, {243, 85, 53, 61, 1, 52, 0}, {244, 95, 6, 6, 0, 0, 0}, {245, 130, 2, 2, 0, 0, 0}, {246, 184, 38, 38, 0, 0, 0}, {247, 81, 19, 19, 0, 0, 0}, {248, 8, 254, 254, 3, 3, 4}, {249, 204, 36, 36, 0, 0, 0}, {250, 49, 30, 30, 0, 0, 0}, {251, 170, 18, 18, 0, 0, 0}, {252, 44, 18, 18, 0, 0, 0}, {253, 83, 51, 54, 0, 0, 0}, {254, 46, 9, 9, 0, 0, 0}, {256, 71, 42, 42, 3, 8, 9}, {257, 131, 9, 9, 0, 0, 0}, {258, 187, 32, 232, 3, 0, 1}, {259, 92, 235, 237, 0, 0, 0}, {260, 146, 5, 14, 0, 0, 0}, {261, 179, 27, 61, 0, 0, 0}, {262, 12, 18, 23, 0, 0, 0}, {263, 133, 255, 255, 0, 0, 0}, {264, 49, 28, 32, 0, 0, 0}, {265, 26, 16, 20, 0, 0, 0}, {266, 193, 255, 255, 3, 2, 3}, {267, 35, 255, 255, 3, 2, 3}, {268, 14, 4, 4, 3, 2, 3}, {269, 109, 213, 215, 0, 0, 0}, {270, 59, 19, 20, 0, 0, 0}, {271, 22, 52, 53, 0, 0, 0}, {275, 126, 31, 32, 0, 0, 0}, {276, 18, 49, 50, 0, 0, 0}, {277, 62, 30, 30, 0, 0, 0}, {280, 70, 33, 33, 0, 0, 0}, {281, 48, 13, 13, 0, 0, 0}, {282, 123, 35, 35, 3, 32, 33}, {283, 74, 144, 145, 0, 0, 0}, {284, 99, 32, 32, 3, 30, 31}, {285, 137, 40, 49, 3, 38, 39}, {286, 210, 53, 57, 3, 50, 51}, {287, 1, 23, 23, 3, 20, 21}, {288, 20, 23, 23, 3, 20, 21}, {290, 251, 46, 46, 0, 0, 0}, {291, 10, 57, 57, 0, 0, 0}, {295, 234, 12, 12, 0, 0, 0}, {299, 19, 96, 98, 0, 0, 0}, {300, 217, 22, 22, 0, 0, 0}, {301, 243, 58, 58, 0, 0, 0}, {310, 28, 17, 17, 0, 0, 0}, {311, 95, 116, 116, 0, 0, 0}, {320, 243, 20, 20, 3, 2, 3}, {321, 88, 2, 2, 3, 0, 1}, {322, 243, 149, 149, 0, 0, 0}, {323, 78, 147, 147, 3, 0, 1}, {324, 132, 146, 146, 0, 0, 0}, {330, 23, 158, 167, 0, 0, 0}, {331, 91, 230, 233, 0, 0, 0}, {332, 236, 239, 239, 0, 0, 0}, {333, 231, 109, 109, 0, 0, 0}, {334, 72, 10, 10, 0, 0, 0}, {335, 225, 24, 24, 0, 0, 0}, {336, 245, 84, 84, 0, 0, 0}, {339, 199, 5, 5, 0, 0, 0}, {340, 99, 70, 70, 0, 0, 0}, {345, 209, 21, 21, 3, 2, 3}, {350, 232, 20, 252, 0, 0, 0}, {360, 11, 25, 25, 0, 0, 0}, {361, 93, 33, 33, 0, 0, 0}, {370, 75, 87, 109, 0, 0, 0}, {371, 10, 26, 26, 0, 0, 0}, {372, 26, 140, 140, 0, 0, 0}, {373, 117, 42, 42, 0, 0, 0}, {375, 251, 140, 140, 0, 0, 0}, {376, 199, 8, 8, 0, 0, 0}, {380, 232, 20, 20, 0, 0, 0}, {385, 147, 133, 133, 3, 2, 3}, {386, 132, 16, 16, 3, 4, 5}, {387, 4, 72, 72, 3, 4, 5}, {388, 8, 37, 37, 3, 32, 33}, {390, 156, 238, 240, 0, 0, 0}, {395, 0, 212, 212, 0, 0, 0}, {396, 50, 160, 160, 0, 0, 0}, {397, 182, 108, 108, 0, 0, 0}, {400, 110, 254, 254, 3, 4, 5}, {401, 183, 6, 6, 3, 4, 5}, {410, 160, 53, 53, 0, 0, 0}, {411, 106, 3, 3, 0, 0, 0}, {412, 33, 6, 6, 3, 4, 5}, {413, 77, 7, 7, 3, 4, 5}, {435, 134, 46, 46, 0, 0, 0}, {436, 193, 9, 9, 0, 0, 0}, {437, 30, 1, 1, 0, 0, 0}, {440, 66, 35, 35, 0, 0, 0}, {9000, 113, 137, 137, 0, 0, 0}, {9005, 117, 34, 34, 0, 0, 0}, {12900, 114, 44, 44, 3, 0, 1}, {12901, 254, 59, 59, 3, 30, 31}, {12902, 140, 53, 53, 3, 4, 5}, {12903, 249, 46, 46, 3, 0, 1}, {12904, 77, 54, 54, 3, 28, 29}, {12905, 49, 43, 43, 3, 0, 1}, {12915, 94, 249, 249, 3, 0, 1}, {12918, 139, 51, 51, 0, 0, 0}, {12919, 7, 18, 18, 3, 16, 17}, {12920, 20, 5, 5, 0, 0, 0}}

*Array of msg\_entry needed for mavlink\_parse\_char() (through mavlink\_get\_msg\_entry())*

- constexpr auto `mavlink::common::MAVLINK_VERSION` = 3  
*MAVLINK VERSION.*
- constexpr auto `mavlink::common::HL_FAILURE_FLAG_ENUM_END` = 8193  
*HL\_FAILURE\_FLAG\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_GOTO_ENUM_END` = 4  
*MAV\_GOTO\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_MODE_ENUM_END` = 221  
*MAV\_MODE\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_SYS_STATUS_SENSOR_ENUM_END` = 2147483649  
*MAV\_SYS\_STATUS\_SENSOR\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_SYS_STATUS_SENSOR_EXTENDED_ENUM_END` = 2  
*MAV\_SYS\_STATUS\_SENSOR\_EXTENDED\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_FRAME_ENUM_END` = 22



- MAV\_FRAME ENUM\_END.*
- constexpr auto [mavlink::common::MAVLINK\\_DATA\\_STREAM\\_TYPE\\_ENUM\\_END](#) = 6  
*MAVLINK\_DATA\_STREAM\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::FENCE\\_BREACH\\_ENUM\\_END](#) = 4  
*FENCE\_BREACH ENUM\_END.*
- constexpr auto [mavlink::common::FENCE\\_MITIGATE\\_ENUM\\_END](#) = 3  
*FENCE\_MITIGATE ENUM\_END.*
- constexpr auto [mavlink::common::FENCE\\_TYPE\\_ENUM\\_END](#) = 9  
*FENCE\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_MOUNT\\_MODE\\_ENUM\\_END](#) = 7  
*MAV\_MOUNT\_MODE ENUM\_END.*
- constexpr auto [mavlink::common::GIMBAL\\_DEVICE\\_CAP\\_FLAGS\\_ENUM\\_END](#) = 8193  
*GIMBAL\_DEVICE\_CAP\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::GIMBAL\\_MANAGER\\_CAP\\_FLAGS\\_ENUM\\_END](#) = 131073  
*GIMBAL\_MANAGER\_CAP\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::GIMBAL\\_DEVICE\\_FLAGS\\_ENUM\\_END](#) = 513  
*GIMBAL\_DEVICE\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::GIMBAL\\_MANAGER\\_FLAGS\\_ENUM\\_END](#) = 513  
*GIMBAL\_MANAGER\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::GIMBAL\\_DEVICE\\_ERROR\\_FLAGS\\_ENUM\\_END](#) = 513  
*GIMBAL\_DEVICE\_ERROR\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::GRIPPER\\_ACTIONS\\_ENUM\\_END](#) = 3  
*GRIPPER\_ACTIONS ENUM\_END.*
- constexpr auto [mavlink::common::WINCH\\_ACTIONS\\_ENUM\\_END](#) = 10  
*WINCH\_ACTIONS ENUM\_END.*
- constexpr auto [mavlink::common::UAVCAN\\_NODE\\_HEALTH\\_ENUM\\_END](#) = 4  
*UAVCAN\_NODE\_HEALTH ENUM\_END.*
- constexpr auto [mavlink::common::UAVCAN\\_NODE\\_MODE\\_ENUM\\_END](#) = 8  
*UAVCAN\_NODE\_MODE ENUM\_END.*
- constexpr auto [mavlink::common::ESC\\_CONNECTION\\_TYPE\\_ENUM\\_END](#) = 6  
*ESC\_CONNECTION\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::ESC\\_FAILURE\\_FLAGS\\_ENUM\\_END](#) = 65  
*ESC\_FAILURE\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::STORAGE\\_STATUS\\_ENUM\\_END](#) = 4  
*STORAGE\_STATUS ENUM\_END.*
- constexpr auto [mavlink::common::STORAGE\\_TYPE\\_ENUM\\_END](#) = 255  
*STORAGE\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::STORAGE\\_USAGE\\_FLAG\\_ENUM\\_END](#) = 9  
*STORAGE\_USAGE\_FLAG ENUM\_END.*
- constexpr auto [mavlink::common::ORBIT\\_YAW\\_BEHAVIOUR\\_ENUM\\_END](#) = 6  
*ORBIT\_YAW\_BEHAVIOUR ENUM\_END.*
- constexpr auto [mavlink::common::WIFI\\_CONFIG\\_AP\\_RESPONSE\\_ENUM\\_END](#) = 6  
*WIFI\_CONFIG\_AP\_RESPONSE ENUM\_END.*
- constexpr auto [mavlink::common::CELLULAR\\_CONFIG\\_RESPONSE\\_ENUM\\_END](#) = 5  
*CELLULAR\_CONFIG\_RESPONSE ENUM\_END.*
- constexpr auto [mavlink::common::WIFI\\_CONFIG\\_AP\\_MODE\\_ENUM\\_END](#) = 4  
*WIFI\_CONFIG\_AP\_MODE ENUM\_END.*
- constexpr auto [mavlink::common::COMP\\_METADATA\\_TYPE\\_ENUM\\_END](#) = 6  
*COMP\_METADATA\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::ACTUATOR\\_CONFIGURATION\\_ENUM\\_END](#) = 6  
*ACTUATOR\_CONFIGURATION ENUM\_END.*



- constexpr auto [mavlink::common::ACTUATOR\\_OUTPUT\\_FUNCTION\\_ENUM\\_END](#) = 49  
*ACTUATOR\_OUTPUT\_FUNCTION\_ENUM\_END.*
- constexpr auto [mavlink::common::AUTOTUNE\\_AXIS\\_ENUM\\_END](#) = 5  
*AUTOTUNE\_AXIS\_ENUM\_END.*
- constexpr auto [mavlink::common::PREFLIGHT\\_STORAGE\\_PARAMETER\\_ACTION\\_ENUM\\_END](#) = 5  
*PREFLIGHT\_STORAGE\_PARAMETER\_ACTION\_ENUM\_END.*
- constexpr auto [mavlink::common::PREFLIGHT\\_STORAGE\\_MISSION\\_ACTION\\_ENUM\\_END](#) = 3  
*PREFLIGHT\_STORAGE\_MISSION\_ACTION\_ENUM\_END.*
- constexpr auto [mavlink::common::REBOOT\\_SHUTDOWN\\_ACTION\\_ENUM\\_END](#) = 5  
*REBOOT\_SHUTDOWN\_ACTION\_ENUM\_END.*
- constexpr auto [mavlink::common::REBOOT\\_SHUTDOWN\\_CONDITIONS\\_ENUM\\_END](#) = 20190227  
*REBOOT\_SHUTDOWN\_CONDITIONS\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_CMD\\_ENUM\\_END](#) = 43004  
*MAV\_CMD\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_DATA\\_STREAM\\_ENUM\\_END](#) = 13  
*MAV\_DATA\_STREAM\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ROI\\_ENUM\\_END](#) = 5  
*MAV\_ROI\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_PARAM\\_TYPE\\_ENUM\\_END](#) = 11  
*MAV\_PARAM\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_PARAM\\_ERROR\\_ENUM\\_END](#) = 9  
*MAV\_PARAM\_ERROR\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_PARAM\\_EXT\\_TYPE\\_ENUM\\_END](#) = 12  
*MAV\_PARAM\_EXT\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_RESULT\\_ENUM\\_END](#) = 11  
*MAV\_RESULT\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_MISSION\\_RESULT\\_ENUM\\_END](#) = 16  
*MAV\_MISSION\_RESULT\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_SEVERITY\\_ENUM\\_END](#) = 8  
*MAV\_SEVERITY\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_POWER\\_STATUS\\_ENUM\\_END](#) = 33  
*MAV\_POWER\_STATUS\_ENUM\_END.*
- constexpr auto [mavlink::common::SERIAL\\_CONTROL\\_DEV\\_ENUM\\_END](#) = 110  
*SERIAL\_CONTROL\_DEV\_ENUM\_END.*
- constexpr auto [mavlink::common::SERIAL\\_CONTROL\\_FLAG\\_ENUM\\_END](#) = 17  
*SERIAL\_CONTROL\_FLAG\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_DISTANCE\\_SENSOR\\_ENUM\\_END](#) = 5  
*MAV\_DISTANCE\_SENSOR\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_SENSOR\\_ORIENTATION\\_ENUM\\_END](#) = 101  
*MAV\_SENSOR\_ORIENTATION\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_MISSION\\_TYPE\\_ENUM\\_END](#) = 256  
*MAV\_MISSION\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ESTIMATOR\\_TYPE\\_ENUM\\_END](#) = 9  
*MAV\_ESTIMATOR\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_BATTERY\\_TYPE\\_ENUM\\_END](#) = 5  
*MAV\_BATTERY\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_BATTERY\\_FUNCTION\\_ENUM\\_END](#) = 5  
*MAV\_BATTERY\_FUNCTION\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_BATTERY\\_CHARGE\\_STATE\\_ENUM\\_END](#) = 8  
*MAV\_BATTERY\_CHARGE\_STATE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_BATTERY\\_MODE\\_ENUM\\_END](#) = 3

- MAV\_BATTERY\_MODE ENUM\_END.*
- `constexpr auto mavlink::common::MAV_BATTERY_FAULT_ENUM_END = 257`  
*MAV\_BATTERY\_FAULT ENUM\_END.*
- `constexpr auto mavlink::common::MAV_FUEL_TYPE_ENUM_END = 3`  
*MAV\_FUEL\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::MAV_GENERATOR_STATUS_FLAG_ENUM_END = 4194305`  
*MAV\_GENERATOR\_STATUS\_FLAG ENUM\_END.*
- `constexpr auto mavlink::common::MAV_VTOL_STATE_ENUM_END = 5`  
*MAV\_VTOL\_STATE ENUM\_END.*
- `constexpr auto mavlink::common::MAV_LANDED_STATE_ENUM_END = 5`  
*MAV\_LANDED\_STATE ENUM\_END.*
- `constexpr auto mavlink::common::ADSB_ALTITUDE_TYPE_ENUM_END = 2`  
*ADSB\_ALTITUDE\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::ADSB_EMITTER_TYPE_ENUM_END = 20`  
*ADSB\_EMITTER\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::ADSB_FLAGS_ENUM_END = 32769`  
*ADSB\_FLAGS ENUM\_END.*
- `constexpr auto mavlink::common::MAV_DO_REPOSITION_FLAGS_ENUM_END = 3`  
*MAV\_DO\_REPOSITION\_FLAGS ENUM\_END.*
- `constexpr auto mavlink::common::SPEED_TYPE_ENUM_END = 4`  
*SPEED\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::ESTIMATOR_STATUS_FLAGS_ENUM_END = 2049`  
*ESTIMATOR\_STATUS\_FLAGS ENUM\_END.*
- `constexpr auto mavlink::common::MOTOR_TEST_ORDER_ENUM_END = 3`  
*MOTOR\_TEST\_ORDER ENUM\_END.*
- `constexpr auto mavlink::common::MOTOR_TEST_THROTTLE_TYPE_ENUM_END = 4`  
*MOTOR\_TEST\_THROTTLE\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::GPS_INPUT_IGNORE_FLAGS_ENUM_END = 129`  
*GPS\_INPUT\_IGNORE\_FLAGS ENUM\_END.*
- `constexpr auto mavlink::common::MAV_COLLISION_ACTION_ENUM_END = 7`  
*MAV\_COLLISION\_ACTION ENUM\_END.*
- `constexpr auto mavlink::common::MAV_COLLISION_THREAT_LEVEL_ENUM_END = 3`  
*MAV\_COLLISION\_THREAT\_LEVEL ENUM\_END.*
- `constexpr auto mavlink::common::MAV_COLLISION_SRC_ENUM_END = 2`  
*MAV\_COLLISION\_SRC ENUM\_END.*
- `constexpr auto mavlink::common::GPS_FIX_TYPE_ENUM_END = 9`  
*GPS\_FIX\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::RTK_BASELINE_COORDINATE_SYSTEM_ENUM_END = 2`  
*RTK\_BASELINE\_COORDINATE\_SYSTEM ENUM\_END.*
- `constexpr auto mavlink::common::LANDING_TARGET_TYPE_ENUM_END = 4`  
*LANDING\_TARGET\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::VTOL_TRANSITION_HEADING_ENUM_END = 5`  
*VTOL\_TRANSITION\_HEADING ENUM\_END.*
- `constexpr auto mavlink::common::CAMERA_CAP_FLAGS_ENUM_END = 8193`  
*CAMERA\_CAP\_FLAGS ENUM\_END.*
- `constexpr auto mavlink::common::VIDEO_STREAM_STATUS_FLAGS_ENUM_END = 5`  
*VIDEO\_STREAM\_STATUS\_FLAGS ENUM\_END.*
- `constexpr auto mavlink::common::VIDEO_STREAM_TYPE_ENUM_END = 4`  
*VIDEO\_STREAM\_TYPE ENUM\_END.*
- `constexpr auto mavlink::common::VIDEO_STREAM_ENCODING_ENUM_END = 3`  
*VIDEO\_STREAM\_ENCODING ENUM\_END.*

- constexpr auto [mavlink::common::CAMERA\\_TRACKING\\_STATUS\\_FLAGS\\_ENUM\\_END](#) = 9  
*CAMERA\_TRACKING\_STATUS\_FLAGS\_ENUM\_END.*
- constexpr auto [mavlink::common::CAMERA\\_TRACKING\\_MODE\\_ENUM\\_END](#) = 3  
*CAMERA\_TRACKING\_MODE\_ENUM\_END.*
- constexpr auto [mavlink::common::CAMERA\\_TRACKING\\_TARGET\\_DATA\\_ENUM\\_END](#) = 5  
*CAMERA\_TRACKING\_TARGET\_DATA\_ENUM\_END.*
- constexpr auto [mavlink::common::CAMERA\\_ZOOM\\_TYPE\\_ENUM\\_END](#) = 5  
*CAMERA\_ZOOM\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::SET\\_FOCUS\\_TYPE\\_ENUM\\_END](#) = 7  
*SET\_FOCUS\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::CAMERA\\_SOURCE\\_ENUM\\_END](#) = 4  
*CAMERA\_SOURCE\_ENUM\_END.*
- constexpr auto [mavlink::common::PARAM\\_ACK\\_ENUM\\_END](#) = 4  
*PARAM\_ACK\_ENUM\_END.*
- constexpr auto [mavlink::common::CAMERA\\_MODE\\_ENUM\\_END](#) = 3  
*CAMERA\_MODE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ARM\\_AUTH\\_DENIED\\_REASON\\_ENUM\\_END](#) = 6  
*MAV\_ARM\_AUTH\_DENIED\_REASON\_ENUM\_END.*
- constexpr auto [mavlink::common::RC\\_TYPE\\_ENUM\\_END](#) = 2  
*RC\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::RC\\_SUB\\_TYPE\\_ENUM\\_END](#) = 3  
*RC\_SUB\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::ENGINE\\_CONTROL\\_OPTIONS\\_ENUM\\_END](#) = 2  
*ENGINE\_CONTROL\_OPTIONS\_ENUM\_END.*
- constexpr auto [mavlink::common::POSITION\\_TARGET\\_TPEMASK\\_ENUM\\_END](#) = 2049  
*POSITION\_TARGET\_TPEMASK\_ENUM\_END.*
- constexpr auto [mavlink::common::ATTITUDE\\_TARGET\\_TPEMASK\\_ENUM\\_END](#) = 129  
*ATTITUDE\_TARGET\_TPEMASK\_ENUM\_END.*
- constexpr auto [mavlink::common::UTM\\_FLIGHT\\_STATE\\_ENUM\\_END](#) = 33  
*UTM\_FLIGHT\_STATE\_ENUM\_END.*
- constexpr auto [mavlink::common::UTM\\_DATA\\_AVAIL\\_FLAGS\\_ENUM\\_END](#) = 129  
*UTM\_DATA\_AVAIL\_FLAGS\_ENUM\_END.*
- constexpr auto [mavlink::common::CELLULAR\\_STATUS\\_FLAG\\_ENUM\\_END](#) = 13  
*CELLULAR\_STATUS\_FLAG\_ENUM\_END.*
- constexpr auto [mavlink::common::CELLULAR\\_NETWORK\\_FAILED\\_REASON\\_ENUM\\_END](#) = 4  
*CELLULAR\_NETWORK\_FAILED\_REASON\_ENUM\_END.*
- constexpr auto [mavlink::common::CELLULAR\\_NETWORK\\_RADIO\\_TYPE\\_ENUM\\_END](#) = 5  
*CELLULAR\_NETWORK\_RADIO\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::PRECISION\\_LAND\\_MODE\\_ENUM\\_END](#) = 3  
*PRECISION\_LAND\_MODE\_ENUM\_END.*
- constexpr auto [mavlink::common::PARACHUTE\\_ACTION\\_ENUM\\_END](#) = 3  
*PARACHUTE\_ACTION\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_TUNNEL\\_PAYLOAD\\_TYPE\\_ENUM\\_END](#) = 213  
*MAV\_TUNNEL\_PAYLOAD\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_ID\\_TYPE\\_ENUM\\_END](#) = 5  
*MAV\_ODID\_ID\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_UA\\_TYPE\\_ENUM\\_END](#) = 16  
*MAV\_ODID\_UA\_TYPE\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_STATUS\\_ENUM\\_END](#) = 5  
*MAV\_ODID\_STATUS\_ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_HEIGHT\\_REF\\_ENUM\\_END](#) = 2

- MAV\_ODID\_HEIGHT\_REF ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_HOR\\_ACC\\_ENUM\\_END](#) = 13  
*MAV\_ODID\_HOR\_ACC ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_VER\\_ACC\\_ENUM\\_END](#) = 7  
*MAV\_ODID\_VER\_ACC ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_SPEED\\_ACC\\_ENUM\\_END](#) = 5  
*MAV\_ODID\_SPEED\_ACC ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_TIME\\_ACC\\_ENUM\\_END](#) = 16  
*MAV\_ODID\_TIME\_ACC ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_AUTH\\_TYPE\\_ENUM\\_END](#) = 6  
*MAV\_ODID\_AUTH\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_DESC\\_TYPE\\_ENUM\\_END](#) = 3  
*MAV\_ODID\_DESC\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_OPERATOR\\_LOCATION\\_TYPE\\_ENUM\\_END](#) = 3  
*MAV\_ODID\_OPERATOR\_LOCATION\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_CLASSIFICATION\\_TYPE\\_ENUM\\_END](#) = 2  
*MAV\_ODID\_CLASSIFICATION\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_CATEGORY\\_EU\\_ENUM\\_END](#) = 4  
*MAV\_ODID\_CATEGORY\_EU ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_CLASS\\_EU\\_ENUM\\_END](#) = 8  
*MAV\_ODID\_CLASS\_EU ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_OPERATOR\\_ID\\_TYPE\\_ENUM\\_END](#) = 1  
*MAV\_ODID\_OPERATOR\_ID\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_ODID\\_ARM\\_STATUS\\_ENUM\\_END](#) = 2  
*MAV\_ODID\_ARM\_STATUS ENUM\_END.*
- constexpr auto [mavlink::common::TUNE\\_FORMAT\\_ENUM\\_END](#) = 3  
*TUNE\_FORMAT ENUM\_END.*
- constexpr auto [mavlink::common::AIS\\_TYPE\\_ENUM\\_END](#) = 100  
*AIS\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::AIS\\_NAV\\_STATUS\\_ENUM\\_END](#) = 16  
*AIS\_NAV\_STATUS ENUM\_END.*
- constexpr auto [mavlink::common::AIS\\_FLAGS\\_ENUM\\_END](#) = 4097  
*AIS\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::FAILURE\\_UNIT\\_ENUM\\_END](#) = 106  
*FAILURE\_UNIT ENUM\_END.*
- constexpr auto [mavlink::common::FAILURE\\_TYPE\\_ENUM\\_END](#) = 8  
*FAILURE\_TYPE ENUM\_END.*
- constexpr auto [mavlink::common::NAV\\_VTOL\\_LAND\\_OPTIONS\\_ENUM\\_END](#) = 3  
*NAV\_VTOL\_LAND\_OPTIONS ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_WINCH\\_STATUS\\_FLAG\\_ENUM\\_END](#) = 8193  
*MAV\_WINCH\_STATUS\_FLAG ENUM\_END.*
- constexpr auto [mavlink::common::MAG\\_CAL\\_STATUS\\_ENUM\\_END](#) = 8  
*MAG\_CAL\_STATUS ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_EVENT\\_ERROR\\_REASON\\_ENUM\\_END](#) = 1  
*MAV\_EVENT\_ERROR\_REASON ENUM\_END.*
- constexpr auto [mavlink::common::MAV\\_EVENT\\_CURRENT\\_SEQUENCE\\_FLAGS\\_ENUM\\_END](#) = 2  
*MAV\_EVENT\_CURRENT\_SEQUENCE\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::HIL\\_SENSOR\\_UPDATED\\_FLAGS\\_ENUM\\_END](#) = 2147483649  
*HIL\_SENSOR\_UPDATED\_FLAGS ENUM\_END.*
- constexpr auto [mavlink::common::HIGHRES\\_IMU\\_UPDATED\\_FLAGS\\_ENUM\\_END](#) = 4097  
*HIGHRES\_IMU\_UPDATED\_FLAGS ENUM\_END.*

- constexpr auto `mavlink::common::CAN_FILTER_OP_ENUM_END` = 3  
*CAN\_FILTER\_OP\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_FTP_ERR_ENUM_END` = 11  
*MAV\_FTP\_ERR\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_FTP_OPCODE_ENUM_END` = 130  
*MAV\_FTP\_OPCODE\_ENUM\_END.*
- constexpr auto `mavlink::common::MISSION_STATE_ENUM_END` = 6  
*MISSION\_STATE\_ENUM\_END.*
- constexpr auto `mavlink::common::SAFETY_SWITCH_STATE_ENUM_END` = 2  
*SAFETY\_SWITCH\_STATE\_ENUM\_END.*
- constexpr auto `mavlink::common::ILLUMINATOR_MODE_ENUM_END` = 3  
*ILLUMINATOR\_MODE\_ENUM\_END.*
- constexpr auto `mavlink::common::ILLUMINATOR_ERROR_FLAGS_ENUM_END` = 5  
*ILLUMINATOR\_ERROR\_FLAGS\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_STANDARD_MODE_ENUM_END` = 9  
*MAV\_STANDARD\_MODE\_ENUM\_END.*
- constexpr auto `mavlink::common::MAV_MODE_PROPERTY_ENUM_END` = 5  
*MAV\_MODE\_PROPERTY\_ENUM\_END.*
- constexpr auto `mavlink::common::HIL_ACTUATOR_CONTROLS_FLAGS_ENUM_END` = 2  
*HIL\_ACTUATOR\_CONTROLS\_FLAGS\_ENUM\_END.*
- constexpr auto `mavlink::common::COMPUTER_STATUS_FLAGS_ENUM_END` = 9  
*COMPUTER\_STATUS\_FLAGS\_ENUM\_END.*
- constexpr auto `mavlink::common::AIRSPEED_SENSOR_FLAGS_ENUM_END` = 3  
*AIRSPEED\_SENSOR\_FLAGS\_ENUM\_END.*

### 6.7.1 Detailed Description

MAVLink comm protocol generated from common.xml.

See also

<http://mavlink.org>

### 6.7.2 Enumeration Type Documentation

#### 6.7.2.1 ACTUATOR\_CONFIGURATION

```
enum mavlink::common::ACTUATOR_CONFIGURATION [strong]
```

Actuator configuration, used to change a setting on an actuator.

Component information metadata can be used to know which outputs support which commands.

### 6.7.2.2 ACTUATOR\_OUTPUT\_FUNCTION

```
enum mavlink::common::ACTUATOR_OUTPUT_FUNCTION [strong]
```

Actuator output function.

Values greater or equal to 1000 are autopilot-specific.

### 6.7.2.3 ADSB\_FLAGS

```
enum mavlink::common::ADSB_FLAGS : uint16_t [strong]
```

These flags indicate status such as data validity of each data source.

Set = data valid

### 6.7.2.4 AIS\_FLAGS

```
enum mavlink::common::AIS_FLAGS : uint16_t [strong]
```

These flags are used in the AIS\_VESSEL.fields bitmask to indicate validity of data in the other message fields.

When set, the data is valid.

### 6.7.2.5 AUTOTUNE\_AXIS

```
enum mavlink::common::AUTOTUNE_AXIS [strong]
```

Axes that will be autotuned by MAV\_CMD\_DO\_AUTOTUNE\_ENABLE.

Note that at least one flag must be set in MAV\_CMD\_DO\_AUTOTUNE\_ENABLE.param2: if none are set, the flight stack will tune its default set of axes.

### 6.7.2.6 COMP\_METADATA\_TYPE

```
enum mavlink::common::COMP_METADATA_TYPE [strong]
```

Supported component metadata types.

These are used in the "general" metadata file returned by COMPONENT\_METADATA to provide information about supported metadata types. The types are not used directly in MAVLink messages.

### 6.7.2.7 FENCE\_TYPE

```
enum mavlink::common::FENCE_TYPE [strong]
```

Fence types to enable or disable when using MAV\_CMD\_DO\_FENCE\_ENABLE.

Note that at least one of these flags must be set in MAV\_CMD\_DO\_FENCE\_ENABLE.param2. If none are set, the flight stack will ignore the field and enable/disable its default set of fences (usually all of them).

### 6.7.2.8 GIMBAL\_MANAGER\_CAP\_FLAGS

```
enum mavlink::common::GIMBAL_MANAGER_CAP_FLAGS : uint32_t [strong]
```

Gimbal manager high level capability flags (bitmap).

The first 16 bits are identical to the GIMBAL\_DEVICE\_CAP\_FLAGS. However, the gimbal manager does not need to copy the flags from the gimbal but can also enhance the capabilities and thus add flags.

### 6.7.2.9 MAV\_BATTERY\_FAULT

```
enum mavlink::common::MAV_BATTERY_FAULT : uint32_t [strong]
```

Smart battery supply status/fault flags (bitmask) for health indication.

The battery must also report either MAV\_BATTERY\_CHARGE\_STATE\_FAILED or MAV\_BATTERY\_CHARGE\_STATE\_UNHEALTHY if any of these are set.

### 6.7.2.10 MAV\_BATTERY\_MODE

```
enum mavlink::common::MAV_BATTERY_MODE : uint8_t [strong]
```

Battery mode.

Note, the normal operation mode (i.e. when flying) should be reported as MAV\_BATTERY\_MODE\_UNKNOWN to allow message trimming in normal flight.

### 6.7.2.11 MAV\_CMD

```
enum mavlink::common::MAV_CMD : uint16_t [strong]
```

Commands to be executed by the MAV.

They can be executed on user request, or as part of a mission script. If the action is used in a mission, the parameter mapping to the waypoint/mission message is as follows: Param 1, Param 2, Param 3, Param 4, X↔ : Param 5, Y:Param 6, Z:Param 7. This command list is similar what ARINC 424 is for commercial aircraft: A data format how to interpret waypoint/mission data. NaN and INT32\_MAX may be used in float/integer params (respectively) to indicate optional/default values (e.g. to use the component's current yaw or latitude rather than a specific value). See [https://mavlink.io/en/guide/xml\\_schema.html#MAV\\_CMD](https://mavlink.io/en/guide/xml_schema.html#MAV_CMD) for information about the structure of the MAV\_CMD entries

### 6.7.2.12 MAV\_DATA\_STREAM

```
enum mavlink::common::MAV_DATA_STREAM : uint8_t [strong]
```

A data stream is not a fixed set of messages, but rather a recommendation to the autopilot software.

Individual autopilots may or may not obey the recommended messages.

### 6.7.2.13 MAV\_FRAME

```
enum mavlink::common::MAV_FRAME : uint8_t [strong]
```

Coordinate frames used by MAVLink.

Not all frames are supported by all commands, messages, or vehicles.

Global frames use the following naming conventions:

- "GLOBAL": Global coordinate frame with WGS84 latitude/longitude and altitude positive over mean sea level (MSL) by default. The following modifiers may be used with "GLOBAL":
  - "RELATIVE\_ALT": Altitude is relative to the vehicle home position rather than MSL.
  - "TERRAIN\_ALT": Altitude is relative to ground level rather than MSL.
  - "INT": Latitude/longitude (in degrees) are scaled by multiplying by 1E7.

Local frames use the following naming conventions:

- "LOCAL": Origin of local frame is fixed relative to earth. Unless otherwise specified this origin is the origin of the vehicle position-estimator ("EKF").
- "BODY": Origin of local frame travels with the vehicle. NOTE, "BODY" does NOT indicate alignment of frame axis with vehicle attitude.
- "OFFSET": Deprecated synonym for "BODY" (origin travels with the vehicle). Not to be used for new frames.

Some deprecated frames do not follow these conventions (e.g. MAV\_FRAME\_BODY\_NED and MAV\_FRAME\_BODY\_OFFSET\_NED).

### 6.7.2.14 MAV\_FTP\_ERR

```
enum mavlink::common::MAV_FTP_ERR [strong]
```

MAV FTP error codes (may be used in FILE\_TRANSFER\_PROTOCOL).

See <https://mavlink.io/en/services/ftp.html>.

### 6.7.2.15 MAV\_FTP\_OPCODE

```
enum mavlink::common::MAV_FTP_OPCODE [strong]
```

MAV FTP opcodes (may be used in FILE\_TRANSFER\_PROTOCOL).

See <https://mavlink.io/en/services/ftp.html>.

### 6.7.2.16 MAV\_FUEL\_TYPE

```
enum mavlink::common::MAV_FUEL_TYPE : uint32_t [strong]
```

Fuel types for use in FUEL\_TYPE.

Fuel types specify the units for the maximum, available and consumed fuel, and for the flow rates.



#### 6.7.2.17 MAV\_GENERATOR\_STATUS\_FLAG

```
enum mavlink::common::MAV_GENERATOR_STATUS_FLAG : uint64_t [strong]
```

Flags to report status/failure cases for a power generator (used in GENERATOR\_STATUS).

Note that FAULTS are conditions that cause the generator to fail. Warnings are conditions that require attention before the next use (they indicate the system is not operating properly).

#### 6.7.2.18 MAV\_MODE

```
enum mavlink::common::MAV_MODE : uint8_t [strong]
```

Predefined OR-combined MAV\_MODE\_FLAG values.

These can simplify using the flags when setting modes. Note that manual input is enabled in all modes as a safety override.

#### 6.7.2.19 MAV\_MOUNT\_MODE

```
enum mavlink::common::MAV_MOUNT_MODE [strong]
```

Enumeration of possible mount operation modes.

This message is used by obsolete/deprecated gimbal messages.

#### 6.7.2.20 MAV\_ROI

```
enum mavlink::common::MAV_ROI [strong]
```

The ROI (region of interest) for the vehicle.

This can be used by the vehicle for camera/vehicle attitude alignment (see MAV\_CMD\_NAV\_ROI).

#### 6.7.2.21 MAV\_SEVERITY

```
enum mavlink::common::MAV_SEVERITY : uint8_t [strong]
```

Indicates the severity level, generally used for status messages to indicate their relative urgency.

Based on RFC-5424 using expanded definitions at: <http://www.kiwisyslog.com/kb/info←:-syslog-message-levels/>.

#### 6.7.2.22 MAV\_STANDARD\_MODE

```
enum mavlink::common::MAV_STANDARD_MODE : uint8_t [strong]
```

Standard modes with a well understood meaning across flight stacks and vehicle types.

For example, most flight stack have the concept of a "return" or "RTL" mode that takes a vehicle to safety, even though the precise mechanics of this mode may differ. The modes supported by a flight stack can be queried using AVAILABLE\_MODES and set using MAV\_CMD\_DO\_SET\_STANDARD\_MODE. The current mode is streamed in CURRENT\_MODE. See [https://mavlink.io/en/services/standard\\_modes.html](https://mavlink.io/en/services/standard_modes.html)

### 6.7.2.23 MISSION\_STATE

```
enum mavlink::common::MISSION_STATE : uint8_t [strong]
```

States of the mission state machine.

Note that these states are independent of whether the mission is in a mode that can execute mission items or not (is suspended). They may not all be relevant on all vehicles.

### 6.7.2.24 PARACHUTE\_ACTION

```
enum mavlink::common::PARACHUTE_ACTION [strong]
```

Parachute actions.

Trigger release and enable/disable auto-release.

### 6.7.2.25 POSITION\_TARGET\_TYEMASK

```
enum mavlink::common::POSITION_TARGET_TYEMASK : uint16_t [strong]
```

Bitmap to indicate which dimensions should be ignored by the vehicle: a value of 0b0000000000000000 or 0b0000000100000000 indicates that none of the setpoint dimensions should be ignored.

If bit 9 is set the floats afx afy afz should be interpreted as force instead of acceleration.

### 6.7.2.26 PREFLIGHT\_STORAGE\_MISSION\_ACTION

```
enum mavlink::common::PREFLIGHT_STORAGE_MISSION_ACTION [strong]
```

Actions for reading and writing plan information (mission, rally points, geofence) between persistent and volatile storage when using MAV\_CMD\_PREFLIGHT\_STORAGE.

(Commonly missions are loaded from persistent storage (flash/EEPROM) into volatile storage (RAM) on startup and written back when they are changed.)

### 6.7.2.27 PREFLIGHT\_STORAGE\_PARAMETER\_ACTION

```
enum mavlink::common::PREFLIGHT_STORAGE_PARAMETER_ACTION [strong]
```

Actions for reading/writing parameters between persistent and volatile storage when using MAV\_CMD\_↔ PREFLIGHT\_STORAGE.

(Commonly parameters are loaded from persistent storage (flash/EEPROM) into volatile storage (RAM) on startup and written back when they are changed.)

### 6.7.2.28 RC\_SUB\_TYPE

```
enum mavlink::common::RC_SUB_TYPE [strong]
```

RC sub-type of types defined in RC\_TYPE.

Used in MAV\_CMD\_START\_RX\_PAIR. Ignored if value does not correspond to the set RC\_TYPE.

### 6.7.2.29 RC\_TYPE

```
enum mavlink::common::RC_TYPE [strong]
```

RC type.

Used in MAV\_CMD\_START\_RX\_PAIR.

## 6.8 single\_hand\_controller/include/mavlink/common/gtestsuite.hpp File Reference

MAVLink comm testsuite protocol generated from common.xml.

```
#include <gtest/gtest.h>
```

```
#include "common.hpp"
```

Include dependency graph for gtestsuite.hpp:



### Functions

- **TEST** (common, SYS\_STATUS)
- **TEST** (common, SYSTEM\_TIME)
- **TEST** (common, PING)
- **TEST** (common, CHANGE\_OPERATOR\_CONTROL)
- **TEST** (common, CHANGE\_OPERATOR\_CONTROL\_ACK)
- **TEST** (common, AUTH\_KEY)
- **TEST** (common, LINK\_NODE\_STATUS)
- **TEST** (common, SET\_MODE)
- **TEST** (common, PARAM\_REQUEST\_READ)
- **TEST** (common, PARAM\_REQUEST\_LIST)
- **TEST** (common, PARAM\_VALUE)
- **TEST** (common, PARAM\_SET)
- **TEST** (common, GPS\_RAW\_INT)
- **TEST** (common, GPS\_STATUS)
- **TEST** (common, SCALED\_IMU)
- **TEST** (common, RAW\_IMU)
- **TEST** (common, RAW\_PRESSURE)
- **TEST** (common, SCALED\_PRESSURE)
- **TEST** (common, ATTITUDE)
- **TEST** (common, ATTITUDE\_QUATERNION)

- **TEST** (common, LOCAL\_POSITION\_NED)
- **TEST** (common, RC\_CHANNELS\_SCALED)
- **TEST** (common, RC\_CHANNELS\_RAW)
- **TEST** (common, SERVO\_OUTPUT\_RAW)
- **TEST** (common, MISSION\_REQUEST\_PARTIAL\_LIST)
- **TEST** (common, MISSION\_WRITE\_PARTIAL\_LIST)
- **TEST** (common, MISSION\_ITEM)
- **TEST** (common, MISSION\_REQUEST)
- **TEST** (common, MISSION\_SET\_CURRENT)
- **TEST** (common, MISSION\_CURRENT)
- **TEST** (common, MISSION\_REQUEST\_LIST)
- **TEST** (common, MISSION\_COUNT)
- **TEST** (common, MISSION\_CLEAR\_ALL)
- **TEST** (common, MISSION\_ITEM\_REACHED)
- **TEST** (common, MISSION\_ACK)
- **TEST** (common, SET\_GPS\_GLOBAL\_ORIGIN)
- **TEST** (common, GPS\_GLOBAL\_ORIGIN)
- **TEST** (common, PARAM\_MAP\_RC)
- **TEST** (common, MISSION\_REQUEST\_INT)
- **TEST** (common, SAFETY\_SET\_ALLOWED\_AREA)
- **TEST** (common, SAFETY\_ALLOWED\_AREA)
- **TEST** (common, ATTITUDE\_QUATERNION\_COV)
- **TEST** (common, NAV\_CONTROLLER\_OUTPUT)
- **TEST** (common, GLOBAL\_POSITION\_INT\_COV)
- **TEST** (common, LOCAL\_POSITION\_NED\_COV)
- **TEST** (common, RC\_CHANNELS)
- **TEST** (common, REQUEST\_DATA\_STREAM)
- **TEST** (common, DATA\_STREAM)
- **TEST** (common, MANUAL\_CONTROL)
- **TEST** (common, RC\_CHANNELS\_OVERRIDE)
- **TEST** (common, MISSION\_ITEM\_INT)
- **TEST** (common, VFR\_HUD)
- **TEST** (common, COMMAND\_INT)
- **TEST** (common, COMMAND\_LONG)
- **TEST** (common, COMMAND\_ACK)
- **TEST** (common, COMMAND\_CANCEL)
- **TEST** (common, MANUAL\_SETPOINT)
- **TEST** (common, SET\_ATTITUDE\_TARGET)
- **TEST** (common, ATTITUDE\_TARGET)
- **TEST** (common, SET\_POSITION\_TARGET\_LOCAL\_NED)
- **TEST** (common, POSITION\_TARGET\_LOCAL\_NED)
- **TEST** (common, SET\_POSITION\_TARGET\_GLOBAL\_INT)
- **TEST** (common, POSITION\_TARGET\_GLOBAL\_INT)
- **TEST** (common, LOCAL\_POSITION\_NED\_SYSTEM\_GLOBAL\_OFFSET)
- **TEST** (common, HIL\_STATE)
- **TEST** (common, HIL\_CONTROLS)
- **TEST** (common, HIL\_RC\_INPUTS\_RAW)
- **TEST** (common, HIL\_ACTUATOR\_CONTROLS)
- **TEST** (common, OPTICAL\_FLOW)
- **TEST** (common, GLOBAL\_VISION\_POSITION\_ESTIMATE)
- **TEST** (common, VISION\_POSITION\_ESTIMATE)
- **TEST** (common, VISION\_SPEED\_ESTIMATE)
- **TEST** (common, VICON\_POSITION\_ESTIMATE)
- **TEST** (common, HIGHRES\_IMU)
- **TEST** (common, OPTICAL\_FLOW\_RAD)

- **TEST** (common, HIL\_SENSOR)
- **TEST** (common, SIM\_STATE)
- **TEST** (common, RADIO\_STATUS)
- **TEST** (common, FILE\_TRANSFER\_PROTOCOL)
- **TEST** (common, TIMESYNC)
- **TEST** (common, CAMERA\_TRIGGER)
- **TEST** (common, HIL\_GPS)
- **TEST** (common, HIL\_OPTICAL\_FLOW)
- **TEST** (common, HIL\_STATE\_QUATERNION)
- **TEST** (common, SCALED\_IMU2)
- **TEST** (common, LOG\_REQUEST\_LIST)
- **TEST** (common, LOG\_ENTRY)
- **TEST** (common, LOG\_REQUEST\_DATA)
- **TEST** (common, LOG\_DATA)
- **TEST** (common, LOG\_ERASE)
- **TEST** (common, LOG\_REQUEST\_END)
- **TEST** (common, GPS\_INJECT\_DATA)
- **TEST** (common, GPS2\_RAW)
- **TEST** (common, POWER\_STATUS)
- **TEST** (common, SERIAL\_CONTROL)
- **TEST** (common, GPS\_RTK)
- **TEST** (common, GPS2\_RTK)
- **TEST** (common, SCALED\_IMU3)
- **TEST** (common, DATA\_TRANSMISSION\_HANDSHAKE)
- **TEST** (common, ENCAPSULATED\_DATA)
- **TEST** (common, DISTANCE\_SENSOR)
- **TEST** (common, TERRAIN\_REQUEST)
- **TEST** (common, TERRAIN\_DATA)
- **TEST** (common, TERRAIN\_CHECK)
- **TEST** (common, TERRAIN\_REPORT)
- **TEST** (common, SCALED\_PRESSURE2)
- **TEST** (common, ATT\_POS\_MOCAP)
- **TEST** (common, SET\_ACTUATOR\_CONTROL\_TARGET)
- **TEST** (common, ACTUATOR\_CONTROL\_TARGET)
- **TEST** (common, ALTITUDE)
- **TEST** (common, RESOURCE\_REQUEST)
- **TEST** (common, SCALED\_PRESSURE3)
- **TEST** (common, FOLLOW\_TARGET)
- **TEST** (common, CONTROL\_SYSTEM\_STATE)
- **TEST** (common, BATTERY\_STATUS)
- **TEST** (common, LANDING\_TARGET)
- **TEST** (common, FENCE\_STATUS)
- **TEST** (common, MAG\_CAL\_REPORT)
- **TEST** (common, EFI\_STATUS)
- **TEST** (common, ESTIMATOR\_STATUS)
- **TEST** (common, WIND\_COV)
- **TEST** (common, GPS\_INPUT)
- **TEST** (common, GPS\_RTCM\_DATA)
- **TEST** (common, HIGH\_LATENCY)
- **TEST** (common, HIGH\_LATENCY2)
- **TEST** (common, VIBRATION)
- **TEST** (common, HOME\_POSITION)
- **TEST** (common, SET\_HOME\_POSITION)
- **TEST** (common, MESSAGE\_INTERVAL)
- **TEST** (common, EXTENDED\_SYS\_STATE)

- **TEST** (common, ADSB\_VEHICLE)
- **TEST** (common, COLLISION)
- **TEST** (common, V2\_EXTENSION)
- **TEST** (common, MEMORY\_VECT)
- **TEST** (common, DEBUG\_VECT)
- **TEST** (common, NAMED\_VALUE\_FLOAT)
- **TEST** (common, NAMED\_VALUE\_INT)
- **TEST** (common, STATUSTEXT)
- **TEST** (common, DEBUG)
- **TEST** (common, SETUP\_SIGNING)
- **TEST** (common, BUTTON\_CHANGE)
- **TEST** (common, PLAY\_TUNE)
- **TEST** (common, CAMERA\_INFORMATION)
- **TEST** (common, CAMERA\_SETTINGS)
- **TEST** (common, STORAGE\_INFORMATION)
- **TEST** (common, CAMERA\_CAPTURE\_STATUS)
- **TEST** (common, CAMERA\_IMAGE\_CAPTURED)
- **TEST** (common, FLIGHT\_INFORMATION)
- **TEST** (common, MOUNT\_ORIENTATION)
- **TEST** (common, LOGGING\_DATA)
- **TEST** (common, LOGGING\_DATA\_ACKED)
- **TEST** (common, LOGGING\_ACK)
- **TEST** (common, VIDEO\_STREAM\_INFORMATION)
- **TEST** (common, VIDEO\_STREAM\_STATUS)
- **TEST** (common, CAMERA\_FOV\_STATUS)
- **TEST** (common, CAMERA\_TRACKING\_IMAGE\_STATUS)
- **TEST** (common, CAMERA\_TRACKING\_GEO\_STATUS)
- **TEST** (common, CAMERA\_THERMAL\_RANGE)
- **TEST** (common, GIMBAL\_MANAGER\_INFORMATION)
- **TEST** (common, GIMBAL\_MANAGER\_STATUS)
- **TEST** (common, GIMBAL\_MANAGER\_SET\_ATTITUDE)
- **TEST** (common, GIMBAL\_DEVICE\_INFORMATION)
- **TEST** (common, GIMBAL\_DEVICE\_SET\_ATTITUDE)
- **TEST** (common, GIMBAL\_DEVICE\_ATTITUDE\_STATUS)
- **TEST** (common, AUTOPILOT\_STATE\_FOR\_GIMBAL\_DEVICE)
- **TEST** (common, GIMBAL\_MANAGER\_SET\_PITCHYAW)
- **TEST** (common, GIMBAL\_MANAGER\_SET\_MANUAL\_CONTROL)
- **TEST** (common, ESC\_INFO)
- **TEST** (common, ESC\_STATUS)
- **TEST** (common, AIRSPEED)
- **TEST** (common, WIFI\_CONFIG\_AP)
- **TEST** (common, PROTOCOL\_VERSION)
- **TEST** (common, AIS\_VESSEL)
- **TEST** (common, UAVCAN\_NODE\_STATUS)
- **TEST** (common, UAVCAN\_NODE\_INFO)
- **TEST** (common, PARAM\_EXT\_REQUEST\_READ)
- **TEST** (common, PARAM\_EXT\_REQUEST\_LIST)
- **TEST** (common, PARAM\_EXT\_VALUE)
- **TEST** (common, PARAM\_EXT\_SET)
- **TEST** (common, PARAM\_EXT\_ACK)
- **TEST** (common, OBSTACLE\_DISTANCE)
- **TEST** (common, ODOMETRY)
- **TEST** (common, TRAJECTORY\_REPRESENTATION\_WAYPOINTS)
- **TEST** (common, TRAJECTORY\_REPRESENTATION\_BEZIER)
- **TEST** (common, CELLULAR\_STATUS)

- **TEST** (common, ISBD\_LINK\_STATUS)
- **TEST** (common, CELLULAR\_CONFIG)
- **TEST** (common, RAW\_RPM)
- **TEST** (common, UTM\_GLOBAL\_POSITION)
- **TEST** (common, PARAM\_ERROR)
- **TEST** (common, DEBUG\_FLOAT\_ARRAY)
- **TEST** (common, ORBIT\_EXECUTION\_STATUS)
- **TEST** (common, SMART\_BATTERY\_INFO)
- **TEST** (common, FIGURE\_EIGHT\_EXECUTION\_STATUS)
- **TEST** (common, FUEL\_STATUS)
- **TEST** (common, BATTERY\_INFO)
- **TEST** (common, GENERATOR\_STATUS)
- **TEST** (common, ACTUATOR\_OUTPUT\_STATUS)
- **TEST** (common, RELAY\_STATUS)
- **TEST** (common, TIME\_ESTIMATE\_TO\_TARGET)
- **TEST** (common, TUNNEL)
- **TEST** (common, CAN\_FRAME)
- **TEST** (common, ONBOARD\_COMPUTER\_STATUS)
- **TEST** (common, COMPONENT\_INFORMATION)
- **TEST** (common, COMPONENT\_INFORMATION\_BASIC)
- **TEST** (common, COMPONENT\_METADATA)
- **TEST** (common, PLAY\_TUNE\_V2)
- **TEST** (common, SUPPORTED\_TUNES)
- **TEST** (common, EVENT)
- **TEST** (common, CURRENT\_EVENT\_SEQUENCE)
- **TEST** (common, REQUEST\_EVENT)
- **TEST** (common, RESPONSE\_EVENT\_ERROR)
- **TEST** (common, AVAILABLE\_MODES)
- **TEST** (common, CURRENT\_MODE)
- **TEST** (common, AVAILABLE\_MODES\_MONITOR)
- **TEST** (common, ILLUMINATOR\_STATUS)
- **TEST** (common, CANFD\_FRAME)
- **TEST** (common, CAN\_FILTER\_MODIFY)
- **TEST** (common, WHEEL\_DISTANCE)
- **TEST** (common, WINCH\_STATUS)
- **TEST** (common, OPEN\_DRONE\_ID\_BASIC\_ID)
- **TEST** (common, OPEN\_DRONE\_ID\_LOCATION)
- **TEST** (common, OPEN\_DRONE\_ID\_AUTHENTICATION)
- **TEST** (common, OPEN\_DRONE\_ID\_SELF\_ID)
- **TEST** (common, OPEN\_DRONE\_ID\_SYSTEM)
- **TEST** (common, OPEN\_DRONE\_ID\_OPERATOR\_ID)
- **TEST** (common, OPEN\_DRONE\_ID\_MESSAGE\_PACK)
- **TEST** (common, OPEN\_DRONE\_ID\_ARM\_STATUS)
- **TEST** (common, OPEN\_DRONE\_ID\_SYSTEM\_UPDATE)
- **TEST** (common, HYGROMETER\_SENSOR)

### 6.8.1 Detailed Description

MAVLink comm testsuite protocol generated from common.xml.

See also

<http://mavlink.org>

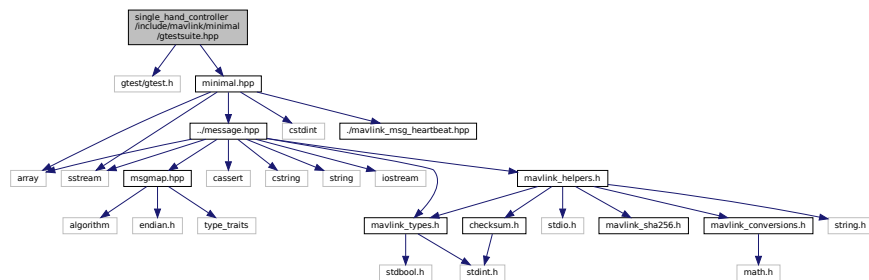
## 6.9 single\_hand\_controller/include/mavlink/minimal/gtestsuite.hpp File Reference

MAVLink comm testsuite protocol generated from minimal.xml.

```
#include <gtest/gtest.h>
```

```
#include "minimal.hpp"
```

Include dependency graph for gtestsuite.hpp:



### Functions

- **TEST** (minimal, HEARTBEAT)

### 6.9.1 Detailed Description

MAVLink comm testsuite protocol generated from minimal.xml.

See also

<http://mavlink.org>

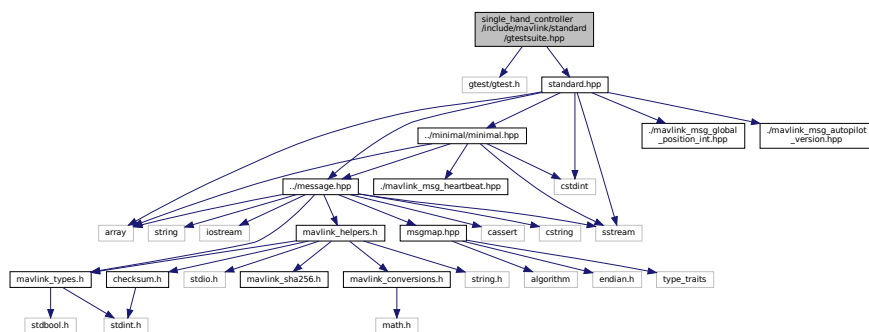
## 6.10 single\_hand\_controller/include/mavlink/standard/gtestsuite.hpp File Reference

MAVLink comm testsuite protocol generated from standard.xml.

```
#include <gtest/gtest.h>
```

```
#include "standard.hpp"
```

Include dependency graph for gtestsuite.hpp:





## Functions

- **TEST** (standard, GLOBAL\_POSITION\_INT)
- **TEST** (standard, AUTOPILOT\_VERSION)

### 6.10.1 Detailed Description

MAVLink comm testsuite protocol generated from standard.xml.

See also

<http://mavlink.org>

## 6.11 single\_hand\_controller/include/mavlink/common/mavlink.h File Reference

MAVLink comm protocol built from common.xml.

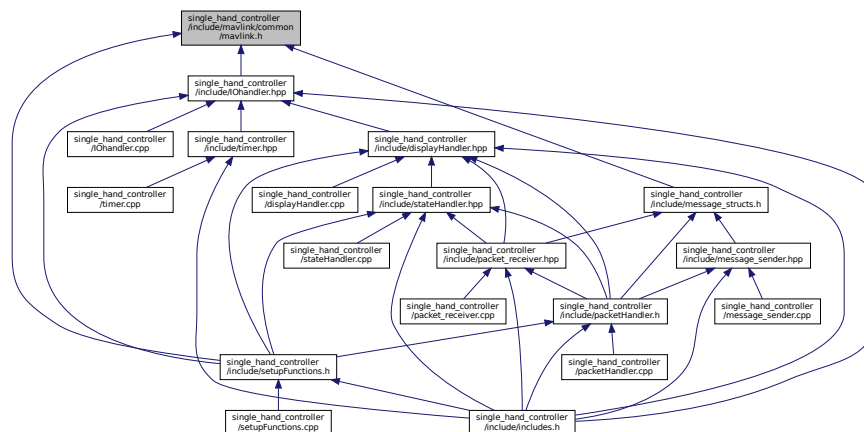
```
#include "version.h"
```

```
#include "common.h"
```

Include dependency graph for mavlink.h:



This graph shows which files directly or indirectly include this file:



## Macros

- **#define MAVLINK\_H**
- **#define MAVLINK\_PRIMARY\_XML\_HASH** -6952494756206127088
- **#define MAVLINK\_STX** 253
- **#define MAVLINK\_ENDIAN** MAVLINK\_LITTLE\_ENDIAN
- **#define MAVLINK\_ALIGNED\_FIELDS** 1
- **#define MAVLINK\_CRC\_EXTRA** 1
- **#define MAVLINK\_COMMAND\_24BIT** 1

### 6.11.1 Detailed Description

MAVLink comm protocol built from common.xml.

See also

<http://mavlink.org>

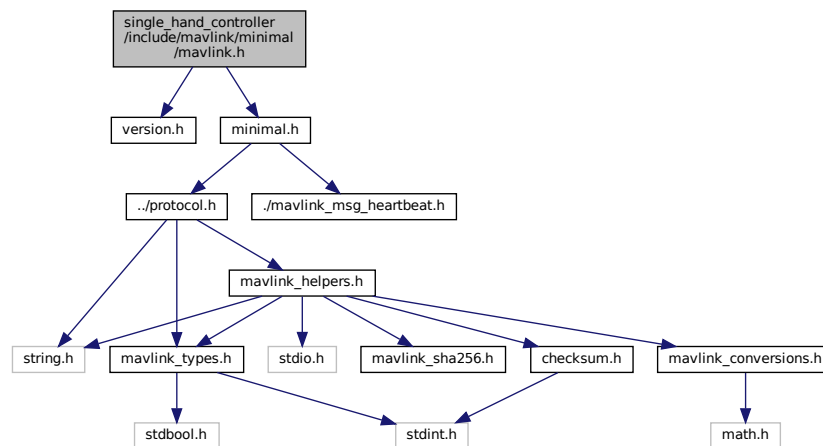
## 6.12 single\_hand\_controller/include/mavlink/minimal/mavlink.h File Reference

MAVLink comm protocol built from minimal.xml.

```
#include "version.h"
```

```
#include "minimal.h"
```

Include dependency graph for mavlink.h:



### Macros

- `#define MAVLINK_H`
- `#define MAVLINK_PRIMARY_XML_HASH 2229504358257161077`
- `#define MAVLINK_STX 253`
- `#define MAVLINK_ENDIAN MAVLINK_LITTLE_ENDIAN`
- `#define MAVLINK_ALIGNED_FIELDS 1`
- `#define MAVLINK_CRC_EXTRA 1`
- `#define MAVLINK_COMMAND_24BIT 1`

### 6.12.1 Detailed Description

MAVLink comm protocol built from minimal.xml.

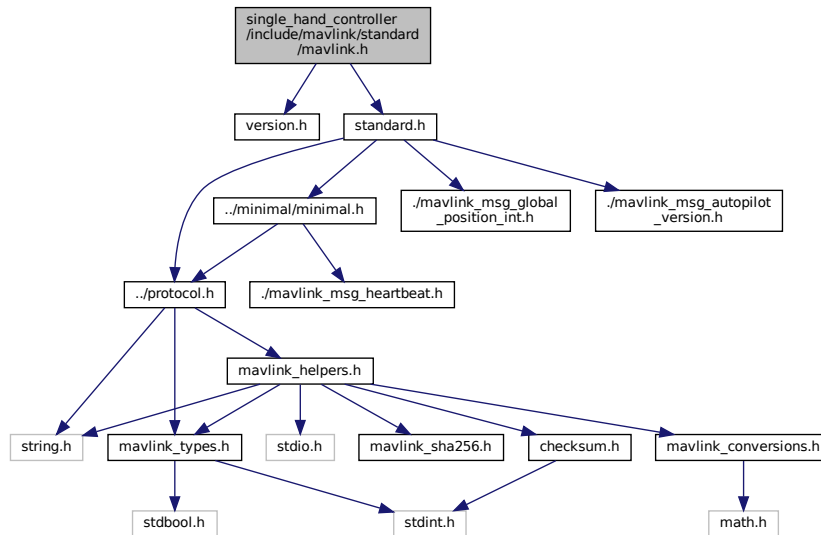
See also

<http://mavlink.org>

## 6.13 single\_hand\_controller/include/mavlink/standard/mavlink.h File Reference

MAVLink comm protocol built from standard.xml.

```
#include "version.h"
#include "standard.h"
Include dependency graph for mavlink.h:
```



### Macros

- `#define MAVLINK_H`
- `#define MAVLINK_PRIMARY_XML_HASH -2768953998991881157`
- `#define MAVLINK_STX 253`
- `#define MAVLINK_ENDIAN MAVLINK_LITTLE_ENDIAN`
- `#define MAVLINK_ALIGNED_FIELDS 1`
- `#define MAVLINK_CRC_EXTRA 1`
- `#define MAVLINK_COMMAND_24BIT 1`

#### 6.13.1 Detailed Description

MAVLink comm protocol built from standard.xml.

See also

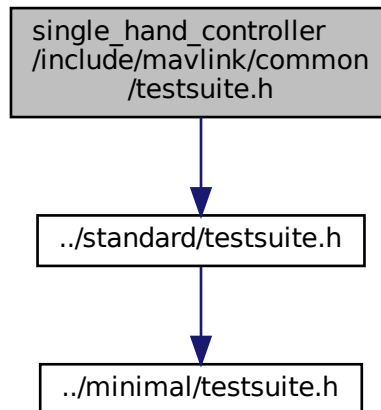
<http://mavlink.org>

## 6.14 single\_hand\_controller/include/mavlink/common/testsuite.h File Reference

MAVLink comm protocol testsuite generated from common.xml.

```
#include "../standard/testsuite.h"
```

Include dependency graph for testsuite.h:



### 6.14.1 Detailed Description

MAVLink comm protocol testsuite generated from common.xml.

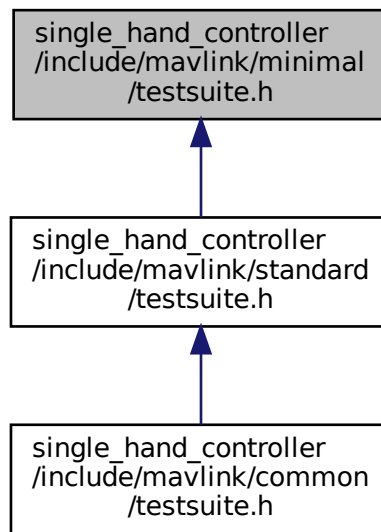
See also

<https://mavlink.io/en/>

## 6.15 single\_hand\_controller/include/mavlink/minimal/testsuite.h File Reference

MAVLink comm protocol testsuite generated from minimal.xml.

This graph shows which files directly or indirectly include this file:



### 6.15.1 Detailed Description

MAVLink comm protocol testsuite generated from minimal.xml.

See also

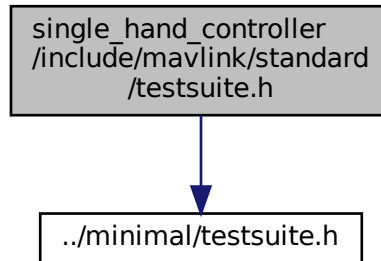
<https://mavlink.io/en/>

## 6.16 single\_hand\_controller/include/mavlink/standard/testsuite.h File Reference

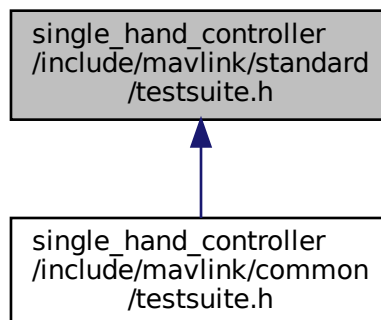
MAVLink comm protocol testsuite generated from standard.xml.

```
#include "../minimal/testsuite.h"
```

Include dependency graph for testsuite.h:



This graph shows which files directly or indirectly include this file:



### 6.16.1 Detailed Description

MAVLink comm protocol testsuite generated from standard.xml.

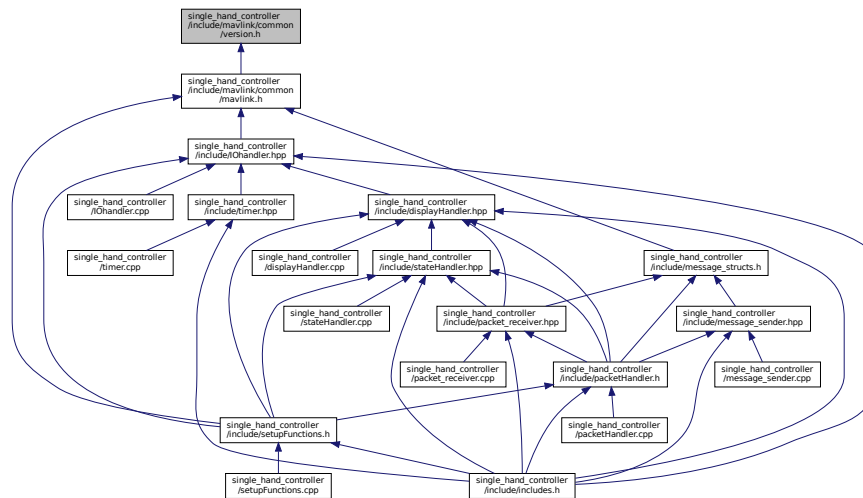
See also

<https://mavlink.io/en/>

## 6.17 single\_hand\_controller/include/mavlink/common/version.h File Reference

MAVLink comm protocol built from common.xml.

This graph shows which files directly or indirectly include this file:



## Macros

- `#define MAVLINK_VERSION_H`
- `#define MAVLINK_BUILD_DATE "Tue Feb 17 2026"`
- `#define MAVLINK_WIRE_PROTOCOL_VERSION "2.0"`
- `#define MAVLINK_MAX_DIALECT_PAYLOAD_SIZE 255`

### 6.17.1 Detailed Description

MAVLink comm protocol built from common.xml.

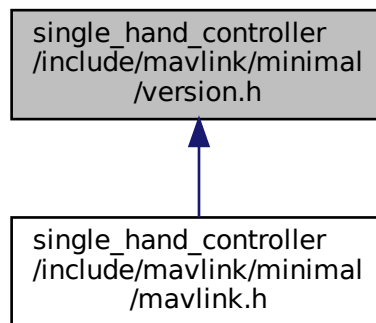
See also

<http://mavlink.org>

## 6.18 single\_hand\_controller/include/mavlink/minimal/version.h File Reference

MAVLink comm protocol built from minimal.xml.

This graph shows which files directly or indirectly include this file:



## Macros

- #define **MAVLINK\_VERSION\_H**
- #define **MAVLINK\_BUILD\_DATE** "Tue Feb 17 2026"
- #define **MAVLINK\_WIRE\_PROTOCOL\_VERSION** "2.0"
- #define **MAVLINK\_MAX\_DIALECT\_PAYLOAD\_SIZE** 9

### 6.18.1 Detailed Description

MAVLink comm protocol built from minimal.xml.

See also

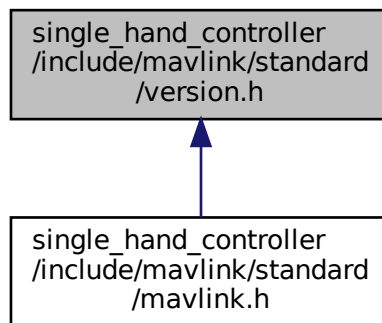
<http://mavlink.org>

## 6.19 single\_hand\_controller/include/mavlink/standard/version.h File Reference

MAVLink comm protocol built from standard.xml.



This graph shows which files directly or indirectly include this file:



## Macros

- #define **MAVLINK\_VERSION\_H**
- #define **MAVLINK\_BUILD\_DATE** "Tue Feb 17 2026"
- #define **MAVLINK\_WIRE\_PROTOCOL\_VERSION** "2.0"
- #define **MAVLINK\_MAX\_DIALECT\_PAYLOAD\_SIZE** 78

### 6.19.1 Detailed Description

MAVLink comm protocol built from standard.xml.

See also

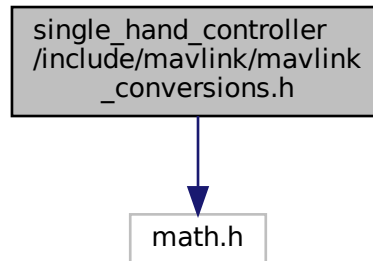
<http://mavlink.org>

## 6.20 single\_hand\_controller/include/mavlink/mavlink\_conversions.h File Reference

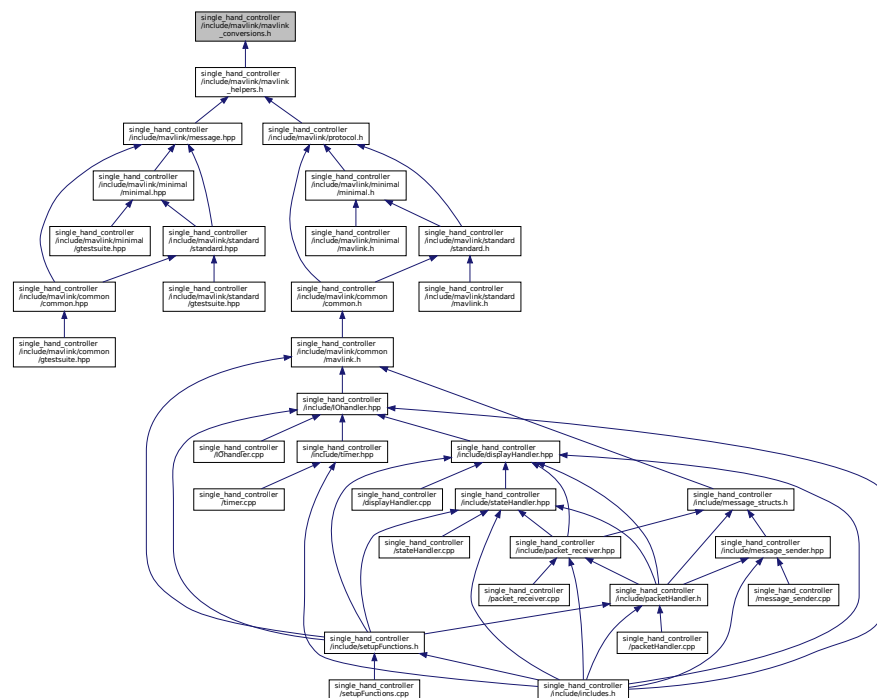
These conversion functions follow the NASA rotation standards definition file available online.

```
#include <math.h>
```

Include dependency graph for mavlink\_conversions.h:



This graph shows which files directly or indirectly include this file:



## Macros

- `#define M_PI_2 ((float)asin(1))`

## Functions

- MAVLINK\_HELPER void [mavlink\\_quaternion\\_to\\_dcm](#) (const float quaternion[4], float dcm[3][3])

*Converts a quaternion to a rotation matrix.*

- MAVLINK\_HELPER void [mavlink\\_dcm\\_to\\_euler](#) (const float dcm[3][3], float \*roll, float \*pitch, float \*yaw)

*Converts a rotation matrix to euler angles.*

- MAVLINK\_HELPER void [mavlink\\_quaternion\\_to\\_euler](#) (const float quaternion[4], float \*roll, float \*pitch, float \*yaw)

*Converts a quaternion to euler angles.*

- MAVLINK\_HELPER void [mavlink\\_euler\\_to\\_quaternion](#) (float roll, float pitch, float yaw, float quaternion[4])

*Converts euler angles to a quaternion.*

- MAVLINK\_HELPER void [mavlink\\_dcm\\_to\\_quaternion](#) (const float dcm[3][3], float quaternion[4])

*Converts a rotation matrix to a quaternion Reference:*

- MAVLINK\_HELPER void [mavlink\\_euler\\_to\\_dcm](#) (float roll, float pitch, float yaw, float dcm[3][3])

*Converts euler angles to a rotation matrix.*

## 6.20.1 Detailed Description

These conversion functions follow the NASA rotation standards definition file available online.

Their intent is to lower the barrier for MAVLink adopters to use gimbal-lock free (both rotation matrices, sometimes called DCM, and quaternions are gimbal-lock free) rotation representations. Euler angles (roll, pitch, yaw) will be phased out of the protocol as widely as possible.

### Author

James Goppert

Thomas Gubler [thomasgubler@gmail.com](mailto:thomasgubler@gmail.com)

## 6.20.2 Function Documentation

### 6.20.2.1 [mavlink\\_dcm\\_to\\_euler\(\)](#)

```
MAVLINK_HELPER void mavlink_dcm_to_euler (
 const float dcm[3][3],
 float * roll,
 float * pitch,
 float * yaw)
```

Converts a rotation matrix to euler angles.

#### Parameters

|              |                            |
|--------------|----------------------------|
| <i>dcm</i>   | a 3x3 rotation matrix      |
| <i>roll</i>  | the roll angle in radians  |
| <i>pitch</i> | the pitch angle in radians |
| <i>yaw</i>   | the yaw angle in radians   |

### 6.20.2.2 mavlink\_dcm\_to\_quaternion()

```
MAVLINK_HELPER void mavlink_dcm_to_quaternion (
 const float dcm[3][3],
 float quaternion[4])
```

Converts a rotation matrix to a quaternion Reference:

- Shoemake, Quaternions, <http://www.cs.ucr.edu/~vbz/resources/quatut.pdf>

#### Parameters

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| <i>dcm</i>        | a 3x3 rotation matrix                                           |
| <i>quaternion</i> | a [w, x, y, z] ordered quaternion (null-rotation being 1 0 0 0) |

### 6.20.2.3 mavlink\_euler\_to\_dcm()

```
MAVLINK_HELPER void mavlink_euler_to_dcm (
 float roll,
 float pitch,
 float yaw,
 float dcm[3][3])
```

Converts euler angles to a rotation matrix.

#### Parameters

|              |                            |
|--------------|----------------------------|
| <i>roll</i>  | the roll angle in radians  |
| <i>pitch</i> | the pitch angle in radians |
| <i>yaw</i>   | the yaw angle in radians   |
| <i>dcm</i>   | a 3x3 rotation matrix      |

### 6.20.2.4 mavlink\_euler\_to\_quaternion()

```
MAVLINK_HELPER void mavlink_euler_to_quaternion (
 float roll,
 float pitch,
 float yaw,
 float quaternion[4])
```

Converts euler angles to a quaternion.

#### Parameters

|             |                           |
|-------------|---------------------------|
| <i>roll</i> | the roll angle in radians |
|-------------|---------------------------|

## Parameters

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| <i>pitch</i>      | the pitch angle in radians                                      |
| <i>yaw</i>        | the yaw angle in radians                                        |
| <i>quaternion</i> | a [w, x, y, z] ordered quaternion (null-rotation being 1 0 0 0) |

**6.20.2.5 mavlink\_quaternion\_to\_dcm()**

```
MAVLINK_HELPER void mavlink_quaternion_to_dcm (
 const float quaternion[4],
 float dcm[3][3])
```

Converts a quaternion to a rotation matrix.

## Parameters

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| <i>quaternion</i> | a [w, x, y, z] ordered quaternion (null-rotation being 1 0 0 0) |
| <i>dcm</i>        | a 3x3 rotation matrix                                           |

**6.20.2.6 mavlink\_quaternion\_to\_euler()**

```
MAVLINK_HELPER void mavlink_quaternion_to_euler (
 const float quaternion[4],
 float * roll,
 float * pitch,
 float * yaw)
```

Converts a quaternion to euler angles.

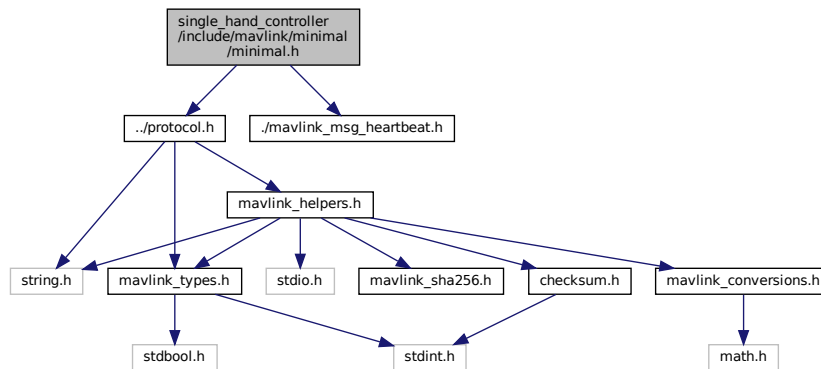
## Parameters

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| <i>quaternion</i> | a [w, x, y, z] ordered quaternion (null-rotation being 1 0 0 0) |
| <i>roll</i>       | the roll angle in radians                                       |
| <i>pitch</i>      | the pitch angle in radians                                      |
| <i>yaw</i>        | the yaw angle in radians                                        |

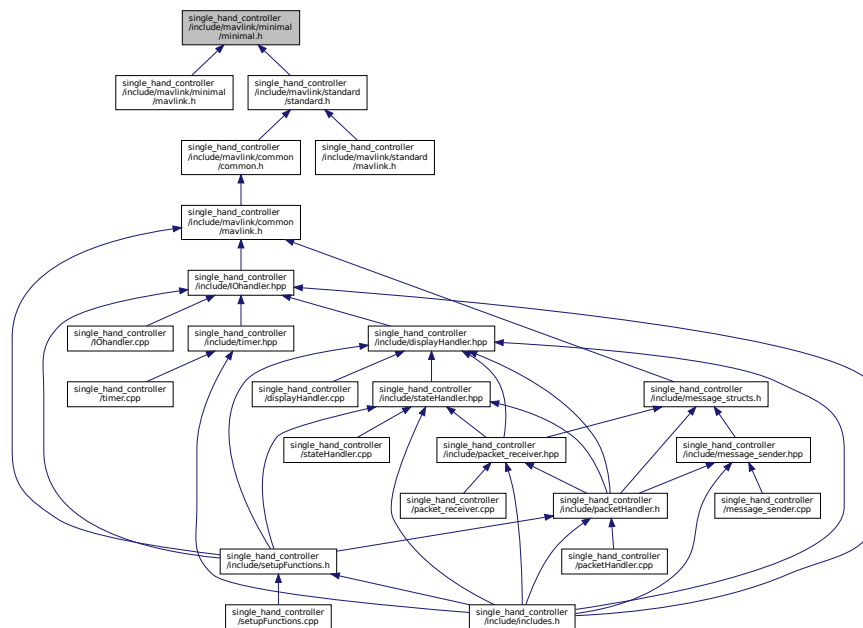
**6.21 single\_hand\_controller/include/mavlink/minimal/minimal.h File Reference**

MAVLink comm protocol generated from minimal.xml.

```
#include "../protocol.h"
#include "../mavlink_msg_heartbeat.h"
Include dependency graph for minimal.h:
```



This graph shows which files directly or indirectly include this file:



## Macros

- `#define MAVLINK_MINIMAL_H`
- `#define MAVLINK_MINIMAL_XML_HASH 2229504358257161077`
- `#define MAVLINK_MESSAGE_LENGTHS {}`
- `#define MAVLINK_MESSAGE_CRCS {{0, 50, 9, 9, 0, 0, 0}}`
- `#define MAVLINK_ENABLED_MINIMAL`
- `#define HAVE_ENUM_MAV_AUTOPILOT`

*Micro air vehicle / autopilot classes.*

- #define **HAVE\_ENUM\_MAV\_TYPE**  
*MAVLink component type reported in HEARTBEAT message.*
- #define **HAVE\_ENUM\_MAV\_MODE\_FLAG**  
*These flags encode the MAV mode, see MAV\_MODE enum for useful combinations.*
- #define **HAVE\_ENUM\_MAV\_MODE\_FLAG\_DECODE\_POSITION**  
*These values encode the bit positions of the decode position.*
- #define **HAVE\_ENUM\_MAV\_STATE**
- #define **HAVE\_ENUM\_MAV\_COMPONENT**  
*Legacy component ID values for particular types of hardware/software that might make up a MAVLink system (autopilot, cameras, servos, avoidance systems etc.).*
- #define **MAVLINK\_VERSION** 3

## Typedefs

- typedef enum MAV\_AUTOPILOT **MAV\_AUTOPILOT**
- typedef enum MAV\_TYPE **MAV\_TYPE**
- typedef enum MAV\_MODE\_FLAG **MAV\_MODE\_FLAG**
- typedef enum MAV\_MODE\_FLAG\_DECODE\_POSITION **MAV\_MODE\_FLAG\_DECODE\_POSITION**
- typedef enum MAV\_STATE **MAV\_STATE**
- typedef enum MAV\_COMPONENT **MAV\_COMPONENT**

## Enumerations

- enum **MAV\_AUTOPILOT** {  
**MAV\_AUTOPILOT\_GENERIC** =0 , **MAV\_AUTOPILOT\_RESERVED** =1 , **MAV\_AUTOPILOT\_SLUGS** =2 ,  
**MAV\_AUTOPILOT\_ARDUPILOTMEGA** =3 ,  
**MAV\_AUTOPILOT\_OPENPILOT** =4 , **MAV\_AUTOPILOT\_GENERIC\_WAYPOINTS\_ONLY** =5 , **MAV\_↵**  
**AUTOPILOT\_GENERIC\_WAYPOINTS\_AND\_SIMPLE\_NAVIGATION\_ONLY** =6 , **MAV\_↵**  
**GENERIC\_MISSION\_FULL** =7 ,  
**MAV\_AUTOPILOT\_INVALID** =8 , **MAV\_AUTOPILOT\_PPZ** =9 , **MAV\_AUTOPILOT\_UBD** =10 , **MAV\_↵**  
**AUTOPILOT\_FP** =11 ,  
**MAV\_AUTOPILOT\_PX4** =12 , **MAV\_AUTOPILOT\_SMACCMPILOT** =13 , **MAV\_AUTOPILOT\_AUTOQUAD**  
=14 , **MAV\_AUTOPILOT\_ARMAZILA** =15 ,  
**MAV\_AUTOPILOT\_AEROB** =16 , **MAV\_AUTOPILOT\_ASLUAV** =17 , **MAV\_AUTOPILOT\_SMARTAP** =18 ,  
**MAV\_AUTOPILOT\_AIRRAILS** =19 ,  
**MAV\_AUTOPILOT\_REFLEX** =20 , **MAV\_AUTOPILOT\_ENUM\_END** =21 }
- enum **MAV\_TYPE** {  
**MAV\_TYPE\_GENERIC** =0 , **MAV\_TYPE\_FIXED\_WING** =1 , **MAV\_TYPE\_QUADROTOR** =2 , **MAV\_↵**  
**\_COAXIAL** =3 ,  
**MAV\_TYPE\_HELICOPTER** =4 , **MAV\_TYPE\_ANTENNA\_TRACKER** =5 , **MAV\_TYPE\_GCS** =6 , **MAV\_↵**  
**TYPE\_AIRSHIP** =7 ,  
**MAV\_TYPE\_FREE\_BALLOON** =8 , **MAV\_TYPE\_ROCKET** =9 , **MAV\_TYPE\_GROUND\_ROVER** =10 ,  
**MAV\_TYPE\_SURFACE\_BOAT** =11 ,  
**MAV\_TYPE\_SUBMARINE** =12 , **MAV\_TYPE\_HEXAROTOR** =13 , **MAV\_TYPE\_OCTOROTOR** =14 , **MAV\_↵**  
**\_TYPE\_TRICOPTER** =15 ,  
**MAV\_TYPE\_FLAPPING\_WING** =16 , **MAV\_TYPE\_KITE** =17 , **MAV\_TYPE\_ONBOARD\_CONTROLLER**  
=18 , **MAV\_TYPE\_VTOL\_TAILSITTER\_DUOROTOR** =19 ,  
**MAV\_TYPE\_VTOL\_TAILSITTER\_QUADROTOR** =20 , **MAV\_TYPE\_VTOL\_TILTROTOR** =21 , **MAV\_↵**  
**TYPE\_VTOL\_FIXEDROTOR** =22 , **MAV\_TYPE\_VTOL\_TAILSITTER** =23 ,  
**MAV\_TYPE\_VTOL\_TILTWING** =24 , **MAV\_TYPE\_VTOL\_RESERVED5** =25 , **MAV\_TYPE\_GIMBAL** =26 ,  
**MAV\_TYPE\_ADSB** =27 ,

```

MAV_TYPE_PARAFOIL =28 , MAV_TYPE_DODECAROTOR =29 , MAV_TYPE_CAMERA =30 , MAV_↵
TYPE_CHARGING_STATION =31 ,
MAV_TYPE_FLARM =32 , MAV_TYPE_SERVO =33 , MAV_TYPE_ODID =34 , MAV_TYPE_DECAROTOR
=35 ,
MAV_TYPE_BATTERY =36 , MAV_TYPE_PARACHUTE =37 , MAV_TYPE_LOG =38 , MAV_TYPE OSD
=39 ,
MAV_TYPE_IMU =40 , MAV_TYPE_GPS =41 , MAV_TYPE_WINCH =42 , MAV_TYPE_GENERIC_↵
MULTIROTOR =43 ,
MAV_TYPE_ILLUMINATOR =44 , MAV_TYPE_SPACECRAFT_ORBITER =45 , MAV_TYPE_GROUND_↵
QUADRUPEL =46 , MAV_TYPE_VTOL_GYRODYNE =47 ,
MAV_TYPE_GRIPPER =48 , MAV_TYPE_RADIO =49 , MAV_TYPE_ENUM_END =50 }
• enum MAV_MODE_FLAG {
MAV_MODE_FLAG_CUSTOM_MODE_ENABLED =1 , MAV_MODE_FLAG_TEST_ENABLED =2 , MAV_↵
_MODE_FLAG_AUTO_ENABLED =4 , MAV_MODE_FLAG_GUIDED_ENABLED =8 ,
MAV_MODE_FLAG_STABILIZE_ENABLED =16 , MAV_MODE_FLAG_HIL_ENABLED =32 , MAV_↵
MODE_FLAG_MANUAL_INPUT_ENABLED =64 , MAV_MODE_FLAG_SAFETY_ARMED =128 ,
MAV_MODE_FLAG_ENUM_END =129 }
• enum MAV_MODE_FLAG_DECODE_POSITION {
MAV_MODE_FLAG_DECODE_POSITION_CUSTOM_MODE =1 , MAV_MODE_FLAG_DECODE_↵
POSITION_TEST =2 , MAV_MODE_FLAG_DECODE_POSITION_AUTO =4 , MAV_MODE_FLAG_↵
DECODE_POSITION_GUIDED =8 ,
MAV_MODE_FLAG_DECODE_POSITION_STABILIZE =16 , MAV_MODE_FLAG_DECODE_POSITION_↵
_HIL =32 , MAV_MODE_FLAG_DECODE_POSITION_MANUAL =64 , MAV_MODE_FLAG_DECODE_↵
POSITION_SAFETY =128 ,
MAV_MODE_FLAG_DECODE_POSITION_ENUM_END =129 }
• enum MAV_STATE {
MAV_STATE_UNINIT =0 , MAV_STATE_BOOT =1 , MAV_STATE_CALIBRATING =2 , MAV_STATE_↵
STANDBY =3 ,
MAV_STATE_ACTIVE =4 , MAV_STATE_CRITICAL =5 , MAV_STATE_EMERGENCY =6 , MAV_STATE_↵
POWEROFF =7 ,
MAV_STATE_FLIGHT_TERMINATION =8 , MAV_STATE_ENUM_END =9 }
• enum MAV_COMPONENT {
MAV_COMP_ID_ALL =0 , MAV_COMP_ID_AUTOPILOT1 =1 , MAV_COMP_ID_USER1 =25 , MAV_↵
COMP_ID_USER2 =26 ,
MAV_COMP_ID_USER3 =27 , MAV_COMP_ID_USER4 =28 , MAV_COMP_ID_USER5 =29 , MAV_↵
COMP_ID_USER6 =30 ,
MAV_COMP_ID_USER7 =31 , MAV_COMP_ID_USER8 =32 , MAV_COMP_ID_USER9 =33 , MAV_↵
COMP_ID_USER10 =34 ,
MAV_COMP_ID_USER11 =35 , MAV_COMP_ID_USER12 =36 , MAV_COMP_ID_USER13 =37 , MAV_↵
COMP_ID_USER14 =38 ,
MAV_COMP_ID_USER15 =39 , MAV_COMP_ID_USER16 =40 , MAV_COMP_ID_USER17 =41 , MAV_↵
COMP_ID_USER18 =42 ,
MAV_COMP_ID_USER19 =43 , MAV_COMP_ID_USER20 =44 , MAV_COMP_ID_USER21 =45 , MAV_↵
COMP_ID_USER22 =46 ,
MAV_COMP_ID_USER23 =47 , MAV_COMP_ID_USER24 =48 , MAV_COMP_ID_USER25 =49 , MAV_↵
COMP_ID_USER26 =50 ,
MAV_COMP_ID_USER27 =51 , MAV_COMP_ID_USER28 =52 , MAV_COMP_ID_USER29 =53 , MAV_↵
COMP_ID_USER30 =54 ,
MAV_COMP_ID_USER31 =55 , MAV_COMP_ID_USER32 =56 , MAV_COMP_ID_USER33 =57 , MAV_↵
COMP_ID_USER34 =58 ,
MAV_COMP_ID_USER35 =59 , MAV_COMP_ID_USER36 =60 , MAV_COMP_ID_USER37 =61 , MAV_↵
COMP_ID_USER38 =62 ,
MAV_COMP_ID_USER39 =63 , MAV_COMP_ID_USER40 =64 , MAV_COMP_ID_USER41 =65 , MAV_↵
COMP_ID_USER42 =66 ,
MAV_COMP_ID_USER43 =67 , MAV_COMP_ID_TELEMETRY_RADIO =68 , MAV_COMP_ID_USER45
=69 , MAV_COMP_ID_USER46 =70 ,
MAV_COMP_ID_USER47 =71 , MAV_COMP_ID_USER48 =72 , MAV_COMP_ID_USER49 =73 , MAV_↵
COMP_ID_USER50 =74 ,

```



```

MAV_COMP_ID_USER51 =75 , MAV_COMP_ID_USER52 =76 , MAV_COMP_ID_USER53 =77 , MAV_↵
COMP_ID_USER54 =78 ,
MAV_COMP_ID_USER55 =79 , MAV_COMP_ID_USER56 =80 , MAV_COMP_ID_USER57 =81 , MAV_↵
COMP_ID_USER58 =82 ,
MAV_COMP_ID_USER59 =83 , MAV_COMP_ID_USER60 =84 , MAV_COMP_ID_USER61 =85 , MAV_↵
COMP_ID_USER62 =86 ,
MAV_COMP_ID_USER63 =87 , MAV_COMP_ID_USER64 =88 , MAV_COMP_ID_USER65 =89 , MAV_↵
COMP_ID_USER66 =90 ,
MAV_COMP_ID_USER67 =91 , MAV_COMP_ID_USER68 =92 , MAV_COMP_ID_USER69 =93 , MAV_↵
COMP_ID_USER70 =94 ,
MAV_COMP_ID_USER71 =95 , MAV_COMP_ID_USER72 =96 , MAV_COMP_ID_USER73 =97 , MAV_↵
COMP_ID_USER74 =98 ,
MAV_COMP_ID_USER75 =99 , MAV_COMP_ID_CAMERA =100 , MAV_COMP_ID_CAMERA2 =101 ,
MAV_COMP_ID_CAMERA3 =102 ,
MAV_COMP_ID_CAMERA4 =103 , MAV_COMP_ID_CAMERA5 =104 , MAV_COMP_ID_CAMERA6 =105
, MAV_COMP_ID_RADIO =110 ,
MAV_COMP_ID_RADIO2 =111 , MAV_COMP_ID_RADIO3 =112 , MAV_COMP_ID_SERVO1 =140 , MAV_↵
COMP_ID_SERVO2 =141 ,
MAV_COMP_ID_SERVO3 =142 , MAV_COMP_ID_SERVO4 =143 , MAV_COMP_ID_SERVO5 =144 ,
MAV_COMP_ID_SERVO6 =145 ,
MAV_COMP_ID_SERVO7 =146 , MAV_COMP_ID_SERVO8 =147 , MAV_COMP_ID_SERVO9 =148 ,
MAV_COMP_ID_SERVO10 =149 ,
MAV_COMP_ID_SERVO11 =150 , MAV_COMP_ID_SERVO12 =151 , MAV_COMP_ID_SERVO13 =152 ,
MAV_COMP_ID_SERVO14 =153 ,
MAV_COMP_ID_GIMBAL =154 , MAV_COMP_ID_LOG =155 , MAV_COMP_ID_ADSB =156 , MAV_↵
COMP_ID_OSD =157 ,
MAV_COMP_ID_PERIPHERAL =158 , MAV_COMP_ID_QX1_GIMBAL =159 , MAV_COMP_ID_FLARM
=160 , MAV_COMP_ID_PARACHUTE =161 ,
MAV_COMP_ID_WINCH =169 , MAV_COMP_ID_GIMBAL2 =171 , MAV_COMP_ID_GIMBAL3 =172 ,
MAV_COMP_ID_GIMBAL4 =173 ,
MAV_COMP_ID_GIMBAL5 =174 , MAV_COMP_ID_GIMBAL6 =175 , MAV_COMP_ID_BATTERY =180 ,
MAV_COMP_ID_BATTERY2 =181 ,
MAV_COMP_ID MAVCAN =189 , MAV_COMP_ID_MISSIONPLANNER =190 , MAV_COMP_ID_↵
ONBOARD_COMPUTER =191 , MAV_COMP_ID_ONBOARD_COMPUTER2 =192 ,
MAV_COMP_ID_ONBOARD_COMPUTER3 =193 , MAV_COMP_ID_ONBOARD_COMPUTER4 =194 ,
MAV_COMP_ID_PATHPLANNER =195 , MAV_COMP_ID_OBSTACLE_AVOIDANCE =196 ,
MAV_COMP_ID_VISUAL_INERTIAL_ODOMETRY =197 , MAV_COMP_ID_PAIRING_MANAGER =198 ,
MAV_COMP_ID_IMU =200 , MAV_COMP_ID_IMU_2 =201 ,
MAV_COMP_ID_IMU_3 =202 , MAV_COMP_ID_GPS =220 , MAV_COMP_ID_GPS2 =221 , MAV_COMP_↵
_ID_ODID_TXRX_1 =236 ,
MAV_COMP_ID_ODID_TXRX_2 =237 , MAV_COMP_ID_ODID_TXRX_3 =238 , MAV_COMP_ID_UDP_↵
BRIDGE =240 , MAV_COMP_ID_UART_BRIDGE =241 ,
MAV_COMP_ID_TUNNEL_NODE =242 , MAV_COMP_ID_ILLUMINATOR =243 , MAV_COMP_ID_↵
SYSTEM_CONTROL =250 , MAV_COMPONENT_ENUM_END =251 }

```

### 6.21.1 Detailed Description

MAVLink comm protocol generated from minimal.xml.

See also

<http://mavlink.org>

### 6.21.2 Macro Definition Documentation

### 6.21.2.1 HAVE\_ENUM\_MAV\_AUTOPILOT

```
#define HAVE_ENUM_MAV_AUTOPILOT
```

Micro air vehicle / autopilot classes.

This identifies the individual model.

### 6.21.2.2 HAVE\_ENUM\_MAV\_COMPONENT

```
#define HAVE_ENUM_MAV_COMPONENT
```

Legacy component ID values for particular types of hardware/software that might make up a MAVLink system (autopilot, cameras, servos, avoidance systems etc.).

Components are not required or expected to use IDs with names that correspond to their type or function, but using an ID that matches the type may slightly reduce the chances of component id clashes, as, for historical reasons. System integration will still need to ensure that all components have unique IDs.

Component IDs are used for addressing messages to a particular component within a system. A component can use any unique ID between 1 and 255 (MAV\_COMP\_ID\_ALL value is the broadcast address, used to send messages to all components).

Historically component IDs were also used for identifying the type of component. New code must not use component IDs to infer the component type, but instead check the MAV\_TYPE in the HEARTBEAT message.

### 6.21.2.3 HAVE\_ENUM\_MAV\_MODE\_FLAG\_DECODE\_POSITION

```
#define HAVE_ENUM_MAV_MODE_FLAG_DECODE_POSITION
```

These values encode the bit positions of the decode position.

These values can be used to read the value of a flag bit by combining the base\_mode variable with AND with the flag position value. The result will be either 0 or 1, depending on if the flag is set or not.

### 6.21.2.4 HAVE\_ENUM\_MAV\_TYPE

```
#define HAVE_ENUM_MAV_TYPE
```

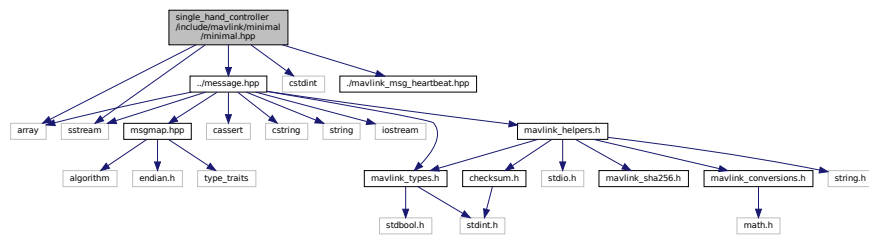
MAVLINK component type reported in HEARTBEAT message.

Flight controllers must report the type of the vehicle on which they are mounted (e.g. MAV\_TYPE\_OCTOROTOR). All other components must report a value appropriate for their type (e.g. a camera must use MAV\_TYPE\_CAMERA).

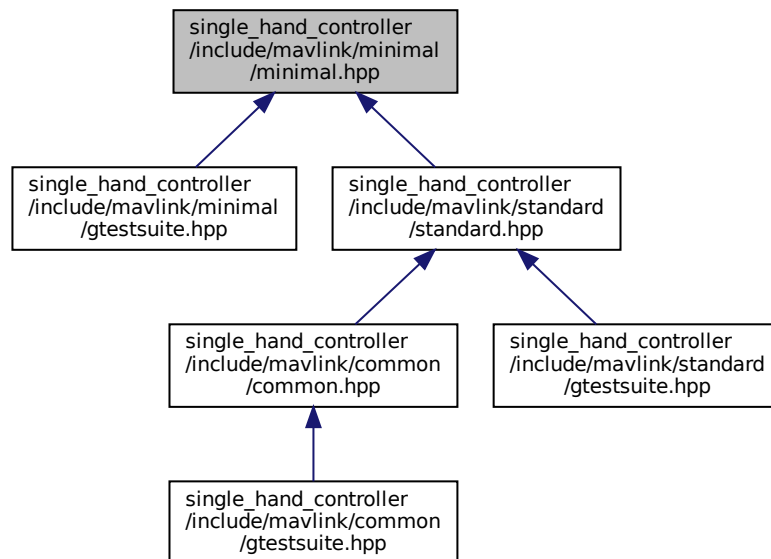
## 6.22 single\_hand\_controller/include/mavlink/minimal/minimal.hpp File Reference

MAVLink comm protocol generated from minimal.xml.

```
#include <array>
#include <cstdint>
#include <sstream>
#include "../message.hpp"
#include "../mavlink_msg_heartbeat.hpp"
Include dependency graph for minimal.hpp:
```



This graph shows which files directly or indirectly include this file:



### Macros

- `#define MAVLINK_STX 253`

## Enumerations

- enum class [mavlink::minimal::MAV\\_AUTOPILOT](#) : uint8\_t {  
**GENERIC** =0 , **RESERVED** =1 , **SLUGS** =2 , **ARDUPILOTMEGA** =3 ,  
**OPENPILOT** =4 , **GENERIC\_WAYPOINTS\_ONLY** =5 , **GENERIC\_WAYPOINTS\_AND\_SIMPLE\_↵**  
**NAVIGATION\_ONLY** =6 , **GENERIC\_MISSION\_FULL** =7 ,  
**INVALID** =8 , **PPZ** =9 , **UDB** =10 , **FP** =11 ,  
**PX4** =12 , **SMACCMPILOT** =13 , **AUTOQUAD** =14 , **ARMAZILA** =15 ,  
**AEROB** =16 , **ASLUAV** =17 , **SMARTAP** =18 , **AIRRAILS** =19 ,  
**REFLEX** =20 }

*Micro air vehicle / autopilot classes.*

- enum class [mavlink::minimal::MAV\\_TYPE](#) : uint8\_t {  
**GENERIC** =0 , **FIXED\_WING** =1 , **QUADROTOR** =2 , **COAXIAL** =3 ,  
**HELICOPTER** =4 , **ANTENNA\_TRACKER** =5 , **GCS** =6 , **AIRSHIP** =7 ,  
**FREE\_BALLOON** =8 , **ROCKET** =9 , **GROUND\_ROVER** =10 , **SURFACE\_BOAT** =11 ,  
**SUBMARINE** =12 , **HEXAROTOR** =13 , **OCTOROTOR** =14 , **TRICOPTER** =15 ,  
**FLAPPING\_WING** =16 , **KITE** =17 , **ONBOARD\_CONTROLLER** =18 , **VTOL\_TAILSITTER\_DUOROTOR**  
=19 ,  
**VTOL\_TAILSITTER\_QUADROTOR** =20 , **VTOL\_TILTROTOR** =21 , **VTOL\_FIXEDROTOR** =22 , **VTOL\_↵**  
**TAILSITTER** =23 ,  
**VTOL\_TILTWING** =24 , **VTOL\_RESERVED5** =25 , **GIMBAL** =26 , **ADSB** =27 ,  
**PARAFOIL** =28 , **DODECAROTOR** =29 , **CAMERA** =30 , **CHARGING\_STATION** =31 ,  
**FLARM** =32 , **SERVO** =33 , **ODID** =34 , **DECAROTOR** =35 ,  
**BATTERY** =36 , **PARACHUTE** =37 , **LOG** =38 , **OSD** =39 ,  
**IMU** =40 , **GPS** =41 , **WINCH** =42 , **GENERIC\_MULTIROTOR** =43 ,  
**ILLUMINATOR** =44 , **SPACECRAFT\_ORBITER** =45 , **GROUND\_QUADRUPE** =46 , **VTOL\_GYRODYNE**  
=47 ,  
**GRIPPER** =48 , **RADIO** =49 }

*MAVLINK component type reported in HEARTBEAT message.*

- enum class [mavlink::minimal::MAV\\_MODE\\_FLAG](#) : uint8\_t {  
**CUSTOM\_MODE\_ENABLED** =1 , **TEST\_ENABLED** =2 , **AUTO\_ENABLED** =4 , **GUIDED\_ENABLED** =8 ,  
**STABILIZE\_ENABLED** =16 , **HIL\_ENABLED** =32 , **MANUAL\_INPUT\_ENABLED** =64 , **SAFETY\_ARMED**  
=128 }

*These flags encode the MAV mode, see MAV\_MODE enum for useful combinations.*

- enum class [mavlink::minimal::MAV\\_MODE\\_FLAG\\_DECODE\\_POSITION](#) {  
**CUSTOM\_MODE** =1 , **TEST** =2 , **AUTO** =4 , **GUIDED** =8 ,  
**STABILIZE** =16 , **HIL** =32 , **MANUAL** =64 , **SAFETY** =128 }

*These values encode the bit positions of the decode position.*

- enum class [mavlink::minimal::MAV\\_STATE](#) : uint8\_t {  
**UNINIT** =0 , **BOOT** =1 , **CALIBRATING** =2 , **STANDBY** =3 ,  
**ACTIVE** =4 , **CRITICAL** =5 , **EMERGENCY** =6 , **POWEROFF** =7 ,  
**FLIGHT\_TERMINATION** =8 }

- enum class [mavlink::minimal::MAV\\_COMPONENT](#) {  
**COMP\_ID\_ALL** =0 , **COMP\_ID\_AUTOPILOT1** =1 , **COMP\_ID\_USER1** =25 , **COMP\_ID\_USER2** =26 ,  
**COMP\_ID\_USER3** =27 , **COMP\_ID\_USER4** =28 , **COMP\_ID\_USER5** =29 , **COMP\_ID\_USER6** =30 ,  
**COMP\_ID\_USER7** =31 , **COMP\_ID\_USER8** =32 , **COMP\_ID\_USER9** =33 , **COMP\_ID\_USER10** =34 ,  
**COMP\_ID\_USER11** =35 , **COMP\_ID\_USER12** =36 , **COMP\_ID\_USER13** =37 , **COMP\_ID\_USER14** =38 ,  
**COMP\_ID\_USER15** =39 , **COMP\_ID\_USER16** =40 , **COMP\_ID\_USER17** =41 , **COMP\_ID\_USER18** =42 ,  
**COMP\_ID\_USER19** =43 , **COMP\_ID\_USER20** =44 , **COMP\_ID\_USER21** =45 , **COMP\_ID\_USER22** =46 ,  
**COMP\_ID\_USER23** =47 , **COMP\_ID\_USER24** =48 , **COMP\_ID\_USER25** =49 , **COMP\_ID\_USER26** =50 ,  
**COMP\_ID\_USER27** =51 , **COMP\_ID\_USER28** =52 , **COMP\_ID\_USER29** =53 , **COMP\_ID\_USER30** =54 ,  
**COMP\_ID\_USER31** =55 , **COMP\_ID\_USER32** =56 , **COMP\_ID\_USER33** =57 , **COMP\_ID\_USER34** =58 ,  
**COMP\_ID\_USER35** =59 , **COMP\_ID\_USER36** =60 , **COMP\_ID\_USER37** =61 , **COMP\_ID\_USER38** =62 ,  
**COMP\_ID\_USER39** =63 , **COMP\_ID\_USER40** =64 , **COMP\_ID\_USER41** =65 , **COMP\_ID\_USER42** =66 ,

```

COMP_ID_USER43 =67 , COMP_ID_TELEMETRY_RADIO =68 , COMP_ID_USER45 =69 , COMP_ID_↵
USER46 =70 ,
COMP_ID_USER47 =71 , COMP_ID_USER48 =72 , COMP_ID_USER49 =73 , COMP_ID_USER50 =74 ,
COMP_ID_USER51 =75 , COMP_ID_USER52 =76 , COMP_ID_USER53 =77 , COMP_ID_USER54 =78 ,
COMP_ID_USER55 =79 , COMP_ID_USER56 =80 , COMP_ID_USER57 =81 , COMP_ID_USER58 =82 ,
COMP_ID_USER59 =83 , COMP_ID_USER60 =84 , COMP_ID_USER61 =85 , COMP_ID_USER62 =86 ,
COMP_ID_USER63 =87 , COMP_ID_USER64 =88 , COMP_ID_USER65 =89 , COMP_ID_USER66 =90 ,
COMP_ID_USER67 =91 , COMP_ID_USER68 =92 , COMP_ID_USER69 =93 , COMP_ID_USER70 =94 ,
COMP_ID_USER71 =95 , COMP_ID_USER72 =96 , COMP_ID_USER73 =97 , COMP_ID_USER74 =98 ,
COMP_ID_USER75 =99 , COMP_ID_CAMERA =100 , COMP_ID_CAMERA2 =101 , COMP_ID_CAMERA3
=102 ,
COMP_ID_CAMERA4 =103 , COMP_ID_CAMERA5 =104 , COMP_ID_CAMERA6 =105 , COMP_ID_↵
RADIO =110 ,
COMP_ID_RADIO2 =111 , COMP_ID_RADIO3 =112 , COMP_ID_SERVO1 =140 , COMP_ID_SERVO2
=141 ,
COMP_ID_SERVO3 =142 , COMP_ID_SERVO4 =143 , COMP_ID_SERVO5 =144 , COMP_ID_SERVO6
=145 ,
COMP_ID_SERVO7 =146 , COMP_ID_SERVO8 =147 , COMP_ID_SERVO9 =148 , COMP_ID_SERVO10
=149 ,
COMP_ID_SERVO11 =150 , COMP_ID_SERVO12 =151 , COMP_ID_SERVO13 =152 , COMP_ID_↵
SERVO14 =153 ,
COMP_ID_GIMBAL =154 , COMP_ID_LOG =155 , COMP_ID_ADSB =156 , COMP_ID_OSD =157 ,
COMP_ID_PERIPHERAL =158 , COMP_ID_QX1_GIMBAL =159 , COMP_ID_FLARM =160 , COMP_ID_↵
PARACHUTE =161 ,
COMP_ID_WINCH =169 , COMP_ID_GIMBAL2 =171 , COMP_ID_GIMBAL3 =172 , COMP_ID_GIMBAL4
=173 ,
COMP_ID_GIMBAL5 =174 , COMP_ID_GIMBAL6 =175 , COMP_ID_BATTERY =180 , COMP_ID_↵
BATTERY2 =181 ,
COMP_ID_MAVCAN =189 , COMP_ID_MISSIONPLANNER =190 , COMP_ID_ONBOARD_COMPUTER
=191 , COMP_ID_ONBOARD_COMPUTER2 =192 ,
COMP_ID_ONBOARD_COMPUTER3 =193 , COMP_ID_ONBOARD_COMPUTER4 =194 , COMP_ID_↵
PATHPLANNER =195 , COMP_ID_OBSTACLE_AVOIDANCE =196 ,
COMP_ID_VISUAL_INERTIAL_ODOMETRY =197 , COMP_ID_PAIRING_MANAGER =198 , COMP_ID_↵
IMU =200 , COMP_ID_IMU_2 =201 ,
COMP_ID_IMU_3 =202 , COMP_ID_GPS =220 , COMP_ID_GPS2 =221 , COMP_ID_ODID_TXRX_1 =236
,
COMP_ID_ODID_TXRX_2 =237 , COMP_ID_ODID_TXRX_3 =238 , COMP_ID_UDP_BRIDGE =240 ,
COMP_ID_UART_BRIDGE =241 ,
COMP_ID_TUNNEL_NODE =242 , COMP_ID_ILLUMINATOR =243 , COMP_ID_SYSTEM_CONTROL
=250 }

```

*Legacy component ID values for particular types of hardware/software that might make up a MAVLink system (autopilot, cameras, servos, avoidance systems etc.).*

## Variables

- constexpr std::array< mavlink\_msg\_entry\_t, 1 > mavlink::minimal::MESSAGE\_ENTRIES {{ {0, 50, 9, 9, 0, 0, 0} }}
- Array of msg\_entry needed for mavlink\_parse\_char() (through mavlink\_get\_msg\_entry())
- constexpr auto mavlink::minimal::MAVLINK\_VERSION = 3
- MAVLINK VERSION.
- constexpr auto mavlink::minimal::MAV\_AUTOPILOT\_ENUM\_END = 21
- MAV\_AUTOPILOT ENUM\_END.
- constexpr auto mavlink::minimal::MAV\_TYPE\_ENUM\_END = 50
- MAV\_TYPE ENUM\_END.
- constexpr auto mavlink::minimal::MAV\_MODE\_FLAG\_ENUM\_END = 129

- `MAV_MODE_FLAG ENUM_END.`  
 • constexpr auto `mavlink::minimal::MAV_MODE_FLAG_DECODE_POSITION_ENUM_END` = 129  
`MAV_MODE_FLAG_DECODE_POSITION ENUM_END.`
- constexpr auto `mavlink::minimal::MAV_STATE_ENUM_END` = 9  
`MAV_STATE ENUM_END.`
- constexpr auto `mavlink::minimal::MAV_COMPONENT_ENUM_END` = 251  
`MAV_COMPONENT ENUM_END.`

### 6.22.1 Detailed Description

MAVLink comm protocol generated from minimal.xml.

See also

<http://mavlink.org>

### 6.22.2 Enumeration Type Documentation

#### 6.22.2.1 MAV\_AUTOPILOT

```
enum mavlink::minimal::MAV_AUTOPILOT : uint8_t [strong]
```

Micro air vehicle / autopilot classes.

This identifies the individual model.

#### 6.22.2.2 MAV\_COMPONENT

```
enum mavlink::minimal::MAV_COMPONENT [strong]
```

Legacy component ID values for particular types of hardware/software that might make up a MAVLink system (autopilot, cameras, servos, avoidance systems etc.).

Components are not required or expected to use IDs with names that correspond to their type or function, but m Using an ID that matches the type may slightly reduce the chances of component id clashes, as, for historical System integration will still need to ensure that all components have unique IDs.

Component IDs are used for addressing messages to a particular component within a system.  
A component can use any unique ID between 1 and 255 (MAV\_COMP\_ID\_ALL value is the broadcast address, used to s

Historically component ID were also used for identifying the type of component.  
New code must not use component IDs to infer the component type, but instead check the MAV\_TYPE in the HEARTBE

### 6.22.2.3 MAV\_MODE\_FLAG\_DECODE\_POSITION

```
enum mavlink::minimal::MAV_MODE_FLAG_DECODE_POSITION [strong]
```

These values encode the bit positions of the decode position.

These values can be used to read the value of a flag bit by combining the base\_mode variable with AND with the flag position value. The result will be either 0 or 1, depending on if the flag is set or not.

### 6.22.2.4 MAV\_TYPE

```
enum mavlink::minimal::MAV_TYPE : uint8_t [strong]
```

MAVLINK component type reported in HEARTBEAT message.

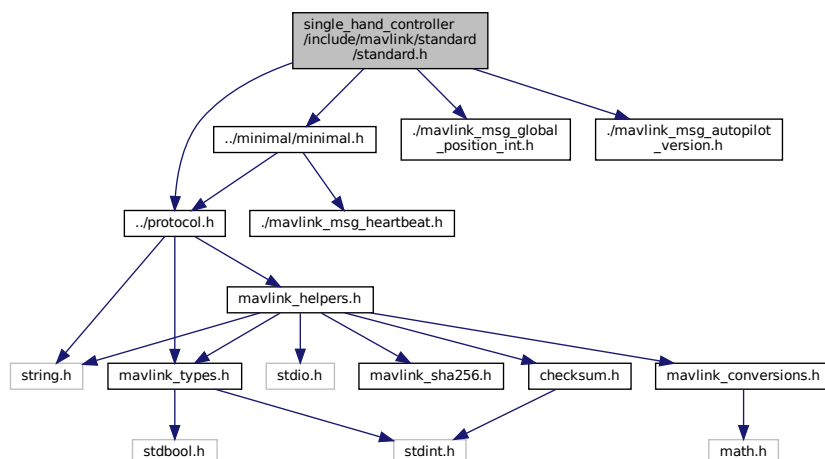
Flight controllers must report the type of the vehicle on which they are mounted (e.g. MAV\_TYPE\_OCTOROTOR). All other components must report a value appropriate for their type (e.g. a camera must use MAV\_TYPE\_CAMERA).

## 6.23 single\_hand\_controller/include/mavlink/standard/standard.h File Reference

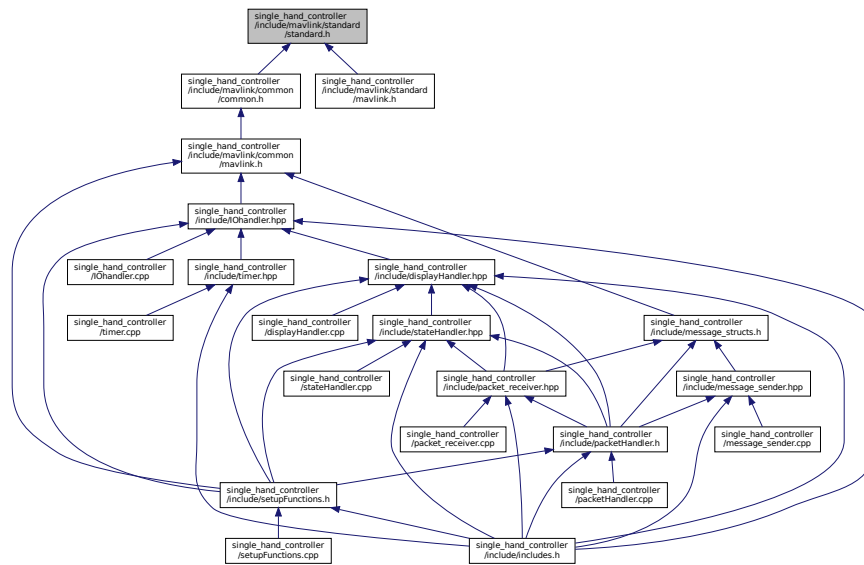
MAVLINK comm protocol generated from standard.xml.

```
#include "../protocol.h"
#include "../mavlink_msg_global_position_int.h"
#include "../mavlink_msg_autopilot_version.h"
#include "../minimal/minimal.h"
```

Include dependency graph for standard.h:



This graph shows which files directly or indirectly include this file:



## Macros

- `#define MAVLINK_STANDARD_H`
- `#define MAVLINK_STANDARD_XML_HASH -2768953998991881157`
- `#define MAVLINK_MESSAGE_LENGTHS {}`
- `#define MAVLINK_MESSAGE_CRCS {{0, 50, 9, 9, 0, 0, 0}, {33, 104, 28, 28, 0, 0, 0}, {148, 178, 60, 78, 0, 0, 0}}`
- `#define MAVLINK_ENABLED_STANDARD`
- `#define HAVE_ENUM_MAV_BOOL`  
*Enum used to indicate true or false (also: success or failure, enabled or disabled, active or inactive).*
- `#define HAVE_ENUM_MAV_PROTOCOL_CAPABILITY`  
*Bitmask of (optional) autopilot capabilities (64 bit).*
- `#define HAVE_ENUM_FIRMWARE_VERSION_TYPE`  
*These values define the type of firmware release.*
- `#define MAVLINK_VERSION 2`

## Typedefs

- `typedef enum MAV_BOOL MAV_BOOL`
- `typedef enum MAV_PROTOCOL_CAPABILITY MAV_PROTOCOL_CAPABILITY`
- `typedef enum FIRMWARE_VERSION_TYPE FIRMWARE_VERSION_TYPE`

## Enumerations

- `enum MAV_BOOL { MAV_BOOL_FALSE =0 , MAV_BOOL_TRUE =1 , MAV_BOOL_ENUM_END =2 }`



- enum **MAV\_PROTOCOL\_CAPABILITY** {  
**MAV\_PROTOCOL\_CAPABILITY\_MISSION\_FLOAT** =1 , **MAV\_PROTOCOL\_CAPABILITY\_PARAM\_**↵  
**FLOAT** =2 , **MAV\_PROTOCOL\_CAPABILITY\_MISSION\_INT** =4 , **MAV\_PROTOCOL\_CAPABILITY\_**↵  
**COMMAND\_INT** =8 ,  
**MAV\_PROTOCOL\_CAPABILITY\_PARAM\_ENCODE\_BYTEWISE** =16 , **MAV\_PROTOCOL\_CAPABILITY\_**↵  
**\_FTP** =32 , **MAV\_PROTOCOL\_CAPABILITY\_SET\_ATTITUDE\_TARGET** =64 , **MAV\_PROTOCOL\_**↵  
**CAPABILITY\_SET\_POSITION\_TARGET\_LOCAL\_NED** =128 ,  
**MAV\_PROTOCOL\_CAPABILITY\_SET\_POSITION\_TARGET\_GLOBAL\_INT** =256 , **MAV\_PROTOCOL\_**↵  
**\_CAPABILITY\_TERRAIN** =512 , **MAV\_PROTOCOL\_CAPABILITY\_RESERVED3** =1024 , **MAV\_**↵  
**PROTOCOL\_CAPABILITY\_FLIGHT\_TERMINATION** =2048 ,  
**MAV\_PROTOCOL\_CAPABILITY\_COMPASS\_CALIBRATION** =4096 , **MAV\_PROTOCOL\_CAPABILITY\_**↵  
**\_MAVLINK2** =8192 , **MAV\_PROTOCOL\_CAPABILITY\_MISSION\_FENCE** =16384 , **MAV\_PROTOCOL\_**↵  
**CAPABILITY\_MISSION\_RALLY** =32768 ,  
**MAV\_PROTOCOL\_CAPABILITY\_RESERVED2** =65536 , **MAV\_PROTOCOL\_CAPABILITY\_PARAM\_**↵  
**\_ENCODE\_C\_CAST** =131072 , **MAV\_PROTOCOL\_CAPABILITY\_COMPONENT\_IMPLEMENTES\_**↵  
**GIMBAL\_MANAGER** =262144 , **MAV\_PROTOCOL\_CAPABILITY\_COMPONENT\_ACCEPTS\_GCS\_**↵  
**CONTROL** =524288 ,  
**MAV\_PROTOCOL\_CAPABILITY\_GRIPPER** =1048576 , **MAV\_PROTOCOL\_CAPABILITY\_ENUM\_END**  
=1048577 }
- enum **FIRMWARE\_VERSION\_TYPE** {  
**FIRMWARE\_VERSION\_TYPE\_DEV** =0 , **FIRMWARE\_VERSION\_TYPE\_ALPHA** =64 , **FIRMWARE\_**↵  
**VERSION\_TYPE\_BETA** =128 , **FIRMWARE\_VERSION\_TYPE\_RC** =192 ,  
**FIRMWARE\_VERSION\_TYPE\_OFFICIAL** =255 , **FIRMWARE\_VERSION\_TYPE\_ENUM\_END** =256 }

## 6.23.1 Detailed Description

MAVLink comm protocol generated from standard.xml.

See also

<http://mavlink.org>

## 6.23.2 Macro Definition Documentation

### 6.23.2.1 HAVE\_ENUM\_FIRMWARE\_VERSION\_TYPE

```
#define HAVE_ENUM_FIRMWARE_VERSION_TYPE
```

These values define the type of firmware release.

These values indicate the first version or release of this type. For example the first alpha release would be 64, the second would be 65.

### 6.23.2.2 HAVE\_ENUM\_MAV\_PROTOCOL\_CAPABILITY

```
#define HAVE_ENUM_MAV_PROTOCOL_CAPABILITY
```

Bitmask of (optional) autopilot capabilities (64 bit).

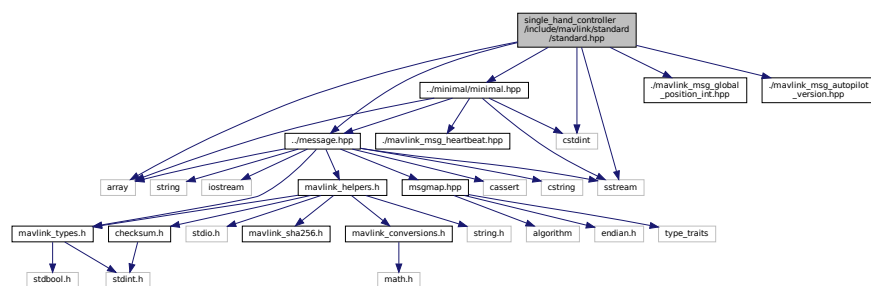
If a bit is set, the autopilot supports this capability.

## 6.24 single\_hand\_controller/include/mavlink/standard/standard.hpp File Reference

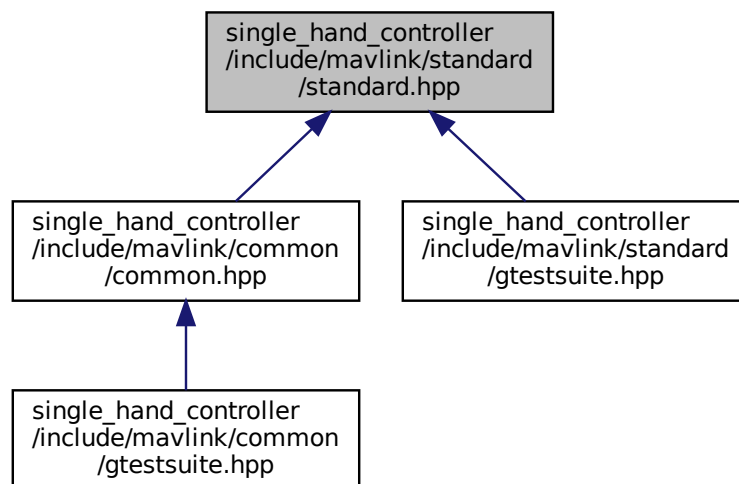
MAVLink comm protocol generated from standard.xml.

```
#include <array>
#include <cstdint>
#include <sstream>
#include "../message.hpp"
#include "../mavlink_msg_global_position_int.hpp"
#include "../mavlink_msg_autopilot_version.hpp"
#include "../minimal/minimal.hpp"
```

Include dependency graph for standard.hpp:



This graph shows which files directly or indirectly include this file:



### Macros

- `#define MAVLINK_STX 253`

## Enumerations

- enum class `mavlink::standard::MAV_BOOL` { **FALSE\_** =0 , **TRUE\_** =1 }  
*Enum used to indicate true or false (also: success or failure, enabled or disabled, active or inactive).*
- enum class `mavlink::standard::MAV_PROTOCOL_CAPABILITY` : uint64\_t {  
**MISSION\_FLOAT** =1 , **PARAM\_FLOAT** =2 , **MISSION\_INT** =4 , **COMMAND\_INT** =8 ,  
**PARAM\_ENCODE\_BYTEWISE** =16 , **FTP** =32 , **SET\_ATTITUDE\_TARGET** =64 , **SET\_POSITION\_**↵  
**TARGET\_LOCAL\_NED** =128 ,  
**SET\_POSITION\_TARGET\_GLOBAL\_INT** =256 , **TERRAIN** =512 , **RESERVED3** =1024 , **FLIGHT\_**↵  
**TERMINATION** =2048 ,  
**COMPASS\_CALIBRATION** =4096 , **MAVLINK2** =8192 , **MISSION\_FENCE** =16384 , **MISSION\_RALLY**  
=32768 ,  
**RESERVED2** =65536 , **PARAM\_ENCODE\_C\_CAST** =131072 , **COMPONENT\_IMPLEMENTS\_GIMBAL\_**↵  
**\_MANAGER** =262144 , **COMPONENT\_ACCEPTS\_GCS\_CONTROL** =524288 ,  
**GRIPPER** =1048576 }  
*Bitmask of (optional) autopilot capabilities (64 bit).*
- enum class `mavlink::standard::FIRMWARE_VERSION_TYPE` {  
**DEV** =0 , **ALPHA** =64 , **BETA** =128 , **RC** =192 ,  
**OFFICIAL** =255 }  
*These values define the type of firmware release.*

## Variables

- constexpr std::array< `mavlink_msg_entry_t`, 3 > `mavlink::standard::MESSAGE_ENTRIES` {{ {0, 50, 9, 9, 0, 0, 0}, {33, 104, 28, 28, 0, 0, 0}, {148, 178, 60, 78, 0, 0, 0} }}
- Array of msg\_entry needed for mavlink\_parse\_char() (through mavlink\_get\_msg\_entry())*
- constexpr auto `mavlink::standard::MAVLINK_VERSION` = 2  
*MAVLINK VERSION.*
- constexpr auto `mavlink::standard::MAV_BOOL_ENUM_END` = 2  
*MAV\_BOOL ENUM\_END.*
- constexpr auto `mavlink::standard::MAV_PROTOCOL_CAPABILITY_ENUM_END` = 1048577  
*MAV\_PROTOCOL\_CAPABILITY ENUM\_END.*
- constexpr auto `mavlink::standard::FIRMWARE_VERSION_TYPE_ENUM_END` = 256  
*FIRMWARE\_VERSION\_TYPE ENUM\_END.*

### 6.24.1 Detailed Description

MAVLink comm protocol generated from standard.xml.

See also

<http://mavlink.org>

### 6.24.2 Enumeration Type Documentation

### 6.24.2.1 FIRMWARE\_VERSION\_TYPE

```
enum mavlink::standard::FIRMWARE_VERSION_TYPE [strong]
```

These values define the type of firmware release.

These values indicate the first version or release of this type. For example the first alpha release would be 64, the second would be 65.

### 6.24.2.2 MAV\_PROTOCOL\_CAPABILITY

```
enum mavlink::standard::MAV_PROTOCOL_CAPABILITY : uint64_t [strong]
```

Bitmask of (optional) autopilot capabilities (64 bit).

If a bit is set, the autopilot supports this capability.

## 6.25 single\_hand\_controller/include/message\_sender.hpp File Reference

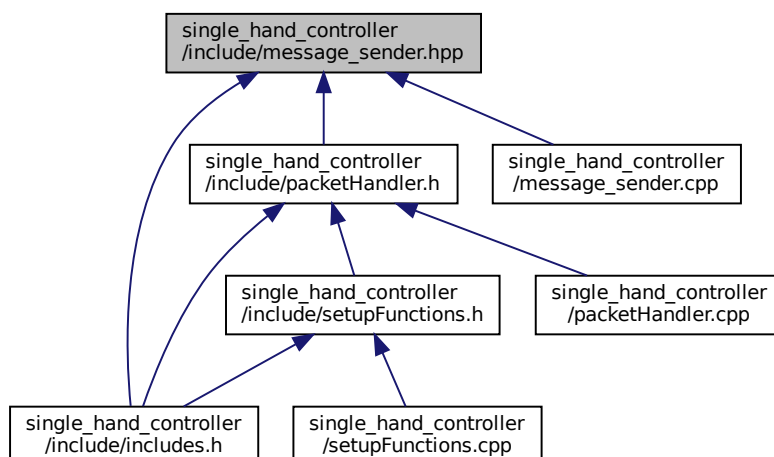
```
#include "message_structs.h"
```

```
#include "definitions.h"
```

Include dependency graph for message\_sender.hpp:



This graph shows which files directly or indirectly include this file:



## Classes

- class [message\\_sender](#)  
*pack data and write MAVLink packets to buffer.*

## Enumerations

- enum [directionToggle](#) { **neutral** , **forward** , **reverse** }  
*directions within hand controller*
- enum [speedToggle](#) { **low** , **mid** , **high** }  
*speeds within hand controller.*

### 6.25.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026 MAVLink messages are packed and placed into the char buffer based on input parameters required for the given message.

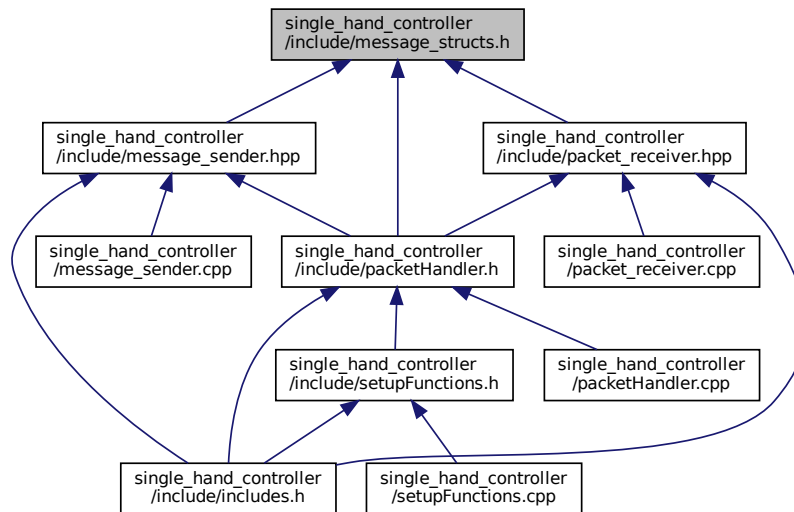
## 6.26 single\_hand\_controller/include/message\_structs.h File Reference

This header file is part of the Single Hand Controller for Scout project; Defines all message structs being sent and received by the Single Hand Controller.

```
#include "definitions.h"
#include "mavlink/common/mavlink.h"
Include dependency graph for message_structs.h:
```



This graph shows which files directly or indirectly include this file:



## Classes

- struct [HC\\_ATLAS\\_HEARTBEAT\\_BC](#)
- struct [HC\\_ATLAS\\_TIMESYNC\\_REQ](#)
- struct [HC\\_ATLAS\\_MANUAL\\_CONTROL\\_BC](#)
- struct [HC\\_ATLAS\\_ARM\\_DISARM\\_CMD](#)
- struct [ATLAS\\_HC\\_HEARTBEAT\\_BC](#)
- struct [ATLAS\\_HC\\_TIMESYNC\\_RESP](#)
- struct [HC\\_RADIO\\_STATUS](#)
- struct [ATLAS\\_HC\\_SYS\\_STAT](#)
- struct [ATLAS\\_HC\\_ARM\\_DISARM\\_ACK](#)

### 6.26.1 Detailed Description

This header file is part of the Single Hand Controller for Scout project; Defines all message structs being sent and received by the Single Hand Controller.

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

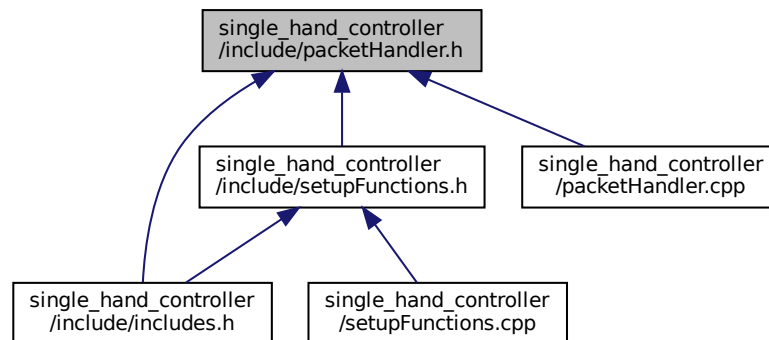
22/04/2026

## 6.27 single\_hand\_controller/include/packetHandler.h File Reference

```
#include "definitions.h"
#include "message_structs.h"
#include "message_sender.hpp"
#include "packet_receiver.hpp"
#include "stateHandler.hpp"
#include "displayHandler.hpp"
Include dependency graph for packetHandler.h:
```



This graph shows which files directly or indirectly include this file:



### Functions

- bool [setupSigning](#) ()  
*sets up signing for all MAVLink packets*
- bool [heartbeat\\_timed\\_out](#) ()  
*check whether heartbeat has not been received within HEARTBEAT\_TIMEOUT microseconds (3 seconds)*
- void [handlePacketReceived](#) ()  
*handle any packet that is received and send to [packet\\_receiver](#)*
- void [sendTimesyncRequest](#) ()  
*send a timesync request*
- void [sendManualControl](#) ()  
*send manual control packet*
- void [initiateSerialComm](#) ()  
*initiates serial communication with radio to send MAVLink packets; baud rate for serial communication is set in [definitions.h](#)*

### 6.27.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026 Handles all incoming communication over UART, from the hand controller radio

### 6.27.2 Function Documentation

#### 6.27.2.1 heartbeat\_timed\_out()

```
bool heartbeat_timed_out ()
```

check whether heartbeat has not been received within HEARTBEAT\_TIMEOUT microseconds (3 seconds)

#### Returns

true when timeout has occurred

#### 6.27.2.2 setupSigning()

```
bool setupSigning ()
```

sets up signing for all MAVLink packets

#### Returns

true when signing is successfully setup

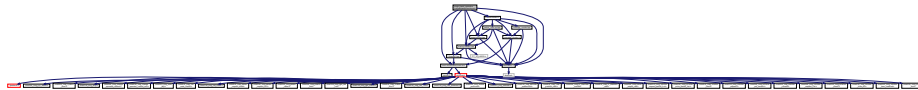


## 6.28 single\_hand\_controller/include/setupFunctions.h File Reference

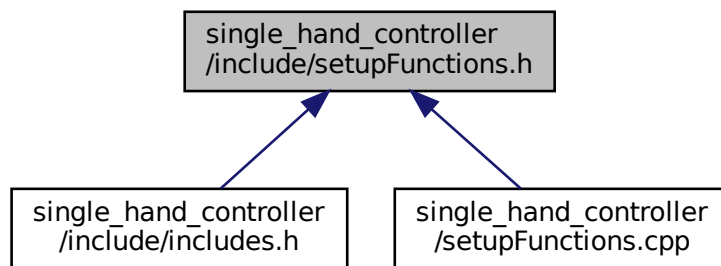
Defines all functions required during setup phase of hand controller.

```
#include "mavlink/common/mavlink.h"
#include "definitions.h"
#include "displayHandler.hpp"
#include "IOhandler.hpp"
#include "stateHandler.hpp"
#include "packetHandler.h"
```

Include dependency graph for setupFunctions.h:



This graph shows which files directly or indirectly include this file:



### Functions

- void [setupIO](#) ()  
*set up microcontroller for all input and output*
- void [initState](#) ()  
*set initial values for states*
- void [setupDisplay](#) ()  
*set up the LCD display (clearing the screen, and displaying necessary startup items)*
- bool [identifyControllerDrift](#) ()  
*obtain the deviation of the thumbstick's neutral position from its expected central position*
- void [initUGV\\_state](#) (ugv\_status s1)  
*set a default UGV state*
- bool [initMAVLink](#) ()  
*initialize variables required for signing and sending MAVLink packets*

### 6.28.1 Detailed Description

Defines all functions required during setup phase of hand controller.

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026

### 6.28.2 Function Documentation

#### 6.28.2.1 identifyControllerDrift()

```
bool identifyControllerDrift ()
```

obtain the deviation of the thumbstick's neutral position from its expected central position

#### Returns

true when the deviation is within permissible limits.

#### 6.28.2.2 initMAVLink()

```
bool initMAVLink ()
```

initialize variables required for signing and sending MAVLink packets

#### Returns

true when initialization is done correctly, false in case of a nullpointer error.

#### 6.28.2.3 initUGV\_state()

```
void initUGV_state (
 ugv_status s1)
```

set a default UGV state

## Parameters

|                 |                                              |
|-----------------|----------------------------------------------|
| <code>s1</code> | enum value indicating the current UGV state. |
|-----------------|----------------------------------------------|

## 6.28.2.4 setupDisplay()

```
void setupDisplay ()
```

set up the LCD display (clearing the screen, and displaying necessary startup items)

1. setup display baud rate
2. clear display
3. display logo/initial messages
4. display text/symbols for status indication

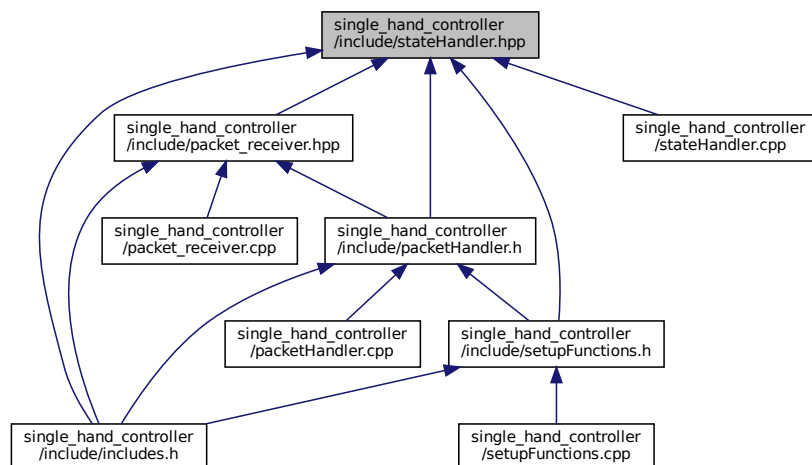
## 6.29 single\_hand\_controller/include/stateHandler.hpp File Reference

```
#include "displayHandler.hpp"
#include "definitions.h"
```

Include dependency graph for stateHandler.hpp:



This graph shows which files directly or indirectly include this file:



## Functions

- bool `setUGV_state` (ugv\_status s1)  
*set the current UGV state.*
- void `increaseSpeed` ()  
*used only within prototype version, to increase speed*
- void `decreaseSpeed` ()  
*used only within prototype version, to decrease speed*

### 6.29.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026

Handling of states within the hand controller is performed within the stateHandler.

currently handles states for ugv state, and speed

### 6.29.2 Function Documentation

#### 6.29.2.1 `setUGV_state()`

```
bool setUGV_state (
 ugv_status s1)
```

set the current UGV state.

#### Returns

true when current UGV state is updated to a new value.

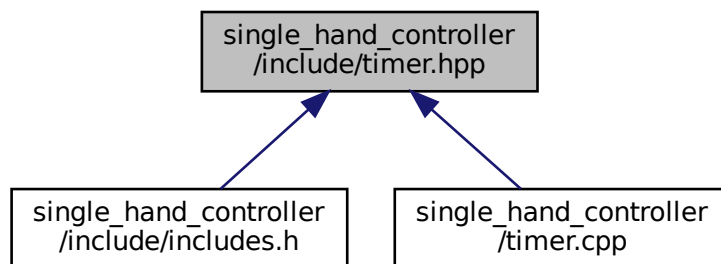
false when current UGV state is same as previous

## 6.30 single\_hand\_controller/include/timer.hpp File Reference

```
#include "definitions.h"
#include "IOhandler.hpp"
Include dependency graph for timer.hpp:
```



This graph shows which files directly or indirectly include this file:



### Functions

- void `startTimer()`  
*start a timer, that keeps count of number of microseconds elapsed*
- bool `timeup` (unsigned long int time\_ms)  
*check whether time\_ms microseconds are completed since start of timer*
- void `resetTimer()`  
*essentially performs same task as `startTimer()`; used to reset the timer to 0 and start again.*
- void `startOFPTimer()`  
*second timer meant to be used only within the OFP loop*
- void `end_OFP_timer` (unsigned long int time\_limit)  
*waits until time\_limit microseconds are completed since beginning of OFP timer*

#### 6.30.1 Detailed Description

Version

0.1

Author

Nikhil Tom Jose

Date

22/04/2026

All timing functions are contained within this header file

## 6.30.2 Function Documentation

### 6.30.2.1 end\_OFP\_timer()

```
void end_OFP_timer (
 unsigned long int time_limit)
```

waits until *time\_limit* microseconds are completed since beginning of OFP timer

#### Parameters

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| <i>time_limit</i> | number of microseconds the OFP loop is meant to last for |
|-------------------|----------------------------------------------------------|

### 6.30.2.2 timeup()

```
bool timeup (
 unsigned long int time_ms)
```

check whether *time\_ms* microseconds are completed since start of timer

#### Parameters

|                |                                   |
|----------------|-----------------------------------|
| <i>time_ms</i> | microseconds since start of timer |
|----------------|-----------------------------------|

#### Returns

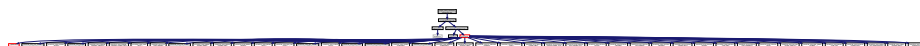
true when *time\_ms* microseconds are completed since start of timer.

check whether *time\_ms* microseconds are completed since start of timer

## 6.31 single\_hand\_controller/IOhandler.cpp File Reference

```
#include "include/IOhandler.hpp"
```

Include dependency graph for IOhandler.cpp:



## Functions

- unsigned long int **micros** ()
- bool **digitalRead** (int k)
- int **analogRead** (int pin)
- void **getXY\_raw** (struct **thumbstickControl** \*control)  
*get thumbstick's raw position (0 to 1023)*
- void **getXY** (struct **thumbstickControl** \*control)  
*get thumbstick's normalized position (-100 to 100)*
- void **checkUserInput** ()  
*update user input*

## Variables

- struct buttons **user\_input**
- int **drift\_x**
- int **drift\_y**

### 6.31.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026

Part of IOhandler library Defines functions and variables used in IOhandler.h

### 6.31.2 Function Documentation

#### 6.31.2.1 getXY()

```
void getXY (
 struct thumbstickControl * control)
```

get thumbstick's normalized position (-100 to 100)

#### Parameters

|                |                                                                       |
|----------------|-----------------------------------------------------------------------|
| <i>control</i> | -> the current position of the thumbstick is written into this struct |
|----------------|-----------------------------------------------------------------------|

### 6.31.2.2 getXRaw()

```
void getXRaw (
 struct thumbstickControl * control)
```

get thumbstick's raw position (0 to 1023)

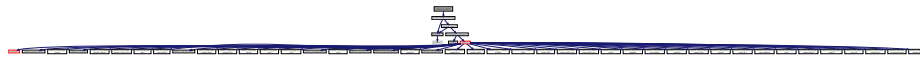
#### Parameters

|                |                                                                       |
|----------------|-----------------------------------------------------------------------|
| <i>control</i> | -> the current position of the thumbstick is written into this struct |
|----------------|-----------------------------------------------------------------------|

## 6.32 single\_hand\_controller/message\_sender.cpp File Reference

```
#include "include/message_sender.hpp"
```

Include dependency graph for message\_sender.cpp:



### Variables

- uint64\_t [UGVTime](#)  
*stores time of the drone at which timesync was received*
- uint64\_t [RecvTime](#)

### 6.32.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026

Part of message sender library. Defines member functions of [message\\_sender](#) class



## 6.33 single\_hand\_controller/packet\_receiver.cpp File Reference

```
#include "include/packet_receiver.hpp"
```

Include dependency graph for packet\_receiver.cpp:



### Variables

- [mavlink\\_signing\\_t](#) **signing**
- [uint64\\_t](#) **UGVTime**  
*stores time of the drone at which timesync was received*
- [uint64\\_t](#) **RecvTime**  
*stores time at which timesync was received.*
- [uint64\\_t](#) **latency**
- [uint64\\_t](#) **last\_heartbeat\_received\_at**

### 6.33.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026

Part of packet receiver library; used in Single Hand Controller for Scout Defines member functions of [message\\_sender](#) class

## 6.34 single\_hand\_controller/packetHandler.cpp File Reference

```
#include "include/packetHandler.h"
```

Include dependency graph for packetHandler.cpp:



## Functions

- [message\\_sender](#) **msgsndr** ([buf](#), &[msg](#))
- bool **is\_unsigned\_message** (const [mavlink\\_status\\_t](#) \*[status](#), uint32\_t [msgId](#))
- bool [setupSigning](#) ()
  - sets up signing for all MAVLink packets*
- void [initiateSerialComm](#) ()
  - initiates serial communication with radio to send MAVLink packets; baud rate for serial communication is set in [definitions.h](#)*
- bool [heartbeat\\_timed\\_out](#) ()
  - check whether heartbeat has not been received within HEARTBEAT\_TIMEOUT microseconds (3 seconds)*
- void **sendBuffer** (int [len](#))
- void [sendTimesyncRequest](#) ()
  - send a timesync request*
- void [sendManualControl](#) ()
  - send manual control packet*
- void [handlePacketReceived](#) ()
  - handle any packet that is received and send to [packet\\_receiver](#)*

## Variables

- [mavlink\\_status\\_t](#) \* **status\_chan**
- [mavlink\\_signing\\_t](#) **signing**
- [mavlink\\_signing\\_streams\\_t](#) **signing\_streams**
- byte [buf](#) [300]
  - buffer to store serialized mavlink data*
- [mavlink\\_status\\_t](#) **status**
  - used to check parsing status*
- [mavlink\\_message\\_t](#) **msg**
- bool [arm\\_request\\_sent](#) = false
  - whether an arm request was sent (COMMAND\_LONG)*
- bool [disarm\\_request\\_sent](#) = false
  - whether a disarm request was sent (COMMAND\_LONG)*
- int **rsi**
- uint64\_t [UGVTime](#) = 0
  - stores time of the drone at which timesync was received*
- uint64\_t [RecvTime](#)
  - stores time at which timesync was received.*
- uint64\_t [latency](#)
  - latency of timesync packet being sent*
- uint64\_t [last\\_heartbeat\\_received\\_at](#) = 0
  - local timestamp of when the last heartbeat was received.*

### 6.34.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026

Part of packet handler library. Defines variables and functions used in [packetHandler.h](#)

## 6.34.2 Function Documentation

### 6.34.2.1 heartbeat\_timed\_out()

```
bool heartbeat_timed_out ()
```

check whether heartbeat has not been received within HEARTBEAT\_TIMEOUT microseconds (3 seconds)

#### Returns

true when timeout has occurred

### 6.34.2.2 setupSigning()

```
bool setupSigning ()
```

sets up signing for all MAVLink packets

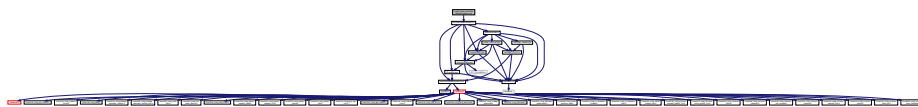
#### Returns

true when signing is successfully setup

## 6.35 single\_hand\_controller/setupFunctions.cpp File Reference

```
#include "include/setupFunctions.h"
```

Include dependency graph for setupFunctions.cpp:



## Functions

- void `setupIO` ()  
*set up microcontroller for all input and output*
- void `initState` ()  
*set initial values for states*
- bool `identifyControllerDrift` ()  
*obtain the deviation of the thumbstick's neutral position from its expected central position*
- bool `initMAVLink` ()  
*initialize variables required for signing and sending MAVLink packets*
- void `setupDisplay` ()  
*set up the LCD display (clearing the screen, and displaying necessary startup items)*

## Variables

- int **drift\_x**
- int **drift\_y**

### 6.35.1 Detailed Description

#### Version

0.1

#### Author

Nikhil Tom Jose

#### Date

22/04/2026

Part of set up functions library; used within the setup() function of single\_hand\_controller.ino. Defines variables and functions used in [setupFunctions.h](#).

### 6.35.2 Function Documentation

#### 6.35.2.1 identifyControllerDrift()

```
bool identifyControllerDrift ()
```

obtain the deviation of the thumbstick's neutral position from its expected central position

#### Returns

true when the deviation is within permissible limits.

#### 6.35.2.2 initMAVLink()

```
bool initMAVLink ()
```

initialize variables required for signing and sending MAVLink packets

#### Returns

true when initialization is done correctly, false in case of a nullpointer error.

### 6.35.2.3 setupDisplay()

```
void setupDisplay ()
```

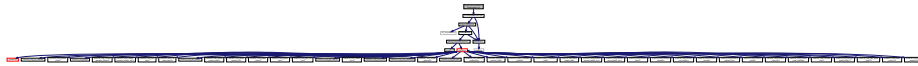
set up the LCD display (clearing the screen, and displaying necessary startup items)

1. setup display baud rate
2. clear display
3. display logo/initial messages
4. display text/symbols for status indication

## 6.36 single\_hand\_controller/stateHandler.cpp File Reference

```
#include "include/stateHandler.hpp"
```

Include dependency graph for stateHandler.cpp:



### Functions

- bool [setUGV\\_state](#) (ugv\_status s1)  
*set the current UGV state.*
- void [increaseSpeed](#) ()  
*used only within prototype version, to increase speed*
- void [decreaseSpeed](#) ()  
*used only within prototype version, to decrease speed*

### Variables

- ugv\_status **current\_state** = unknown
- ugv\_status **prev\_state** = unknown
- int **speed** = 0

### 6.36.1 Detailed Description

Version

0.1

Author

Nikhil Tom Jose

Date

22/04/2026

Part of stateHandler library. Defines variables and functions used in [stateHandler.hpp](#).

currently maintains -

1. state for current\_UGV\_state (active, standby, and disconnected); and the previous UGV state
2. speed, since the single hand controller prototype doesn't have a tristate speed toggle that maintains state by itself

## 6.36.2 Function Documentation

### 6.36.2.1 setUGV\_state()

```
bool setUGV_state (
 ugv_status s1)
```

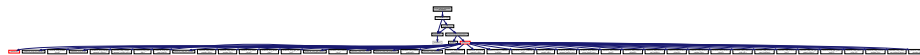
set the current UGV state.

#### Returns

true when current UGV state is updated to a new value.  
false when current UGV state is same as previous

## 6.37 single\_hand\_controller/timer.cpp File Reference

```
#include "include/timer.hpp"
Include dependency graph for timer.cpp:
```



## Functions

- void [startTimer](#) ()  
*start a timer, that keeps count of number of microseconds elapsed*
- bool [timeup](#) (unsigned long int time\_ms)  
*check whether the given amount of time (in ms) is completed or not*
- void [resetTimer](#) ()  
*essentially performs same task as [startTimer\(\)](#); used to reset the timer to 0 and start again.*
- void [startOFPTimer](#) ()  
*second timer meant to be used only within the OFP loop*
- void [end\\_OFP\\_timer](#) (unsigned long int time\_limit)  
*waits until time\_limit microseconds are completed since beginning of OFP timer*

## Variables

- long unsigned int **current\_time** = 0
- long unsigned int **OFP\_timer** = 0

## 6.37.1 Detailed Description

### Version

0.1

### Author

Nikhil Tom Jose

### Date

22/04/2026

Part of timer library; Meant to be used only within the main source file to time intervals between 2 events. Defines variables and functions used in [setupFunctions.h](#).

## 6.37.2 Function Documentation

### 6.37.2.1 end\_OFP\_timer()

```
void end_OFP_timer (
 unsigned long int time_limit)
```

waits until time\_limit microseconds are completed since beginning of OFP timer

#### Parameters

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| <i>time_limit</i> | number of microseconds the OFP loop is meant to last for |
|-------------------|----------------------------------------------------------|

### 6.37.2.2 timeup()

```
bool timeup (
 unsigned long int time_ms)
```

check whether the given amount of time (in ms) is completed or not

check whether time\_ms microseconds are completed since start of timer





# Index

[\\_\\_mavlink\\_actuator\\_control\\_target\\_t](#), [37](#)  
[\\_\\_mavlink\\_actuator\\_output\\_status\\_t](#), [37](#)  
[\\_\\_mavlink\\_adsb\\_vehicle\\_t](#), [38](#)  
[\\_\\_mavlink\\_airspeed\\_t](#), [38](#)  
[\\_\\_mavlink\\_ais\\_vessel\\_t](#), [38](#)  
[\\_\\_mavlink\\_altitude\\_t](#), [39](#)  
[\\_\\_mavlink\\_att\\_pos\\_mocap\\_t](#), [39](#)  
[\\_\\_mavlink\\_attitude\\_quaternion\\_cov\\_t](#), [40](#)  
[\\_\\_mavlink\\_attitude\\_quaternion\\_t](#), [40](#)  
[\\_\\_mavlink\\_attitude\\_t](#), [40](#)  
[\\_\\_mavlink\\_attitude\\_target\\_t](#), [41](#)  
[\\_\\_mavlink\\_auth\\_key\\_t](#), [41](#)  
[\\_\\_mavlink\\_autopilot\\_version\\_t](#), [41](#)  
[\\_\\_mavlink\\_available\\_modes\\_monitor\\_t](#), [42](#)  
[\\_\\_mavlink\\_available\\_modes\\_t](#), [42](#)  
[\\_\\_mavlink\\_battery\\_info\\_t](#), [42](#)  
[\\_\\_mavlink\\_bitfield](#), [43](#)  
[\\_\\_mavlink\\_button\\_change\\_t](#), [43](#)  
[\\_\\_mavlink\\_camera\\_fov\\_status\\_t](#), [43](#)  
[\\_\\_mavlink\\_camera\\_image\\_captured\\_t](#), [44](#)  
[\\_\\_mavlink\\_camera\\_information\\_t](#), [44](#)  
[\\_\\_mavlink\\_camera\\_thermal\\_range\\_t](#), [45](#)  
[\\_\\_mavlink\\_camera\\_tracking\\_geo\\_status\\_t](#), [45](#)  
[\\_\\_mavlink\\_camera\\_tracking\\_image\\_status\\_t](#), [46](#)  
[\\_\\_mavlink\\_camera\\_trigger\\_t](#), [46](#)  
[\\_\\_mavlink\\_can\\_filter\\_modify\\_t](#), [46](#)  
[\\_\\_mavlink\\_can\\_frame\\_t](#), [47](#)  
[\\_\\_mavlink\\_canfd\\_frame\\_t](#), [47](#)  
[\\_\\_mavlink\\_cellular\\_config\\_t](#), [47](#)  
[\\_\\_mavlink\\_cellular\\_status\\_t](#), [48](#)  
[\\_\\_mavlink\\_change\\_operator\\_control\\_ack\\_t](#), [48](#)  
[\\_\\_mavlink\\_change\\_operator\\_control\\_t](#), [48](#)  
[\\_\\_mavlink\\_collision\\_t](#), [49](#)  
[\\_\\_mavlink\\_command\\_ack\\_t](#), [49](#)  
[\\_\\_mavlink\\_command\\_cancel\\_t](#), [49](#)  
[\\_\\_mavlink\\_command\\_int\\_t](#), [50](#)  
[\\_\\_mavlink\\_command\\_long\\_t](#), [50](#)  
[\\_\\_mavlink\\_component\\_information\\_basic\\_t](#), [51](#)  
[\\_\\_mavlink\\_component\\_information\\_t](#), [51](#)  
[\\_\\_mavlink\\_component\\_metadata\\_t](#), [51](#)  
[\\_\\_mavlink\\_control\\_system\\_state\\_t](#), [52](#)  
[\\_\\_mavlink\\_current\\_event\\_sequence\\_t](#), [52](#)  
[\\_\\_mavlink\\_current\\_mode\\_t](#), [52](#)  
[\\_\\_mavlink\\_data\\_stream\\_t](#), [53](#)  
[\\_\\_mavlink\\_data\\_transmission\\_handshake\\_t](#), [53](#)  
[\\_\\_mavlink\\_debug\\_float\\_array\\_t](#), [53](#)  
[\\_\\_mavlink\\_debug\\_t](#), [54](#)  
[\\_\\_mavlink\\_debug\\_vect\\_t](#), [54](#)  
[\\_\\_mavlink\\_encapsulated\\_data\\_t](#), [54](#)  
[\\_\\_mavlink\\_esc\\_info\\_t](#), [55](#)  
[\\_\\_mavlink\\_esc\\_status\\_t](#), [55](#)  
[\\_\\_mavlink\\_estimator\\_status\\_t](#), [55](#)  
[\\_\\_mavlink\\_event\\_t](#), [56](#)  
[\\_\\_mavlink\\_extended\\_sys\\_state\\_t](#), [56](#)  
[\\_\\_mavlink\\_fence\\_status\\_t](#), [56](#)  
[\\_\\_mavlink\\_field\\_info](#), [57](#)  
[\\_\\_mavlink\\_figure\\_eight\\_execution\\_status\\_t](#), [57](#)  
[\\_\\_mavlink\\_file\\_transfer\\_protocol\\_t](#), [57](#)  
[\\_\\_mavlink\\_flight\\_information\\_t](#), [58](#)  
[\\_\\_mavlink\\_follow\\_target\\_t](#), [58](#)  
[\\_\\_mavlink\\_fuel\\_status\\_t](#), [58](#)  
[\\_\\_mavlink\\_generator\\_status\\_t](#), [59](#)  
[\\_\\_mavlink\\_gimbal\\_device\\_attitude\\_status\\_t](#), [59](#)  
[\\_\\_mavlink\\_gimbal\\_device\\_information\\_t](#), [60](#)  
[\\_\\_mavlink\\_gimbal\\_device\\_set\\_attitude\\_t](#), [60](#)  
[\\_\\_mavlink\\_gimbal\\_manager\\_information\\_t](#), [61](#)  
[\\_\\_mavlink\\_gimbal\\_manager\\_set\\_attitude\\_t](#), [61](#)  
[\\_\\_mavlink\\_gimbal\\_manager\\_set\\_manual\\_control\\_t](#), [61](#)  
[\\_\\_mavlink\\_gimbal\\_manager\\_set\\_pitchyaw\\_t](#), [62](#)  
[\\_\\_mavlink\\_gimbal\\_manager\\_status\\_t](#), [62](#)  
[\\_\\_mavlink\\_global\\_position\\_int\\_cov\\_t](#), [62](#)  
[\\_\\_mavlink\\_global\\_position\\_int\\_t](#), [63](#)  
[\\_\\_mavlink\\_global\\_vision\\_position\\_estimate\\_t](#), [63](#)  
[\\_\\_mavlink\\_gps2\\_rtk\\_t](#), [63](#)  
[\\_\\_mavlink\\_gps\\_inject\\_data\\_t](#), [64](#)  
[\\_\\_mavlink\\_gps\\_rtc\\_data\\_t](#), [64](#)  
[\\_\\_mavlink\\_gps\\_rtk\\_t](#), [64](#)  
[\\_\\_mavlink\\_gps\\_status\\_t](#), [65](#)  
[\\_\\_mavlink\\_heartbeat\\_t](#), [65](#)  
[\\_\\_mavlink\\_high\\_latency2\\_t](#), [65](#)  
[\\_\\_mavlink\\_high\\_latency\\_t](#), [66](#)  
[\\_\\_mavlink\\_highres\\_imu\\_t](#), [67](#)  
[\\_\\_mavlink\\_hil\\_actuator\\_controls\\_t](#), [67](#)  
[\\_\\_mavlink\\_hil\\_controls\\_t](#), [67](#)  
[\\_\\_mavlink\\_hil\\_optical\\_flow\\_t](#), [68](#)  
[\\_\\_mavlink\\_hil\\_rc\\_inputs\\_raw\\_t](#), [68](#)  
[\\_\\_mavlink\\_hil\\_sensor\\_t](#), [69](#)  
[\\_\\_mavlink\\_hil\\_state\\_quaternion\\_t](#), [69](#)  
[\\_\\_mavlink\\_hil\\_state\\_t](#), [70](#)  
[\\_\\_mavlink\\_hygrometer\\_sensor\\_t](#), [70](#)  
[\\_\\_mavlink\\_illuminator\\_status\\_t](#), [70](#)  
[\\_\\_mavlink\\_isbd\\_link\\_status\\_t](#), [71](#)  
[\\_\\_mavlink\\_link\\_node\\_status\\_t](#), [71](#)  
[\\_\\_mavlink\\_local\\_position\\_ned\\_cov\\_t](#), [71](#)  
[\\_\\_mavlink\\_local\\_position\\_ned\\_system\\_global\\_offset\\_t](#), [72](#)  
[\\_\\_mavlink\\_local\\_position\\_ned\\_t](#), [72](#)  
[\\_\\_mavlink\\_log\\_data\\_t](#), [72](#)

- [\\_\\_mavlink\\_log\\_entry\\_t, 73](#)
- [\\_\\_mavlink\\_log\\_erase\\_t, 73](#)
- [\\_\\_mavlink\\_log\\_request\\_data\\_t, 73](#)
- [\\_\\_mavlink\\_log\\_request\\_end\\_t, 73](#)
- [\\_\\_mavlink\\_log\\_request\\_list\\_t, 74](#)
- [\\_\\_mavlink\\_logging\\_ack\\_t, 74](#)
- [\\_\\_mavlink\\_logging\\_data\\_acked\\_t, 74](#)
- [\\_\\_mavlink\\_logging\\_data\\_t, 75](#)
- [\\_\\_mavlink\\_manual\\_setpoint\\_t, 75](#)
- [\\_\\_mavlink\\_memory\\_vect\\_t, 75](#)
- [\\_\\_mavlink\\_message\\_info, 76](#)
- [\\_\\_mavlink\\_message\\_interval\\_t, 76](#)
- [\\_\\_mavlink\\_mission\\_ack\\_t, 76](#)
- [\\_\\_mavlink\\_mission\\_clear\\_all\\_t, 77](#)
- [\\_\\_mavlink\\_mission\\_item\\_int\\_t, 77](#)
- [\\_\\_mavlink\\_mission\\_item\\_reached\\_t, 77](#)
- [\\_\\_mavlink\\_mission\\_item\\_t, 78](#)
- [\\_\\_mavlink\\_mission\\_request\\_int\\_t, 78](#)
- [\\_\\_mavlink\\_mission\\_request\\_list\\_t, 78](#)
- [\\_\\_mavlink\\_mission\\_request\\_partial\\_list\\_t, 79](#)
- [\\_\\_mavlink\\_mission\\_request\\_t, 79](#)
- [\\_\\_mavlink\\_mission\\_set\\_current\\_t, 79](#)
- [\\_\\_mavlink\\_mission\\_write\\_partial\\_list\\_t, 80](#)
- [\\_\\_mavlink\\_mount\\_orientation\\_t, 80](#)
- [\\_\\_mavlink\\_msg\\_entry, 80](#)
- [\\_\\_mavlink\\_named\\_value\\_float\\_t, 81](#)
- [\\_\\_mavlink\\_named\\_value\\_int\\_t, 81](#)
- [\\_\\_mavlink\\_nav\\_controller\\_output\\_t, 81](#)
- [\\_\\_mavlink\\_odometry\\_t, 82](#)
- [\\_\\_mavlink\\_onboard\\_computer\\_status\\_t, 82](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_arm\\_status\\_t, 83](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_authentication\\_t, 83](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_basic\\_id\\_t, 83](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_location\\_t, 84](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_message\\_pack\\_t, 84](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_operator\\_id\\_t, 85](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_self\\_id\\_t, 85](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_system\\_t, 85](#)
- [\\_\\_mavlink\\_open\\_drone\\_id\\_system\\_update\\_t, 86](#)
- [\\_\\_mavlink\\_optical\\_flow\\_rad\\_t, 86](#)
- [\\_\\_mavlink\\_orbit\\_execution\\_status\\_t, 86](#)
- [\\_\\_mavlink\\_param\\_error\\_t, 87](#)
- [\\_\\_mavlink\\_param\\_ext\\_ack\\_t, 87](#)
- [\\_\\_mavlink\\_param\\_ext\\_request\\_list\\_t, 87](#)
- [\\_\\_mavlink\\_param\\_ext\\_request\\_read\\_t, 87](#)
- [\\_\\_mavlink\\_param\\_ext\\_set\\_t, 88](#)
- [\\_\\_mavlink\\_param\\_ext\\_value\\_t, 88](#)
- [\\_\\_mavlink\\_param\\_map\\_rc\\_t, 88](#)
- [\\_\\_mavlink\\_param\\_request\\_list\\_t, 89](#)
- [\\_\\_mavlink\\_param\\_request\\_read\\_t, 89](#)
- [\\_\\_mavlink\\_param\\_set\\_t, 89](#)
- [\\_\\_mavlink\\_param\\_value\\_t, 89](#)
- [\\_\\_mavlink\\_ping\\_t, 90](#)
- [\\_\\_mavlink\\_play\\_tune\\_t, 90](#)
- [\\_\\_mavlink\\_play\\_tune\\_v2\\_t, 90](#)
- [\\_\\_mavlink\\_position\\_target\\_global\\_int\\_t, 91](#)
- [\\_\\_mavlink\\_position\\_target\\_local\\_ned\\_t, 91](#)
- [\\_\\_mavlink\\_power\\_status\\_t, 92](#)
- [\\_\\_mavlink\\_protocol\\_version\\_t, 92](#)
- [\\_\\_mavlink\\_radio\\_status\\_t, 92](#)
- [\\_\\_mavlink\\_raw\\_pressure\\_t, 93](#)
- [\\_\\_mavlink\\_raw\\_rpm\\_t, 93](#)
- [\\_\\_mavlink\\_rc\\_channels\\_override\\_t, 93](#)
- [\\_\\_mavlink\\_rc\\_channels\\_raw\\_t, 94](#)
- [\\_\\_mavlink\\_rc\\_channels\\_scaled\\_t, 94](#)
- [\\_\\_mavlink\\_rc\\_channels\\_t, 95](#)
- [\\_\\_mavlink\\_relay\\_status\\_t, 95](#)
- [\\_\\_mavlink\\_request\\_data\\_stream\\_t, 96](#)
- [\\_\\_mavlink\\_request\\_event\\_t, 96](#)
- [\\_\\_mavlink\\_resource\\_request\\_t, 96](#)
- [\\_\\_mavlink\\_response\\_event\\_error\\_t, 97](#)
- [\\_\\_mavlink\\_safety\\_allowed\\_area\\_t, 97](#)
- [\\_\\_mavlink\\_safety\\_set\\_allowed\\_area\\_t, 97](#)
- [\\_\\_mavlink\\_scaled\\_imu2\\_t, 98](#)
- [\\_\\_mavlink\\_scaled\\_imu3\\_t, 98](#)
- [\\_\\_mavlink\\_scaled\\_imu\\_t, 99](#)
- [\\_\\_mavlink\\_scaled\\_pressure2\\_t, 99](#)
- [\\_\\_mavlink\\_scaled\\_pressure3\\_t, 99](#)
- [\\_\\_mavlink\\_scaled\\_pressure\\_t, 100](#)
- [\\_\\_mavlink\\_serial\\_control\\_t, 100](#)
- [\\_\\_mavlink\\_set\\_actuator\\_control\\_target\\_t, 100](#)
- [\\_\\_mavlink\\_set\\_mode\\_t, 101](#)
- [\\_\\_mavlink\\_set\\_position\\_target\\_global\\_int\\_t, 101](#)
- [\\_\\_mavlink\\_set\\_position\\_target\\_local\\_ned\\_t, 102](#)
- [\\_\\_mavlink\\_setup\\_signing\\_t, 102](#)
- [\\_\\_mavlink\\_signing, 103](#)
- [\\_\\_mavlink\\_signing\\_streams, 104](#)
- [\\_\\_mavlink\\_signing\\_streams::\\_\\_mavlink\\_signing\\_stream, 103](#)
- [\\_\\_mavlink\\_sim\\_state\\_t, 105](#)
- [\\_\\_mavlink\\_status, 106](#)
- [\\_\\_mavlink\\_storage\\_information\\_t, 107](#)
- [\\_\\_mavlink\\_supported\\_tunes\\_t, 107](#)
- [\\_\\_mavlink\\_system\\_time\\_t, 108](#)
- [\\_\\_mavlink\\_terrain\\_check\\_t, 108](#)
- [\\_\\_mavlink\\_terrain\\_data\\_t, 108](#)
- [\\_\\_mavlink\\_terrain\\_report\\_t, 108](#)
- [\\_\\_mavlink\\_terrain\\_request\\_t, 109](#)
- [\\_\\_mavlink\\_time\\_estimate\\_to\\_target\\_t, 109](#)
- [\\_\\_mavlink\\_timesync\\_t, 109](#)
- [\\_\\_mavlink\\_trajectory\\_representation\\_bezier\\_t, 110](#)
- [\\_\\_mavlink\\_trajectory\\_representation\\_waypoints\\_t, 110](#)
- [\\_\\_mavlink\\_tunnel\\_t, 110](#)
- [\\_\\_mavlink\\_uavcan\\_node\\_info\\_t, 111](#)
- [\\_\\_mavlink\\_uavcan\\_node\\_status\\_t, 111](#)
- [\\_\\_mavlink\\_utm\\_global\\_position\\_t, 111](#)
- [\\_\\_mavlink\\_v2\\_extension\\_t, 112](#)
- [\\_\\_mavlink\\_vfr\\_hud\\_t, 112](#)
- [\\_\\_mavlink\\_vibration\\_t, 113](#)
- [\\_\\_mavlink\\_vicon\\_position\\_estimate\\_t, 113](#)
- [\\_\\_mavlink\\_video\\_stream\\_information\\_t, 113](#)
- [\\_\\_mavlink\\_video\\_stream\\_status\\_t, 114](#)
- [\\_\\_mavlink\\_vision\\_position\\_estimate\\_t, 114](#)
- [\\_\\_mavlink\\_vision\\_speed\\_estimate\\_t, 114](#)
- [\\_\\_mavlink\\_wheel\\_distance\\_t, 115](#)
- [\\_\\_mavlink\\_wifi\\_config\\_ap\\_t, 115](#)

\_\_mavlink\_winch\_status\_t, 115  
 \_\_mavlink\_wind\_cov\_t, 116

## ACTUATOR\_CONFIGURATION

common.hpp, 829

## ACTUATOR\_OUTPUT\_FUNCTION

common.hpp, 829

## ADSB\_FLAGS

common.hpp, 830

## AIS\_FLAGS

common.hpp, 830

## ATLAS\_HC\_ARM\_DISARM\_ACK, 132

## ATLAS\_HC\_HEARTBEAT\_BC, 132

## ATLAS\_HC\_SYS\_STAT, 133

## ATLAS\_HC\_TIMESYNC\_RESP, 133

## AUTOTUNE\_AXIS

common.hpp, 830

## buffer\_arm\_disarm\_cmd

message\_sender, 453

## buffer\_heartbeat

message\_sender, 453

## buffer\_manual\_control

message\_sender, 453

## common.h

HAVE\_ENUM\_ACTUATOR\_CONFIGURATION,  
 797

HAVE\_ENUM\_ACTUATOR\_OUTPUT\_FUNCTION,  
 798

HAVE\_ENUM\_ADSB\_FLAGS, 798

HAVE\_ENUM\_AIS\_FLAGS, 798

HAVE\_ENUM\_AUTOTUNE\_AXIS, 798

HAVE\_ENUM\_COMP\_METADATA\_TYPE, 798

HAVE\_ENUM\_FENCE\_TYPE, 798

HAVE\_ENUM\_GIMBAL\_MANAGER\_CAP\_FLAGS,  
 799

HAVE\_ENUM\_MAV\_BATTERY\_FAULT, 799

HAVE\_ENUM\_MAV\_BATTERY\_MODE, 799

HAVE\_ENUM\_MAV\_CMD, 799

HAVE\_ENUM\_MAV\_DATA\_STREAM, 799

HAVE\_ENUM\_MAV\_FRAME, 800

HAVE\_ENUM\_MAV\_FTP\_ERR, 800

HAVE\_ENUM\_MAV\_FTP\_OPCODE, 800

HAVE\_ENUM\_MAV\_FUEL\_TYPE, 800

HAVE\_ENUM\_MAV\_GENERATOR\_STATUS\_FLAG,  
 801

HAVE\_ENUM\_MAV\_MODE, 801

HAVE\_ENUM\_MAV\_MOUNT\_MODE, 801

HAVE\_ENUM\_MAV\_ROI, 801

HAVE\_ENUM\_MAV\_SEVERITY, 801

HAVE\_ENUM\_MAV\_STANDARD\_MODE, 801

HAVE\_ENUM\_MISSION\_STATE, 802

HAVE\_ENUM\_PARACHUTE\_ACTION, 802

HAVE\_ENUM\_POSITION\_TARGET\_TYEMASK,  
 802

HAVE\_ENUM\_PREFLIGHT\_STORAGE\_MISSION\_ACTION,  
 802

HAVE\_ENUM\_PREFLIGHT\_STORAGE\_PARAMETER\_ACTION,  
 802

HAVE\_ENUM\_RC\_SUB\_TYPE, 803

HAVE\_ENUM\_RC\_TYPE, 803

## common.hpp

ACTUATOR\_CONFIGURATION, 829

ACTUATOR\_OUTPUT\_FUNCTION, 829

ADSB\_FLAGS, 830

AIS\_FLAGS, 830

AUTOTUNE\_AXIS, 830

COMP\_METADATA\_TYPE, 830

FENCE\_TYPE, 830

GIMBAL\_MANAGER\_CAP\_FLAGS, 830

MAV\_BATTERY\_FAULT, 831

MAV\_BATTERY\_MODE, 831

MAV\_CMD, 831

MAV\_DATA\_STREAM, 831

MAV\_FRAME, 831

MAV\_FTP\_ERR, 832

MAV\_FTP\_OPCODE, 832

MAV\_FUEL\_TYPE, 832

MAV\_GENERATOR\_STATUS\_FLAG, 832

MAV\_MODE, 833

MAV\_MOUNT\_MODE, 833

MAV\_ROI, 833

MAV\_SEVERITY, 833

MAV\_STANDARD\_MODE, 833

MISSION\_STATE, 833

PARACHUTE\_ACTION, 834

POSITION\_TARGET\_TYEMASK, 834

PREFLIGHT\_STORAGE\_MISSION\_ACTION, 834

PREFLIGHT\_STORAGE\_PARAMETER\_ACTION,  
 834

RC\_SUB\_TYPE, 834

RC\_TYPE, 835

## COMP\_METADATA\_TYPE

common.hpp, 830

## deserialize

mavlink::common::msg::ACTUATOR\_CONTROL\_TARGET,  
 118

mavlink::common::msg::ACTUATOR\_OUTPUT\_STATUS,  
 120

mavlink::common::msg::ADSB\_VEHICLE, 122

mavlink::common::msg::AIRSPEED, 125

mavlink::common::msg::AIS\_VESSEL, 127

mavlink::common::msg::ALTITUDE, 131

mavlink::common::msg::ATT\_POS\_MOCAP, 135

mavlink::common::msg::ATTITUDE, 137

mavlink::common::msg::ATTITUDE\_QUATERNION,  
 140

mavlink::common::msg::ATTITUDE\_QUATERNION\_COV,  
 142

mavlink::common::msg::ATTITUDE\_TARGET, 145

mavlink::common::msg::AUTH\_KEY, 147

mavlink::common::msg::AUTOPILOT\_STATE\_FOR\_GIMBAL\_DEVIC  
 150

mavlink::common::msg::AVAILABLE\_MODES, 156

- mavlink::common::msg::AVAILABLE\_MODES\_MONITOR, 252
- 158
- mavlink::common::msg::BATTERY\_INFO, 161
- mavlink::common::msg::BATTERY\_STATUS, 164
- mavlink::common::msg::BUTTON\_CHANGE, 167
- mavlink::common::msg::CAMERA\_CAPTURE\_STATUS, 169
- mavlink::common::msg::CAMERA\_FOV\_STATUS, 172
- mavlink::common::msg::CAMERA\_IMAGE\_CAPTURED, 174
- mavlink::common::msg::CAMERA\_INFORMATION, 178
- mavlink::common::msg::CAMERA\_SETTINGS, 181
- mavlink::common::msg::CAMERA\_THERMAL\_RANGE, 183
- mavlink::common::msg::CAMERA\_TRACKING\_GEO\_STATUS, 186
- mavlink::common::msg::CAMERA\_TRACKING\_IMAGE\_STATUS, 188
- mavlink::common::msg::CAMERA\_TRIGGER, 191
- mavlink::common::msg::CAN\_FILTER\_MODIFY, 193
- mavlink::common::msg::CAN\_FRAME, 196
- mavlink::common::msg::CANFD\_FRAME, 198
- mavlink::common::msg::CELLULAR\_CONFIG, 201
- mavlink::common::msg::CELLULAR\_STATUS, 204
- mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL, 206
- mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL\_ACK, 209
- mavlink::common::msg::COLLISION, 211
- mavlink::common::msg::COMMAND\_ACK, 214
- mavlink::common::msg::COMMAND\_CANCEL, 216
- mavlink::common::msg::COMMAND\_INT, 219
- mavlink::common::msg::COMMAND\_LONG, 222
- mavlink::common::msg::COMPONENT\_INFORMATION, 225
- mavlink::common::msg::COMPONENT\_INFORMATION\_BASIC, 227
- mavlink::common::msg::COMPONENT\_METADATA, 230
- mavlink::common::msg::CONTROL\_SYSTEM\_STATE, 232
- mavlink::common::msg::CURRENT\_EVENT\_SEQUENCE, 235
- mavlink::common::msg::CURRENT\_MODE, 237
- mavlink::common::msg::DATA\_STREAM, 240
- mavlink::common::msg::DATA\_TRANSMISSION\_HANDSHAKE, 242
- mavlink::common::msg::DEBUG, 245
- mavlink::common::msg::DEBUG\_FLOAT\_ARRAY, 247
- mavlink::common::msg::DEBUG\_VECT, 250
- mavlink::common::msg::DISTANCE\_SENSOR, 252
- mavlink::common::msg::EFI\_STATUS, 255
- mavlink::common::msg::ENCAPSULATED\_DATA, 258
- mavlink::common::msg::ESC\_INFO, 260
- mavlink::common::msg::ESC\_STATUS, 263
- mavlink::common::msg::ESTIMATOR\_STATUS, 266
- mavlink::common::msg::EVENT, 270
- mavlink::common::msg::EXTENDED\_SYS\_STATE, 273
- mavlink::common::msg::FENCE\_STATUS, 275
- mavlink::common::msg::FIGURE\_EIGHT\_EXECUTION\_STATUS, 278
- mavlink::common::msg::FILE\_TRANSFER\_PROTOCOL, 280
- mavlink::common::msg::FLIGHT\_INFORMATION, 283
- mavlink::common::msg::FOLLOW\_TARGET, 286
- mavlink::common::msg::FUEL\_STATUS, 289
- mavlink::common::msg::GENERATOR\_STATUS, 292
- mavlink::common::msg::GIMBAL\_DEVICE\_ATTITUDE\_STATUS, 295
- mavlink::common::msg::GIMBAL\_DEVICE\_INFORMATION, 298
- mavlink::common::msg::GIMBAL\_DEVICE\_SET\_ATTITUDE, 301
- mavlink::common::msg::GIMBAL\_MANAGER\_INFORMATION, 305
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_ATTITUDE, 308
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_MANUAL\_CONTROL, 310
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_PITCHYAW, 313
- mavlink::common::msg::GIMBAL\_MANAGER\_STATUS, 316
- mavlink::common::msg::GLOBAL\_POSITION\_INT\_COV, 321
- mavlink::common::msg::GLOBAL\_VISION\_POSITION\_ESTIMATE, 324
- mavlink::common::msg::GPS2\_RAW, 327
- mavlink::common::msg::GPS2\_RTK, 329
- mavlink::common::msg::GPS\_GLOBAL\_ORIGIN, 332
- mavlink::common::msg::GPS\_INJECT\_DATA, 334
- mavlink::common::msg::GPS\_INPUT, 337
- mavlink::common::msg::GPS\_RAW\_INT, 339
- mavlink::common::msg::GPS\_RTCM\_DATA, 342
- mavlink::common::msg::GPS\_RTK, 344
- mavlink::common::msg::GPS\_STATUS, 347
- mavlink::common::msg::HIGH\_LATENCY, 355
- mavlink::common::msg::HIGH\_LATENCY2, 357
- mavlink::common::msg::HIGHRES\_IMU, 360
- mavlink::common::msg::HIL\_ACTUATOR\_CONTROLS, 363
- mavlink::common::msg::HIL\_CONTROLS, 365

- mavlink::common::msg::HIL\_GPS, [368](#)
- mavlink::common::msg::HIL\_OPTICAL\_FLOW, [371](#)
- mavlink::common::msg::HIL\_RC\_INPUTS\_RAW, [374](#)
- mavlink::common::msg::HIL\_SENSOR, [377](#)
- mavlink::common::msg::HIL\_STATE, [379](#)
- mavlink::common::msg::HIL\_STATE\_QUATERNION, [383](#)
- mavlink::common::msg::HOME\_POSITION, [386](#)
- mavlink::common::msg::HYGROMETER\_SENSOR, [388](#)
- mavlink::common::msg::ILLUMINATOR\_STATUS, [391](#)
- mavlink::common::msg::ISBD\_LINK\_STATUS, [394](#)
- mavlink::common::msg::LANDING\_TARGET, [396](#)
- mavlink::common::msg::LINK\_NODE\_STATUS, [399](#)
- mavlink::common::msg::LOCAL\_POSITION\_NED, [402](#)
- mavlink::common::msg::LOCAL\_POSITION\_NED\_COV, [404](#)
- mavlink::common::msg::LOCAL\_POSITION\_NED\_SYSTEM\_OFFSET, [407](#)
- mavlink::common::msg::LOG\_DATA, [410](#)
- mavlink::common::msg::LOG\_ENTRY, [412](#)
- mavlink::common::msg::LOG\_ERASE, [415](#)
- mavlink::common::msg::LOG\_REQUEST\_DATA, [417](#)
- mavlink::common::msg::LOG\_REQUEST\_END, [420](#)
- mavlink::common::msg::LOG\_REQUEST\_LIST, [422](#)
- mavlink::common::msg::LOGGING\_ACK, [425](#)
- mavlink::common::msg::LOGGING\_DATA, [427](#)
- mavlink::common::msg::LOGGING\_DATA\_ACKED, [430](#)
- mavlink::common::msg::MAG\_CAL\_REPORT, [432](#)
- mavlink::common::msg::MANUAL\_CONTROL, [435](#)
- mavlink::common::msg::MANUAL\_SETPOINT, [438](#)
- mavlink::common::msg::MEMORY\_VECT, [441](#)
- mavlink::common::msg::MESSAGE\_INTERVAL, [451](#)
- mavlink::common::msg::MISSION\_ACK, [456](#)
- mavlink::common::msg::MISSION\_CLEAR\_ALL, [458](#)
- mavlink::common::msg::MISSION\_COUNT, [460](#)
- mavlink::common::msg::MISSION\_CURRENT, [463](#)
- mavlink::common::msg::MISSION\_ITEM, [466](#)
- mavlink::common::msg::MISSION\_ITEM\_INT, [469](#)
- mavlink::common::msg::MISSION\_ITEM\_REACHED, [471](#)
- mavlink::common::msg::MISSION\_REQUEST, [473](#)
- mavlink::common::msg::MISSION\_REQUEST\_INT, [476](#)
- mavlink::common::msg::MISSION\_REQUEST\_LIST, [479](#)
- mavlink::common::msg::MISSION\_REQUEST\_PARTIAL\_LIST, [482](#)
- mavlink::common::msg::MISSION\_SET\_CURRENT, [484](#)
- mavlink::common::msg::MISSION\_WRITE\_PARTIAL\_LIST, [487](#)
- mavlink::common::msg::MOUNT\_ORIENTATION, [489](#)
- mavlink::common::msg::NAMED\_VALUE\_FLOAT, [493](#)
- mavlink::common::msg::NAMED\_VALUE\_INT, [495](#)
- mavlink::common::msg::NAV\_CONTROLLER\_OUTPUT, [498](#)
- mavlink::common::msg::OBSTACLE\_DISTANCE, [500](#)
- mavlink::common::msg::ODOMETRY, [503](#)
- mavlink::common::msg::ONBOARD\_COMPUTER\_STATUS, [506](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_ARM\_STATUS, [509](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_AUTHENTICATION, [511](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_BASIC\_ID, [514](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_LOCATION, [517](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_MESSAGE\_PACK, [519](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_OPERATOR\_ID, [522](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SELF\_ID, [524](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM, [527](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM\_UPDATE, [530](#)
- mavlink::common::msg::OPTICAL\_FLOW, [532](#)
- mavlink::common::msg::OPTICAL\_FLOW\_RAD, [535](#)
- mavlink::common::msg::ORBIT\_EXECUTION\_STATUS, [538](#)
- mavlink::common::msg::PARAM\_ERROR, [542](#)
- mavlink::common::msg::PARAM\_EXT\_ACK, [545](#)
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_LIST, [547](#)
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_READ, [550](#)
- mavlink::common::msg::PARAM\_EXT\_SET, [552](#)
- mavlink::common::msg::PARAM\_EXT\_VALUE, [555](#)
- mavlink::common::msg::PARAM\_MAP\_RC, [557](#)
- mavlink::common::msg::PARAM\_REQUEST\_LIST, [560](#)
- mavlink::common::msg::PARAM\_REQUEST\_READ, [562](#)
- mavlink::common::msg::PARAM\_SET, [565](#)

- mavlink::common::msg::PARAM\_VALUE, [567](#)
- mavlink::common::msg::PING, [570](#)
- mavlink::common::msg::PLAY\_TUNE, [573](#)
- mavlink::common::msg::PLAY\_TUNE\_V2, [576](#)
- mavlink::common::msg::POSITION\_TARGET\_GLOBAL\_INT, [575](#)
- mavlink::common::msg::POSITION\_TARGET\_LOCAL\_NED, [581](#)
- mavlink::common::msg::POWER\_STATUS, [584](#)
- mavlink::common::msg::PROTOCOL\_VERSION, [586](#)
- mavlink::common::msg::RADIO\_STATUS, [589](#)
- mavlink::common::msg::RAW\_IMU, [591](#)
- mavlink::common::msg::RAW\_PRESSURE, [594](#)
- mavlink::common::msg::RAW\_RPM, [596](#)
- mavlink::common::msg::RC\_CHANNELS, [599](#)
- mavlink::common::msg::RC\_CHANNELS\_OVERRIDE, [602](#)
- mavlink::common::msg::RC\_CHANNELS\_RAW, [605](#)
- mavlink::common::msg::RC\_CHANNELS\_SCALED, [608](#)
- mavlink::common::msg::RELAY\_STATUS, [610](#)
- mavlink::common::msg::REQUEST\_DATA\_STREAM, [613](#)
- mavlink::common::msg::REQUEST\_EVENT, [615](#)
- mavlink::common::msg::RESOURCE\_REQUEST, [618](#)
- mavlink::common::msg::RESPONSE\_EVENT\_ERROR, [620](#)
- mavlink::common::msg::SAFETY\_ALLOWED\_AREA, [623](#)
- mavlink::common::msg::SAFETY\_SET\_ALLOWED\_AREA, [626](#)
- mavlink::common::msg::SCALED\_IMU, [628](#)
- mavlink::common::msg::SCALED\_IMU2, [631](#)
- mavlink::common::msg::SCALED\_IMU3, [634](#)
- mavlink::common::msg::SCALED\_PRESSURE, [636](#)
- mavlink::common::msg::SCALED\_PRESSURE2, [639](#)
- mavlink::common::msg::SCALED\_PRESSURE3, [641](#)
- mavlink::common::msg::SERIAL\_CONTROL, [644](#)
- mavlink::common::msg::SERVO\_OUTPUT\_RAW, [647](#)
- mavlink::common::msg::SET\_ACTUATOR\_CONTROL\_TARGET, [650](#)
- mavlink::common::msg::SET\_ATTITUDE\_TARGET, [653](#)
- mavlink::common::msg::SET\_GPS\_GLOBAL\_ORIGIN, [656](#)
- mavlink::common::msg::SET\_HOME\_POSITION, [658](#)
- mavlink::common::msg::SET\_MODE, [661](#)
- mavlink::common::msg::SET\_POSITION\_TARGET\_GLOBAL\_INT, [663](#)
- mavlink::common::msg::SET\_POSITION\_TARGET\_LOCAL\_NED, [667](#)
- mavlink::common::msg::SETUP\_SIGNING, [670](#)
- mavlink::common::msg::SIM\_STATE, [673](#)
- mavlink::common::msg::SMART\_BATTERY\_INFO, [675](#)
- mavlink::common::msg::STATUSTEXT, [678](#)
- mavlink::common::msg::STORAGE\_INFORMATION, [681](#)
- mavlink::common::msg::SUPPORTED\_TUNES, [684](#)
- mavlink::common::msg::SYS\_STATUS, [686](#)
- mavlink::common::msg::SYSTEM\_TIME, [690](#)
- mavlink::common::msg::TERRAIN\_CHECK, [692](#)
- mavlink::common::msg::TERRAIN\_DATA, [695](#)
- mavlink::common::msg::TERRAIN\_REPORT, [697](#)
- mavlink::common::msg::TERRAIN\_REQUEST, [700](#)
- mavlink::common::msg::TIME\_ESTIMATE\_TO\_TARGET, [703](#)
- mavlink::common::msg::TIMESYNC, [705](#)
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_BEZIER, [708](#)
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_WAYPOINTS, [710](#)
- mavlink::common::msg::TUNNEL, [713](#)
- mavlink::common::msg::UAVCAN\_NODE\_INFO, [716](#)
- mavlink::common::msg::UAVCAN\_NODE\_STATUS, [719](#)
- mavlink::common::msg::UTM\_GLOBAL\_POSITION, [724](#)
- mavlink::common::msg::V2\_EXTENSION, [726](#)
- mavlink::common::msg::VFR\_HUD, [729](#)
- mavlink::common::msg::VIBRATION, [731](#)
- mavlink::common::msg::VICON\_POSITION\_ESTIMATE, [734](#)
- mavlink::common::msg::VIDEO\_STREAM\_INFORMATION, [737](#)
- mavlink::common::msg::VIDEO\_STREAM\_STATUS, [739](#)
- mavlink::common::msg::VISION\_POSITION\_ESTIMATE, [742](#)
- mavlink::common::msg::VISION\_SPEED\_ESTIMATE, [745](#)
- mavlink::common::msg::WHEEL\_DISTANCE, [747](#)
- mavlink::common::msg::WIFI\_CONFIG\_AP, [749](#)
- mavlink::common::msg::WINCH\_STATUS, [752](#)
- mavlink::common::msg::WIND\_COV, [754](#)
- mavlink::Message, [444](#)
- mavlink::minimal::msg::HEARTBEAT, [352](#)
- mavlink::standard::msg::AUTOPILOT\_VERSION, [153](#)
- mavlink::standard::msg::GLOBAL\_POSITION\_INT, [318](#)
- displayHandler.cpp
- AppAndDisplay, [758](#)
- displayHandler.hpp
- AppAndDisplay, [762](#)



- updateDisplay, [762](#)
- end\_OFP\_timer
  - timer.cpp, [887](#)
  - timer.hpp, [878](#)
- FENCE\_TYPE
  - common.hpp, [830](#)
- FIRMWARE\_VERSION\_TYPE
  - standard.hpp, [867](#)
- get\_name
  - mavlink::common::msg::ACTUATOR\_CONTROL\_TARGET, [118](#)
  - mavlink::common::msg::ACTUATOR\_OUTPUT\_STATUS, [120](#)
  - mavlink::common::msg::ADSB\_VEHICLE, [123](#)
  - mavlink::common::msg::AIRSPEED, [125](#)
  - mavlink::common::msg::AIS\_VESSEL, [129](#)
  - mavlink::common::msg::ALTITUDE, [131](#)
  - mavlink::common::msg::ATT\_POS\_MOCAP, [135](#)
  - mavlink::common::msg::ATTITUDE, [137](#)
  - mavlink::common::msg::ATTITUDE\_QUATERNION, [140](#)
  - mavlink::common::msg::ATTITUDE\_QUATERNION\_COV, [143](#)
  - mavlink::common::msg::ATTITUDE\_TARGET, [145](#)
  - mavlink::common::msg::AUTH\_KEY, [148](#)
  - mavlink::common::msg::AUTOPILOT\_STATE\_FOR\_GIMBAL\_DEVICE, [151](#)
  - mavlink::common::msg::AVAILABLE\_MODES, [156](#)
  - mavlink::common::msg::AVAILABLE\_MODES\_MONITOR, [159](#)
  - mavlink::common::msg::BATTERY\_INFO, [162](#)
  - mavlink::common::msg::BATTERY\_STATUS, [164](#)
  - mavlink::common::msg::BUTTON\_CHANGE, [167](#)
  - mavlink::common::msg::CAMERA\_CAPTURE\_STATUS, [169](#)
  - mavlink::common::msg::CAMERA\_FOV\_STATUS, [172](#)
  - mavlink::common::msg::CAMERA\_IMAGE\_CAPTURED, [176](#)
  - mavlink::common::msg::CAMERA\_INFORMATION, [178](#)
  - mavlink::common::msg::CAMERA\_SETTINGS, [181](#)
  - mavlink::common::msg::CAMERA\_THERMAL\_RANGE, [183](#)
  - mavlink::common::msg::CAMERA\_TRACKING\_GEO\_STATUS, [186](#)
  - mavlink::common::msg::CAMERA\_TRACKING\_IMAGE\_STATUS, [189](#)
  - mavlink::common::msg::CAMERA\_TRIGGER, [191](#)
  - mavlink::common::msg::CAN\_FILTER\_MODIFY, [194](#)
  - mavlink::common::msg::CAN\_FRAME, [196](#)
  - mavlink::common::msg::CANFD\_FRAME, [199](#)
  - mavlink::common::msg::CELLULAR\_CONFIG, [201](#)
  - mavlink::common::msg::CELLULAR\_STATUS, [204](#)
  - mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL, [207](#)
  - mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL\_ACK, [209](#)
  - mavlink::common::msg::COLLISION, [212](#)
  - mavlink::common::msg::COMMAND\_ACK, [214](#)
  - mavlink::common::msg::COMMAND\_CANCEL, [217](#)
  - mavlink::common::msg::COMMAND\_INT, [220](#)
  - mavlink::common::msg::COMMAND\_LONG, [222](#)
  - mavlink::common::msg::COMPONENT\_INFORMATION, [225](#)
  - mavlink::common::msg::COMPONENT\_INFORMATION\_BASIC, [227](#)
  - mavlink::common::msg::COMPONENT\_METADATA, [230](#)
  - mavlink::common::msg::CONTROL\_SYSTEM\_STATE, [233](#)
  - mavlink::common::msg::CURRENT\_EVENT\_SEQUENCE, [235](#)
  - mavlink::common::msg::CURRENT\_MODE, [238](#)
  - mavlink::common::msg::DATA\_STREAM, [240](#)
  - mavlink::common::msg::DATA\_TRANSMISSION\_HANDSHAKE, [243](#)
  - mavlink::common::msg::DEBUG, [245](#)
  - mavlink::common::msg::DEBUG\_FLOAT\_ARRAY, [248](#)
  - mavlink::common::msg::DEBUG\_VECT, [250](#)
  - mavlink::common::msg::DISTANCE\_SENSOR, [253](#)
  - mavlink::common::msg::EFI\_STATUS, [256](#)
  - mavlink::common::msg::ENCAPSULATED\_DATA, [258](#)
  - mavlink::common::msg::ESC\_INFO, [261](#)
  - mavlink::common::msg::ESC\_STATUS, [263](#)
  - mavlink::common::msg::ESTIMATOR\_STATUS, [268](#)
  - mavlink::common::msg::EVENT, [270](#)
  - mavlink::common::msg::EXTENDED\_SYS\_STATE, [273](#)
  - mavlink::common::msg::FENCE\_STATUS, [275](#)
  - mavlink::common::msg::FIGURE\_EIGHT\_EXECUTION\_STATUS, [278](#)
  - mavlink::common::msg::FILE\_TRANSFER\_PROTOCOL, [281](#)
  - mavlink::common::msg::FLIGHT\_INFORMATION, [283](#)
  - mavlink::common::msg::FOLLOW\_TARGET, [287](#)
  - mavlink::common::msg::FUEL\_STATUS, [290](#)
  - mavlink::common::msg::GENERATOR\_STATUS, [292](#)
  - mavlink::common::msg::GIMBAL\_DEVICE\_ATTITUDE\_STATUS, [295](#)
  - mavlink::common::msg::GIMBAL\_DEVICE\_INFORMATION, [298](#)
  - mavlink::common::msg::GIMBAL\_DEVICE\_SET\_ATTITUDE, [303](#)

mavlink::common::msg::GIMBAL\_MANAGER\_INFORMATION, [417](#)  
[305](#)  
 mavlink::common::msg::GIMBAL\_MANAGER\_SET\_ATTITUDE, [420](#)  
[308](#)  
 mavlink::common::msg::GIMBAL\_MANAGER\_SET\_MANUAL\_CONTROL, [402](#)  
[311](#)  
 mavlink::common::msg::GIMBAL\_MANAGER\_SET\_PITCH\_YAW, [417](#)  
[313](#)  
 mavlink::common::msg::GIMBAL\_MANAGER\_STATUS, [430](#)  
[316](#)  
 mavlink::common::msg::GLOBAL\_POSITION\_INT\_COV, [435](#)  
[321](#)  
 mavlink::common::msg::GLOBAL\_VISION\_POSITION\_ESTIMATE, [435](#)  
[324](#)  
 mavlink::common::msg::GPS2\_RAW, [327](#)  
 mavlink::common::msg::GPS2\_RTK, [329](#)  
 mavlink::common::msg::GPS\_GLOBAL\_ORIGIN, [332](#)  
 mavlink::common::msg::GPS\_INJECT\_DATA, [334](#)  
 mavlink::common::msg::GPS\_INPUT, [337](#)  
 mavlink::common::msg::GPS\_RAW\_INT, [340](#)  
 mavlink::common::msg::GPS\_RTCM\_DATA, [342](#)  
 mavlink::common::msg::GPS\_RTK, [345](#)  
 mavlink::common::msg::GPS\_STATUS, [347](#)  
 mavlink::common::msg::HIGH\_LATENCY, [355](#)  
 mavlink::common::msg::HIGH\_LATENCY2, [358](#)  
 mavlink::common::msg::HIGHRES\_IMU, [360](#)  
 mavlink::common::msg::HIL\_ACTUATOR\_CONTROLS, [363](#)  
 mavlink::common::msg::HIL\_CONTROLS, [365](#)  
 mavlink::common::msg::HIL\_GPS, [368](#)  
 mavlink::common::msg::HIL\_OPTICAL\_FLOW, [372](#)  
 mavlink::common::msg::HIL\_RC\_INPUTS\_RAW, [374](#)  
 mavlink::common::msg::HIL\_SENSOR, [377](#)  
 mavlink::common::msg::HIL\_STATE, [381](#)  
 mavlink::common::msg::HIL\_STATE\_QUATERNION, [383](#)  
 mavlink::common::msg::HOME\_POSITION, [386](#)  
 mavlink::common::msg::HYGROMETER\_SENSOR, [389](#)  
 mavlink::common::msg::ILLUMINATOR\_STATUS, [391](#)  
 mavlink::common::msg::ISBD\_LINK\_STATUS, [394](#)  
 mavlink::common::msg::LANDING\_TARGET, [397](#)  
 mavlink::common::msg::LINK\_NODE\_STATUS, [399](#)  
 mavlink::common::msg::LOCAL\_POSITION\_NED, [402](#)  
 mavlink::common::msg::LOCAL\_POSITION\_NED\_COV, [405](#)  
 mavlink::common::msg::LOCAL\_POSITION\_NED\_SYSTEM\_OFFSET, [407](#)  
 mavlink::common::msg::LOG\_DATA, [410](#)  
 mavlink::common::msg::LOG\_ENTRY, [412](#)  
 mavlink::common::msg::LOG\_ERASE, [415](#)  
 mavlink::common::msg::LOG\_REQUEST\_DATA, [417](#)  
 mavlink::common::msg::LOG\_REQUEST\_END, [420](#)  
 mavlink::common::msg::LOG\_REQUEST\_LIST, [425](#)  
 mavlink::common::msg::LOGGING\_ACK, [425](#)  
 mavlink::common::msg::LOGGING\_DATA, [427](#)  
 mavlink::common::msg::LOGGING\_DATA\_ACKED, [430](#)  
 mavlink::common::msg::MAG\_CAL\_REPORT, [433](#)  
 mavlink::common::msg::MANUAL\_CONTROL, [435](#)  
 mavlink::common::msg::MANUAL\_SETPOINT, [435](#)  
 mavlink::common::msg::MEMORY\_VECT, [441](#)  
 mavlink::common::msg::MESSAGE\_INTERVAL, [451](#)  
 mavlink::common::msg::MISSION\_ACK, [456](#)  
 mavlink::common::msg::MISSION\_CLEAR\_ALL, [458](#)  
 mavlink::common::msg::MISSION\_COUNT, [461](#)  
 mavlink::common::msg::MISSION\_CURRENT, [463](#)  
 mavlink::common::msg::MISSION\_ITEM, [466](#)  
 mavlink::common::msg::MISSION\_ITEM\_INT, [469](#)  
 mavlink::common::msg::MISSION\_ITEM\_REACHED, [471](#)  
 mavlink::common::msg::MISSION\_REQUEST, [474](#)  
 mavlink::common::msg::MISSION\_REQUEST\_INT, [476](#)  
 mavlink::common::msg::MISSION\_REQUEST\_LIST, [480](#)  
 mavlink::common::msg::MISSION\_REQUEST\_PARTIAL\_LIST, [482](#)  
 mavlink::common::msg::MISSION\_SET\_CURRENT, [485](#)  
 mavlink::common::msg::MISSION\_WRITE\_PARTIAL\_LIST, [487](#)  
 mavlink::common::msg::MOUNT\_ORIENTATION, [490](#)  
 mavlink::common::msg::NAMED\_VALUE\_FLOAT, [493](#)  
 mavlink::common::msg::NAMED\_VALUE\_INT, [495](#)  
 mavlink::common::msg::NAV\_CONTROLLER\_OUTPUT, [498](#)  
 mavlink::common::msg::OBSTACLE\_DISTANCE, [501](#)  
 mavlink::common::msg::ODOMETRY, [504](#)  
 mavlink::common::msg::ONBOARD\_COMPUTER\_STATUS, [507](#)  
 mavlink::common::msg::OPEN\_DRONE\_ID\_ARM\_STATUS, [509](#)  
 mavlink::common::msg::OPEN\_DRONE\_ID\_AUTHENTICATION, [512](#)  
 mavlink::common::msg::OPEN\_DRONE\_ID\_BASIC\_ID, [514](#)  
 mavlink::common::msg::OPEN\_DRONE\_ID\_LOCATION, [517](#)



- mavlink::common::msg::OPEN\_DRONE\_ID\_MESSAGE\_ACK, [641](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_OPERATOR\_ID, [520](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SELF\_ID, [522](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM, [525](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM\_UPDATE, [527](#)
- mavlink::common::msg::OPTICAL\_FLOW, [530](#)
- mavlink::common::msg::OPTICAL\_FLOW\_RAD, [533](#)
- mavlink::common::msg::ORBIT\_EXECUTION\_STATUS, [535](#)
- mavlink::common::msg::PARAM\_ERROR, [538](#)
- mavlink::common::msg::PARAM\_EXT\_ACK, [542](#)
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_LIST, [545](#)
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_READ, [547](#)
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_READ, [550](#)
- mavlink::common::msg::PARAM\_EXT\_SET, [552](#)
- mavlink::common::msg::PARAM\_EXT\_VALUE, [555](#)
- mavlink::common::msg::PARAM\_MAP\_RC, [557](#)
- mavlink::common::msg::PARAM\_REQUEST\_LIST, [560](#)
- mavlink::common::msg::PARAM\_REQUEST\_READ, [562](#)
- mavlink::common::msg::PARAM\_SET, [565](#)
- mavlink::common::msg::PARAM\_VALUE, [566](#)
- mavlink::common::msg::PING, [570](#)
- mavlink::common::msg::PLAY\_TUNE, [574](#)
- mavlink::common::msg::PLAY\_TUNE\_V2, [576](#)
- mavlink::common::msg::POSITION\_TARGET\_GLOBAL\_INT, [579](#)
- mavlink::common::msg::POSITION\_TARGET\_GLOBAL\_INT, [579](#)
- mavlink::common::msg::POSITION\_TARGET\_LOCAL\_NED, [581](#)
- mavlink::common::msg::POWER\_STATUS, [584](#)
- mavlink::common::msg::PROTOCOL\_VERSION, [586](#)
- mavlink::common::msg::RADIO\_STATUS, [589](#)
- mavlink::common::msg::RAW\_IMU, [592](#)
- mavlink::common::msg::RAW\_PRESSURE, [594](#)
- mavlink::common::msg::RAW\_RPM, [597](#)
- mavlink::common::msg::RC\_CHANNELS, [600](#)
- mavlink::common::msg::RC\_CHANNELS\_OVERRIDE, [603](#)
- mavlink::common::msg::RC\_CHANNELS\_RAW, [605](#)
- mavlink::common::msg::RC\_CHANNELS\_SCALED, [608](#)
- mavlink::common::msg::RELAY\_STATUS, [611](#)
- mavlink::common::msg::REQUEST\_DATA\_STREAM, [613](#)
- mavlink::common::msg::REQUEST\_EVENT, [616](#)
- mavlink::common::msg::RESOURCE\_REQUEST, [618](#)
- mavlink::common::msg::RESPONSE\_EVENT\_ERROR, [621](#)
- mavlink::common::msg::SAFETY\_ALLOWED\_AREA, [623](#)
- mavlink::common::msg::SAFETY\_SET\_ALLOWED\_AREA, [626](#)
- mavlink::common::msg::SCALED\_IMU, [629](#)
- mavlink::common::msg::SCALED\_IMU2, [631](#)
- mavlink::common::msg::SCALED\_IMU3, [634](#)
- mavlink::common::msg::SCALED\_PRESSURE, [637](#)
- mavlink::common::msg::SCALED\_PRESSURE2, [639](#)
- mavlink::common::msg::SCALED\_PRESSURE3, [642](#)
- mavlink::common::msg::SERIAL\_CONTROL, [644](#)
- mavlink::common::msg::SERVO\_OUTPUT\_RAW, [647](#)
- mavlink::common::msg::SET\_ACTUATOR\_CONTROL\_TARGET, [651](#)
- mavlink::common::msg::SET\_ATTITUDE\_TARGET, [653](#)
- mavlink::common::msg::SET\_GPS\_GLOBAL\_ORIGIN, [656](#)
- mavlink::common::msg::SET\_HOME\_POSITION, [659](#)
- mavlink::common::msg::SET\_MODE, [661](#)
- mavlink::common::msg::SET\_POSITION\_TARGET\_GLOBAL\_INT, [665](#)
- mavlink::common::msg::SET\_POSITION\_TARGET\_LOCAL\_NED, [668](#)
- mavlink::common::msg::SETUP\_SIGNING, [670](#)
- mavlink::common::msg::SIM\_STATE, [673](#)
- mavlink::common::msg::SMART\_BATTERY\_INFO, [676](#)
- mavlink::common::msg::STATUSTEXT, [678](#)
- mavlink::common::msg::STORAGE\_INFORMATION, [682](#)
- mavlink::common::msg::SUPPORTED\_TUNES, [684](#)
- mavlink::common::msg::SYS\_STATUS, [688](#)
- mavlink::common::msg::SYSTEM\_TIME, [690](#)
- mavlink::common::msg::TERRAIN\_CHECK, [693](#)
- mavlink::common::msg::TERRAIN\_DATA, [695](#)
- mavlink::common::msg::TERRAIN\_REPORT, [698](#)
- mavlink::common::msg::TERRAIN\_REQUEST, [700](#)
- mavlink::common::msg::TIME\_ESTIMATE\_TO\_TARGET, [703](#)
- mavlink::common::msg::TIMESYNC, [705](#)
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_BEZIER, [708](#)
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_WAYPOINTS, [711](#)
- mavlink::common::msg::TUNNEL, [713](#)
- mavlink::common::msg::UAVCAN\_NODE\_INFO, [717](#)
- mavlink::common::msg::UAVCAN\_NODE\_STATUS, [717](#)

- 719
- mavlink::common::msg::UTM\_GLOBAL\_POSITION, 724
- mavlink::common::msg::V2\_EXTENSION, 727
- mavlink::common::msg::VFR\_HUD, 729
- mavlink::common::msg::VIBRATION, 732
- mavlink::common::msg::VICON\_POSITION\_ESTIMATE, 734
- mavlink::common::msg::VIDEO\_STREAM\_INFORMATION, 737
- mavlink::common::msg::VIDEO\_STREAM\_STATUS, 740
- mavlink::common::msg::VISION\_POSITION\_ESTIMATE, 742
- mavlink::common::msg::VISION\_SPEED\_ESTIMATE, 745
- mavlink::common::msg::WHEEL\_DISTANCE, 747
- mavlink::common::msg::WIFI\_CONFIG\_AP, 750
- mavlink::common::msg::WINCH\_STATUS, 752
- mavlink::common::msg::WIND\_COV, 755
- mavlink::Message, 446
- mavlink::minimal::msg::HEARTBEAT, 352
- mavlink::standard::msg::AUTOPILOT\_VERSION, 153
- mavlink::standard::msg::GLOBAL\_POSITION\_INT, 319
- getXY
  - IOhandler.cpp, 879
  - IOhandler.hpp, 764
- getXY\_raw
  - IOhandler.cpp, 880
  - IOhandler.hpp, 766
- GIMBAL\_MANAGER\_CAP\_FLAGS
  - common.hpp, 830
- HAVE\_ENUM\_ACTUATOR\_CONFIGURATION
  - common.h, 797
- HAVE\_ENUM\_ACTUATOR\_OUTPUT\_FUNCTION
  - common.h, 798
- HAVE\_ENUM\_ADSB\_FLAGS
  - common.h, 798
- HAVE\_ENUM\_AIS\_FLAGS
  - common.h, 798
- HAVE\_ENUM\_AUTOTUNE\_AXIS
  - common.h, 798
- HAVE\_ENUM\_COMP\_METADATA\_TYPE
  - common.h, 798
- HAVE\_ENUM\_FENCE\_TYPE
  - common.h, 798
- HAVE\_ENUM\_FIRMWARE\_VERSION\_TYPE
  - standard.h, 865
- HAVE\_ENUM\_GIMBAL\_MANAGER\_CAP\_FLAGS
  - common.h, 799
- HAVE\_ENUM\_MAV\_AUTOPILOT
  - minimal.h, 857
- HAVE\_ENUM\_MAV\_BATTERY\_FAULT
  - common.h, 799
- HAVE\_ENUM\_MAV\_BATTERY\_MODE
  - common.h, 799
- HAVE\_ENUM\_MAV\_CMD
  - common.h, 799
- HAVE\_ENUM\_MAV\_COMPONENT
  - minimal.h, 858
- HAVE\_ENUM\_MAV\_DATA\_STREAM
  - common.h, 799
- HAVE\_ENUM\_MAV\_FRAME
  - common.h, 800
- HAVE\_ENUM\_MAV FTP\_ERR
  - common.h, 800
- HAVE\_ENUM\_MAV FTP\_OPCODE
  - common.h, 800
- HAVE\_ENUM\_MAV\_FUEL\_TYPE
  - common.h, 800
- HAVE\_ENUM\_MAV\_GENERATOR\_STATUS\_FLAG
  - common.h, 801
- HAVE\_ENUM\_MAV\_MODE
  - common.h, 801
- HAVE\_ENUM\_MAV\_MODE\_FLAG\_DECODE\_POSITION
  - minimal.h, 858
- HAVE\_ENUM\_MAV\_MOUNT\_MODE
  - common.h, 801
- HAVE\_ENUM\_MAV\_PROTOCOL\_CAPABILITY
  - standard.h, 865
- HAVE\_ENUM\_MAV\_ROI
  - common.h, 801
- HAVE\_ENUM\_MAV\_SEVERITY
  - common.h, 801
- HAVE\_ENUM\_MAV\_STANDARD\_MODE
  - common.h, 801
- HAVE\_ENUM\_MAV\_TYPE
  - minimal.h, 858
- HAVE\_ENUM\_MISSION\_STATE
  - common.h, 802
- HAVE\_ENUM\_PARACHUTE\_ACTION
  - common.h, 802
- HAVE\_ENUM\_POSITION\_TARGET\_TYPEMASK
  - common.h, 802
- HAVE\_ENUM\_PREFLIGHT\_STORAGE\_MISSION\_ACTION
  - common.h, 802
- HAVE\_ENUM\_PREFLIGHT\_STORAGE\_PARAMETER\_ACTION
  - common.h, 802
- HAVE\_ENUM\_RC\_SUB\_TYPE
  - common.h, 803
- HAVE\_ENUM\_RC\_TYPE
  - common.h, 803
- HC\_ATLAS\_ARM\_DISARM\_CMD, 349
- HC\_ATLAS\_HEARTBEAT\_BC, 349
- HC\_ATLAS\_MANUAL\_CONTROL\_BC, 349
- HC\_ATLAS\_TIMESYNC\_REQ, 350
- HC\_RADIO\_STATUS, 350
- heartbeat\_timed\_out
  - packetHandler.cpp, 883
  - packetHandler.h, 872
- identifyControllerDrift
  - setupFunctions.cpp, 884
  - setupFunctions.h, 874
- initMAVLink

- setupFunctions.cpp, [884](#)
  - setupFunctions.h, [874](#)
- initUGV\_state
  - setupFunctions.h, [874](#)
- IOhandler.cpp
  - getXY, [879](#)
  - getXY\_raw, [880](#)
- IOhandler.hpp
  - getXY, [764](#)
  - getXY\_raw, [766](#)
- MAV\_AUTOPILOT
  - minimal.hpp, [862](#)
- MAV\_BATTERY\_FAULT
  - common.hpp, [831](#)
- MAV\_BATTERY\_MODE
  - common.hpp, [831](#)
- MAV\_CMD
  - common.hpp, [831](#)
- MAV\_COMPONENT
  - minimal.hpp, [862](#)
- MAV\_DATA\_STREAM
  - common.hpp, [831](#)
- MAV\_FRAME
  - common.hpp, [831](#)
- MAV\_FTP\_ERR
  - common.hpp, [832](#)
- MAV\_FTP\_OPCODE
  - common.hpp, [832](#)
- MAV\_FUEL\_TYPE
  - common.hpp, [832](#)
- MAV\_GENERATOR\_STATUS\_FLAG
  - common.hpp, [832](#)
- MAV\_MODE
  - common.hpp, [833](#)
- MAV\_MODE\_FLAG\_DECODE\_POSITION
  - minimal.hpp, [862](#)
- MAV\_MOUNT\_MODE
  - common.hpp, [833](#)
- MAV\_PROTOCOL\_CAPABILITY
  - standard.hpp, [868](#)
- MAV\_ROI
  - common.hpp, [833](#)
- MAV\_SEVERITY
  - common.hpp, [833](#)
- MAV\_STANDARD\_MODE
  - common.hpp, [833](#)
- MAV\_TYPE
  - minimal.hpp, [863](#)
- mavlink::common::msg::ACTUATOR\_CONTROL\_TARGET,
  - [116](#)
  - deserialize, [118](#)
  - get\_name, [118](#)
  - serialize, [118](#)
- mavlink::common::msg::ACTUATOR\_OUTPUT\_STATUS,
  - [119](#)
  - deserialize, [120](#)
  - get\_name, [120](#)
  - serialize, [120](#)
- mavlink::common::msg::ADSB\_VEHICLE, [121](#)
  - deserialize, [122](#)
  - get\_name, [123](#)
  - serialize, [123](#)
- mavlink::common::msg::AIRSPEED, [123](#)
  - deserialize, [125](#)
  - get\_name, [125](#)
  - serialize, [125](#)
- mavlink::common::msg::AIS\_VESSEL, [126](#)
  - deserialize, [127](#)
  - get\_name, [129](#)
  - serialize, [129](#)
- mavlink::common::msg::ALTITUDE, [129](#)
  - deserialize, [131](#)
  - get\_name, [131](#)
  - serialize, [131](#)
- mavlink::common::msg::ATT\_POS\_MOCAP, [133](#)
  - deserialize, [135](#)
  - get\_name, [135](#)
  - serialize, [135](#)
- mavlink::common::msg::ATTITUDE, [136](#)
  - deserialize, [137](#)
  - get\_name, [137](#)
  - serialize, [138](#)
- mavlink::common::msg::ATTITUDE\_QUATERNION,
  - [138](#)
  - deserialize, [140](#)
  - get\_name, [140](#)
  - serialize, [140](#)
- mavlink::common::msg::ATTITUDE\_QUATERNION\_COV,
  - [141](#)
  - deserialize, [142](#)
  - get\_name, [143](#)
  - serialize, [143](#)
- mavlink::common::msg::ATTITUDE\_TARGET, [143](#)
  - deserialize, [145](#)
  - get\_name, [145](#)
  - serialize, [145](#)
- mavlink::common::msg::AUTH\_KEY, [146](#)
  - deserialize, [147](#)
  - get\_name, [148](#)
  - serialize, [148](#)
- mavlink::common::msg::AUTOPILOT\_STATE\_FOR\_GIMBAL\_DEVICE,
  - [148](#)
  - deserialize, [150](#)
  - get\_name, [151](#)
  - serialize, [151](#)
- mavlink::common::msg::AVAILABLE\_MODES, [154](#)
  - deserialize, [156](#)
  - get\_name, [156](#)
  - serialize, [156](#)
- mavlink::common::msg::AVAILABLE\_MODES\_MONITOR,
  - [157](#)
  - deserialize, [158](#)
  - get\_name, [159](#)
  - serialize, [159](#)
- mavlink::common::msg::BATTERY\_INFO, [159](#)
  - deserialize, [161](#)

- get\_name, 162
  - serialize, 162
- mavlink::common::msg::BATTERY\_STATUS, 162
  - deserialize, 164
  - get\_name, 164
  - serialize, 165
- mavlink::common::msg::BUTTON\_CHANGE, 165
  - deserialize, 167
  - get\_name, 167
  - serialize, 167
- mavlink::common::msg::CAMERA\_CAPTURE\_STATUS, 168
  - deserialize, 169
  - get\_name, 169
  - serialize, 170
- mavlink::common::msg::CAMERA\_FOV\_STATUS, 170
  - deserialize, 172
  - get\_name, 172
  - serialize, 172
- mavlink::common::msg::CAMERA\_IMAGE\_CAPTURED, 173
  - deserialize, 174
  - get\_name, 176
  - serialize, 176
- mavlink::common::msg::CAMERA\_INFORMATION, 176
  - deserialize, 178
  - get\_name, 178
  - serialize, 179
- mavlink::common::msg::CAMERA\_SETTINGS, 179
  - deserialize, 181
  - get\_name, 181
  - serialize, 181
- mavlink::common::msg::CAMERA\_THERMAL\_RANGE, 182
  - deserialize, 183
  - get\_name, 183
  - serialize, 184
- mavlink::common::msg::CAMERA\_TRACKING\_GEO\_STATUS, 184
  - deserialize, 186
  - get\_name, 186
  - serialize, 186
- mavlink::common::msg::CAMERA\_TRACKING\_IMAGE\_STATUS, 187
  - deserialize, 188
  - get\_name, 189
  - serialize, 189
- mavlink::common::msg::CAMERA\_TRIGGER, 189
  - deserialize, 191
  - get\_name, 191
  - serialize, 191
- mavlink::common::msg::CAN\_FILTER\_MODIFY, 192
  - deserialize, 193
  - get\_name, 194
  - serialize, 194
- mavlink::common::msg::CAN\_FRAME, 194
  - deserialize, 196
  - get\_name, 196
- serialize, 196
- mavlink::common::msg::CANFD\_FRAME, 197
  - deserialize, 198
  - get\_name, 199
  - serialize, 199
- mavlink::common::msg::CELLULAR\_CONFIG, 199
  - deserialize, 201
  - get\_name, 201
  - serialize, 202
- mavlink::common::msg::CELLULAR\_STATUS, 202
  - deserialize, 204
  - get\_name, 204
  - serialize, 204
- mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL, 205
  - deserialize, 206
  - get\_name, 207
  - serialize, 207
- mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL\_ACK, 207
  - deserialize, 209
  - get\_name, 209
  - serialize, 209
- mavlink::common::msg::COLLISION, 210
  - deserialize, 211
  - get\_name, 212
  - serialize, 212
- mavlink::common::msg::COMMAND\_ACK, 212
  - deserialize, 214
  - get\_name, 214
  - serialize, 214
- mavlink::common::msg::COMMAND\_CANCEL, 215
  - deserialize, 216
  - get\_name, 217
  - serialize, 217
- mavlink::common::msg::COMMAND\_INT, 217
  - deserialize, 219
  - get\_name, 220
  - serialize, 220
- mavlink::common::msg::COMMAND\_LONG, 220
  - deserialize, 222
  - get\_name, 222
  - serialize, 223
- mavlink::common::msg::COMPONENT\_INFORMATION, 223
  - deserialize, 225
  - get\_name, 225
  - serialize, 225
- mavlink::common::msg::COMPONENT\_INFORMATION\_BASIC, 226
  - deserialize, 227
  - get\_name, 227
  - serialize, 228
- mavlink::common::msg::COMPONENT\_METADATA, 228
  - deserialize, 230
  - get\_name, 230
  - serialize, 230

- mavlink::common::msg::CONTROL\_SYSTEM\_STATE,
  - 231
  - deserialize, 232
  - get\_name, 233
  - serialize, 233
- mavlink::common::msg::CURRENT\_EVENT\_SEQUENCE,
  - 233
  - deserialize, 235
  - get\_name, 235
  - serialize, 235
- mavlink::common::msg::CURRENT\_MODE, 236
  - deserialize, 237
  - get\_name, 238
  - serialize, 238
- mavlink::common::msg::DATA\_STREAM, 238
  - deserialize, 240
  - get\_name, 240
  - serialize, 240
- mavlink::common::msg::DATA\_TRANSMISSION\_HANDSHAKE,
  - 241
  - deserialize, 242
  - get\_name, 243
  - serialize, 243
- mavlink::common::msg::DEBUG, 243
  - deserialize, 245
  - get\_name, 245
  - serialize, 245
- mavlink::common::msg::DEBUG\_FLOAT\_ARRAY, 246
  - deserialize, 247
  - get\_name, 248
  - serialize, 248
- mavlink::common::msg::DEBUG\_VECT, 248
  - deserialize, 250
  - get\_name, 250
  - serialize, 250
- mavlink::common::msg::DISTANCE\_SENSOR, 251
  - deserialize, 252
  - get\_name, 253
  - serialize, 253
- mavlink::common::msg::EFI\_STATUS, 253
  - deserialize, 255
  - get\_name, 256
  - serialize, 256
- mavlink::common::msg::ENCAPSULATED\_DATA, 256
  - deserialize, 258
  - get\_name, 258
  - serialize, 258
- mavlink::common::msg::ESC\_INFO, 259
  - deserialize, 260
  - get\_name, 261
  - serialize, 261
- mavlink::common::msg::ESC\_STATUS, 261
  - deserialize, 263
  - get\_name, 263
  - serialize, 263
- mavlink::common::msg::ESTIMATOR\_STATUS, 265
  - deserialize, 266
  - get\_name, 268
  - serialize, 268
- mavlink::common::msg::EVENT, 268
  - deserialize, 270
  - get\_name, 270
  - serialize, 271
- mavlink::common::msg::EXTENDED\_SYS\_STATE, 271
  - deserialize, 273
  - get\_name, 273
  - serialize, 273
- mavlink::common::msg::FENCE\_STATUS, 274
  - deserialize, 275
  - get\_name, 275
  - serialize, 276
- mavlink::common::msg::FIGURE\_EIGHT\_EXECUTION\_STATUS,
  - 276
  - deserialize, 278
  - get\_name, 278
  - serialize, 278
- mavlink::common::msg::FILE\_TRANSFER\_PROTOCOL,
  - 279
  - deserialize, 280
  - get\_name, 281
  - serialize, 281
- mavlink::common::msg::FLIGHT\_INFORMATION, 281
  - deserialize, 283
  - get\_name, 283
  - serialize, 283
- mavlink::common::msg::FOLLOW\_TARGET, 285
  - deserialize, 286
  - get\_name, 287
  - serialize, 287
- mavlink::common::msg::FUEL\_STATUS, 287
  - deserialize, 289
  - get\_name, 290
  - serialize, 290
- mavlink::common::msg::GENERATOR\_STATUS, 290
  - deserialize, 292
  - get\_name, 292
  - serialize, 293
- mavlink::common::msg::GIMBAL\_DEVICE\_ATTITUDE\_STATUS,
  - 293
  - deserialize, 295
  - get\_name, 295
  - serialize, 295
- mavlink::common::msg::GIMBAL\_DEVICE\_INFORMATION,
  - 296
  - deserialize, 298
  - get\_name, 298
  - serialize, 298
- mavlink::common::msg::GIMBAL\_DEVICE\_SET\_ATTITUDE,
  - 300
  - deserialize, 301
  - get\_name, 303
  - serialize, 303
- mavlink::common::msg::GIMBAL\_MANAGER\_INFORMATION,
  - 303
  - deserialize, 305
  - get\_name, 305

- serialize, 306
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_ATTITUDE, 306
  - deserialize, 308
  - get\_name, 308
  - serialize, 308
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_MANUAL\_CONTROL, 309
  - deserialize, 310
  - get\_name, 311
  - serialize, 311
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_PITCHYAW, 311
  - deserialize, 313
  - get\_name, 313
  - serialize, 314
- mavlink::common::msg::GIMBAL\_MANAGER\_STATUS, 314
  - deserialize, 316
  - get\_name, 316
  - serialize, 316
- mavlink::common::msg::GLOBAL\_POSITION\_INT\_COV, 319
  - deserialize, 321
  - get\_name, 321
  - serialize, 322
- mavlink::common::msg::GLOBAL\_VISION\_POSITION\_ESTIMATE, 322
  - deserialize, 324
  - get\_name, 324
  - serialize, 324
- mavlink::common::msg::GPS2\_RAW, 325
  - deserialize, 327
  - get\_name, 327
  - serialize, 327
- mavlink::common::msg::GPS2\_RTK, 328
  - deserialize, 329
  - get\_name, 329
  - serialize, 330
- mavlink::common::msg::GPS\_GLOBAL\_ORIGIN, 330
  - deserialize, 332
  - get\_name, 332
  - serialize, 332
- mavlink::common::msg::GPS\_INJECT\_DATA, 333
  - deserialize, 334
  - get\_name, 334
  - serialize, 334
- mavlink::common::msg::GPS\_INPUT, 335
  - deserialize, 337
  - get\_name, 337
  - serialize, 337
- mavlink::common::msg::GPS\_RAW\_INT, 338
  - deserialize, 339
  - get\_name, 340
  - serialize, 340
- mavlink::common::msg::GPS\_RTCM\_DATA, 340
  - deserialize, 342
  - get\_name, 342
- serialize, 342
- mavlink::common::msg::GPS\_RTK, 343
  - deserialize, 344
  - get\_name, 345
  - serialize, 345
- mavlink::common::msg::GPS\_STATUS, 345
  - deserialize, 347
  - get\_name, 347
  - serialize, 347
- mavlink::common::msg::HIGH\_LATENCY, 353
  - deserialize, 355
  - get\_name, 355
  - serialize, 355
- mavlink::common::msg::HIGH\_LATENCY2, 356
  - deserialize, 357
  - get\_name, 358
  - serialize, 358
- mavlink::common::msg::HIGHRES\_IMU, 358
  - deserialize, 360
  - get\_name, 360
  - serialize, 361
- mavlink::common::msg::HIL\_ACTUATOR\_CONTROLS, 361
  - deserialize, 363
  - get\_name, 363
  - serialize, 363
- mavlink::common::msg::HIL\_CONTROLS, 364
  - deserialize, 365
  - get\_name, 365
  - serialize, 366
- mavlink::common::msg::HIL\_GPS, 366
  - deserialize, 368
  - get\_name, 368
  - serialize, 368
- mavlink::common::msg::HIL\_OPTICAL\_FLOW, 370
  - deserialize, 371
  - get\_name, 372
  - serialize, 372
- mavlink::common::msg::HIL\_RC\_INPUTS\_RAW, 372
  - deserialize, 374
  - get\_name, 374
  - serialize, 375
- mavlink::common::msg::HIL\_SENSOR, 375
  - deserialize, 377
  - get\_name, 377
  - serialize, 377
- mavlink::common::msg::HIL\_STATE, 378
  - deserialize, 379
  - get\_name, 381
  - serialize, 381
- mavlink::common::msg::HIL\_STATE\_QUATERNION, 381
  - deserialize, 383
  - get\_name, 383
  - serialize, 384
- mavlink::common::msg::HOME\_POSITION, 384
  - deserialize, 386
  - get\_name, 386

- serialize, 386
- mavlink::common::msg::HYGROMETER\_SENSOR, 387
  - deserialize, 388
  - get\_name, 389
  - serialize, 389
- mavlink::common::msg::ILLUMINATOR\_STATUS, 389
  - deserialize, 391
  - get\_name, 391
  - serialize, 392
- mavlink::common::msg::ISBD\_LINK\_STATUS, 392
  - deserialize, 394
  - get\_name, 394
  - serialize, 394
- mavlink::common::msg::LANDING\_TARGET, 395
  - deserialize, 396
  - get\_name, 397
  - serialize, 397
- mavlink::common::msg::LINK\_NODE\_STATUS, 397
  - deserialize, 399
  - get\_name, 399
  - serialize, 400
- mavlink::common::msg::LOCAL\_POSITION\_NED, 400
  - deserialize, 402
  - get\_name, 402
  - serialize, 402
- mavlink::common::msg::LOCAL\_POSITION\_NED\_COV, 403
  - deserialize, 404
  - get\_name, 405
  - serialize, 405
- mavlink::common::msg::LOCAL\_POSITION\_NED\_SYSTEM\_GLOBAL\_OFFSET, 405
  - deserialize, 407
  - get\_name, 407
  - serialize, 408
- mavlink::common::msg::LOG\_DATA, 408
  - deserialize, 410
  - get\_name, 410
  - serialize, 410
- mavlink::common::msg::LOG\_ENTRY, 411
  - deserialize, 412
  - get\_name, 412
  - serialize, 413
- mavlink::common::msg::LOG\_ERASE, 413
  - deserialize, 415
  - get\_name, 415
  - serialize, 415
- mavlink::common::msg::LOG\_REQUEST\_DATA, 416
  - deserialize, 417
  - get\_name, 417
  - serialize, 418
- mavlink::common::msg::LOG\_REQUEST\_END, 418
  - deserialize, 420
  - get\_name, 420
  - serialize, 420
- mavlink::common::msg::LOG\_REQUEST\_LIST, 421
  - deserialize, 422
- get\_name, 422
  - serialize, 423
- mavlink::common::msg::LOGGING\_ACK, 423
  - deserialize, 425
  - get\_name, 425
  - serialize, 425
- mavlink::common::msg::LOGGING\_DATA, 426
  - deserialize, 427
  - get\_name, 427
  - serialize, 428
- mavlink::common::msg::LOGGING\_DATA\_ACKED, 428
  - deserialize, 430
  - get\_name, 430
  - serialize, 430
- mavlink::common::msg::MAG\_CAL\_REPORT, 431
  - deserialize, 432
  - get\_name, 433
  - serialize, 433
- mavlink::common::msg::MANUAL\_CONTROL, 433
  - deserialize, 435
  - get\_name, 435
  - serialize, 436
- mavlink::common::msg::MANUAL\_SETPOINT, 436
  - deserialize, 438
  - get\_name, 438
  - serialize, 438
- mavlink::common::msg::MEMORY\_VECT, 439
  - deserialize, 441
  - get\_name, 441
  - serialize, 441
- mavlink::common::msg::MESSAGE\_INTERVAL, 449
  - get\_name, 451
  - serialize, 451
- mavlink::common::msg::MISSION\_ACK, 454
  - deserialize, 456
  - get\_name, 456
  - serialize, 456
- mavlink::common::msg::MISSION\_CLEAR\_ALL, 457
  - deserialize, 458
  - get\_name, 458
  - serialize, 458
- mavlink::common::msg::MISSION\_COUNT, 459
  - deserialize, 460
  - get\_name, 461
  - serialize, 461
- mavlink::common::msg::MISSION\_CURRENT, 461
  - deserialize, 463
  - get\_name, 463
  - serialize, 464
- mavlink::common::msg::MISSION\_ITEM, 464
  - deserialize, 466
  - get\_name, 466
  - serialize, 466
- mavlink::common::msg::MISSION\_ITEM\_INT, 467
  - deserialize, 469
  - get\_name, 469
  - serialize, 469



- mavlink::common::msg::MISSION\_ITEM\_REACHED, 470
  - deserialize, 471
  - get\_name, 471
  - serialize, 471
- mavlink::common::msg::MISSION\_REQUEST, 472
  - deserialize, 473
  - get\_name, 474
  - serialize, 474
- mavlink::common::msg::MISSION\_REQUEST\_INT, 474
  - deserialize, 476
  - get\_name, 476
  - serialize, 476
- mavlink::common::msg::MISSION\_REQUEST\_LIST, 478
  - deserialize, 479
  - get\_name, 480
  - serialize, 480
- mavlink::common::msg::MISSION\_REQUEST\_PARTIAL\_LIST, 480
  - deserialize, 482
  - get\_name, 482
  - serialize, 482
- mavlink::common::msg::MISSION\_SET\_CURRENT, 483
  - deserialize, 484
  - get\_name, 485
  - serialize, 485
- mavlink::common::msg::MISSION\_WRITE\_PARTIAL\_LIST, 485
  - deserialize, 487
  - get\_name, 487
  - serialize, 487
- mavlink::common::msg::MOUNT\_ORIENTATION, 488
  - deserialize, 489
  - get\_name, 490
  - serialize, 490
- mavlink::common::msg::NAMED\_VALUE\_FLOAT, 491
  - deserialize, 493
  - get\_name, 493
  - serialize, 493
- mavlink::common::msg::NAMED\_VALUE\_INT, 494
  - deserialize, 495
  - get\_name, 495
  - serialize, 495
- mavlink::common::msg::NAV\_CONTROLLER\_OUTPUT, 496
  - deserialize, 498
  - get\_name, 498
  - serialize, 498
- mavlink::common::msg::OBSTACLE\_DISTANCE, 499
  - deserialize, 500
  - get\_name, 501
  - serialize, 501
- mavlink::common::msg::ODOMETRY, 501
  - deserialize, 503
  - get\_name, 504
  - serialize, 504
- mavlink::common::msg::ONBOARD\_COMPUTER\_STATUS, 504
  - deserialize, 506
  - get\_name, 507
  - serialize, 507
- mavlink::common::msg::OPEN\_DRONE\_ID\_ARM\_STATUS, 507
  - deserialize, 509
  - get\_name, 509
  - serialize, 509
- mavlink::common::msg::OPEN\_DRONE\_ID\_AUTHENTICATION, 510
  - deserialize, 511
  - get\_name, 512
  - serialize, 512
- mavlink::common::msg::OPEN\_DRONE\_ID\_BASIC\_ID, 512
  - deserialize, 514
  - get\_name, 514
  - serialize, 514
- mavlink::common::msg::OPEN\_DRONE\_ID\_LOCATION, 515
  - deserialize, 517
  - get\_name, 517
  - serialize, 517
- mavlink::common::msg::OPEN\_DRONE\_ID\_MESSAGE\_PACK, 518
  - deserialize, 519
  - get\_name, 520
  - serialize, 520
- mavlink::common::msg::OPEN\_DRONE\_ID\_OPERATOR\_ID, 520
  - deserialize, 522
  - get\_name, 522
  - serialize, 522
- mavlink::common::msg::OPEN\_DRONE\_ID\_SELF\_ID, 523
  - deserialize, 524
  - get\_name, 525
  - serialize, 525
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM, 525
  - deserialize, 527
  - get\_name, 527
  - serialize, 528
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM\_UPDATE, 528
  - deserialize, 530
  - get\_name, 530
  - serialize, 530
- mavlink::common::msg::OPTICAL\_FLOW, 531
  - deserialize, 532
  - get\_name, 533
  - serialize, 533
- mavlink::common::msg::OPTICAL\_FLOW\_RAD, 533
  - deserialize, 535
  - get\_name, 535
  - serialize, 536



- mavlink::common::msg::ORBIT\_EXECUTION\_STATUS,
  - 536
  - deserialize, 538
  - get\_name, 538
  - serialize, 538
- mavlink::common::msg::PARAM\_ERROR, 541
  - deserialize, 542
  - get\_name, 542
  - serialize, 543
- mavlink::common::msg::PARAM\_EXT\_ACK, 543
  - deserialize, 545
  - get\_name, 545
  - serialize, 545
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_LIST,
  - 546
  - deserialize, 547
  - get\_name, 547
  - serialize, 547
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_READ,
  - 548
  - deserialize, 550
  - get\_name, 550
  - serialize, 550
- mavlink::common::msg::PARAM\_EXT\_SET, 551
  - deserialize, 552
  - get\_name, 552
  - serialize, 553
- mavlink::common::msg::PARAM\_EXT\_VALUE, 553
  - deserialize, 555
  - get\_name, 555
  - serialize, 555
- mavlink::common::msg::PARAM\_MAP\_RC, 556
  - deserialize, 557
  - get\_name, 557
  - serialize, 558
- mavlink::common::msg::PARAM\_REQUEST\_LIST, 558
  - deserialize, 560
  - get\_name, 560
  - serialize, 560
- mavlink::common::msg::PARAM\_REQUEST\_READ,
  - 561
  - deserialize, 562
  - get\_name, 562
  - serialize, 563
- mavlink::common::msg::PARAM\_SET, 563
  - deserialize, 565
  - get\_name, 565
  - serialize, 565
- mavlink::common::msg::PARAM\_VALUE, 566
  - deserialize, 567
  - get\_name, 568
  - serialize, 568
- mavlink::common::msg::PING, 568
  - deserialize, 570
  - get\_name, 570
  - serialize, 570
- mavlink::common::msg::PLAY\_TUNE, 572
  - deserialize, 573
  - get\_name, 574
  - serialize, 574
- mavlink::common::msg::PLAY\_TUNE\_V2, 574
  - deserialize, 576
  - get\_name, 576
  - serialize, 576
- mavlink::common::msg::POSITION\_TARGET\_GLOBAL\_INT,
  - 577
  - deserialize, 578
  - get\_name, 579
  - serialize, 579
- mavlink::common::msg::POSITION\_TARGET\_LOCAL\_NED,
  - 579
  - deserialize, 581
  - get\_name, 581
  - serialize, 582
- mavlink::common::msg::POWER\_STATUS, 582
  - deserialize, 584
  - get\_name, 584
  - serialize, 584
- mavlink::common::msg::PROTOCOL\_VERSION, 585
  - deserialize, 586
  - get\_name, 586
  - serialize, 587
- mavlink::common::msg::RADIO\_STATUS, 587
  - deserialize, 589
  - get\_name, 589
  - serialize, 589
- mavlink::common::msg::RAW\_IMU, 590
  - deserialize, 591
  - get\_name, 592
  - serialize, 592
- mavlink::common::msg::RAW\_PRESSURE, 592
  - deserialize, 594
  - get\_name, 594
  - serialize, 594
- mavlink::common::msg::RAW\_RPM, 595
  - deserialize, 596
  - get\_name, 597
  - serialize, 597
- mavlink::common::msg::RC\_CHANNELS, 597
  - deserialize, 599
  - get\_name, 600
  - serialize, 600
- mavlink::common::msg::RC\_CHANNELS\_OVERRIDE,
  - 600
  - deserialize, 602
  - get\_name, 603
  - serialize, 603
- mavlink::common::msg::RC\_CHANNELS\_RAW, 603
  - deserialize, 605
  - get\_name, 605
  - serialize, 606
- mavlink::common::msg::RC\_CHANNELS\_SCALED,
  - 606
  - deserialize, 608
  - get\_name, 608
  - serialize, 608

mavlink::common::msg::RELAY\_STATUS, 609  
     deserialize, 610  
     get\_name, 611  
     serialize, 611  
 mavlink::common::msg::REQUEST\_DATA\_STREAM, 611  
     deserialize, 613  
     get\_name, 613  
     serialize, 613  
 mavlink::common::msg::REQUEST\_EVENT, 614  
     deserialize, 615  
     get\_name, 616  
     serialize, 616  
 mavlink::common::msg::RESOURCE\_REQUEST, 616  
     deserialize, 618  
     get\_name, 618  
     serialize, 618  
 mavlink::common::msg::RESPONSE\_EVENT\_ERROR, 619  
     deserialize, 620  
     get\_name, 621  
     serialize, 621  
 mavlink::common::msg::SAFETY\_ALLOWED\_AREA, 621  
     deserialize, 623  
     get\_name, 623  
     serialize, 623  
 mavlink::common::msg::SAFETY\_SET\_ALLOWED\_AREA, 624  
     deserialize, 626  
     get\_name, 626  
     serialize, 626  
 mavlink::common::msg::SCALED\_IMU, 627  
     deserialize, 628  
     get\_name, 629  
     serialize, 629  
 mavlink::common::msg::SCALED\_IMU2, 629  
     deserialize, 631  
     get\_name, 631  
     serialize, 632  
 mavlink::common::msg::SCALED\_IMU3, 632  
     deserialize, 634  
     get\_name, 634  
     serialize, 634  
 mavlink::common::msg::SCALED\_PRESSURE, 635  
     deserialize, 636  
     get\_name, 637  
     serialize, 637  
 mavlink::common::msg::SCALED\_PRESSURE2, 637  
     deserialize, 639  
     get\_name, 639  
     serialize, 639  
 mavlink::common::msg::SCALED\_PRESSURE3, 640  
     deserialize, 641  
     get\_name, 642  
     serialize, 642  
 mavlink::common::msg::SERIAL\_CONTROL, 642  
     deserialize, 644  
     get\_name, 644  
     serialize, 645  
 mavlink::common::msg::SERVO\_OUTPUT\_RAW, 645  
     deserialize, 647  
     get\_name, 647  
     serialize, 647  
 mavlink::common::msg::SET\_ACTUATOR\_CONTROL\_TARGET, 649  
     deserialize, 650  
     get\_name, 651  
     serialize, 651  
 mavlink::common::msg::SET\_ATTITUDE\_TARGET, 651  
     deserialize, 653  
     get\_name, 653  
     serialize, 654  
 mavlink::common::msg::SET\_GPS\_GLOBAL\_ORIGIN, 654  
     deserialize, 656  
     get\_name, 656  
     serialize, 656  
 mavlink::common::msg::SET\_HOME\_POSITION, 657  
     deserialize, 658  
     get\_name, 659  
     serialize, 659  
 mavlink::common::msg::SET\_MODE, 659  
     deserialize, 661  
     get\_name, 661  
     serialize, 661  
 mavlink::common::msg::SET\_POSITION\_TARGET\_GLOBAL\_INT, 662  
     deserialize, 663  
     get\_name, 665  
     serialize, 665  
 mavlink::common::msg::SET\_POSITION\_TARGET\_LOCAL\_NED, 665  
     deserialize, 667  
     get\_name, 668  
     serialize, 668  
 mavlink::common::msg::SETUP\_SIGNING, 668  
     deserialize, 670  
     get\_name, 670  
     serialize, 670  
 mavlink::common::msg::SIM\_STATE, 671  
     deserialize, 673  
     get\_name, 673  
     serialize, 673  
 mavlink::common::msg::SMART\_BATTERY\_INFO, 674  
     deserialize, 675  
     get\_name, 676  
     serialize, 676  
 mavlink::common::msg::STATUSTEXT, 676  
     deserialize, 678  
     get\_name, 678  
     serialize, 678  
 mavlink::common::msg::STORAGE\_INFORMATION, 680  
     deserialize, 681  
     get\_name, 682

- serialize, 682
- mavlink::common::msg::SUPPORTED\_TUNES, 682
  - deserialize, 684
  - get\_name, 684
  - serialize, 684
- mavlink::common::msg::SYS\_STATUS, 685
  - deserialize, 686
  - get\_name, 688
  - serialize, 688
- mavlink::common::msg::SYSTEM\_TIME, 688
  - deserialize, 690
  - get\_name, 690
  - serialize, 690
- mavlink::common::msg::TERRAIN\_CHECK, 691
  - deserialize, 692
  - get\_name, 693
  - serialize, 693
- mavlink::common::msg::TERRAIN\_DATA, 693
  - deserialize, 695
  - get\_name, 695
  - serialize, 695
- mavlink::common::msg::TERRAIN\_REPORT, 696
  - deserialize, 697
  - get\_name, 698
  - serialize, 698
- mavlink::common::msg::TERRAIN\_REQUEST, 698
  - deserialize, 700
  - get\_name, 700
  - serialize, 700
- mavlink::common::msg::TIME\_ESTIMATE\_TO\_TARGET, 701
  - deserialize, 703
  - get\_name, 703
  - serialize, 703
- mavlink::common::msg::TIMESYNC, 704
  - deserialize, 705
  - get\_name, 705
  - serialize, 706
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_Bezier, 706
  - deserialize, 708
  - get\_name, 708
  - serialize, 708
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_WAYPOINTS, 709
  - deserialize, 710
  - get\_name, 711
  - serialize, 711
- mavlink::common::msg::TUNNEL, 711
  - deserialize, 713
  - get\_name, 713
  - serialize, 713
- mavlink::common::msg::UAVCAN\_NODE\_INFO, 715
  - deserialize, 716
  - get\_name, 717
  - serialize, 717
- mavlink::common::msg::UAVCAN\_NODE\_STATUS, 717
  - deserialize, 719
- get\_name, 719
- serialize, 719
- mavlink::common::msg::UTM\_GLOBAL\_POSITION, 722
  - deserialize, 724
  - get\_name, 724
  - serialize, 724
- mavlink::common::msg::V2\_EXTENSION, 725
  - deserialize, 726
  - get\_name, 727
  - serialize, 727
- mavlink::common::msg::VFR\_HUD, 727
  - deserialize, 729
  - get\_name, 729
  - serialize, 729
- mavlink::common::msg::VIBRATION, 730
  - deserialize, 731
  - get\_name, 732
  - serialize, 732
- mavlink::common::msg::VICON\_POSITION\_ESTIMATE, 732
  - deserialize, 734
  - get\_name, 734
  - serialize, 734
- mavlink::common::msg::VIDEO\_STREAM\_INFORMATION, 735
  - deserialize, 737
  - get\_name, 737
  - serialize, 737
- mavlink::common::msg::VIDEO\_STREAM\_STATUS, 738
  - deserialize, 739
  - get\_name, 740
  - serialize, 740
- mavlink::common::msg::VISION\_POSITION\_ESTIMATE, 740
  - deserialize, 742
  - get\_name, 742
  - serialize, 743
- mavlink::common::msg::VISION\_SPEED\_ESTIMATE, 743
  - deserialize, 745
  - get\_name, 745
- mavlink::common::msg::WHEEL\_DISTANCE, 746
  - deserialize, 747
  - get\_name, 747
  - serialize, 747
- mavlink::common::msg::WIFI\_CONFIG\_AP, 748
  - deserialize, 749
  - get\_name, 750
  - serialize, 750
- mavlink::common::msg::WINCH\_STATUS, 750
  - deserialize, 752
  - get\_name, 752
  - serialize, 752
- mavlink::common::msg::WIND\_COV, 753
  - deserialize, 754

- get\_name, 755
- serialize, 755
- mavlink::impl::UIntBuffer< \_T >, 721
- mavlink::impl::UIntBufferHelper< 1 >, 721
- mavlink::impl::UIntBufferHelper< 2 >, 721
- mavlink::impl::UIntBufferHelper< 4 >, 722
- mavlink::impl::UIntBufferHelper< 8 >, 722
- mavlink::impl::UIntBufferHelper< \_N >, 721
- mavlink::Message, 442
  - deserialize, 444
  - get\_name, 446
  - serialize, 447
- mavlink::Message::Info, 392
- mavlink::minimal::msg::HEARTBEAT, 350
  - deserialize, 352
  - get\_name, 352
  - serialize, 352
- mavlink::MsgMap, 490
- mavlink::standard::msg::AUTOPILOT\_VERSION, 151
  - deserialize, 153
  - get\_name, 153
  - serialize, 154
- mavlink::standard::msg::GLOBAL\_POSITION\_INT, 317
  - deserialize, 318
  - get\_name, 319
  - serialize, 319
- mavlink\_conversions.h
  - mavlink\_dcm\_to\_euler, 851
  - mavlink\_dcm\_to\_quaternion, 851
  - mavlink\_euler\_to\_dcm, 852
  - mavlink\_euler\_to\_quaternion, 852
  - mavlink\_quaternion\_to\_dcm, 853
  - mavlink\_quaternion\_to\_euler, 853
- mavlink\_dcm\_to\_euler
  - mavlink\_conversions.h, 851
- mavlink\_dcm\_to\_quaternion
  - mavlink\_conversions.h, 851
- mavlink\_euler\_to\_dcm
  - mavlink\_conversions.h, 852
- mavlink\_euler\_to\_quaternion
  - mavlink\_conversions.h, 852
- mavlink\_quaternion\_to\_dcm
  - mavlink\_conversions.h, 853
- mavlink\_quaternion\_to\_euler
  - mavlink\_conversions.h, 853
- mavlink\_sha256\_ctx, 439
- message\_sender, 452
  - buffer\_arm\_disarm\_cmd, 453
  - buffer\_heartbeat, 453
  - buffer\_manual\_control, 453
  - message\_sender, 452
- minimal.h
  - HAVE\_ENUM\_MAV\_AUTOPILOT, 857
  - HAVE\_ENUM\_MAV\_COMPONENT, 858
  - HAVE\_ENUM\_MAV\_MODE\_FLAG\_DECODE\_POSITION, 858
  - HAVE\_ENUM\_MAV\_TYPE, 858
- minimal.hpp
  - MAV\_AUTOPILOT, 862
  - MAV\_COMPONENT, 862
  - MAV\_MODE\_FLAG\_DECODE\_POSITION, 862
  - MAV\_TYPE, 863
- MISSION\_STATE
  - common.hpp, 833
- packet\_receiver, 539
  - receive\_ack, 539
  - receive\_heartbeat, 539
  - receive\_radio\_status, 540
  - receive\_sys\_status, 540
  - receive\_timesync, 540
- packetHandler.cpp
  - heartbeat\_timed\_out, 883
  - setupSigning, 883
- packetHandler.h
  - heartbeat\_timed\_out, 872
  - setupSigning, 872
- PARACHUTE\_ACTION
  - common.hpp, 834
- POSITION\_TARGET\_TYEMASK
  - common.hpp, 834
- PREFLIGHT\_STORAGE\_MISSION\_ACTION
  - common.hpp, 834
- PREFLIGHT\_STORAGE\_PARAMETER\_ACTION
  - common.hpp, 834
- RC\_SUB\_TYPE
  - common.hpp, 834
- RC\_TYPE
  - common.hpp, 835
- receive\_ack
  - packet\_receiver, 539
- receive\_heartbeat
  - packet\_receiver, 539
- receive\_radio\_status
  - packet\_receiver, 540
- receive\_sys\_status
  - packet\_receiver, 540
- receive\_timesync
  - packet\_receiver, 540
- serialize
  - mavlink::common::msg::ACTUATOR\_CONTROL\_TARGET, 118
  - mavlink::common::msg::ACTUATOR\_OUTPUT\_STATUS, 120
  - mavlink::common::msg::ADSB\_VEHICLE, 123
  - mavlink::common::msg::AIRSPEED, 125
  - mavlink::common::msg::AIS\_VESSEL, 129
  - mavlink::common::msg::ALTITUDE, 131
  - mavlink::common::msg::ATT\_POS\_MOCAP, 135
  - mavlink::common::msg::ATTITUDE, 138
  - mavlink::common::msg::ATTITUDE\_QUATERNION, 140
  - mavlink::common::msg::ATTITUDE\_QUATERNION\_COV, 143
  - mavlink::common::msg::ATTITUDE\_TARGET, 145

- mavlink::common::msg::AUTH\_KEY, [148](#)
- mavlink::common::msg::AUTOPILOT\_STATE\_FOR\_GIMBAL\_DEVICE, [151](#)
- mavlink::common::msg::AVAILABLE\_MODES, [156](#)
- mavlink::common::msg::AVAILABLE\_MODES\_MONITOR, [159](#)
- mavlink::common::msg::BATTERY\_INFO, [162](#)
- mavlink::common::msg::BATTERY\_STATUS, [165](#)
- mavlink::common::msg::BUTTON\_CHANGE, [167](#)
- mavlink::common::msg::CAMERA\_CAPTURE\_STATUS, [170](#)
- mavlink::common::msg::CAMERA\_FOV\_STATUS, [172](#)
- mavlink::common::msg::CAMERA\_IMAGE\_CAPTURED, [176](#)
- mavlink::common::msg::CAMERA\_INFORMATION, [179](#)
- mavlink::common::msg::CAMERA\_SETTINGS, [181](#)
- mavlink::common::msg::CAMERA\_THERMAL\_RANGE, [184](#)
- mavlink::common::msg::CAMERA\_TRACKING\_GEO\_STATUS, [186](#)
- mavlink::common::msg::CAMERA\_TRACKING\_IMAGE\_STATUS, [189](#)
- mavlink::common::msg::CAMERA\_TRIGGER, [191](#)
- mavlink::common::msg::CAN\_FILTER\_MODIFY, [194](#)
- mavlink::common::msg::CAN\_FRAME, [196](#)
- mavlink::common::msg::CANFD\_FRAME, [199](#)
- mavlink::common::msg::CELLULAR\_CONFIG, [202](#)
- mavlink::common::msg::CELLULAR\_STATUS, [204](#)
- mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL, [207](#)
- mavlink::common::msg::CHANGE\_OPERATOR\_CONTROL\_ACK, [209](#)
- mavlink::common::msg::COLLISION, [212](#)
- mavlink::common::msg::COMMAND\_ACK, [214](#)
- mavlink::common::msg::COMMAND\_CANCEL, [217](#)
- mavlink::common::msg::COMMAND\_INT, [220](#)
- mavlink::common::msg::COMMAND\_LONG, [223](#)
- mavlink::common::msg::COMPONENT\_INFORMATION, [225](#)
- mavlink::common::msg::COMPONENT\_INFORMATION\_BASIC, [228](#)
- mavlink::common::msg::COMPONENT\_METADATA, [230](#)
- mavlink::common::msg::CONTROL\_SYSTEM\_STATE, [233](#)
- mavlink::common::msg::CURRENT\_EVENT\_SEQUENCE, [235](#)
- mavlink::common::msg::CURRENT\_MODE, [238](#)
- mavlink::common::msg::DATA\_STREAM, [240](#)
- mavlink::common::msg::DATA\_TRANSMISSION\_HANDSHAKE, [243](#)
- mavlink::common::msg::DEBUG, [245](#)
- mavlink::common::msg::DEBUG\_FLOAT\_ARRAY, [248](#)
- mavlink::common::msg::DEBUG\_VECT, [250](#)
- mavlink::common::msg::DISTANCE\_SENSOR, [253](#)
- mavlink::common::msg::EFI\_STATUS, [256](#)
- mavlink::common::msg::ENCAPSULATED\_DATA, [258](#)
- mavlink::common::msg::ESC\_INFO, [261](#)
- mavlink::common::msg::ESC\_STATUS, [263](#)
- mavlink::common::msg::ESTIMATOR\_STATUS, [268](#)
- mavlink::common::msg::EVENT, [271](#)
- mavlink::common::msg::EXTENDED\_SYS\_STATE, [273](#)
- mavlink::common::msg::FENCE\_STATUS, [276](#)
- mavlink::common::msg::FIGURE\_EIGHT\_EXECUTION\_STATUS, [278](#)
- mavlink::common::msg::FILE\_TRANSFER\_PROTOCOL, [281](#)
- mavlink::common::msg::FLIGHT\_INFORMATION, [283](#)
- mavlink::common::msg::FOLLOW\_TARGET, [287](#)
- mavlink::common::msg::FUEL\_STATUS, [290](#)
- mavlink::common::msg::GENERATOR\_STATUS, [293](#)
- mavlink::common::msg::GIMBAL\_DEVICE\_ATTITUDE\_STATUS, [295](#)
- mavlink::common::msg::GIMBAL\_DEVICE\_INFORMATION, [298](#)
- mavlink::common::msg::GIMBAL\_DEVICE\_SET\_ATTITUDE, [303](#)
- mavlink::common::msg::GIMBAL\_MANAGER\_INFORMATION, [306](#)
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_ATTITUDE, [308](#)
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_MANUAL\_CONTROL, [311](#)
- mavlink::common::msg::GIMBAL\_MANAGER\_SET\_PITCHYAW, [314](#)
- mavlink::common::msg::GIMBAL\_MANAGER\_STATUS, [316](#)
- mavlink::common::msg::GLOBAL\_POSITION\_INT\_COV, [322](#)
- mavlink::common::msg::GLOBAL\_VISION\_POSITION\_ESTIMATE, [324](#)
- mavlink::common::msg::GPS2\_RAW, [327](#)
- mavlink::common::msg::GPS2\_RTK, [330](#)
- mavlink::common::msg::GPS\_GLOBAL\_ORIGIN, [332](#)
- mavlink::common::msg::GPS\_INJECT\_DATA, [334](#)
- mavlink::common::msg::GPS\_INPUT, [337](#)
- mavlink::common::msg::GPS\_RAW\_INT, [340](#)
- mavlink::common::msg::GPS\_RTCM\_DATA, [342](#)
- mavlink::common::msg::GPS\_RTK, [345](#)
- mavlink::common::msg::GPS\_STATUS, [347](#)
- mavlink::common::msg::HIGH\_LATENCY, [355](#)
- mavlink::common::msg::HIGH\_LATENCY2, [358](#)

- mavlink::common::msg::HIGHRES\_IMU, [361](#)
- mavlink::common::msg::HIL\_ACTUATOR\_CONTROLS, [363](#)
- mavlink::common::msg::HIL\_CONTROLS, [366](#)
- mavlink::common::msg::HIL\_GPS, [368](#)
- mavlink::common::msg::HIL\_OPTICAL\_FLOW, [372](#)
- mavlink::common::msg::HIL\_RC\_INPUTS\_RAW, [375](#)
- mavlink::common::msg::HIL\_SENSOR, [377](#)
- mavlink::common::msg::HIL\_STATE, [381](#)
- mavlink::common::msg::HIL\_STATE\_QUATERNION, [384](#)
- mavlink::common::msg::HOME\_POSITION, [386](#)
- mavlink::common::msg::HYGROMETER\_SENSOR, [389](#)
- mavlink::common::msg::ILLUMINATOR\_STATUS, [392](#)
- mavlink::common::msg::ISBD\_LINK\_STATUS, [394](#)
- mavlink::common::msg::LANDING\_TARGET, [397](#)
- mavlink::common::msg::LINK\_NODE\_STATUS, [400](#)
- mavlink::common::msg::LOCAL\_POSITION\_NED, [402](#)
- mavlink::common::msg::LOCAL\_POSITION\_NED\_COV, [405](#)
- mavlink::common::msg::LOCAL\_POSITION\_NED\_SYSTEM\_OFFSET, [408](#)
- mavlink::common::msg::LOG\_DATA, [410](#)
- mavlink::common::msg::LOG\_ENTRY, [413](#)
- mavlink::common::msg::LOG\_ERASE, [415](#)
- mavlink::common::msg::LOG\_REQUEST\_DATA, [418](#)
- mavlink::common::msg::LOG\_REQUEST\_END, [420](#)
- mavlink::common::msg::LOG\_REQUEST\_LIST, [423](#)
- mavlink::common::msg::LOGGING\_ACK, [425](#)
- mavlink::common::msg::LOGGING\_DATA, [428](#)
- mavlink::common::msg::LOGGING\_DATA\_ACKED, [430](#)
- mavlink::common::msg::MAG\_CAL\_REPORT, [433](#)
- mavlink::common::msg::MANUAL\_CONTROL, [436](#)
- mavlink::common::msg::MANUAL\_SETPOINT, [438](#)
- mavlink::common::msg::MEMORY\_VECT, [441](#)
- mavlink::common::msg::MESSAGE\_INTERVAL, [451](#)
- mavlink::common::msg::MISSION\_ACK, [456](#)
- mavlink::common::msg::MISSION\_CLEAR\_ALL, [458](#)
- mavlink::common::msg::MISSION\_COUNT, [461](#)
- mavlink::common::msg::MISSION\_CURRENT, [464](#)
- mavlink::common::msg::MISSION\_ITEM, [466](#)
- mavlink::common::msg::MISSION\_ITEM\_INT, [469](#)
- mavlink::common::msg::MISSION\_ITEM\_REACHED, [471](#)
- mavlink::common::msg::MISSION\_REQUEST, [474](#)
- mavlink::common::msg::MISSION\_REQUEST\_INT, [476](#)
- mavlink::common::msg::MISSION\_REQUEST\_LIST, [480](#)
- mavlink::common::msg::MISSION\_REQUEST\_PARTIAL\_LIST, [482](#)
- mavlink::common::msg::MISSION\_SET\_CURRENT, [485](#)
- mavlink::common::msg::MISSION\_WRITE\_PARTIAL\_LIST, [487](#)
- mavlink::common::msg::MOUNT\_ORIENTATION, [490](#)
- mavlink::common::msg::NAMED\_VALUE\_FLOAT, [493](#)
- mavlink::common::msg::NAMED\_VALUE\_INT, [495](#)
- mavlink::common::msg::NAV\_CONTROLLER\_OUTPUT, [498](#)
- mavlink::common::msg::OBSTACLE\_DISTANCE, [501](#)
- mavlink::common::msg::ODOMETRY, [504](#)
- mavlink::common::msg::ONBOARD\_COMPUTER\_STATUS, [507](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_ARM\_STATUS, [509](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_AUTHENTICATION, [512](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_BASIC\_ID, [514](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_LOCATION, [517](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_MESSAGE\_PACK, [520](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_OPERATOR\_ID, [522](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SELF\_ID, [525](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM, [528](#)
- mavlink::common::msg::OPEN\_DRONE\_ID\_SYSTEM\_UPDATE, [530](#)
- mavlink::common::msg::OPTICAL\_FLOW, [533](#)
- mavlink::common::msg::OPTICAL\_FLOW\_RAD, [536](#)
- mavlink::common::msg::ORBIT\_EXECUTION\_STATUS, [538](#)
- mavlink::common::msg::PARAM\_ERROR, [543](#)
- mavlink::common::msg::PARAM\_EXT\_ACK, [545](#)
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_LIST, [547](#)
- mavlink::common::msg::PARAM\_EXT\_REQUEST\_READ, [550](#)
- mavlink::common::msg::PARAM\_EXT\_SET, [553](#)
- mavlink::common::msg::PARAM\_EXT\_VALUE, [555](#)
- mavlink::common::msg::PARAM\_MAP\_RC, [558](#)
- mavlink::common::msg::PARAM\_REQUEST\_LIST, [561](#)



- 560
- mavlink::common::msg::PARAM\_REQUEST\_READ, 563
- mavlink::common::msg::PARAM\_SET, 565
- mavlink::common::msg::PARAM\_VALUE, 568
- mavlink::common::msg::PING, 570
- mavlink::common::msg::PLAY\_TUNE, 574
- mavlink::common::msg::PLAY\_TUNE\_V2, 576
- mavlink::common::msg::POSITION\_TARGET\_GLOBAL\_INT, 579
- mavlink::common::msg::POSITION\_TARGET\_LOCAL\_NED, 582
- mavlink::common::msg::POWER\_STATUS, 584
- mavlink::common::msg::PROTOCOL\_VERSION, 587
- mavlink::common::msg::RADIO\_STATUS, 589
- mavlink::common::msg::RAW\_IMU, 592
- mavlink::common::msg::RAW\_PRESSURE, 594
- mavlink::common::msg::RAW\_RPM, 597
- mavlink::common::msg::RC\_CHANNELS, 600
- mavlink::common::msg::RC\_CHANNELS\_OVERRIDE, 603
- mavlink::common::msg::RC\_CHANNELS\_RAW, 606
- mavlink::common::msg::RC\_CHANNELS\_SCALED, 608
- mavlink::common::msg::RELAY\_STATUS, 611
- mavlink::common::msg::REQUEST\_DATA\_STREAM, 613
- mavlink::common::msg::REQUEST\_EVENT, 616
- mavlink::common::msg::RESOURCE\_REQUEST, 618
- mavlink::common::msg::RESPONSE\_EVENT\_ERROR, 621
- mavlink::common::msg::SAFETY\_ALLOWED\_AREA, 623
- mavlink::common::msg::SAFETY\_SET\_ALLOWED\_AREA, 626
- mavlink::common::msg::SCALED\_IMU, 629
- mavlink::common::msg::SCALED\_IMU2, 632
- mavlink::common::msg::SCALED\_IMU3, 634
- mavlink::common::msg::SCALED\_PRESSURE, 637
- mavlink::common::msg::SCALED\_PRESSURE2, 639
- mavlink::common::msg::SCALED\_PRESSURE3, 642
- mavlink::common::msg::SERIAL\_CONTROL, 645
- mavlink::common::msg::SERVO\_OUTPUT\_RAW, 647
- mavlink::common::msg::SET\_ACTUATOR\_CONTROL\_TARGET, 651
- mavlink::common::msg::SET\_ATTITUDE\_TARGET, 654
- mavlink::common::msg::SET\_GPS\_GLOBAL\_ORIGIN, 656
- mavlink::common::msg::SET\_HOME\_POSITION, 659
- mavlink::common::msg::SET\_MODE, 661
- mavlink::common::msg::SET\_POSITION\_TARGET\_GLOBAL\_INT, 665
- mavlink::common::msg::SET\_POSITION\_TARGET\_LOCAL\_NED, 668
- mavlink::common::msg::SETUP\_SIGNING, 670
- mavlink::common::msg::SIM\_STATE, 673
- mavlink::common::msg::SMART\_BATTERY\_INFO, 676
- mavlink::common::msg::STATUSTEXT, 678
- mavlink::common::msg::STORAGE\_INFORMATION, 682
- mavlink::common::msg::SUPPORTED\_TUNES, 684
- mavlink::common::msg::SYS\_STATUS, 688
- mavlink::common::msg::SYSTEM\_TIME, 690
- mavlink::common::msg::TERRAIN\_CHECK, 693
- mavlink::common::msg::TERRAIN\_DATA, 695
- mavlink::common::msg::TERRAIN\_REPORT, 698
- mavlink::common::msg::TERRAIN\_REQUEST, 700
- mavlink::common::msg::TIME\_ESTIMATE\_TO\_TARGET, 703
- mavlink::common::msg::TIMESYNC, 706
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_BEZIER, 708
- mavlink::common::msg::TRAJECTORY\_REPRESENTATION\_WAYPOINTS, 711
- mavlink::common::msg::TUNNEL, 713
- mavlink::common::msg::UAVCAN\_NODE\_INFO, 717
- mavlink::common::msg::UAVCAN\_NODE\_STATUS, 719
- mavlink::common::msg::UTM\_GLOBAL\_POSITION, 724
- mavlink::common::msg::V2\_EXTENSION, 727
- mavlink::common::msg::VFR\_HUD, 729
- mavlink::common::msg::VIBRATION, 732
- mavlink::common::msg::VICON\_POSITION\_ESTIMATE, 734
- mavlink::common::msg::VIDEO\_STREAM\_INFORMATION, 737
- mavlink::common::msg::VIDEO\_STREAM\_STATUS, 740
- mavlink::common::msg::VISION\_POSITION\_ESTIMATE, 743
- mavlink::common::msg::VISION\_SPEED\_ESTIMATE, 745
- mavlink::common::msg::WHEEL\_DISTANCE, 747
- mavlink::common::msg::WIFI\_CONFIG\_AP, 750
- mavlink::common::msg::WINCH\_STATUS, 752
- mavlink::common::msg::WIND\_COV, 755
- mavlink::Message, 447
- mavlink::minimal::msg::HEARTBEAT, 352
- mavlink::standard::msg::AUTOPILOT\_VERSION, 154
- mavlink::standard::msg::GLOBAL\_POSITION\_INT, 319

setUGV\_state  
     stateHandler.cpp, 886  
     stateHandler.hpp, 876  
 setupDisplay  
     displayHandler.hpp, 762  
     setupFunctions.cpp, 884  
     setupFunctions.h, 875  
 setupFunctions.cpp  
     identifyControllerDrift, 884  
     initMAVLink, 884  
     setupDisplay, 884  
 setupFunctions.h  
     identifyControllerDrift, 874  
     initMAVLink, 874  
     initUGV\_state, 874  
     setupDisplay, 875  
 setupSigning  
     packetHandler.cpp, 883  
     packetHandler.h, 872  
 single\_hand\_controller/displayHandler.cpp, 757  
 single\_hand\_controller/include/definitions.h, 759  
 single\_hand\_controller/include/displayHandler.hpp, 761  
 single\_hand\_controller/include/includes.h, 763  
 single\_hand\_controller/include/IOhandler.hpp, 763  
 single\_hand\_controller/include/mavlink/common/common.h, 766  
 single\_hand\_controller/include/mavlink/common/common.hpp, setUGV\_state, 886  
     803  
 single\_hand\_controller/include/mavlink/common/gtestsuite.hpp, setUGV\_state, 876  
     835  
 single\_hand\_controller/include/mavlink/common/mavlink.h, thumbstickControl, 701  
     841  
         timer.cpp  
 single\_hand\_controller/include/mavlink/common/testsuite.h, end\_OFP\_timer, 887  
     844  
         timeup, 887  
 single\_hand\_controller/include/mavlink/common/version.h, timer.hpp  
     846  
         end\_OFP\_timer, 878  
 single\_hand\_controller/include/mavlink/mavlink\_conversions.h, timeup, 878  
     849  
         timeup  
 single\_hand\_controller/include/mavlink/minimal/gtestsuite.hpp, timer.cpp, 887  
     840  
         timer.hpp, 878  
 single\_hand\_controller/include/mavlink/minimal/mavlink.h, updateDisplay  
     842  
         displayHandler.cpp, 758  
 single\_hand\_controller/include/mavlink/minimal/minimal.h, displayHandler.hpp, 762  
     853  
 single\_hand\_controller/include/mavlink/minimal/minimal.hpp,  
     859  
 single\_hand\_controller/include/mavlink/minimal/testsuite.h,  
     844  
 single\_hand\_controller/include/mavlink/minimal/version.h,  
     847  
 single\_hand\_controller/include/mavlink/standard/gtestsuite.hpp,  
     840  
 single\_hand\_controller/include/mavlink/standard/mavlink.h,  
     843  
 single\_hand\_controller/include/mavlink/standard/standard.h,  
     863  
 single\_hand\_controller/include/mavlink/standard/standard.hpp,  
     866  
 single\_hand\_controller/include/mavlink/standard/testsuite.h,  
     845  
 single\_hand\_controller/include/mavlink/standard/version.h,  
     848  
 single\_hand\_controller/include/message\_sender.hpp,  
     868  
 single\_hand\_controller/include/message\_structs.h, 869  
 single\_hand\_controller/include/packetHandler.h, 871  
 single\_hand\_controller/include/setupFunctions.h, 873  
 single\_hand\_controller/include/stateHandler.hpp, 875  
 single\_hand\_controller/include/timer.hpp, 877  
 single\_hand\_controller/IOhandler.cpp, 878  
 single\_hand\_controller/message\_sender.cpp, 880  
 single\_hand\_controller/packet\_receiver.cpp, 881  
 single\_hand\_controller/packetHandler.cpp, 881  
 single\_hand\_controller/setupFunctions.cpp, 883  
 single\_hand\_controller/stateHandler.cpp, 885  
 single\_hand\_controller/timer.cpp, 886  
 standard.h  
     HAVE\_ENUM\_FIRMWARE\_VERSION\_TYPE, 865  
     HAVE\_ENUM\_MAV\_PROTOCOL\_CAPABILITY,  
         865  
     standard.hpp  
         FIRMWARE\_VERSION\_TYPE, 867  
         MAV\_PROTOCOL\_CAPABILITY, 868  
     stateHandler.cpp  
     stateHandler.hpp  
     updateDisplay  
         displayHandler.cpp, 758  
         displayHandler.hpp, 762